

of CSF by air). Endoscopic choroid plexus coagulation was originally described in 1910 and was recently resurrected⁴

25.4 Endoscopic third ventriculostomy

25.4.1 Indications

Endoscopic third ventriculostomy (ETV) may be used in patients with obstructive HCP. May also be an option in managing shunt infection (as a means to remove all hardware without subjecting the patient to increased ICP). ETV has also been proposed as an option for patients who developed subdural hematomas after shunting (the shunt is removed before the ETV is performed). ETV may also be indicated for slit ventricle syndrome (p.424).

25.4.2 Contraindications

Communicating hydrocephalus has traditionally been considered a contraindication to ETV. However, it has been occasionally used for NPH.⁵ Relative contraindications to ETV would be the presence of any of the conditions associated with a low success rate (see below).

25.4.3 Complications

- hypothalamic injury: may result in hyperphagia
- injury to pituitary stalk or gland: may result in hormonal abnormalities including diabetes insipidus, amenorrhea
- transient 3rd and 6th nerve palsies
- injury to basilar artery, p-comm, or PCA: a fixed endoscope sheath seated just distal to the foramen of Monro within the third ventricle may allow for safe egress of blood extracranially)
- uncontrollable bleeding
- cardiac arrest⁶
- traumatic basilar artery aneurysm⁷: possibly related to thermal injury from use of laser in performing ETV

25.4.4 Technique

See dedicated section on technique (p.1517).

25.4.5 Success rate

Overall success rate is $\approx 56\%$ (range of 60–94% for nontumoral aqueductal stenosis⁷ (AqS)). Highest maintained patency rate is with previously untreated acquired AqS. Success rate in infants may be poor because they may not have a normally developed subarachnoid space. There is a low success rate (only $\approx 20\%$ of TVs will remain patent) if there is pre-existing pathology including:

1. tumor
2. previous shunt
3. previous SAH
4. previous whole brain radiation (success with focal stereotactic radiosurgery is not known)
5. significant adhesions visible when perforating through the floor of the third ventricle at the time of performance of ETV

The ETV Success Score (see □ Table 25.1)^{8,9} is a validated^{10,11} means of predicting the likelihood of success of ETV and therefore may assist in selecting appropriate patients for the procedure.

Total of the 3 scores (1 from each category: age, etiology and shunt history) expressed as a percent is the approximate chance of an ETV lasting 6 months without failure. Scores $<40\%$ correlated with a very low chance of success. Scores $>80\%$ correlated with a better chance of success compared to shunting from the outset.

Intermediate scores (50-70%) ETV had a higher initial failure rate compared to shunting, but after 3-6 months the balance shifted in favor of ETV.⁹

In one series, clinical improvement after ETV was achieved in 76% (72 of 95 patients), including 6 patients requiring second ETVs (three of which had partially functioning shunts that were left in place at the time of ETV).

Table 25.1 ETV Success Score			
Category	Description	Value	Score
Age	<1 month	0%	____%
	1 to <6 months	10%	
	6 months to <1 year	30%	
	1 to >10 years	40%	
	≥10 years	50%	
Etiology	• post infectious	0%	____%
	• myelomeningocele • post IVH • non-tectal brain tumor	20%	
	• aqueductal stenosis • tectal tumor • other	30%	
Shunt history	• previous shunt	0%	____%
	• no previous shunt	10%	
		Total (range 0-90%)	____%

25.5 Shunts

25.5.1 Types of shunts

1. ventriculoperitoneal (VP) shunt:
 - a) most commonly used shunt in modern era
 - b) lateral ventricle is the usual proximal location
 - c) intraperitoneal pressure: normal is near atmospheric
2. ventriculo-atrial (VA) shunt (“vascular shunt”):
 - a) shunts ventricles through jugular vein to superior vena cava, so-called “ventriculo-atrial” shunt because it shunts the cerebral ventricles to the vascular system with the catheter tip in the region of the right cardiac atrium)
 - b) treatment of choice when abdominal abnormalities are present (extensive abdominal surgery, peritonitis, morbid obesity, in preemies who have had NEC and may not tolerate VP shunt...)
 - c) shorter length of tubing results in lower distal pressure and less siphon effect than VP shunt, however pulsatile pressures may alter CSF hydrodynamics
3. Torkildsen shunt:
 - a) shunts ventricle to cisternal space
 - b) rarely used
 - c) effective only in acquired obstructive HCP, as patients with congenital HCP frequently do not develop normal subarachnoid CSF pathways
4. miscellaneous: various distal projections used historically or in patients who have had significant problems with traditional shunt locations (e.g. peritonitis with VP shunt, SBE with vascular shunts):
 - a) pleural space (ventriculopleural shunt): not a first choice, but a viable alternative if the peritoneum is not available.¹² To avoid symptomatic hydrothorax necessitating relocating distal end, it is recommended only for patients >7 yrs age (although some feel that these may be placed as young as 2 yrs age, and that hydrothorax is primarily a sign of infection regardless of age). Pressure in pleural space is less than atmospheric
 - b) gall bladder
 - c) ureter or bladder: causes electrolyte imbalances due to losses through urine
5. lumboperitoneal (LP) shunt; see insertion technique (p.1517)
 - a) only for communicating HCP: primarily pseudotumor cerebri or CSF fistula.¹³ Useful in situations with small ventricles
 - b) over age 2 yrs, percutaneous insertion with Tuohy needle is preferred

6. cyst or subdural shunt: from arachnoid cyst or subdural hygroma cavity, usually to peritoneum

25.5.2 Disadvantages/complications of various shunts

Complications that may occur with any shunt

1. obstruction: the most common cause of shunt malfunction
 - a) proximal: ventricular catheter (the most common site)
 - b) valve mechanism
 - c) distal: reported incidence of 12–34%¹⁴ Occurs in peritoneal catheter in VP shunt (see below), in atrial catheter in VA shunt
2. disconnection at a junction, or break at any point
3. infection: may produce obstruction
4. hardware erosion through skin, usually only in debilitated patients (especially preemies with enlarged heads and thin scalp from chronic HCP, who lay on one side of head due to elongated cranium). May also indicate silicone allergy (see below)
5. seizures (ventricular shunts only): there is $\approx 5.5\%$ risk of seizures in the first year after placement of a shunt which drops to $\approx 1.1\%$ after the 3rd year¹⁵ (NB: this does not mean that the shunt was the cause of all of these seizures). Seizure risk is questionably higher with frontal catheters than with parieto-occipital
6. act as a conduit for extraneural metastases of certain tumors (e.g. medulloblastoma). This is probably a relatively low risk¹⁶
7. silicone allergy¹⁷: rare (if it occurs at all). May resemble shunt infection with skin breakdown and fungating granulomas. CSF is initially sterile but later infections may occur. May require fabrication of a custom silicone-free device (e.g. polyurethane)

Disadvantages/complications with VP shunt

1. inguinal hernia: incidence = 17% (many shunts are inserted while processus vaginalis is patent)¹⁸
2. need to lengthen catheter with growth: may be obviated by using long peritoneal catheter (p.1515)
3. obstruction of peritoneal catheter:
 - a) may be more likely with distal slit openings (“slit valves”) due to occlusion by omentum or by trapping debris from the shunt system¹⁴
 - b) by peritoneal cyst (or pseudocyst)¹⁹: usually associated with infection, may also be due to reaction to talc from surgical gloves (the omentum tends to “wall off” a nidus of irritation). It may rarely be necessary to differentiate a CSF collection from a urine collection in patients with overdistended bladders that have ruptured (e.g. secondary to neurogenic bladder). Fluid can be aspirated percutaneously and analyzed for BUN and creatinine (which should be absent in CSF)
 - c) severe peritoneal adhesions: reduce surface area for CSF resorption
 - d) malposition of catheter tip:
 - at time of surgery: e.g. in preperitoneal fat
 - tubing may pull out of peritoneal cavity with growth
4. peritonitis from shunt infection
5. hydrocele
6. CSF ascites
7. tip migration
 - a) into scrotum²⁰
 - b) perforation of a viscus²¹: stomach,²² bladder... More common with older spring-reinforced (Raimondi) shunt tubing
 - c) through the diaphragm²³
8. intestinal obstruction (as opposed to perforation): rare
9. volvulus²⁴
10. intestinal strangulation: occurred only in patients in whom attempt was made to remove peritoneal tubing using traction on the catheter applied at the cephalad incision with subsequent breakage of the tubing leaving a residual intraabdominal segment (immediate peritoneal exploration is recommended under these circumstances)²⁵
11. overshunting (p.424): more likely than with VA shunt. Some recommend LP shunt for communicating hydrocephalus

Disadvantages/complications with VA shunt

1. requires repeated lengthening in growing child
2. higher risk of infection, septicemia
3. possible retrograde flow of blood into ventricles if valve malfunctions (rare)
4. shunt embolus
5. vascular complications: perforation, thrombophlebitis, pulmonary micro-emboli may cause pulmonary hypertension²⁶ (incidence $\approx 0.3\%$)

Disadvantages/complications with LP shunt

1. if at all possible, should not be used in growing child unless ventricular access is unavailable (e.g. due to slit ventricles) because of:
 - a) laminectomy in children causes scoliosis in 14%²⁷
 - b) risk of progressive cerebellar tonsillar herniation (Chiari I malformation)²⁸ in up to 70% of cases^{29,30}
2. overshunting harder to control when it occurs (a special horizontal-vertical (H-V) valve increases resistance when upright, see below)
3. difficult access to proximal end for revision or assessment of patency; see Lumboperitoneal (LP) shunt evaluation (p.1518).
4. lumbar nerve root irritation (radiculopathy)
5. leakage of CSF around catheter
6. pressure regulation is difficult
7. bilateral 6th and even 7th cranial nerve dysfunction from overshunting
8. high incidence of arachnoiditis and adhesions

Programmable shunt valves

A number of externally programmable shunts available in the U.S., including:

- Strata by Medtronic (p.429)
- Polaris by Sophysa (p.431)
- Codman Hakim (p.429)
- Certas Plus by Codman
- proGav by Aesculap

All are programmed externally with a magnet, and can potentially be inadvertently reprogrammed by external magnetic fields including those encountered during an MRI (the Polaris valve and the Certas Plus valve are promoted as being less susceptible to inadvertent reprogramming. □ Therefore, valve settings should be rechecked after an MRI scan performed for any reason, or if there is ever a concern about shunt function. The pressure setting on all of these valves can be checked on a plain x-ray taken perpendicular to the shunt valve (see □ Fig. 25.1 to identify the programmable valve type, then see the corresponding section for that valve to determine the shunt pressure setting). Some can also be checked using a special hand-held compass-like device provided by the manufacturer to most hospitals and clinics that deal with their valves.

In all systems on the market, increasing the programmed number results in higher valve opening pressures and therefore less CSF drainage at any given CSF pressure.

X-ray appearance of some shunt valves

□ Fig. 25.1 depicts idealized x-ray appearances of some common shunts. It is intended to help differentiate shunt systems on x-ray, and is not to scale. Appearance may vary with orientation relative to the x-ray beam. Manufacturer's diagrams of these shunts appears in section 25.7.

Miscellaneous shunt hardware

1. tumor filter: used to prevent peritoneal or vascular seeding in tumors that may metastasize through CSF (e.g. medulloblastoma,³¹ PNETs, ependymoma); may eventually become occluded by tumor cells and need replacement; may be able to radiate tumor filter to "sterilize" it. Not often used despite the fact that the risk of "shunt mets" appears to be low¹⁶
2. devices to avoid overdrainage when the patient is upright
 - a) antisiphon devices (ASD): prevents siphoning effect when patient is erect. Some valves have ASDs integrated into the valve. ASDs always increase the resistance of the shunt
 - b) "horizontal-vertical valve" (H-V valve) (p.433) used primarily with LP shunts

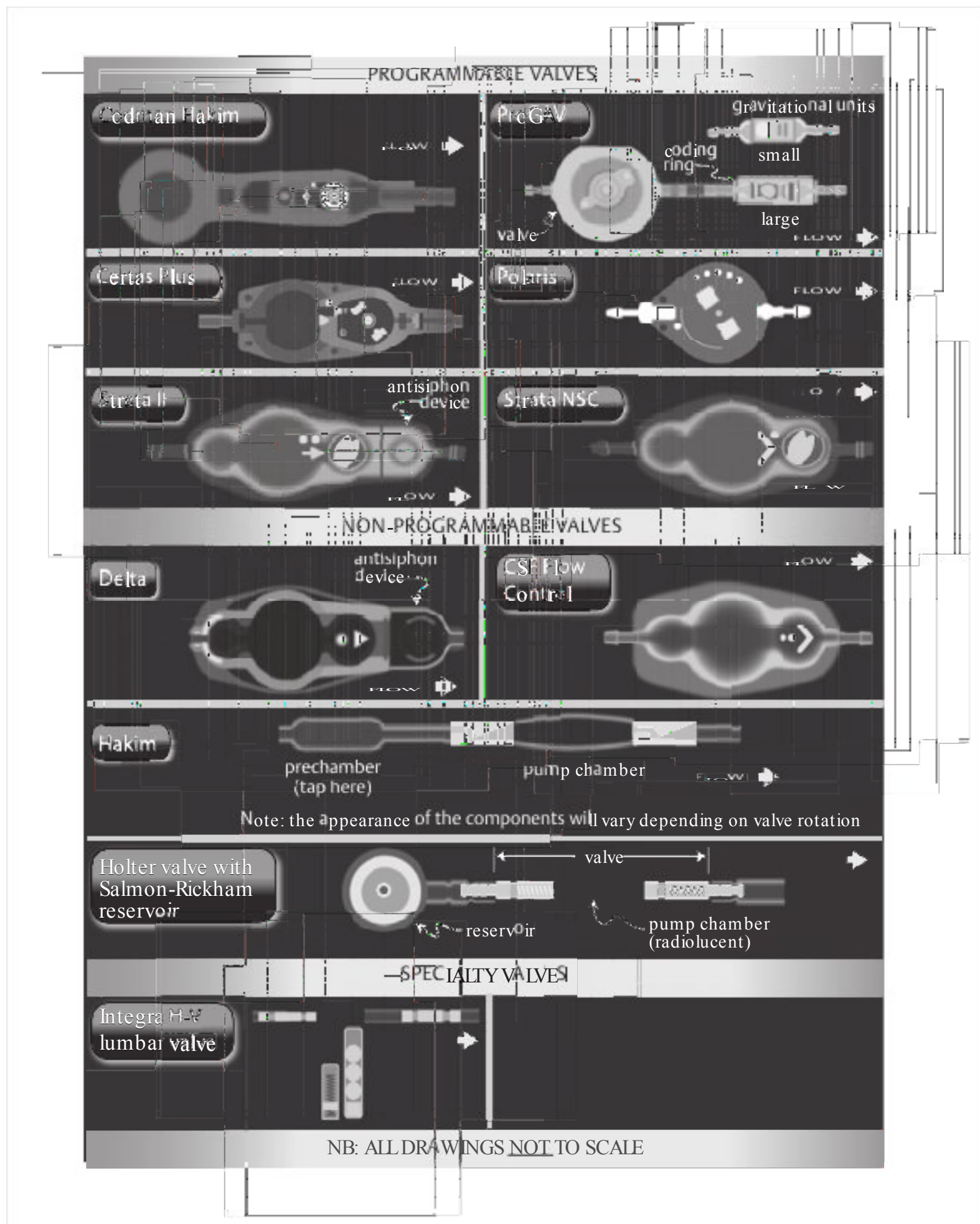


Fig. 25.1 X-ray appearance of common shunts

For x-ray appearance of programmable valves and the corresponding pressures, see the individual valve.

3. with a standard 90 cm long 1.2 mm inner diameter distal catheter, the hydrodynamic resistance of most shunts increases by 2–2.5 mm Hg/(ml/min)³²

25.6 Shunt problems

25.6.1 Risks associated with shunt insertion

1. intraparenchymal or intraventricular hemorrhage: risk $\approx$ 4% (in the absence of coagulopathy³³)
2. seizures

3. malposition
 - a) of ventricular catheter
 - b) of distal catheter
4. infection

25.6.2 Problems in patients with established CSF shunt

Shunt “problems” usually involve one or more of the following (undershunting and infection account for most common shunt problems):

1. undershunting (see below): obstruction (rate: $\approx 10\%$ per year), breakage...
2. infection (p.339): range 1–40% A serious complication. Often associated with obstruction.
3. Having a shunt infection decrease IQ
4. overshunting:
 - a) slit ventricle syndrome
 - b) subdural hematomas... (p.424)
 - c) “spinal headache”
5. seizures (p.417)
6. problems related to the distal catheter
 - a) peritoneal (p.417)
 - b) atrial (p.418)
7. skin breakdown over hardware (p.417): may indicate infection or silicone allergy

25.6.3 Evaluation of the patient with a shunt

History and physical

1. history directed at determining presence of shunt-related symptoms
 - a) acute symptoms of increased ICP
 - H/A: effect of posture, position, activity, migraine-like symptoms (visual aura...)
 - N/V
 - diplopia
 - lethargy
 - ataxia
 - infants: apnea and/or bradycardia; irritability; poor feeding
 - seizures: either new onset, or if pre-existing, an increase in frequency, or difficulty in control
 - b) symptoms of infection: fever, chills, night-sweats, erythema and/or tenderness over shunt tubing. Diarrhea may indicate infection unrelated to shunt. Exposure to other sick individuals
2. physical exam: the following includes signs of increased intracranial pressure
 - a) for children: OFC (p.395). Plot on graph of normal curves (use existing chart for that patient if available) loof for OFC crossing curves
 - b) fontanelle tension (if open): a soft pulsating fontanelle varying with respirations is normal, a tense bulging fontanelle suggests obstruction, a sunken fontanelle may be normal or may represent overshunting
 - c) upward gaze palsy: “setting sun sign”, Parinaud’s syndrome (p.99)
 - d) abducens palsy (p.567): false localizing sign
 - e) field cut, or blindness; see Blindness from hydrocephalus (p.396)
 - f) swelling around shunt tubing: caused by CSF dissecting along tract of an obstructed shunt
3. shunt history
 - a) type of shunt: VP, VA, pleural, LP
 - b) initial insertion of shunt: reason (MM, post-meningitis, etc.) and patient age
 - c) date of last revision and reason for revision
 - d) presence of accessory hardware in system (e.g. antisiphon device, etc.)
4. ability of shunt to pump and refill
 - a) ☐ caution: may exacerbate obstruction, especially if shunt is occluded by ependyma due to overshunting: controversial
 - b) difficult to depress: suggests distal occlusion
 - c) slow to refill (generally, any valve should refill in 15–30 secs): suggests proximal (ventricular) occlusion or slit ventricles
5. evidence of CSF dissecting along tract outside of shunt tubing
6. in children presenting only with vomiting, especially those with cerebral palsy and feeding gastrostomy tubes, rule-out gastroesophageal reflux

Radiographic evaluation

1. “shunt series” (plain x-rays to visualize entire shunt)
 - a) Purpose: R/O disconnection/breakage or migration of tip (NB: a disconnected shunt may continue to function by CSF flow through a subcutaneous fibrous tract)
 - b) For a VP shunt: AP & lateral skull, CXR and abdominal x-ray
 - c) the following hardware may be radiolucent and can mimic disconnection:
 - the central silastic part of older Holter style valves
 - connectors (“Y” & “T” as well as straight)
 - antisiphon devices
 - tumor filters
 - d) obtain most recent x-rays available to compare for breaks (essential for “complicated” shunts involving multiple ventricular or cyst ends or accessory hardware)
2. in infants with open fontanelles, ultrasound is optimal method of evaluation (especially if previous U/S available)
3. CT required if fontanelles closed, may be desirable in complicated shunt systems (e.g. cyst shunts). Minimize the number of CTs in pediatric patients
4. MRI: shunt hardware is difficult to see. May show transependymal absorption of CSF, loculations... Programmable valves must be evaluated and reprogrammed after MRI
5. “shunt-o-gram” if it is still unclear if shunt is functioning
 - a) radionuclide: see below
 - b) x-ray: using iodinated contrast: see below

25

“Shunt-o-gram”

Indications

When shunt function cannot be reliably ascertained using other methods.

Procedure

Shave hair over reservoir & prep (e.g. with Betadine). With patient supine tap the shunt by inserting a 25 gauge butterfly needle into the reservoir. Measure the pressure with a manometer. Patients with multiple ventricular catheters need to have each injected to verify its patency.

Radionuclide “shunt-o-gram” AKA radionuclide shuntography³⁴: after tapping the shunt, drain 2–3 ml of CSF and send 1 ml of CSF for C&S. Inject radio-isotope (e.g. for VP shunt in an adult, use 1 mCi of 99m-Tc (technetium) pertechnetate (usable range: 0.5 to 3 mCi) in 1 cc of fluid) while occluding distal flow (by compressing valve or occluding ports). Flush in isotope with remaining CSF.

Immediately image the abdomen with the gamma camera to rule out direct injection into distal tubing. Image the cranium to verify flow into ventricles (proximal patency). for diffusion of the isotope within the abdomen to rule out pseudocyst formation around catheter.

Interpretation: If spontaneous flow into abdomen occurs within 20 minutes, the shunt is patent. If there is no flow on delayed imaging, it is occluded. The valve can be pumped to look for diffusion of isotope within the abdomen to rule out pseudocyst formation around the catheter tip. If it takes > 20 minutes, or if the patient has to be stood up to get flow, this is indeterminate and you should use other information to decide whether or not to revise the shunt.

X-ray “shunt-o-gram”: after tapping the shunt, drain ≈ 1 ml of CSF and send for C&S. Inject e.g. iohexol (Omnipaque 180) (p.219) while occluding distal flow (by compressing valve or occluding ports).

Pseudocyst (peritoneal) with VP shunt

An abdominal pseudocyst is usually an indication of infection.

Treatment algorithm

One of many viable surgical protocols to deal with this:

1. open abdominal incision over tubing, and divide tubing at this site
2. verify which cut end is the peritoneal end and which is the distal shunt (with a working shunt, pumping the valve should cause CSF to come out the distal shunt)
3. attempt to drain the cyst through the remaining peritoneal end
 - a) when you can't draw any more fluid, or if you don't get any to begin with, withdraw the catheter a little at a time and aspirate at each step
 - b) send any fluid obtained for culture
 - c) if tubing does not pull out smoothly, the abdomen may need to be opened (consider consulting general surgeon)

4. verify function of remaining shunt
 - a) if the remaining shunt is functioning
 - connect it to sterile collection system
 - monitor output volumes & send surveillance cultures of CSF qod
 - after 3 consecutive cultures are negative, internalize distal end of shunt (using fresh distal catheter). The choice of target for distal end (peritoneum, pleura, vein) depends on whether abdominal cyst fluid is infected and if the peritoneal cavity still seems suitable)
 - b) if the shunt is not functioning, a new external ventricular catheter should be inserted and connected to a collection system
 - monitor output volumes & send surveillance cultures of CSF qod
 - after 3 consecutive cultures are negative, remove the old shunt and place a totally new shunt. The choice of target for distal end (peritoneum, pleura, vein) depends on whether abdominal cyst fluid is infected and if the peritoneal cavity still seems suitable)
5. shunt tap: indications vary, generally performed if occlusion suspected or if surgical exploration is considered or if infection is strongly suspected; see Tapping a shunt (p.422)
6. shunt exploration: sometimes even after thorough evaluation the only means to definitively prove or disprove the functioning of various shunt components is to operate and isolate and test each part of the system independently. Even when infection is not suspected, CSF and any removed hardware should be cultured.

Tapping a shunt

Indications

Indications to tap a shunt or ventricular access device (e.g. Ommaya reservoir) include:

1. to obtain CSF specimen
 - a) to evaluate for shunt infection
 - b) for cytology: e.g. in PNET to look for malignant cells in CSF
 - c) to remove blood: e.g. in intraventricular hemorrhage
2. to evaluate shunt function
 - a) measuring pressures
 - b) contrast studies:
 - proximal injection of contrast (iodinated or radio-labeled)
 - distal injection of contrast
3. as a temporizing measure to allow function of a distally occluded shunt^{35,36}
4. to inject medication
 - a) antibiotics: for shunt infection or ventriculitis
 - b) chemotherapeutic (antineoplastic) agents
5. for catheters placed within tumor cyst (not a true shunt):
 - a) periodic withdrawal of accumulated fluid
 - b) for injection of radioactive liquid (usually phosphorous) for ablation

Technique

For LP shunt, see Lumboperitoneal (LP) shunt evaluation (p.1518).

There is a risk of introducing infection with every entry into the shunt system. With care, this may be kept to a minimum.

1. shave area
2. prep: e.g. povidone iodine solution × 5 minutes
3. use 25 gauge butterfly needle or smaller (ideally a noncoring needle should be used): for routine taps, the needle should only be introduced into shunt components specifically designed to be tapped

To measure pressures

A 25 Ga butterfly needle and an LP tray (with manometer, specimen tubes, 3-way stopcock valve...) are helpful. Steps are outlined in □ Table 25.2.

25.6.4 Undershunting

General information

The shunt malfunction rate is ≈ 17% during the first year of placement in the pediatric population.

Table 25.2 Steps in tapping a shunt	
Step	Information provided
• After shave and prep, insert 25 Ga butterfly needle into the reservoir and look for spontaneous flow into butterfly tubing • measure pressure in manometer	• spontaneous flow indicates proximal end not completely occluded • pressure is that of ventricular system (normal is <15 cm of CSF in relaxed recumbent patient)
• then measure the pressure with distal occluder compressed (if present)	• rise in pressure indicates some function of valve and distal shunt
• if no spontaneous flow, try to aspirate CSF with syringe	• if CSF is easily aspirated, it may be that pressure seen by ventricular system is very low. There should be spontaneous flow after that when the end of the tubing is lowered • if no CSF obtained or if difficult to aspirate, indicates proximal occlusion
• if still no flow, carefully inject 1–2 ml of preservative free saline into ventricular catheter and see if spontaneous flow of more than the amount injected occurs	• may dislodge clot or debris from catheter • if only the 1–2 ml that were injected returns, indicates there is not a patent catheter in an open CSF space (possibilities include: occluded catheter, tip lodged in brain, slit ventricles)
• send any CSF for: C&S, protein/glucose, cell count	• checks for infection
• fill manometer with sterile saline with the valve turned off to the shunt • compress proximal (inlet) occluder if present • Open valve to shunt and measure runoff pressure after ≈ 60 seconds	• measures forward transmission pressure (through valve and peritoneal catheter in valves with a proximal occluder); forward pressure should be less than ventricular pressure (and absolute pressure should be <8 cm H₂O)
• if no distal flow, keep inlet occluder compressed and inject 3–5 ml of saline into distal shunt and recheck distal runoff pressure • ☐ do not inject more than ≈ 1–2 ml into ventricles to avoid increasing ICP	• if peritoneal catheter is in a loculated compartment the pressure will be considerably higher after injection

Etiologies

May be due to one or a number of the following:

1. blockage (occlusion)
 - a) possible causes of occlusion:
 - proximal obstruction by choroid plexus
 - buildup of proteinaceous accretions
 - blood
 - cells (inflammatory or tumor)
 - secondary to infection
 - b) site of blockage
 - blockage of ventricular end (most common): usually by choroid plexus, may also be due to glial adhesions, intraventricular blood
 - blockage of intermediate hardware (valves, connectors, etc., tumor filters may become obstructed by tumor cells, antisiphon devices may close due to variable overlying subcutaneous tissue pressures³⁷)
 - blocked distal end, also see VP shunt (p.417)
 - c) disconnection, kinking or breakage of system at any point: with age, silicone elastomers used in catheters calcify and break down, and become more rigid and fragile which may promote subcutaneous attachment.³⁸ Barium impregnation may accelerate this process. Tube fractures often occur near the clavicle, presumably due to the increased motion there

Signs and symptoms of undershunting

Signs and symptoms are those of active hydrocephalus. See “Evaluation of patient with shunt” (p.420).

25.6.5 Shunt infection

See Shunt infection (p.339) for evaluation and treatment.

25.6.6 “Overshunting”

General information

Possible complications of overshunting include³⁹:

1. slit ventricles: including slit ventricle syndrome (see below)
2. intracranial hypotension: see below
3. subdural hematomas (p.426)
4. craniosynostosis and microcephaly (p.427): controversial
5. stenosis or occlusion of sylvian aqueduct

10–12% of long-term ventricular shunt patients will develop one of the above problems within 6.5 yrs of initial shunting.³⁹ Some experts feel that problems related to overshunting could be reduced by utilizing LP shunts for communicating hydrocephalus, and reserving ventricular shunts for obstructive HCP.³⁹ VP shunts may also be more likely to overdrain than VA shunts because of the longer tubing → greater siphoning effect.

Intracranial hypotension

AKA low ICP syndrome. Very rare. Symptoms similar to those of spinal H/A (postural in nature, relieved by recumbency). Although usually not associated with the following symptoms,⁴⁰ they may occur³⁹: N/V, lethargy, or neurologic signs (e.g. diplopia, upgaze palsy). Sometimes the symptoms resemble those of high ICP except that they are relieved when prostrate. Acute effects that may occur include³⁹: tachycardia, loss of consciousness, other brain stem deficits due to a rostral shift of the intracranial contents or to low ICP.

Etiology is a siphoning effect due to the column of CSF in the shunt tube when the patient is erect.⁴¹ Ventricles may be slit-like (as in slit ventricle syndrome (SVS)) or may be normal in appearance. Sometimes it is necessary to document a drop in ICP when going from supine to erect to diagnose this condition. These patients may also develop shunt occlusion and then the distinction from SVS blurs (see below).

With short-term symptoms, an antisiphon device (ASD) is the treatment of choice. However, patients with long-standing overshunting may not tolerate efforts to return intraventricular pressures to normal levels.^{39,42}

Slit ventricles

“Slit ventricles” refers to complete collapse of the ventricles. In a survey, a frontal-occipital horn ratio⁴³ <0.2 was most often interpreted as representing SVS. May be seen in:

1. overshunting
2. with entrapped (isolated) fourth ventricle (p.402)
3. some patients with idiopathic intracranial hypertension (AKA pseudotumor cerebri) (p.766) have slit-like ventricles with consistently elevated ICP

Slit ventricles may be:

1. asymptomatic:
 - a) slit ventricles (totally collapsed lateral ventricles) may be seen on CT in 3 -80% of patients after shunting,^{40,44} most are asymptomatic
 - b) these patients may occasionally present with symptoms unrelated to the shunt, e.g. true migraine
2. slit ventricle syndrome (SVS): seen in $<12\%$ of all shunted patients. Subtypes:
 - a) intermittent shunt occlusion: overshunting leads to ventricular collapse (slit ventricles) which causes the ependymal lining to occlude the inlet ports of the ventricular catheter (by coaptation) producing shunt obstruction. With time, many of these patients develop low ventricular compliance,⁴⁵ where even minimal dilatation results in high pressure which produces symptoms. Expansion then eventually reopens the inlet ports allowing resumption of drainage (hence the intermittent symptoms). Symptoms may resemble shunt malfunction: intermittent headaches unrelated to posture, often with N/V, drowsiness, irritability and impaired

- mentation. Signs may include 6th cranial nerve palsy. Incidence in shunted patients: 2–5%⁴⁰.
⁴⁶ CT or MRI scans may also show evidence of transependymal absorption of CSF
- b) total shunt malfunction (AKA normal volume hydrocephalus⁴⁵): may occur and yet ventricles remain slit-like if the ventricles cannot expand because of subependymal gliosis, or due to the law of Laplace (which states that the pressure required to expand a large container is lower than the pressure required to expand a small container)
 - c) venous hypertension with normal shunt function: may result from partial venous occlusion that occurs in some conditions (e.g. at the level of the jugular foramen in Crouzon's syndrome). Usually subsides by adulthood
3. intracranial hypotension: symptoms often relieved by recumbency (see above)

Evaluation of slit ventricles

The shunt valve fills slowly if pumped when the ventricles are collapsed.

Monitoring CSF pressure: either via lumbar drain, or with a butterfly inserted into the shunt reservoir (with this method, pressure can be followed during postural changes to look for negative pressure when upright; possibly higher risk of infection with this). These patients are also monitored for pressure spikes, especially during sleep.

Alternatively, these patients may be evaluated by “shunt-o-gram” (see above).

Treatment

In treating a patient with slit ventricles in imaging studies, it is important to ascertain into which of the categories (see above) the patient falls. If the patient can be categorized, then the specific treatment listed below should be employed. Otherwise, it is probably most common to initially treat the patient empirically as intracranial hypotension, and then to move on to other methods for treatment failures.

Treatment of asymptomatic slit ventricles

Prophylactic upgrading to a higher pressure valve or insertion of an antisiphon device as initially advocated has largely been abandoned. However, this may be appropriate at the time of shunt revision when done for other reasons.⁴⁴

Treatment of intracranial hypotension

Postural H/A due to intracranial hypotension (true overshunting) is usually self limited, however, if symptoms persist after ≈ 3 days of bed-rest and analgesics and a trial with a tight abdominal binder, the valve should be checked for proper closing pressure. If it is low, replace with a higher pressure valve. If it is not low, an ASD (which, by itself, also increases the resistance of the system) alone or together with a higher pressure valve may be needed.⁴⁷

Treatment of slit ventricle syndrome

Patients with symptoms of SVS are actually suffering from intermittent high pressure. If total shunt malfunction is the cause, then shunt revision is indicated. For intermittent occlusion, treatment options include:

1. if symptoms occur early after shunt insertion or revision, initial expectant management may be indicated since symptoms will spontaneously resolve in many patients as they equilibrate to the new intracranial pressure
2. revision of the proximal shunt. This may be difficult due to the small size of the ventricles. One can attempt to follow the existing tract and insert a longer or shorter length of tubing based on the pre-op imaging studies. Some advocate the placement of a second ventricular catheter, leaving the first one in place
3. patients may “respond” (fortuitously) to either of the following interventions because the slight ventricular enlargement elevates the ependyma off of the inlet ports (this may not always be the therapy of choice):
 - a) valve upgrade⁴⁸ or
 - b) ASD insertion^{40,47}: the procedure of choice in some opinions.³⁹ First described in 1973⁴⁹
4. subtemporal decompression^{50,51,52} sometimes with dural incision.⁵⁰ This results in dilatation of the temporal horns (evidence for elevated pressure) in most, but not all⁵² cases
5. third ventriculostomy (p.415)⁵³

25.6.7 Problems unrelated to shunting

For H/A consistent with migraine that are not postural, a trial with migraine-specific medications is warranted (Fiorinal®). See treatment of idiopathic intracranial hypertension (pseudotumor cerebri) (p.771).

25.6.8 Subdural hematomas

General information

May be due to collapse of brain with tearing of bridging veins. Incidence: 4–23% in adults, 2.8–5.4% in children, and is higher with normal pressure hydrocephalus (20–46%) than with “hypertensive hydrocephalus” (0.4–5%).^{54,55} The risk of SDH is higher in the setting of longstanding hydrocephalus with a large head and little brain parenchyma (craniocerebral disproportion) with a thin cerebral mantle, as usually occurs in children with macrocephaly and large ventricles on initial evaluation. These patients have an “extremely delicate balance between subdural and intraventricular pressure”.⁵⁴ By the same token, SDH can also follow shunting in elderly patients who have severe brain atrophy. The development of SDH may also be facilitated by negative pressures in the ventricles as a result of a siphoning effect when the patient is upright.^{55,56} There is also a low risk of epidural hematoma following CSF shunting.⁵⁵

25

Characteristics of the fluid

The collections may be on the same side as the shunt in 32% on the opposite side in 21% and bilateral in 47%.⁵⁵

At the time of discovery, the SDHs are usually subacute to chronic, and the previously large ventricles are usually collapsed. Only 1 of 19 cases showed colorless fluid.⁵⁵ In all cases tested (even the 1 with clear fluid), the protein was elevated compared to CSF.

Treatment

Indications for treatment

Small (<1–2 cm thick) asymptomatic collections in patients with closed cranial sutures may be followed with serial imaging studies. SDH were symptomatic in $\approx 40\%$ of cases (symptoms often resemble those of shunt malfunction), and these require treatment. Treatment of SDH in children with open sutures has been advocated⁵⁵ to prevent later symptoms and/or development of macrocrania. The controversy arises with large asymptomatic SDH in older children or adults. Many authors recommend not treating asymptomatic lesions regardless of appearance,^{54,57} whereas others vary their recommendations based on diverse criteria including size, appearance (chronic, acute, mixed...), etc.

Treatment techniques

A number of techniques have been described. Most involve evacuation of the SDHs by any of the usual methods (e.g. burr holes for chronic collections, craniotomy for acute collections) together with:

1. reducing the degree of shunting (i.e. to establish a lower pressure in the subdural space than in the intraventricular space, to cause the ventricles to re-expand and to prevent reaccumulation of the SDH)
 - a) in shunt dependent cases
 - replacing the valve with a higher pressure unit (upgrading the valve)
 - increasing the pressure on a programmable pressure valve^{58,59}
 - using a Portnoy device that can be turned off and on externally. Be sure that care providers can reliably open the device in an emergency
 - b) in non-shunt dependent cases
 - any of the methods outlined above for shunt dependent cases, or
 - temporarily tying off the shunt⁶⁰
 - c) insertion of an anti-siphon device⁴⁹
2. drainage of the subdural space to
 - a) the cisterna magna⁶¹
 - b) to the peritoneum (subdural peritoneal shunt) with a low pressure valve (or no valve⁵⁵). Some authors have the care-giver frequently pump the subdural valve

The goal is to achieve a delicate balance between undershunting (producing symptoms of active hydrocephalus) and overshunting (promoting the return of the SDH). Following surgery the patient should be mobilized slowly to prevent recurrence of the SDH.

25.6.9 Miscellaneous shunt issues

Craniosynostosis, microcephaly and skull deformities

See also Craniosynostosis (p.252). A number of skull changes have been described in infants after shunting, including⁶²: thickening and inward growth of the bone of the skull base and cranial vault, decrease in size of the sella turcica, reduction in size of the cranial foramina, and craniosynostosis. The most common skull deformity was dolichocephaly from sagittal synostosis.⁶³ Microcephaly accounted for $\approx 6\%$ of skull deformities after shunting (about half of these had sagittal synostosis). Some of these changes were reversible (except when complete synostosis was present) if intracranial hypertension recurred.

Laparoscopic surgery in patients with VP shunts

Issues regarding safety of laparoscopic surgery in patients with VP shunts:

1. laparoscopic surgery: abdominal insufflation with CO₂ is used to create a pneumoperitoneum permitting the general surgeon to work. Typical insufflation pressure: 15 mm Hg (see conversion factors between mm Hg and cm of water (p.861)). In thin patients, 10 mm Hg may suffice. Transient additional increases in pressure may occur, e.g. when the surgeon leans on the patient's abdomen
2. concerns for patients with VP shunts:
 - a) in some cases insufflation $\rightarrow \uparrow$ ICP⁶⁴ which may be due to:
 - compression of vena cava $\rightarrow$ reduced venous return from head, as in valsalva maneuver (independent of presence of a shunt)
 - absorption of CO₂ from the peritoneum $\rightarrow \uparrow$ in arterial CO₂ causing cerebral arterial dilatation thereby increasing ICP
 - $\downarrow$ CSF drainage due to $\uparrow$ pressure against which CSF must flow
 - retrograde passage of air/debris into intracranial compartment through an incompetent shunt valve (this also has potential for infection in the presence of peritonitis). This risk is minimal even with in vitro back-pressures up to 80 mm Hg.⁶⁵ Retrograde flow may also occur with a valveless shunt (rarely used)
 - in one case report monitoring TCDs,⁶⁶ there was no change during laparoscopic surgery in a patient with a VP shunt (except during periods of very high pressure)
 - b) occlusion of the distal catheter by air, debris⁶⁷ or soft tissue
 - c) extremely high intraabdominal pressures (>80 mm Hg in vitro) may damage the valve,⁶⁵ which could cause malfunction after the laparoscopy

Prophylactic management options:

1. very controversial, special precautions may not be necessary⁶⁸
2. one can temporarily occlude the peritoneal catheter (e.g. by a hemoclip applied by the general surgeon through the laparoscope under minimal initial insufflation pressure; the clip is removed at the end of the procedure), or, temporary externalization of the shunt by the neurosurgeon, with internalization at the end of the procedure (this engenders an increased risk of infection)
3. ICP monitoring during laparoscopy
4. using low insufflation pressures (e.g. <10 mm Hg)

25.7 Specific shunt systems

The following describes the salient features of some commonly used shunt systems. Diagrams are for general information only, and are not to scale.

25.7.1 Comparison of nonprogrammable shunt valves

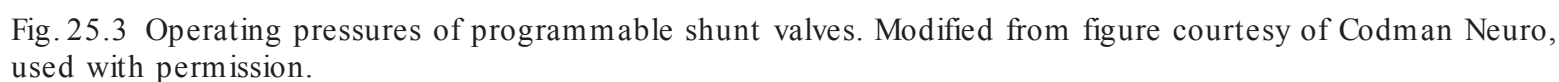
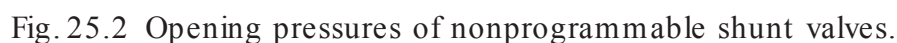
□ Fig. 25.2 compares opening pressures for some common nonprogrammable shunt valves. Symbols superimposed on the pressure bars show the x-ray markings on the valves.

25.7.2 Comparison of programmable shunt valves

□ Fig. 25.3 shows a comparison of operating pressures of some common programmable shunt valves.

25.7.3 PS Medical/Medtronic CSF flow controlled valve

Manufactured by Medtronic Inc.



A single one-way membrane valve design. The radio-opaque arrowhead points in the direction of flow (□ Fig. 25.4).

Pumping the valve

To pump the shunt in the “forward” direction, first occlude the inlet port (□ Fig. 25.5) with pressure from one finger on the “inlet occluder” (prevents back-flow into the ventricle during the next step). Then while maintaining this pressure, depress the reservoir dome with a second finger. Release both fingers, and repeat. The one-way valve regulates shunt pressure and prevents reflux of CSF during normal use and during the release phase of shunt pumping.

X-ray characteristics

The three available valve pressures are indicated by radio-opaque dots on the valve (allows x-ray identification of valve pressure): one dot =low pressure, two dots =medium, three dots =high.

25.7.4 Strata® programmable valve

The Medtronic Strata valve is an externally adjustable valve that is programmed (using a magnet) to one of five performance level (“P/L”) settings (□ Fig. 25.6).

Also, see general information regarding programmable valves (p.418).

25.7.5 Codman Hakim programmable valve

Manufactured by Codman Inc.

18 pressure settings. Programmed by an AC-powered programming unit that requires confirmatory x-ray after re-programming. Newer programming units with acoustic monitoring may obviate

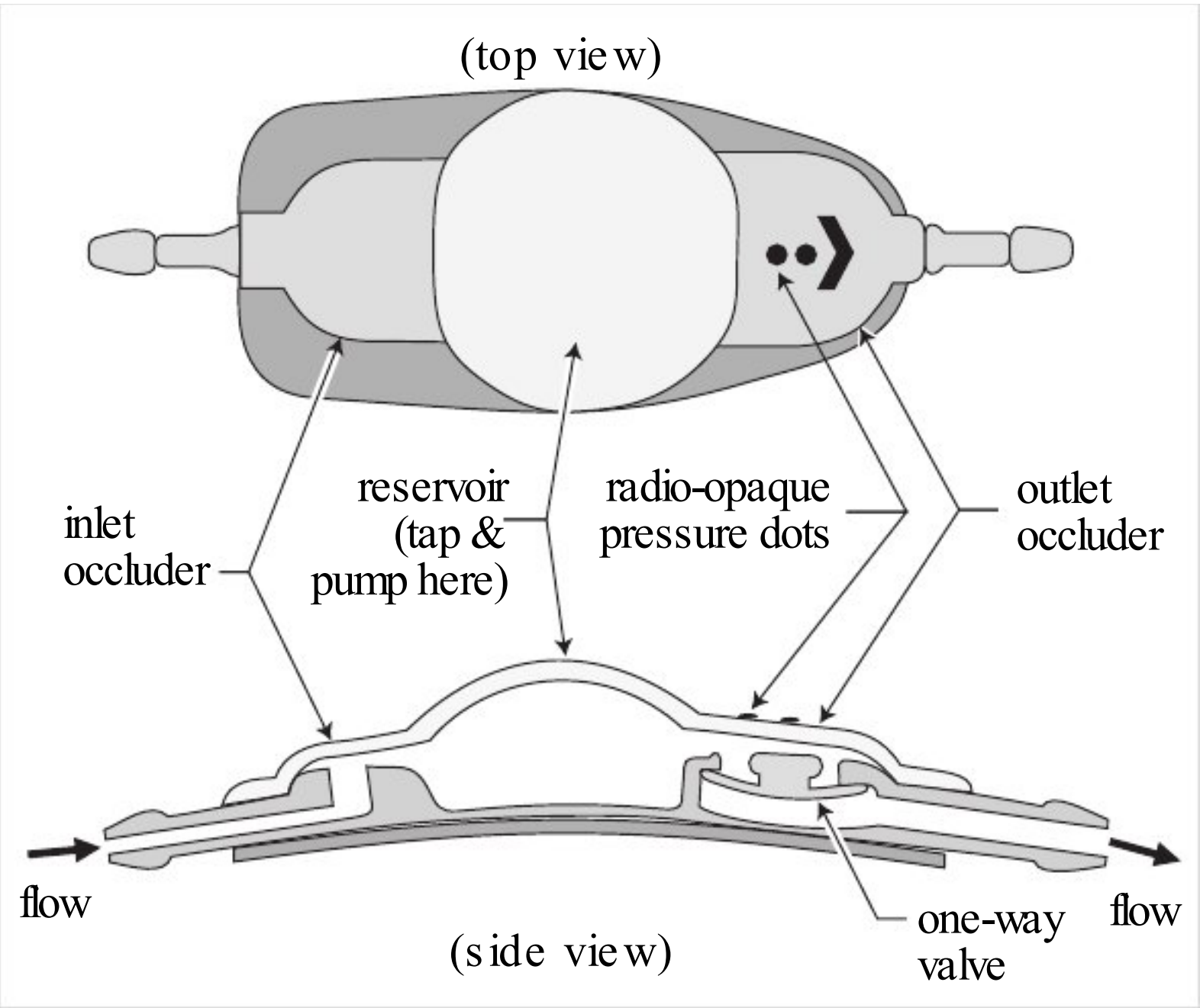


Fig. 25.4 PS Medical standard contoured valve

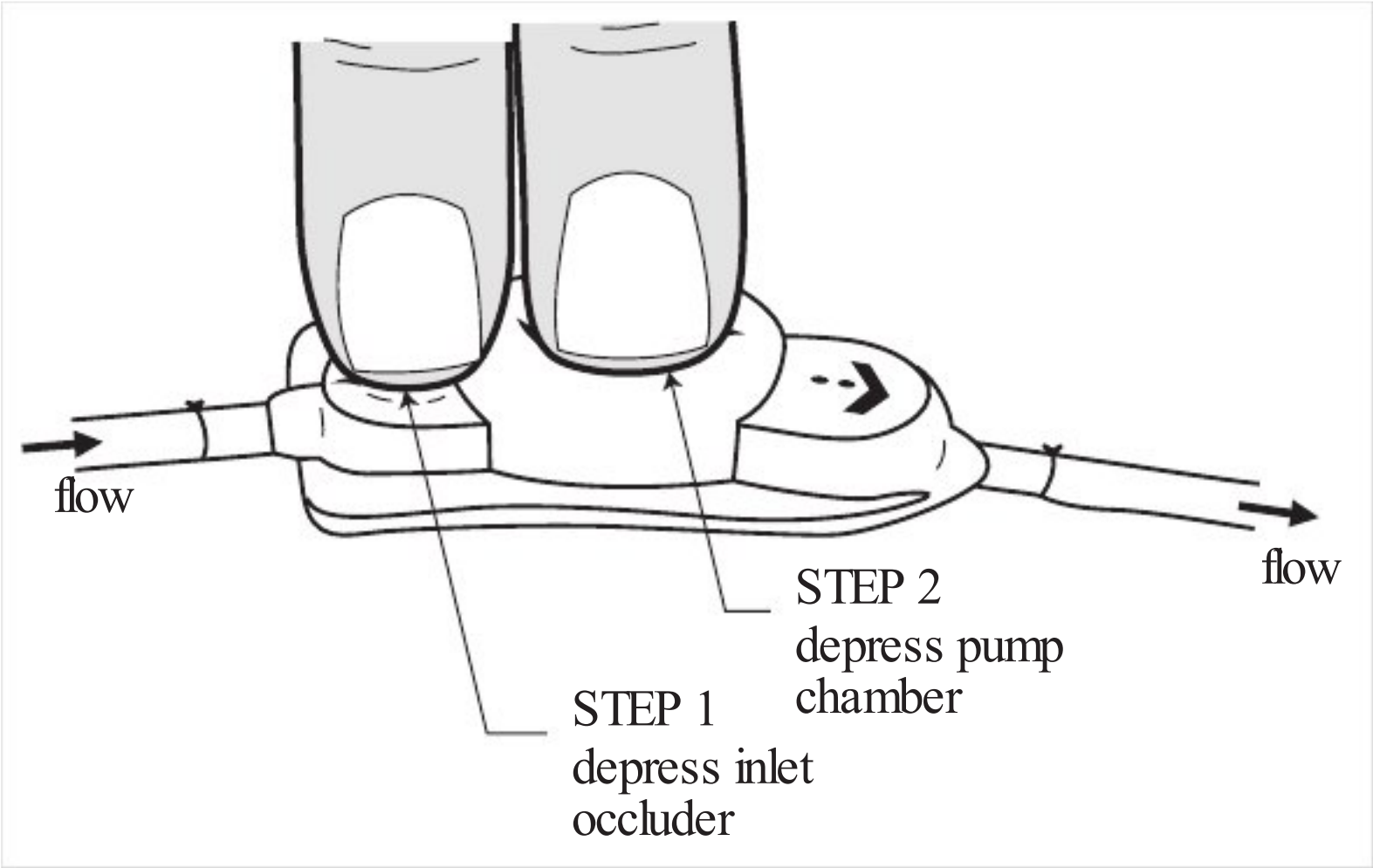


Fig. 25.5 Pumping the PS Medical valve

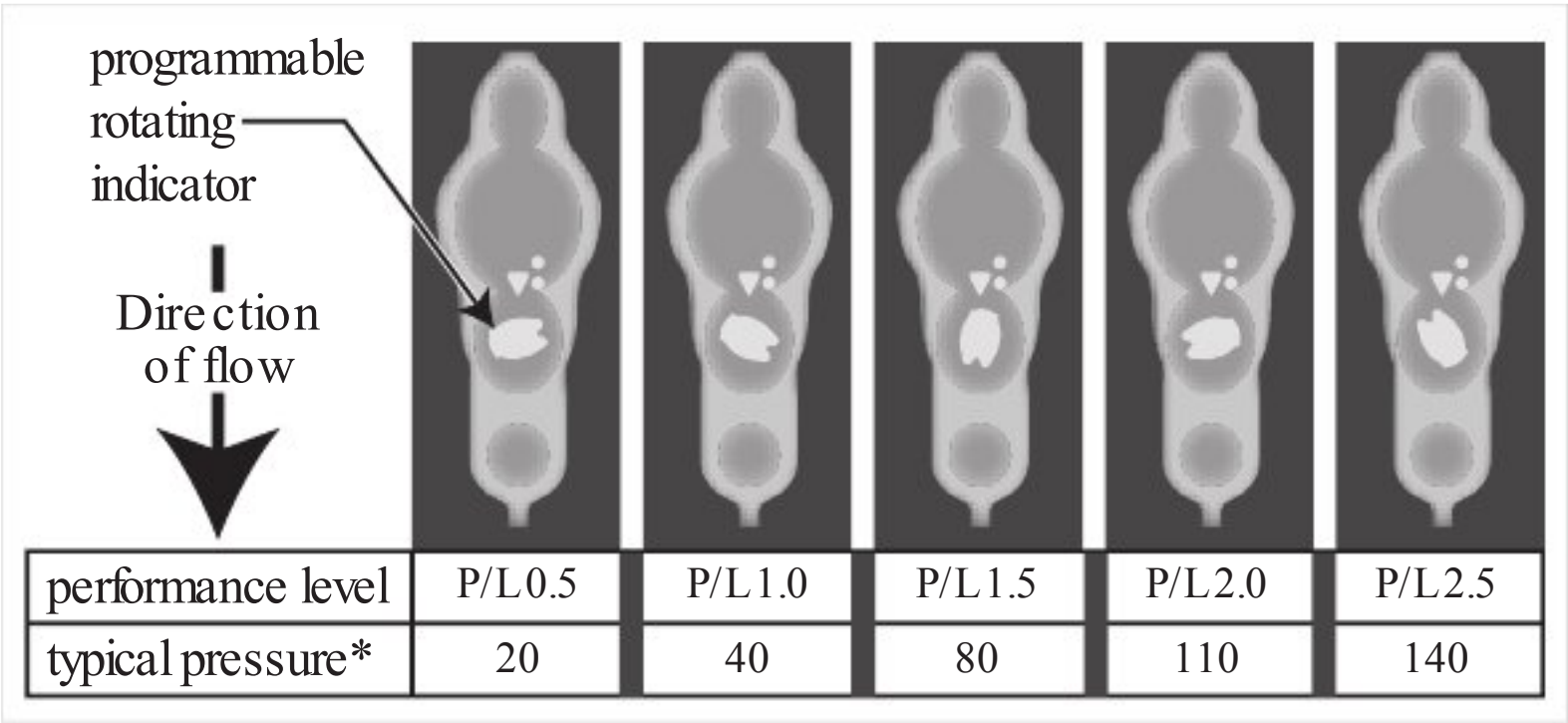


Fig. 25.6 Performance level (P/L) settings for the regular size Strata valve as seen on x-ray
* pressures in mm H₂O at flow rate of 20 ml/hr with patient recumbent

the need for x-ray. The manufacturer advises not to increase the pressure by >40 mm H₂O in a 24-hour period.

X-ray appearance for various settings are shown in □ Fig. 25.7. (note: settings of 70, 120 & 170 mm H₂O align with an arm of the central cross of the valve). NB: when x-rayed correctly, the x-ray beam passes first through the valve and then the patient, which causes the radio-opaque marker to appear as a solid circle to the right of center as shown in □ Fig. 25.7. If the marker is on the left side, the beam is passing from the bottom of the valve, and the actual pressure reading should be based on a mirror image of the x-ray.

25.7.6 Certas Plus programmable valve

Manufactured by Codman Inc.

The x-ray appearance is shown in □ Fig. 25.8, and the pressure settings appear in □ Table 25.2.

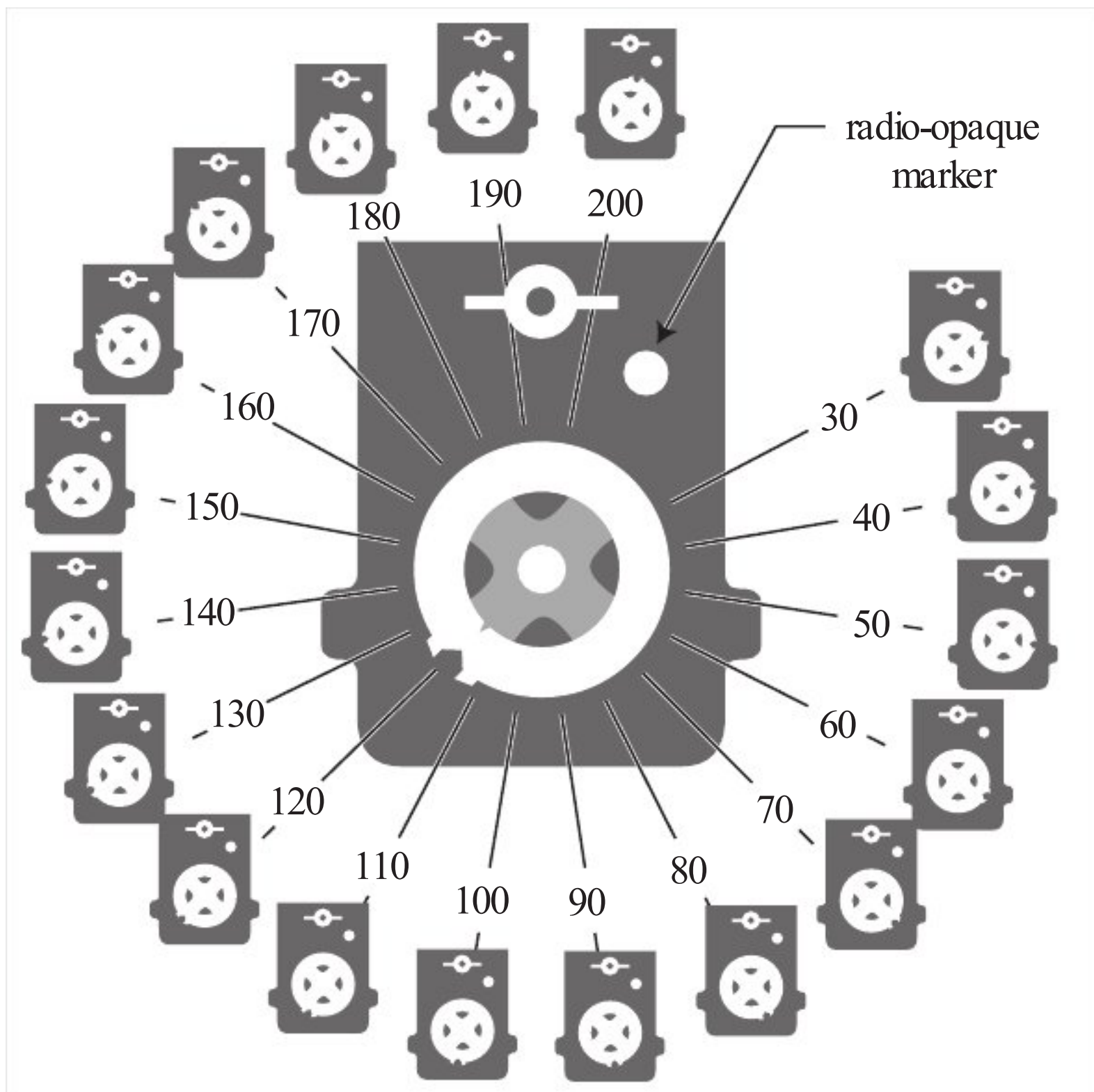


Fig. 25.7 X-ray appearance of Codman Hakim programmable valve at its various settings in mm Hg (e.g. the large central image shows a setting of 120 mm H₂O)

25.7.7 Polaris programmable valve

Manufactured by Sophysa.

The Polaris valve is an externally programmable valve that uses two attracting Samarium-Cobalt magnets to lock the pressure setting and to resist inadvertent reprogramming by environmentally encountered magnets such as MRI scanners, cell phones, headphones...

Available in 4 models (different pressure ranges, each identified by a unique number of radio-opaque dots), each with 5 externally adjustable positions. The x-ray appearance and corresponding pressures are shown in □ Fig. 25.9.

25.7.8 Heyer-Schulte

Distributed by Integra Neurosciences.

The LPV® II valve is shown in □ Fig. 25.10. To pump the shunt, occlude inlet port with one finger, then depress reservoir with another finger (as for the PS Medical valve, see above). This valve may be injected in either direction by depressing the appropriate occluder while injecting into the reservoir.

25.7.9 Hakim (Cordis) shunt

Distributed by Integra Neurosciences.

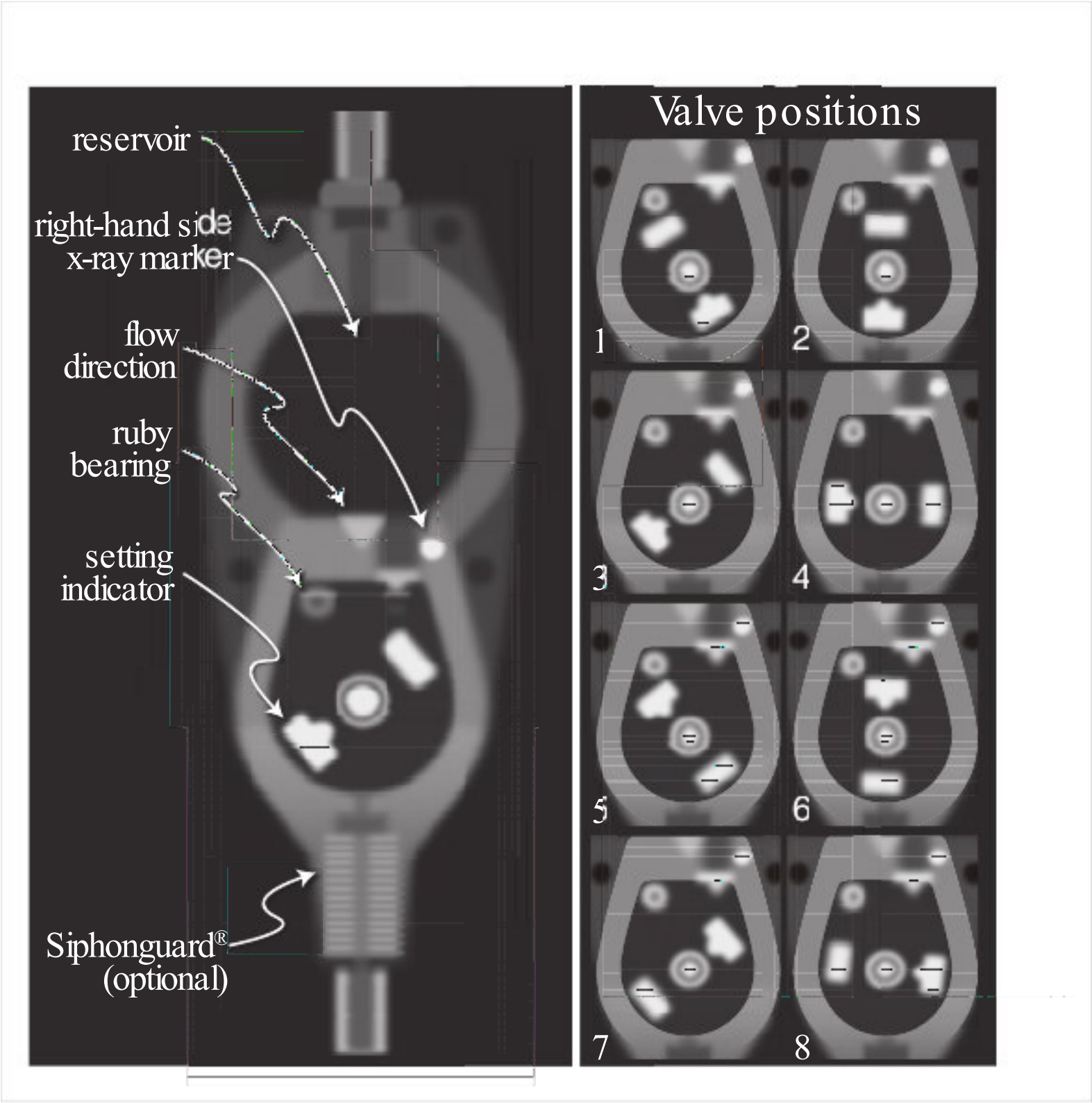


Fig. 25.8 Certas Plus Valve, x-ray appearance

Table 25.3 Certas pressure settings		
Setting number	Mean pressure (mm H ₂ O) measured at 20 ml/hr flow rate	
	Certas (discontinued)	Certas Plus
1	36	25
2	71	50
3	109	80
4	146	110
5	178	145
6	206	180
7	238	215
8	>400	(virtual off)

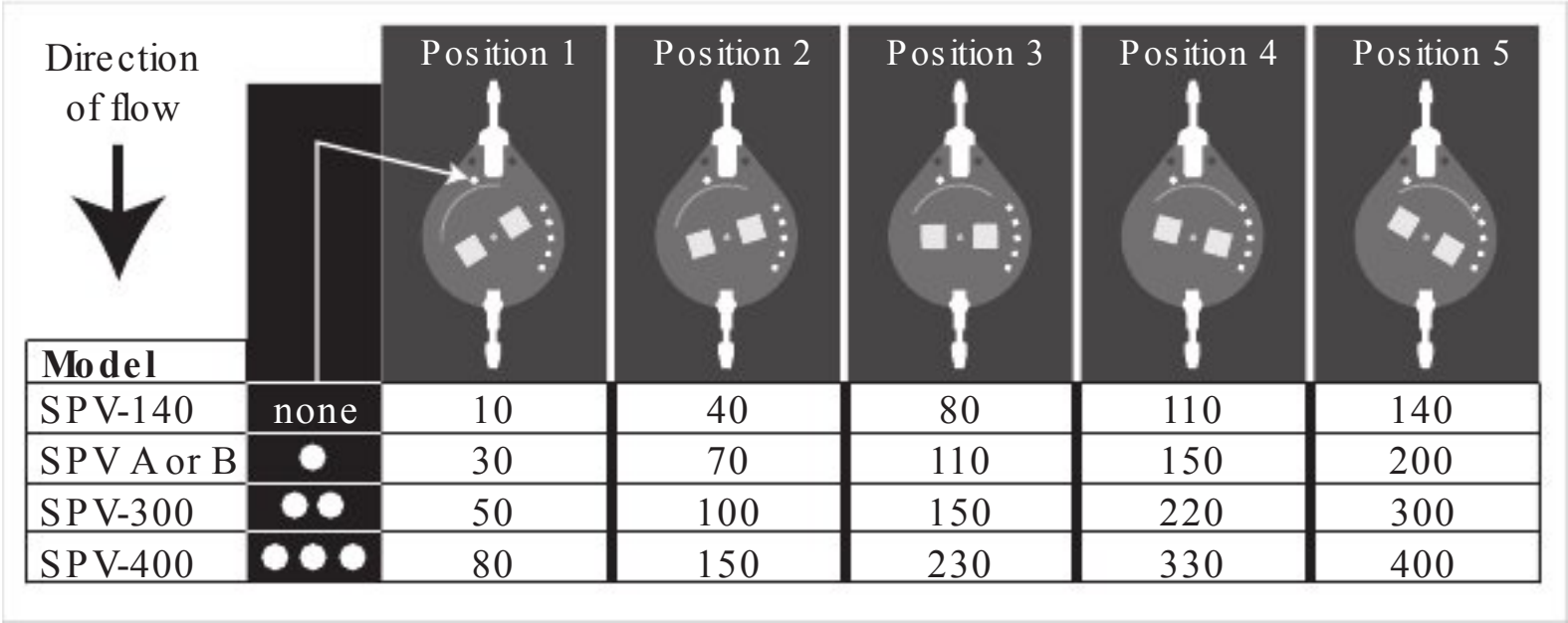


Fig. 25.9 Programmable settings for Polaris valve models as seen on x-ray (pressures in mm H₂O)

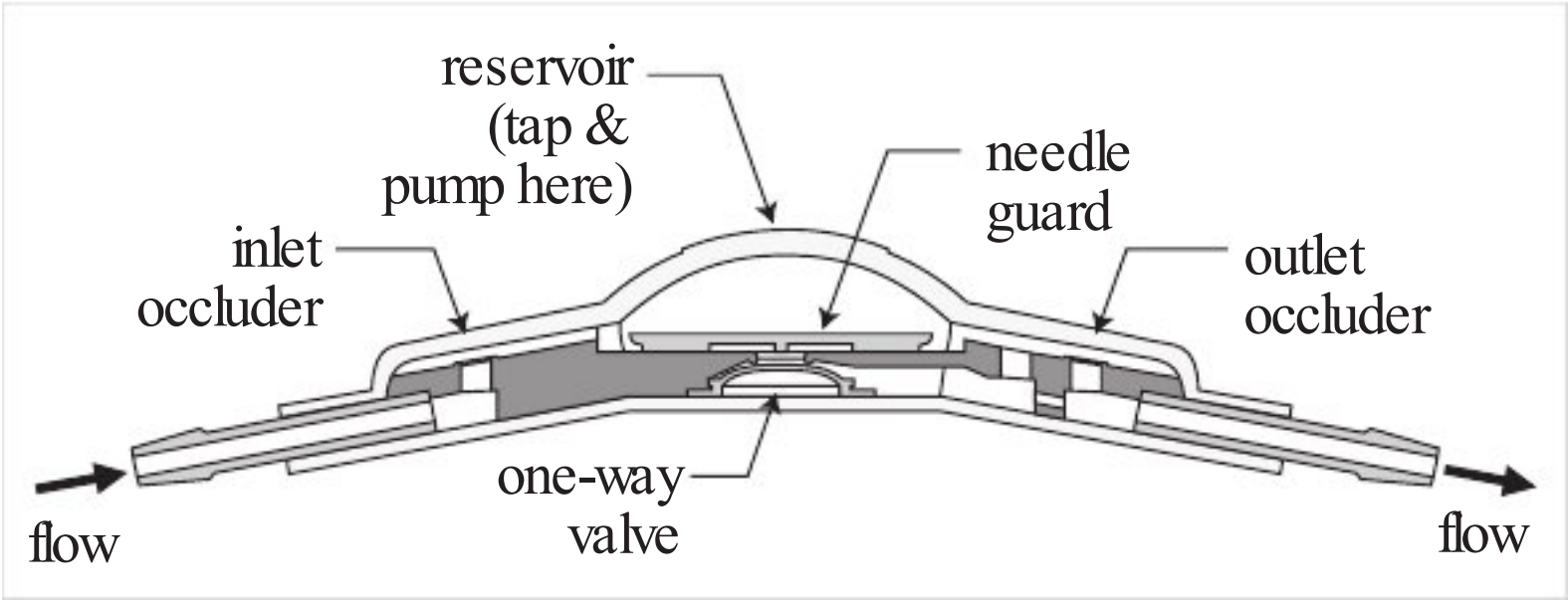


Fig. 25.10 Heyer-Schulte IPV® II(low-profile) valve (side view)

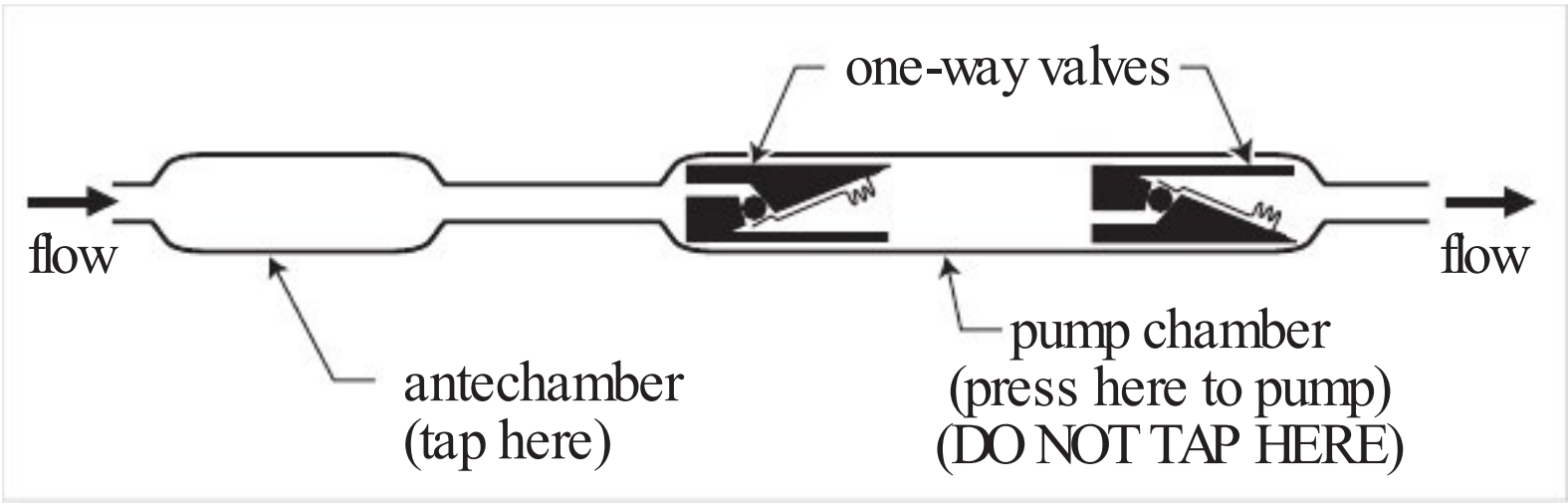


Fig. 25.11 Hakim Standard Mechanism

A dual ball-valve mechanism (□ Fig. 25.11). To pump shunt, depress the indicated portion of the valve. NB: do not tap here, as the silicone elastomer housing is not self-sealing. The antechamber is provided for this type of access.

25.7.10 Integra (Cordis) horizontal-vertical lumbar valve

- Fig. 25.12. May be used in lumboperitoneal shunt to increase the transmission pressure when the patient is upright to prevent overshunting. Markings used to orient the device during implantation:
1. an arrow on the inlet side of the unit indicates direction of flow
 2. inlet tubing is clear

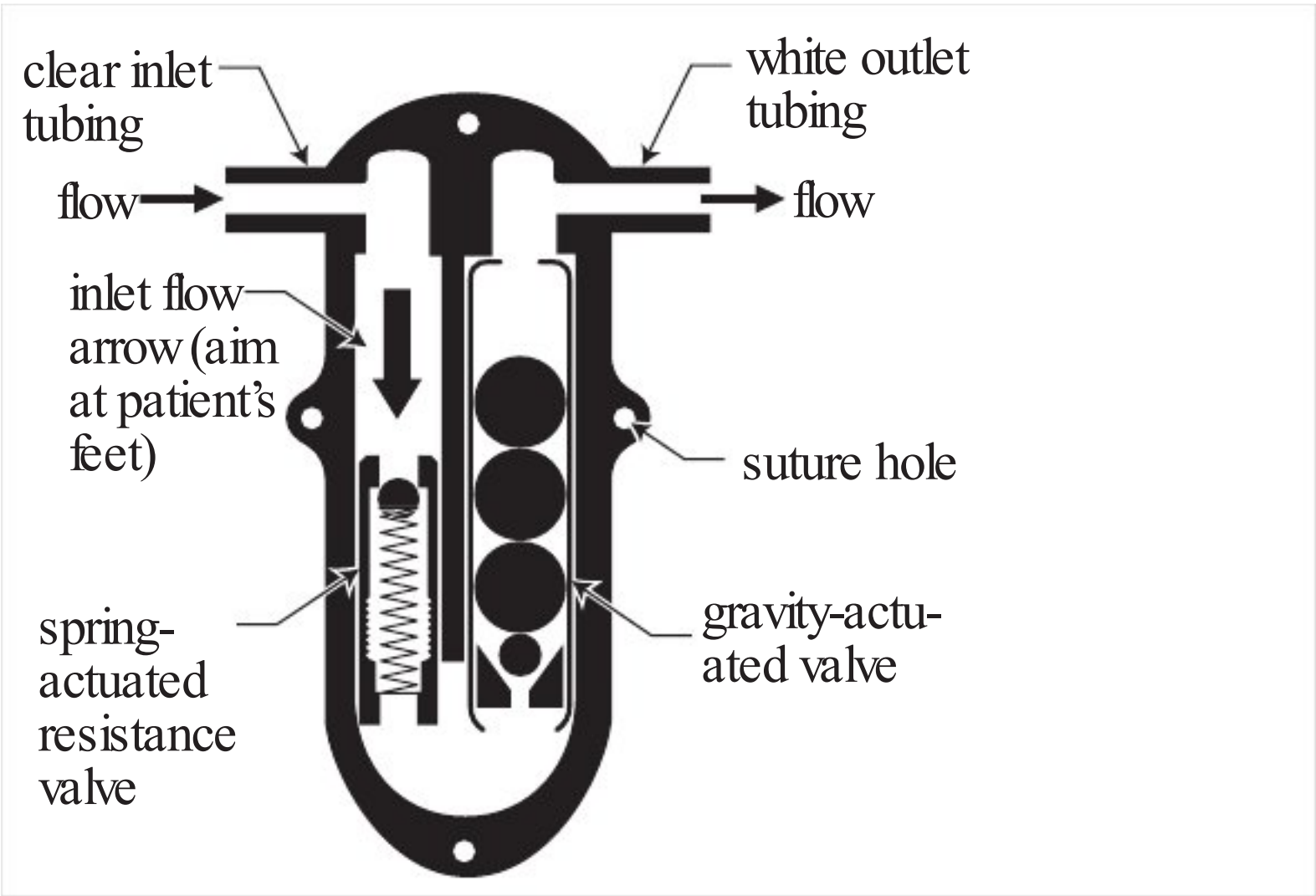


Fig. 25.12 Cordis H-V valve

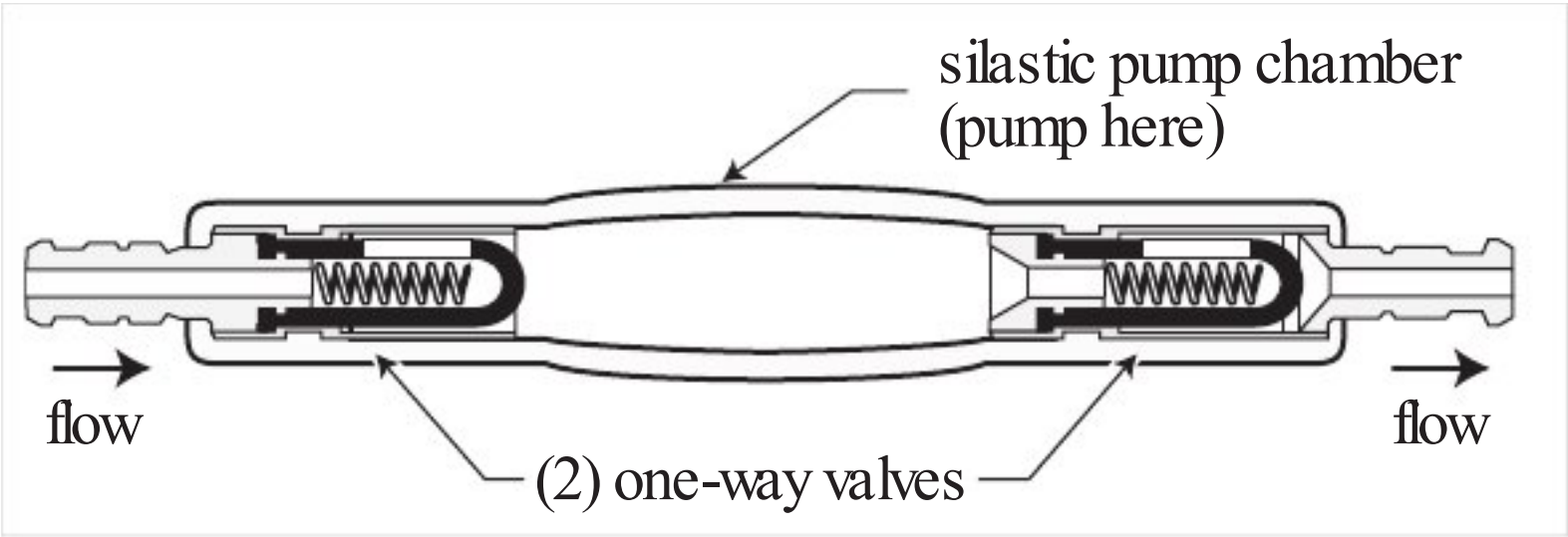


Fig. 25.13 Holter valve

- 3. inlet tubing has smaller diameter than outlet tubing
- 4. outlet tubing is white
- 5. before positioning the valve and fastening it to the fascia with permanent suture, the valve should be connected to both the subarachnoid catheter (inlet) and the peritoneal catheter (outlet). The arrow on the inlet valve should point towards the patient's feet

25.7.11 Holter valve

A dual slit valve mechanism (□ Fig. 25.13). Usually used in combination with a Rickham or Salmon-Rickham reservoir (□ Fig. 25.14).

To pump the shunt, simply depress the indicated portion of the valve.

X-ray characteristics

The silastic tube between the two one-way valves is radiolucent (□ Fig. 25.1).

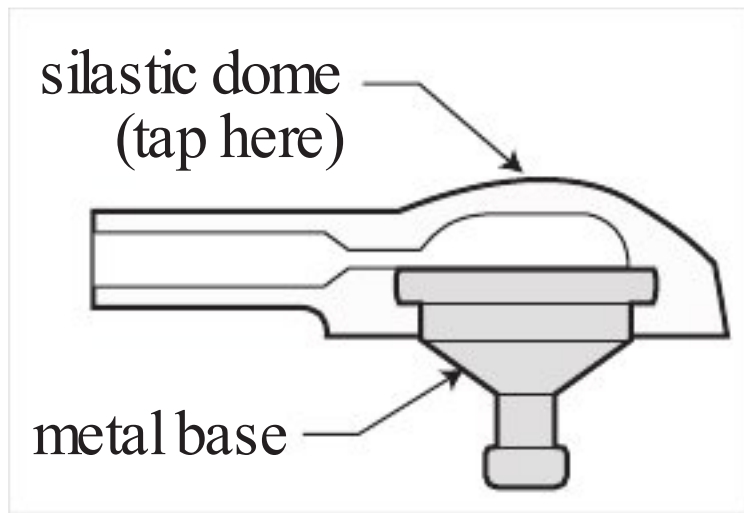


Fig. 25.14 Salmon-Rickham Reservoir

25.7.12 Salmon-Rickham reservoir

Similar to standard Rickham reservoir except for lower profile (□ Fig. 25.14).

25.8 Surgical insertion techniques

For surgical techniques, refer to the section on insertion of Ventricular shunts (p.1514).

25.9 Instructions to patients

All patients and families of patients with hydrocephalus should be instructed regarding the following:

1. signs and symptoms of shunt malfunction or infection
2. not to pump the shunt unless instructed to do so for a specific purpose
3. prophylactic antibiotics: for the following situations (mandatory in vascular shunts, sometimes recommended in other shunts)
 - a) dental procedures other than routine cleaning
 - b) instrumentation of the bladder: not practical for patients who catheterize to void. Important for cystoscopy, CMG, etc. where septicemia may occur
4. in a growing child: the need for periodic evaluation, including assessment of distal shunt length

References

- [1] Shinnar S, Gammon K, Bergman EW, et al. Management of Hydrocephalus in Infancy: Use of Acetazolamide and Furosemide to Avoid Cerebrospinal Fluid Shunts. *J Pediatr.* 1985; 107:31–37
- [2] Kreusser KL, Tarby TJ, Kovnar E, et al. Serial LPs for at Least Temporary Amelioration of Neonatal Post-hemorrhagic Hydrocephalus. *Pediatrics.* 1985; 75:719–724
- [3] Dandy WE. Extirpation of the Choroid Plexus of the Lateral Ventricle in Communicating Hydrocephalus. *Ann Surg.* 1918; 68:569–579
- [4] Griffith HB, Jamjoom AB. The Treatment of Childhood Hydrocephalus by Choroid Plexus Coagulation and Artificial Cerebrospinal Fluid Perfusion. *Br J Neurosurg.* 1990; 4:95–100
- [5] Gangemi M, Maiuri F, Naddeo M, Godano U, Mascari C, Broggi G, Ferroli P. Endoscopic third ventriculostomy in idiopathic normal pressure hydrocephalus: an Italian multicenter study. *Neurosurgery.* 2008; 63:62–7; discussion 67–9
- [6] Handler MH, Abbott R, Lee M. A Near-Fatal Complication of Endoscopic Third Ventriculostomy: Case Report. *Neurosurgery.* 1994; 35:525–528
- [7] McLaughlin MR, Wahlig JB, Kaufmann AM, Albright AL. Traumatic Basilar Aneurysm After Endoscopic Third Ventriculostomy: Case Report. *Neurosurgery.* 1997; 41:1400–1404
- [8] Kulkarni AV, Drake JM, Mallucci CL, Sgouros S, Roth J, Constantini S. Endoscopic third ventriculostomy in the treatment of childhood hydrocephalus. *J Pediatr.* 2009; 155:254–9 e1
- [9] Kulkarni AV, Drake JM, Kestle JR, Mallucci CL, Sgouros S, Constantini S. Predicting who will benefit from endoscopic third ventriculostomy compared with shunt insertion in childhood hydrocephalus using the ETV Success Score. *J Neurosurg Pediatr.* 2010; 6:310–315
- [10] Naftel RP, Reed GT, Kulkarni AV, Wellons JC. Evaluating the Children's Hospital of Alabama endoscopic third ventriculostomy experience using the Endoscopic Third Ventriculostomy Success Score: an external validation study. *J Neurosurg Pediatr.* 2011; 8:494–501
- [11] Durnford AJ, Kirkham FJ, Mathad N, Sparrow OC. Endoscopic third ventriculostomy in the treatment of childhood hydrocephalus: validation of a success score that predicts long-term outcome. *J Neurosurg Pediatr.* 2011; 8:489–493
- [12] Jones RFC, Currie BG, Kwok BCT. Ventriculopleural Shunts for Hydrocephalus: A Useful Alternative. *Neurosurgery.* 1988; 23:753–755
- [13] James HE, Tibbs PA. Diverse Clinical Application of Percutaneous Lumboperitoneal Shunts. *Neurosurgery.* 1981; 8:39–42
- [14] Cozzens JW, Chandler JP. Increased Risk of Distal Ventriculoperitoneal Shunt Obstruction Associated With Slit Valves or Distal Slits in the Peritoneal Catheter. *J Neurosurg.* 1997; 87:682–686
- [15] Dan NG, Wade MJ. The Incidence of Epilepsy After Ventricular Shunting Procedures. *J Neurosurg.* 1986; 65:19–21
- [16] Berger MS, Baumeister B, Geyer JR, Milstein J, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. *J Neurosurg.* 1991; 74:872–877

- [17] Jimenez DF, Keating R, Goodrich JT. Silicone Allergy in Ventriculoperitoneal Shunts. *Childs Nerv Syst.* 1994; 10:59–63
- [18] Moazam F, Glenn JD, Kaplan BJ, et al. Inguinal Hernias After Ventriculoperitoneal Shunt Procedures in Pediatric Patients. *Surg Gynecol Obstet.* 1984; 159:570–572
- [19] Bryant MS, Bremer AM, Tepas JJ, et al. Abdominal Complications of Ventriculoperitoneal Shunts. *Am Surg.* 1988; 54:50–55
- [20] Ram Z, Findler G, Guttman I, et al. Ventriculoperitoneal Shunt Malfunction due to Migration of the Abdominal Catheter into the Scrotum. *J Pediatr Surg.* 1987; 22:1045–1046
- [21] Rush DS, Walsh JW. Abdominal Complications of CSF-Peritoneal Shunts. *Monogr Neural Sci.* 1982; 8:52–54
- [22] Alonso-Vanegas M, Alvarez JL, Delgado L, et al. Gastric Perforation due to Ventriculo-Peritoneal Shunt. *Pediatr Neurosurg.* 1994; 21:192–194
- [23] Lourie H, Bajwa S. Transdiaphragmatic Migration of a Ventriculoperitoneal Catheter. *Neurosurgery.* 1985; 17:324–326
- [24] Sakoda TH, Maxwell JA, Brackett CE. Intestinal Volvulus Secondary to a Ventriculoperitoneal Shunt. Case Report. *J Neurosurg.* 1971; 35:95–96
- [25] Couldwell WT, LeMay DR, McComb JG. Experience with Use of Extended Length Peritoneal Shunt Catheters. *J Neurosurg.* 1996; 85:425–427
- [26] Pascual JMS, Prakash UBS. Development of Pulmonary Hypertension After Placement of a Ventriculoatrial Shunt. *Mayo Clin Proc.* 1993; 68:1177–1182
- [27] Chumas PD, Kulkarni AV, Drake JM, Hoffman HJ, Humphreys RP, Rutka JT. Lumboperitoneal Shunting: A Retrospective Study in the Pediatric Population. *Neurosurgery.* 1993; 32:376–383
- [28] Welch K, Shillito J, Strand R, Fischer EG, Winston KR. Chiari I "malformation": An acquired disorder? *J Neurosurg.* 1982; 55:604–609
- [29] Chumas PD, Armstrong DC, Drake JM, et al. Tonsillar Herniation: The Rule Rather than the Exception After Lumboperitoneal Shunting in the Pediatric Population. *J Neurosurg.* 1993; 78:568–573
- [30] Payner TD, Prenger E, Berger TS, Crone KR. Acquired Chiari Malformations: Incidence, Diagnosis, and Management. *Neurosurgery.* 1994; 34:429–434
- [31] Kessler LA, Dugan P, Concannon JP. Systemic Metastases of Medulloblastoma Promoted by Shunting. *Surg Neurol.* 1975; 3:147–152
- [32] Czosnyka Z, Czosnyka M, Richards HK, Pickard JD. Laboratory testing of hydrocephalus shunts – conclusion of the U.K. Shunt evaluation programme. *Acta Neurochir (Wien).* 2002; 144:525–38; discussion 538
- [33] Savitz MH, Bobroff IM. Low incidence of delayed intracerebral hemorrhage secondary to ventriculoperitoneal shunt insertion. *J Neurosurg.* 1999; 91:32–34
- [34] French BN, Swanson M. Radionuclide Imaging Shuntography for the Evaluation of Shunt Patency. *Monogr Neural Sci.* 1982; 8:39–42
- [35] Chan KH, Mann KS. Prolonged Therapeutic External Ventricular Drainage: A Prospective Study. *Neurosurgery.* 1988; 23:436–438
- [36] Mann KS, Yue CP, Ong GB. Percutaneous Sump Drainage: A Palliation for Oft-Recurring Intracranial Cystic Lesions. *Surg Neurol.* 1983; 19:86–90
- [37] Hassan M, Higashi S, Yamashita J. Risks in Using Siphon-Reducing Devices in Adult Patients with Normal-Pressure Hydrocephalus: Bench Test Investigations with Delta valves. *J Neurosurg.* 1996; 84:634–641
- [38] Boch A-L, Hermelin É, Sainte-Rose C, Sgouros S. Mechanical Dysfunction of Ventriculoperitoneal Shunts Caused by Calcification of the Silicone Rubber Catheter. *J Neurosurg.* 1998; 88:975–982
- [39] Pudenz RH, Foltz EL. Hydrocephalus: Overdrainage by Ventricular Shunts. A Review and Recommendations. *Surg Neurol.* 1991; 35:200–212
- [40] McLaurin RL, Olivi A. Slit-Ventricle Syndrome: Review of 15 Cases. *Pediatr Neurosci.* 1987; 13:118–124
- [41] Gruber R, Jenny P, Herzog B. Experiences with the Anti-Siphon Device (ASD) in Shunt Therapy of Pediatric Hydrocephalus. *J Neurosurg.* 1984; 61:156–162
- [42] Foltz EL, Blanks JP. Symptomatic Low Intraventricular Pressure in Shunted Hydrocephalus. *J Neurosurg.* 1988; 68:401–408
- [43] O'Hayon BB, Drake JM, Ossip MG, Tuli S, Clarke M. Frontal and Occipital Horn Ratio: A Linear Estimate of Ventricular Size for Multiple Imaging Modalities in Pediatric Hydrocephalus. *Pediatric Neurosurgery.* 1998; 29:245–249
- [44] Teo C, Morris W. Slit Ventricle Syndrome. *Contemp Neurosurg.* 1999; 21:1–4
- [45] Engel M, Carmel PW, Chutorian AM. Increased Intraventricular Pressure Without Ventriculomegaly in Children with Shunts: "Normal Volume" Hydrocephalus. *Neurosurgery.* 1979; 5:549–552
- [46] Kiekens R, Mortier W, Pothmann R. The Slit-Ventricle Syndrome After Shunting in Hydrocephalic Children. *Neuropediatrics.* 1982; 13:190–194
- [47] Hyde-Rowan MD, Rekeate HL, Nulsen FE. Reexpansion of Previously Collapsed Ventricles: The Slit Ventricle Syndrome. *J Neurosurg.* 1982; 56:536–539
- [48] Salmon JH. The Collapsed Ventricle: Management and Prevention. *Surg Neurol.* 1978; 9:349–352
- [49] Portnoy HD, Schult RR, Fox JL, et al. Anti-Siphon and Reversible Occlusion Valves for Shunting in Hydrocephalus and Preventing Postshunt Subdural Hematoma. *J Neurosurg.* 1973; 38:729–738
- [50] Epstein FJ, Fleischer AS, Hochwald GM, et al. Subtemporal Craniectomy for Recurrent Shunt Obstruction Secondary to Small Ventricles. *J Neurosurg.* 1974; 41:29–31
- [51] Holness RO, Hoffman HJ, Hendrick EB. Subtemporal Decompression for the Slit-Ventricle Syndrome After Shunting in Hydrocephalic Children. *Childs Brain.* 1979; 5:137–144
- [52] Linder M, Diehl J, Sklar FH. Subtemporal Decompressions for Shunt-Dependent Ventricles: Mechanism of Action. *Surg Neurol.* 1983; 19:520–523
- [53] Reddy K, Fewer HD, West M, Hill NC. Slit Ventricle Syndrome with Aqueduct Stenosis: Third Ventriculostomy as Definitive Treatment. *Neurosurgery.* 1988; 23:756–759
- [54] Puca A, Fernandez E, Colosimo C, et al. Hydrocephalus and Macrocrania: Surgical or Non-Surgical Treatment of Postshunting Subdural Hematoma. *Surg Neurol.* 1996; 45:76–82
- [55] Hoppe-Hirsch E, Sainte Rose C, Renier D, Hirsch J-F. Pericerebral Collections After Shunting. *Childs Nerv Syst.* 1987; 3:97–102
- [56] McCulloch DC, Fox JL. Negative Intracranial Pressure Hydrocephalus in Adults with Shunts and its Relationship to the Production of Subdural Hematoma. *J Neurosurg.* 1974; 40:372–375
- [57] Schut L. Comment on Puca A, et al.: Hydrocephalus and Macrocrania: Surgical or Non-Surgical Treatment of Postshunting Subdural Hematoma. *Surg Neurol.* 1996; 45
- [58] Dietrich U, Lumenta C, Sprick C, et al. Subdural Hematoma in a Case of Hydrocephalus and Macrocrania: Experience with a Pressure-Adjustable Valve. *Childs Nerv Syst.* 1987; 3:242–244
- [59] Kamano S, Nakano Y, Imanishi T, Hattori M. Management with a Programmable Pressure Valve of Subdural Hematomas Caused by a Ventriculoperitoneal Shunt: Case Report. *Surg Neurol.* 1991; 35:381–383
- [60] Illingworth RD. Subdural Hematoma After the Treatment of Chronic Hydrocephalus by Ventriculocaval Shunts. *J Neurol Neurosurg Psychiatry.* 1970; 33:95–99
- [61] Davidoff LM, Feiring EH. Subdural Hematoma Occurring in Surgically Treated Hydrocephalic Children with a Note on a Method of Handling Persistent Accumulations. *J Neurosurg.* 1963; 10:557–563
- [62] Kaufman B, Weiss MH, Young HF, Nulsen FE. Effects of Prolonged Cerebrospinal Fluid Shunting on the Skull and Brain. *J Neurosurg.* 1973; 38:288–297

- [63] Faulhauer K, Schmitz P. Overdrainage Phenomena in Shunt Treated Hydrocephalus. *Acta Neurochir.* 1978; 45:89–101
- [64] Al-Mufarrej F, Nolan C, Sookhai S, Broe P. Laparoscopic procedures in adults with ventriculoperitoneal shunts. *Surg Laparosc Endosc Percutan Tech.* 2005; 15:28–29
- [65] Neale ML, Falk GL. In vitro assessment of back pressure on ventriculoperitoneal shunt valves. Is laparoscopy safe? *Surg Endosc.* 1999; 13:512–515
- [66] Ravaoherisoa J, Meyer P, Afriat R, Meyer Y, Sauvanet E, Tricot A, Carli P. Laparoscopic surgery in a patient with ventriculoperitoneal shunt: monitoring of shunt function with transcranial Doppler. *Br J Anaesth.* 2004; 92:434–437
- [67] Baskin JJ, Vishteh AG, Wesche DE, Rekate HL, Carrión CA. Ventriculoperitoneal shunt failure as a complication of laparoscopic surgery. *J Soc Laparoendosc Surg.* 1998; 2:177–180
- [68] Collure DW, Bumpers HL, Luchette FA, Weaver WL, Hoover EL. Laparoscopic cholecystectomy in patients with ventriculoperitoneal (VP) shunts. *Surg Endosc.* 1995; 9:409–410

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Part VIII

Seizures

26	Seizure Classification and Anti-Convulsant Pharmacology	440
27	Special Types of Seizures	461

VIII

26 Seizure Classification and Anti-Convulsant Pharmacology

26.1 Seizure classification

Definition of a seizure: an abnormal paroxysmal cerebral neuronal discharge that results in alteration of sensation, motor function, behavior or consciousness. Seizures may be classified by type, etiology, and by epileptic syndromes.

26.1.1 Classification of major seizure types

Generalized seizures

26

1. Tonic–clonic (in any combination)
2. Absence
 - Typical
 - Atypical
 - Absence with special features: Myoclonic absence, Eyelid myoclonia
3. Myoclonic
 - Myoclonic
 - Myoclonic atonic
 - Myoclonic tonic
4. Clonic
5. Tonic
6. Atonic
7. Focal seizures
8. Unknown
 - Epileptic spasms

Primary generalized seizures

Primary generalized: bilaterally symmetrical and synchronous involving both cerebral hemispheres at the onset, no local onset, consciousness lost from the start. Represents $\approx 40\%$ of all seizures

1. generalized tonic-clonic (GTC) (née: grand-mal seizure): generalized seizure that evolves from tonic to clonic motor activity. This is a specific type and does NOT include partial seizures that generalize secondarily
2. clonic seizures: fairly symmetric, bilateral synchronous semirhythmic jerking of the UE & LE, usually with elbow flexion and knee extension
3. tonic seizures: sudden sustained increased tone with a characteristic guttural cry or grunt as air is forced through adducted vocal cords
4. absence (née: petit-mal seizure): impaired consciousness with mild or no motor involvement (see below)
 - a) typical absences
 - b) atypical absences: more heterogeneous with more variable EEG pattern than typical absence. Seizures may last longer
5. myoclonic seizures: shocklike body jerks (1 or more in succession) with generalized EEG discharges
6. atonic seizures (AKA astatic seizures or “drop attacks”): sudden brief loss of tone that may cause falls

Partial seizures

Partial (née focal seizure): implies one hemisphere involved at onset. About 57% of all seizures. A new onset of partial seizure represents a structural lesion until proven otherwise

1. simple partial seizure (no impairment of consciousness)
 - a) with motor signs (including Jacksonian)
 - b) with sensory symptoms (special sensory or somatosensory)
 - c) with autonomic signs or symptoms
 - d) with psychic symptoms (disturbance of higher cerebral function)

2. complex partial seizure (many used to be classified as psychomotor seizure, often attributed to temporal lobe but they can arise from any cortical area): any alteration of consciousness, usually LOC or automatisms (including lip smacking, chewing, or picking with the fingers) with autonomic aura (usually an epigastric rising sensation)
 - a) simple partial onset followed by impairment of consciousness (may have premonitory aura)
 - without automatisms
 - with automatisms
 - b) with impairment of consciousness at onset
 - without automatisms (impairment of consciousness only)
 - with automatisms
 - c) partial seizure with secondary generalization
 - simple partial evolving to generalized
 - complex partial evolving to generalized
 - simple partial evolving to complex partial evolving to generalized

Unclassified epileptic seizures

≈ 3% of all seizures

26

26.1.2 Epileptic syndromes

General information

This list is not all inclusive (see reference^{1,2}).

1. symptomatic (AKA “secondary”): seizures of known etiology (e.g. stroke, tumor...)
 - a) temporal lobe epilepsies:
 - mesial temporal sclerosis: see below
2. idiopathic (AKA “primary”): no underlying cause. Includes:
 - a) juvenile myoclonic epilepsy: see below
3. cryptogenic: seizures presumed to be symptomatic but with unknown etiology
 - a) West syndrome (infantile spasms, Blitz-Nick-Salaam Krämpfe): see below
 - b) Lennox-Gastaut syndrome: see below
4. special syndromes: situation-related seizures
 - a) febrile seizures (p. 467)
 - b) seizures occurring only with acute metabolic or toxic event: e.g. alcohol

KEY distinctions (having therapeutic implications)

In generalized tonic-clonic seizures: primary generalized vs. partial with secondary generalization (often, local onset may not be observed).

In staring spells: absence vs. complex partial.

Epilepsy

A disorder, not a single disease. Characterized by recurrent (2 or more), unprovoked seizures.

Absence seizure

Formerly called petit-mal seizure. Impaired consciousness with mild or no motor involvement (automatisms occur more commonly with bursts lasting > 7 secs). No post-ictal confusion. Aura rare. May be induced by hyperventilation × 2–3 mins. EEG shows spike and wave at exactly 3 per second.

Uncinate seizures

Obsolete term: “uncal fits.” Seizures originating in the inferior medial temporal lobe, usually in the hippocampal region. May produce olfactory hallucinations (kakosmia or cacosmia: the perception of bad odors where none exist).

Mesial temporal sclerosis

See references.^{3,4}

The most common cause of intractable temporal lobe epilepsy. Specific pathologic basis: hippocampal sclerosis (cell loss in hippocampus on one side). Characteristics are shown in □ Table 26.1. See also differential diagnosis (p. 1386).

Table 26.1 Syndrome of mesial temporal-lobe epilepsy ⁵
History
• higher incidence of complicated febrile seizures than in other types of epilepsy • common family history of epilepsy • onset in latter half of first decade of life • auras in isolation are common • infrequent secondarily generalized seizures • seizures often remit for several years until adolescence or early adulthood • seizures often become medically refractory • common interictal behavioral disturbances (especially depression)
Clinical features of seizures
• most have aura (especially epigastric, emotional, olfactory or gustatory) × several secs • CPS often begin with arrest & stare; oroalimentary & complex automatisms are common. Posturing of contralateral arm may occur. Seizure usually lasts 1–2 mins • postictal disorientation, recent-memory deficit, amnesia of ictus and (in dominant hemisphere) aphasia usually lasts several mins
Neurologic and laboratory features
• neuro exam: normal except memory deficit • MRI: hippocampal atrophy and signal alteration with ipsilateral dilatation of temporal horn of lateral ventricle • unilateral or bilateral independent anterior temporal EEG spikes with maximal amplitude in basal electrodes • external ictal EEG activity only with CPS, usually initial or delayed focal rhythmic onset pattern of 5–7 Hz, maximal in 1 basal temporal derivation • interictal fluorodeoxyglucose PETscan: hypometabolism in temporal lobe and possibly ipsilateral thalamus and basal ganglia • neuropsychological testing: memory dysfunction specific to involved temporal lobe • Wada test (p.1553): amnesia with contralateral amobarbital injection

Adult seizures are initially responsive to medical therapy but become more varied and refractory, and may respond to seizure surgery.

Juvenile myoclonic epilepsy

See reference.⁶

Sometimes called bilateral myoclonus. 5–10% of cases of epilepsy. An idiopathic generalized epilepsy syndrome with age-related onset consisting of 3 seizure types:

1. myoclonic jerks: predominantly after waking
2. generalized tonic-clonic seizures
3. absence

EEG → polyspike discharges. Strong family history (some studies showing linkage to the HLA region on the short arm of chromosome 6).

Most are responsive to depakene.

West syndrome

This term is being used less frequently as it appears not to be a homogeneous group and as specific etiologies for infantile spasms are identified. Classically a seizure disorder that usually appears in first year of life, and consists of recurrent, gross flexion and occasionally extension of the trunk and limbs (massive myoclonus, AKA infantile spasms, AKA salaam seizures, AKA jackknife spasms). Seizures tend to diminish with age, often abating by 5 yrs. Usually associated with mental retardation. 50% may develop complex-partial seizures, some of the rest may develop Lennox-Gastaut syndrome (see below). An associated brain lesion may be found in some.

EEG → the majority show either interictal hypsarrhythmia (huge spike/wave plus slow wave resembling muscle artifact) or modified hypsarrhythmia at some point.

Usually dramatic response of seizures and EEG findings to ACTH or corticosteroids.

Lennox-Gastaut syndrome

Rare condition that begins in childhood as atonic seizures (“drop attacks”). Often develops into tonic seizures with mental retardation. Seizures are often polymorphic, difficult to treat medically, and may occur as often as 50 per day. May also present with status epilepticus. Approximately 50% of patients have reduced seizures with valproic acid. Corpus callosotomy may reduce the number of atonic seizures.

26.1.3 Miscellaneous seizure information

Factors that lower the seizure threshold

Factors that lower the seizure threshold (i.e. make it easier to provoke a seizure) in individuals with or without a prior seizure history include many items listed under Etiologies of New onset seizures (see below) as well as:

1. sleep deprivation
2. hyperventilation
3. photic stimulation (in some)
4. infection: systemic (febrile seizures (p.467)), CNS (meningitis...)
5. metabolic disturbances: electrolyte imbalance (especially profound hypoglycemia), pH disturbance (especially alkalosis), drugs... (see below)
6. head trauma: closed head injury, penetrating trauma (p.463)
7. cerebral ischemia: stroke (see below)
8. “kindling”: a concept that repeated seizures may facilitate the development of later seizures

Todd’s paralysis

A post-ictal phenomena in which there is partial or total paralysis usually in areas involved in a partial seizure. More common in patients with structural lesions as the source of the seizure. The paralysis usually resolves slowly over a period of an hour or so. Thought to be due to depletion of neurons in the wake of the extensive electrical discharges of a seizure. Other similar phenomena include post-ictal aphasia and hemianopsia.

26.2 Antiepileptic drugs

26.2.1 General information

The goal of antiepileptic drugs (AEDs) is seizure control (a contentious term, usually taken as reduction of seizure frequency and severity to the point to permit the patient to live a normal lifestyle without epilepsy-related limitations) with minimal or no drug toxicity. $\approx 75\%$ of epileptics can achieve satisfactory seizure control with medical therapy.⁷

26.2.2 Classification of AEDs

AEDs can be grouped as shown in □ Table 26.2.

The following agents are considered “broad spectrum” (treat a variety of seizure types):

1. valproic acid
2. lamotrigine (Lamictal®)
3. levetiracetam (Keppra®)

These agents are not considered broad spectrum:

1. phenytoin (Dilantin® and others)
2. carbamazepine (Tegretol®)

Agents that interfere with platelet function and may increase the risk of bleeding complications:

1. valproic acid
2. phenytoin (Dilantin® and others)

Table 26.2 Classification of AEDs	
Drug	Indications ^a
Barbiturates	
pentobarbital (Nembutal®)	status
phenobarbital	status, GTC, partial Sz, febrile Sz, neonatal Sz
primidone (Mysoline®)	
Benzodiazepines	
clonazepam (Klonopin®)	Lennox-Gastaut, akinetic, myoclonic
clorazepate (Tranxene-SD®)	adj – partial Sz
diazepam (Valium®)	status
lorazepam (Ativan®)	status
GABA analogues	
gabapentin (Neurontin®)	adj – partial Sz
tiagabine (Gabitril®)	adj – partial Sz
Hydantoins	
fosphenytoin (Cerebyx®)	status, Sz during neurosurgery, short-term replacement for oral PHT
phenytoin (Dilantin®)	GTC, CP, Sz during or after neurosurgery
Phenyltriazenes	
lamotrigine (Lamictal®)	adj – partial Sz, adj – Lennox-Gastaut
Succinimides	
ethosuximide (Zarontin®)	ABS
methsuximide (Celontin®)	ABS refractory to other drugs
Miscellaneous	
acetazolamide (Diamox®)	
carbamazepine (Tegretol®, Carbatrol®)	partial Sz+complex symptomology, GTC, mixed Sz, □ not for absence
felbamate (Felbatol®)	use only with extreme caution – see text
levetiracetam (Keppra®)	adj – partial Sz
oxcarbazepine (Trileptal®)	mono or adj – partial Sz
topiramate (Topamax®)	adj – partial Sz or primarily GTC
valproate (Depakene®...)	CP (alone or with other types), ABS, adj – multiple Sz types
zonisamide (Zonegran®)	adj – partial Sz
^a Indications for seizure types (does not include other uses, e.g. for chronic pain). FDA approved indications are in bold, off-label indications appear in plain text. Abbreviations: ABS=absence, adj=adjunctive therapy, CP=complex partial, GTC=generalized tonic-clonic, PHT=phenytoin, Sz=seizure, status=status epilepticus	

26.2.3 Choice of antiepileptic drug

Antiepileptic drugs (AED) for various seizure types

Boldface drugs are drug of choice (DOC).

1. primary generalized
 - a) GTC (generalized tonic-clonic):
 - valproic acid (VA) (p.451): if no evidence of focality some studies show fewer side effects and better control than PHT
 - carbamazepine (p.449)
 - phenytoin (PHT) (p.446)
 - phenobarbital (PB) (p.451)
 - primidone (PRM) (p.452)
 - b) absence:
 - ethosuximide
 - valproic acid (VA)
 - clonazepam
 - methsuximide (p.453)
 - c) myoclonic → benzodiazepines
 - d) tonic or atonic:
 - benzodiazepines
 - felbamate (p.453)
 - vigabatrin (p.456)
2. partial (simple or complex, with or without secondary generalization): the well controlled Veterans Administration Cooperative Study⁸ ranked the following drugs (based on seizure control and side effects) in this order (VA may compare favorably with CBZ for secondarily GTC, but is less effective for complex partial seizures⁹):
 - a) carbamazepine (CBZ): most effective, least side effects
 - b) phenytoin (PHT): ↓
 - c) phenobarbital (PB): ↓
 - d) primidone (PRM): slightly less effective, more side effects
3. second line drugs for any of the above seizure types:
 - a) valproate
 - b) lamotrigine (p.456): effective for many types of generalized seizures, but are not FDA approved for this yet
 - c) topiramate (p.456): effective for many types of generalized seizures, but are not FDA approved for this yet

26.2.4 Anticonvulsant pharmacology

See reference.¹⁰

General guidelines

Monotherapy versus polytherapy

1. increase a given medication until seizures are controlled or side effects become intolerable (do not rely solely on therapeutic levels, which is only the range in which most patients have seizure control without side effects)
2. try monotherapy with a different drugs before resorting to two drugs together. 80% of epileptics can be controlled on monotherapy, however, failure of monotherapy indicates an 80% chance that the seizures will not be controllable pharmacologically. Only ≈ 10% benefit significantly from the addition of a second drug.⁹ When > 2 AEDs are required, consider nonepileptic seizures (p.464)
3. when first evaluating patients on multiple drugs, withdraw the most sedating ones first (usually barbiturates and clonazepam)

Generally, dosing intervals should be less than one half-life. Without loading dose, it takes about 5 half-lives to reach steady state.

Many AEDs affect liver function tests (LFTs), however, only rarely do the drugs cause enough hepatic dysfunction to warrant discontinuation. Guideline: discontinue an AED if the GGT exceeds twice normal.

Specific anticonvulsants

Table 26.3 Anticonvulsants: Abbreviations	
AED	antiepileptic drug
ABS	absence
EC	enteric coated
DIV	divided
DOC	drug of choice
GTC	generalized tonic-clonic seizure
S/C-P	simple or complex partial
Pharmacokinetics: Unless otherwise specified, numbers are given for oral dosing form.	
t _{1/2}	half-life
t _{PEAK}	time to peak serum level
t _{SS}	time to steady state (approximately 5 × t _{1/2})
t _{D /C}	time to discontinue (recommended withdrawal period over which drug should be tapered)
MDF	minimum dosing frequency. “Therapeutic level” is the average therapeutic range.

Drug info: Phenytoin (PHT) (Dilantin®)

Indications

GTC, S/C-P, occasionally in ABS.

Pharmacokinetics

Pharmacokinetics are complicated: at low concentrations, kinetics are 1st order (elimination proportional to concentration), metabolism saturates near the therapeutic level resulting in zero-order kinetics (elimination at a constant rate). ≈ 90%of total drug is protein bound. Oral bioavailability is ≈ 90%whereas IV bioavailability is ≈ 95% this small difference may be significant when patients are near limits of therapeutic range (due to zero-order kinetics).

Table 26.4 Pharmacokinetics of phenytoin

t _{1/2} (half-life)	t _{PEAK} (peak serum levels)	t _{SS} (steady state)	t _{D/C} (discontinue)	Therapeutic level ^a
≈ 24 hrs (range: 9–140 hrs) ^b	oral suspension: 1.5–3 hrs regular capsules: 1.5–3 hrs extended release capsules: 4–12 hrs	7–21 days	4 wks	10–20 mcg/ml

^atherapeutic level as measured in most labs: 10–20 mcg/ml (NB: it is the free PHT that is the important moiety; this is usually ≈ 1% of total PHT, thus therapeutic free PHT levels are 1–2 mcg/ml; some labs are able to measure free PHT directly).

^b t_{1/2} for phenytoin

Renal failure: dosage adjustment not needed. However, serum protein binding may be altered in uremia which can obfuscate interpretation of serum phenytoin levels. Eq (26.1) may be used to convert serum PHT concentration in a uremic patient C (observed), to the expected PHT level in nonuremic patients C (nonuremic).

$$C_{\text{nonuremic}} = \frac{C_{\text{observed}}}{\frac{0.1 \text{ } \Delta \text{ albumin } 0.1}{0.26:1}}$$

Oral dose

□ Adult: usual maintenance dose= 300–600 mg/d divided BID or TID (MDF=q d, for single daily dosing, either the phenytoin-sodium capsules or the extended release form should be used). Oral loading dose: 300 mg PO q 4 hrs until 17 mg/kg are given. Peds: oral maintenance: 4–7 mg/kg/d (MDF= BID). Supplied : (oral forms): 100 mg tablets of phenytoin-sodium (sodium-salt); 30 & 100 mg Kapseals® (extended release); 50 mg chewable Infatabs® (phenytoin-acid); oral suspension 125 mg/5-ml in 8 oz. (240 ml) bottles or individual 5 ml unit dose packs; pediatric suspension 30 mg/5-ml. Phenytek® 200 & 300 mg capsules.

Dosage changes

Table 26.5 Guidelines for changing phenytoin dosage

Present level (mg/dl)	Change to make
<6	100 mg/day
6–8	50 mg/day
>8	25–30 mg/day

Because of zero-order kinetics, at near-therapeutic levels a small dosage change can cause large level changes. Although computer models are necessary for a high degree of accuracy, the dosing change guidelines in □ Table 26.5 or the nomogram in □ Fig. 26.1 ¹¹ may be used as a quick approximation.

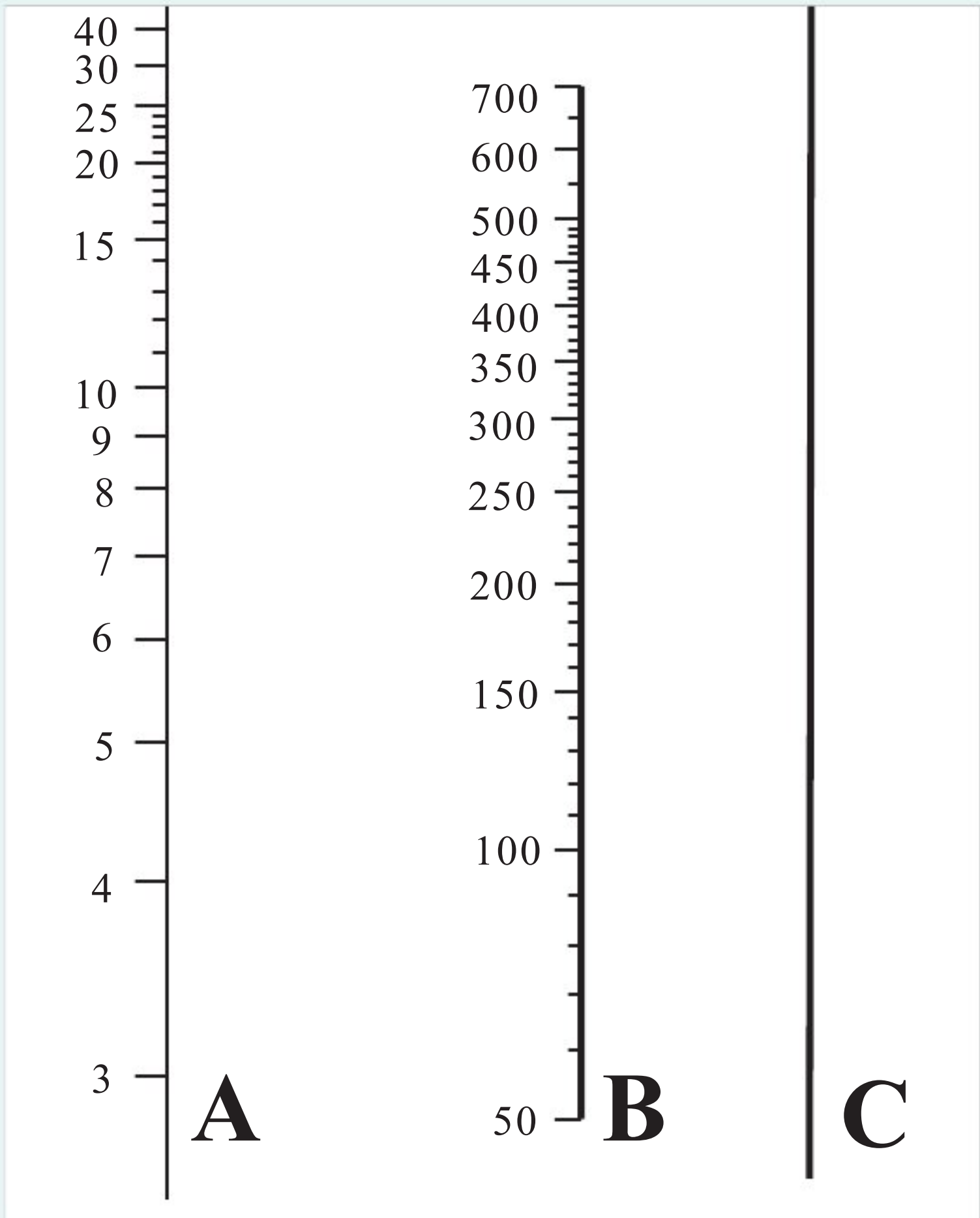


Fig. 26.1 Nomogram for adjusting phenytoin dose.
Directions for using nomogram (assumes steady state)
(a) draw line connecting serum level on line A with current dose on line B
(b) mark point where this line intersects line C
(c) connect point on C to the desired serum level on A
(d) read new dosage on line B (Reproduced from Therapeutic Drug Monitoring, “Predicting Phenytoin Dose - A Revised Nomogram”, Rambeck B, et al, Vol. 1, pp. 325-33, 1979, with permission)

GI absorption of phenytoin suspension or capsules may be decreased by up to 70% when given with nasogastric feedings of Osmolyte® or Isocal®,^{12,13} and the suspension has been reported to have erratic absorption. Hold NG feeding for 2 hrs before and 1 hour after phenytoin dose.

Parenteral dose

Phenytoin is a negative inotrope and can cause hypotension.

Conventional phenytoin may be given slow IVP or by IV drip (see below). The IM route should NOT be used (unreliable absorption, crystallization and sterile abscesses may develop). IV must be given

slowly to reduce risk of arrhythmias and hypotension, viz. Adult: <50 mg/min, Peds: <1–3 mg/kg/min. The only compatible solution is NS, inject at site nearest vein to avoid precipitation.

□ loading. Adult: 18 mg/kg slow IV. Peds: 20 mg/kg slow IV.

□ maintenance. Adult: 200–500 mg/d (MDF=q d). Most adults have therapeutic levels on 100 mg PO TID. Peds: 4–7 mg/kg/d (MDF=BID).

Drip loading method:

Requires cardiac monitoring, and BP check q 5 minutes.

□ Add 500 mg PHT to 50 ml NS to yield 10 mg/ml, run at 2 ml/min (20 mg/min) long enough to give 18 mg/kg (for 70 kg patient: 1200 mg over 60 mins). For more rapid administration, up to 40 mg/min may be used, or use fosphenytoin (see below). Decrease rate if hypotension occurs.

Fosphenytoin sodium injection

Fosphenytoin sodium (FOS) injection (Cerebyx®) is a newer formulation for administering IV phenytoin, and is indicated for short term use (≤ 5 days) when the enteral route is not usable. It is completely converted in vivo to phenytoin by organ and blood phosphatases with a conversion half-life of 10 minutes. Product labeling is given in terms of phenytoin equivalents (PE). Safety in pediatric patients has not been established. Supplied: 50 mg PE/ml in 2 & 10 ml vials (100 mg PE and 500 mg PE respectively).

Advantages of FOS (over conventional IV phenytoin):

1. less venous irritation (due to lower pH of 8.6–9 compared to 12 for phenytoin) resulting in less pain and IV extravasation
2. FOS is water soluble and therefore may be infused with dextrose or saline
3. tolerated well by IM injection (however, the IM route should not be used for status epilepticus)
4. does not come combined with propylene glycol (which can cause cardiac arrhythmias and/or hypotension itself)
5. the maximum administration rate is $3 \times$ as fast (i.e. 150 mg PE/min)

Side effects of phenytoin

May interfere with cognitive function. May produce SLE-like syndrome, hepatic granulomas, megaloblastic anemia, cerebellar degeneration (chronic doses), hirsutism, gingival hypertrophy, hemorrhage in newborn if mother on PHT, toxic epidermal necrolysis (Stevens-Johnson variant). PHT antagonizes vitamin D $\rightarrow$ osteomalacia and rickets. Most hypersensitivity reactions occur within 2 months of initiating therapy.¹² In cases of maculopapular erythematous rash, the drug may be stopped and the patient may be rechallenged; often the rash will not recur the second time. Teratogenic (fetal hydantoin syndrome¹⁴).

Signs of phenytoin toxicity may develop at concentrations above 20 mcg/ml (toxicity is more common at levels >30 mcg/ml) and include nystagmus (may also occur at therapeutic levels), diplopia, ataxia, asterixis, slurred speech, confusion, and CNS depression.

Drug-drug interactions: fluoxetine (Prozac®) results in elevated phenytoin levels (ave: 161% above baseline).¹⁵ Phenytoin may impair the efficacy of: corticosteroids, warfarin, digoxin, doxycycline, estrogens, furosemide, oral contraceptives, quinidine, rifampin, theophylline, vitamin D.

Drug info: Carbamazepine (CBZ) (Tegretol®)

Indications

Partial seizures with or without secondary generalization. Trigeminal neuralgia. An IV form for use in e.g. status epilepticus is in development.

Dose

□ oral route. Adult range: 600–2000 mg/d. Peds: 20–30 mg/kg/d. MDF=BID.

Before starting, check: CBC & platelet count (consider reticulocyte count) & serum Fe. Package insert says “recheck at frequent intervals, perhaps q week $\times$ 3 mos, then q month $\times$ 3 yrs.”

Do not start CBZ (or discontinue it if patient already on CBZ) if: WBC $<4K$, RBC $<3 \times 10^6$, Hct $<32\%$, platelets $<100K$, reticulocytes $<0.3\%$, Fe >150 mcg%

Start low and increment slowly: 200 mg PO q d $\times$ 1 wk, BID $\times$ 1 wk, TID $\times$ 1 wk. As an inpatient, dosage changes may be made every 3 days, monitoring for signs of side effects. As an outpatient, changes should be made only $\approx$ weekly, with levels after each change. Carbatrol® (extended release CBZ) is usually dosed BID.

Supplied : oral form. Scored tabs 200 mg. Chewable scored tabs 100 mg. Suspension 100 mg/5-ml. IV form: not available in the U.S. at the time of this writing. Carbatrol® (extended release CBZ): 200 & 300 mg tablets.

Caveats with oral forms: oral absorption is erratic, and smaller more frequent doses are preferred.¹⁶ Oral suspension is absorbed more readily, and also □ should not be administered simultaneously with other liquid medicinal agents as it may result in the precipitation of a rubbery, orange mass. □ May aggravate hyponatremia by SIADH-like effect.

Pharmacokinetics

Table 26.6 Pharmacokinetics of carbamazepine				
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	t _{D/C} (discontinue)	Therapeutic level(mcg/ml) ^a
single dose: 20–55 hrs after chronic therapy: 10–30 hrs (adults), 8–20 hrs (peds)	4–24 hrs	up to 10 days ^b	4 wks	6–12
^a may be misleading since the active metabolite carbamazepine-10,11-epoxide may cause toxicity and must be assayed separately				
^b t _{SS} may subsequently fall due to autoinduction which plateaus at 4–6 wks				

CBZ induces hepatic enzymes that result in increased metabolism of itself (autoinduction) as well as other drugs over a period of ≈ 3–4 weeks.

Side effects

□ Drug-drug interaction: caution, cimetidine, erythromycin and isoniazid may cause dramatic elevation of CBZ levels due to inhibition of hepatic cytochrome oxidase that degrades CBZ.¹⁷ Side effects include:

- 1. drowsiness and GI upset: minimized by slow dose escalation
- 2. relative leukopenia in many: usually does not require discontinuing drug
- 3. transient diplopia
- 4. ataxia
- 5. less effect on cognitive function than PHT
- 6. hematological toxicity: rare. May be serious → agranulocytosis & aplastic anemia
- 7. Stevens-Johnson syndrome
- 8. SIADH
- 9. hepatitis (occasionally fatal) reported

Drug info: Oxcarbazepine (Trileptal®)

Very similar efficacy profile to carbamazepine with the following differences:

- 1. there is no autoinduction (C-P450 is not involved in metabolism) and therefore minimal drug-drug interactions
- 2. no blood testing is required since:
 - a) there is no liver toxicity
 - b) there is no hematologic toxicity
 - c) there is no need to check drug levels
- 3. dosing is BID
- 4. kinetics are linear
- 5. more expensive

Dose

□: starting dose for pain control is 150 mg PO BID, for seizures it is 300 mg PO BID. Maximum dose 2400 mg/day total. Supplied: 150, 300 & 600 mg scored tablets. 300 mg/5 ml oral suspension.

Drug info: Valproate

Available as valproic acid (Depakene®) and divalproex sodium (Depakote®).

Indications

Effective in primary GTC. Also useful in ABS with GTC, juvenile myoclonic epilepsy, and partial seizures (not FDA approved for latter). Also FDA approved for migraine prophylaxis. Note: severe GI upset and short half-life make valproic acid much less useful than Depakote® (divalproex sodium).

Dose

Adult range: 600–3000 mg/d. Peds range: 15–60 mg/kg/d. MDF=q d.
□ Start at 15 mg/kg/d, increment at 1 wk intervals by 5–10 mg/kg/d. Max recommended adult dose: 60 mg/kg/d. If daily dose >250 mg is required, it should be divided. Supplied: Oral: capsules 250 mg. Syrup 250 mg/5-ml. Depakote® (enteric coated) tabs: 125, 250, & 500 mg; sprinkle capsules 125 mg. IV: Depacon® for IV injection 500 mg/5 ml vial.

Pharmacokinetics

Table 26.7 Pharmacokinetics of valproate				
t _{1/2} (half-life)	t _{PEAK} (peak serum levels)	t _{SS} (steady state)	t _{D/C} (discontinue)	Therapeutic level(mcg/ml)
8–20 hrs	(uncoated) 1–4 hrs	2–4 days	4 wks	50–100

Valproic acid (VA) is 90%protein bound. ASA displaces VA from serum proteins.

Side effects

Serious side effects are rare. Pancreatitis has been reported, sometimes life-threatening. Fatal liver failure has occurred especially if age <2 yrs and in combination with other AEDs. Teratogenic (below). Drowsiness (temporary), minimal cognitive deficits, N/V (minimized with Depakote), liver dysfunction, hyperammonemia (even without liver dysfunction), weight gain, mild hair loss, tremor (dose related; similar to benign familial tremor; if severe and valproic acid is absolutely necessary, the tremor may be treated with beta blockers). May interfere with platelet function, caution with surgery on these patients.

Contraindications

□ Pregnancy: causes neural tube defects (NTD) in ≈ 1–2%of patients.¹⁸ Since a correlation between peak VA levels and the risk of NTDs has been found, if VA must be used, some experts recommend changing from BID to TID dosing. □ Patients ≤ 2 yrs age (risk of hepatotoxicity).

Drug info: Phenobarbital

Indications

Used as alternative in GTC and partial (not DOC). Had been DOC for febrile seizures, dubious benefit.¹⁹ About as effective as PHT, but very sedating. Also used for status epilepticus (p. 469).

Dose

Same dose PO, IV, or IM. MDF=q d.^{20,21} Start slowly to minimize sedation.
□ Adult loading: 20 mg/kg slow IV (administer at rate <100 mg/min). Maintenance: 30–250 mg/d (usually divided BID-TID). Peds loading: 15–20 mg/kg. Maintenance: 2–6 mg/kg/d (usually divided BID). Supplied: tabs 15 mg, 30 mg, 60 mg, 100 mg; elixir 20 mg/5-ml.

Pharmacokinetics

Table 26.8 Pharmacokinetics of phenobarbital				
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	t _{D/C} (discontinue)	Therapeutic level
adult: 5 d (range: 50–160 hrs) peds: 30–70 hrs	PO & IM: 1–6 hrs	16–21 days (may take up to 30 days)	≈ 6–8 wks (reduce ≈ 25% per week)	15–30 mcg/ml

Phenobarbital is a potent inducer of hepatic enzymes that metabolize other AEDs.

Side effects

Cognitive impairment (may be subtle and may outlast administration of the drug by at least several months¹⁹), thus avoid in peds; sedation; paradoxical hyperactivity (especially in peds); may cause hemorrhage in newborn if mother is on phenobarbital.

Drug info: Primidone (Mysoline®)

Indications

Same as phenobarbital (not DOC). NB: when used in combination therapy, low doses (50–125 mg/day) may add significant seizure control to the primary AED with few side effects.

Dose

□ Adult: 250–1500 mg/d. Peds: 15–30 mg/kg/d; MDF= BID.
Start at 125 mg/d × 1 wk, and inc. slowly to avoid sedation. Supplied: (oral only): scored tabs 50 mg, 250 mg; suspension 250 mg/5-ml.

Pharmacokinetics

Metabolites include phenylethylmalonamide (PEMA) and phenobarbital. Therefore always check phenobarbital level at same time as primidone level.

Table 26.9 Pharmacokinetics of primidone				
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	t _{D/C} (discontinue)	Therapeutic level (mcg/ml)
Primidone: 4–12 hrs derived phenobarbital: 50–160 hrs	2–5 hrs	up to 30 days	same as phenobarbital	primidone: 1–15 derived phenobarbital: 10–30

Side effects

Same as phenobarbital, plus: loss of libido, rare macrocytic anemia.

Drug info: Ethosuximide (Zarontin®)

Indications

DOC in ABS.

Dose

□ Adult: 500–1500 mg/d. Peds: 10–40 mg/kg/d; MDF=q d. Supplied: oral only; capsules 250 mg; syrup 250 mg/5-ml.

Pharmacokinetics

Table 26.10 Pharmacokinetics of ethosuximide			
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	Therapeutic level (mcg/ml)
adult: 40–70 hrs peds: 20–40 hrs	1–4 hrs	adult: up to 14 days peds: up to 7 days	40–100

Side effects

N/V; lethargy; hiccups; H/A; rarely: eosinophilia, leukopenia, erythema multiforme, Stevens-Johnson syndrome, SLE-like syndrome. Toxic levels→psychotic behavior.

Drug info: Methsuximide (Celontin®)

Indications
Indicated for absence seizures refractory to other drugs.

Dose
□ optimum dose must be determined by trial. Start with 300 mg PO q d, increase by 300 mg at weekly intervals PRN to a maximum of 1200 mg/d. Supplied: 150 & 300 mg capsules.

Drug info: Felbamate (Felbatol®)

□ CAUTION: Due to an unacceptably high rate of aplastic anemia and hepatic failure, felbamate (FBM) should not be used except in those circumstances where the benefit clearly outweighs the risk; then, hematologic consultation is recommended by the manufacturer. See Side effects below (also for drug-drug interactions).
FBM is efficacious for monotherapy and adjunctive therapy for partial seizures (complex and secondary generalization), and reduces the frequency of atonic and GTC seizures in Lennox-Gastaut syndrome.

Pharmacokinetics

Table 26.11 Pharmacokinetics of felbamate			
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	Therapeutic level (mcg/ml)
20–23 hrs	1–3 hrs	5–7 days	not established

Dose

Table 26.12 Effect of felbamate on other AED levels		
AED	Change in level	Recommended dosing change
phenytoin	↑ 30–50%	↓ 20–33%
carbamazepine	↓ 30% total ↑ 50–60% epoxide	↓ 20–33%
valproic acid	↑ 25–100%	↓ 33%

□: CAUTION see above. Felbamate is not to be used as a first-line drug. Patient or guardian should sign informed consent release. Start with 1200 mg/d divided BID, TID, or QID, and decrease other AEDS by one–third. Increase felbamate biweekly in 600 mg increments to usual dose of 1600–3600 mg/d (max: 45 mg/kg/d). Slow down increments and/or reduce other AEDs further if side effects become severe. Administer at upper end of range when used as monotherapy. Supplied: (oral only) 400 & 600 mg scored tablets; suspension 600 mg/5-ml.

Side effects

Felbamate has been associated with aplastic anemia (usually discovered after 5–30 wks of therapy) in ≈ 2–5 cases per million persons per yr, and hepatic failure (some fatal, necessitating baseline and serial LFTs every 1–2 wks). Other side effects: insomnia, anorexia, N/V, H/A. Felbamate is a potent metabolic inhibitor, thus it is necessary to reduce the dose of phenytoin, valproate or carbamazepine when used with felbamate²² (□ Table 26.12, general rule: drop dose by one third).

Drug info: Levetiracetam (Keppra®)

No identified drug-drug interactions. Less than 10%protein bound. Linear pharmacokinetics, no level monitoring needed.

Indications

Adjunctive therapy for partial onset Sz with secondary generalization in patients 4 years of age and older. Myoclonic seizures (juvenile myoclonic epilepsy). Generalized tonic-clonic.

Dose

□ start with 500 mg PO BID. Increment by 1000 mg/d q 2 weeks PRN to a maximum of 3000 mg/d. Keppra XR: the same dose of levetiracetam can be converted to Keppra XR for q d dosing.

IV: 500–1500 mg diluted in 100 ml of diluent (IR, D5 W, normal saline) infused over 15 minutes BID.

Supplied: 250, 500, 750 & 1000 mg scored film-coated tabs; 100 mg/ml oral solution. Keppra XR (extended release) 500 mg.

IV: 1 vial (5 ml) contains 500 mg.

Side effects

PO or IV: somnolence and fatigue in 15% Dizziness in 9% Asthenia 15%and infection 13%(nasopharyngitis and influenza may or may not have been related).

Keppra XR: somnolence 8% irritability 6%

Drug info: Clonazepam (Klonopin®)

A benzodiazepine derivative.

Indications

□ Not a recommended drug for seizures (see below).

Used for myoclonic, atonic, and absence seizures (in absence, less effective than valproate or ethosuximide, and tolerance may develop).

NB: clonazepam usually works very well for several months, and then tends to become less effective, leaving only the sedating effects. Also, many cases have been reported of patients having seizures during withdrawal, including status epilepticus (even in patients with no history of status). Thus, may need to taper this drug over 3–6 months.

Dose

□ Adult: start at 1.5 mg/d DIV TID, increase by 0.5–1 mg q 3 d, usual dosage range is 1–12 mg/d (max 20 mg/d); MDF=q d. Peds: start at 0.01–0.03 mg/kg/d DIV BID or TID, increase by 0.25–0.5 mg/kg/d q 3 d; usual dosage range is 0.01–0.02 mg/kg/d; MDF=q d. Supplied: oral only; scored tabs: 0.5 mg, 1 mg, 2 mg.

Pharmacokinetics

Table 26.13 Pharmacokinetics of clonazepam				
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	t _{D/C} (discontinue)	Therapeutic level (mcg/ml)
20–60 hrs	1–3 hrs	up to 14 days	≈ 3–6 months ^a	0.013–0.072
^a CAUTION: withdrawal seizures are common, see text above				

Side effects

Ataxia; drowsiness; behavior changes.

Drug info: Zonisamide (Zonegran®)

Indications
Adjunctive therapy for partial Sz in adults.

Drug info: Acetazolamide (Diamox®)

The anti-epileptic effect may be either due to direct inhibition of CNS carbonic anhydrase (which also reduces CSF production rate) or due to the slight CNS acidosis that results.

Indications
Centricephalic epilepsies (absence, nonfocal seizures). Best results are in absence seizures; however benefit has also been observed in GTC, myoclonic jerk.

Side effects
Do not use in first trimester of pregnancy (may be teratogenic). The diuretic effect causes renal loss of HCO₃ which may lead to an acidotic state with long-term therapy. A sulfonamide, therefore any typical reaction to this class may occur (anaphylaxis, fever, rash, Stevens-Johnson syndrome, toxic epidermal necrolysis...). Paresthesias: medication should be discontinued.

Dose
□ Adult: 8–30 mg/kg/d in divided doses (max 1 gm/d, higher doses do not improve control). When given with another AED, the suggested starting dose is 250 mg once daily, and this is gradually increased. Supplied: tablets 125, 250 mg. Diamox sequels® are sustained release 500 mg capsules. Sterile cryodessicated powder is also available in 500 mg vials for parenteral (IV) use.

Drug info: Gabapentin (Neurontin®)

Although developed to be a GABA agonist, it does not interact at any known GABA receptor. Efficacious for primary generalized seizures and partial seizures (with or without secondary generalization). Ineffective for absence seizures. Very low incidence of known side-effects. No known drug interactions (probably because it is renally excreted). Also used for central pain

Dose
□ Adult: 300 mg PO × 1 day 1; 300 mg BID day 2; 300 mg TID day 3; then increase rapidly up to usual doses of ≈ 800–1800 mg per day. Doses of 1800–3600 may be needed in intractable patients. □ Dosage must be reduced in patients with renal insufficiency or on dialysis, see Eq (7.1) to estimate. Supplied: 100, 300, 400, 600, 800 mg capsules. 50 mg/ml suspension.

Pharmacokinetics
Gabapentin is not metabolized, and 93% is excreted unchanged renally with plasma clearance directly proportional to creatinine clearance.²³ Does not affect hepatic microsomal enzymes, and does not affect metabolism of other AEDs. Antacids decrease bioavailability by ≈ 20%, therefore give gabapentin > 2 hrs after the antacid.²⁴

Table 26.14 Pharmacokinetics of gabapentin			
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	Therapeutic level (mcg/ml)
5–7 hrs ^a	2–3 hrs	1–2 days	not established
^a with normal renal function			

Side effects
Somnolence, dizziness, ataxia, fatigue, nystagmus; all reduce after 2–3 weeks of drug therapy. Increased appetite. Not known to be teratogenic.

Drug info: Lamotrigine (Lamictal®)

Anticonvulsant effect may be due to presynaptic inhibition of glutamate release.²³
Efficacious as adjunctive therapy for partial seizures (with or without secondary generalization) and Lennox-Gastaut syndrome. Preliminary data suggest it may also be useful as an adjunct for refractory generalized seizures, or as monotherapy for newly diagnosed partial or generalized seizures.²⁵ Also FDA approved for bipolar disorder.

Side effects
Somnolence, dizziness, diplopia. □ Serious rashes requiring hospitalization and discontinuation of therapy have been reported (rash usually begins 2 weeks after initiating therapy and may be severe and potentially life-threatening, including Stevens-Johnson syndrome (more of a concern with simultaneous use of valproate), and rarely, toxic epidermal necrolysis (TEN)). Incidence of significant epidermal reaction may be decreased by a slow ramping-up of dosage. May increase seizure frequency in some patients with severe myoclonic epilepsy of infancy.²⁶ Metabolism of lamotrigine is affected by concurrent use of other AEDs.

Dose
□ Adult: In adults receiving enzyme-inducing AEDs (PHT, CBZ, or phenobarbital), start with 50 mg PO q d x 2 wks, then 50 mg BID x 2 wks, then ↑ by 100 mg/d q week until the usual maintenance dose of 200–700 mg/d (divided into 2 doses) is reached. For patients on valproic acid (VA) alone, the maintenance dose was 100–200 mg/d (divided into 2 doses), and VA levels drop by ≈ 25% within a few weeks of starting lamotrigine. For patients on both enzyme-inducing AEDs and VA, the starting dose is 25 mg PO qod x 2 wks, then 25 mg qd x 2 wks, then ↑ by 25–50 mg/d q 1–2 wks up to a maintenance of 100–150 mg/d (divided into 2 doses). Instruct patients that rash, fever or lymphadenopathy may herald a serious reaction and that a physician should be contacted immediately. Peds: not indicated for use in patients <16 yrs old due to higher incidence of potentially life-threatening rash in the pediatric population.²³ Supplied: 25, 100, 150 & 200 mg tablets. 2, 5 & 25 mg chewable dispersible tablets.

Pharmacokinetics²⁵

Table 26.15 Pharmacokinetics of lamotrigine			
t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	Therapeutic level (mcg/ml)
24 hrs ^a	1.5–5 hrs	4–7 days	controversial ²⁷
^a half-life is shortened to ≈ 15 hrs by PHT and CBZ, whereas valproic acid increases it to 59 hrs			

Drug info: Vigabatrin

Indications
Effective in treating partial seizures. Less so for generalized seizures.

Dose
□ Adult: 1500–3000 mg/d.

Drug info: Topiramate (Topamax®)

May block voltage-sensitive sodium channels and enhance GABA activity at GABAA receptors and attenuate some glutamate receptors.²³

Indications²⁸
As an oral adjunct to other drugs in treating refractory partial onset seizures.

Dose

□ Adult: start with 50 mg/d and increase slowly up to 200–400 mg/d,²⁹ with no significant benefit noted at dosages > 600 mg/d.³⁰ Supplied: 25, 100, & 200 mg tabs.

Pharmacokinetics

30% is metabolized in the liver, the rest is excreted unchanged in the urine.

Table 26.16 Pharmacokinetics of topiramate		
t _{1/2} (half-life)	t _{ss} (steady state)	Therapeutic level
19–25 hrs	5–7 days	not established

Side effects

May increase phenytoin concentration by up to 25% Levels of topiramate are reduced by other AEDs (phenytoin, carbamazepine, valproic acid and possibly others).

Cognitive impairment (word finding difficulty, problems with concentration...), weight loss, dizziness, ataxia, diplopia, paresthesias, nervousness and confusion have been troublesome. ≈ 1.5% incidence of renal stones which usually pass spontaneously²³

Oligohidrosis (reduced sweating) and hyperthermia, primarily in children in association with elevated environmental temperatures and/or vigorous physical activity.

Drug info: Tiagabine (Gabitril®)

A GABA uptake inhibitor, with cognitive problems of a similar frequency to that with topiramate.³¹

□ Adult: start with 4 mg/d, increase weekly by 4–8 mg to a maximum of 32–56 mg (divided BID to QID). Supplied: 4, 12, 16 & 20 mg tablets.

Drug info: Lacosamide (Vimpat®)

Enhances slow inactivation of voltage gated sodium channels, affecting only neurons that are depolarized of active over a prolonged period (as in a seizure).

Indications

Partial onset seizures. Painful diabetic neuropathy.

Dose

□ Adult: 200–400 mg. Supplied: mg tabs.

26.2.5 Withdrawal of antiepileptic drugs

General information

Most seizure recurrences develop during the first 6 months after AED withdrawal.³²

Indications for AED withdrawal

There is no agreement on how long a patient should be seizure-free before withdrawal of anticonvulsants, nor is there agreement on the prognostic value of EEGs and on the best time period over which to withdraw AEDs.

The following is based on a study of 92 patients with idiopathic epilepsy, who had been free of seizures for two years.³³ Generalization, e.g. to posttraumatic seizures, may not be appropriate. Taper was by 1 “unit” q 2 weeks (where a unit is defined as 200 mg for CBZ or valproic acid, or 100 mg for PHT). Follow-up: mean = 26 mos (range: 6–62).

- 31 patients (34%) relapsed, with the average time to relapse being 8 mos (range: 1–36). Using actuarial methods, the risk for recurrence is 5.9%/month for 3 months, then 2.7%/month for 3 months, then 0.5%/month for 3 months. Factors found to affect the likelihood of relapse include:
1. seizure type: 37%relapse rate for generalized seizures; 16%for complex or simple partial; 54%for complex partial with secondary generalization
 2. number of seizures before control attained: those with ≥ 100 seizures before control had statistically significant higher relapse rate than those with < 100
 3. the number of drugs that had to be tried before single drug therapy successfully controlled seizures: 29%if 1st drug worked, 40%if a change to a 2nd drug was needed, and 80%if a change to a 3rd drug was required
 4. EEG class (□ Table 26.17): class 4 had worst prognosis for relapse. Epileptiform discharges on EEG serves to discourage AED withdrawal³⁴

- In a larger randomized study,³⁵ the most important factors identified to predict freedom from recurrent seizures were:
1. longer seizure-free period
 2. use of only one AED (vs. multiple AEDs)
 3. seizures other than tonic-clonic seizures

Withdrawal times

The recommended withdrawal times in □ Table 26.18 should be used only as guidelines.

26.2.6 Pregnancy and antiepileptic drugs

General information

Women of childbearing potential with epilepsy should undergo counseling regarding pregnancy.³⁶

Birth control

AEDs that induce liver microsomal cytochrome P450 enzymes (□ Table 26.19) increase the failure rate of oral contraceptives up to fourfold.³⁷ Patients desiring to use BCPs should employ barrier contraceptive measures until ovulation is consistently suppressed, and they should watch for breakthrough bleeding which may indicate a need for a change in the hormone dosage.³² Non-oral hormonal contraceptives (e.g. levonorgestrel implant (Norplant®)) circumvents first pass liver degradation but should combined with a barrier method because of declining effectiveness with time.

Table 26.17 EEG class and seizure relapse rate				
Class	EEG description		Relapse rate	No. of relapses/ patients at risk
	Before treatment	Before withdrawal		
1	normal	normal	34%	11/31
2	abnormal	normal	11%	4/35
3	abnormal	improved	50%	2/4
4	abnormal	unchanged	74%	14/19

Table 26.18 Recommended AED withdrawal times	
AED	Recommended withdrawal period
phenytoin, valproic acid, carbamazepine	2–4 weeks
phenobarbital	6–8 weeks (25% per week)
clonazepam	3–6 months; see CAUTION (p.454)

Table 26.19 Effect of AEDs on liver cytochrome P ₄₅₀ ^a	
Inducers	Noninducers
carbamazepine phenobarbital phenytoin felbamate primidone	valproic acid benzodiazepines gabapentin lamotrigine
^a references ^{32,38}	

Table 26.20 Changes in free AED levels during pregnancy ³⁹	
Drug	Change
carbamazepine	↓ 11%
phenobarbital	↓ 50%
phenytoin	↓ 31%
valproic acid	↑ 25%

Complications during pregnancy

Women with epilepsy have more complications with pregnancy than mothers without epilepsy, but >90%of pregnancies have a favorable outcome.³²

There is an increase in the number of gravid seizures in ≈ 17%(reported range: 17–30%) of epileptic women, which may be due to noncompliance or to changes of free drug levels of AEDs during pregnancy (□ Table 26.20). Isolated seizures can occasionally be deleterious, but usually cause no problem. Status epilepticus poses serious risk to mother and fetus during pregnancy and should be treated aggressively.

There is also a slightly increased risk of toxemia (HTN of pregnancy) and fetal loss.

Birth defects

The incidence of fetal malformations in offspring of patients with a known seizure disorder is ≈ 4–5% or approximately double that of the general population.⁴⁰ The degree to which this is due to the use of AEDs vs. genetic and environmental factors is unknown. All AEDs have the potential to cause deleterious effects on the infant. Polytherapy is associated with an increased risk over monotherapy in a more than additive manner.

Generally, the risk of seizures (with possible concomitant maternal and fetal hypoxia and acidosis) is felt to outweigh the teratogenic risk of most AEDs, but this must be evaluated on a case-by-case basis. Occasionally patients may be weaned off AEDs.

Specific drugs

Carbamazepine (CBZ) produced an increased incidence of “minor” malformations (but not of “major” malformations) in one study⁴¹ (this study may have had methodologic problems), and may increase the incidence of neural tube defects (NTD).⁴² In utero exposure to phenytoin may lead to the fetal hydantoin syndrome^{14,43} and a child with an IQ lower by ≈ 10 points.⁴⁴ Phenobarbital produced the highest incidence of major malformations (9.1%) in one prospective study⁴⁵ and was also associated with most of the increase in fetal death or anomalies in another study.⁴⁶ Valproate (VA) causes the highest incidence of NTD (1-2%⁸), which can be detected with amniocentesis and allow an abortion if desired. TID dosing may reduce the risk of NTD (p.451). Benzodiazepines given shortly before delivery can produce the “floppy infant syndrome”.⁴⁷ Similar effects may occur with other sedating AEDs such as phenobarbital.

Drug recommendations

A general consensus is that for most women of childbearing potential who require AEDs, that monotherapy with the lowest dose of CBZ that is effective is the method of choice if the seizure disorder is responsive to it.⁴⁸ If ineffective, then monotherapy with valproic acid (with TID dosing) is currently the recommended second choice. Folate supplementation (after confirming normal B12 levels) should be used in all.

References

[1] Commission on Classification and Terminology of the International League Against Epilepsy. Guidelines for Epidemiologic Studies on Epilepsy. *Epilepsia*. 1989; 30:389–399

[2] Mosewich RK, So EL. A Clinical Approach to the Classification of Seizures and Epileptic Syndromes. *Mayo Clin Proc*. 1996; 71:405–414

[3] French JA, Williamson PD, Thadani VM, et al. Characteristics of Medial Temporal Lobe Epilepsy. I. Results of History and Physical Examination. *Ann Neurol*. 1993; 34:774–780

[4] Williamson PD, French JA, Thadani VM, et al. Characteristics of Medial Temporal Lobe Epilepsy. II. Interictal and Ictal Scalp Electroencephalography, Neuropsychological Testing, Neuroimaging, Surgical Results, and Pathology. *Ann Neurol*. 1993; 34:781–787

[5] Engel JJ. Surgery for Seizures. *N Engl J Med*. 1996; 334:647–652

[6] Grunewald RA, Panayiotopoulos CP. Juvenile Myoclonic Epilepsy: A Review. *Arch Neurol*. 1993; 50:594–598

[7] Brodie MJ, Dichter MA. Antiepileptic Drugs. *N Engl J Med*. 1996; 334:168–175

[8] Mattson RH, Cramer JA, Collins JF, et al. Comparison of Carbamazepine, Phenobarbital, Phenytoin, and Primidone in Partial and Secondarily Generalized Tonic-Clonic Seizures. *N Engl J Med*. 1985; 313:145–151

[9] Mattson RH, Cramer JA, Collins JF, et al. A Comparison of Valproate with Carbamazepine for the Treatment of Complex Partial Seizures and Secondarily Generalized Tonic-Clonic Seizures in Adults. *N Engl J Med*. 1992; 327:765–771

[10] Drugs for Epilepsy. *Med Letter*. 1986; 28:91–93

[11] Rambeck B, Boenigk HE, Dunlop A, et al. Predicting Phenytoin Dose - A Revised Nomogram. *Ther Drug Monit*. 1979; 1:325–333

[12] Saklad JJ, Graves RH, Sharp WP. Interaction of Oral Phenytoin with Enteral Feedings. *J Parent Ent Nutr*. 1986; 10:322–323

[13] Worden JP, Wood CA, Workman CH. Phenytoin and Nasogastric Feedings. *Neurology*. 1984; 34

[14] Buehler BA, Delimont D, van Waes M, et al. Prenatal Prediction of Risk of the Fetal Hydantoin Syndrome. *N Engl J Med*. 1990; 322:1567–1572

[15] Public Health Service. Fluoxetine-Phenytoin Interaction. *FDA Medical Bulletin*. 1994; 24:3–4

[16] Winkler SR, Luer MS. Antiepileptic Drug Review: Part 1. *Surg Neurol*. 1998; 49:449–452

[17] Oles KS, Waqar M, Penry JK. Catastrophic Neurologic Signs due to Drug Interaction: Tegretol and Darvon. *Surg Neurol*. 1989; 32:144–151

[18] Oakeshott P, Hunt GM. Valproate and Spina Bifida. *Br Med J*. 1989; 298:1300–1301

[19] Farwell JR, Lee YI, Hirtz DG, Sulzbacher SI, et al. Phenobarbital for Febrile Seizures - Effects on Intelligence and on Seizure Recurrence. *N Engl J Med*. 1990; 322:364–369

[20] Wroblewski BA, Garvin WH. Once-Daily Administration of Phenobarbital in Adults: Clinical Efficacy and Benefit. *Arch Neurol*. 1985; 42:699–700

[21] Davis AG, Mutchie KD, Thompson JA, Myers GG. Once-Daily Dosing with Phenobarbital in Children with Seizure Disorders. *Pediatrics*. 1981; 68:824–827

[22] Felbamate. *Med Letter*. 1993; 35:107–109

[23] Winkler SR, Luer MS. Antiepileptic Drug Review: Part 2. *Surg Neurol*. 1998; 49:566–568

[24] Gabapentin - A new anticonvulsant. *Med Letter*. 1994; 36:39–40

[25] Lamotrigine for Epilepsy. *Med Letter*. 1995; 37:21–23

[26] Guerrini R, Dravet C, Genton P, et al. Lamotrigine and Seizure Aggravation in Severe Myoclonic Epilepsy. *Epilepsia*. 1998; 39:508–512

[27] Sondergaard Khinchi M, Nielsen KA, Dahl M, Wolf P. Lamotrigine therapeutic thresholds. *Seizure*. 2008; 17:391–395

[28] Topiramate for Epilepsy. *Med Letter*. 1997; 39:51–52

[29] Faught E, Wilder BJ, Ramsay RE, et al. Topiramate Placebo-Controlled Dose-Ranging Trial in Refractory Partial Epilepsy using 200-, 400-, and 600-mg Daily Dosages. *Neurology*. 1996; 46:1684–1690

[30] Privitera M, Fincham R, Penry J, et al. Topiramate Placebo-Controlled Dose-Ranging Trial in Refractory Partial Epilepsy using 600-, 800-, and 1,000-mg Daily Dosages. *Neurology*. 1996; 46:1678–1683

[31] Tiagabine for Epilepsy. *Med Letter*. 1998; 40:45–46

[32] Shuster EA. Epilepsy in Women. *Mayo Clin Proc*. 1996; 71:991–999

[33] Callaghan N, Garrett A, Goggin T. Withdrawal of Anticonvulsant Drugs in Patients Free of Seizures for Two Years. *N Engl J Med*. 1988; 318:942–946

[34] Anderson T, Braathen G, Persson A, et al. A Comparison Between One and Three Years of Treatment in Uncomplicated Childhood Epilepsy: A Prospective Study. II. The EEG as Predictor of Outcome After Withdrawal of Treatment. *Epilepsia*. 1997; 38:225–232

[35] Medical Research Council Antiepileptic Drug Withdrawal Study Group. Randomized study of antiepileptic drug withdrawal in patients in remission. *Lancet*. 1991; 337:1175–1180

[36] Delgado-Escueta A, Janz D. Consensus Guidelines: Preconception Counseling, Management, and Care of the Pregnant Woman with Epilepsy. *Neurology*. 1992; 42:149–160

[37] Mattson RH, Cramer JA, Darney PD, Naftolin F. Use of Oral Contraceptives by Women with Epilepsy. *JAMA*. 1986; 256:238–240

[38] Perucca E, Hedges A, Makki KA, et al. A Comparative Study of the Relative Enzyme Inducing Properties of Anticonvulsant Drugs in Epileptic Patients. *Br J Clin Pharmacol*. 1984; 18:401–410

[39] Yerby MS, Freil PN, McCormick K. Antiepileptic Drug Disposition During Pregnancy. *Neurology*. 1992; 42:12–16

[40] Dias MS, Sekhar LN. Intracranial Hemorrhage from Aneurysms and Arteriovenous Malformations during Pregnancy and the Puerperium. *Neurosurgery*. 1990; 27:855–866

[41] Jones KL, Lacro RV, Johnson KA, Adams J. Patterns of Malformations in the Children of Women Treated With Carbamazepine During Pregnancy. *N Engl J Med*. 1989; 310:1661–1666

[42] Rosa FW. Spina Bifida in Infants of Women Treated with Carbamazepine During Pregnancy. *N Engl J Med*. 1991; 324:674–677

[43] Hanson JW, Smith DW. The Fetal Hydantoin Syndrome. *J Pediatr*. 1975; 87:285–290

[44] Scolnik D, Nulman I, Rovet J, et al. Neurodevelopment of Children Exposed In Utero to Phenytoin and Carbamazepine Monotherapy. *JAMA*. 1994; 271:767–770

[45] Nakane Y, Okuma T, Takahashi R, et al. Multi-Institutional Study of the Teratogenicity and Fetal Toxicity of Antiepileptic Drugs: A Report of a Collaborative Study Group in Japan. *Epilepsia*. 1980; 21:663–680

[46] Waters CH, Belai Y, Gott PS, et al. Outcomes of Pregnancy Associated with Antiepileptic Drugs. *Arch Neurol*. 1994; 51:250–253

[47] Kanto JH. Use of Benzodiazepines During Pregnancy, Labor, and Lactation, with Particular Reference to Pharmacokinetic Considerations. *Drugs*. 1982; 23:354–380

[48] Saunders M. Epilepsy in Women of Childbearing Age: If Anticonvulsants Cannot be Avoided, Use Carbamazepine. *Br Med J*. 1989; 199

27 Special Types of Seizures

27.1 New onset seizures

27.1.1 General information

The age adjusted incidence of new onset seizures in Rochester, Minnesota was 44 per 100,000 person years.¹

27.1.2 Etiologies

In patients presenting with a first-time seizure, etiologies include (modified²):

1. following neurologic insult: either acutely (i.e. <1 week) or remotely (>1 week, and usually <3 mos from insult)
 - a) stroke: 4.2% had a seizure within 14 days of a stroke. Risk increased with severity of stroke³
 - b) head trauma: closed head injury, penetrating trauma (p.463)
 - c) CNS infection: meningitis, cerebral abscess, subdural empyema
 - d) febrile seizures (p.467)
 - e) birth asphyxia
2. underlying CNS abnormality
 - a) congenital CNS abnormalities
 - b) degenerative CNS disease
 - c) CNS tumor: metastatic or primary
 - d) hydrocephalus
 - e) AVM
3. acute systemic metabolic disturbance
 - a) electrolyte disorders: uremia, hyponatremia, hypoglycemia (especially profound hypoglycemia), hypercalcemia
 - b) drug related, including:
 - alcohol-withdrawal (p.464)
 - cocaine toxicity (p.207)
 - opioids (narcotics)
 - phenothiazine antiemetics (p.143)
 - with administration of flumazenil (Romazicon®) to treat benzodiazepine (BDZ) overdose (especially when BDZs are taken with other seizure lowering drugs such as tricyclic antidepressants or cocaine)
 - phencyclidine (PCP): originally used as an animal tranquilizer
 - cyclosporine: can affect Mg^{++} levels
 - c) eclampsia
4. idiopathic

In 166 pediatric patients presenting to an emergency department with either a chief complaint of, or a discharge diagnosis of a first-time seizure⁴:

1. 110 were found to actually have either a recurrent seizure or a non-ictal event
2. of the 56 patients actually thought to have had a first-time seizure
 - a) 71% were febrile seizures
 - b) 21% were idiopathic
 - c) 7% were “symptomatic” (hyponatremia, meningitis, drug intoxication...)

In a prospective study of 244 patients with a new-onset unprovoked seizure, only 27% had further seizures during follow-up.^{2,5} Recurrent seizures were more common in patients with a family seizure history, spike-and-waves on EEG, or a history of a CNS insult (stroke, head injury...). No patient seizure-free for 3 years had a recurrence. Following a second seizure, the risk of further seizures was high.

27.1.3 Evaluation

Adults

A new-onset seizure in an adult in the absence of obvious cause (e.g. alcohol withdrawal) should prompt a search for an underlying basis (the onset of idiopathic seizures, i.e. epilepsy, is most

common before or during adolescence). A CT or MRI (without and with enhancement) should be performed. A systemic work-up should be done to identify the presence of any factors listed previously (see above). If all this is negative, then an MRI should be performed if not already done. If this is negative also, a repeat study (CT or MRI) should be done in $\approx$ 6 months and at 1 and possibly 2 years to rule-out a tumor which might not be evident on the initial study.

Pediatrics

Among pediatric patients with first-time seizures, laboratory and radiologic evaluations were often costly and not helpful.⁴ A detailed history and physical exam were more helpful.

Management

Management of an adult with the new onset of idiopathic seizures (i.e. no abnormality found on CT or MRI, no evidence of drug withdrawal) is controversial. In one study, an EEG was performed, which if normal was followed by a sleep deprived EEG with the following observations⁶:

1. there is substantial interobserver variation in interpreting such EEGs
2. if both EEGs were normal, the 2-yr recurrence rate of seizures was 12%
3. if one or both EEGs showed epileptic discharges, the 2-yr recurrence rate was 83%
4. the presence of nonepileptic abnormalities in one or both EEGs had a 41% 2-yr recurrence rate
5. the recurrence rate with focal epileptic discharges (87%) was slightly higher than for generalized epileptic discharges (78%)

The conclusion is that EEGs thus obtained have moderate predictive value, and may be factored into the decision of whether or not to treat such seizures with AEDs.

27

27.2 Posttraumatic seizures

27.2.1 General information

Key concepts

- 2 categories: early ($\leq$ 7 days) and late ($>$ 7 days) after head trauma
- anticonvulsants (AEDs) may be used to prevent early posttraumatic seizures (PTS) in patients at high risk for seizures
- prophylactic AEDs do NOT reduce the frequency of late PTS
- discontinue AEDs after 1 week except for cases meeting specific criteria (see text)

Posttraumatic seizures (PTS) are often divided (arbitrarily) into: early (occurring within 1 week of injury) and late (thereafter).⁷ There may be justification for a third category: “immediate”, i.e. within minutes to an hour or so.

27.2.2 Early PTS ($\leq$ 7 days after head trauma)

30% incidence in severe head injury (“severe” defined as: LOC $>$ 24 hrs, amnesia $>$ 24 hrs, focal neuro deficit, documented contusion, or intracranial hematoma) and $\approx$ 1% in mild to moderate injuries. Occurs in 2.6% of children $<$ 15 yrs age with head injury causing at least brief LOC or amnesia.⁸ Early PTS may precipitate adverse events as a result of elevation of ICP, alterations in BP, changes in oxygenation, and excess neurotransmitter release.⁹

27.2.3 7 days after head trauma), 3, 7, 2, 0, 0mm, 0mm, 0mm, 0mm > Late onset PTS ($>$ 7 days after head trauma)

Estimated incidence 10–13% within 2 yrs after “significant” head trauma (includes LOC $>$ 2 mins, GCS $<$ 8 on admission, epidural hematoma...) for all age groups.^{10,11} Relative risk: 3.6 times control population. Incidence in severe head injury $\gg$ moderate $>$ mild.⁸ The incidence of early PTS is higher in children than adults, but late seizures are much less frequent in children (in children who have PTS, 94.5% develop them within 24 hrs of the injury¹²). Most patients who have not had a seizure within 3 yrs of penetrating head injury will not develop

seizures.¹³ Risk of late PTS in children does not appear related to the occurrence of early PTS (in adults: only true for mild injuries). Risk of developing late PTS may be higher after repeated head injuries.

27.2.4 Penetrating trauma

The incidence of PTS is higher with penetrating head injuries than with closed head injuries (occurs in 50% of penetrating trauma cases followed 15 yrs¹⁴).

27.2.5 Treatment

General information

Some early retrospective studies suggested that early administration of PHT prevents early PTS, and reduces the risk of late PTS even after discontinuation of the drug. Later prospective studies disputed this but were criticized for not maintaining satisfactory levels and for lacking statistical power.^{7,11} A prospective double blind study of patients at high risk of PTS (excluding penetrating trauma) showed a 73% reduction of risk of early PTS by administering 20 mg/kg loading dose of PHT within 24 hrs of injury and maintaining high therapeutic levels; but after 1 week there was no benefit in continuing the drug (based on intention to treat).¹⁵ Carbamazepine (Tegretol®) has also been shown to be effective in reducing the risk of early PTS.

Phenytoin has adverse cognitive effects when given long-term as prophylaxis against PTS.¹⁶

Treatment guidelines

Based on available information (see above) it appears that:

- 1. no treatment studied effectively impedes epileptogenesis (i.e. neuronal changes that ultimately lead to late PTS)
- 2. in high-risk patients (□ Table 27.1), AEDS reduces the incidence of early PTS
- 3. however, no study has shown that reducing early PTS improves outcome¹⁷
- 4. once epilepsy has developed, continued AEDs reduces the recurrence of further seizures

The following are therefore offered as guidelines.

Initiation of AEDs

AEDs may be considered for short term use especially if a seizure could be detrimental. Early post-traumatic seizures were effectively reduced when phenytoin was used for 2 weeks following head injury with no significant increased risk of adverse effects.¹⁸

Acutely, seizures may elevate ICP, and may adversely affect blood pressure and oxygen delivery, and may worsen other injuries (e.g. spinal cord injury in the setting of an unstable cervical spine). There may also be negative psychological effects on the family, loss of driving privileges, and possibly deleterious effects of excess neurotransmitters.⁹

Option: begin AEDs (usually levetiracetam, phenytoin or carbamazepine) within 24 hrs of injury in the presence of any of the high risk criteria shown in □ Table 27.1 (modified^{9,12,15,19}). When using PHT, load with 20 mg/kg and maintain high therapeutic levels. Switch to phenobarbital if PHT not tolerated.

Discontinuation of AEDs

- 1. taper AEDs after 1 week of therapy except in the following:
 - a) penetrating brain injury
 - b) development of late PTS (i.e. a seizure > 7 days following head trauma)

Table 27.1 High risk criteria for PTS
1. acute subdural, epidural, or intracerebral hematoma SDH, EDH or ICH
2. open-depressed skull fracture with parenchymal injury
3. seizure within the first 24 hrs after injury
4. Glasgow Coma Scale score <10
5. penetrating brain injury
6. history of significant alcohol abuse
7. ± cortical (hemorrhagic) contusion on CT

- c) prior seizure history
- d) patients undergoing craniotomy²⁰
- 2. for patients not meeting the criteria to discontinue AEDs after 1 week (see above):
 - a) maintain $\approx$ 6–12 mos of therapeutic AED levels
 - b) recommend EEG to rule-out presence of a seizure focus before discontinuing AEDs (shown to have poor predictive value is poor, however probably advisable for legal purposes) for the following:
 - repeated seizures
 - presence of high risk criteria shown in □ Table 27.1.

27.3 Alcohol withdrawal seizures

27.3.1 General information

Also, see Alcohol withdrawal syndrome (p.204). The withdrawal syndrome may begin hours after the EtOH peak; see also prevention and treatment (p.205). Ethanol withdrawal seizures are classically seen in up to 33% (some say 75%) of habituated drinkers within 7–30 hours of cessation or reduction of ethanol intake. They typically consist of 1–6 tonic-clonic generalized seizures without focality within a 6 hour period.²¹ Seizures usually occur before delirium develops. They may also occur during intoxication (without withdrawal).

The seizure risk persists for 48 hrs (risk of delirium tremens (DTs) continues beyond that), thus a single loading dose of PHT is frequently adequate for prophylaxis. However, since most EtOH withdrawal seizures are single, brief, and self-limited, PHT has not been shown to be of benefit in uncomplicated cases and is thus usually not indicated. Chlordiazepoxide (Librium®) or other benzodiazepines (p.205) administered during detoxification reduces the risk withdrawal seizures.²²

27.3.2 Evaluation

The following patients should have a CT scan of the brain, and should be admitted for further evaluation as well as for observation for additional seizures or for DTs:

1. those with their first EtOH withdrawal seizure
2. those with focal findings
3. those having more than 6 seizures in 6 hrs
4. those with evidence of trauma

Other causes of seizure should also be considered, e.g. a febrile patient may require an LP to rule-out meningitis.

27.3.3 Treatment

A brief single seizure may not warrant treatment, except as outlined below. A seizure that continues beyond 3–4 minutes may be treated with diazepam or lorazepam, with further measures used as in status epilepticus (p.468) if seizures persist. Loading with phenytoin (18 mg/kg = 1200 mg/70 kg) and long-term treatment is indicated for:

1. a history of previous alcohol withdrawal seizures
2. recurrent seizures after admission
3. history of a prior seizure disorder unrelated to alcohol
4. presence of other risk factors for seizure (e.g. subdural hematoma)

27.4 Nonepileptic seizures

27.4.1 General information

AKA pseudoseizures (some prefer not to use this term since it may connote voluntary feigning of seizures), with the term psychogenic seizures being preferred for nonepileptic seizures (NES) with a psychologic etiology (psychogenic seizures are real events and may not be under voluntary control).²³

One of the hazards of NES is that patients may end up needlessly taking AEDs, which in some cases may worsen NES. Possible etiologies of NES are given in □ Table 27.2. Most NES are psychogenic.

Table 27.2 Differential diagnosis of nonepileptic seizures ²³
psychologic disorders (psychogenic seizure) a) somatoform disorders: especially conversion disorder b) anxiety disorders: especially panic attack and posttraumatic stress disorder c) dissociative disorders d) psychotic disorders e) impulse control disorders f) attention-deficit disorders ^a g) factitious disorders: including Munchausen’s syndrome
cardiovascular disorders a) syncope b) cardiac arrhythmias c) transient ischemic attacks d) breath-holding spells ^a
migraine syndromes a) complicated migraines ^a b) basilar migraines
movement disorders a) tremors b) dyskinesias c) tics ^a , spasms d) other (including shivering)
parasomnias & sleep-related disorders a) night terrors ^a , nightmares ^a , somnambulism ^a b) narcolepsy, cataplexy c) rapid eye movement behavior disorder d) nocturnal paroxysmal dystonia
gastrointestinal disorders a) episodic nausea or colic ^a b) cyclic vomiting syndrome ^a
other a) malingering b) cognitive disorders with episodic behavioral or speech symptoms c) medication effects or toxicity d) daydreams ^a
^a usually encountered in children

- DDx for seizures:
1. psychogenic: 20–90%of patients with intractable seizures referred to epilepsy centers. These patients carry the diagnosis of seizures from 5–7 years. Up to 50%of these may have legitimate seizures at some time as well.²⁴
 2. tic: can be suppressed, is not repetitive (if repetitive, may be hemifacial spasm)
 3. movement disorder: myoclonus (can be epileptic or non-epileptic)
 - a) cataplexy: e.g. with narcolepsy often provoked by laughter or other emotional stimulus (can rarely be caught on EEG, when it is it shows REM intrusion into wakefulness)
 - b) parasomnia: a sleep movement disorder (occurs during sleep). Includes: night terrors (occurs in slow wave sleep, vs. nightmare which occurs in REM), sleep walking, REM behavior disorders (usually occurs in older men, and there is a high probability they will go on to have degenerative brain disease (used to be called paroxysmal nocturnal PNT). Head banging is a benign parasomnia
 4. syncope: 90%of the time people who faint have myoclonic jerks or shaking²⁵
 5. TIA

27.4.2 Differentiating NES from epileptic seizures

General information

Distinguishing between epileptic seizures (ES) and NES is a common clinical dilemma. There are unusual seizures that may fool experts.²⁶ Some frontal lobe and temporal lobe complex partial

seizures may produce bizarre behaviors that do not correspond to classic ES findings and may not produce discernible abnormalities with scalp-electrode EEG (and therefore may be misdiagnosed even with video-EEG monitoring, although this is more likely with partial seizures than with generalized). A multidisciplinary team approach may be required.

□ Table 27.3 contrasts some features of true seizures vs. NES, and □ Table 27.4 lists some features often associated with NES, however, no characteristics are definitively diagnostic of NES since a number of them may also occur with ES.

Features common to both true seizures and NES: verbal unresponsiveness, rarity of automatisms and whole-body flaccidity, rarity of urinary incontinence. Reminder: some seizures can be bizarre and can resemble NES (sometimes called pseudo-pseudoseizures). 10%of patients with psychogenic seizures actually have epilepsy.

Table 27.3 Features of ES vs. NES ²⁴		
Feature	Epileptic seizure	NES
% males	72%	20%
Clonic UE movement		
in-phase	96%	20%
out-of-phase	0	56%
Clonic LE movement		
in-phase	88%	16%
out-of-phase	0	56%
Vocalizations		
none	16%	56%
start of seizure	24%	44%
middle	60% “epileptic cry”	0
types	only sounds of tonic or clonic respiratory muscle contraction	moans, screams, grunts, snorts, gagging, retching, understandable statements, gasps
Head turning		
unilateral	64%	16%
side-to-side	8% (slow, low amplitude)	36% (violent, high amplitude)

Table 27.4 Features often associated with NES ²³	
• frequent seizures despite therapeutic AEDs • multiple different-physician visits • lingering prodrome or gradual ictal onset (over minutes) • prolonged duration (>5 mins) • manifestations altered by distraction • suggestible or inducible seizures • intermittent arrhythmic and out-of-phase convulsive activity • fluctuating intensity and severity during Sz • side-to-side rolling, pelvic thrusting, wild movements • bilateral motor activity with preserved consciousness • nonphysiologic spread of neurologic signs • absence of labored breathing or drooling after generalized convulsion • expression of relief or indifference • crying or whimpering • no postictal confusion or lethargy • disproportionate postictal mental status changes • absence of stereotypy	

Features suggestive of non-epileptic seizures:

1. arching of the back: 90% specific for NES
2. asynchronous movement
3. stop & go: seizures usually build and then gradually subside
4. forced eye closing during entire seizure
5. provoked with stimuli that would not cause a seizure (e.g. tuning fork to the head, alcohol pad to the neck, IV saline...)
6. bilateral shaking with preserved awareness. Exception: supplementary motor area seizures (mesial frontal area) these seizures are usually tonic (not clonic)
7. weeping (whining): highly specific
8. multiple or variable seizure types (ES is usually stereotypical), fluctuating level of consciousness, denial of correlation of Sz with stress

If any two of the following are demonstrated, 96% of time this will be NES:

1. out-of-phase clonic UE movement
2. out-of-phase clonic LE movement
3. no vocalization or vocalization at start of event

Lateral tongue laceration is very specific for seizures.

History

Attempt to document: prodromal symptoms, precipitating factors, time and environment of Sz, mode and duration of progression, ictal and postictal events, frequency and stereotypy of manifestations. Determine if patient has history of psychiatric conditions, and if they are acquainted with individuals who have ES.

Psychological testing

May help. Differences occur in ES and NES on the Minnesota Multiphasic Personality Inventory (MMPI) scales in hypochondriasis, depression hysteria, and schizophrenia.²⁷

Prolactin levels after seizures

Transient elevations in human serum prolactin (HSP) levels occur following 80% of generalized motor, 45% of complex partial, and only 15% of simple partial seizures.²⁸ Peak levels are reached in 15–20 minutes, and gradually return to baseline over the subsequent hour.^{29,30,31} It has been suggested that drawing a serum prolactin level shortly after a questionable seizure may be helpful in differentiating NES (which may have elevated cortisol levels but normal HSP levels³²).

Repetitive seizures are associated with progressively smaller HSP elevations,³³ and no rise follows absence seizures or status epilepticus (whether convulsive or absence).³⁴ Greater than twofold HSP elevations consistently follow seizures that produce intense widespread high frequency mesial temporal lobe discharges; whereas such elevations do not occur in seizures not involving these limbic structures.³⁵ Furthermore, there may be higher baseline HSP levels in cases with right-sided interictal EEG discharges compared to those with left-sided,³⁶ and the presence of psychopathology may affect postictal HSP elevations.³⁷

Therefore, the presence of HSP peaks may be strongly indicative of true seizures, but the absence may be due to a variety of complex phenomena.³⁸ The overall classification accuracy is $\approx 72\%$ ³¹

27.5 Febrile seizures

27.5.1 Definitions

See reference.³⁹

- ☐ Febrile seizure. A seizure in infants or children associated with fever with no defined cause and unaccompanied by acute neurologic illness (includes seizures during vaccination fevers)
- ☐ Complex febrile seizure. A convulsion that lasts longer than 15 minutes, is focal, or multiple (more than one convulsion per episode of fever)
- ☐ Simple febrile seizure. Not complex
- ☐ Recurrent febrile seizure. More than one episode of fever associated with seizures

27.5.2 Epidemiology

See reference.³⁹

Febrile convulsions are the most common type of seizure. Excluding children with pre-existing neurologic or developmental abnormalities, the prevalence of febrile seizures is ≈ 2.7%(range: 2–5% in U.S. children aged 6 mos-6 yrs). The risk for developing epilepsy after a simple febrile seizure is ≈ 1% and for a complex febrile seizure is 6%(9% for prolonged seizure, 29% for focal seizure). An underlying neurological or developmental abnormality or a family history of epilepsy increases the risk of developing epilepsy. The notion that the younger the child with a febrile seizure the greater the risk of epilepsy is unproven.

27.5.3 Treatment

In one study, the IQ in the group treated with phenobarbital was 8.4 points lower (95%confidence interval) than the placebo group, and there remained a significant difference several months after discontinuing the drug.⁴⁰ Furthermore, there was no significant reduction in seizures in the phenobarbital group. And yet, no other drug really appears well suited to treating this entity: carbamazepine and phenytoin appear ineffective, valproate may be effective but has serious risks in the <2 yrs age group. Given the low incidence (1%) of having afebrile seizures (i.e. epilepsy) after a simple febrile seizure and the fact that AEDs probably do not prevent this development, there is little support for prescribing anticonvulsants in these cases. The recurrence rate of febrile seizures in children with a history of one or more febrile seizure can be reduced by administering diazepam 0.33 mg/kg PO q 8 hrs during a febrile episode (temp >38.1° C) and continuing until 24 hrs after the fever subsides.⁴¹

27.6 Status epilepticus

27.6.1 General information

Key concepts

- definition: Sz > 5 mins, or persistent Sz after 1st & 2nd line AEDs
- morbidity and mortality are high in untreated status epilepticus (SE)
- most common etiology: patient with known Sz disorder with low AED levels
- de novo SE in acute illness is considered a manifestation of the illness which should be treated at the same time as the SE
- see □ Table 27.5 for treatment measures

□ Definition. A seizure lasting > 5 minutes or persistent seizure activity after sequential administration of appropriate first and second-line AEDs.⁴²

- Features important to management:
- 61% of seizures that persist > 5 mins will continue > 1 hour⁴³
 - in patients with no prior Sz history, status epilepticus (SE) is usually a manifestation of illness-related cortical irritation or injury⁴² and treatment of the underlying disorder (in addition to treating the SE) is critical
 - a relapse of Sz. in a patient with a known Sz. disorder and subtherapeutic AED levels usually responds to a bolus of the maintenance AEDs. However, SE should be treated by the standard protocol⁴²
 - most cases of convulsive status in adults start as partial seizures that generalize secondarily
 - the choice of 1st and 2nd-line AEDs is arbitrary, dose⁴² and starting treatment in <30 minutes⁴⁴ are more important determinants of success in aborting SE

27.6.2 Types of status epilepticus

See reference.⁴⁵

- generalized status
 1. convulsive (tonic-clonic, tonic-clonic-tonic, or clonic): generalized convulsive tonic-clonic status epilepticus (SE) is the most frequent type.⁴⁶ A medical emergency
 2. absence (note: in status, this may present in twilight state)

Table 27.5 Summary of initial steps for status epilepticus (adults and children > 13 kg; see text for details) ⁵⁰
ABC's. Start O ₂ . Turn patient on their side. Check VS. Do a neuro exam
Monitor/labs: Pulse oximetry. EKG/telemetry. ☐ Fingerstick glucose. Blood tests (do not wait for results to begin ☐): ☐ electrolytes, ☐ CBC, ☐ ABG, ☐ AED levels, ☐ IFTs, ☐ Mg ⁺⁺ , ☐ Ca ⁺⁺ , ☐ head CT
Large bore IV X 2. Start IV fluids
• thiamine 100 mg IV and/or 50 ml of 50% dextrose (if needed based on fingerstick glucose)
First-line AED: • lorazepam (Ativan®) 4 mg IV for adults, 2 mg IV for children > 13 kg @<2 mg/min - OR • midazolam (Versed®) 10 mg IM for adults, 5 mg IM for children > 13 kg OR (if no IV access or midazolam injections not available) • diazepam can be given rectally in Diastat® gel formulation (0.2-0.5 mg/kg) Repeat loading dose of benzodiazepine if necessary
Second-line AED: given with failure of (or simultaneously with administration of) repeat dose of benzodiazepine • fosphenytoin: 15-20 mg PE/kg IV @<150 mg PE/min (preferred drug: faster infusion rate, less irritation) - OR • phenytoin: 15-20 mg/kg IV @<50 mg/min (less expensive) If no response to loading dose, an additional 10 mg/kg IV may be given after 20 min. NB: following infusion rate guidelines is imperative. Significant cardiovascular risk is associated with rapid infusion of phenytoin/fosphenytoin.
☐ phenytoin level ≈ 10 min after PHT loading dose; repeat 10 min later additional dose if required
Alternative second-line AEDs: • sodium valproate: 20-30 mg/kg IV bolus (max rate: 100 mg/min) – has been shown to be equal or superior to phenytoin in a few small studies - OR • phenobarbital: 20 mg/kg IV (start infusing @50-100 mg/min) – commonly used 2nd or 3rd line AED. A repeat dose of 25-30 mg/kg can be given 10 min after first dose. - OR • Levetiracetam (Keppra®): 20 mg/kg IV bolus of over 15 minutes – evidence for Keppra as a first or second line drug is less clear
If seizures continue >30 mins and are refractory to 1st and 2nd line AEDs: intubate in ICU and begin continuous infusion therapy (CIT) of: • Midazolam: 0.2 mg/kg IV loading dose followed by 0.2-0.6 mg/kg/hr - OR • Propofol: 2 mg/kg IV loading dose followed by 2-5 mg/kg/hr
If Sz persist, ensure that correctable conditions have been ruled-out and/or treated
novel therapeutic options (not systematically studied): shock therapy...

- 3. secondarily generalized: accounts for ≈ 75% of generalized SE
- 4. myoclonic
- 5. atonic (drop attack): especially in Lennox-Gastaut syndrome (p.443)
- partial status (usually related to an anatomic abnormality)
 - 1. simple (AKA epilepsy partialis continua)
 - 2. complex (note: in status, this may present in twilight state) most often from frontal lobe focus. Urgent treatment is required (several case reports of permanent deficits following this)
 - 3. secondarily generalized
- non-convulsive SE
 - 1. benign variants (typical absence SE, complex partial SE)
 - 2. electrical SE during sleep
 - 3. atypical absence SE
 - 4. tonic SE (associated with learning disability in children) SE in coma

Alternatively, SE can be broken down as follows:

- with prominent motor effects
- without prominent motor effects

- boundary syndromes (syndromes which combine encephalopathy, behavioral disturbances, delirium, or psychosis with SE-like EEG findings)

27.6.3 Epidemiology

Incidence is $\approx 150,000$ new cases/year in the U.S. in the outpatient setting.⁴² Most cases occur in young children (among children, 73% were <5 yrs old⁴⁷), the next most affected group is patients >60 yrs age. In $>50\%$ of cases, SE is the patient's first seizure.⁴⁶ One out of six patients presenting with a first time seizure will present in SE

27.6.4 Etiologies

Most common causes: low level of AED (34%), remote symptomatic cause (24%), cerebrovascular accident (22%), metabolic disturbances (15%), and hypoxia (13%).

A more comprehensive list:

1. a patient with a known seizure disorder having low AED levels for any reason (non-compliance, intercurrent infection preventing PO intake of meds, drug-drug interactions $\rightarrow$ lowering effectiveness of AEDs...)
2. febrile seizures: a common precipitator in young patients. 5–6% of patients presenting with SE have a history of prior febrile seizures
3. stroke: the most commonly identified cause in the elderly
4. CNS infection: in children, most are bacterial, the most common organisms were *H. influenza* and *S. pneumoniae*
5. idiopathic: accounts for $\approx$ one-third (in children, usually associated with fever)
6. epilepsy: is present or is subsequently diagnosed in $\approx 50\%$ of patients presenting with SE. About 10% of adults ultimately diagnosed as having epilepsy will present in SE
7. electrolyte imbalance: hyponatremia (most common in children, usually due to water intoxication⁴⁷), hypoglycemia, hypocalcemia, uremia, hypomagnesemia...
8. illicit drug intoxication: especially cocaine, amphetamines
9. precipitous drug withdrawal: barbiturates, benzodiazepines, alcohol or narcotics
10. proconvulsant drugs, including: β -lactam antibiotics (penicillins, cephalosporins), certain antidepressants (bupropion), clonazapine, bronchodilators, immunosuppressants
11. traumatic brain injury: acute as well as old
12. hypoxia/ischemia
13. tumor

In children <1 yr age, 75% had an acute cause: 28% were secondary to CNS infection, 30% due to electrolyte disorders, 19% associated with fever.⁴⁷ In adults, a structural lesion is more likely. In an adult, the most common cause of SE is subtherapeutic AED levels in a patient with a known seizure disorder.

27.6.5 Morbidity and mortality from SE

Outcomes are related to underlying cause and duration of SE. Mean duration of SE in patients without neurologic sequelae is 1.5 hrs (therefore, proceed to pentobarbital anesthesia before ≈ 1 hour of SE). Recent mortality: <10 –12% (only $\approx 2\%$ of deaths are directly attributable to SE or its complications; the rest are due to the underlying process producing the SE). Mortality is lowest amongst children ($\approx 6\%$ ⁴⁷), patients with SE related to subtherapeutic AEDs, and patients with unprovoked SE.⁴⁸ The highest mortality occurs in elderly patients and those with SE resulting from anoxia or stroke.⁴⁸ 1% of patients die during the episode itself.

Morbidity and mortality is due to⁴⁹:

1. CNS injury from repetitive electric discharges: irreversible changes begin to appear in neurons after as little as 20 minutes of convulsive activity. Cell death is very common after 60 mins
2. systemic stress from the seizure (cardiac, respiratory, renal, metabolic)
3. CNS damage by the acute insult that provoked the SE

27.6.6 Treatment

General treatment measures for status epilepticus

Treatment success, like morbidity/mortality, may be time dependent. One review showed that first-line AED therapy aborted SE in 60% of patients if initiated within the first 30 minutes, and efficacy

decreased as seizure duration increased [Lowenstein 1993]. As such, treatment should be initiated as soon as possible and should be directed at stabilizing the patient, stopping the seizure, and identifying the cause (determining if there is an acute insult to the brain) and if possible also treating the underlying process. Treatment often must be initiated prior to the availability of test results to confirm the diagnosis and may even be initiated in the pre-hospital setting.

1. “ABC’s”
 - a) Airway: oral airway if feasible. Turn patient on their side to avoid aspiration
 - b) Breathing: O₂ by nasal cannula or bag-valve-mask. Consider intubation if respirations compromised or if seizure persists > 30 min
 - c) Circulation: CPR if needed. Large bore proximal IV access (2 if possible: 1 for phenytoin (PHT) (Dilantin®), not necessary if fosphenytoin is available): start with NS KVO
2. Simultaneous with ABCs, AEDs should be prepared and/or given if SE suspected
3. neurologic exam
4. monitor: EKG & baseline vital signs. Pulse oximeter. Frequent blood pressure checks
5. bloodwork: STAT capillary blood (fingerstick) glucose (to R/O hypoglycemia), electrolytes (including glucose), CBC, LFTs, Mg⁺⁺, Ca⁺⁺, AED levels, ABG
6. head CT (usually without contrast)
7. correct any electrolyte imbalance (SE due to electrolyte imbalance responds more readily to correction than to AEDs⁴⁷)
8. if CNS infection is a major consideration, perform LP for CSF analysis (especially in febrile children) unless contraindicated (p.1504). WBC pleocytosis up to 80 × 10⁶/L can occur following SE (benign postictal pleocytosis), and these patients should be treated with antibiotics until infection can be ruled out by negative cultures
9. general meds for unknown patient:
 - a) glucose:
 - in patients with poor nutrition (e.g. alcoholics): giving glucose in thiamine deficiency can precipitate Wernicke’s encephalopathy (p.206) □ prior to glucose bolus give thiamine 50–100 mg IV
 - if fingerstick glucose can be obtained immediately and it shows hypoglycemia, or if no fingerstick glucose can be done: give 25–50 ml of D50 IV push for adults (2 ml/kg of 25% glucose for peds). If at all possible, draw blood for definitive serum glucose first
 - b) naloxone (Narcan®) 0.4 mg IVP (in case of narcotics)
 - c) ± bicarbonate to counter acidosis (1–2 amps depending on length of seizure)
 - d) for neonate < 2 years: consider pyridoxine 100 mg IV push (pyridoxine-dependent seizures constitute a rare autosomal recessive condition that generally presents in the early neonatal period⁴³)
10. administer specific anticonvulsants for seizures lasting > 5- 10 mins (see below)
11. EEG monitor if possible
12. if paralytics are used (e.g. to intubate), use short acting agents and be aware that muscle paralysis alone may stop visible seizure manifestations, but does not stop the electrical seizure activity in the brain, which can lead to permanent neurologic damage if prolonged (see above).

Medications for generalized convulsive status epilepticus

General information

There are no randomized trials for refractory status epilepticus, although there is published data regarding specific treatment options. Numerous protocols exist. □ Table 27.5 shows a summary of medications for a status epilepticus protocol that is outlined in further detail below (adapted^{42,45,50,51}). “Peds dosing” refers to patients < 40 kg or approximately 12 yrs of age. Rapid treatment is indicated as delays are associated with neuronal injury and reduced response to medications.

Prehospital phase

1. impending SE: may be heralded by a crescendo in Sz. A 1–3 d course of lorazepam may preempt the development of SE
2. SE treatment may be initiated in the home setting with buccal midazolam or rectal diazepam

Hospital phase

Start IV drugs at half the maximal rate, and titrate up to maximal rate if VS stable.

1. First line drugs
 - a) benzodiazepine⁵² (main side effect: respiratory depression in ≈ 12% be prepared to intubate). Onset of action is rapid (1–2 mins):
 - lorazepam (Ativan®) 4 mg IV for adults, 2 mg IV for children @ < 2 mg/min

- OR midazolam (Versed®) 10 mg IM for adults, 5 mg IM for children > 13 kg—Repeat dose of benzodiazepine if necessary after 10 min.
 - If no IV access or if midazolam injections not available, diazepam can be given rectally in Diastat® gel formulation (0.2-0.5 mg/kg)
2. If seizures persist after first dose of benzodiazepine, initiate second line agent in a different IV.
 - a) load with fosphenytoin (Cerebyx®) or phenytoin (Dilantin®) as listed below. Do not worry about acutely overdosing, but do follow dosing rates, monitor BP for hypotension and EKG for arrhythmias. After giving the following loading dose, start on maintenance. Fosphenytoin has the advantage of being less irritating and able to infuse at a faster rate, but phenytoin is less expensive and does not need to be metabolized.
 - fosphenytoin: 15-20 mg PE/kg IV @ 150 mg PE/min
 - OR phenytoin: 15-20 mg/kg IV @ 50 mg/min
 - if no response to loading dose, an additional 10 mg/kg IV may be given after 20 min.
 - if pt is on PHT and a recent level is known: a rule of thumb is giving 0.74 mg/kg to an adult raises the level by ≈ 1 mcg/ml
 - if on PHT and level not known: adult: give 500 mg @ < 50 mg/min
 - b) There are several good alternatives to fosphenytoin/phenytoin as second-line AEDs:
 - Sodium valproate: 20-30 mg/kg IV bolus (max rate: 100 mg/min) – has been shown to be equal or superior to phenytoin in a few small studies
 - Phenobarbital: 20 mg/kg IV (start infusing @ 50-100 mg/min) – commonly used 2nd or 3rd line AED. A repeat dose of 25-30 mg/kg can be given 10 min after first dose.
 - Levetiracetam (Keppra®): 20 mg/kg IV bolus of over 15 minutes – evidence for Keppra as a first or second line drug is less clear
 3. Traditionally, a third line agent was given prior to continuous infusion therapy (CIT), however was successful in only 7%⁵² As such, most new protocols proceed directly to anesthetic administration. If seizures are continuing after above therapies have been administered (15-30 min after initial presentation), begin CIT as follows:
 - Midazolam: 0.2 mg/kg IV loading dose followed by 0.2-0.6 mg/kg/hr
 - OR Propofol: 2 mg/kg IV loading dose followed by 2-10 mg/kg/hr
 4. At this time, lab results and tests should be available. Ensure that all reversible etiologies have been addressed and that a CT head has been performed.
 5. Pentobarbital is often reserved for SE that is refractory to all of the above interventions. If necessary, Pentobarbital is administered as follows:
 - Pentobarbital: 5 mg/kg IV followed by 1-5 mg/kg/hr
 6. While some practitioners will try additional drugs (carbamazepine, oxcarbazepine, topiramate, levetiracetam, lamotrigine, gabapentin), these are likely to be of limited utility.
 7. Experimental interventions include: lidocaine infusion, inhalational anesthesia, direct brain stimulation, transcranial magnetic stimulation, electroconvulsive therapy (shock therapy), surgical intervention if a seizure focus is identified

Remember: Paralytics stop the visible manifestations of the seizure and they may be useful for intubation and/or in order to obtain head imaging; however, they do not stop the abnormal electrical brain activity or the neurological damage that results.

Efficacy of drug therapy

Studies vary widely, but it appears that approximately 2/3 of patients will respond to initial therapy with the other 1/3 progressing to refractory SE.⁵⁰

Medications to avoid in status epilepticus

1. narcotics
2. phenothiazines: including promethazine (Phenergan®)
3. neuromuscular blocking agents in the absence of AED therapy: seizures may continue and cause neurologic injury but would not be clinically evident

27.6.7 Medications for non-convulsive status epilepticus

In non-convulsive status epilepticus, the first and second line AEDs listed in □ Table 27.5 should be utilized. However, many practitioners avoid escalating to the anesthetic options (CIT, pentobarbital) instead opting for trials of additional AEDs first (carbamazepine, oxcarbazepine, topiramate, lamotrigine, etc).

27.6.8 Miscellaneous status epilepticus

Myoclonic status

Treatment: valproic acid (drug of choice). Place NG, give 20 mg/kg per NG loading dose. Maintenance: 40 mg/kg/d divided.

Can add lorazepam (Ativan®) or clonazepam (Klonopin®) to help with acute control.

Absence status epilepticus

Almost always responds to diazepam.

References

- [1] Hauser WA, Annegers JF, Kurland LT. Incidence of Epilepsy and Unprovoked Seizures in Rochester, Minnesota, 1935-1984. *Epilepsia*. 1993; 34:453-468
- [2] Hauser WA, Anderson VE, Loewenson RB, McRoberts SM. Seizure Recurrence After a First Unprovoked Seizure. *New Engl J Med*. 1982; 307:522-528
- [3] Reith J, Jorgensen HS, Nakayama H, et al. Seizures in acute stroke: Predictors and prognostic significance. The Copenhagen stroke study. *Stroke*. 1997; 28:1585-1589
- [4] Landfish N, Gieron-Korthals M, Weibley RE, Panzarino V. New Onset Childhood Seizures: Emergency Department Experience. *J Fla Med Assoc*. 1992; 79:697-700
- [5] Hauser WA, Rich SS, Jacobs MP, Anderson VE. Patterns of Seizure Occurrence and Recurrence Risks in Patients with Newly Diagnosed Epilepsy. *Epilepsia*. 1983; 24:516-517
- [6] van Donselaar C, Schimsheimer R-J, Geerts AT, Declerck AC. Value of the Electroencephalogram in Adult Patients With Untreated Idiopathic First Seizures. *Arch Neurol*. 1992; 49:231-237
- [7] Young B, Rapp RP, Norton JA, et al. Failure of Prophylactically Administered Phenytoin to Prevent Late Posttraumatic Seizures. *J Neurosurg*. 1983; 58:236-241
- [8] Annegers JF, Grabow JD, Groover RV, et al. Seizures After Head Trauma: A Population Study. *Neurology*. 1980; 30:683-689
- [9] Bullock R, Chesnut RM, Clifton G, et al. Guidelines for the Management of Severe Head Injury. 1995
- [10] McQueen JK, Blackwood DHR, Harris P, et al. Low Risk of Late Posttraumatic Seizures Following Severe Head Injury. *J Neurol Neurosurg Psychiatry*. 1983; 46:899-904
- [11] Young B, Rapp RP, Norton JA, et al. Failure of Prophylactically Administered Phenytoin to Prevent Early Posttraumatic Seizures. *J Neurosurg*. 1983; 58:231-235
- [12] Hahn YS, Fuchs S, Flannery AM, Barthel MJ, McLone DG. Factors Influencing Posttraumatic Seizures in Children. *Neurosurgery*. 1988; 22:864-867
- [13] Weiss GH, Salazar AM, Vance SC, et al. Predicting Posttraumatic Epilepsy in Penetrating Head Injury. *Arch Neurol*. 1986; 43:771-773
- [14] Temkin NR, Dikmen SS, Winn HR. Posttraumatic Seizures. *Neurosurg Clin North Amer*. 1991; 2:425-435
- [15] Temkin NR, Dikmen SS, Wilensky AJ, Keihm J, et al. A Randomized, Double-Blind Study of Phenytoin for the Prevention of Post-Traumatic Seizures. *N Engl J Med*. 1990; 323:497-502
- [16] Dikmen SS, Temkin NR, Miller B, Machamer J, et al. Neurobehavioral Effects of Phenytoin Prophylaxis of Posttraumatic Seizures. *JAMA*. 1991; 265:1271-1277
- [17] Brain Trauma Foundation, Povlishock JT, Bullock MR. Antiseizure prophylaxis. *J Neurotrauma*. 2007; 24:S83-S86
- [18] Haltiner AM, Newell DW, Temkin NR, et al. Side Effects and Mortality Associated with Use of Phenytoin for Early Posttraumatic Seizure Prophylaxis. *J Neurosurg*. 1999; 91:588-592
- [19] Yablon SA. Posttraumatic Seizures. *Arch Phys Med Rehabil*. 1993; 74:983-1001
- [20] North JB, Penhall RK, Hanieh A, Frewin DB, et al. Phenytoin and Postoperative Epilepsy: A Double-Blind Study. *J Neurosurg*. 1983; 58:672-677
- [21] Charness ME, Simon RP, Greenberg DA. Ethanol and the Nervous System. *N Engl J Med*. 1989; 321:442-454
- [22] Lechtenberg R, Worner TM. Seizure Risk With Recurrent Alcohol Detoxification. *Arch Neurol*. 1990; 47:535-538
- [23] Chabolla DR, Krahn LE, So EL, Rummans TA. Psychogenic Nonepileptic Seizures. *Mayo Clin Proc*. 1996; 71:493-500
- [24] Gates JR, Ramani V, Whalen S, et al. Ictal Characteristics of Pseudoseizures. *Arch Neurol*. 1985; 42:1183-1187
- [25] Lempert T, Bauer M, Schmidt D. Syncope: a videometric analysis of 56 episodes of transient cerebral hypoxia. *Ann Neurol*. 1994; 36:233-237
- [26] King DW, Gallagher BB, Marvin AJ, et al. Pseudoseizures: Diagnostic Evaluation. *Neurology*. 1982; 32:18-23
- [27] Henrichs TF, Tucker DM, Farha J, Novelly RA. MMPI Indices in the Identification of Patients Evidencing Pseudoseizures. *Epilepsia*. 1988; 29:184-187
- [28] Wyllie E, Luders H, MacMillan JP, et al. Serum Prolactin Levels After Epileptic Seizures. *Neurology*. 1984; 34:1601-1604
- [29] Dana-Haeri J, Trimble MR, Oxley J. Prolactin and Gonadotropin Changes Following Generalized and Partial Seizures. *J Neurol Neurosurg Psychiatry*. 1983; 46:331-335
- [30] Prichard PB, Wannamaker BB, Sagel J, et al. Serum Prolactin and Cortisol Levels in Evaluation of Pseudoepileptic Seizures. *Ann Neurol*. 1985; 18:87-89
- [31] Laxer KD, Mullooly JP, Howell B. Prolactin Changes After Seizures Classified by EEG Monitoring. *Neurology*. 1985; 35:31-35
- [32] Abbott RJ, Browning MCK, Davidson DLW. Serum Prolactin and Cortisol Concentrations After Grand Mal Seizures. *J Neurol Neurosurg Psychiatry*. 1980; 43:163-167
- [33] Jackel RA, Malkowicz D, Trivedi R, Sussman NM, et al. Reduction of Prolactin Response with Repetitive Seizures. *Epilepsia*. 1987; 28
- [34] Tomson T, Lindbom U, Nilsson BY, Svanborg E, et al. Serum Prolactin During Status Epilepticus. *J Neurol Neurosurg Psychiatry*. 1989; 52:1435-1437
- [35] Sperling MR, Pritchard PB, Engel J, et al. Prolactin in Partial Epilepsy: An Indicator of Limbic Seizures. *Ann Neurol*. 1986; 20:716-722
- [36] Meierkord H, Shorvon S, Lightman S, Trimble M. Comparison of the Effects of Frontal and Temporal Lobe Partial Seizures on Prolactin Levels. *Arch Neurol*. 1992; 49:225-230
- [37] Dana-Haeri J, Trimble MR. Prolactin and Gonadotropin Changes Following Partial Seizures in Epileptic Patients With and Without Psychopathology. *Biol Psychiatry*. 1984; 19:329-336
- [38] Herzog AG. Prolactin: Quo Vadis? *Arch Neurol*. 1992; 49:223-224
- [39] Verity CM, Golding J. Risk of Epilepsy After Febrile Convulsions: A National Cohort Study. *BMJ*. 1991; 303:1373-1376

- [40] Farwell JR, Lee YJ, Hirtz DG, Sulzbacher SI, et al. Phenobarbital for Febrile Seizures - Effects on Intelligence and on Seizure Recurrence. *N Engl J Med*. 1990; 322:364–369
- [41] Rosman NP, Colton T, Labazzo J, et al. A Controlled Trial of Diazepam Administered During Febrile Illnesses to Prevent Recurrence of Febrile Seizures. *N Engl J Med*. 1993; 329:79–84
- [42] Costello DJ, Cole AJ. Treatment of acute seizures and status epilepticus. *J Intensive Care Med*. 2007; 22:319–347
- [43] Abend NS, Dlugos DJ. Treatment of refractory status epilepticus: literature review and a proposed protocol. *Pediatr Neurol*. 2008; 38:377–390
- [44] Eriksson K, Metsaranta P, Huhtala H, Auvinen A, Kuusela AL, Koivikko M. Treatment delay and the risk of prolonged status epilepticus. *Neurology*. 2005; 65:1316–1318
- [45] Varelas PN, Spanaki MV, Mirski MA. Status epilepticus: an update. *Curr Neurol Neurosci Rep*. 2013; 13. DOI: 10.1007/s11910-013-0357-0
- [46] Hauser WA. Status Epilepticus: Epidemiologic Considerations. *Neurology*. 1990; 40:9–13
- [47] Phillips SA, Shanahan RJ. Etiology and Mortality of Status Epilepticus in Children. *Arch Neurol*. 1989; 46:74–76
- [48] Delorenzo RJ, Pellock JM, Towne AR, Boggs JG. Epidemiology of Status Epilepticus. *J Clin Neurophysiol*. 1995; 12:312–325
- [49] Fountain NB, Lothman EW. Pathophysiology of Status Epilepticus. *J Clin Neurophysiol*. 1995; 12:326–342
- [50] Betjemann JP, Lowenstein DH. Status epilepticus in adults. *Lancet Neurol*. 2015; 14:615–624
- [51] Kinney M, Craig J. Grand Rounds: An Update on Convulsive Status Epilepticus. *Ulster Med J*. 2015; 84:88–93
- [52] Treiman DM, Meyers PD, Walton NY, Collins JF, Colling C, Rowan AJ, Handforth A, Faught E, Calabrese VP, Uthman BM, Ramsay RE, Mamdani MB. A comparison of four treatments for generalized convulsive status epilepticus. Veterans Affairs Status Epilepticus Cooperative Study Group. *N Engl J Med*. 1998; 339:792–798

Part IX

Pain

28 Pain

476

IX

28 Pain

28.1 General information

For pain medication, see Analgesics (p.136).

Major types of pain:

- 1. nociceptive
 - a) somatic: well localized. Described as sharp, stabbing, aching or cramping. Results from tissue injury or inflammation, or from nerve or plexus compression. Responds to treating the underlying pathology or by interrupting the nociceptive pathway
 - b) visceral: poorly localized. Poor response to primary pain medications
- 2. deafferentation: poorly localized. Described as crushing, tearing, tingling or numbness. Also causes burning dysesthesia numbness often with lancinating pain, and hyperpathia. Unaffected by ablative procedures
- 3. “sympathetically maintained” pain and the likes, e.g. causalgia (p.497)

28.2 Neuropathic pain syndromes

28.2.1 General information

Definition: Neuropathic pain: pain caused by a lesion of the peripheral and/or central nervous system manifesting with sensory symptoms and signs (Backonja¹ modified from the International Association for the Study of Pain²).

Neuropathic pain syndromes (NPS) are typified by painful diabetic neuropathy (PDN) and post-herpetic neuralgia (PHN). Common chronic NPSs are shown in □ Table 28.1,³ divided into central or peripheral nervous system origin of the pain. The pain of PDN and PHN is typically burning and aching, and is continuous. and is characteristically refractory to medical and surgical treatment.

Table 28.1 Common neuropathic pain syndromes
Peripheral neuropathic pain
acute & chronic inflammatory demyelinating polyradiculoneuropathy (CIDP)
alcoholic polyneuropathy
chemotherapy induced polyneuropathy
complex regional pain syndrome (CRPS)
entrapment neuropathies
HIV sensory neuropathy
iatrogenic neuralgias (e.g. postthoracotomy pain)
idiopathic sensory neuropathy
neoplastic nerve compression or infiltration
nutritional-deficiency neuropathies
painful diabetic neuropathy (PDN)
phantom limb pain
postherpetic neuralgia (PHN)
postradiation plexopathy
radiculopathy
toxic exposure-related neuropathies
trigeminal neuralgia
posttraumatic neuralgias
Central neuropathic pain
cervical spondylotic myelopathy
HIV myelopathy
multiple sclerosis-related pain
Parkinson disease-related pain
postischemic myelopathy
postradiation myelopathy
poststroke pain
posttraumatic spinal cord injury pain
syringomyelia

28.2.2 Medical treatment of neuropathic pain

General information

Treatment traditionally includes narcotic analgesics,⁴ and tricyclic antidepressants (see below). For further details and other treatment measures, see PHN (p.494).

Tricyclic antidepressants

Use is often limited by anticholinergic and central effects and by limited pain relief.^{5,6} Possibly because serotonin potentiates the analgesic effect of endorphins and elevates pain thresholds, serotonin re-uptake blockers are more effective than norepinephrine re-uptake blockers, e.g. trazodone (Desyrel®) blocks only serotonin. Also useful: amitriptyline (Elavil®) 75 mg daily; desipramine (Norpramin®) 10–25 mg/d; doxepin (Sinequan®) 75–150 mg/d. Some benefit may also derive from the fact that many patients with chronic pain are depressed. Side effects: anticholinergic effects and orthostatic hypotension, especially in the elderly. □ Not recommended for use in patients with ischemic heart disease.

Gabapentin

Effective in postherpetic neuralgia (PHN) (p.495) and painful diabetic neuropathy. Benefit also reported in pain associated with: trigeminal neuralgia, cancer,⁷ multiples sclerosis, HIV-related sensory neuropathy, CRPS, spinal cord injury, post-operative state,⁸ migraine⁹ (a number of these studies may have been sponsored by the manufacturer¹⁰). See also side effects, dosing & availability... (p.455)

Lidocaine patch (Lidoderm®)

May be effective.³ □: apply patch for up to 12 hrs/day up to a maximum of 3 patches at a time to the intact skin over the most painful area (may trim patch to appropriate size). Supplied: 5% lidocaine (p.496).

Tramadol (Ultram®)

A centrally acting analgesic (p.140).³

28.3 Craniofacial pain syndromes

28.3.1 General information

Possible pathways for facial pain include: trigeminal nerve (portio major as well as portio minor (motor root)), facial nerve (usually deep facial pain), and eighth nerve.¹¹ Etiologies (adapted¹² (p.2328),¹³).

1. cephalic neuralgias
 - a) trigeminal neuralgia (see below)
 - vascular compression of Cr. N. V at root entry zone: the most common cause
 - MS: plaque within Cr. N. V nucleus
 - b) glossopharyngeal neuralgia (p.492): pain usually in base of tongue and adjacent pharynx
 - c) geniculate neuralgia (p.493): otalgia and deep prosopalgia
 - d) tic convulsif (p.493): geniculate neuralgia with hemifacial spasm
 - e) occipital neuralgia (p.515)
 - f) superior laryngeal neuralgia: a branch of the vagus, results primarily in laryngeal pain and occasionally pain on the auricle
 - g) sphenopalatine neuralgia
 - h) herpes zoster: pain is continuous (not paroxysmal). Characteristic vesicles and crusting usually follow pain, most often in distribution of V1 (isolated V1 TGN is rare). In rare cases without vesicles, diagnosis may be difficult
 - i) postherpetic neuralgia (Ramsay-Hunt syndrome) (p.493)
 - j) supraorbital neuralgia (SON) (p.491)
 - k) trigeminal neuropathic pain (AKA trigeminal deafferentation pain)¹³: may follow injuries from sinus or dental surgery, head trauma
 - l) trigeminal deafferentation pain: follows trigeminal denervation including therapeutic measures to treat trigeminal neuralgia¹³

- m) short-lasting unilateral neuralgiform headache with conjunctival injection and tearing (SUNCT)¹⁴: rare. Usually affects males 23–77 years old. Brief (<2 minutes) pain (burning, stabbing or shock-like) usually near the eye, occurring multiple times per day. Associated autonomic findings (the “hallmark of SUNCT”): ptosis, conjunctival injection, lacrimation, rhinorrhea, hyperemia. May be due to CPA AVM. Microvascular decompression or trigeminal rhizotomy may be effective in some cases refractory to medical treatment with AEDs or corticosteroids. Note: lacrimation (the most common) or other autonomic signs may occur in V1 trigeminal neuralgia but are usually mild, and appear only in the later stages of the condition and with long lasting attacks.¹⁵ Dramatic lacrimation and conjunctival injection from the onset of symptoms with SUNCT are the best characteristics to distinguish this from trigeminal neuralgia.¹⁶ May also occur in cluster headache (p.175).
2. ophthalmic pain
 - a) Tolosa-Hunt syndrome (p.569): painful ophthalmoplegia
 - b) (Raeder’s) paratrigeminal neuralgia (p.569): unilateral Horner’s syndrome + trigeminal neuralgia
 - c) orbital pseudotumor (p.569): proptosis, pain, and EOM dysfunction
 - d) diabetic (oculomotor) neuritis
 - e) optic neuritis
 - f) iritis
 - g) glaucoma
 - h) anterior uveitis
 3. otalgia (see below)
 4. masticatory disorders
 - a) dental or periodontal disease
 - b) nerve injury (inferior and/or superior alveolar nerves)
 - c) temporo-mandibular joint (TMJ) dysfunction
 - d) elongated styloid process
 - e) temporal & masseter myositis
 5. vascular pain syndromes
 - a) migraine headaches: see Migraine (p.175)
 - simple migraine: includes classic migraine, common migraine
 - complicated migraine: includes hemiplegic migraine, ophthalmoplegic migraine
 - b) cluster H/A (p.175); subtypes: episodic, chronic, chronic paroxysmal hemicrania
 - c) giant cell arteritis (p.195) (temporal arteritis). Tenderness over STA
 - d) toxic or metabolic vascular H/A (fever, hypercapnia, EtOH, nitrites, hypoxia, hypoglycemia, caffeine withdrawal)
 - e) hypertensive H/A
 - f) aneurysm or AVM (due either to mass effect or hemorrhage)
 - g) carotidynia: e.g. with carotid dissection (p.1324)
 - h) basilar dolichoectasia with fifth n. compression or indentation of the pons
 6. sinusitis (maximally, frontal, ethmoidal, sphenoidal)
 7. dental disease
 8. neoplasm: may cause referred pain or fifth nerve compression
 - a) extracranial
 - b) intracranial tumor: primarily posterior fossa lesions, neoplastic compression of trigeminal nerve usually causes sensory deficit (p.481)
 9. atypical facial pain (AFP) (prosopalgia): traditionally a “wastebasket” category used for many things. It has been proposed¹³ to reserve this term for a psychogenic disorder. May be suspected by
 10. primary (nonvascular) H/A: including
 - a) tension (muscle contraction) H/A
 - b) post-traumatic H/A

28.3.2 Otagia

General information

Because of redundant innervation of the region of the ear, primary otalgia may have its source in the 5th, 7th, 9th, or 10th cranial nerves or the occipital nerves.¹⁷ As a result, sectioning of the 5th, 9th or 10th nerve or a component of the 7th (nervus intermedius, chorda tympani, geniculate ganglion) has been performed with varying results.¹⁸ Also, microvascular decompression (MVD) of the corresponding nerve may also be done.¹⁹

Work-up includes: neurotologic evaluation to rule out causes of secondary otalgia (otitis media or externa, temporal bone neoplasms...). CT or MRI should be done in any case where no cause is found.

Primary otalgia

Primary otalgia is unilateral in most (~ 80%). Trigger mechanisms are identified in slightly more than half, with cold air or water being the most common.¹⁸ About 75% have associated aural symptoms: hearing loss, tinnitus, vertigo. Pain relief upon cocaineization or nerve block of the pharyngeal tonsils suggests glossopharyngeal neuralgia (p.492), however, the overlap of innervation limits the certainty.

An initial trial with medications used in trigeminal neuralgia (carbamazepine, phenytoin, baclofen...) (p.481) is the first line of defense. In intractable cases not responding to pharyngeal anesthesia, suboccipital exploration of the 7th (nervus intermedius) and lower cranial nerves may be indicated. If significant vascular compression is found, one may consider MVD alone. If MVD fails, or if no significant vessels are found, Rupa et al. recommend sectioning the nervus intermedius, the 9th and upper 2 fibers of 10th nerve, and a geniculate ganglionectomy (or, if glossopharyngeal neuralgia is strongly suspected, just 9th and upper 2 fibers of 10th).¹⁸

28.3.3 Trigeminal neuralgia

General information

Key concepts

- sharp, electric shock-like paroxysmal lancinating pain in the distribution of one or more branches of the trigeminal nerve on one side
- characterized by periods of remission and initial response to carbamazepine
- neurologic exam must be intact (only exception: mild sensory loss)
- 80–90% of cases are caused by compression of the trigeminal nerve at the root entry zone by the superior cerebellar artery. In MS patients, may be due to MS plaque (MS patients are usually less responsive to procedures)
- 75% will ultimately fail medical therapy and require a procedure (main options: microvascular decompression, percutaneous rhizotomy or radiosurgery). Choice of modality depends on patient age, location of symptoms, prior treatment and the side effect profile of the treatment modality

Trigeminal neuralgia (TGN) (AKA tic douloureux): paroxysmal lancinating electric-like pain lasting a few seconds, often triggered by sensory stimuli, confined to the distribution of one or more branches of the trigeminal nerve (□ Fig. 28.1) on one side of the face, with no neurologic deficit. The term “atypical facial pain” (AFP) is sometimes used to describe any other type of facial pain.

Rarely, TGN manifests as status trigeminus, a rapid succession of tic-like spasms triggered by seemingly any stimulus. IV carbamazepine (where available) or phenytoin may be effective for this.

Epidemiology

See □ Table 28.2. Annual incidence 4/100,000. There is no correlation with herpes simplex infection.²⁰ There is a tendency for spontaneous remission with pain free intervals of weeks or months being characteristic, regardless of treatment. 2% of patients with MS have TGN,²¹ whereas ~ 18% of patients with bilateral trigeminal neuralgia have MS.²²

Pathophysiology

Probably due to ephaptic transmission in trigeminal nerve from large-diameter partially demyelinated A fibers to thinly myelinated A-delta and C (nociceptive) fibers. Pathogenesis may be due to:

1. vascular compression of the trigeminal nerve at the root entry zone (NB: compression may be seen in up to 50% of autopsies in patients without TGN²⁵):
 - a) most commonly (80%) by the SCA; see Neurovascular compression syndromes (p.1534) for more details
 - b) persistent primitive trigeminal artery (p.83) ²⁶
 - c) dolichoectatic basilar artery^{27 (p 1108)}

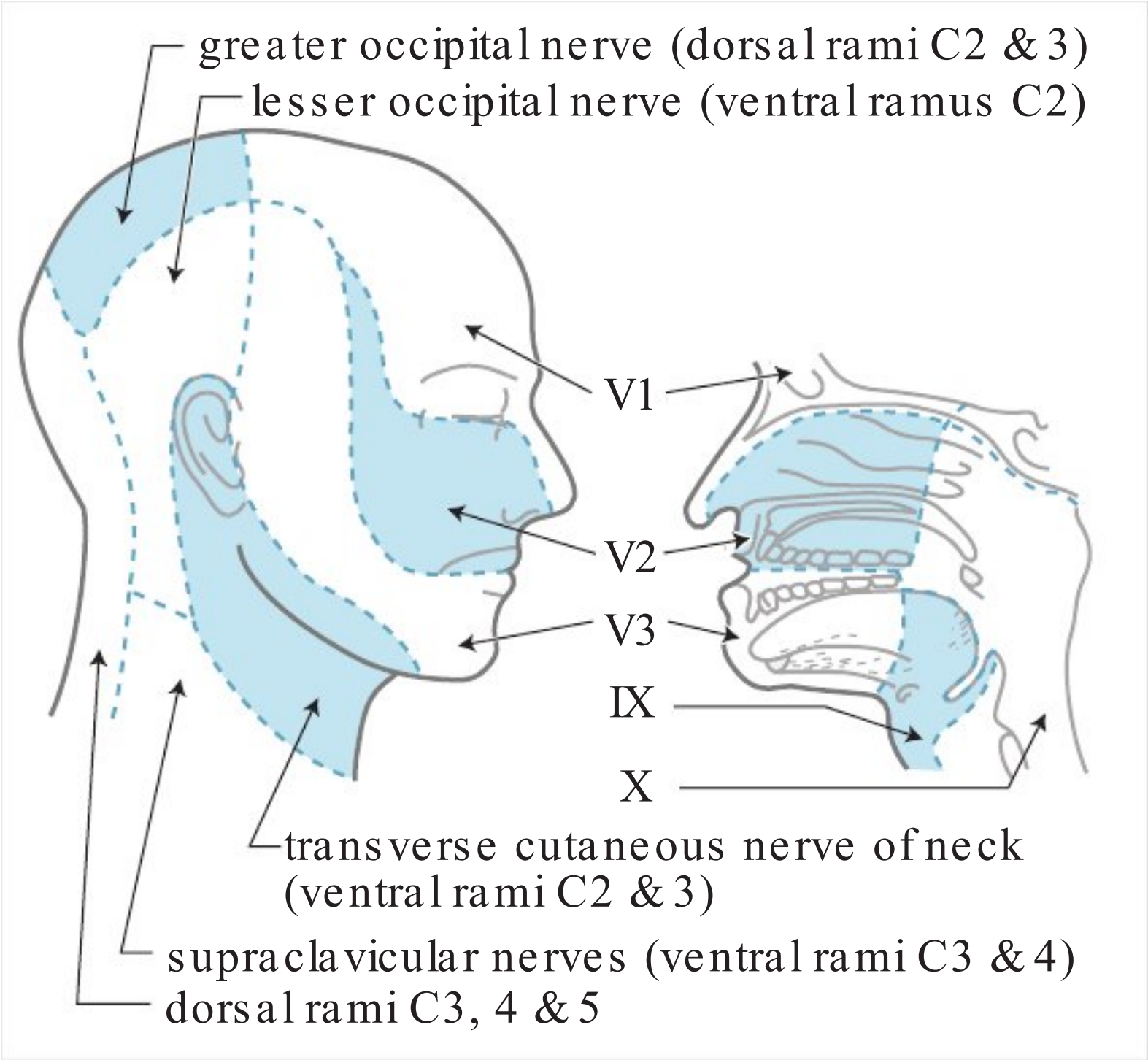


Fig. 28.1 Pain/temperature innervation of the head*
* KEY: V1 = ophthalmic nerve; V2 = maxillary nerve; V3 = mandibular nerve; IX = glossopharyngeal nerve; X = vagus

Table 28.2 Epidemiology of trigeminal neuralgia ^{23,24}	
age (years)	typically >50 (average 63)
female: male	1.8:1
Laterality	
right	60%
left	39%
both	1%
Division involved	
V1 only	2%
V2 only	20%
V3 only	17%
V1 & V2	14%
V2 & V3	42%
all three	5%

2. posterior fossa tumor (below)
3. in MS, plaque within brainstem may cause TGN that is often poorly responsive to microvascular decompression

In addition to the sensory division of the trigeminal nerve, other possible pain pathways include¹¹: the motor branch of the 5th nerve (portio minor), or the 7th or 8th nerve.

Tumors and trigeminal neuralgia

In >2000 patients with facial pain seen over 10 yrs, only 16 harbored tumor (<0.8% incidence).²⁸ 3 tumors outside cranial vault included nasal carcinoma and skull base mets; all had hypalgesia and atypical facial pain (AFP). 6 middle fossa tumors included 2 meningiomas, 2 schwannomas (1 primary tumor of Gasserian ganglion), and 1 pituitary adenoma. Posterior fossa tumors are the most likely to cause symptoms that most closely resemble true TGN; of these, vestibular schwannoma (VS) is most common. 2 of 7 VSs had tumors contralateral to the neuralgia (presumably due to brainstem shift). Patients with true TGN initially responded to carbamazepine, none with AFP did.

When facial pain is caused by tumor, especially with peripheral tumors, the pain is frequently atypical (usually constant), neurologic abnormalities are often present (usually sensory loss, although some are neurologically normal at first), and the age is often younger than typical TGN.

Differential diagnosis

See Craniofacial pain syndromes (p.477).

28

Evaluation

History and physical (in addition to routine)

1. history
 - a) accurate description of pain localization to determine which divisions of trigeminal nerve need to be treated
 - b) determine time of onset of TGN, trigger mechanisms
 - c) ascertain presence and length of pain-free intervals (lack of any pain-free interval is atypical for TGN)
 - d) determine duration, side-effects, dosages, and responses to medications tried
 - e) inquire about symptoms that may indicate the presence of conditions other than TGN: e.g. history of herpetic vesicles, excessive tearing of the eye (may indicate SUNCT (p.478), facial twitching (tic convulsif), tongue pain (glossopharyngeal neuralgia), sensory loss (tumor...), progressive relentless pain (tumor, herpes...), symptoms that suggest MS
2. physical exam: the exam should be normal in TGN, any neurologic deficit (except very mild sensory loss) in previously unoperated patient should prompt search for structural cause, e.g. tumor (see below). This exam also serves as a baseline for post-op comparison
 - a) assess sensation in all 3 divisions of trigeminal nerve bilaterally (include corneal reflexes)
 - b) assess masseter function (bite) and pterygoid function (on opening mouth, chin deviates to weak side)
 - c) assess EOM function

Imaging

MRI is often used to evaluate these patients for possible intracranial tumors or MS plaques, especially in cases with atypical features. The yield in typical cases is low.

Medical therapy for trigeminal neuralgia

Drug info: Carbamazepine (Tegretol®)

Complete or acceptable relief in 69%(if 600–800 mg/d are tolerated and give no relief, diagnosis of TGN is suspect²¹). Side effects: Drowsiness. Rash in 5–10% Possible Stevens-Johnson syndrome. Relative leukopenia is common (usually does not require discontinuing drug). See precautions under carbamazepine (CBZ, Tegretol®) (p.450).

□ 100 mg PO BID, increase by 200 mg/d up to maximum of 1200 mg/d divided TID. Supplied: See supply information (p. 450).

Drug info: Carbamazepine (Tegretol®)

Rapidly metabolized to carbamazepine, similar efficacy, often tolerated at higher doses than carbamazepine. Side effects : symptomatic hyponatremia.

□ for trigeminal neuralgia: 300 mg PO BID, increase by 600 mg/d q week. Usual dose: 450–1200 mg. Maximum of 2400 mg/d. Supplied : 150, 300, 600 mg tablets; 500 mg/5-ml suspension.

Drug info: Baclofen (Lioresal®)

2nd DOC (not as effective as carbamazepine, but fewer side-effects). Caution: teratogenic in rats. Avoid abrupt withdrawal (can cause hallucinations and seizures). May be more effective if used in conjunction with low dose carbamazepine.

□ Start low, 5 mg PO TID, increase q 3 d by 5 mg/dose; not to exceed 20 mg QID (80 mg/d); use smallest effective dose.

Drug info: Gabapentin (Neurontin®)

An anticonvulsant, may act synergistically with carbamazepine and baclofen. Side effects : include ataxia, sedation and rash.

□ start with 100 mg po BID, titrate to 5–7 mg/kg/day (3600 mg/d max).

Miscellaneous drugs

Also possibly effective:

1. phenytoin (Dilantin®): may be useful IV in patients in too much pain to open their mouths to take carbamazepine orally
2. capsaicin (Zostrix®): 1 gm applied TID for several days resulted in remission of symptoms in 10 of 12 patients (4 relapsed in <4 mos, but remained pain free for 1 yr after 2nd course)²⁹
3. clonazepam (Klonopin®) (p.454): works in 25%
4. lamotrigine (Lamictal®)
5. amitriptyline (Elavil®): more commonly used for atypical facial pain
6. botulinum toxin (Botox®): reduces transmission of CGRP producing a direct effect on the sensory nerve fibers

Surgical therapy for trigeminal neuralgia

Indications for surgery

Reserved for cases refractory to medical management, or when side effects of medications exceed risks and drawbacks of surgery.

Surgical options

1. peripheral trigeminal nerve branch procedures to block or ablate the division involved with pain, or can be used to block the trigger³⁰:
 - a) means of blocking
 - local blocks (phenol, alcohol)
 - neurectomy of trigeminal branch involved
 - b) nerve branches:
 - V1 (ophthalmic division) at the supraorbital, supratrochlear, or infraorbital nerves
 - V2 (maxillary division) at the foramen rotundum
 - V3 (mandibular division) block at the foramen ovale, or neurectomy of inferior dental nerve

2. blocking the trigger: either via percutaneous rhizotomy or alcohol block
3. percutaneous trigeminal rhizotomy (PTR): AKA percutaneous (stereotactic) rhizotomy (PSR) of trigeminal (Gasserian) ganglion (see below) (not truly a stereotactic procedure in the current sense of the word, therefore the term percutaneous trigeminal rhizotomy is preferred). Objective is to selectively destroy A-delta and C fibers (nociceptive) while preserving A-alpha and beta fibers (touch). Ideally, a retrogasserian lesion (not a ganglionic lesion). May also be used to block trigger. Lesioning techniques include (see below for comparison of techniques):
 - a) radiofrequency rhizotomy (RFR) (originated by Sweet and Wespis³¹). Uses radiofrequency energy to thermocoagulate the pain fibers. Requires the patient to be awake at intervals during the procedure
 - b) glycerol injection into Meckel's cave^{32,33}: possibly lower incidence of sensory loss and anesthesia dolorosa than with radiofrequency lesion.³⁴ Water soluble contrast cisternography was recommended in original description, may not be essential³⁵
 - c) mechanotrauma (percutaneous microcompression (PMC) rhizolysis): via inflation of No. 4 Fogarty catheter balloon.^{36,37,38} Does not require the patient to be awake
 - d) injection of sterile boiling water
4. Spiller-Frazier subtemporal extradural approach with retrogasserian rhizotomy (rarely used today). Original Spiller-Frazier technique involved avulsion of the nerve, which was met with unacceptably high risk of bleeding. The approach may be used to expose the ganglion and then mildly traumatize it
5. intradural retrogasserian trigeminal nerve section (sensory portion $\pm$ motor root, see below): may be performed during MVD if no vascular compression is identified
6. cutting descending trigeminal tract in lower medulla (99.5% success): rarely used
7. microvascular decompression (MVD)³⁹: (see below) microsurgical exploration of root entry zone, usually via posterior fossa craniectomy, and displacement of vessel impinging on nerve (if such a vessel is found). Usually with the placement of a non-absorbable "insulator" (Ivalon® sponge or shredded Teflon felt; see relative merits of Ivalon® vs. Teflon felt (p. 1536)
8. complete section of the nerve proximal to the ganglion via a p-fossa crani
9. stereotactic radiosurgery: see below
10. motor cortex stimulation⁴⁰: (somewhat analogous to spinal cord stimulation for spinal or extremity pain). Better for neuropathic trigeminal pain (as distinct from trigeminal neuralgia)

Selection of surgical option

Some pearls that influence treatment option choices (expert opinion⁴¹):

1. V3 neuralgia: RF. Can selectively treat V3 without involving other divisions
2. V1 or V2: balloon compression. Causes numbness in all 3 divisions, but unlike the lesion with RF the corneal numbness is better tolerated and the corneal reflex is often preserved
3. bilateral pain: glycerol. It has the shortest duration of effect, which is an advantage if you think you may need to treat the other side at some point
4. SRS: due to latency until pain relief, suboptimal for patients who need immediate pain relief

Peripheral nerve ablation and neurectomies

Limited to pain or trigger points in territory of supraorbital/supratrochlear, infraorbital, or inferior dental nerves. Neurectomy may be a consideration especially for elderly patients who are not candidates for MVD (neurectomy may be done under local anesthesia) with pain in the forehead (to avoid anesthesia of the eye, as could occur with RFR). Disadvantages include sensory loss in the distribution of the nerve and a high rate of pain recurrence due to nerve regeneration (usually in 18–36 months) which often responds to repeat neurectomy.⁴² May also be used following PTR.

Supraorbital and supratrochlear nerve: See also information on supraorbital neuralgia (SON) or supratrochlear neuralgia (STN) (p. 491). SON may be treated with rhizotomy (e.g. with alcohol or radiofrequency) or with neurectomy. Alcohol injection is used with caution for STN because of risk of injury to the superior oblique muscle. For neurectomy, these nerves are exposed through a 2 cm incision parallel to and just above the medial portion of the eyebrow (never through the eyebrow as this can create an unsightly "bi-brow"; shaving the eyebrow is also discouraged since it occasionally does not grow back). The incision is carried down to the bone and the periosteum is elevated caudally towards the supraorbital foramen or notch. The nerves will be visible on the undersurface of the periosteal flap. The supraorbital nerve is freed in its foramen/notch, and is then avulsed by grasping it with a mosquito hemostat and twisting the clamp. The nerve avulses "like pulling a worm out of a hole." The distal portion of the nerve should be located at the site where the periosteum was incised and it, too, should be avulsed. The process can be repeated for the more medially situated supratrochlear nerve.

Other nerves: Not covered here, other nerve branches that may be cut or avulsed include: infra-trochlear, lacrimal (the branch of VI at the lateral edge of the orbit), infraorbital nerve, inferior alveolar, lingual and mental nerves.^{43 (p 290)}

Percutaneous trigeminal rhizotomy (PTR)

Recommended for patients who: are poor risk for general anesthesia (elderly or those with increased risk for general anesthesia), wish to avoid “major” surgery, have unresectable intracranial tumors, have MS, have impaired hearing on the other side, or have limited life expectancy (<5 yrs).³⁴ For “atypical facial pain”, denervating the painful region of the face benefits <20% of patients, and worsens 20%.⁴⁴ Recurrences are easily treated by repeat procedures. May be used to treat failures of peripheral nerve ablation.

□ Choice of lesion technique. Recurrence rates and incidence of dysesthesias are comparable among the various lesioning techniques. Incidence of intraoperative hypertension is less with PMC than with radiofrequency rhizotomy (RFR) lesion³⁸ (no reports of intracerebral hemorrhage). Bradycardia occurs regularly with PMC which may not be harmful (some prophylax with atropine⁴⁵). RFR requires a patient who is able to cooperate; PMC can be done with the patient asleep. Paralysis of ipsilateral trigeminal motor root (e.g. pterygoids) is more common after PMC (usually temporary) than RFR, and so PMC should not be done if there is already contralateral paralysis from a previous procedure. See also description of technique (p.486).

Complications with percutaneous radiofrequency (note: some “numbness” is actually expected in most successful PTRs and occurs in 98% of cases,²⁴ and is not considered a complication here)

- 1. mortality: only 17 deaths in over 22,000 procedures (includes lesser experienced neurosurgeons and patients often considered poor surgical risks)²¹
- 2. dysesthesias²⁴ (sometimes called “annoying paresthesias”): higher rate in more complete lesions
 - a) minor: 9%
 - b) major (requiring medical treatment): 2%
 - c) anesthesia dolorosa (severe, constant, burning aching pain that is refractory to all treatment): 0.2–4%
- 3. meningitis²³: 0.3%
- 4. alterations in salivation⁴⁶: 20%(increased in 17% decreased in 3%)
- 5. partial masseter weakness (usually not perceived by patient, □ Table 28.3)
- 6. oculomotor paresis (usually temporary, □ Table 28.3)
- 7. reduced hearing (secondary to paresis of tensor tympani, □ Table 28.3)
- 8. neuroparalytic keratitis (keratitis due to fifth nerve deficit which impairs sensation, □ Table 28.3)
- 9. intracranial hemorrhage: personal report of 7 cases (6 fatal) in > 14,000 procedures, probably due to transient HTN (SBP up to 300 torr)
- 10. alterations in lacrimation⁴⁶: 20%(increased in 17% decreased in 3%)
- 11. herpes simplex eruption: prescribe antiherpetic drug if patient develops symptoms, e.g. Acyclovir® (p.366).
- 12. bradycardia and hypotension: 1% with RFR compared to up to 15% with glycerol injection)
- 13. rare^{47,48}:
 - a) temporal lobe abscess
 - b) aseptic meningitis
 - c) intracerebral abscess: 0.1%

Table 28.3 Complications with percutaneous radiofrequency			
Complication	850 cases ⁵⁶		315 cases ⁴⁶
	straight electrode (N=700)	curved electrode (N=150)	
partial masseter weakness (usually not perceived by patient)	15–24%	7%	50%
oculomotor paresis (usually temporary)	2%	0	
reduced hearing (secondary to paresis of tensor tympani)	0	0	27%
neuroparalytic keratitis (keratitis due to fifth nerve deficit which impairs sensation)	4%	2%	0

- d) trigeminal trophic syndrome (TTS)⁴⁹: triad of unilateral crescentic nasal alar ulceration with anesthesia and paresthesia of the trigeminal dermatome (may present with severe pruritus and self-induced skin lesions from scratching). A result of trigeminal nerve injury. Treatment has included: carbamazepine, diazepam, amitriptyline, chlorpromazine, clonazepam or pimozide⁵⁰
- e) complications related to needle placement⁵¹:
 - carotid cavernous fistula (CCF): may occur with any percutaneous technique⁵² (including balloon microcompression⁵³)
 - blindness: from penetration of inferior orbital fissure⁵⁴
 - injury to other cranial nerves: II, III, IV, VI⁵⁵
- f) subarachnoid hemorrhage
- g) seizures

Microvascular decompression (MVD)

See also more detailed information (p.488).

Recommended for patients with inadequate medical control of pain with >5 years anticipated survival and able to tolerate a small craniotomy³⁴ (surgical morbidity increases with age). Relief is often long lived, persevering 10 yrs in 70% Incidence of facial anesthesia is much less than with PTR, and anesthesia dolorosa does not occur. Mortality: <1% Incidence of aseptic meningitis (AKA hemogenic meningitis): 20% 1–10% major neurologic morbidity. Failure rate: 20–25%

1–2% of patients with MS will have a demyelinating plaque at the root entry zone, this usually does not respond to MVD, and one should attempt a PTR.

28

Stereotactic radiosurgery (SRS)

The first use of SRS by Leksell was for the treatment of TGN. Initially, this was reserved for refractory cases following multiple operations,⁵⁷ now becoming more widely practiced. The least invasive procedure. Generally recommended for patients with co-morbidities, high-risk medical illness, pain refractory to prior surgical procedures, or those on anticoagulants (anticoagulation does not have to be reversed to have SRS).

Treatment plan: 4–5 mm isocenter in the trigeminal nerve root entry zone identified on MRI. Use 70–80 Gy at the center, keeping the 80% isodose curve outside of the brainstem.

Results: Significant pain reduction after initial SRS: 80–96%^{58,59,60,61} but only ≈ 65% become pain free. Median latency to pain relief: 3 months (range: 1 d–13 months).⁶² Recurrent pain occurs within three years in 10–25% Patients with TN and multiple sclerosis are less likely to respond to SRS than those without MS. SRS can be repeated, but only after four months following the original procedure.

Favorable prognosticators: higher radiation doses, previously unoperated patient, absence of atypical pain component, normal pre-treatment sensory function.⁶³

Side effects: Hypesthesia occurred in 20% after initial SRS, and in 32% of those requiring repeat treatment⁶² (higher rates associated with higher radiation doses⁵⁹).

Management of surgical treatment failures

90% of recurrences are in distribution of previously involved divisions; 10% are in new division and may represent progression of the underlying process. Some treatment failures are not persistent TN, but rather represent trigeminal neuropathic pain (AKA trigeminal deafferentation pain).

PTR may be repeated in patients who have a recurrence with some preservation of facial sensation. Attempted repeat PTR is often productive, and failures can be managed as below.

MVD may be performed in patients failing PTR, but the success rate may be reduced⁶⁴ (91% for patients undergoing MVD first, vs. 43% for those having MVD following PTR; note: 91% may be an unrealistically high success rate, and taking patients that fail PTR may select for a more difficult subgroup). Repeat MVD may also be performed, with attention given to possible slippage of the insulating sponge, or the fact that the true offending vessel may be “artificially” moved away from the nerve secondary to the surgical positioning.

SRS can be repeated, using the same dose, with reported significant reduction in pain in 89% and complete relief in 58%⁶²

Intradural retrogasserian trigeminal nerve section

May be used as a measure of last resort in patients who have recurrent TGN following one or more PTRs in the presence of total facial anesthesia, or in patients undergoing posterior-fossa craniectomy for the purpose of MVD when no impinging vessel can be identified. In the latter case, a partial rhizotomy is performed by sectioning 2/3 of nerve, with resulting partial anesthesia. In the case of patients with facial anesthesia pre-op, consideration should be given to sectioning the motor division (portio minor) as an alternate pain pathway.¹¹

Percutaneous trigeminal rhizotomy (PTR)

Due to concerns about hemorrhage, check coagulation profile (PT/PTT, consider bleeding time), and discontinue ASA and NSAIDs, preferably 10 days pre-op. Procedure may be performed in OR with fluoro, or in angiography suite in x-ray department.

Booking the case: Percutaneous trigeminal rhizotomy

(For any of the percutaneous methods: balloon, glycerol, RFR)
Also see defaults & disclaimers (p.27) and pre-op orders (see below).

1. position: supine

2. anesthesia: MAC with sedation

3. equipment:

a) lesion generator and needle kit for radiofrequency rhizotomy

b) C-arm fluoroscopy (2 C-arms for balloon compression)

c) calibrated inflatable balloons (as in kyphoplasty) for balloon rhizotomy

4. consent (in lay terms for the patient – not all-inclusive):

a) procedure: put a needle into the cheek to numb the nerve to the face

b) alternatives: medical treatment, surgery through the back of the skull (microvascular decompression), radiation (stereotactic radiosurgery)

c) complications: facial numbness is anticipated, rarely: stroke, bleeding, blindness

28

Pre-op orders (RFR)

1. NPO after MN except meds
2. continue Tegretol® & other meds PO with sips of water
3. AM of procedure: IV NS @KVO in arm contralateral to neuralgia
4. atropine 0.4 mg IM PRN (□ contraindications include rapid a-fib)
5. non-disposable LP tray to accompany patient

Technique: Percutaneous trigeminal radiofrequency rhizotomy (RFR)

Adapted technique.⁶⁵

NB: needle insertion and/or lesioning may cause HTN, consider monitoring BP. Use either a straight electrode (bare 5 mm for 1 division, 7.5 mm for 2 divisions, or 10 mm for total lesions) or a curved electrode.⁵⁶

- Electrode insertion
1. attach ground electrode to patient’s upper arm

2. prep the cheek on the involved side with Betadine

3. entry point: under short-acting anesthetic agent – e.g. propofol (Diprivan®) (p.133) or methohexital (Brevitol®) (p.132) -, insert electrode-needle 2.5–3 cm lateral to oral commissure

4. trajectory:

a) palpate the buccal mucosa with a gloved finger inside the mouth (lateral to the teeth) and with the other hand pass the electrode medial to the coronoid process of the mandible (keeping the needle deep to the oral mucosa, i.e. outside the oral cavity) initially aiming towards the plane intersecting a point 3 cm anterior to EAM and the medial aspect of the pupil when the eye is directed forward. Be careful not to contaminate the field with the hand that was in the patient’s mouth

b) as insertion progresses, use fluoroscopy to direct the tip towards the intersection of the top of the petrous bone with the clivus (5–10 mm below floor of sella along clivus)

c) upon entering foramen ovale the masseter often contracts, causing the jaw to briefly close. Remove the stylet, look for CSF to verify location (may not occur in re-do cases), and insert electrode through needle

In difficult cases, intraoperative fluoroscopy may assist in localizing the needle to Meckel’s cave and to R/O e.g. entry into superior orbital fissure (which can cause blindness after lesioning), or entry into foramen spinosum (middle meningeal artery). If necessary to visualize (e.g. when there is difficulty entering), the foramen ovale is optimally seen on a submental x-ray by hyperextending neck 20° and rotating head 15–20° away from side of pain.⁶⁶

Impedance measurements: from the tip of the electrode when available may help indicate location of needle tip. Impedance: CSF (or any fluid) low ($\approx 40\text{--}120\ \Omega$); connective tissue, muscle, or nerve is usually $200\text{--}300\ \Omega$ (may be up to $400\ \Omega$); if $>400\ \Omega$ this likely indicates electrode is contacting periosteum or bone. After starting the lesion, impedance often goes down by $30\ \Omega$ transiently, and then as the lesioning continues it gradually returns to baseline or $\approx 20\ \Omega$ above it. If char develops on the electrode tip, the impedance will read higher than where it started.

Stimulation and repositioning

Once the foramen ovale is entered, the needle is positioned with the following guidelines: for V3 division lesion the curved electrode should be just short of the clivus and pointing down, for V2 it is at the clivus and directed up, for V1 it is 5 mm beyond clivus and pointing up. □ At no time should the needle tip extend >8 mm beyond clival line (to avoid Cr. N. III or VI complications).

The patient is allowed to wake up and is stimulated through the electrode with the following settings: frequency = $50\text{--}75$ Hz, 1 mS duration, start at 0.1 V amplitude and slowly increase (usually 0.2–0.5 V is adequate, higher voltages may indicate that the needle is not near the target and that stimulation is due to far-field currents, however, in previously lesioned patients up to 4 V may sometimes be necessary). If stimulation does not reproduce pain in the distribution of the patient's TGN, then the amplitude is returned to 0, the electrode is repositioned (straight electrode: advance needle <5 mm at a time, until the tip is in the vicinity of the clival line; curved tip electrode: advance and/or rotate) and then slowly elevate the voltage again from 0 and repeat the repositioning-stimulating process until stimulation reproduces the distribution of tic pain. If previous lesions have produced analgesia and the patient cannot feel the stimulating current, one may stimulate at 2 Hz. and watch for masseter twitch (requires preserved motor root).

Lesioning

When stimulation produces pain in the involved distribution of the TGN, perform the first lesion under short-acting anesthesia at $60\text{--}70^\circ\text{C} \times 90$ sec. A facial flush may be noted.⁶⁶ After every lesion, perform a post-lesion assessment (see below). The goal is analgesia (but not anesthesia) in the areas of tic pain and hypalgesia in areas of trigger points. An average of three lesions are necessary at the first sitting, each $\approx 5^\circ\text{C}$ higher than the previous for 90 seconds. Anesthetic may not be needed after the first lesion if moderate analgesia has been produced by previous lesions.

Post-lesion assessment

After each lesion and at completion of procedure, assess:

1. sensitivity to pinprick and light touch in all three divisions of trigeminal nerve (grading: normal, hypalgesic, analgesic, anesthetic)
2. corneal reflex bilaterally
3. EOM function
4. masseter muscle strength (patient clenches teeth, palpate cheeks for contraction)
5. pterygoid muscle strength (ask patient to open mouth, chin deviates towards side of pterygoid weakness)

Post-op care (PTR)

Include in post-op orders:

1. ice pack to face on side of procedure for 4 hrs
2. soft diet
3. routine activity when alert
4. avoid narcotics (usually not necessary)
5. if corneal reflex impaired: risk of neuroparalytic keratitis. Natural tears 2 gtt q 2 hrs while awake to eye on affected side. Lacrilube® to eye & tape eye shut q hs

Prior to discharge from hospital, repeat post-lesion assessment (see above). Patients are then weaned off of carbamazepine as tolerated.

Percutaneous microcompression rhizolysis balloon (PMC)

Via inflation of No. 4 Fogarty catheter balloon.

Technique

1. the needle is placed as with RFR (p.486).
2. aim for balloon placement in the medial foramen ovale (to avoid entering the middle fossa). After placing the balloon, insert the stylet to visualize where the balloon will go. Use Omnipaque 240 to fill the balloon
3. inflate to 1.4 atmospheres of pressure

Table 28.4 Comparison of outcomes of percutaneous techniques to MVD				
Parameter	Percutaneous techniques (PTR)			MVD
	RFR ^a	Glycerol	Balloon	
initial success rate ^{11,24}	91–99%	91%	93%	85–98%
medium-term recurrence rate	19% at 6 yrs ²³	54% at 4 yrs	21% at 2 yrs	15% in 5 yrs
long-term recurrence rate	80% at 12 yrs ^{46b}			30% at 10 yrs
facial numbness ²⁴	98%	60%	72%	2%
^a abbreviations: RFR=radiofrequency rhizotomy; MVD=microvascular decompression; balloon = balloon micro-compression				
^b this author included initial failures to PTR requiring repeat procedures during same hospitalization				

Results

Results of various PTR techniques compared to microvascular decompression (MVD) are shown in Table 28.4. Recurrence rate is higher in patients with multiple sclerosis (50% at 3 yrs mean F/U).⁶⁷

Microvascular decompression (MVD) for trigeminal neuralgia

Indications

- 1. patients unable to achieve adequate medical control of trigeminal neuralgia with ≥ 5 yrs anticipated survival, without significant medical or surgical risk factors³⁴ (although a small p-fossa exploration is usually well tolerated, surgical morbidity increases with age)
- 2. may be used in patients who do not fit the above criteria, but have intractable pain and fail PTR
- 3. patient with tic involving V1 for whom the risk of exposure keratitis due to corneal anesthesia would be unacceptable (e.g. already blind in contralateral eye) or patient wishing to avoid facial anesthesia for any reason
- 4. □ patients with MS are usually not considered candidates for MVD due to low response rate

Booking the case: Microvascular decompression

Also see defaults & disclaimers (p.27) and pre-op preparation (see below).

- 1. position: park bench
- 2. equipment: microscope
- 3. implants: Ivalon sponge or shredded Teflon
- 4. intra-op monitoring: (optional) BAER, facial EMG (monitors VII and portio minor (motor) of VII), VIII (CNAP (compound nerve action potential) using Cueva electrode placed directly on VIII nerve referenced to ipsilateral earlobe)
- 5. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery behind the ear to move a blood vessel from the sensory nerve of the face, if no offending vessel can be identified then possible partial sectioning of the appropriate part of the trigeminal nerve with associated numbness)
 - b) alternatives: needle procedures through the cheek (percutaneous rhizotomy), radiation (stereotactic radiosurgery)
 - c) complications: (in addition to usual craniotomy complications), CSF leak, hearing loss (≈ 10%), facial numbness, pain near incision (occipital neuralgia or lesser occipital neuralgia), rarely: diplopia, facial paralysis, failure of the procedure

Technique

Also see Paramedian suboccipital craniectomy (p.1447), for important pointers, including use of armored endotracheal tube.

Preoperative preparation

An MRI is recommended (with FIESTA sequence or equivalent, if available) to rule-out mass lesion or vascular abnormality. Baseline BAER are performed by some⁶⁸ (see below for intra-op monitoring).

O.R. setup

Setup for lateral oblique suboccipital (posterior-fossa) craniotomy (p.1446). Microscope: observer's eyepiece is placed on the side opposite to that of the tic.

Positioning

See reference.⁶⁹

1. lateral oblique position (p.1446), symptomatic side up, axillary roll
2. thorax elevated 10–15° to reduce venous pressure
3. 3-pin skull fixation. Head position:
 - a) head rotation: head rotated 10–15° away from the affected side. Do not exceed 30° rotation
 - b) lateral head tilt
 - for trigeminal neuralgia or VIII nerve approach: the head is parallel to the floor (if it is lower, nerves VII & VIII will obscure view of V)
 - for VII nerve or lower, the vertex is tilted 15° down from the horizontal
 - c) flex neck: leave 2 fingerbreadths room between the chin and the sternum
4. upper shoulder retracted caudally with adhesive tape
5. option: lumbar spinal drain. Drain 20–30 cc during craniotomy, then drain off small amounts from time-to-time during the case to keep the field mostly dry, but occasionally letting CSF build up to bathe cranial nerves

Intra-operative monitoring

Option: intra-operative monitoring of facial EMG and BAER (assesses acoustic nerve).⁶⁸

Approach

1. skin incision⁶⁹: vertical incision 3–5 cm in length, 5 mm medial to mastoid notch, a small “5–6–4” incision (p.1448); in thick or short-necked patients, a slightly longer incision that angles infero-medially is used. 75% of the incision is inferior to the transverse sinus, 25% superior
2. burr hole:
 - a) 1 cm inferior and 1 cm medial to the asterion⁷⁰ (p.60)
 - b) if the asterion is not easily identified or if there are concerns about the reliability of the asterion as a landmark for the junction of the transverse and sigmoid sinuses,⁷¹ place the burr hole directly over the mastoid emissary vein which drains superolaterally into the sigmoid sinus
3. craniotomy: top of bone opening as close as possible to transverse sinus. The position of the transverse sinus can be approximated by a line drawn from the posterior base of the zygomatic process to the inion, or roughly $\approx$ 2 finger-breadths above the upper end of the mastoid notch. Lateral limit of bone opening is sigmoid sinus. A triangular bony opening with a leg along each sinus works well. Craniectomy diameter needs to be only $\approx$ 3 cm. Apply bone wax liberally (blocks off any possible opening into the mastoid air cells)
4. dural opening: either a curvilinear with each end at a sinus and the convexity away from the junction (Jannetta) or an inverted “T” (with one incision towards each sinus and the third towards junction of sinuses)
5. minimal or no retraction of cerebellum is usually required
6. allow CSF to drain before proceeding: this may require gentle advancement of a cottonoid in the CPA. A lumbar drain should be placed if CSF cannot be drained
7. follow the junction of tentorium with temporal bone deep. Place a retractor that both medially displaces the cerebellum and slightly “lifts” the cerebellum towards the surgeon (medial displacement alone is not as effective)
8. petrosal vein: coagulate and divide the petrosal vein complex (usually 2–3 veins connecting to the tentorial dura). If the vein is torn, the dural side is tamponaded (sometimes up to 30 minutes is needed) while the free end is coagulated
9. V is deeper than the VII/VIII complex, which should not even be seen with this approach. If VII/VIII are seen, move the retractor superiorly as even gentle traction may cause hearing loss (□ Fig. 1.9). There is often a hillock of bone just posterior to Meckel's cave obscuring the site where the fifth nerve enters the cave

Decompression of nerve

1. arachnoid overlying the fifth nerve is sharply divided (caution re Cranial Nerve IV which follows the tentorial opening in the arachnoid rostral to the fifth nerve). Intra-op changes in BAER are often attributed to retraction of arachnoid that is tethered to the VII/VIII complex
2. the fifth nerve may be markedly atrophic if previous PTRs have been done
3. identify the smaller motor root (portio minor) of the fifth nerve
4. arteries and/or veins compressing V should be dissected off the nerve. NB: vessels located proximally are the most likely offenders, however, the dorsal root entry zone (which is the sensitive part of the nerve) may be variable in location and peripheral vessels may be culpable. The nerve should be inspected and freed of vessels from its origin at the brainstem all the way to its entrance into Meckel's cave.⁶⁹ Veins may be coagulated and then should be divided (to prevent recanalization)
5. the most common cause of compression is the superior cerebellar artery (SCA)
6. check the nerve at the junction with brainstem for any residual compression prior to the next step
7. insulating material is interposed between nerve and vessel to prevent re-compression. Options include:
 - a) e.g. Ivalon®, (polyvinyl formyl alcohol) sponge (Ivalon Surgical Products, 1040 OCL Parkway, Eudora, KS, 66025, U.S.A. distributed by Fabco in the U.S.A. (860) 536-8499, toll free: (888) 813-8214, <http://fabco.net/catalog/ivalon-ophthalmic/>) cut in a saddle shape. Note: if an Ivalon block is used instead of pre-packaged sterile pads, it must be rinsed thoroughly to remove formalin, then autoclaved. Ivalon should be hydrated in NS for 10 minutes prior to cutting it
 - b) shredded Teflon felt; see merits of Ivalon® vs. Teflon or muscle (p.1536)
8. Wilson recommends performing a partial sensory rhizotomy of the inferior one-half to two-thirds of the portio major for the following: cases where no vascular contact with the nerve or no deformity of the nerve is identified, in most cases of patients undergoing a repeat MVD, or for cases with duration of symptoms >8-9 yrs as this latter group tends to have a lower success rate with MVD alone⁷²
9. if the procedure is for a failed MVD and it is desired to partially divide the nerve, the nerve is organized somatotopically with V1 fibers superiorly, and V3 inferiorly. If the goal is total elimination of pain pathways and there is concern about pain conduction through ancillary pathways, consider also dividing the motor root (portio minor)

Closure

1. bone wax should be applied liberally to the exposed lateral bone edges (to paraphrase Dr. Jannetta⁶⁹ and Mr. Miyagi,⁷³ "Wax in, wax out.")
2. irrigate gently with warm saline (avoid "jet" irrigation which can damage the VIII nerve)
3. intra-op BAER decline may occur on dural closure and should prompt re-opening of the dura and checking for tension on the VIII nerve from a vessel or Telfa
4. perform several Valsalva maneuvers to ensure watertight closure of dura
5. the bone defect should be covered e.g. with burr hole cover to reduce chance of pain associated with uncovered craniectomy
6. after fascial closure, Valsalva maneuver is performed again to ensure watertight closure
7. use 4-0 running locked nylon to approximate skin in watertight fashion (avoid excessive tension)

Post-op care following MVD

Include in post-op orders

1. admit to ICU
2. arterial-line for continuous BP monitoring
3. analgesics (e.g. codeine 30-60 mg IM q 3 hrs)
4. anti-emetics (e.g. ondansetron 4 mg IV q 6 hrs)
5. medication to aggressively treat HTN (viz. SBP > 160 mm Hg)

Post-op H/A, nausea and pain

Patients routinely have H/A and nausea for 2-3 days (there tends to be less intracranial air and less "pneumoencephalogram sickness" if the park-bench position is used instead of the sitting position). However, severe H/A should prompt a STAT CT to R/O bleeding. If the CT is negative, severe H/A may be due to transient elevation of CSF pressure that occurs in some, and which usually responds to 1, or at most 2, LPs to halve the pressure. Aseptic meningitis usually responds to steroids. Some patients have continued but lessened tic douloureux pain for several days post-op, this usually subsides.⁶⁹

Complications

The short list:

1. cerebellar injury
2. hearing loss
3. CSF leak

The long list:

1. mortality: 0.22–2% in experienced hands (>900 procedures)^{74,75}
2. meningitis
 - a) aseptic meningitis (AKA hemogenic meningitis): H/A, meningismus, mild fever, culture negative CSF, pleocytosis. Incidence: $\approx$ 2% (up to 20% has been reported). Usually occurs 3–7 days post-op. Responds to LP + steroids
 - b) bacterial meningitis: 0.9%
3. major neurologic morbidity: 1–10% (higher rates with less experienced surgeons), including:
 - a) deafness: 1%
 - b) vestibular nerve dysfunction
 - c) facial nerve dysfunction
4. mild facial sensory loss: 25%
5. cranial nerve palsies⁷⁶:
 - a) fourth nerve (diplopia): 4.3% (only $\approx$ 0.1% are permanent)
 - b) facial nerve: 1.6% (most are transient)
 - c) eighth nerve (hearing loss): 3%
6. postoperative hemorrhage⁷⁷: subdural, intracerebral (1%²⁴), subarachnoid
7. seizures: including status epilepticus⁷⁷
8. infarction⁷⁷: including posterior cerebral artery distribution, brain stem
9. CSF leak: resolves with lumbar drainage in most cases
10. pneumonia: 0.6%

Outcome

1. success rate: 75–80% (rates may be lower in patients having prior destructive procedure); good but not total relief in an additional $\approx$ 10%
2. recurrence rate in large series is difficult to ascertain from literature; in a series of 40 patients followed 8.5 yrs mean⁷⁵:
 - a) major recurrence (recurrent tic not controlled by medications) rate: 31%
 - b) minor recurrence (mild or controlled by medications) rate: 17%
 - c) using Kaplan-Meier curve, expect 70% to be either pain free or have minor recurrence by 8.5 years (or $\approx$ 80% at 5 years)
 - the risk for a major recurrence after MVD is 3.5% annually
 - the risk for a minor recurrence after MVD is 1.5% annually
 - d) major recurrence rate is lower for patients having major arterial cross-compression of the nerve discovered at the time of surgery (patients with venous compression had much higher rate)
 - e) this study found no correlation between previous destructive surgery and major recurrence rate (in 11 patients)

Some feel that the longer one waits before performing a MVD, the lower the success rate.

28.3.4 Supraorbital and supratrochlear neuralgia

Anatomy

The supraorbital and supratrochlear nerves arise from the frontal nerve and are 2 of the 5 branches of V1 (ophthalmic division of the trigeminal nerve). The supraorbital nerve is the largest branch. It exits the orbit through the supraorbital notch or foramen, usually within the medial third of the orbital roof (mean distance from exit to medial angle of orbit: 20 mm (range: 5–47)⁷⁸). The supratrochlear nerve exits the orbit without a foramen or notch 3–38 mm medial to the supraorbital nerve (mean: 15.3 mm)⁷⁸, the most medial branch varies from 8–30 mm lateral to the patient's midline.⁷⁸

Supraorbital neuralgia characteristics

Trigeminal neuralgia (TGN) may present with pain in the distribution of the supraorbital nerve, however, the supraorbital nerve may be involved in supraorbital neuralgia (SON), a distinct syndrome

with different clinical characteristics. SON is a rare condition slightly more common in women, with onset typically 40–50 years of age.⁷⁹ Characteristics⁸⁰: 1) unilateral pain in the distribution of the supraorbital nerve (□ Fig. 93.2), 2) tenderness in the region of the supraorbital notch or along the distribution of the nerve, and 3) temporary relief with nerve block.

The pain is usually chronic-continuous or remitting-intermittent.⁷⁹

SON may be:

1. primary (no identifiable etiology): these cases lack any sensory loss
2. secondary (e.g. due to trauma to the area, or resulting from chronic pressure such as with wearing swim goggles): more common than primary SON. Most cases remit within one year⁷⁹ with elimination of the offending pressure

Supratrochlear neuralgia

Cases of pain isolated to the supratrochlear nerve appear to exist. Supratrochlear neuralgia (STN) may be differentiated from SON by restriction of pain in the more medial forehead, and with relief on blockade of the supratrochlear nerve alone.

Differential diagnosis

1. migraine: suggested by nausea, vomiting and photophobia
2. associated autonomic activity is rare with SON, and should prompt consideration of cluster H/A (p.175) or SUNCT (p.478)
3. TGN: typical TGN features lacking in SON include characteristic triggers and pain consisting exclusively of paroxysmal/ultra-brief electric shock-like pain
4. hemicrania continua: continuous unilateral pain that tends to be located more posteriorly and is absolutely responsive to indomethacin⁸⁰
5. trochleitis: inflammation of the trochlea/superior-oblique muscle complex, may mimic supratrochlear neuralgia with pain of the medial upper orbit extending a short distance to the forehead.⁸¹ The pain is typically exacerbated by supraduction of the eye and to palpation of the trochlea, and is relieved with injection of local anesthetic or, by the usually definitive treatment of infiltration of corticosteroids close to the trochlea. Diplopia is rare and minimal
6. nummular (coin-like) H/A⁸²: round or oval 2–6 cm diameter area of pressure-like continuous head pain without underlying structural abnormality. In 13 patients, 9 (70%) the area was located at the parieto-occipital junction. 9 (70%) demonstrated hypoesthesia and touch provoked paresthesias in the affected area

Treatment

Gabapentin (800–2400 mg/d) or pregabalin (150 mg/d) is helpful for some.⁸³

Topical capsaicin (p.482) applied to the symptomatic area may help.

Refractory cases may respond to rhizotomy with alcohol (providing an average of 8.5 months relief⁸⁴) or with radiofrequency ablation.

Persistent cases may require exploration and decompression of the nerve by lysing bands overlying the supraorbital notch,⁸⁵ or, ultimately, to neurectomy (p.483) which provides an average of 33.2 months relief.⁸⁶

28.3.5 Glossopharyngeal neuralgia

Epidemiology

Incidence: 1 case for every 70 of trigeminal neuralgia.⁸⁷ (p 3604–5)

Clinical

Severe, lancinating pain in the distribution of the glossopharyngeal and vagus nerves (throat & base of tongue most commonly involved, radiates to ear (otalgia), occasionally to neck), occasionally with salivation and coughing. Rarely: hypotension,⁸⁸ syncope,⁸⁹ cardiac arrest and convulsions may accompany. May be triggered by swallowing, talking, chewing. Trigger zones are rare.

Treatment

Pain may be reduced by cocaineization of tonsillar pillars and fossa. Usually, the persistence and severity of pain requires surgical intervention. One may either perform microvascular decompression,

or nerve division via extra- or intra-cranial approach (latter may be required for permanent relief).

Intracranial approach: Section of preganglionic glossopharyngeal nerve (IX) and upper one-third or two fibers (whichever is larger) of vagus (X). IX is readily identified at its dural exit zone where it is separated from X by a dural septum. The upper third of X is usually composed of a single rootlet, or less commonly, multiple small rootlets. Initial post-op dysphagia usually resolves. Cardiovascular complications following vagal section have been reported, warrants close monitoring $\times 24$ hrs.

28.3.6 Geniculate neuralgia

General information

Geniculate neuralgia (GeN) AKA Hunt's neuralgia AKA nervus intermedius neuralgia: a very rare neuralgia affecting the nervus intermedius (the somatic sensory branch of the facial nerve primarily innervating mechanoreceptors of the hair follicles on the inner surface of the pinna and deep mechanoreceptors of nasal and buccal cavities and chemoreceptors in the taste buds on the anterior 2/3 of the tongue).

Symptoms: unilateral paroxysmal otalgia (lancinating pain experienced deep within the ear, often described as an "ice pick in the ear") radiating to the auricle, with occasional burning sensations around the ipsilateral eye and cheek, and prosopalgia (pain referred to deep facial structures, including orbit, posterior nasal and palatal regions). During pain attacks, some patients have: salivation, bitter taste, tinnitus, or vertigo.

GeN occasionally has cutaneous trigger points in the anterior EAC and tragus, and pain may also be triggered by cold, noise, or swallowing.

Work-up includes neuro-otologic evaluation with audiometry and ENG. Some patients may require imaging (MRI or high-resolution CT) and angio (to R/O aneurysm).

Variants

Tic convulsif (AKA convulsive tic): GeN combined with hemifacial spasm, usually due to neurovascular compression of both the sensory and motor roots of the facial nerve,¹⁷ most often by AICA. First described by Cushing in 1920.

GeN may be associated with herpetic infections of the geniculate ganglion (AKA herpetic ganglionitis, AKA Ramsay Hunt syndrome (RHS)) in which case herpetic lesions appear on pinna, in EAC, and possibly on TM. May include facial palsy, decreased auditory acuity, tinnitus or vertigo. Unlike idiopathic GeN, RHS is more chronic and less paroxysmal, tends to remit with time, and is usually refractory to carbamazepine. Idiopathic GeN tends to be more painful than RHS, and does not remit spontaneously.

Treatment

1. medical therapy
 - a) mild cases may respond to carbamazepine, sometimes in combination with phenytoin
 - b) may respond to valproate (Depakote®) 250 mg PO BID
 - c) topical antibiotics for secondary infections of herpetic lesions
 - d) local anesthetic to EAC
2. surgery: for severe cases where medical treatment fails or is not tolerated
 - a) microvascular decompression together with division of the nervus intermedius (nerve of Wrisberg)⁹⁰. To find the nerve, a micro nerve hook is hooked around the VII nerve, the hook is rotated 90°, and the nerve is fished out from in front of VII. Operating under local anesthesia allows verification by stimulating nerve
 - b) geniculate ganglion section⁹¹

28.4 Postherpetic neuralgia

28.4.1 General information

Herpes zoster (HZ) (Greek: zoster – girdle) (shingles in lay terms): painful vesicular cutaneous eruptions caused by the herpes varicella zoster virus (VZV) (the etiologic agent of chickenpox, a herpesvirus that is distinct from herpes simplex virus). It occurs in a dermatomal distribution over one side of the thorax in $\approx 65\%$ of cases (rarely, infections occur without vesicles, called zoster sine herpete). In 20% of cases it involves the trigeminal nerve (with a predilection for the ophthalmic division, called herpes zoster ophthalmicus). Pain usually resolves after 2–4 weeks. When the pain persists > 1

month after the vesicular eruption has healed, this pain syndrome is known as postherpetic neuralgia (PHN). PHN can follow a herpes varicella infection in any site and is difficult to treat by any means (medical or surgical). It can occasionally be seen in a limb, and follows a dermatomal distribution (not a peripheral nerve distribution). PHN may remit spontaneously, but if it hasn't done so by 6 mos this is unlikely.

28.4.2 Epidemiology

Incidence of herpes zoster is $\approx 125/100,000/\text{year}$ in the general population, or about 850,000 cases per year in the U.S.⁹² Both sexes are equally affected. There is no seasonal variance. HZ is also more common in those with reduced immunity and in those with a coexistent malignancy (especially lympho-proliferative).^{93,94} PHN occurs in $\approx 10\%$ of cases of HZ.⁹² Both HZ and PHN are more common in older patients (PHN is rare in age <40 yrs, and usually occurs in age >60) and in those with diabetes mellitus. PHN is more likely after ophthalmic HZ than after spinal segmental involvement.

28.4.3 Etiology

It is postulated that the VZV lies dormant in the sensory ganglia (dorsal root ganglia of the spine, trigeminal (semilunar) ganglion for facial involvement) until such time that the patient's immune system is weakened and then the virus erupts. Inflammatory changes within the nerve are present early and are later replaced by fibrosis.

28.4.4 Clinical

PHN is usually described as a constant burning and aching. There may be superimposed shocks or jabs. It rarely produces throbbing or cramping pain. Pain may be spontaneous, or may be triggered by light cutaneous stimulation (allodynia) (e.g. by clothing), and may be relieved by constant pressure. The pain is present to some degree at all times with no pain-free intervals. Scars and pigmentary changes from the acute vesicular eruption are usually visible. It is not known if PHN can follow zoster sine herpete. The involved area may demonstrate hypesthesia, hypalgesia, paresthesias and dysesthesias.

28.4.5 Medical treatment

For herpes zoster

Varicella vaccination of older individuals can increase immunity to herpes zoster, but it will be several years before it can be determined if this will reduce PHN.⁹²

Treatment for the pain of the acute attack of herpes zoster may be accomplished with epidural or paravertebral somatic (intercostal) nerve block.^{95 (p 4018)}

Oral antiherpetic drugs: Also effective (they shorten the duration of pain) and also reduce the incidence of PHN. They may cause thrombotic thrombocytopenic purpura/hemolytic uremic syndrome (TTP/HUS) when used in severely immunocompromised patients at high doses. These drugs include:

Acyclovir (Zovirax®): poorly absorbed from the GI tract (15–30% bioavailability). □ 800 mg PO q 4 hrs 5 times/d $\times$ 7 d.

Valacyclovir (Valtrex®)⁹⁶ is a prodrug of acyclovir and is more completely absorbed and should be equally as effective with fewer daily doses. □ 1,000 mg PO TID starting within 72 hrs of onset of the rash $\times$ 7 days.

Famciclovir (Famvir®): □ 500 mg PO TID $\times$ 7 d.

For post-herpetic neuralgia

General information

Most drugs useful for trigeminal neuralgia (p.481) are less effective for PHN. Some treatment alternatives for PHN are summarized in □ Table 28.5. Details of some drugs follows. It is suggested to initiate therapy with lidocaine skin patches (p.496) since this modality has the lowest potential for serious side effects.⁹²

Table 28.5 Medical treatments for PHN ^a	
Treatment	Efficacy
PHN treatments that appear effective	
tricyclic antidepressants	widely used for PHN (see text)
lidocaine patch (Lidoderm®) ⁹⁷	effective, few side effects (p.496)
intrathecal steroids + lidocaine (see text)	appears very effective, larger studies & long-term follow-up needed
Gabapentin	proven efficacy (see text)
oxycodone CR 10 mg PO BID ⁴	proven efficacy
Treatments of questionable efficacy	
SSRIs ^b	may be effective
SNRIs	may be effective
Tramadol	may be effective
topical capsaicin	controversial (see text)
Iontophoresis	insufficient evidence
nonsteroidal creams	questionable
aspirin suspended in acetone, ether or chloroform	questionable
EMLA cream	questionable
Treatments that are not useful	
dextromethorphan, benzodiazepines, acyclovir, acupuncture	no benefit ⁹⁸
ketamine (NMDA receptor antagonist)	may be beneficial, but hepatotoxic
Preventative treatment	
oral antiherpetic drugs given during HZ infection	shortens length of HZ, may reduce incidence of PHN
varicella vaccination of older patients	trials of this strategy are in progress ⁹²
^a modified with permission from Rubin M, Relief for postherpetic neuralgia, Neurology Alert, 6: 33–4, 2001 ^b abbreviations: oxycodone CR=controlled release (Oxycontin®); HZ=herpes zoster; PHN=postherpetic neuralgia; SNRIs=serotonin-norepinephrine reuptake inhibitor; SSRIs=selective serotonin reuptake inhibitors (e. g. Prozac®),	

Antiepileptic drugs

Drug info: Gabapentin (Neurontin®)

FDA approved only for partial seizures and postherpetic neuralgia (PHN).

Side effects : dizziness and somnolence (usually during titration, often diminish with time). Ataxia, fatigue, peripheral edema, confusion and depression may occur.

Rx For PHN, start with 300 mg on Day 1, 300 mg BID on Day 2, and 300 mg TID on Day 3. Dose may be titrated up to 1800 mg/d divided TID. To limit daytime drowsiness, patients may need to start with 100 mg at hs and increase slowly over 3–8. Although doses up to 3600 mg/day (the antiseizure dose) were studied⁹⁹ there was no significant benefit for PHN over 1800 mg/d. Lower doses are required for renal insufficiency. Supplied: 100, 300 & 400 mg capsules; 600 & 800 mg scored tabs. 50 mg/ml suspension.

Drug info: Oxcarbazepine (Trileptal®)

□ 150 mg PO BID.

Drug info: Zonisamide (Zonegran®)

□ Initiate therapy with 100 mg PO q PM×2 wks, then increase dose by 100 mg/d q 2 wks up to 400 mg/d. Bioavailability is not affected by food. Steady state is achieved within 14 days of dosage changes. Supplied : 100 mg capsules.

Tricyclic antidepressants (TCA)

Drug info: Amitriptyline (Elavil®)

Helpful in ≈ 66%of patients at a mean dose of 75 mg/d even without antidepressant effect.⁵ Side effects: see Amitriptyline, side effects (p.545), minimized by starting low and slowly incrementing dose.

□ Start with 12.5–25 mg PO q hs, and increase by the same amount q 2–5 days to a maximum of 150 mg/d.

Drug info: Nortriptyline (Pamelor®)

Fewer side effects than amitriptyline.

□ Start with 10–20 mg PO q hs, and increase gradually.

Topical treatment

Drug info: Capsaicin (Zostrix®)

A vanillyl alkaloid derived from hot peppers, available without prescription for topical treatment of the pain of herpes zoster and diabetic neuropathy. Beneficial in some patients with either of these conditions (response rate at 8 weeks was 90%for PHN, 71%for diabetic neuropathy, vs. 50%with placebo in either group), although the high placebo response rate is disturbing and many authorities are skeptical.¹⁰⁰ Expensive. Side effects: include burning and erythema at the application site (usually subsides by 2–4 weeks).

□ Manufacturer recommends massaging the medication into the affected area of the skin TID-QID (apply a very thin coat). Some authorities recommend q 2 hr application. Avoid contact with eyes or damaged skin. Supplied as Zostrix® (0.25%capsaicin) or Zostrix-HP® (0.75%).

Drug info: Lidocaine patch 5% (Lidoderm®)

Often better tolerated by elderly patients than TCAs (due to pre-existing cognitive impairments, cardiac disease, or systemic illness).

□ Apply up to 3 patches of 5%lidocaine (to cover a maximum of 420 cm²) to intact skin q 12 hrs to cover as much of the area of greatest pain as possible.⁹⁷

Intrathecal steroids

Over 90% of patients receiving intrathecal methylprednisolone (60 mg)+3% lidocaine (3 ml) given once per week for up to 4 weeks, reported good to excellent pain relief for up to 2 years.¹⁰¹ This technique was not studied for use in PHN involving the trigeminal nerve. Further clinical trials are needed to verify the efficacy and safety⁹² (potential long-term side effects include adhesive arachnoiditis).

Surgical treatment

There is no operation that is uniformly successful in treating PHN. Numerous operations have been shown to work occasionally. Procedures that have been tried include:

1. nerve blocks: once PHN is established, nerve blocks provide only temporary relief¹⁰²
2. cordotomy: although percutaneous cordotomy (p.1542) may work when the level of PHN is at least 3–4 segments below the cordotomy, this procedure is not recommended for pain of benign etiology because of possible complications and the high likelihood of pain recurrence
3. rhizotomy: including retrogasserian for facial involvement
4. neurectomies
5. sympathectomy
6. DREZ (p.1550)¹⁰³: often offers good early relief, but recurrence rate is high.
7. acupuncture¹⁰⁴
8. TENS
9. spinal cord stimulation (p.1547)
10. undermining the skin
11. motor cortex stimulation: for facial PHN

28.5 Complex regional pain syndrome (CRPS)

28.5.1 General information

The terminology is confusing. Formerly also called causalgia (reflex sympathetic dystrophy). The term causalgia (Greek: kausis – burning, algos – pain) was introduced by Weir Mitchell in 1864. It was used to describe a rare syndrome that followed a minority of partial peripheral nerve injuries in the American civil war. Triad: burning pain, autonomic dysfunction and trophic changes.

CRPS Type II (AKA major causalgia) follows nerve injury (originally described after high velocity missile injuries). CRPS Type I (AKA reflex sympathetic dystrophy or causalgia minor) denoted less severe forms, and has been described after non-penetrating trauma.¹⁰⁵ Shoulder-hand syndrome and Sudeck's atrophy are other variant designations. In 1916, the autonomic nervous system was implicated by René Leriche, and the term reflex sympathetic dystrophy (RSD) later came into use¹⁰⁶ (but RSD may be distinct from causalgia¹⁰⁷).

Post-op CRPS has been described following carpal tunnel surgery as well as surgery on the lumbar¹⁰⁸ and cervical spine.

At best, CRPS must be regarded as a symptom complex, and not as a discrete syndrome nor medical entity (for a cogent editorial on the subject, see the essay by Ochoa¹⁰⁹). Patients exhibiting CRPS phenomenology are not a homogeneous group, and include¹¹⁰:

1. actual CRPS (for these, Mailis proposes the term “physiogenic RSD”): a complex set of neuropathic phenomena that may occur with or without nerve injury
2. medical conditions distinct from CRPS but with signs and symptoms that mimic CRPS: vascular, inflammatory, neurologic...
3. the product of mere immobilization: as in severe pain avoidance behavior, or at times psychiatric disorders
4. part of a factitious disorder with either a psychological basis (e.g. Munchausen's syndrome) or for secondary gain (financial, drug seeking...) i.e. malingering

28.5.2 Pathogenesis

Early theories invoked ephaptic transmission between sympathetics and afferent pain fibers. This theory is rarely cited currently. Another more recent postulate involves nor-epinephrine released at sympathetic terminals together with hypersensitivity secondary to denervation or sprouting. Many modern hypotheses do not even embrace involvement of the autonomic nervous system in all cases.^{106,107,110}

Thus, many of the alterations seen in CRPS may simply be epiphenomena rather than part of the etiopathogenetic mechanism.

28.5.3 Clinical

CRPS may be described as a phenomenology, i.e. a variable complex of signs and symptoms due to multiple etiologies included in this nonhomogeneous group.¹¹⁰ No diagnostic criteria for the condition have been established, and various investigators select different factors to include or exclude patients from their studies.

28.5.4 Symptoms

Pain: affecting a limb, usually burning, and prominent in the hand or foot. Onset in the majority is within 24 hrs of injury (unless injury causes anesthesia, then hrs or days may intervene); however, CRPS may take days to weeks to develop. Median, ulnar and sciatic nerves are the most commonly cited involved nerves. However, it is not always possible to identify a specific nerve that has been injured. Almost any sensory stimulus worsens the pain (allodynia is pain induced by a nonnoxious stimulus).

28.5.5 Signs

The physical exam is often difficult due to pain.

Vascular changes: either vasodilator (warm and pink) or vasoconstrictor (cold, mottled blue). Trophic changes (may be partly or wholly due to immobility): dry/scaly skin, stiff joints, tapering fingers, ridged uncut nails, either long/course hair or loss of hair, sweating alterations (varies from anhidrosis to hyperhidrosis).

28.5.6 Diagnostic aids

In the absence of an agreed upon etiology or pathophysiology, there can be no basis for specific tests, and the lack of a “gold-standard” diagnostic criteria makes it impossible to verify the authenticity of any diagnostic marker. Numerous tests have been presented as aids to the diagnosis of CRPS, and essentially all have eventually been refuted. Candidates have included:

1. thermography: discredited in clinical practice
2. three-phase bone scan: typical CRPS changes also occur after sympathectomy,¹¹¹ which has traditionally been considered curative of CRPS
3. osteoporosis on x-ray,¹¹² particularly periarticular demineralization: nonspecific
4. response to sympathetic block (once thought to be the sine qua non for causalgia major and minor, the response sought was relief (complete or significant) with sympathetic block of appropriate trunk (stellate for UE, lumbar for LE)): has failed to hold up once stringent placebo-controlled trials were executed
5. various autonomic tests¹¹³: resting sweat output, resting skin temperature, quantitative sudomotor axon reflex test

28.5.7 Treatment

In the absence of a delineated pathophysiology, treatment is judged purely by subjective impression of improvement. CRPS treatment studies have had an unusually high placebo response rate.¹¹⁴ Medical therapy is usually ineffective. Proposed treatments include:

1. tricyclic antidepressants
2. 18–25% have satisfactory long-lasting relief after a series of sympathetic blocks, see Stellate ganglion block (p.1521) and Lumbar sympathetic block (p.1521), although one report found no long-lasting benefit in any of 30 patients¹¹⁵
3. intravenous regional sympathetic block, particularly for UE CRPS: agents used include guanethidine¹¹⁶ 20 mg, reserpine, bretylium..., injected IV with arterial tourniquet (sphygmomanometer cuff) inflated for 10 min. If no relief, repeat in 3–4 wks. No better than placebo in several trials^{117, 118}
4. surgical sympathectomy (p.1537): some purport that this relieves pain in >90% of patients (with a few retaining some tenderness or hyperpathia). Others opine that there is no rational reason to consider sympathectomy since sympathetic blocks have been shown to be no more effective than placebo¹⁰⁶
5. spinal cord stimulation: some success has been reported

References

- [1] Backonja MM. Defining neuropathic pain. *Anesth Analg*. 2003; 97:785–790
- [2] Merskey H, Bogduk N. Classification of Chronic Pain: Descriptions of Chronic Pain Syndromes and Definitions of Pain Terms. 2nd ed. Seattle, WA: IASP Press; 1994
- [3] Dworkin RH, Backonja M, Rowbotham MC, Allen RR, Argoff CR, Bennett GJ, Bushnell MC, Farrar JT, Galer BS, Haythornthwaite JA, Hewitt DJ, Loeser JD, Max MB, Saltarelli M, Schmader KE, Stein C, Thompson D, Turk DC, Wallace MS, Watkins LR, Weinstein SM. Advances in neuropathic pain: diagnosis, mechanisms, and treatment recommendations. *Arch Neurol*. 2003; 60:1524–1534
- [4] Watson CPN, Babul N. Efficacy of oxycodone in neuropathic pain: a randomized trial in postherpetic neuralgia. *Neurology*. 1998; 50:1837–1841
- [5] Watson CP, Evans RJ, Reed K, Merserkey H, et al. Amitriptyline versus Placebo in Postherpetic Neuralgia. *Neurology*. 1982; 32:671–673
- [6] Max MB, Lynch SA, Muir J, Shoaf SE, et al. Effects of Desipramine, Amitriptyline, and Fluoxetine on Pain in Diabetic Neuropathy. *N Engl J Med*. 1992; 326:1250–1256
- [7] Bennett MI, Simpson KH. Gabapentin in the treatment of neuropathic pain. *Palliat Med*. 2004; 18:5–11
- [8] Dierking G, Duedahl TH, Rasmussen ML, Fomsgaard JS, Moiniche S, Romsing J, Dahl JB. Effects of gabapentin on postoperative morphine consumption and pain after abdominal hysterectomy: a randomized, double-blind trial. *Acta Anaesthesiol Scand*. 2004; 48:322–327
- [9] Mathew NT, Rapoport A, Saper J, Magnus L, Klapper J, Ramadan N, Stacey B, Tepper S. Efficacy of gabapentin in migraine prophylaxis. *Headache*. 2001; 41:119–128
- [10] Gabapentin (Neurontin®) for chronic pain. *Med Letter*. 2004; 46:29–31
- [11] Keller JT, van Loveren H. Pathophysiology of the Pain of Trigeminal Neuralgia and Atypical Facial Pain: A Neuroanatomical Perspective. *Clin Neurosurg*. 1985; 32:275–293
- [12] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
- [13] Burchiel KJ. A new classification for facial pain. *Neurosurgery*. 2003; 91:1164–1167
- [14] Pareja JA, Sjaastad O. SUNCT syndrome. A clinical review. *Headache*. 1997; 37:195–195
- [15] Sjaastad O, Pareja JA, Zukerman E, Jansen J, Kruszewski P. Trigeminal neuralgia. Clinical manifestations of first division involvement. *Headache*. 1997; 37:346–357
- [16] Pareja JA, Baron M, Gili P, Yanguela J, Caminero AB, Dobato JL, Barriga FJ, Vela L, Sanchez-del-Rio M. Objective assessment of autonomic signs during triggered first division trigeminal neuralgia. *Cephalalgia*. 2002; 22:251–255
- [17] Yeh HS, Tew JM. Tic Convulsif, the Combination of Geniculate Neuralgia and Hemifacial Spasm Relieved by Vascular Decompression. *Neurology*. 1984; 34:682–683
- [18] Rupa V, Saunders RL, Weider DJ. Geniculate Neuralgia: The Surgical Management of Primary Otagia. *J Neurosurg*. 1991; 75:505–511
- [19] Young RF. Geniculate Neuralgia. *J Neurosurg*. 1992; 76
- [20] Wepsic JG. Tic Douloureux: Etiology, Refined Treatment. *N Engl J Med*. 1973; 288:680–681
- [21] Sweet WH. The Treatment of Trigeminal Neuralgia (Tic Douloureux). *N Engl J Med*. 1986; 315:174–177
- [22] Brisman R. Bilateral Trigeminal Neuralgia. *J Neurosurg*. 1987; 67:44–48
- [23] van Loveren H, Tew JM, Keller JT, et al. A 10-Year Experience in the Treatment of Trigeminal Neuralgia: Comparison of Percutaneous Stereotaxic Rhizotomy and Posterior Fossa Exploration. *J Neurosurg*. 1982; 57:757–764
- [24] Taha JM, Tew JM. Comparison of Surgical Treatments for Trigeminal Neuralgia: Reevaluation of Radiofrequency Rhizotomy. *Neurosurgery*. 1996; 38:865–871
- [25] Hardy DG, Rhoton AL. Microsurgical Relationships of the Superior Cerebellar Artery and the Trigeminal Nerve. *J Neurosurg*. 1978; 49:669–678
- [26] Morita A, Fukushima T, Miyazaki S, et al. Tic Douloureux Caused by Primitive Trigeminal Artery or its Variant. *J Neurosurg*. 1989; 70:415–419
- [27] Apfelbaum RI, Carter LP, Spetzler RF, Hamilton MG. In: Trigeminal Neuralgia: Vascular Decompression. *Neurovascular Surgery*. New York: McGraw-Hill; 1995:1107–1117
- [28] Bullitt E, Tew JM, Boyd J. Intracranial Tumors in Patients with Facial Pain. *J Neurosurg*. 1986; 64:865–871
- [29] Fusco BM, Alessandri M. Analgesic Effect of Capsaicin in Idiopathic Trigeminal Neuralgia. *Anesth Analg*. 1992; 74:375–377
- [30] Poppen JL. *An Atlas of Neurosurgical Techniques*. Philadelphia: W. B. Saunders; 1960
- [31] Sweet WH, Wepsic JG. Controlled Thermocoagulation of Trigeminal Ganglion and Rootlets for Differential Destruction of Pain Fibers. Part I. Trigeminal Neuralgia. *J Neurosurg*. 1974; 40:143–156
- [32] Hakanson S. Trigeminal Neuralgia Treated by the Injection of Glycerol into the Trigeminal Cistern. *Neurosurgery*. 1981; 9:638–646
- [33] Sweet WH, Poletti CE, Macon JB. Treatment of Trigeminal Neuralgia and Other Facial Pains by the Retrogasserian Injection of Glycerol. *Neurosurgery*. 1981; 9:647–653
- [34] Lunsford LD, Apfelbaum RI. Choice of Surgical Therapeutic Modalities for Treatment of Trigeminal Neuralgia. *Clin Neurosurg*. 1985; 32:319–333
- [35] Young RF. Glycerol Rhizolysis for Treatment of Trigeminal Neuralgia. *J Neurosurg*. 1988; 69:39–45
- [36] Mullan S, Lichtor T. Percutaneous Microcompression of the Trigeminal Ganglion for Trigeminal Neuralgia. *J Neurosurg*. 1983; 59:1007–1012
- [37] Belber CJ, Rak RA. Balloon Compression Rhizolysis in the Surgical Management of Trigeminal Neuralgia. *Neurosurgery*. 1987; 20:908–913
- [38] Lichtor T, Mullan JF. A 10-Year Follow-Up Review of Percutaneous Microcompression of the Trigeminal Ganglion. *J Neurosurg*. 1990; 72:49–54
- [39] Taarnhøj P. Decompression of the Posterior Trigeminal Root in Trigeminal Neuralgia. *J Neurosurg*. 1982; 57:14–17
- [40] Henderson JM, Lad SP. Motor cortex stimulation and neuropathic facial pain. *Neurosurg Focus*. 2006; 21
- [41] van Loveren H, Greenberg MS. Tampa 2009
- [42] Murali R, Rovit RL. Are peripheral neurectomies of value in the treatment of trigeminal neuralgia? An analysis of new cases and cases involving previous radiofrequency Gasserian thermocoagulation. *J Neurosurg*. 1996; 85:435–437
- [43] Wilkins RH, Burchiel KJ. In: Trigeminal neuralgia: Historical overview, with emphasis on surgical treatment. *Surgical Management of Pain*. New York: Thieme Medical Publishers, Inc.; 2002:288–301
- [44] Tew JM, van Loveren H, Schmidek HH, Sweet WH. In: Percutaneous Rhizotomy in the Treatment of Intractable Facial Pain (Trigeminal, Glossopharyngeal, and Vagal Nerves). *Operative Neurosurgical Techniques*. 2nd ed. Philadelphia: W B Saunders; 1988:1111–1123
- [45] Brown JA, Preul MC. Percutaneous Trigeminal Ganglion Compression for Trigeminal Neuralgia. Experience in 22 Cases and Review of the Literature. *J Neurosurg*. 1989; 70:900–904
- [46] Menzel J, Piotrowski W, Penzholtz H. Long-Term Results of Gasserian Ganglion Electrocoagulation. *J Neurosurg*. 1975; 42:140–143
- [47] Wepsic JG. Complications of Percutaneous Surgery for Pain. *Clin Neurosurg*. 1976; 23:454–464
- [48] Tew JM, Keller JT. The Treatment of Trigeminal Neuralgia by Percutaneous Radiofrequency Technique. *Clin Neurosurg*. 1977; 24:557–578

- [49] Luksic I, Sestan-Crnek S, Virag M, Macan D. Trigeminal trophic syndrome of all three nerve branches: an underrecognized complication after brain surgery. *J Neurosurg.* 2008; 108:170–173
- [50] Setyadi HG, Cohen PR, Schulze KE, Mason SH, Martinelli PT, Alford EL, Taffet GE, Nelson BR. Trigeminal trophic syndrome. *South Med J.* 2007; 100:43–48
- [51] Kaplan M, Erol FS, Ozveren MF, Topsakal C, Sam B, Tekdemir I. Review of complications due to foramen ovale puncture. *J Clin Neurosci.* 2007; 14:563–568
- [52] Sekhar L, Heros RC, Kerber CW. Carotid-Cavernous Fistula Following Percutaneous Retrogasserian Procedures. *J Neurosurg.* 1979; 51:700–706
- [53] Kuether TA, O'Neill OR, Nesbit GM, Barnwell SL. Direct Carotid Cavernous Fistula After Trigeminal Balloon Microcompression Gangliolysis: Case Report. *Neurosurgery.* 1996; 39:853–856
- [54] Agazzi S, Chang S, Drucker MD, Youssef AS, Van Loveren HR. Sudden blindness as a complication of percutaneous trigeminal procedures: mechanism analysis and prevention. *J Neurosurg.* 2009; 110:638–641
- [55] Kanpolat Y, Savas A, Bekar A, Berk C. Percutaneous controlled radiofrequency trigeminal rhizotomy for the treatment of idiopathic trigeminal neuralgia: 25-year experience with 1,600 patients. *Neurosurgery.* 2001; 48:524–32; discussion 532–4
- [56] Tobler WD, Tew JM, Cosman E, Keller J, et al. Improved outcome in the treatment of trigeminal neuralgia by percutaneous stereotactic rhizotomy with a new, curved tip electrode. *Neurosurgery.* 1983; 12:313–317
- [57] Lunsford LD. Comment on Taha JM and Tew JM: Comparison of Surgical Treatments for Trigeminal Neuralgia: Reevaluation of Radiofrequency Rhizotomy. *Neurosurgery.* 1996; 38
- [58] Brisman R. Gamma knife surgery with a dose of 75 to 76.8 Gray for trigeminal neuralgia. *J Neurosurg.* 2004; 100:848–854
- [59] Pollock BE, Phuong LK, Foote RL, Stafford SL, Gorman DA. High-dose trigeminal neuralgia radiosurgery associated with increased risk of trigeminal nerve dysfunction. *Neurosurgery.* 2001; 49:58–62; discussion 62–4
- [60] Kondziolka D, Lunsford LD, Flickinger JC. Stereotactic radiosurgery for the treatment of trigeminal neuralgia. *Clin J Pain.* 2002; 18:42–47
- [61] Massager N, Lorenzoni J, Devriendt D, Desmedt F, Brotchi J, Levivier M. Gamma knife surgery for idiopathic trigeminal neuralgia performed using a far-anterior cisternal target and a high dose of radiation. *J Neurosurg.* 2004; 100:597–605
- [62] Urgosik D, Liscak R, Novotny J, Jr, Vymazal J, Vladyska V. Treatment of essential trigeminal neuralgia with gamma knife surgery. *J Neurosurg.* 2005; 102 Suppl:29–33
- [63] Maesawa S, Salame C, Flickinger JC, Pirris S, Kondziolka D, Lunsford LD. Clinical outcomes after stereotactic radiosurgery for idiopathic trigeminal neuralgia. *J Neurosurg.* 2001; 94:14–20
- [64] Barba D, Alksne JF. Success of Microvascular Decompression with and without Prior Surgical Therapy for Trigeminal Neuralgia. *J Neurosurg.* 1984; 60:104–107
- [65] Schmidek HH, Sweet WH. *Operative Neurosurgical Techniques.* New York 1982
- [66] Onofrio BM. Radiofrequency Percutaneous Gasserian Ganglion Lesions: Results in 140 Patients with Trigeminal Pain. *J Neurosurg.* 1975; 42:132–139
- [67] Kondziolka D, Lunsford LD, Bissonette DJ. Long-Term Results After Glycerol Rhizotomy for Multiple Sclerosis-Related Trigeminal Neuralgia. *Can J Neurol Sci.* 1994; 21:137–140
- [68] Burchiel KJ, Favre J. Current Techniques for Pain Control. *Contemp Neurosurg.* 1997; 19:1–6
- [69] McLaughlin MR, Jannetta PJ, Clyde BL, Subach BR, Comey CH, Resnick DK. Microvascular decompression of cranial nerves: Lessons learned after 4400 operations. *J Neurosurg.* 1999; 90:1–8
- [70] Tew JM, van Loveren HR. *Atlas of Operative Micro-neurosurgery.* Philadelphia: W. B. Saunders; 1994; 1: Aneurysms and Arteriovenous Malformations
- [71] Day JD, Tschabitscher M. Anatomic position of the asterion. *Neurosurgery.* 1998; 42:198–199
- [72] Bederson JB, Wilson CB. Evaluation of Microvascular Decompression and Partial Sensory Rhizotomy in 252 Cases of Trigeminal Neuralgia. *J Neurosurg.* 1989; 71:359–367
- [73] Avildsen JG. *The Karate Kid.* 1984
- [74] Jannetta PJ. Microsurgical Management of Trigeminal Neuralgia. *Arch Neurol.* 1985; 42
- [75] Burchiel KJ, Clarke H, Haglund M, et al. Long-Term Efficacy of Microvascular Decompression in Trigeminal Neuralgia. *J Neurosurg.* 1988; 69:35–38
- [76] Schmidek HH, Sweet WH. *Operative Neurosurgical Techniques.* Philadelphia 1988
- [77] Hanakita J, Kondo A. Serious Complications of Microvascular Decompression Operations for Trigeminal Neuralgia and Hemifacial Spasm. *Neurosurgery.* 1988; 22:348–352
- [78] Andersen NB, Bovim G, Sjaastad O. The frontotemporal peripheral nerves. Topographic variations of the supraorbital, supratrochlear and auriculotemporal nerves and their possible clinical significance. *Surg Radiol Anat.* 2001; 23:97–104
- [79] Pareja JA, Caminero AB. Supraorbital neuralgia. *Curr Pain Headache Rep.* 2006; 10:302–305
- [80] Headache Classification Committee of the International Headache Society. Classification and diagnostic criteria for headache disorders, cranial neuralgias, and facial pain, 2nd edition. *Cephalalgia.* 2004; 24:9–160
- [81] Pareja JA, Pareja J, Yanguela J. Nummular headache, trochleitis, supraorbital neuralgia, and other epicranial headaches and neuralgias: the epicranias. *J Headache Pain.* 2003; 4:125–131
- [82] Pareja JA, Caminero AB, Serra J, Barriga FJ, Baron M, Dobato JL, Vela L, Sanchez del Rio M. Nummular headache: a coin-shaped cephalgia. *Neurology.* 2002; 58:1678–1679
- [83] Caminero AB, Pareja JA. Supraorbital neuralgia: a clinical study. *Cephalalgia.* 2001; 21:216–223
- [84] Stookey B, Ransohoff J. *Trigeminal Neuralgia: Its History and Treatment.* Springfield, IL: Charles C Thomas; 1959
- [85] Sjaastad O, Stolt-Nielsen A, Pareja JA, Vincent M. Supraorbital neuralgia: on the clinical manifestations and a possible therapeutic approach. *Headache.* 1999; 39:204–212
- [86] Grantham EG, Segerberg LH. An evaluation of palliative surgical procedures in trigeminal neuralgia. *J Neurosurg.* 1952; 9:390–394
- [87] Youmans JR. *Neurological Surgery.* Philadelphia 1982
- [88] Weinstein RE, Herec D, Friedman JH. Hypotension due to Glossopharyngeal Neuralgia. *Arch Neurol.* 1986; 43:90–92
- [89] Ferrante L, Artico M, Nardacci B, et al. Glossopharyngeal Neuralgia with Cardiac Syncope. *Neurosurgery.* 1995; 36:58–63
- [90] Lovely TJ, Jannetta PJ. Surgical management of geniculate neuralgia. *Am J Otol.* 1997; 18:512–517
- [91] Pulec JL. Geniculate neuralgia: diagnosis and surgical management. *Laryngoscope.* 1976; 86:955–964
- [92] Watson CPN. A new treatment for postherpetic neuralgia. *N Engl J Med.* 2000; 343:1563–1565
- [93] Loeser JD. Herpes Zoster and Postherpetic Neuralgia. *Pain.* 1986; 25:149–164
- [94] Schimpff S, Serpick A, Stoler B, Rumack B, et al. Varicella-Zoster Infection in Patients with Cancer. *Ann Intern Med.* 1972; 76:241–254
- [95] Youmans JR. *Neurological Surgery.* Philadelphia 1990
- [96] Valacyclovir. *Med Letter.* 1996; 38:3–4
- [97] Rowbotham MC, Davies PS, Verkempinck C, et al. Lidocaine patch: double-blind controlled trial of a new treatment method for postherpetic neuralgia. *Pain.* 1996; 65:39–44
- [98] Alper BS, Lewis PR. Treatment of postherpetic neuralgia: a systematic review of the literature. *J Fam Pract.* 2002; 51:121–128

- [99] Rowbotham MC, Harden N, Stacey B, et al. Gabapentin for the treatment of postherpetic neuralgia: A randomized controlled trial. *JAMA*. 1998; 280:1837–1842
- [100] Capsaicin - A Topical Analgesic. *Med Letter*. 1992; 34:62–63
- [101] Kotani N, Kushikata T, Hashimoto H, et al. Intrathecal methylprednisolone for intractable postherpetic neuralgia. *N Engl J Med*. 2000; 343:1514–1519
- [102] Dan K, Higa K, Noda B, Fields H, Dubner R, Cervero F. In: Nerve block for herpetic pain. *Advances in Pain Research and Therapy*. New York: Raven Press; 1985:831–838
- [103] Friedman AH, Nashold BS. Dorsal Root Entry Zone Lesions for the Treatment of Postherpetic Neuralgia. *Neurosurgery*. 1984; 15:969–970
- [104] Lewith GT, Field J, Machin D. Acupuncture Compared with Placebo in Post-Herpetic Pain. *Pain*. 1983; 17:361–368
- [105] Sternschein MJ, Myers SJ, Frewin DB, et al. Causalgia. *Arch Phys Med Rehabil*. 1975; 56:58–63
- [106] Schott GD. An Unsympathetic View of Pain. *Lancet*. 1995; 345:634–636
- [107] Ochoa JL, Verdugo RJ. Reflex Sympathetic Dystrophy: A Common Clinical Avenue for Somatoform Expression. *Neurol Clin*. 1995; 13:351–363
- [108] Sachs BL, Zindrick MR, Beasley RD. Reflex Sympathetic Dystrophy After Operative Procedures on the Lumbar Spine. *J Bone Joint Surg*. 1993; 75A:721–725
- [109] Ochoa JL. Reflex? Sympathetic? Dystrophy? Triple Questioned Again. *Mayo Clin Proc*. 1995; 70:1124–1125
- [110] Mailis A. Is Diabetic Autonomic Neuropathy Protective Against Reflex Sympathetic Dystrophy? *Clin JPain*. 1995; 11:77–81
- [111] Mailis A, Meindok H, Papagapiou M, Pham D. Alterations of the Three-Phase Bone Scan After Sympathectomy. *Clin JPain*. 1994; 10:146–155
- [112] Kozin F, Genant HK, Bekerman C, et al. The Reflex Sympathetic Dystrophy Syndrome. *Am J Med*. 1976; 60:332–338
- [113] Chelimsky TC, Low PA, Naessens JM, et al. Value of Autonomic Testing in Reflex Sympathetic Dystrophy. *Mayo Clin Proc*. 1995; 70:1029–1040
- [114] Ochoa JL. Pain Mechanisms in Neuropathy. *Curr Opin Neurol*. 1994; 7:407–414
- [115] Dotson R, Ochoa JL, Cline M, Yarnitsky D. A Reassessment of Sympathetic Blocks as Long Term Therapeutic Modality for "RSD". *Pain*. 1990; 5
- [116] Hannington-Kiff JG. Relief of Sudek's Atrophy by Regional Intravenous Guanethidine. *Lancet*. 1977; 1:1132–1133
- [117] Blanchard J, Ramamurthy W, Walsh N, et al. Intravenous Regional Sympatholysis: A Double-Blind Comparison of Guanethedine, Reserpine, and Normal Saline. *J Pain Symptom Manage*. 1990; 5:357–361
- [118] Jadad AR, Carroll D, Glynn CJ, McQuay HJ. Intravenous Regional Sympathetic Blockade for Pain Relief in Reflex Sympathetic Dystrophy: A Systematic Review and a Randomized, Double-Blind Crossover Study. *J Pain Symptom Manage*. 1995; 10:13–20

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Part X

Peripheral Nerves

29	Peripheral Nerves	504
30	Entrapment Neuropathies	515
31	Non-Entrapment Peripheral Neuropathies	541



29 Peripheral Nerves

29.1 General information

29.1.1 Peripheral nervous system definition

The peripheral nervous system (PNS) consists of those structures (including cranial nerves III-XII, spinal nerves, nerves of the extremities, and the cervical, brachial and lumbosacral plexi) containing nerve fibers or axons that connect the central nervous system (CNS) with motor and sensory, somatic and visceral, end organs.¹ □ Table 29.1 shows classification of motor and sensory nerves.

29.1.2 Grading strength and reflexes

Muscle strength grading most commonly employs the Royal Medical Research Council of Great Britain (MRC) scale,² a common modification of which is shown in □ Table 29.2. Muscle stretch reflexes may be graded as shown in □ Table 29.3.²

29.1.3 Upper motor neuron vs. lower motor neuron

Lower motor neurons (LMN) (first-order motor neuron): cell bodies (soma) reside in spinal cord (in anterior gray matter) or in brainstem (for cranial nerve motor nuclei). Axons connect directly to neuromuscular junction of muscles.

Upper motor neurons (UMN) (second-order motor neurons): some soma reside in the primary motor cortex (precentral gyrus) of the brain. Axons project to LMNs.

See □ Table 29.4 for comparison of weakness due to UMN vs. LMN.

29

Table 29.1 Motor & sensory classification of nerves					
Sensory	Sensory & motor	Greatest fiber diameter (mcm)	Greatest conduction velocity (m/sec)	Motor/ sensory	Comments
Ia	A-alpha	22	120	motor	large alpha motor neurons of lamina IX (extrafusal) sensory-primary afferents (anulospiral of muscle spindles for proprioception)
Ib	A-alpha	22	120	sensory	Golgi tendon organs, touch & pressure receptors
II	A-beta	13	70	sensory	secondary afferents (flower spray) of muscle spindles, crude touch, pressure receptors, Pacinian corpucles (vibratory) (to posterior columns ^a)
	A-gamma	8	40	motor	small γ motor neurons of lamina IX (intrafusal)
III	A-delta	5	15	sensory	small, lightly myelinated, fine touch, pressure, pain & temperature
	B	3	14	motor	small, lightly myelinated preganglionic autonomic fibers
IV	C	1	2	motor	all post-ganglionic autonomic
				sensory	unmyelinated pain & temp (to spinothalamic tract)
^a these fibers to the posterior columns are more medial in the root entry zone than the C-fibers (important in DREZ lesions where the goal is to lesion C-fibers and spare A-beta)					

Table 29.2 Muscle grading (modified Medical Research Council system)		
Grade	Strength	
0	no contraction (total paralysis)	
1	flicker or trace contraction (palpable or visible)	
2	active movement with gravity eliminated	
3	active movement through full ROM against gravity	
4	active movement against resistance; subdivisions →	4– Slight resistance 4 Moderate resistance 4 + Slight resistance
5	normal strength (against full resistance)	
NT	not testable	

Table 29.3 Muscle stretch reflex (deep tendon reflex) grading scale	
Grade	Definition
0	no contraction (total paralysis)
0.5 +	elicitable only with reinforcement ^a
1 +	low normal
2 +	normal
3 +	more brisk than normal (hyperreflexic)
4 +	hyperreflexic with clonus
5 +	sustained clonus
^a In the IEs, reinforcement consists of having the patient hook the tips of the fingers of the left hand into the tips of the hooked fingers of the right hand and pulling (Jendrassik maneuver). Reinforcement in the UEs consists of having the patient clench their teeth	

Table 29.4 Upper vs. lower motor neuron paralysis		
	Upper motor neuron paralysis	Lower motor neuron paralysis
possible etiologies	stroke (motor strip, internal capsule...), spinal cord injury, cervical spondylotic myelopathy	herniated intervertebral disk, nerve entrapment syndrome, polio, progressive muscular atrophy (PMA)
muscle tone	initially flaccid; later spastic with clasp-knife resistance	flaccid
tendon reflexes	hyperactive; clonus may be present	absent
pathologic reflexes (e.g. Babinski, Hoffman)	present (after days to weeks)	absent
muscle manifestations	spontaneous spasms may occur; some atrophy of disuse may occur	fibrillations, fasciculations Atrophy after days to weeks due to trophic influence

29.1.4 Fasciculations vs. fibrillations

Fasciculations are coarse muscle contractions that are visible to the naked eye, whereas fibrillations are not visible and require EMG to detect; AKA fibrillation potentials (p.242).

Fasciculations represent discharge of a group of muscle fibers (all or part of an entire motor unit), and occur most often in diseases involving anterior horn cells, including:

1. amyotrophic lateral sclerosis (ALS) (p.183)
2. spinal muscular atrophy (p.1413)
3. polio
4. syringomyelia

29.2 Muscle innervation

29.2.1 Muscles, roots, trunks, cords and nerves of the upper extremities

Table 29.5 Muscle innervation – shoulder & upper extremity ^a						
Muscle		Action to test	Roots ^b	Trunk ^c	Cord ^d	Nerve
	deep neck	flex, ext, rotation of neck	C1–4	–	–	cervical
	trapezius	elevates shoulder, abducts arm >90°	XI, C3, 4			(spinal acc + roots)
	diaphragm	inspiration	C3–5			phrenic
•	serratus anterior	forward shoulder thrust	C5–7	–	–	long thoracic
	levator scapulae	elevate scapula	C3, 4, 5			dorsal scapular
	rhomboids	adduct & elevate scapula	C4, 5			
	supraspinatus	abduct arm (15–30°)	C4, 5, 6	S		suprascapular
•	infraspinatus	exorotation of humerus	C5, 6	S	–	
	latissimus dorsi	adduct arm	C5, 6, 7, 8			thoracodorsal
	teres major, subscapularis		C5–7			subscapular
•	deltoid	abduct arm (30–90°)	C5, 6	S	P	axillary
	teres minor	exorotate & adduct humerus	C4,5			
•	biceps brachii	flex forearm (with hand supinated), & supinate forearm	C5, C6	S	L	musculocutaneous
	coracobrachialis	flex humerus at shoulder	C5–7			
	brachialis	flex forearm	C5, 6			
•	flexor carpi ulnaris	ulnar flexion of wrist	C7, 8, T1	M, I	M	ulnar
•	flexor digitorum profundus III & IV (ulnar part)	flex distal phalanx of Dig 4–5	C7, 8, T1	M, I	M	
	adductor pollicis	thumb adduction	C8, T1		M	
	abductor digiti minimi	abduction Dig 5	C8, T1		M	
	opponens digiti minimi	opposition Dig 5	C7, 8, T1			
	flexor digiti minimi brevis	flexion Dig 5	C7, 8, T1		M	
•	interossei	flex proximal phalanx, extend 2 distal phalanges, abduct or adduct fingers	C8, T1	I	M	

Table 29.5 continued

Muscle		Action to test	Roots ^b	Trunk ^c	Cord ^d	Nerve
	lumbricals 3 & 4	flex proximal phalanges & extend 2 distal phalanges of Dig 4–5	C7, 8			
•	pronator teres	forearm pronation	C6,7	S,M	L	median
•	flexor carpi radialis	radial flexion of wrist	" "	S,M	L	
	palmaris longus	wrist flexion	C7, 8, T1			
•	flexor digitorum superficialis	flexion middle phalanx Dig 2–5, flex wrist	C7, 8, T1	M, I	M	
•	abductor pollicis brevis	abduct thumb metacarpal	C8, T1	I	M	
	flexor pollicis brevis	flex prox phalanx thumb	C8, T1			
•	opponens pollicis	opposes thumb metacarp	C8, T1	I	M	
	lumbricals 1 & 2	flex proximal phalanx & extend 2 distal phalanges Dig 2–3	C8, T1			
•	flexor digitorum profundus I & II (radial part)	flex distal phalanx of Dig 2–3; flex wrist	C7, 8, T1	M, I	M	anterior interosseous
•	flexor pollicis longus	flex distal phalanx thumb	C7, 8, T1			
•	triceps brachii	forearm extension	C6, 7, 8	all	P	radial
•	brachioradialis	forearm flexion (with thumb pointed up)	C5, 6	S	P	
•	extensor carpi radialis	radial wrist extension	C5, 6	S,M	P	
•	supinator	forearm supination	C6, 7	S	P	
•	extensor digitorum	extension of wrist & phalanges of Dig 2–5	C7, C8	M, I	P	posterior interosseous (PIN)
	extensor carpi ulnaris	ulnar wrist extension	C7, C8			
•	abductor pollicis longus	abduction thumb metacarpal & radial wrist extens.	C7, C8	M, I	P	
	extensor pollicis brevis & longus	thumb extension & radial wrist extension	C7, C8			
	extensor indicis proprius	extension Dig 2 & wrist extension	C7, C8			
	pectoralis major: clavicular head	push arm forward against resistance	C5, 6			lateral pectoral
	pectoralis major: sternal-costal head	adduct arm	C6, 7, 8			lateral & medial pectoral

^aNB: items marked with a bullet (•) are clinically important muscle/nerves.
NB: Dig ® U.S. digit numbering convention: 1=thumb, 2=index finger, 3=middle, 4=ring, 5=little.
^bMajor innervation is indicated in boldface. Differing opinions exist, most shown are based on reference.³
^cTrunk (trunks of brachial plexus): S=superior, M=middle, I=inferior, all=all three.
^dCord (cords of brachial plexus): P=posterior, L=lateral, M=medial.

29.2.2 Thumb innervation/movement

Table 29.6 The 3 innervations of the thumb		
Action	Nerve	Muscle(s)
abduction, flexion, opposition ^a	median	abductor pollicis brevis, flexor pollicis brevis, opponens pollicis
adduction	ulnar	adductor pollicis
extension	radial ^b	extensor pollicis brevis & longus
^a occasional anomalous innervation by ulnar nerve		
^b via the posterior interosseous nerve		

Flexion/extension: occurs in the plane of the palm.
Abduction/adduction: occur in a plane at right angles to palm.
Opposition: bringing the thumb across the hand.

29.2.3 Muscles, roots, trunks, cords and nerves of the lower extremities

Table 29.7 Muscle innervation – hip & lower extremity ^a					
Muscle		Action	Roots ^b	Plexus ^c	Nerves
•	iliopsoas ^d	hip flexion	L1, 2, 3	L	femoral & L1, 2, 3
	sartorius	hip flex & thigh evert	L2, 3		femoral
•	quadriceps femoris	leg (knee) extension	L2, 3, 4	L	
	pectineus	thigh adduction	L2, 3		obturator
•	adductor longus		L2, 3, 4	L	
	adductor brevis		L2–4		
	adductor magnus		L3, 4		
	gracilis		L2–4		
	obturator externus	thigh adduction & lateral rotation	L3, 4		
•	gluteus medius/minimus	thigh abduction & medial rotation	L4, 5, S1	S	superior gluteal
	tensor fasciae lata	thigh flexion	L4, 5		
	piriformis	lateral thigh rotation	L5, S1		
•	gluteus maximus	thigh abduction (patient prone)	L5, S1, 2	S	inferior gluteal
	obturator internus	lateral thigh rotation	L5, S1	S	muscular branches
	gemelli		L4, 5, S1	S	
	quadratus femoris		L4, 5, S1	S	
•	biceps femoris ^e	leg flexion (& assist thigh extension)	L5, S1, 2		sciatic (trunk)
•	semitendinosus ^e		L5, S1, 2		
•	semimembranosus ^e		L5, S1, 2		
•	tibialis anterior	foot dorsiflexion & supination	L4, 5 ^f	S	deep peroneal
•	extensor digitorum longus	extension toes 2–5 & foot dorsiflexion	L5, S1		

Table 29.7 continued

Muscle		Action	Roots ^b	Plexus ^c	Nerves
•	extensor hallucis longus (EHL) ^g	great toe extension & foot dorsi-flexion	L5 ^f , S1	S	
•	extensor digitorum brevis	extension great toe & toes 2–5	L5, S1	S	
•	peroneus longus & brevis	P-flex pronated foot & eversion	L5, S1	L/S	superficial peroneal
•	posterior tibialis	P-flex supinated foot & inversion	L4, 5	S	tibial
	flexor digitorum longus	P-flex sup foot, flex terminal phalanx toes 2–5	L5, S1, 2		
	flexor hallucis longus	P-flex sup foot, flex terminal phalanx great toe	L5, S1, 2		
	flexor digitorum brevis	flex mid phalanx toes 2–5	S2, 3		
	flexor hallucis brevis	flex proximal phalanx great toe	L5, S1, 2		
•	gastrocnemius	knee flexion, ankle P-flex	S1, 2	S	
	plantaris		S1, 2		
•	soleus	ankle P-flex	S1, 2	S	
•	abductor hallicus ^h	(cannot test ^h)	S1, 2	S	
	perineal & sphincters	voluntary contract pelvic floor	S2–4		pudendal

NB: items marked with a bullet (•) are clinically important muscle/nerves.
^aAbbreviations: P-flex=plantarflexion, D-flex=dorsiflexion, phlnx=phalanx.
^bMajor innervation is indicated in boldface type. Eg. when roots are shown as L4, 5, this indicates L5 is the main innervation, but both L4 & L5 contribute.
^cPlexus: L=lumbar, S=sacral
^diliopsoas is the term for the combined iliacus and psoas major muscles
^e “hamstrings”: familiar term for the grouped: semitendinosus and semimembranosus (together, the medial hamstrings) and the biceps femoris (lateral hamstrings)
^falthough many references, including some venerable ones, cite AT as being primarily L4, many clinicians agree that L5 innervation is probably more significant
^g EHL is the best L5 muscle to test clinically (although S1 radiculopathy can also weaken this muscle)
^habductor hallicus cannot be tested clinically, but is important for EMG

29.3 Peripheral nerve injury/surgery

29.3.1 Nerve action potentials

Stimulating a healthy nerve fiber with an electrical stimulus of an amplitude and duration that exceeds its threshold will produce a conducted impulse, or nerve action potential (NAP).^{4 (p 103)} Medium-sized axons (fibers) have a lower threshold than large ones which have lower threshold than small or fine axons.^{4 (p 103)}

29.3.2 Use of NAP with lesion in continuity

There is some degree of continuity in ≥60%of nerve injuries.^{4 (p 104)}
For a lesion in continuity (LIC), if surgical repair is needed, it may be too late if one waits until there is failure of anticipated clinical improvement. Presence of a NAP (regardless of amplitude or latency) distal to an LIC in the first few months after an injury usually indicates that operative intervention will not be needed. For recommended timing to obtain NAP recording, □ Table 29.8.^{4 (p 106)}

Table 29.8 Recommended timing to obtain NAP recording	
Injury	Timing
relatively focal contusions	2–4 months
stretch injuries (esp. brachial plexus)	4–5 months
partial injuries & entrapments, compressive lesions and tumors	any time
to identify an area of conduction block (regardless if lesion is from neuropraxia, axonotmesis, or neurotmesis)	acutely

29.3.3 Timing of surgical repair

The longer the distance from the injury site to the functional unit to be reinnervated, the earlier surgical intervention should be considered.^{4 (p 74)}

24 month rule^{4 (p 74)}: after 24 months of denervation, most muscles cannot recover useful function even with reinnervation. Exceptions: facial muscles, large bulky muscles such as biceps, brachialis, gastrocs, and some lesions in continuity with some preserved innervation.

29.3.4 Brachial plexus

General information

Formed by ventral rami (the dorsal rami innervate the paraspinal muscles), most commonly of nerve roots C5-T1 (schematically depicted in □ Fig. 29.1).

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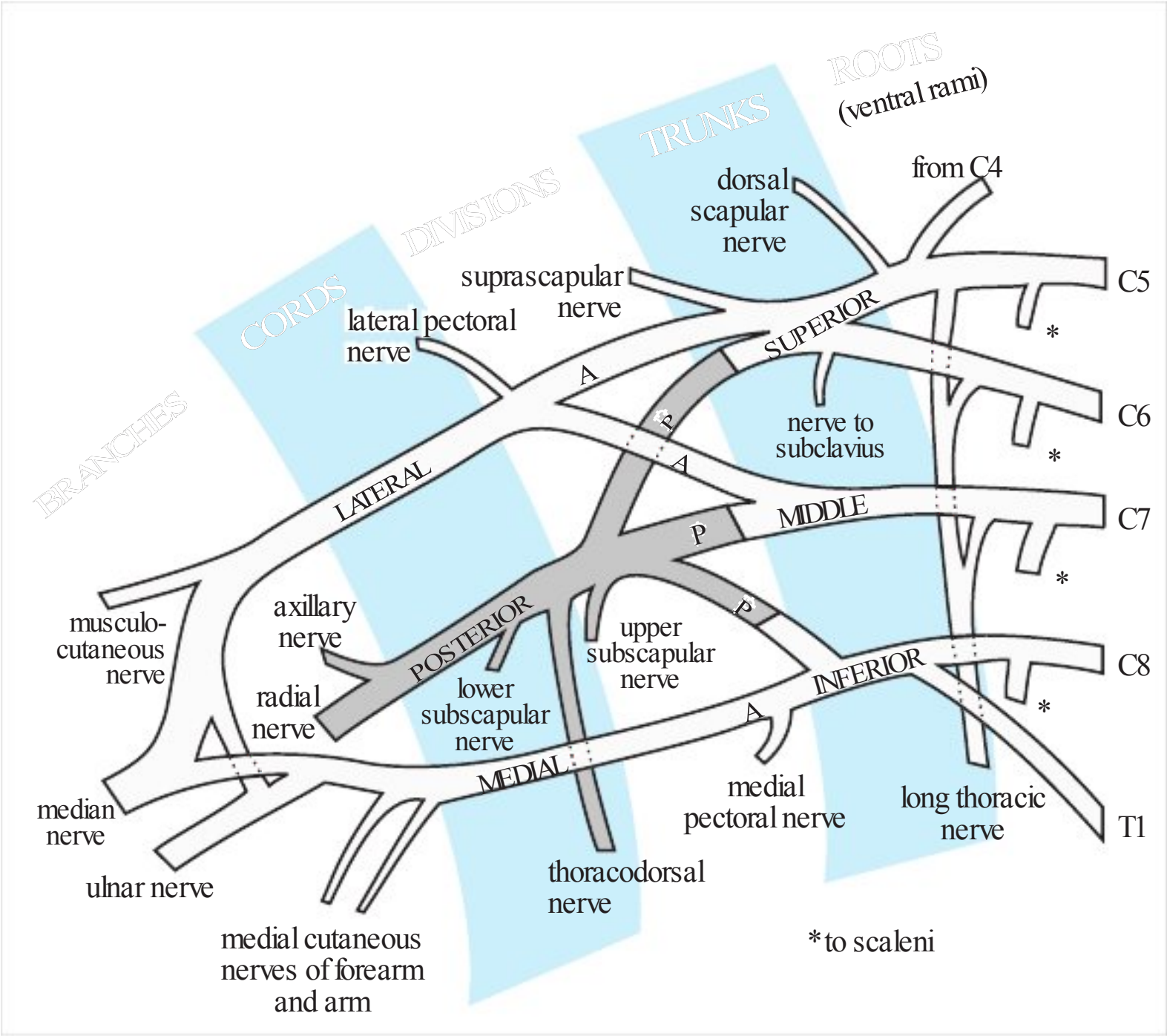


Fig. 29.1 Schematic diagram of the brachial plexus (By Permission: Churchill Livingstone, Edinburgh, 1973, R. Warwick & P. Williams: Gray’s Anatomy 35th Edition © Longman Group UK Limited)

□ Table 29.5 shows action, etc. of specific muscles. Also see □ Fig. 29.1. □ indicates that the nerve supplies the muscles listed; □ denotes a branch of the preceding nerve.

Nerves arising from the brachial plexus

Radial nerve (C5-C8)

See □ Fig. 29.2.

Radial nerve (and its branches) innervate the extensors of arm and forearm:

- □ triceps (all 3 heads)
- □ anconeus
- □ brachioradialis
- □ extensor carpi radialis longus & brevis (latter originates ≈ at terminal branch)
- □ supinator (originates near the terminal branch)
- □ continues into forearm as posterior interosseous nerve (C7, C8)
 - □ extensor carpi ulnaris
 - □ extensor digitorum
 - □ extensor digiti minimi
 - □ extensor pollicis brevis & longus
 - □ abductor pollicis longus
 - □ extensor indicis

Axillary nerve (C5, C6)

See □ Fig. 29.2.

- □ teres minor
- □ deltoid

Median nerve (C5-T1)

See □ Fig. 29.3, also see Martin-Gruber anastomosis (p.514).

- nothing in arm
- all forearm pronators and flexors except the two supplied by ulnar nerve
 - □ pronator teres
 - □ flexor carpi radialis
 - □ palmaris longus
 - □ flexor digitorum superficialis
- in the hand □ only the “LOAF muscles”
 - □ Lumbricals 1 & 2
 - □ Opponens pollicis
 - □ Abductor pollicis brevis
 - □ Flexor pollicis brevis (C8, T1)
- □ branch at or just distal to elbow anterior interosseous nerve (purely motor)
- □ flexor digitorum profundus I & II
- □ flexor pollicis longus
 - □ pronator quadratus

Ulnar nerve (C8, T1)

□ Fig. 29.3. (Note: This is classic winging of the scapula. A variant of winging can occur with loss of trapezius muscle, e.g. with accessory nerve injury, and typically manifests when the patient pushes forward with the elbow held at the side of the thorax.)

- nothing in arm
- only 2 muscles in forearm:
 - □ flexor carpi ulnaris
 - □ half of flexor digitorum profundus (parts III & IV)
- all hand muscles excluding “LOAF” muscles (see above), viz.:
 - □ adductor pollicis
 - □ all interossei (4 dorsal & 3 palmar)
 - □ lumbricals 3 & 4
 - □ 3 hypothenar muscles: abductor, opponens & flexor digiti minimi
 - □ deep part of flexor pollicis brevis (by deep branch of ulnar nerve)
 - □ palmaris brevis (by the superficial branch of the ulnar nerve)

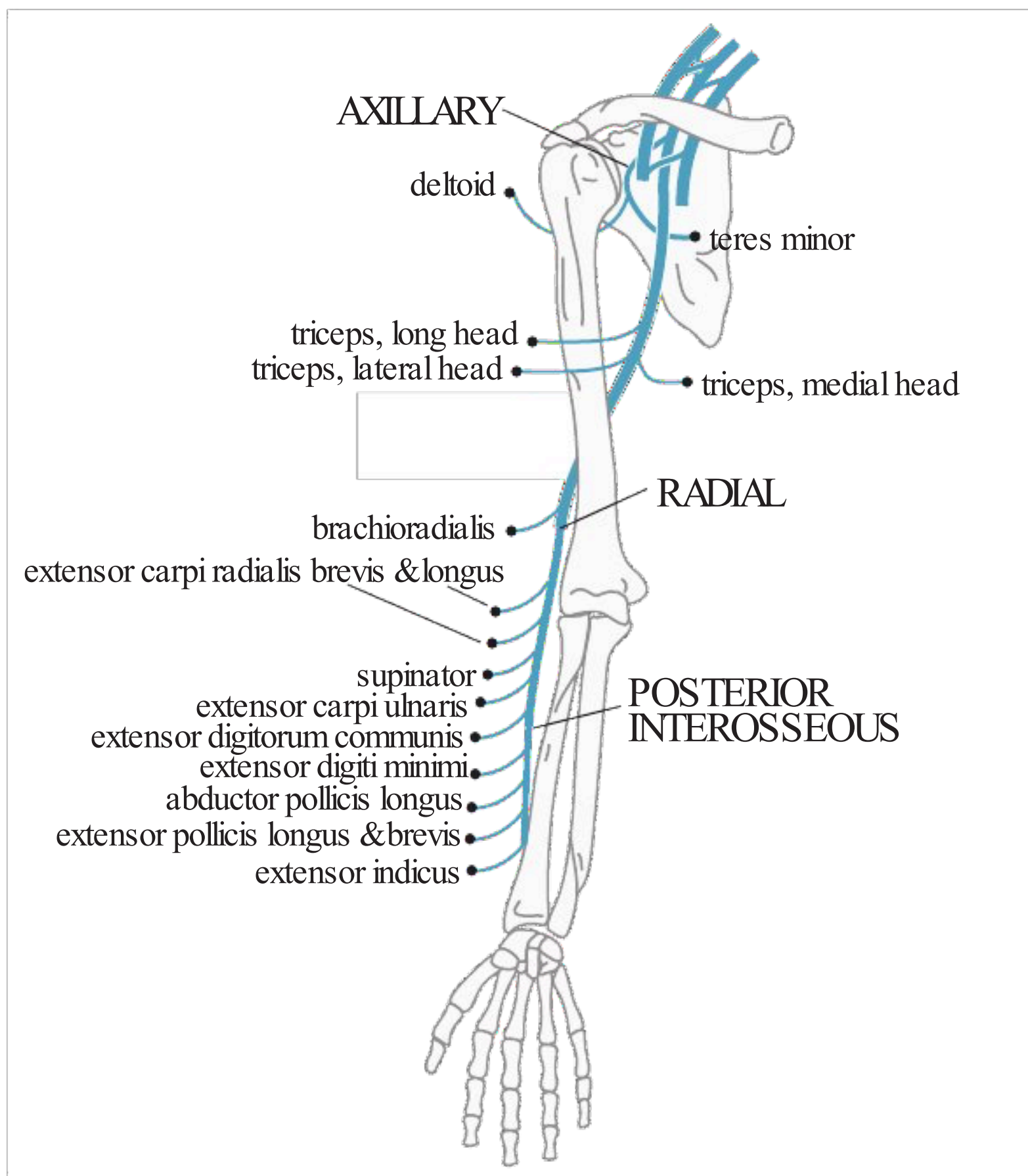


Fig. 29.2 Muscles of the radial and axillary nerves

Musculocutaneous nerve (C5, C6)

Supplies arm flexors

- ☐ Coracobrachialis
- ☐ biceps
- ☐ brachialis

☐ lateral cutaneous nerve of the forearm (terminal branch) supplies cutaneous sensation to radial aspect of forearm

Dorsal scapular nerve (C4, C5)

- ☐ rhomboids (major & minor)
- ☐ levator scapulae

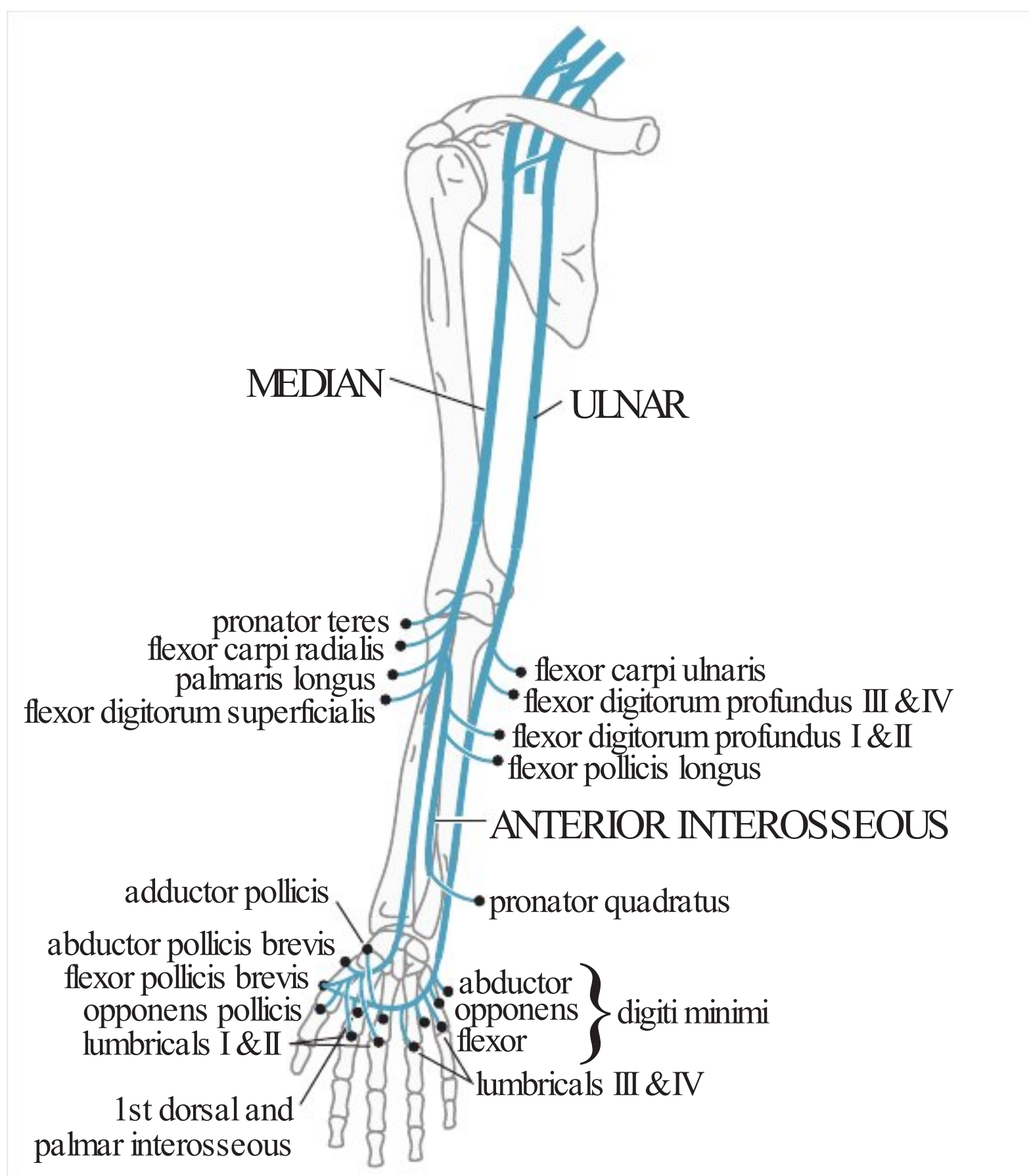


Fig. 29.3 Muscles of the median and ulnar nerves

Suprascapular nerve (C5, C6)

- ☐ Supraspinatus
- ☐ Infraspinatus

Subscapular nerve (C5–7)

- ☐ teres major
- ☐ subscapularis

Thoracodorsal nerve (C6, C7, C8)

- ☐ latissimus dorsi

Long thoracic nerve (C5–7)

Originates off of proximal nerve roots

□ serratus anterior (holds scapula to chest wall): lesion → winging of the scapula. To test: patient leans forward against wall with arms outstretched, scapula separates from posterior chest wall if the serratus anterior is not contracting. (Note: This is classic winging of the scapula. A variant of winging can occur with loss of trapezius muscle, e.g. with accessory nerve injury, and typically manifests when the patient pushes forward with the elbow held at the side of the thorax.)

Anatomic variants

Martin-Gruber anastomosis

See reference.⁵

Anastomosis between median and ulnar nerves in the forearm found in 16 of 70 (23%) cadavers, bilateral in 3 (19%). Pattern I (90%): 1 anastomotic branch, Pattern II (10%) had 2.

Classification based on the origin from the median nerve: Type a (47.3%) from the branch to the superficial forearm flexor muscles, Type b (10.6%) from the common trunk, and Type c (31.6%) from the anterior interosseous nerve. Pattern II was a duplication of Type c (10.5%). The anastomotic branch was undivided in 15 cases, and divided into two branches in four cases. The anastomosis took an oblique angle or arched course to the ulnar nerve and passed superficial to the ulnar artery in four cases, deep to it in six, and in nine cases it was related to the anterior ulnar recurrent artery.⁵

Richie-Cannieu anastomosis

Motor connections from median to ulnar nerve at the palm. Found in 70% of patients.

References

- [1] Fernandez E, Pallini R, La Marca F, et al. Neurosurgery of the Peripheral Nervous System - Part I: Basic Anatomic Concepts. Surg Neurol. 1996; 46:47–48
- [2] Dyck PJ, Boes CJ, Mulder D, Millikan C, Windebank AJ, Espinosa R. History of standard scoring, notation, and summation of neuromuscular signs. A current survey and recommendation. J Peripher Nerv Syst. 2005; 10:158–173
- [3] Medical Research Council. Aids to the Examination of the Peripheral Nervous System. London: Her Majesty's Stationery Office; 1976
- [4] Kline DG, Hudson AR. Nerve Injuries: Operative Results for Major Nerve Injuries, Entrapments, and Tumors. Philadelphia: W. B. Saunders; 1995
- [5] Rodriguez-Niedenfuhr M, Vazquez T, Parkin I, Logan B, Sanudo JR. Martin-Gruber anastomosis revisited. Clin Anat. 2002; 15:129–134

30 Entrapment Neuropathies

30.1 General information

Entrapment neuropathy is a peripheral nerve injury resulting from compression either by external forces or from nearby anatomic structures. Mechanism can vary from one or two significant compressive insults to many localized, repetitive mild compressions of a nerve. Certain nerves are particularly vulnerable at specific locations by virtue of being superficial, fixed in position, traversing a confined space, or in proximity to a joint. The most common symptom is pain (frequently at rest, more severe at night, often with retrograde radiation causing more proximal lesion to be suspected) with tenderness at the point of entrapment. Referred pain is so common that Frank Mayfield once said that patients with nerve entrapment don't know where the problem is located. Always consider the possibility of a systemic cause. Entrapment neuropathies may be associated with:

1. diabetes mellitus
2. hypothyroidism: due to glycogen deposition in Schwann cells
3. acromegaly
4. amyloidosis: primary or secondary (as in multiple myeloma)
5. carcinomatosis
6. polymyalgia rheumatica (p.198)
7. rheumatoid arthritis: 45% incidence of 1 or more entrapment neuropathies
8. gout

30.2 Mechanism of injury

Brief compression primarily affects myelinated fibers, and classically spares unmyelinated fibers (except in cases of severe acute compression). Acute compression compromises axoplasmic flow which can reduce membrane excitability. Chronic compression affects both myelinated and unmyelinated fibers and can produce segmental demyelination in the former, and if the insult persists, axolysis and wallerian degeneration will occur in both types. The issue of ischemia is more controversial.¹ Some contend that simultaneous venous stasis at the site of compression can produce ischemia which can lead to edema outside the axonal sheath which may further exacerbate the ischemia. Eventually, fibrosis, neuroma formation, and progressive neuropathy can occur.

30

30.3 Occipital nerve entrapment

30.3.1 General information

Greater occipital nerve (nerve of Arnold) is a sensory branch of C2 (□ Fig. 28.1 for dermatome). Entrapment presents as occipital neuralgia: pain in occiput usually with a trigger point near the superior nuchal line. Pressure here reproduces pain radiating up along back of head towards vertex.

More common in women.

30.3.2 Differential diagnosis

1. headache
 - a) may be mimicked by migraine headache
 - b) may be part of muscle contraction (tension) headache
2. myofascial pain²: the pain may be widely separated from the trigger point
3. vertebrobasilar disease including aneurysm and SAH
4. cervical spondylosis
5. pain from Chiari I malformation (p.277)

30.3.3 Possible causes of entrapment

1. trauma
 - a) direct trauma (including iatrogenic placement of suture through the nerve during surgical procedures, e.g. in closing a posterior fossa craniectomy)
 - b) following traumatic cervical extension³ which may crush the C2 root and ganglion between the C1 arch and C2 lamina
 - c) fractures of the upper cervical spine

- 2. atlanto-axial subluxation (AAS) (e.g. in rheumatoid arthritis) or arthrosis
- 3. entrapment by hypertrophic C1–2 (epistrophic) ligament⁴
- 4. neuromas
- 5. arthritis of the C2–3 zygapophyseal joint

30.3.4 Treatment

General information

Σ

For idiopathic occipital neuralgia: available evidence is from small, retrospective, case series studies and is insufficient to conclude that either local injection or surgery are effective. Nerve blocks with steroids and local anesthetics provide only temporary relief. Surgical procedures such as nerve root decompression or neurectomy may provide effective pain relief for some patients; however, patient-selection criteria for these procedures have not been defined, and recurrence is common.

In idiopathic cases with no neurologic deficit, the condition is usually self limited.

Non surgical treatment

- 1. greater occipital nerve block with local anesthetic and steroids (see below)
 - a) may provide relief typically lasting ≈ 1 month⁵
 - b) is no longer considered diagnostic because it is not sufficiently specific
- 2. physical therapy: massage and daily stretching exercises
- 3. TENS unit: provided ≥ 50% relief in 13 patients for up to 5 yrs⁶
- 4. oral antiinflammatory agents
- 5. centrally acting pain medications: Neurontin, Paxil, Elavil...
- 6. botulinum toxin injection⁷: although this study had quite a few placebo responders

If these measures do not provide permanent relief in disabling cases, surgical treatment may be considered, although caution is advised by many due to poor results.^{2,8} Alcohol neurolysis may be tried. A collar is not indicated as it may irritate the condition.

Occipital nerve block

Inject trigger point(s) if one or more can be identified (there is usually a trigger point near the superior nuchal line). The nerve may also be blocked at the point where it emerges from the dorsal neck muscles.

If the pathology is more proximal (e.g. at C2 spinal ganglion), then block of the ganglion may be required. Technique⁹ (done under fluoroscopy): shave hair below the mastoid process; prep with iodine; infiltrate with local; insert a 20 gauge spinal needle midway between C1 and C2, halfway between the midline and the lateral margin of the dorsal neck muscles. Aim rostrally, the final target is the midpoint of the C1–2 joint on AP fluoro, and almost but not touching the inferior articular process of C1. Infiltrate 1–3 ml of anesthetic and check for analgesia in the C2 distribution.

Surgical treatment

- 1. decompression of C2 nerve root if compressed between C1 and C2⁴
- 2. in cases of AAS, decompression and atlanto-axial fusion (p.1479) may work

Surgical treatment options for idiopathic occipital neuralgia:

- 1. peripheral occipital nerve procedures: these may not be effective for proximal compression of the C2 root or ganglion:
 - a) occipital neurectomy (see below)
 - peripheral avulsion of the nerve
 - avulsion of the greater occipital nerve as it exits between the transverse process of C2 and the inferior oblique muscle
 - b) alcohol injection of greater occipital nerve

2. occipital nerve stimulators
3. release of the nerve within the trapezius muscle. Immediate results: relief in 46% improvement in 36% Only 56% reported improvement at 14.5 mos¹⁰
4. intradural division of the C2 dorsal root via a posterior intradural approach
5. ganglionectomy

Occipital neurectomy: The occipital nerve usually pierces the cervical muscles ≈ 2.5 cm lateral to the midline, just below the inion. Palpation or doppler localization of the pulse of the accompanying greater occipital artery sometimes helps to locate the nerve. However, relief only occurs in $\approx 50\%$ and recurrence, usually within a year, is common.

30.4 Median nerve entrapment

30.4.1 General information

The two most common sites of entrapment of the median nerve:

1. at the wrist by the transverse carpal ligament: carpal tunnel syndrome (see below)
2. in upper forearm by pronator teres muscle: pronator teres syndrome (p.521)

30.4.2 Anatomy

Contributing nerve roots: C5 through T1. The median nerve arises from the medial and lateral cords of the brachial plexus (□ Fig. 29.1), and descends the upper arm adjacent to the lateral side of the brachial artery. It crosses to the medial side of the artery at the level of the coracobrachialis. In the cubital fossa, the median nerve passes behind the lacertus fibrosus (bicipital aponeurosis) and enters the upper forearm between the two heads of the pronator teres and supplies this muscle.

Just beyond this point, it branches to form the purely motor anterior interosseous nerve which supplies all but 2 muscles of finger and wrist flexion. It descends adherent to deep surface of flexor digitorum superficialis (FDS), lying on the flexor digitorum profundus. Near the wrist, it emerges from the lateral edge of FDS becoming more superficial, lying medial to the tendon of flexor carpi radialis, just lateral to and partially under the cover of the palmaris longus tendon. It passes under the transverse carpal ligament (TCL) through the carpal tunnel which also contains the tendons of the flexor digitorum profundus and superficialis deep to the nerve (9 tendons total, 2 to each finger, 1 to the thumb¹¹). The motor branch arises deep to the TCL, but may anomalously pierce the TCL. It supplies the “LOAF muscles” (Lumbricals 1 & 2, Opponens pollicis, and Abductor and Flexor pollicis brevis).

The TCL attaches medially to pisiform and hook of hamate, laterally to trapezium and tubercles of scaphoid. TCL is continuous proximally with fascia over FDS and antebrachial fascia, distally with the flexor retinaculum of the hand. The TCL extends distally into the palm to ≈ 3 cm beyond the distal wrist crease. The palmaris longus tendon, which is absent in 10% of population, partially attaches to the TCL.

Palmar cutaneous branch (PCB) of median nerve: arises from the radial aspect of the median nerve approximately 5.5 cm proximal to styloid process of the radius, underneath the cover of FDS of the middle finger. It crosses the wrist above the TCL to provide sensory innervation to the base of the thenar eminence (and is thus spared in carpal tunnel syndrome).

The sensory distribution of the average median nerve is shown in □ Fig. 30.1.

30.4.3 Injuries to the main trunk of the median nerve

General information

Above the elbow, the median nerve may rarely be compressed by Struther’s ligament (see below). At the elbow and forearm, the median nerve may rarely be trapped at any of three sites: 1) lacertus fibrosus (bicipital aponeurosis),¹² 2) pronator teres, 3) sublimis bridge. Neuropathy may also result from direct or indirect trauma or external pressure (“honeymoon paralysis”).¹² Longstanding compression of the main trunk of the median nerve produces a “benediction hand” when trying to make a fist (index finger extended, middle finger partially flexed; due to weakness of flexor digitorum profundus I & II).

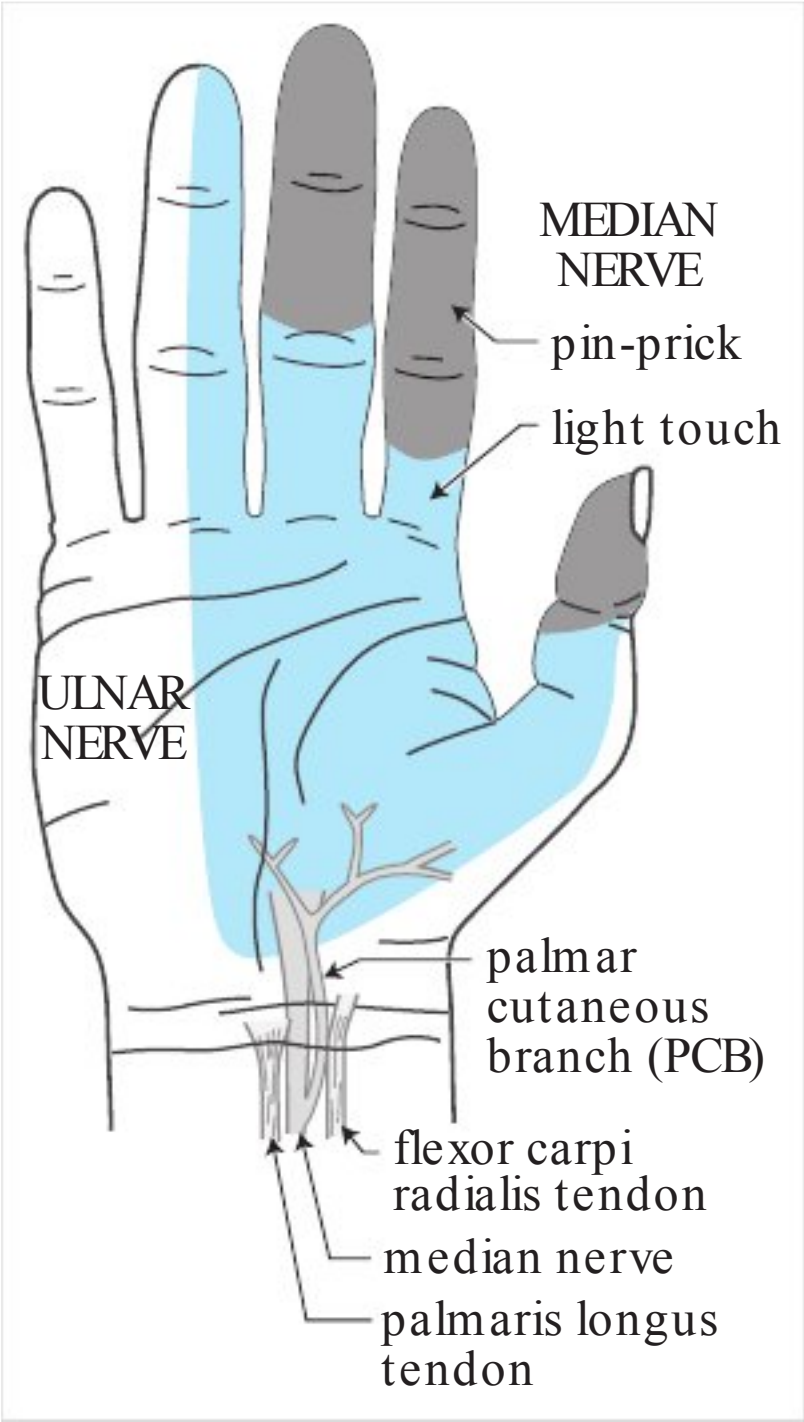


Fig. 30.1 Sensory distribution of the median and ulnar nerves in the hand palmar surface.

30

Struther’s ligament

Distinct from struthers arcade (p.527) which is a normal finding. The supracondylar process (SCP) is an anatomical variant located 5–7 cm above medial epicondyle, present in 0.7–2.7% of population. Struther’s ligament bridges the SCP to the medial epicondyle. The median nerve and brachial artery pass underneath, the ulnar nerve may also. Usually asymptomatic, but occasionally may cause typical median nerve syndrome.

Pronator (teres) syndrome

From direct trauma or repeated pronation with tight hand-grip. Trapped where nerve dives between 2 heads of pronator teres. Causes vague aching and easy fatiguing of forearm muscles with weak grip and poorly localized paresthesias in index finger and thumb. Nocturnal exacerbation is absent. Pain in palm distinguishes this from carpal tunnel syndrome (CTS) since the median palmar cutaneous branch (PCB) exits before the TCL and is spared in CTS.

Treat with resting forearm. Surgical decompression indicated for cases that progress while on rest or when continued trauma is unavoidable.

Anterior interosseous neuropathy

General information

Key concepts

- weakness of 3 muscles: FDP I & II, FPL, & pronator quadratus. No sensory loss
- loss of flexion of the distal phalanges of the thumb and index finger (pinch sign)

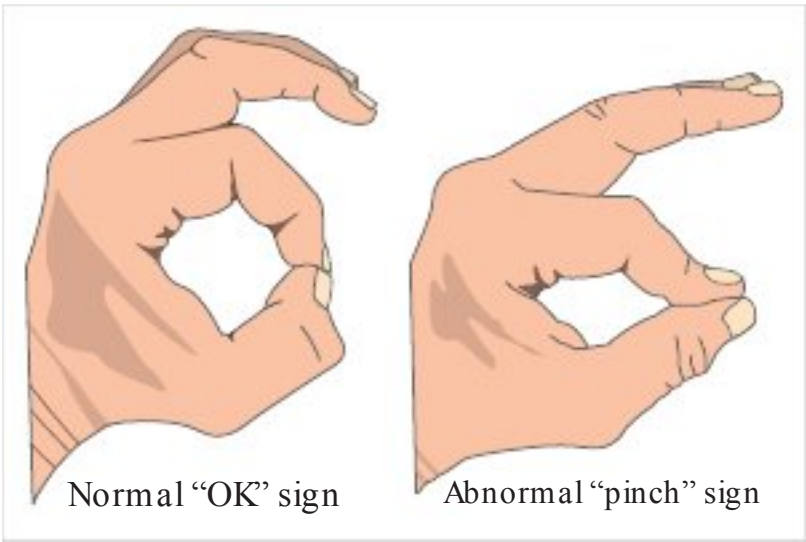


Fig. 30.2 "Pinch sign" seen with AIN

The anterior interosseous nerve is a purely motor branch of the median nerve that arises in the upper forearm. Anterior interosseous neuropathy (AIN) produces no sensory loss and weakness of the 3 muscles supplied by the nerve:

- 1. flexor digitorum profundus (FDP) I & II: flexion of distal phalanx of digits 2 & 3
- 2. flexor pollicis longus (FPL): flexion of distal phalanx of thumb
- 3. pronator quadratus (in the distal forearm): difficult to isolate clinically

Etiologies of AIN

Include: idiopathic, amyotrophy, ulna/radius fractures, penetrating injuries, forearm lacerations.

Clinical

Symptoms: Patients complain of difficulty grasping small objects between the thumb and the index finger. Idiopathic cases may be preceded with forearm aching.

Physical exam: Sensory: no sensory loss.

Strength: digits 1, 2 & 3 are examined individually. The proximal interphalangeal joints are stabilized by the examiner and the patient is asked to flex the DIP. With AIN, there is no significant flexion of the DIP.

Pinch sign: the patient attempts to forcefully pinch the tips of the index finger and thumb as in making an "OK" sign (□ Fig. 30.2, left), with AIN the terminal phalanges extend and the pulps touch instead of the tips¹³ (□ Fig. 30.2, right).

Diagnosis

In addition to the physical exam, EMG may be helpful.

EMG: primarily assesses pronator quadratus & flexor pollicis longus (FDP I & II is difficult on EMG because it has dual innervation with the ulnar nerve innervated portion being more superficial than the median nerve innervated portion). Important to evaluate pronator teres (abnormalities suggest involvement more proximal than forearm).

Management

In the absence of an identifiable cause of nerve injury, expectant management is recommended for 8–12 weeks, following which exploration is indicated which may reveal a constricting band near the origin.

30.4.4 Carpal tunnel syndrome

General information

Key concepts

- the most common compression neuropathy. Involves median nerve in the wrist
- symptoms: tingling in the hand, worse at night and with elevation of hands
- physical exam is not very sensitive:
 - sensory: decreased pinprick in digits 1–3 and the radial half of 4
 - sensitivity: Tinel's (tapping on wrist) 60% Phalen's (flexion of wrist) 80%
- electrodiagnostics: sensory latency @ wrist >3.7 ms is the most sensitive test

- treatment:
 - mild cases: nonsurgical treatment (NSAIDs, neutral position splint...)
 - unresponsive or severe cases (neurologic deficits, duration > 1 year): surgical neurolysis (decompression) of the median nerve at the wrist has 70% satisfaction rate

Carpal tunnel syndrome (CTS) is the most common entrapment neuropathy in the upper extremity.^{14,15,16} Carpal tunnel release (CTR) is one of the most frequently performed hand procedures.^{17,18} The majority of patients have a satisfactory outcome from surgical treatment; see Outcome of surgical treatment (carpal tunnel release) (p.526). The median nerve is compressed within its course through the carpal tunnel just distal to the wrist crease. □ Table 30.1 shows the effect of pressure within the carpal tunnel.

Epidemiology

Usually occurs in middle aged patients. Ratio of female:male=4:1. It is bilateral in over 50% of cases, but is usually worse in the dominant hand. The prevalence of CTS and UNE is increased in diabetics.

Common etiologies

See reference.¹⁹

In most cases, no specific etiology can be identified. CTS is very common in the geriatric population without any additional risk factors. The following etiologies tend to be more common in younger patients:

1. “classic” CTS: chronic time course, usually over a period of months to years
 - a) trauma: often job-related (may also be associated with avocations)
 - repetitive movements of hand or /wrist: e.g. carpenter’s
 - repeated forceful grasping or pinching of tools or other objects
 - awkward positions of hand and/or wrist, including wrist extension, ulnar deviation, or especially forced wrist flexion
 - direct pressure over carpal tunnel
 - use of vibrating hand tools
 - b) systemic conditions: in addition to systemic causes listed for entrapment neuropathies (p.515) – especially rheumatoid arthritis, diabetes -, also consider:
 - obesity
 - local trauma
 - pregnancy: 54% remained symptomatic 1 year post-partum, and patients with onset early in pregnancy were less likely to improve²⁰
 - mucopolysaccharidosis V
 - tuberculous tenosynovitis
 - multiple myeloma (p.714) (amyloid deposition in flexor retinaculum)
 - c) patients with A-V dialysis shunts in the forearm have an increased incidence of CTS, possibly on an ischemic basis (steal and/or venous stasis) or possibly from the underlying renal disorder
2. “acute” CTS: an uncommon condition where the symptoms of CTS appear suddenly and severely, usually following some type of exertion or trauma. Etiologies:
 - a) median artery thrombosis: <10% of individuals have a persistent median artery
 - b) hemorrhage or hematoma in the transverse carpal ligament

Table 30.1 Pressure within carpal tunnel	
Pressure (mm Hg)	Description
<20	normal
20–30	venular flow retarded
30	axonal transport impaired
40	sensory & motor dysfunction
60–80	blood flow ceases

Signs and symptoms

The physical exam for CTS is fairly insensitive. Signs and symptoms may include:

1. dysesthesias:
 - a) characteristically patients are awakened at night by a painful numbness in the hand (often described as the “hand falling asleep”) that often subjectively feels like a loss of circulation of blood. They often seek relief by: shaking or dangling or swinging the hand, opening and closing or rubbing the fingers, running hot or cold water over the hand, or pacing the floor. It may radiate up the arm, occasionally as far as shoulder
 - b) daytime activities that characteristically elicit symptoms usually involve prolonged hand elevation: holding a book or newspaper to read, driving a car, holding a telephone receiver, brushing the hair
 - c) distribution of symptoms:
 - on palmar side in radial 3.5 fingers (palmar side of thumb, index finger, middle finger, and radial half of ring finger)
 - dorsal side of these same fingers distal to the PIP joint
 - radial half of palm
 - subjective involvement of little finger occurs not infrequently for reasons that are not clear
2. weakness of hand, especially grip. Characteristically manifests as difficulty opening jars. May be associated with thenar atrophy (late change, severe atrophy is seldom seen with current awareness of CTS by most physicians). An occasional patient may present with severe atrophy and no history of pain
3. clumsiness of the hand and/or difficulty with fine motor skills: probably due mostly to numbness more than a motor deficit. Often presents as difficulties buttoning buttons or zipping zippers, putting on earrings, fastening bra straps...
4. hypesthesia in median nerve sensory distribution: usually best appreciated in finger tips, loss of 2-point discrimination may be more sensitive test
5. Phalen’s test: 30–60 secs wrist flexion to a 90° angle exaggerates or reproduces pain or tingling. Positive in 80% of cases (80% sensitive)²¹
6. Tinel’s sign at the wrist: paresthesias or pain in median nerve distribution produced by gently percussing over the carpal tunnel. Positive in 60% of cases. May also be present in other conditions. Reverse Tinel’s sign: produces symptoms radiating up the forearm for variable distance
7. ischemic testing: place blood pressure cuff proximal to wrist, inflation x 30–60 seconds may reproduce CTS pain

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Differential diagnosis

Differential diagnosis includes (modified²²):

1. cervical radiculopathy: coexists in 70% of patients with either median or ulnar neuropathy (C6 radiculopathy may mimic CTS). Usually relieved by rest, and exacerbated by neck movement. Sensory impairment has dermatomal distribution. It has been postulated that cervical nerve root compression may interrupt axoplasmic flow and predispose the nerve to compressive injury distally (the term double-crush syndrome was coined to describe this²³), and although this has been challenged²⁴ it has not been disproven
2. thoracic outlet syndrome (p.554): loss of bulk in hand muscles other than thenar. Sensory impairment in ulnar side of hand and forearm
3. pronator teres syndrome (p.518): more prominent palmar pain than with CTS (median palmar cutaneous branch does not pass through carpal tunnel)
4. de Quervain’s syndrome: tenosynovitis of the abductor pollicis longus and extensor pollicis brevis tendons often caused by repetitive hand movements. Results in pain and tenderness in the wrist near the thumb. Onset in 25% of cases is during pregnancy, and many in 1st postpartum year. Usually responds to wrist splints and/or steroid injections. NCVs should be normal. Finkelstein’s test: the thumb is passively abducted while thumb abductors are palpated, positive if this aggravates the pain²⁵
5. reflex sympathetic dystrophy: may respond to sympathetic block (p.1521)
6. tenosynovitis of any of the flexor ligaments: may occasionally be due to TB or fungus. Usually a long, indolent course. Fluid accumulation may be present

Diagnostic tests

Electrodiagnostics (EDX)

Electromyogram (EMG) and nerve conduction study (NCS) which includes measurement of nerve conduction velocities (NCV): may help confirm the diagnosis of CTS and distinguish it from cervical root abnormalities and from tendonitis.

CTS is predominantly a demyelinating injury although it can progress to axonal loss.²⁶ Two sensory comparison techniques that clearly agree (either normal or abnormal) are adequate to confirm or refute the diagnosis. For borderline abnormalities additional sensory comparison testing or the combined sensory index (CSI) can clarify the diagnosis. If sensory responses are absent, the median motor latency in comparison to the ulnar latency can help localize a focal abnormality.²⁷

Practice guideline: Electrodiagnostic criteria for CTS

The practice guideline for CTS recommends diagnostic examination strategies^{28,29,30}:

1. Standard: perform a median sensory nerve conduction study (NCS) across the wrist with conduction distance of 13 to 14 cm. If abnormal, compare to an adjacent sensory nerve in the symptomatic limb
2. Standard: if the initial median sensory NCS across the wrist is normal then additional comparison studies are recommended
3. Guideline: motor NCS of the median nerve recording from the thenar muscle and of 1 other nerve in the symptomatic limb
4. Option: supplementary NCS
5. Option: needle electromyography (EMG) of cervical root screen muscles including a thenar muscle

NCV: Electrophysiologic studies support a diagnosis of carpal tunnel syndrome (CTS) using median nerve conduction studies across the transverse carpal ligament. Characteristic abnormalities: prolongation of sensory and motor distal latencies, slowing of the conduction velocity and decreased amplitudes of sensory and motor responses. Guidelines regarding recommended studies are published by the American Association of Neuromuscular and Electrodiagnostic Medicine (AANEM), AAPM&R, and AAN. Adhering to the guidelines, sensitivity is greater than 85% and specificity is greater than 95%²⁸ Sensory latencies are more sensitive than motor. (Note: although up to 15% of cases may have normal electrodiagnostic studies, great reservation should be exercised in considering operating on CTS with normal sensory NCV and amplitude.)

Normal findings are shown in □ Table 30.2. Abnormal values also listed are a rough guide, but the correlation between severity of EDX findings and symptoms in CTS is not well established.²⁷ Nevertheless, classification as follows can in part predict the outcome of carpal tunnel release (surgery) with normal and very severe NCS abnormalities having a worse prognosis than patients with moderate NCS abnormalities.^{27,31}

EDX interpretation (reports may also include the degree of slowing in the summary of findings or interpretation^{27,32,33}):

- mild: Prolonged (relative or absolute) median nerve sensory latencies with normal motor studies. No evidence of axonal loss
- moderate: Prolonged (relative or absolute) median nerve sensory latencies with prolongation of motor distal latency. No evidence of axonal loss.
- severe: Any of the aforementioned NCS abnormalities with evidence of axonal loss on EMG.

Table 30.2 Distal conduction latencies through carpal tunnel ^a				
Degree of involvement ^b	Sensory		Motor	
	latency ^c (mSec)	amplitude (mV)	latency ^d (mSec)	amplitude (mV)
normal	<3.7	>25	<4.5	>4
mild ^b	3.7–4.0		4.4–6.9	
moderate ^b	4.1–5.0		7.0–9.9	
severe ^b	>5 or unobtainable		>10	

^aassumes normal proximal NCV
^bseverity does not reliably correlate with latency (see text)
^cto index finger. Sensory latency is measured to the peak of the waveform
^dto abductor pollicis brevis

Additional comparison studies for uncertain cases compare median nerve sensory conduction velocity to that of the ulnar nerve (or radial nerve): normal median nerve should be at least 4 m/sec faster than the ulnar, reversal of this pattern suggests median nerve injury. Alternatively, the sensory latencies for the palmar median and ulnar nerves can be compared; the median nerve latency should not be ≥ 0.3 mS longer than the ulnar.

EMG: normal in up to 31% of cases of CTS. In relatively advanced CTS, it may show increased polyphasicity, positive waves, fibrillation potentials, and decreased motor unit numbers on maximal voluntary thenar muscle contraction. EMG may detect cervical radiculopathy if motor involvement is present.

With severe “end stage” CTS, sensory and motor potentials may not be recordable, and EMG is not helpful in localizing (i.e. differentiating CTS from other etiologies).

Laboratory tests

Recommended in cases where an underlying peripheral neuropathy is suspected (e.g. unclear etiology in a young individual with no risk factors such as repetitive hand use). This same protocol is a useful initial workup for any case of peripheral neuropathy:

- 1. thyroid hormone levels (T4 (total or free) & TSH): to R/O myxedema
- 2. CBC: anemia is common in multiple myeloma, also to R/O amyloidosis
- 3. electrolytes:
 - a) to R/O chronic renal failure that could cause uremic neuropathy
 - b) blood glucose: R/O diabetes
- 4. in cases suspicious for multiple myeloma: see full details (p.714)
 - a) 24 hour urine for kappa Bence-Jones protein
 - b) bloodwork: serum protein electrophoresis (SPEP) and immune electrophoresis (IEP) (looking for IgG kappa band)
 - c) skeletal radiologic survey
 - d) anemia is common on the CBC

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Imaging studies

Not routinely done unless a mass lesion is suspected.

Wrist MRI: very sensitive. Findings with CTS include: flattening or swelling of the nerve, palmar bowing of the flexor retinaculum. May also demonstrate ganglion cysts, lipomas... Enhancement may occur with hypervascular edema.

Diagnostic ultrasound: faster and less expensive than MRI, and can assess bloodflow and changes with different wrist positions. 18 MHz probes may improve images.

Management of CTS

Practice guideline: Management of CTS

American Association of Orthopedic Surgeons (AAOS) Clinical Practice Guideline endorsed by American Association of Neurological Surgeons, Congress of Neurological Surgeons, American Society of Plastic Surgeons, American Academy of PM&R and AANEM³⁴

- 1. A course of non-operative treatment is an option in patients diagnosed with CTS. Early surgery is an option when there is clinical evidence of median nerve denervation or the patient elects to proceed directly to surgical treatment (Grade C, Level V)
- 2. another non-operative treatment or surgery is suggested when the current treatment fails to resolve the symptoms within 2- 7 weeks (Grade B, Level I and II)
- 3. There is not sufficient evidence to provide specific treatment recommendations for CTS when found in association with diabetes, coexisting cervical radiculopathy, hypothyroidism, polyneuropathy, pregnancy, rheumatoid arthritis and CTS in the workplace. (inconclusive)
- 4. Management specifics
 - local steroid injection or splinting is suggested when treating patients with CTS before considering surgery (Grade B, Level I and II)
 - oral steroids or ultrasound are options for treatment of CTS (Grade C, Level II)
 - carpal tunnel release is recommended for treatment of CTS (Grade A, level I)

Notwithstanding the AAOS recommendations, multiple studies report the results of carpal tunnel release in diabetics are good even when polyneuropathy is present.^{35,36}

Non-surgical management

Options include:

1. rest
2. medications: non-steroidal anti-inflammatory drugs (NSAIDs), diuretics, and pyridoxine (vitamin B6) have been studied with no evidence of efficacy¹¹
3. treatment of associated conditions (e.g. hypothyroidism or DM) is appropriate, but there is no data as to whether this relieves CTS¹¹
4. neutral position splints: alleviates symptoms in >80% of patients³⁷ (usually within a few days) and reduces prolonged sensory latencies.³⁸ Relapse is common (works best when patient does not return to heavy manual labor). A trial of at least 2–4 weeks is recommended
5. steroid injection: symptoms improve in >75% of patients.¹¹ 33% relapse within 15 mos. Repeat injections are possible, but most clinicians limit to 3/year
 - a) use 10–25 mg hydrocortisone. Avoid local anesthetics (may mask symptoms of intra-neural injection)
 - b) inject into carpal tunnel (deep to transverse carpal ligament) to ulnar side of palmaris longus to avoid median nerve (in patients without palmaris longus, inject in line with fourth digit)
 - c) median nerve injuries have been reported with this technique,³⁹ primarily due to intra-neural injection (all steroids are neurotoxic upon intrafascicular injection, and so are some of the carrier agents)
 - d) risk factors for recurrence: severe electrodiagnostic abnormalities, constant numbness, impaired sensation, & weakness or atrophy of thenar muscles¹¹

Surgical treatment

General information

The operation is commonly called a carpal tunnel release (CTR), AKA neurolysis or neuroplasty of the median nerve at the wrist.

Indications

Surgical intervention is recommended for: constant numbness, symptoms > 1 year duration, sensory loss, or thenar weakness/atrophy.¹¹ Surgical treatment of cases due to amyloidosis from multiple myeloma is also effective.

With bilateral CTS, in general one operates on the more painful hand first. However if the condition is severe in both hands (on EMG) and if it has progressed beyond the painful stage and is only causing weakness and/or numbness, it may be best to operate on the “better” hand first in order to try and maximize recovery of the median nerve at least on that side. Simultaneous bilateral procedures may also be done.⁴⁰ In severe cases, nerve recovery may not occur, it may be necessary to wait up to a year to determine extent of recovery.

Success rate

>70% of patients report satisfaction with their surgical results,⁴¹ with 70–90% being free of nocturnal pain.^{41,42}

Surgical techniques

A number of techniques are popular, including: incision through palm of hand, transverse incision through wrist crease (with or without a retinaculotome⁴³), and endoscopic techniques (using single or dual incisions). The efficacies of the various approaches have not been compared in an adequately powered randomized study¹¹ and there is no consensus on the superiority of any one technique^{14,44,45,46} including endoscopic vs. open CTR.

Transpalmar approach (□ Fig. 30.3): For a right-handed surgeon, sit in patient’s “axilla” (facing head of patient) for left sided CTS. Sit above patient’s arm (facing feet) for right sided CTS. Usually performed under local or regional anesthesia on outpatient basis. Magnification (operating loupes) are helpful.

Incision along imaginary line extending proximally from space between digits 3 and 4 (usually stay just to the ulnar side of the interthenar crease to avoid the PCB). The location of the median nerve may also be estimated by the palmaris longus tendon (stay slightly to ulnar side of tendon). Incision starts at distal wrist flexion crease, and the length depends on thickness of hand (it may extend as far distally as a line even with the crotch of the thumb). Optionally: curve ulnarward at proximal wrist flexion crease (to facilitate retraction).

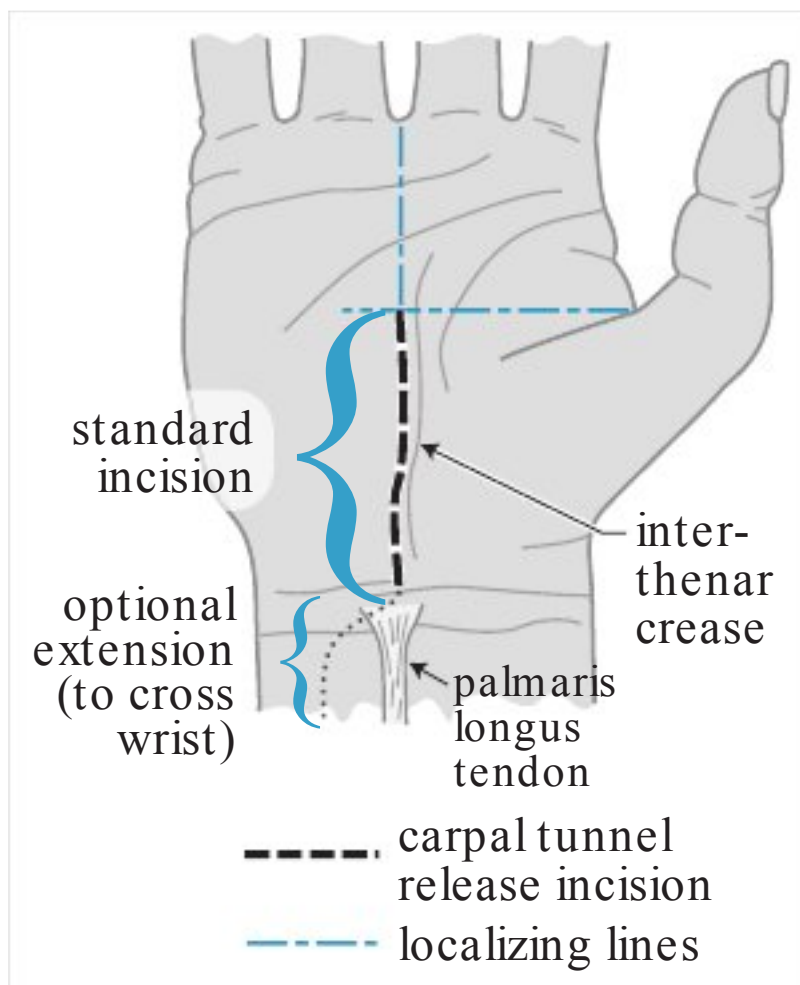


Fig. 30.3 Transpalmar incision for carpal tunnel syndrome

The median nerve is carefully approached through the TCL with progressively deepening incisions made with a 15 blade. All approaches to CTS surgery require complete division of TCL at and distal to the wrist. If tendons of the flexor digitorum superficialis are encountered, you need to look radially (towards the thumb) to find the nerve. In selected cases, the epineurium may be opened; however, internal neurolysis probably does more harm than good and in general should be avoided.

Close with absorbable 4-0 inverted sutures. Approximate skin edges with 4-0 nylon running or interrupted vertical mattress. Pad palm with several fluffs (opened dressing sponge). Cover with Kerlix®.

Post-op: wrap hand with thumb exposed. Wrist elevation and rest is recommended for several days. Analgesics for mild-to-moderate pain (e.g. acetaminophen with codeine) for 3-4 days. Sutures are removed at 7-10 days. No heavy work with hand for 2-3 weeks.

Complications of carpal tunnel surgery

See reference.⁴⁷

1. pain due to neuroma formation following transection of palmar cutaneous branch (PCB) of the median nerve
 - a) branches of PCB may cross interthenar crease
 - b) avoid by: using magnification and making incision slightly to ulnar side of interthenar crease
 - c) treated by ligating this branch where it originates from median nerve in forearm (results in small area of numbness at base of thenar eminence)
2. neuroma of dorsal sensory branch of radial nerve
 - a) caused by extending incision proximally and radially
 - b) may be treated by neurolysis of neuroma
3. injury to recurrent thenar (motor) branch of median nerve
 - a) anomaly may cause nerve to lie above or to pierce TCL
 - b) avoided by: staying to ulnar side of midline
4. direct injury to median nerve
5. volar displacement and entrapment of median nerve in healing edges of TCL
6. hypertrophic scar causing compression of median nerve
 - a) usually caused by incision crossing wrist perpendicular to flexion crease
 - b) avoid by not crossing flexion crease, or in cases where necessary (e.g. in releasing Guyon's canal entrapment of ulnar nerve, in tenosynovectomy for rheumatoid arthritis, or in dealing with an anomalous superficialis or palmaris muscle) by crossing wrist obliquely at 45° angle directed toward ulnar side⁴⁷ (see optional extension line in □ Fig. 30.3)
7. failure to improve symptoms
 - a) incorrect diagnosis: if EMG or NCV not done pre-op, they should be done after surgical failure (to R/O e.g. cervical root involvement (look for posterior myotome involvement), or generalized peripheral neuropathy)

- b) incomplete transection of TCL: the most common cause for failure if diagnosis is correct (also possibility of accessory ligament or fascial band proximal to TCL in cases where division was complete). When this is identified on re-exploration, 75% of patients will be cured or improved after division is completed
- 8. joint stiffness
 - a) caused by excessively long immobilization of wrist and fingers
- 9. injury to superficial palmar arch (arterial): usually results from “blind” distal division of TCL
- 10. bowstringing of flexor tendons
- 11. complex regional pain syndrome AKA reflex sympathetic dystrophy: exact incidence is unknown, reported in 4 of 132 patients in one series (probably too high, most surgeons will see only one or two cases in their career). Treatment with IV phentolamine has been suggested, but most cases are self limited after ≈ 2 weeks
- 12. infection: usually causes exquisite tenderness
- 13. hematoma: also usually quite painful and tender

Outcome of surgical treatment (carpal tunnel release)

75-90% of patients have symptom resolution or are improved to a satisfactory state following carpal tunnel release.^{17,44,48} Clinical improvement peaks at 6 months post-op^{49,50,51} although paresthesias may take ≥ 9 months to resolve.^{44,52,53,54}

The results of decompression are good in diabetics with CTS even when a generalized peripheral neuropathy is present.⁵⁵ In comparison, ulnar neuropathy at the elbow is often severe in diabetics and predominantly motor with axonal injury. These patients usually do not respond well to surgery.³⁵

Managing surgical treatment failures

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Less than satisfactory outcomes following CTR should be categorized as:

1. new symptoms: may include neuropathic pain out of proportion to the surgery, new areas of numbness/ paresthesias or marked weakness of the thenar muscles.⁵⁶ When present immediately post-op, this suggests iatrogenic injury to branches of the median nerve
2. persistent symptoms (primary failure or failure to improve) defined as symptoms which remain unchanged compared to preop. Etiologies include: incorrect initial diagnosis, incomplete release of the transverse carpal ligament and severe (i.e. irreversible) CTS at initial diagnosis.¹⁴
3. or recurrent symptoms: requires symptom free interval before return of symptoms (there is no standardization of either the level of recurrent symptoms or length of interval,⁴⁴ although 6 months has been used in some studies⁵⁶). Etiologies include: circumferential fibrosis around the median nerve, soft tissue adhesions, synovial proliferation, tenosynovitis, ganglions, amyloid deposits and subtle palmar subluxation.^{14,56}

□ Electrodiagnostic (EDX) studies. Following CTR, distal motor latency improves after 3 months and 6 months and may continue to improve for up to 2 years.^{50,57,58} Electrophysiologic abnormalities may improve but may not return to normal range after CTR even with clinical improvement.^{44,51} Electrophysiologic studies are most helpful when it is possible to compare to pre-op studies.^{14,17,56,59,60} There are no guidelines or standard recommendations for when to obtain postoperative studies for surgical failures. It is reasonable to obtain repeat studies at 3 to 6 months following carpal tunnel release for persistent symptoms and at onset for new or recurrent symptoms. If the repeat nerve conduction studies are worse or if the EMG needle exam has findings of denervation (fibrillation potentials and positive sharp waves) not previously present, then repeat surgery is indicated.^{17,56} If the preoperative study is not available, repeating the EDX study with comparison at 2 points in time to evaluate for improvement or worsening is advised. Prolonged latencies alone are not an indication for reoperation.⁶¹

30.5 Ulnar nerve entrapment

30.5.1 General information

Ulnar nerve has components of C7, C8 and T1 nerve roots. Even though this is the second most prevalent entrapment neuropathy after CTS, it is still relatively uncommon. Potential sites of compression:

1. above elbow: possibly by the arcade of Struthers
2. at the elbow: retroepicondylar groove (“ulnar groove”): between the medial epicondyle and the olecranon process. Compression by fascia or by dynamic compression or repetitive trauma

3. cubital: just distal to the ulnar groove, under the aponeurosis between the heads of the flexor carpi ulnaris (FCU)
4. at the point of exit from the FCU
5. wrist: Guyon's canal

Etiologies: structural, mechanical or idiopathic.

Motor findings include:

1. wasting of the interossei may occur, and is most evident in the first dorsal interosseous (in the thumb web space)
2. Wartenberg's sign: one of the earliest findings of ulnar nerve entrapment (abducted little finger due to weakness of the third palmar interosseous muscle—patient may complain that the little finger doesn't make it in when they reach into their pocket)
3. Froment's prehensile thumb sign: grasping a sheet of paper between thumb and the extended index finger results in extension of the proximal phalanx of the thumb and flexion of the distal phalanx as a result of substituting flexor pollicis longus (which is spared since it is innervated by anterior interosseous nerve) for the weak adductor pollicis^{62 (p 18)}
4. claw deformity of the hand (main en griffe): in severe ulnar nerve injuries on attempted finger extension (some have called this "benediction hand", which differs from that with the same name in median nerve injury where the named sign occurs on trying to make a fist. Fingers 4 and 5 and to a lesser extent 3 are hyperextended at the MCP joints (extensor digitorum is unopposed by interossei and "ulnar" lumbricals III & IV) and flexed at the interphalangeal joints (due to pull of long flexor muscles). NB: C8 radiculopathy can also cause benediction sign⁶³)

Sensory findings: Disturbance of sensation involving:

1. The little finger and ulnar half of the ring finger
2. Sensory loss over the ulnar side of the dorsum of the hand. This will be spared in ulnar nerve entrapment at the wrist (dorsal ulnar cutaneous nerve branches proximal to the wrist)

30.5.2 Injury above elbow

May occur with injury to the medial cord of the brachial plexus.

In the upper arm, the ulnar nerve descends anterior to the medial head of triceps; in 70% of people it passes under arcade of Struthers – distinct from Struther's ligament (p.518) – a flat, thin, aponeurotic band. This is not normally a point of entrapment, but may cause kinking after ulnar nerve transposition if not adequately divided.^{64 (p 1781)}

30.5.3 Ulnar nerve entrapment at elbow (UNE)

General information

Entrapment at or just distal to the elbow produces the cubital tunnel syndrome. (Technically, the cubital tunnel is formed by the fibrous arch between the two heads of the FCU,^{65 (p 877)} the proximal entrance to which is just distal to the retrocondylar groove. However, common vernacular usually includes entrapment within the groove itself as being "cubital tunnel syndrome.")

Can also present as so-called tardy ulnar palsy because of delayed onset following bony injury at the elbow, initial case reports occurred ≥ 12 years later, with the majority commencing >10 years following the original injury. The elbow is the most vulnerable point of ulnar nerve: here the nerve is superficial, fixed, and crosses a joint. Most cases are idiopathic, although there may be a history of elbow fracture (especially lateral condyle of the humerus, with associated cubitus valgus deformity), dislocation, arthritis, or repeated minor trauma. The aponeurotic arch extending over the ulnar groove and attaching on the medial epicondyle may become thickened and can compress the nerve, especially with elbow flexion.^{65 (p 884)} The ulnar nerve may also be injured during anesthesia (p.548).⁶⁶ In contrast to CTS which is predominantly demyelinating, UNE has more axonal loss even when chronic.³¹

Presentation

Typically presents with discomfort (pain, numbness and/or tingling) in little finger and ulnar half of ring finger, elbow pain, and hand weakness. Early symptoms may be purely motor (see Froment's sign and claw deformity above) – unlike the median nerve where sensory involvement is almost always present. Symptoms may be exacerbated by the cold, and are often somewhat vague and may be described as a loss of finger coordination or clumsiness. Cramping and easy fatiguing of the ulnar

Table 30.3 Stewart classification system for severity of ulnar nerve injury(Stewart ⁶⁷ after Bartels ⁶⁸)	
Grade	Description
1 (mild)	sensory symptoms ± motor symptoms; ± sensory loss; no muscle atrophy or weakness
2 (moderate)	sensory symptoms with detectable sensory loss. Mild atrophy; 4 or 4 + muscle strength
3 (severe)	usually constant sensory symptoms with detectable sensory loss. Moderate to marked atrophy; 4- or less muscle strength.

innervated muscles of the hand may occur. Pain may not be a significant feature, but if present tends to be aching in nature along the ulnar aspect of the elbow or forearm. Atrophy of interossei is common by the time of presentation.

The ulnar nerve is usually tender and may be palpably enlarged in the ulnar groove. Tinel’s sign may be positive over the elbow, but this is not very specific.

Grading: the classification system proposed by Stewart⁶⁷ is shown in □ Table 30.3.

Evaluation

Electrodiagnostic studies

The literature review from AANEM Practice Parameters for EDX studies in ulnar neuropathy at the elbow (UNE) report sensitivities to range from 37%to 86%with specificities of 95%^{28,29,30}

EDX interpretation: Reports should include localization and may comment on whether the lesion is predominantly demyelinating or axonal. Reports may use grading classification.⁶⁹

The following suggest a focal lesion involving the ulnar nerve at the elbow. Multiple internally consistent abnormalities are more convincing than isolated abnormalities. These are listed in order of strength of evidence.

Sensory abnormalities of the distal sensory or mixed nerve action potential (NAP), especially loss of amplitude, are not localizing for ulnar neuropathy (in contradistinction to median nerve/carpal tunnel), the motor component of the exam for the ulnar nerve is more useful for localization of the site of entrapment.

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Practice guideline: Electrodiagnostic criteria for ulnar neuropathy at the elbow

- Not all criteria need to be present and an EMG needle exam is not required^{29,70}:
1. absolute motor nerve conduction velocity (NCV) < 50 m/sec from below elbow (BE) to above elbow (AE)
 2. drop of NCV > 10 m/sec comparing BE to wrist segment to the AE to BE segment
 3. amplitude of compound motor action potential (CMAP) normally decreases with distance, but a drop > 20% from BE to AE is abnormal (in the absence of anomalous innervation, e.g. Martin-Gruber anastomosis (p. 514))
 4. if ulnar motor studies with stimulation at the wrist, above and below the elbow recording from ADQ are inconclusive, the following may be of benefit:
 - NCS recorded from the FDI (first dorsal interosseus muscle, innervated by ulnar nerve)
 - inching study recording latencies and amplitudes at cm increments.
 - with severe UNE and wallerian degeneration -comparison of AE to BE with axilla to AE
 - needle EMG should always include FDI and ulnar innervated forearm muscles (flexor digitorum profundus (FDP) to ring or little finger and/or abductor digiti quinti/minimi (ADQ)). If abnormal then extend to include non-ulnar innervated C8, medial cord, lower trunk muscles and cervical paraspinals to exclude brachial plexopathy/cervical radiculopathy.

The 2 most important parameters that predict a good outcome are preserved compound muscle action potential (CMAP) amplitudes in ulnar hand muscles and conduction block (CB) with slowed conduction velocity across the elbow which is consistent with demyelination and has a better prognosis.^{26,71} Poor prognosis correlates with small or absent CMAP and no CB consistent with axonal loss.²⁶

Diagnostic ultrasound

Localizing ulnar nerve lesions with electrodiagnostic studies can be difficult. There has been a recent renewed interest in diagnostic ultrasound using high frequency (18 MHz) probes to help with localization, and also for identification of pathology, including nerve swelling, transection,⁷² and neuroma, that exceeds MRI in some aspects and at a lower cost and with faster acquisition time.

Management of ulnar neuropathy at the elbow

There are no Clinical Practice Guidelines for treatment of UNE endorsed by AANEM, AAOS, CNS, AANS, AAPM&R, or Am Society of Plastic Surgeons. A fundamental difficulty in treating UNE are the multiple etiologies and locations so that the natural history and responses to treatment vary widely. The primary management decision is conservative versus surgical treatment.

A Cochrane Database Review concluded the available evidence is not sufficient to identify the best treatment for UNE on the basis of clinical, neurophysiologic and imaging characteristics.^{73,74}

Nonsurgical treatment (see below) may be considered for the patient with intermittent symptoms, no atrophy and mild EDX findings. Surgical intervention has been advised for patients who fail conservative management, although the best nonsurgical management and duration of treatment are not well defined.⁷⁴

Suggested protocol:

- mild or moderate UNE (grade 1 and 2, table 30.3): treat conservatively as studies report improvement or complete recovery in 30-90%^{31,67} Follow these patients every 2 months to detect deterioration. If worsening occurs image with CT or MRI. Surgical exploration is indicated regardless of imaging results
- severe (grade 3) UNE: initiate conservative treatment, obtain imaging and f/u in 1 month. If there is worsening or if structural abnormality found or both then surgical intervention. If stable or improving and imaging normal then follow at monthly intervals. Surgical intervention for worsening.⁶⁷

There is a probable increased prevalence of UNE in diabetics. The UNE is often more severe with predominant axonal injury and these patients do not respond well to surgery.³⁵

Non-surgical treatment

Patient education on positions to avoid (prolonged elbow bending to $\geq 90^\circ$ flexion). Avoid trauma to the elbow, including resting it on firm surfaces (tables, rigid armrests in motor vehicles...), an elbow pad may help. Results are often better when definite traumatic etiology can be identified and eliminated.

Surgical treatment

General approach

Most operations utilize a “lazy omega” skin incision centered over the medial epicondyle, extending at least ≈ 6 cm proximal and distal to elbow with the central “hump” directed anteriorly. The ulnar nerve is most constant and therefore most easily found immediately at the entrance to the ulnar groove. It may then be followed proximally and distally. Nerve branches that should be preserved include: posterior branches of the medial antebrachial cutaneous nerve (or else numbness or dysesthesias along medial forearm may occur) and branches to the flexor carpi ulnaris (which may branch early). Small articular branches at or proximal to the elbow joint can be preserved with simple decompression but may need to be sacrificed in transposition if they cannot be dissected far enough along the ulnar nerve. Internal neurolysis should be avoided as it may promote intraneural fibrosis.

The choice of one of the options below will determine subsequent steps.

Surgical options primarily consist of:

1. simple nerve decompression without transposition⁷⁵ (see below). Includes all of the following:
 - a) at the elbow: division of the cubital tunnel retinaculum
 - b) distal to the elbow: dividing the aponeurosis connecting the two heads of the flexor carpi ulnaris, some advocate resuturing the aponeurosis underneath the nerve
 - c) proximal to the elbow: dividing the medial intermuscular septum (between distal biceps and triceps muscles) and the arcade of Struthers (if present)
 - d) preservation of the branch to the flexor carpi ulnaris and the dorsal cutaneous branch to the hand (arises 5 cm proximal to wrist)

- 2. nerve decompression and transposition (extent of surgery differs because degree of entrapment varies; all forms of transposition require fashioning a sling to retain the nerve in its new location). Transposition may be to:
 - a) subcutaneous tissue: this leaves the nerve fairly superficial and vulnerable to further trauma
 - b) within the flexor carpi ulnaris muscle (intramuscular transposition): some contend this actually worsens the condition due to intramuscular fibrosis
 - c) a submuscular position: below
- 3. medial epicondylectomy. Usually combined with decompression. Normally reserved for patients with a bony deformity
- 4. sometimes excision of neuroma and possibly jump graft may be required

Submuscular transposition

Placement under pronator teres, within a groove fashioned in the flexor carpi ulnaris (FCU). Usually requires general anesthesia (endotracheal or laryngeal mask airway).

Some key concepts^{76 (p 247, 260–5),77}:

- 1. the skin incision must extend at least ≈ 8 cm distal and proximal to the medial epicondyle to mobilize the nerve (spare the medial antecubital cutaneous nerve in the subcutaneous fatty tissue just distal to the elbow)
- 2. the nerve is mobilized, sparing branches to flexor carpi ulnaris (FCU) and the ulnar flexor profundus branch(es) (usually arise 2–4 cm distal to olecranon)
- 3. the medial intermuscular septum (between distal biceps and triceps muscles) must be cut in the distal arm to prevent the nerve from being kinked over it
- 4. the pronator teres muscle must be sectioned completely through just distal to the medial epicondyle
 - a) start by undermining the muscle just distal to the medial epicondyle
 - b) a mosquito hemostat may be passed under the muscle to assist
 - c) the muscle is cut sharply, cautery is used to treat bleeding areas
- 5. a trough is cut in the volar aspect of the FCU to accommodate the nerve
- 6. after the pronator teres is reattached over the nerve, make sure that the nerve can slide back-and-forth easily under the muscle
- 7. test the elbow through a range of motion after the transposition to look for snapping of the medial portion of the triceps over the medial epicondyle⁷⁸

Transposition vs. decompression

Σ

For most cases, simple decompression is recommended over transposition. Possible exceptions include: bony deformity, nerve subluxation

Randomized studies have shown similar success but lower complication rate with simple decompression vs. transposition.^{73,79,80} Advantages of simple decompression include^{68,81}: shorter operation that can be done more easily under local anesthesia, avoidance of nerve kinking and muscular fibrosis around the transposed nerve, reduced risk of wound infection⁷³ and scar formation,⁶⁷ and preservation of cutaneous branches, ulnar branches, and nourishing blood vessels (vasa nervorum⁶⁷) that are sometimes sacrificed with transposition, rendering portions of the nerve ischemic.

Arguments against simple decompression: Continued dynamic compression with elbow flexion, possible nerve subluxation (if present pre-op, simple decompression may make this worse; to avoid nerve subluxation and loss of vascular supply with simple decompression, avoid a 360° freeing of the nerve) and incomplete release of pressure points.

Results with surgery

Not as good as with CTS, possibly due in part to the fact that patients tend to present much later. Overall, a good to excellent result is obtained in 60%, fair result in 25% and a poor result (no improvement or worsening) in 15%^{82 (p 2530)} These results may be worse in patients with symptoms

present >1 year, with only 30% of these symptomatically improved in one series.⁷⁵ Lower success rate is also seen in older patients and those with certain medical conditions (diabetes, alcoholism...). Pain and sensory changes respond better than muscle weakness and atrophy.

30.5.4 Entrapment in the forearm

Very rare. Just distal to the elbow, the ulnar nerve passes from the retroepicondylar groove to pass under the fascial band connecting the two heads of the flexor carpi ulnaris (FCU), superficial to the flexor superficialis and pronator teres. Findings with entrapment in the forearm are similar to tardy ulnar nerve palsy (see above).

Surgical treatment consists of steps outlined for nerve distal to the elbow in ulnar nerve decompression (see above). A technique for locating the course of the ulnar nerve distal to the elbow: the surgeon takes the little finger of his/her own hand (using the hand contralateral to the patient's side that is being decompressed) and places the proximal phalanx in the ulnar groove aiming it toward the ulnar side of the wrist.⁷⁶ (p 262)

30.5.5 Entrapment in the wrist or hand

At the wrist, the terminal ulnar nerve enters Guyon's canal, the roof of which is the palmar fascia and palmaris brevis, the floor is the flexor retinaculum of the palm and the pisohamate ligament.

□ Guyon's canal. Is superficial to the transverse carpal ligament (which overlies the carpal tunnel and compresses the median nerve in carpal tunnel syndrome).

The canal contains no tendons, only the ulnar nerve and artery. At the middle of the canal the nerve divides into a deep and superficial branch. The superficial branch is mostly sensory (except for the branch to palmaris brevis), and supplies hypothenar eminence and ulnar half of ring finger. The deep (muscular) branch innervates hypothenar muscles, lumbricals 3 & 4, and all interossei. Occasionally the abductor digiti minimi branch arises from the main trunk or superficial branch.

Shea and McClain⁸³ divided lesions of the ulnar nerve in Guyon's canal into 3 types shown in □ Table 30.4. Injury to the distal motor branch can also occur in the palm and produces findings similar to a Type II injury.

Injury is most often due to a ganglion of the wrist,⁸⁴ but also may be due to trauma (use of pneumatic drill, pliers, repetitively slamming a stapler, leaning on palm while riding bicycle). Symptoms are similar to those of ulnar nerve involvement at the elbow, except there will never be sensory loss in the dorsum of the hand in the ulnar nerve territory because the dorsal cutaneous branch leaves the nerve in the forearm 5–8 cm proximal to the wrist (sparing of flexor carpi ulnaris and flexor digitorum profundus III & IV is not helpful in localizing because these are so rarely involved even in proximal lesions). Electrodiagnostics are usually helpful in localizing the site of the lesion. Pain, when present, may be exacerbated by tapping over pisiform (Tinel's sign). It may also radiate up the forearm.

Surgical decompression may be indicated in refractory cases. To locate: find the ulnar artery, and the nerve is on the ulnar side of the artery. Controversial whether simple decompression vs. subcutaneous transposition is best, the outcome is similar but there may be more complications in the transposition group^{79,80} but studies are small.

Table 30.4 Types of ulnar nerve lesions in Guyon's canal			
Type	Location of compression	Weakness	Sensory deficit
Type I	just proximal to or within Guyon's canal	all intrinsic hand muscles innervated by ulnar n.	palmar ulnar distribution ^a
Type II	along deep branch	muscles innervated by deep branch ^b	none
Type III	distal end of Guyon's canal	none	palmar ulnar distribution ^a
^a palmar ulnar distribution: the hypothenar eminence and ulnar half of ring finger, both on the palmar surface only (the dorsum is innervated by the dorsal cutaneous nerve)			
^b depending on the location, may spare hypothenar muscles			

30.6 Radial nerve injuries

See reference.⁸⁵ (p 1443–45)

30.6.1 Applied anatomy

The radial nerve arises from the posterior divisions of the 3 trunks of the brachial plexus. Receives contributions from C5 to C8. The nerve winds laterally along the spiral groove of the humerus where it is vulnerable to compression or injury from fracture.

Distinguish radial nerve injury from injury of posterior cord of brachial plexus by sparing of deltoid (axillary nerve) and latissimus dorsi (thoracodorsal nerve).

30.6.2 Axillary compression

Etiologies: crutch misuse; poor arm position during (drunken) sleep.

30.6.3 Mid-upper arm compression

Etiologies

1. “Saturday night palsy”: improper positioning of arm in sleep (especially when drunk and therefore less likely to self-reposition in response to the accompanying discomfort, e.g. due to a bed-mate’s head resting on the arm)
2. from positioning under general anesthesia
3. from callus due to old humeral fracture

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Clinical: weakness of wrist extensors (wrist drop) and finger extensors. □ Key: triceps is normal because takeoff of nerve to triceps is proximal to spiral groove. Involvement of distal nerve is variable, may include thumb extensor palsy and paresthesias in radial nerve distribution. Differential diagnosis: isolated wrist and finger extensor weakness can also occur in lead poisoning (usually bilateral, more common in adults).

Forearm compression

General information

The radial nerve enters the anterior compartment of the arm just above the elbow. It gives off branches to brachialis, brachioradialis, and extensor carpi radialis (ECR) longus before dividing into the posterior interosseous nerve and the superficial radial nerve. The posterior interosseous nerve dives into the supinator muscle through a fibrous band known as the arcade of Frohse.

Posterior interosseous neuropathy

Posterior interosseous neuropathy (often referred to by its acronym “PIN”) may result from: lipomas, ganglia, fibromas, rheumatoid arthritis changes at elbow, entrapment at the arcade of Frohse (rare), and occasionally from strenuous use of the arm.

Treatment: cases that do not respond to 4–8 weeks of expectant management should be explored, and any constrictions lysed (including arcade of Frohse).

Radial tunnel syndrome

AKA supinator syndrome. Controversial. The “radial tunnel” extends from just above the elbow to just distal to it, and is composed of different structures (muscles, fibrous bands...) depending on the level.⁸⁶ It contains the radial nerve and its two main branches (posterior interosseous and superficial radial nerves). Repeated forceful supination or pronation or inflammation of supinator muscle attachments (as in tennis elbow) may traumatize the nerve (sometimes by ECR brevis). Characteristic finding: pain in the region of the common extensor origin at the lateral epicondyle on resisted extension of the middle finger which tightens the ECR brevis. May be mistakenly diagnosed as resistant “tennis elbow” (lateral epicondylitis must be excluded). There may also be paresthesias in the distribution of the superficial radial nerve and local tenderness along the radial nerve anterior to the radial head. Even though the site of entrapment is similar to PIN, unlike PIN, there is usually no muscle weakness. Surgery: rarely required, consists of nerve decompression.⁸⁶

Injury in the hand

The distal cutaneous branches of the superficial radial nerve cross the extensor pollicis longus tendon, and can often be palpated at this point with the thumb in extension. Injury to the medial branch of this nerve occurs commonly e.g. with handcuff injuries, and causes a small area of sensory loss in the dorsal web-space of the thumb.

30.7 Axillary nerve injuries

Isolated neuropathy of the axillary nerve may occur in the following situations⁸⁷:

1. shoulder dislocation: the nerve is tethered to the joint capsule⁸⁸
2. sleeping in the prone position with the arms abducted above the head
3. compression from a thoracic harness
4. injection injury in the high posterior aspect of the shoulder
5. entrapment of the nerve in the quadrilateral space (bounded by the teres major and minor muscles, long head of triceps, and neck of humerus) which contains the axillary nerve and the posterior humeral circumflex artery. Arteriogram may show loss of filling of the artery with the arm abducted and externally rotated

30.8 Suprascapular nerve

□ General information. The suprascapular nerve is a mixed peripheral nerve arising from the superior trunk of the brachial plexus, with contributions from C5 & C6. There is often a history of shoulder trauma or frozen shoulder. Entrapment results in weakness & atrophy of infra- and supraspinatus (IS & SS) and deep, poorly localized (referred) shoulder pain (the sensory part of the nerve innervates the posterior joint capsule but has no cutaneous representation).

□ Etiologies

1. nerve entrapment within the suprascapular notch beneath the transverse scapular (suprascapular) ligament (TSL)⁸⁹
2. repetitive shoulder trauma: may be bilateral when the injury is from activities such as weight-lifting
3. ganglion or tumor⁹⁰ (MRI is the test of choice for imaging these)
4. paralabral cyst from labral tear (the tendon of the long head of the biceps attaches to the superior glenoid labrum; test of choice for labral tears is MR arthrography)

□ Differential diagnosis⁸⁹

1. pathology in or around shoulder joint
 - a) rotator cuff injuries (distinction may be very difficult)
 - b) adhesive capsulitis
 - c) bicipital tenosynovitis
 - d) arthritis
2. cases of Parsonage-Turner syndrome limited to the suprascapular nerve; see Neuralgic amyotrophy (p.543)
3. cervical radiculopathy ($\approx$ C5)
4. upper brachial plexus lesion

NB: cervical radiculopathy and upper brachial plexus lesion will also produce rhomboid and deltoid weakness and, usually, cutaneous sensory loss.

Diagnosis requires temporary relief with nerve block, and EMG abnormalities of SS & IS (in rotator cuff tears, fibrillation potentials will be absent). Transient pain relief with a suprascapular nerve block helps verify the diagnosis.⁹¹

□ Treatment. In cases where a mass is not the underlying cause, initial treatment consists of resting the affected UE, PT (including gentle conditioning), NSAIDs, topical capsaicin cream, and sometimes corticosteroid injection.

Surgical treatment is indicated for documented cases that fail to improve with conservative treatment (PT, NSAIDs, steroid/local anesthetic injection...). Position: lateral decubitus. Incision: 2 cm above and parallel to the scapular spine (atrophy of SS facilitates this). Only the trapezius needs to be split along its fibers (caution re spinal accessory nerve). To locate suprascapular notch, follow omohyoid to where it attaches to scapula and palpate just lateral to this. The suprascapular artery

and vein pass over the TSL and should be preserved. Elevate the TSL with a dull nerve hook and divide it (exposure of the nerve and/or resection of the bony notch are not necessary).

30.9 Meralgia paresthetica

30.9.1 General information

AKA Originally known as the Bernhardt-Roth syndrome, and sometimes called “swashbuckler’s disease”, meralgia paresthetica (MP) (Greek: meros – thigh, algos – pain) is a condition often caused by entrapment of the lateral femoral cutaneous nerve (LFCN) of the thigh, (a purely sensory branch with contributions from L2 and L3 nerve roots, see □ Fig. 1.14 for distribution) where it enters the thigh through the opening between the inguinal ligament and its attachment to the anterior superior iliac spine (ASIS). Anatomic variation is common, and the nerve may actually pass through the ligament, and as many as four branches may be found. May also be an initial manifestation of diabetes (diabetic neuropathy).

30.9.2 Signs and symptoms

Burning dysesthesias in the lateral aspect of the upper thigh, occasionally just above the knee, usually with increased sensitivity to clothing (hyperpathia). There may be decreased sensation in this distribution. Spontaneous rubbing or massaging the area in order to obtain relief is very characteristic.⁹² MP may be bilateral in up to 20% of cases. Sitting or lying prone usually ameliorates the symptoms.

There may be point tenderness at the site of entrapment (where pressure may reproduce the pain), which is often located where the nerve exits the pelvis medial to the ASIS. Hip extension may also cause pain.

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30.9.3 Occurrence

Usually seen in obese patients, may be exacerbated by wearing tight belts or girdles, and by prolonged standing or walking. Recently found in long distance runners. Higher incidence in diabetics. May also occur post-op in slender patients positioned prone, tends to be bilateral (p.549) .

Possible etiologies are too numerous to list, more common ones include: tight clothing or belts, surgical scars post-abdominal surgery, cardiac catheterization (p.549), pregnancy, iliac crest bone graft harvesting, ascites, obesity, metabolic neuropathies, and abdominal or pelvic mass.

30.9.4 Differential diagnosis

1. femoral neuropathy: sensory changes tend to be more anteromedial than MP
2. L2 or L3 radiculopathy: look for motor weakness (thigh flexion or knee extension)
3. nerve compression by abdominal or pelvic tumor (suspected if concomitant GI or GU symptoms)

The condition can usually be diagnosed on clinical grounds. When it is felt to be necessary, confirmatory tests may help (but frequently are disappointing), including:

1. EMG: may be difficult, the electromyographer cannot always find the nerve)
2. MRI or CT/myelography: when disc disease is suspected
3. pelvic imaging (MRI or CT)
4. somatosensory evoked potentials
5. response to local anesthetic injections
6. recent promise of diagnostic ultrasound using high frequency (18 MHz) probes

30.9.5 Treatment

Nonsurgical management

Tends to regress spontaneously, but recurrence is common. Nonsurgical measures achieve relief in ≈ 91% of cases and should be tried prior to considering surgery⁹³:

1. remove offending articles (constricting belts, braces, casts, tight garments...)
2. in obese patients: weight loss and exercises to strengthen the abdominal muscles is usually effective, but is rarely achieved by the patient
3. elimination of activities involving hip extension

4. application of ice to the area of presumed constriction $\times$ 30 minutes TID
5. NSAID of choice $\times$ 7–10 days
6. capsaicin ointment applied TID (p.496)
7. lidoderm patches (p.477) in areas of hyperesthesia may help⁹⁴
8. centrally acting pain medications (e.g. gabapentin, carbamazepine...) are rarely effective
9. if the above measures fail, injection of 5–10 ml of local anesthetic (with or without steroids) at the point of tenderness, or medial to the ASIS may provide temporary or sometimes long lasting relief, and confirms the diagnosis

Surgical treatment

Options include:

1. surgical decompression (neurolysis) of the nerve: higher failure and recurrence rate than neurectomy
2. decompression and transposition
3. selective L2 nerve stimulation
4. division of the nerve (neurectomy) may be more effective, but risks denervation pain, and leaves an anesthetic area (usually a minor nuisance). May be best reserved for treatment failures

Technique

See references.^{93,95}

The operation is best performed under general anesthesia. A 4–6 cm oblique incision is centered 2 cm distal to the point of tenderness. Since the course of the nerve is variable, the operation is exploratory in nature, and generous exposure is required. If the nerve can't be located, it is usually because the exposure is too superficial. If the nerve still cannot be found, a small abdominal muscle incision can be made and the nerve may be located in the retroperitoneal area. CAUTION: cases have occurred where the femoral nerve has erroneously been divided.

If neurectomy instead of neurolysis is elected, electrical stimulation should be performed prior to sectioning to rule out a motor component (which would disqualify the nerve as the LFCN). If the nerve is to be divided, it should be placed on stretch and then cut to allow the proximal end to retract back into the pelvis. Any segment of apparent pathology should be resected for microscopic analysis. Neurectomy results in anesthesia in the distribution of the LFCN that is rarely distressing and gradually reduces in size.

A supra-inguinal ligament approach has also been described.⁹⁵

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30.10 Obturator nerve entrapment

Controversial if this exists. The obturator nerve is composed of L2–4 roots. It courses along the pelvic wall to provide sensation to the inner thigh, and motor to the thigh adductors (gracilis and adductors longus, brevis, and magnus). It may be compressed by pelvic tumors, also from the pressure of the fetal head or forceps during parturition.

The result is numbness of the medial thigh and weak thigh adduction.

30.11 Femoral nerve entrapment

Composed of roots L2–4. Entrapment is a rare cause of femoral neuropathy. More commonly due to fracture or surgery. See Femoral neuropathy (p.546).

30.12 Common peroneal nerve palsy

30.12.1 General information and applied anatomy

The peroneal nerve is the most common nerve to develop acute compression palsy.

Functional anatomy: the sciatic nerve (L4–S3) consists of 2 separate nerves within a common sheath that separate at a variable location in the thigh (the peroneal division of the sciatic nerve is more vulnerable to injury than the tibial division); see □ Fig. 92.1 for diagram:

1. posterior tibial nerve, or, just tibial nerve (AKA medial popliteal nerve) which provides for foot inversion among other motor functions
2. common peroneal nerve (CPN), or, just peroneal nerve (AKA lateral popliteal nerve): high injuries may involve the lateral hamstring (short head of the biceps femoris) in addition to the following.

- The CPN passes behind the fibular head where it is superficial and fixed, making it vulnerable to pressure or trauma (e.g. from crossing the legs at the knee). Just distal to this, the CPN divides into:
- a) deep peroneal nerve (AKA anterior tibial nerve): primarily motor
 - motor: foot and toe extension (extensor hallicus longus (EHL), anterior tibialis (AT), extensor digitorum longus (EDL))
 - sensory: very small area between great toe and second toe
 - b) superficial peroneal nerve (AKA musculocutaneous nerve)
 - motor: foot eversion (peroneus longus and brevis)
 - sensory: lateral distal leg and dorsum of foot

30.12.2 Causes of common peroneal nerve injury

- The most frequent cause of serious peroneal nerve injury is knee injury ± fracture; see also causes of foot drop other than peroneal nerve palsy (p.1417).
- 1. entrapment as it crosses the fibular neck or as it penetrates the peroneus longus
 - 2. diabetes mellitus and other metabolic peripheral neuropathies
 - 3. inflammatory neuropathy: including Hansen’s disease (leprosy)
 - 4. traumatic: e.g. clipping injury in football players, stretch injury due to dislocating force applied to the knee, fibular fracture, injury during hip or knee replacement surgery
 - 5. penetrating injury
 - 6. masses in the area of the fibular head/proximal lower leg: popliteal fossa cysts (Baker cyst), anterior tibial artery aneurysm⁹⁶ (rare)
 - 7. pressure at fibular head: e.g. from crossing the legs at the knee, casts, obstetrical stirrups...
 - 8. traction injuries: severe inversion sprains of the ankle
 - 9. intraneural tumors: neurofibroma, schwannoma, neurogenic sarcoma, ganglion cysts
 - 10. vascular: venous thrombosis
 - 11. weight loss

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30.12.3 Findings in peroneal nerve palsy

General information

- 1. sensory changes (uncommon): involves lateral aspect of lower half of leg
- 2. muscle involvement: See □ Table 30.5

Common peroneal nerve palsy (most common) produces weak ankle dorsiflexion (foot drop) due to anterior tibialis palsy, weak foot eversion, and sensory impairment in areas innervated by deep and superficial peroneal nerve (lateral calf and dorsum of foot). There may be a Tinel’s sign with percussion over the nerve near the fibular neck. Occasionally, only the deep peroneal nerve is involved, resulting in foot drop with minimal sensory loss. Must differentiate from other causes of foot drop (p.1416).

Examination/clinical correlation

- See reference.⁷⁶ (p 293)
- buttock level injury: unless the injury is one that permits spontaneous regeneration, prognosis is poor for return of peroneal nerve function even with surgery

Table 30.5 Muscle involvement in peroneal nerve palsy			
Muscle	Nerve	Action	Involvement
EHL	deep peroneal	great toe dorsiflexion	Most commonly involved ↓ ↓ ↓ Least commonly involved (often spared)
anterior tibialis		ankle dorsiflexion	
EDL		toe extension	
peroneus longus & brevis	superficial peroneal	foot eversion	

- thigh level injury: also difficult to get improvement with surgical repair. Some peroneus function may occur at ≥ 6 mos, early contraction of AT may take ≥ 1 yr
- knee level injury: with successful regeneration, peroneus contraction may begin by 3–5 months. First signs: quivering of muscle lateral to the proximal fibula on attempted foot eversion, or tightening of tendon posterior and behind the lateral malleolus on attempted ankle dorsiflexion

30.12.4 Evaluation

EMG

EMG takes 2–4 weeks from the onset of symptoms to become positive. Stimulate above and below fibular head for prognostic information: if absent in both sites, the prognosis is poor (indicates retrograde degeneration has occurred). Wallerian degeneration takes ≈ 5 days to cause deterioration.

In addition to the expected findings of denervation – PSWs & fibs (p.242) – in the anterior tibialis, evaluate:

1. L5 innervated muscles outside the distribution of the common peroneal nerve:
 - a) posterior tibialis
 - b) flexor digitorum longus
2. L5 muscles whose nerve originates above the knee (these muscles are spared in cases of compression of the peroneal nerve at the fibular head due to the fact that the nerve takeoff is proximal to the popliteal fossa):
 - a) biceps femoris (short or long head)
 - b) tensor fascia lata
3. paraspinal muscles: signs of denervation solidifies the location of the lesion as nerve root; not helpful if negative

MRI

May demonstrate causes such as tumor or a ganglion cyst arising from the superior tibiofibular articulation.

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30.12.5 Treatment

General information

When treatment can eliminate a reversible cause, the outcome is usually good. Surgical exploration and decompression may be considered when there is no reversible cause or when improvement does not occur.

Nonsurgical management

Bracing: ankle-foot-orthosis (AFO) compensates for loss of ankle dorsiflexion which inserts unobtrusively into a shoe. If this is inadequate, or to stabilize the ankle, a spring-loaded kick-up foot brace built into a shoe may be used. The patient should be instructed in techniques to avoid contracture of the Achilles tendon (heel cord) which would impair ankle dorsiflexion if nerve function returns.

Surgical management

At the level of the popliteal fossa the skin incision is made just medial to the tendon of the short head of the biceps femoris (lateral hamstring) as the peroneal nerve is best located deep to or slightly medial to this tendon. The incision is carried distally slightly laterally along the surgical neck of the fibula. The biceps femoris is retracted laterally and the nerve is isolated and tagged with a Penrose drain. The sensory sural nerve branches off the peroneal nerve at variable sites ranging from the sciatic portion of the nerve (proximal to the flexor crease) or distal to this.

In cases of compression, the fascia from the lateral gastrocnemius and soleus overlying the nerve distal to the fibular head is lysed and the nerve is exposed in 360°. As the nerve crosses the fibular neck it divides into superficial and deep branches. The superficial branch travels directly distally to supply the peroneus longus and brevis (foot evertors). The deep branches curve anteriorly to the anterior tibialis, EHL, and toe extensors.

If a graft is needed, the contralateral sural nerve is usually used, which may be supplemented with the ipsilateral sural nerve if needed.

30.13 Tarsal tunnel

30.13.1 General information

Entrapment of (posterior) tibial nerve may occur in the tarsal tunnel, posterior and inferior to medial malleolus. The tunnel is covered by the flexor retinaculum (lacinate ligament) which extends downward from the medial malleolus to the tubercle of the calcaneus. There is often (but not necessarily) a history of old ankle dislocation or fracture. The nerve may be trapped at the retinacular ligament. This results in pain and paresthesias in the toes and sole of foot (often sparing the heel because the sensory branches often originate proximal to the tunnel), typically worse at night. May cause clawing of toes secondary to weakness of intrinsic foot muscles. Often caused by fracture or dislocation, also rheumatoid arthritis, rarely tumors.

30.13.2 Exam

Percussion of nerve at medial malleolus produces paresthesias that radiate distally (Tinel's sign). Maximal inversion and eversion of the foot tend to exacerbate. Dorsiflexion-eversion test: examiner maximally everts and dorsiflexes the ankle while dorsiflexing the toes at the MTP joints for 5–10 seconds. Positive test reproduces the pain.

30.13.3 Diagnosis

EMG and NCV studies may help.

30.13.4 Nonsurgical management

External ankle support to improve foot mechanics.

30.13.5 Surgical management

Surgical decompression is indicated for confirmed cases that fail to improve. A curvilinear incision is used, ≈ 1.5 cm posterior and inferior to the medial malleolus. The flexor retinaculum is divided, as are any septa underneath, and the distal branches should be followed until they dive into the muscle.

References

- [1] Neary D, Ochoa JL, Gilliatt RW. Subclinical Entrapment Neuropathy in Man. *J Neurol Sci.* 1975; 24:283–298
- [2] Graff-Radford SB, Jaeger B, Reeves JL. Myofascial Pain May Present Clinically as Occipital Neuralgia. *Neurosurgery.* 1986; 19:610–613
- [3] Hunter CR, Mayfield FH. Role of the Upper Cervical Roots in the Production of Pain in the Head. *Am J Surg.* 1949; 48:743–751
- [4] Poletti CE, Sweet WH. Entrapment of the C2 Root and Ganglion by the Atlanto-Epistropheic Ligament: Clinical Syndrome and Surgical Anatomy. *Neurosurgery.* 1990; 27:288–291
- [5] Anthony M. Headache and the greater occipital nerve. *Clin Neurol Neurosurg.* 1992; 94:297–301
- [6] Weiner RL, Reed KL. Peripheral neurostimulation for control of intractable occipital neuralgia. *Neuromodulation.* 1999; 2:217–221
- [7] Freund BJ, Schwartz M. Treatment of chronic cervical-associated headache with botulinum toxin A: a pilot study. *Headache.* 2000; 40:231–236
- [8] Weinberger LM. Cervico-Occipital Pain and Its Surgical Treatment. *Am J Surg.* 1978; 135:243–247
- [9] Bogduk N. Local Anesthetic Blocks of the Second Cervical Ganglion. A Technique with Application in Occipital Headache. *Cephalalgia.* 1981; 1:41–50
- [10] Bovim G, Fredriksen TA, Stolt-Nielsen A, Sjaastad O. Neurolysis of the greater occipital nerve in cervicogenic headache. A follow up study. *Headache.* 1992; 32:175–179
- [11] Katz JN, Simmons BP. Clinical practice. Carpal tunnel syndrome. *N Engl J Med.* 2002; 346:1807–1812
- [12] Laha RK, Lunsford LD, Dujovny M. Lacertus Fibrosus Compression of the Median Nerve. *J Neurosurg.* 1978; 48:838–841
- [13] Nakano KK, Lundergan C, Okihiro M. Anterior Interosseous Nerve Syndromes: Diagnostic Methods and Alternative Treatments. *Arch Neurol.* 1977; 34:477–480
- [14] Stewart JD. In: Median Nerve. Focal Peripheral Neuropathies . 4th ed. West Vancouver, Canada: JBJ Publishing; 2010:214–239
- [15] Yasargil MG, Antic J, Laciga R, et al. Microsurgical Pterional Approach to Aneurysms of the Basilar Bifurcation. *Surg Neurol.* 1976; 6
- [16] Atroshi I, Gummesson C, Johnsson R, Ornstein E, Ranstam J, Rosen I. Prevalence of carpal tunnel syndrome in a general population. *JAMA.* 1999; 282:153–158
- [17] Mosier BA, Hughes TB. Recurrent carpal tunnel syndrome. *Hand Clin.* 2013; 29:427–434
- [18] Jain NB, Higgins LD, Losina E, Collins J, Blazar PE, Katz JN. Epidemiology of musculoskeletal upper extremity ambulatory surgery in the United States. *BMC Musculoskelet Disord.* 2014; 15. DOI: 10.1186/1471-2474-15-4
- [19] Feldman RG, Goldman R, Keyserling WM. Classical Syndromes in Occupational Medicine: Peripheral Nerve Entrapment Syndromes and Ergonomic Factors. *Am J Ind Med.* 1983; 4:661–681
- [20] Padua L, Aprile I, Caliandro P, et al. Carpal tunnel syndrome in pregnancy: multiperspective follow-up of untreated cases. *Neurology.* 2002; 59:1643–1646

- [21] Phalen GS. The Carpal Tunnel Syndrome. *Clinical Evaluation of 598 Hands*. Clin Ortho Rel Res. 1972; 83
- [22] Sandzen SC. Carpal Tunnel Syndrome. *Am Fam Physician*. 1981; 24:190–204
- [23] Upton RM, McComas AJ. The Double Crush in Nerve Entrapment Syndromes. *Lancet*. 1973; 11:359–362
- [24] Wilbourn AJ, Gilliatt RW. Double-Crush Syndrome: A Critical Analysis. *Neurology*. 1997; 49:21–29
- [25] Rempel DM, Harrison RJ, Barnhart S. Work-Related Cumulative Trauma Disorders of the Upper Extremity. *JAMA*. 1992; 267:838–842
- [26] Robinson LR. How electrodiagnosis predicts clinical outcome of focal peripheral nerve lesions. *Muscle Nerve*. 2015; 52:321–333
- [27] Werner RA, Andary M. Electrodiagnostic evaluation of carpal tunnel syndrome. *Muscle Nerve*. 2011; 44:597–607
- [28] Jablecki CK, Andary MT, Floeter MK, Miller RG, Quartly CA, Vennix MJ, Wilson JR, American Association of Electrodiagnostic Medicine, American Academy of Neurology, American Academy of Physical Medicine, Rehabilitation. Practice parameter: Electrodiagnostic studies in carpal tunnel syndrome. Report of the American Association of Electrodiagnostic Medicine, American Academy of Neurology, and the American Academy of Physical Medicine and Rehabilitation. *Neurology*. 2002; 58:1589–1592
- [29] Campbell WW. Guidelines in electrodiagnostic medicine. Practice parameter for electrodiagnostic studies in ulnar neuropathy at the elbow. *Muscle Nerve Suppl*. 1999; 8:S171–S205
- [30] American Association of Electrodiagnostic Medicine. Chapter 9: Practice parameter for needle electromyographic evaluation of patients with suspected cervical radiculopathy: Summary statement. *Muscle Nerve*. 1999; 22:S209–S211
- [31] Bland JD. A neurophysiological grading scale for carpal tunnel syndrome. *Muscle Nerve*. 2000; 23:1280–1283
- [32] Robinson L, Kliot M. Stop using arbitrary grading schemes in carpal tunnel syndrome. *Muscle Nerve*. 2008; 37. DOI: 10.1002/mus.21012
- [33] Bland JD. Stop using arbitrary grading schemes in carpal tunnel syndrome. *Muscle Nerve*. 2008; 38:1527; author reply 1527–1527; author reply 1528
- [34] Keith MW, Masear V, Chung KC, Amadio PC, Andary M, Barth RW, Maupin K, Graham B, Watters WC, 3rd, Turkelson CM, Haralson RH, 3rd, Wies JL, McGowan R. American Academy of Orthopaedic Surgeons clinical practice guideline on the treatment of carpal tunnel syndrome. *J Bone Joint Surg Am*. 2010; 92:218–219
- [35] Stewart JD. Mononeuropathies in Diabetics. 2012
- [36] Thomsen NO, Cederlund R, Rosen I, Bjork J, Dahlin LB. Clinical outcomes of surgical release among diabetic patients with carpal tunnel syndrome: prospective follow-up with matched controls. *J Hand Surg Am*. 2009; 34:1177–1187
- [37] Burke DT, Burke MM, Stewart GW, Cambre A. Splinting for carpal tunnel syndrome: in search of the optimal angle. *Arch Phys Med Rehabil*. 1994; 75:1241–1244
- [38] Walker WC, Metzler M, Cifu DX, Swartz Z. Neutral wrist splinting in carpal tunnel syndrome: a comparison of night-only versus full-time wear instructions. *Arch Phys Med Rehabil*. 2000; 81:424–429
- [39] Linskey ME, Segal R. Median Nerve Injury from Local Steroid Injection in Carpal Tunnel Syndrome. *Neurosurgery*. 1990; 26:512–515
- [40] Pagnanelli DM, Barrer SJ. Bilateral Carpal Tunnel Release at One Operation: Report of 228 Patients. *Neurosurgery*. 1992; 31:1030–1034
- [41] Katz JN, Keller RB, Simmons BP, Rogers WD, Besette L, Fossel AH, Mooney NA. Maine Carpal Tunnel Study: outcomes of operative and nonoperative therapy for carpal tunnel syndrome in a community-based cohort. *J Hand Surg Am*. 1998; 23:697–710
- [42] Brown RA, Gelberman RH, Seiler JG, III, Abrahamson SO, Weiland AJ, Urbaniak JR, Schoenfeld DA, Furcolo D. Carpal tunnel release. A prospective, randomized assessment of open and endoscopic methods. *J Bone Joint Surg*. 1993; 75:1265–1275
- [43] Pagnanelli DM, Barrer SJ. Carpal Tunnel Syndrome: Surgical Treatment Using the Paine Retinaculotome. *J Neurosurg*. 1991; 75:77–81
- [44] Louie D, Earp B, Blazar P. Long-term outcomes of carpal tunnel release: a critical review of the literature. *Hand (N Y)*. 2012; 7:242–246
- [45] Atroshi I, Larsson GU, Ornstein E, Hofer M, Johnsson R, Ranstam J. Outcomes of endoscopic surgery compared with open surgery for carpal tunnel syndrome among employed patients: randomised controlled trial. *BMJ*. 2006; 332. DOI: 10.1136/bmj.38863.632789.1F
- [46] Atroshi I, Hofer M, Larsson GU, Ornstein E, Johnsson R, Ranstam J. Open compared with 2-portal endoscopic carpal tunnel release: a 5-year follow-up of a randomized controlled trial. *J Hand Surg Am*. 2009; 34:266–272
- [47] Louis DS, Greene TL, Noellert RC. Complications of Carpal Tunnel Surgery. *J Neurosurg*. 1985; 62:352–356
- [48] Louie DL, Earp BE, Collins JE, Losina E, Katz JN, Black EM, Simmons BP, Blazar PE. Outcomes of open carpal tunnel release at a minimum of ten years. *J Bone Joint Surg Am*. 2013; 95:1067–1073
- [49] Guyette TM, Wilgis EF. Timing of improvement after carpal tunnel release. *J Surg Orthop Adv*. 2004; 13:206–209
- [50] Zyluk A, Puchalski P. A comparison of the results of carpal tunnel release in patients in different age groups. *Neurol Neurochir Pol*. 2013; 47:241–246
- [51] Padua L, Lo Monaco M, Padua R, Tamburrelli F, Gregori B, Tonali P. Carpal tunnel syndrome: neurophysiological results of surgery based on preoperative electrodiagnostic testing. *J Hand Surg Br*. 1997; 22:599–601
- [52] Nancollas MP, Peimer CA, Wheeler DR, Sherwin FS. Long-term results of carpal tunnel release. *J Hand Surg Br*. 1995; 20:470–474
- [53] Katz JN, Fossel KK, Simmons BP, Swartz RA, Fossel AH, Koris MJ. Symptoms, functional status, and neuromuscular impairment following carpal tunnel release. *J Hand Surg Am*. 1995; 20:549–555
- [54] Pensy RA, Burke FD, Bradley MJ, Dubin NH, Wilgis EF. A 6-year outcome of patients who cancelled carpal tunnel surgery. *J Hand Surg Eur Vol*. 2011; 36:642–647
- [55] Thomsen NO, Rosen I, Dahlin LB. Neurophysiologic recovery after carpal tunnel release in diabetic patients. *Clin Neurophysiol*. 2010; 121:1569–1573
- [56] Jones NF, Ahn HC, Eo S. Revision surgery for persistent and recurrent carpal tunnel syndrome and for failed carpal tunnel release. *Plast Reconstr Surg*. 2012; 129:683–692
- [57] Ginanneschi F, Milani P, Reale F, Rossi A. Short-term electrophysiological conduction change in median nerve fibres after carpal tunnel release. *Clin Neurol Neurosurg*. 2008; 110:1025–1030
- [58] Shurr DG, Blair WF, Bassett G. Electromyographic changes after carpal tunnel release. *J Hand Surg Am*. 1986; 11:876–880
- [59] Schrijver HM, Gerritsen AA, Strijers RL, Uitdehaag BM, Scholten RJ, de Vet HC, Bouter LM. Correlating nerve conduction studies and clinical outcome measures on carpal tunnel syndrome: lessons from a randomized controlled trial. *J Clin Neurophysiol*. 2005; 22:216–221
- [60] Rotman MB, Enkvetchakul BV, Megerian JT, Gozani SN. Time course and predictors of median nerve conduction after carpal tunnel release. *J Hand Surg Am*. 2004; 29:367–372
- [61] Stolp KA. Upper extremity Focal Neuropathies. 2013
- [62] Brazis PW, Masdeu JC, Biller J. Localization in Clinical Neurology. 2nd ed. Boston: Little Brown and Company; 1990
- [63] Harrop JS, Hanna A, Silva MT, Sharan A, Benzel EC, Stewart TJ. Neurological manifestations of cervical spondylosis: an overview of signs, symptoms, and pathophysiology. *Neurosurgery*. 2007; 60:S1–14–20

- [64] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
- [65] Dumitru D. *Electrodiagnostic Medicine*. Philadelphia: Hanley and Belfus; 1995
- [66] Bonney G. Iatrogenic Injuries of Nerves. *J Bone Joint Surg*. 1986; 68B:9–13
- [67] Stewart JD. In: *Ulnar Nerve. Focal Peripheral Neuropathies*. 4th ed. West Vancouver, Canada: JBJ Publishing; 2010:258–313
- [68] Bartels RHMA, Menovsky T, Van Overbeeke JJ, Verhagen WIM. Surgical Management of Ulnar Nerve Compression at the Elbow: An Analysis of the Literature. *J Neurosurg*. 1988; 89:722–727
- [69] Padua L, Aprile I, Mazza O, Padua R, Pietracci E, Caliendo P, Pauri F, D'Amico P, Tonali P. Neurophysiological classification of ulnar entrapment across the elbow. *Neurol Sci*. 2001; 22:11–16
- [70] Practice parameter: electrodiagnostic studies in ulnar neuropathy at the elbow. American Association of Electrodiagnostic Medicine, American Academy of Neurology, and American Academy of Physical Medicine and Rehabilitation. *Neurology*. 1999; 52:688–690
- [71] Beekman R, Wokke JH, Schoemaker MC, Lee ML, Visser LH. Ulnar neuropathy at the elbow: follow-up and prognostic factors determining outcome. *Neurology*. 2004; 63:1675–1680
- [72] Cartwright MS, Chloros GD, Walker FO, Wiesler ER, Campbell WW. Diagnostic ultrasound for nerve transection. *Muscle Nerve*. 2007; 35:796–799
- [73] Caliendo P, La Torre G, Padua R, Giannini F, Padua L. Treatment for ulnar neuropathy at the elbow. *Cochrane Database Syst Rev*. 2012; 7. DOI: 10.1002/14651858.CD006839.pub3
- [74] Elhassan B, Steinmann SP. Entrapment neuropathy of the ulnar nerve. *J Am Acad Orthop Surg*. 2007; 15:672–681
- [75] Le Roux PD, Ensign TD, Burchiel KJ. Surgical Decompression Without Transposition for Ulnar Neuropathy: Factors Determining Outcome. *Neurosurgery*. 1990; 27:709–714
- [76] Kline DG, Hudson AR. *Nerve Injuries: Operative Results for Major Nerve Injuries, Entrapments, and Tumors*. Philadelphia: W. B. Saunders; 1995
- [77] Janjua RM, Fernandez J, Tender G, Kline DG. Submuscular transposition of the ulnar nerve for the treatment of cubital tunnel syndrome. *Neurosurgery*. 2008; 63:321–4; discussion 324–5
- [78] Spinner RJ, O'Driscoll SW, Jupiter JB, Goldner RD. Unrecognized dislocation of the medial portion of the triceps: another cause of failed ulnar nerve transposition. *J Neurosurg*. 2000; 92:52–57
- [79] Bartels RH, Verhagen WI, van der Wilt GJ, Meulstee J, van Rossum LG, Grotenhuis JA. Prospective randomized controlled study comparing simple decompression versus anterior subcutaneous transposition for idiopathic neuropathy of the ulnar nerve at the elbow: Part 1. *Neurosurgery*. 2005; 56:522–30; discussion 522–30
- [80] Biggs M, Curtis JA. Randomized, prospective study comparing ulnar neurolysis in situ with submuscular transposition. *Neurosurgery*. 2006; 58:296–304; discussion 296–304
- [81] Tindall SC. Comment on LeRoux P D, et al.: Surgical Decompression without Transposition for Ulnar Neuropathy: Factors Determining Outcome. *Neurosurgery*. 1990; 27
- [82] Youmans JR. *Neurological Surgery*. Philadelphia 1990
- [83] Shea JD, McClain EJ. Ulnar-Nerve Compression Syndromes at and Below the Wrist. *J Bone Joint Surg*. 1969; 51A:1095–1103
- [84] Cavallo M, Poppi M, Martinelli P, Gaist G. Distal Ulnar Neuropathy from Carpal Ganglia: A Clinical and Electrophysiological Study. *Neurosurgery*. 1988; 22:902–905
- [85] Dyck PJ, Thomas PK. *Peripheral Neuropathy*. 2nd ed. Philadelphia: W. B. Saunders; 1984
- [86] Roles NC, Maudsley RH. Radial Tunnel Syndrome: Resistant Tennis Elbow as a Nerve Entrapment. *J Bone Joint Surg*. 1972; 54B:499–508
- [87] McKowen HC, Voorhies RM. Axillary Nerve Entrapment in the Quadrilateral Space: Case Report. *J Neurosurg*. 1987; 66:932–934
- [88] de Laat EAT, Visser CPJ, Coene LNJEM, Pahlplatz PVM, Tavy DLJ. Nerve Lesions in Primary Shoulder Dislocations and Humeral Neck Fractures. *J Bone Joint Surg*. 1994; 76B:381–383
- [89] Hadley MN, Sonntag VKH, Pittman HW. Suprascapular Nerve Entrapment: A Summary of Seven Cases. *J Neurosurg*. 1986; 64:843–848
- [90] Fritz RC, Helms CA, Steinbach LS, et al. Suprascapular nerve entrapment: Evaluation with MR imaging. *Radiology*. 1992; 182:437–444
- [91] Callahan JD, Scully TB, Shapiro SA, Worth RM. Suprascapular Nerve Entrapment: A Series of 27 Cases. *J Neurosurg*. 1991; 74:893–896
- [92] Stevens HI. Meralgia Paresthetica. *Arch Neurol Psychiatry*. 1957; 77:557–574
- [93] Williams PH, Trzil KP. Management of Meralgia Paresthetica. *J Neurosurg*. 1991; 74:76–80
- [94] Devers A, Galer BS. Topical lidocaine patch relieves a variety of neuropathic pain conditions: an open-label study. *Clin J Pain*. 2000; 16:205–208
- [95] Aldrich EF, Van den Heever C. Suprainguinal Ligament Approach for Surgical Treatment of Meralgia Paresthetica. Technical Note. *J Neurosurg*. 1989; 70:492–494
- [96] Kars HZ, Topaktas S, Dogan K. Aneurysmal Peroneal Nerve Compression. *Neurosurgery*. 1992; 30:930–931

31 Non-Entrapment Peripheral Neuropathies

31.1 Definitions

- Peripheral neuropathy. (The term polyneuropathy is also sometimes used.) Diffuse lesions of peripheral nerves producing weakness, sensory disturbance, and/or reflex changes
- Mononeuropathy. A disorder of a single nerve, often due to trauma or entrapment
- Mononeuropathy multiplex. Involvement of 2 or more nerves, usually due to a systemic abnormality (e.g. vasculitis, rheumatoid arthritis, DM...). Treatment is directed at the underlying disorder

31.2 Etiologies of peripheral neuropathy

A mnemonic for etiologies of peripheral neuropathies is “GRAND THERAPIST” See □ Table 31.1). Diabetes, alcoholism, and Guillain-Barré (italicized in table) account for 90% of cases. Other etiologies include: arteritis/vasculitis, monoclonal gammopathy (p.547), hepatitis C virus-associated cryoglobulinemia, acute idiopathic polyneuritis, Sjögren’s syndrome (disease).

31.3 Classification

1. inherited neuropathies
 - a) Charcot-Marie-Tooth (CMT) (AKA peroneal muscular atrophy, AKA Hereditary Motor and Sensory Neuropathy (HMSN)): Up to 7 types, (the most common form is autosomal dominant, but X-linked recessive forms also exist). CMT Types 1 & 2 together make up the most common inherited disorder of peripheral nerves (up to 40/100,000). The most common forms involve demyelination. Progressive loss of motor (primarily distal LE) and, to a lesser degree, sensory function (predominantly proprioception and vibration), with atrophy in UEs & LEs. Earliest findings: pes cavus with hammer toes, foot drop and frequent ankle sprains. Patients are more susceptible to entrapment neuropathies due to underlying compromise of peripheral nerves. Patients with Type 1 usual maintain ability to ambulate, whereas Type 2 usually lose ambulation by their teenage years
 - b) hereditary neuropathy with liability to pressure palsies (HNPP): similar to CMT but due to focal areas of irregular thickening of myelin sheaths (“tomaculous” changes), mild trauma or pressure can produce nerve palsies that may last for months
2. acquired neuropathies: see sections below for details
 - a) acquired pure sensory neuropathies (in the absence of autonomic dysfunction) are rare. May be seen with pyridoxine therapy or paraneoplastic syndromes (see below)
 - b) entrapment neuropathies (p.515)
3. pseudoneuropathy
 - a) definition: psychogenic somatoform disorders or malingering, reproducing the pains, paresthesias, hyperalgesia, weakness, and even objective findings such as changes in color and temperature which may mimic neuropathic symptoms²

Table 31.1 Mnemonic for etiologies of peripheral neuropathy	
G-R-A-N-D	T-H-E-R-A-P-I-S-T
Guillain-Barré (p. 184) Renal; uremic neuropathy (p. 549) Alcoholism (see below) Nutritional (B12 deficiency...) Diabetes; see below or Drugs (p. 546)	Traumatic Hereditary Endocrine or Entrapment Radiation Amyloid (p. 549) or AIDS (p. 547) Porphyria (should be under hereditary) or Psychiatric or Paraneoplastic (see below) or Pseudoneuropathy (see below) or PMR (p. 198) or Polycythema vera ¹ Infectious/post-infectious (e.g. Hansen’s disease) Sarcoidosis, see neurosarcoidosis (p. 189), or “Systemic” Toxins, including heavy metals, e.g. lead toxicity (plumbism) (p. 1017)

31.4 Clinical

31.4.1 Presentation

Peripheral neuropathies can present as loss of sensation, pain, weakness, incoordination and difficulty ambulating.

31.4.2 Evaluation

Initial (screening) work-up for peripheral neuropathies of unknown etiology:

1. bloodwork: Hgb-A1C, TSH, ESR and vitamin B12
2. EMG

31.5 Syndromes of peripheral neuropathy

31.5.1 Critical illness polyneuropathy (CIP)

AKA neuropathy of critical illness, ICU neuropathy... See DDx under Guillain-Barré syndrome (p.186).

May occur in up to 70% of septic patients (not all are significantly symptomatic). Affects primarily distal muscles.

Diagnostic criteria:

1. presence of sepsis, multi-organ failure, respiratory failure, or septic inflammatory response syndrome (SIRS)
2. difficulty weaning from ventilator or extremity weakness
3. EMG: ↓ amplitudes of compound muscle action potentials (CMAP) & SNAP
4. widespread muscle denervation potentials
5. normal or only mild increase in serum CPK levels

31

Recovery occurs in weeks to months (faster than Guillain-Barré).

Treatment is supportive. Complete recovery occurs in 50%

31.5.2 Paraneoplastic syndromes affecting the nervous system

Occurs in <1% of cancer patients. Peripheral sensory neuropathy of unknown etiology has been associated with cancer since its earliest description.³ Therefore, However, in patients with sensory neuropathy of unknown etiology, occult neoplasms should be ruled out. If the work-up is negative, the patient should be followed since up to 35% of patients will be found to have cancer after a mean interval of 28 months after the onset of neuropathy (range: 3–72 months)⁴ (no one particular cancer type predominated, in spite of the fact that historically lung cancer is the most frequent neoplasm associated with sensory neuropathy⁵).

31.5.3 Alcohol neuropathy

Characteristically produces a diffuse sensory neuropathy, with absent achilles reflexes.

31.5.4 Brachial plexus neuropathy

See reference.⁶ (p 918)

Evaluation

When the etiology is unclear, check CXR (with apical lordotic view), glucose, ESR and ANA.

If no improvement by ≈ 4 weeks, obtain MRI of the plexus (idiopathic brachial plexitis will usually start to show some improvement by this time, therefore tumor should be ruled out if no improvement).

Differential diagnosis of etiologies of brachial plexopathy

1. Pancoast syndrome or Pancoast tumor AKA superior sulcus tumor. Clinical: various combinations of pain in the shoulder radiating into the upper extremity in the ulnar nerve distribution from

- involvement of the lower brachial plexus, atrophy of hand muscles, Horner’s syndrome (p.564), UE edema. Etiologies:
- a) neoplasms:
 - most common: bronchogenic cancer, usually non-small cell (NSCLC) (squamous cell or adenocarcinoma) arising in the pulmonary apex
 - metastases
 - b) infections
 - c) inflammatory: granulomas, amyloid
 2. (idiopathic) brachial plexitis AKA neuralgic amyotrophy: most commonly upper plexus or diffuse (see below)
 3. cervical rib
 4. viral
 5. following radiation treatment: often diffuse (see below)
 6. diabetes
 7. vasculitis
 8. inherited: dominant genetics
 9. trauma (p.550)

Neuralgic amyotrophy of the upper extremity

General information

AKA idiopathic brachial plexus neuropathy AKA (paralytic) brachial neuritis, AKA brachial plexitis, AKA Parsonage-Turner syndrome,⁷ AKA immune-mediated brachial plexus neuropathy, among others. Idiopathic. Not clearly infectious or inflammatory; allergic mechanism possible. Prognosis is generally good. Common patterns: single or multiple mononeuropathy, plexopathy, or some combination. Demographics are shown in □ Table 31.2.

In a review of 99 cases⁸: predominant symptom is acute onset of intense pain, with weakness developing simultaneously or after a variable period (70% occur within 2 weeks of pain) usually as the pain lessened.^{7,9} Weakness never preceded pain, onset of weakness was sudden in 80% Pain was usually constant, and described as “sharp”, “stabbing”, “throbbing” or “aching.” Arm movement exacerbated pain, and muscle soreness was noted in 15% Pain lasted hours to several weeks. Paresthesias occurred in 35% Pain usually lacked radicular features. When bilateral, weakness is usually asymmetric.

Exam

Weakness or paralysis in 96% confined to shoulder girdle in 50% In descending order of involvement: deltoid, spinati, serratus anterior, biceps brachii, and triceps. Winging of the scapula occurred in 20% Sensory loss occurred in 60% of plexus lesions, of mixed variety (superficial cutaneous and proprioceptive). Sensory loss most common in outer surface of upper arm (circumflex nerve distribution) and radial aspect of forearm. Reflexes were variable.

Table 31.2 Neuralgic amyotrophy	
incidence	1.64 per 100,000 population
male:female	2.4:1
age range at onset	3 mos – 75 years
prodrome	• ≈ 45% had viral prodrome (URI in 25%) • may follow vaccination
onset	rapid onset of pain or paralysis/paresis
initial symptom	pain in 95%
weakness	• 50% confined to shoulder girdle • 10% confined to a single peripheral nerve
sensory deficit	67% usually axillary and antebrachial cutaneous
laterality	• 66% unilateral (right side 54%) • 34% bilateral
lab tests	normal

Overall distribution judged to predominantly involve upper plexus in 56% diffuse plexus in 38% and lower in 6%

EMG/NCV

May help localize the portion of the plexus involved, and may detect subclinical involvement of the contralateral extremity. Must wait ≥ 3 weeks from onset for findings. Differentiating from cervical radiculopathy: SNAP should be normal in radiculopathy whereas some involvement usually occurs in plexitis. Cervical paraspinals will usually be normal in plexitis (except for very severe cases where there can be some retrograde involvement), and will be abnormal (fibrillations) in radiculopathy (except in cases where there has been enough time that significant recovery has occurred).

Outcome

Functional recovery is better in patients with primarily upper plexus involvement. After 1 year, 60% of upper plexus lesions were functioning normally, whereas none with lower involvement were (latter took 1.5–3 years). Rate of recovery estimated to be 36% within 1 year, 75% within 2, and 89% by 3 years. Recurrence was seen in only 5%. No evidence that steroids altered the course of the disease although it is still often prescribed in the acute phase.

Radiation induced brachial plexus neuropathy

Often follows external beam irradiation in the region of the axilla for breast carcinoma. Produces sensory loss with or without weakness. CT or MRI or biopsy may be needed to rule-out tumor invasion of the brachial plexus.

31.5.5 Lumbosacral plexus neuropathy

See reference.¹⁰

31

General information

Analogous to idiopathic brachial plexitis (see above). It is controversial whether this actually exists in isolation without diabetes. Often starts with LE pain of abrupt onset, followed in days or a few weeks by weakness with or without muscle atrophy. Sensory symptoms are less prominent, and usually involve paresthesias. Objective sensory loss is only occasionally seen. There may be tenderness over the femoral nerve.

Differential diagnosis

May be confused with femoral neuropathy or L4 radiculopathy when quadriceps weakness and wasting occurs. Similarly, L5 radiculopathy or peroneal neuropathy may be erroneously suspected when foot drop is seen. Straight leg raising may occasionally be positive. Conspicuously absent are: back pain, exacerbation of pain by Valsalva maneuver or back motion, and significant sensory involvement. See differential diagnosis of foot drop (p.1417) and other causes of sciatica (p.1410).

Etiologies

Other etiologies are similar to that for brachial plexus neuropathy (see above) except that under tumor, a pelvic mass should also be included (check prostate on rectal exam).

Evaluation

Evaluation is as for brachial plexus neuropathy (see above), except that instead of a brachial plexus MRI, a lumbar MRI and pelvic CT should be done to rule out masses.

EMG is key to diagnosis: evidence of patchy denervation (fibrillation potentials, and motor unit potentials that are either decreased in number or increased in amplitude or duration and polyphasic) involving at least 2 segmental levels with sparing of the paraspinal muscles is highly diagnostic (once diabetes, etc. have been ruled-out).

Outcome

Recovery from pain precedes return of strength. Improvement is generally monophasic, slow (years), and incomplete.

31.5.6 Diabetic neuropathy

General information

≈ 50% of patients with DM develop neuropathic symptoms or show slowing of nerve conduction velocities on electrodiagnostic testing. Neuropathy may sometimes be the initial manifestation of diabetes. Diabetic neuropathy is reduced by tight control of blood glucose.¹¹

Syndromes

Disagreement exists over the number of distinct clinical syndromes; there is probably a continuum¹² and they likely occur in various combinations. Some of the more readily identified syndromes include:

1. primary sensory polyneuropathy: symmetric, affecting feet and legs more than hands. Chronic, slowly-progressive. Often with accelerated loss of distal vibratory sense (normal loss with aging is ≈ 1% per year after age 40). Presents as pain, paresthesias, and dysesthesias. Soles of feet may be tender to pressure. Meralgia paresthetica (p.534) may be first manifestation
2. autonomic neuropathy: involving bladder, bowel, and circulatory reflexes (resulting in orthostatic hypotension). May produce impotence, impaired micturition, diarrhea, constipation, impaired pupillary light response
3. diabetic plexus neuropathy¹³ or proximal neuropathy: possibly secondary to vascular injury to nerves (similar to a diabetic mononeuritis):
 - a) one that occurs in patients >50 years old with mild diabetes type II that is often confused with femoral neuropathy. Causes severe pain in the hip, anterior thigh, knee, and sometimes medial calf. Weakness of the quadriceps, iliopsoas, and occasionally thigh adductors. Loss of knee jerk. Possible sensory loss over medial thigh and lower leg. Pain usually improves in weeks, the weakness in months
 - b) □ diabetic amyotrophy: occurs in similar patient population often with recently diagnosed DM. Alternative names include¹⁴: Bruns-Garland syndrome, ischemic mononeuropathy multiplex....¹⁵ Abrupt onset of asymmetric pain (usually deep aching/burning with superimposed lancinating paroxysms, most severe at night) in back, hip, buttocks, thigh, or leg. Progressive weakness in proximal or proximal and distal muscles, often preceded by weight loss. Patellar reflexes are absent or reduced. Sensory loss is minimal. Proximal muscles (especially thigh) may atrophy. EMG findings consistent with demyelination invariably accompanied by axonal degeneration, with involvement of paraspinals and no evidence of myopathy. Symptoms may progress steadily or stepwise for weeks or even up to 18 months, and then gradually resolve. Opposite extremity may become involved during the course or may occur months or years later. Sural nerve biopsy may suggest demyelination
 - c) diabetic proximal neuropathy (DPN): fairly similar findings to diabetic amyotrophy except for subacute onset of symmetric LE involvement that usually start with weakness may be a variant.¹⁶ □ Table 31.3 (adapted¹⁶) compares DPN to diabetic amyotrophy and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP)

Treatment

Treatment of Bruns-Garland syndrome is primarily expectant, although immunotherapy (steroids, immune globulin, or plasma exchange) may be considered in severe or progressive cases (efficacy is unproven).¹⁶

For sensory polyneuropathy, good control of blood sugar contributes to reduction of symptoms. Adjunctive agents that have been used include:

1. mexiletine (Mexitil®): start at 150 mg q 8 hrs, and titrate to symptoms to a maximum of 10 mg/kg/d
2. amitriptyline (Elavil®) and fluphenazine (Prolixin®): □: start with 25 mg amitriptyline PO q hs and 1 mg fluphenazine PO TID; and work up to 75 mg amitriptyline PO q hs¹⁷ (≈ 100 mg qd amitriptyline alone may also be effective¹⁸). Usefulness has been challenged,¹⁹ but many studies do show benefit.^{18,20} Side effects: that may limit use include sedation, confusion, fatigue, malaise, hypomania, rash, urinary retention, and orthostatic hypotension
3. desipramine (Norpramin®): more selective blocker of norepinephrine reuptake (which seems more effective for this condition than serotonin reuptake blockers). Effectiveness at mean doses of 110 mg/day ≈ same as amitriptyline and therefore may be useful for patients unable to tolerate amitriptyline.¹⁸ Side effects: include insomnia (may be minimized by AM dosing), orthostatic hypotension, rash, bundle branch block, tremor, pyrexia. Supplied : 10, 25, 50, 75, 100 & 150 mg tablets
4. capsaicin (Zostrix®) (p.496): effective in some

Table 31.3 Comparison of diabetic amyotrophy, diabetic proximal neuropathy (DPN), & CIDP			
Description	Diabetic amyotrophy	DPN	CIDP
Onset	acute	subacute	gradual
Initial symptoms	asymmetric pain→ weakness	symmetric weakness	symmetric weakness
UE weakness	no	uncommon	yes
Sensory loss	minimal	minimal	moderate
Areflexia	IE	IE	generalized
CSF protein	variable	increased	increased
Axonal pathologic changes	common	typical	uncommon
Conduction slowing	patchy	patchy	diffuse
Prognosis	good	good	poor without treatment
Response to immunotherapy	unknown	possible	yes
Course	monophasic	monophasic	progressive

- 31
5. paroxetine (Paxil®): a selective serotonin reuptake inhibitor (SSRI) antidepressant. □: 20 mg PO q AM. If necessary, increase by 10 mg/d q week up to a maximum of 50 mg/day (except in elderly, debilitated, or renal or hepatic failure where maximum is 40 mg/day). Supplied : 20 mg (scored) & 30 mg tablets

6. gabapentin (Neurontin®) doses of 1800–3600 mg/d produces at least moderate pain relief from painful diabetic neuropathy in 60%of patients²¹ and was ≈ as efficacious as amitriptyline.²² Dosage must be reduced with renal insufficiency. See details (p.455)

7. pregabalin (Lyrica®) □: start with 50 mg TID and increase up to a maximum of 100 mg PO TID within 1 week in patients with creatinine clearance ≥ 60 ml/min, see Eq (7.1) to estimate. Dosage must be reduced with renal insufficiency. Supplied: 25, 50, 75, 100, 150, 200, 225, 300 mg capsules

31.5.7 Drug-induced neuropathy

Many drugs have been implicated as possible causes of peripheral neuropathy. Those that are better established or more notorious include:

1. thalidomide: neuropathy may occur with chronic use, and may be irreversible²³

2. metronidazole (Flagyl®)

3. phenytoin (Dilantin®)

4. amitriptyline (Elavil®)

5. dapsone: a rare complication reported with use in nonleprosy patients is a reversible peripheral neuropathy that may be due to axonal degeneration, producing a Guillain-Barré-like syndrome (p.184).

6. nitrofurantoin (Macrochantin®): may additionally cause optic neuritis

7. cholesterol lowering drugs: e.g. lovastatin (Mevacor®), indapamide (Lozol®), gemfibrozil (Lopid®)

8. thallium: may produce tremors, leg pains, paresthesias in the hands and feet, polyneuritis in the LE, psychosis, delirium, seizures, encephalopathy

9. arsenic: may produce numbness, burning and tingling of the extremities

10. chemotherapy: cisplatin, vincristine...

31.5.8 Femoral neuropathy

Clinical findings

1. motor deficits:

a) wasting and weakness of the quadriceps femoris (knee extension)

b) weakness of iliopsoas (hip flexion): if present, indicates very proximal pathology (lumbar root or plexus lesion) as the branches to the iliopsoas arise just distal to the neural foramina

2. diminution of the patellar (knee jerk) reflex
3. sensory findings:
 - a) sensory loss over the anterior thigh and medial calf
 - b) pain in same distribution may occur
4. mechanical signs: positive femoral stretch test (p.1048)

Etiologies

1. diabetes: the most frequent cause
2. femoral nerve entrapment: rare
 - a) may occur secondary to inguinal hernia or may be injured by deep sutures placed during herniorrhaphy
 - b) secondary to prolonged pelvic surgery from retractor compression (usually bilateral)
3. intraabdominal tumor
4. femoral arterial catheterization: see below
5. retroperitoneal hematoma (e.g. in hemophiliac or on anticoagulants)
6. during surgery (p.549)

Differential diagnosis

1. L4 radiculopathy: L4 radiculopathy should not cause iliopsoas weakness; see L4 involvement (p.1412).
2. diabetic plexus neuropathy (see above)
3. (idiopathic) lumbosacral plexus neuropathy (see above)

31.5.9 AIDS neuropathy

General information

3.3% of patients with AIDS will develop peripheral nerve disorders²⁴ (whereas none who were just HIV positive developed neuropathy). The most common disorder is distal symmetric polyneuropathy (DSP), usually consisting of vague numbness and tingling, and sometimes painful feet (although it may also be painless). There may be subtle reduction of light touch and vibratory sense. Other neuropathies include mononeuropathies – usually meralgia paresthetica (p.534) – mononeuropathy multiplex, or lumbar polyradiculopathy. Drugs used to treat HIV can also cause neuropathies (see below).

The DSP in AIDS patients is often associated with CMV infection, Mycobacterium avium intracellulare infection, or may be due to lymphomatous invasion of the nerve or lymphomatous meningitis. May demonstrate a mixed axonal demyelinating type of neuropathy on electrodiagnostic testing.

Neuropathies associated with drugs used to treat HIV

1. nucleoside reverse transcriptase inhibitors
 - a) zidovudine (Retrovir®) (formerly AZT)
 - b) didanosine (ddI; Videx®): (formerly dideoxyinosine) can cause a painful dose-related neuropathy²⁵
 - c) stavudine (d4T; Zerit®): can cause sensory neuropathy which usually improves when d4 T is discontinued, and may not recur if restarted at lower dose²⁵
 - d) zalcitabine (ddC; Hivid®): the least potent of the nucleoside analogs, therefore rarely used; dose-related neuropathy can be severe and persistent. More common in patients with DM or didanosine treatment²⁵
2. protease inhibitors
 - a) ritonavir (Norvir®): can cause peripheral paresthesias
 - b) amprenavir (Agenerase®): can cause perioral paresthesias

31.5.10 Neuropathy associated with monoclonal gammopathy

General information

Monoclonal gammopathies include myeloma (p.714), Waldenstrom's macroglobulinemia, and non-malignant entities such as monoclonal gammopathy of undetermined significance (MGUS). Much effort has gone into determining which benign gammopathies are or are not likely to progress, and will not be addressed here.

≈ 10%of patients with neuropathy with no apparent etiology will be determined to have a mono-clonal gammopathy (malignant or otherwise).

Etiologies

- 1. antibodies directed primarily against oligosaccharides of peripheral nerves, e.g. myelin associated glycoprotein (MAG), producing demyelinating neuropathy
- 2. cryoglobulins may damage vaso-nervorum (small blood vessels nourishing peripheral nerves)
- 3. in malignant gammopathies, tumor cells can invade the peripheral nerves (lymphomatosis)
- 4. amyloidosis (p.549): deposition of amyloid in peripheral nerves
- 5. thalidomide (p.546) used to treat some myelomas may cause neuropathy

Treatment

- 1. IgM monoclonal gammopathies: reduce the IgM antibody concentration
- 2. IgG or IgA monoclonal gammopathies:
 - a) treatment for myeloma related neuropathy is directed at treating the myeloma
 - b) solitary plasmacytoma: excision or XRT can improve the neuropathy

31.5.11 Perioperative neuropathies

General information

Also, below. Represent ≈ 1/3 of all anesthesia-associated malpractice claims in the U.S.²⁶ Most often involves ulnar nerve or brachial plexus. In many cases, a nerve that is abnormal but asymptomatic may become symptomatic as a result of any of the following factors: stretch or compression of the nerve, generalized ischemia or metabolic derangement. The injury may be permanent or temporary. Occurs almost exclusively in adults.²⁷

Types of perioperative neuropathies

Examples include:

- 1. ulnar neuropathy: controversial. Often blamed on external nerve compression or stretch as a result of malpositioning. Although this may be true in some cases, in one series this was felt to be a factor in only ≈ 17%of cases.²⁸ Patient-related characteristics associated with these neuropathies are shown in □ Table 31.4.²⁹ Many of these patients have abnormal contralateral nerve conduction, suggesting a possible predisposing condition.³⁰ Many patients do not complain of symptoms until >48 hours post-op^{29,30,31} (if it were due to compression, deficit would be maximal immediately post-op). Risk may be reduced by padding the arm at, and especially distal to, the elbow, and avoiding flexion of the elbow (especially avoiding >110° flexion which tightens the cubital tunnel retinaculum) and by reducing the amount of time spent convalescing in the recumbent position with leaning on the elbows³¹
- 2. brachial plexus neuropathy: may be mistaken for ulnar neuropathy. May be associated with:
 - a) median sternotomy (most common with internal mammary dissection). Posterior sternal retraction displaces the upper ribs and may stretch or compress the C6 through T1 roots (which are major contributors to the ulnar nerve)
 - b) head-down (Trendelenburg) positions where the patient is stabilized with a shoulder brace. The brace should be placed over the acromioclavicular joint(s), and non-slip mattresses and flexion of the knees may be used as adjuncts²⁷
 - c) prone position (rare): especially with shoulder abduction and elbow flexion with contralateral head rotation²⁷
- 3. median neuropathy: perioperative median nerve injury may result from stretch of the nerve. Rare. Seems to occur primarily in middle-aged muscular males with reduced extension of the elbows due to muscle mass. This may result in stretching of the nerve after muscle relaxants are

Table 31.4 Patient-related characteristics in anesthesia-related ulnar neuropathy
male gender
obesity (body mass index≥38)
prolonged post-op bed rest

given. Padding should be placed under the forearms and hands of these patients to maintain mild elbow flexion²⁷

4. lower extremity neuropathies: most occur in patients undergoing procedures in the lithotomy position.²⁷ Frequency of involvement in a large series of patients undergoing procedures in the lithotomy position³²: common peroneal 81% sciatic 15% and femoral 4% Risk factors other than position: prolonged duration of procedure, extremely thin body habitus, and cigarette smoking in the preoperative period
 - a) common peroneal neuropathy: susceptible to injury in the posterior popliteal fossa where it wraps around the fibular head. May be compressed by leg holders, which should be padded in this area
 - b) femoral neuropathy: compression of the nerve by self-retaining abdominal wall retractor or rendering the nerve ischemic by occlusion of the external iliac artery.²⁷ Hemorrhage into the iliopsoas muscle may also compress the nerve. Cutaneous branches of the femoral nerve may be injured during labor and/or delivery³³ (most are transient)
 - c) sciatic neuropathy: stretch injuries may occur with hyperflexion of the hip and extension of the knee as may occur in some variants of the lithotomy position
 - d) meralgia paresthetica³⁴: tends to occur bilaterally in young, slender males positioned prone, with operations lasting 6–10+ hours. Onset: 1–8 days post-op. Spontaneous recovery typically occurs over an average of 5.8 months

Management

Once a neuropathy is detected, determine if it is sensory, motor, or both. Pure sensory neuropathies are more often temporary than motor,²⁹ and expectant management for ≈ 5 days is suggested (have the patient avoid postures or activities that may further injure the nerve). Neurologic consultation should be requested for all motor neuropathies and for sensory neuropathies persisting >5 days²⁷ (EMG evaluation will not usually be helpful earlier than ≈ 3 weeks after onset).

31.5.12 Other neuropathies

Amyloid neuropathy

Amyloid is an insoluble extracellular protein aggregate that can be deposited in peripheral nerves. Amyloidosis occurs in a number of conditions, e.g. in $\approx 15\%$ of patients with multiple myeloma (p.714). The neuropathy predominantly produces a progressive autonomic neuropathy and symmetric dissociated sensory loss (reduced pain and temperature, preserved vibratory sense). There is usually less prominent motor involvement. May predispose to pressure injury of nerves (especially carpal tunnel syndrome, see laboratory tests (p.523)).

Uremic neuropathy

Occurs in chronic renal failure. Early symptoms include calf cramps (“Charlie horses”), dysesthetic pain in feet (similar to painful diabetic neuropathy) and “restless legs.” Achilles reflexes are lost. A stocking sensory loss is followed later by LE weakness that starts distally and ascends. The offending toxin is not known. Dialysis or renal transplantation relieves the symptoms.

Neuropathy after cardiac catheterization

In a series of $\approx 10,000$ patients followed after femoral artery catheterization³⁵ (e.g. for coronary angiography or angioplasty), neuropathy occurred in 0.2% (with an estimated range in the literature up to $\approx 3\%$). Risk factors identified include: patients developing retroperitoneal hematomas or pseudoaneurysms after the procedure, procedures requiring larger introducer sheaths (e.g. angioplasty & stent placement $>$ diagnostic catheterization), excessive anticoagulation (PTT >90 for at least 12 hours).

Two groups of patients were identified and are shown in □ Table 31.5.

Excruciating pain after the catheterization procedure often preceded the development or recognition of neuropathy.

Treatment

After considering available information, the recommendation is to repair pseudoaneurysms surgically, but to treat the neuropathy conservatively. A case could not be made that surgical drainage of hematoma reduced the risk of neuropathy. Weakness from femoral or obturator neuropathy was treated with inpatient rehabilitation.

Table 31.5 Neuropathy after cardiac catheterization (N=9585) ³⁵	
Catheterization complication	Neurologic complication
Group I (4 patients)	
groin hematoma or pseudoaneurysm	sensory neuropathy in all 4 cases • in distribution of medial & intermediate femoral cutaneous nerves → isolated sensory neuropathy (dysesthesia & sensory loss) of the anterior and medial thigh • no motor deficit
Group II (16 patients)	
large retroperitoneal hematoma	femoral neuropathy • sensory in all 16 cases: dysesthesia of the anterior/medial thigh & medial calf • motor in 13 cases: iliopsoas & quadriceps weakness
	obturator neuropathy in 4 cases • sensory: upper medial thigh • motor: obturator weakness
	lateral femoral cutaneous nerve → meralgia paresthetica

Outcome

Group I patients all had resolution in <5 mos. In group II, 50% had complete resolution in 2 mos. 6 patients had persistent symptoms, 5 had mild femoral sensory neuropathy (1 of whom felt it was at least somewhat disabling), 1 had mild persistent quadriceps weakness and occasionally walks with a cane.

31 31.6 Peripheral nerve injuries

31.6.1 General information

Anatomy of peripheral nerves

See □ Fig. 31.1. Endoneurium surrounds myelinated and unmyelinated axons. These bundles are gathered into fascicles surrounded by perineurium. The epineurium encases the nerve trunk, containing fascicles separated by interfascicular epineurium or mesoneurium.

Nerve regeneration

Peripheral nerves regenerate ≈1 mm/day (about 1 inch/month). Divide this figure into distance that the nerve has to traverse (from knowledge of anatomy) for guide as to how long to wait before considering failure of therapy (either operative or non-operative). However, this rule may not be applicable to long distances (>≈ 12 inches), and it may take longer to traverse regions of entrapment, scar or nerve injury. There may also be fibrosis of the muscle beyond salvage.

Peripheral nerve injury classification

□ Table 31.6
There are numerous classification systems. The Seddon classification is an older 3-tiered system, The Sunderland system has 5 tiers, essentially dividing axonotmesis into 3 subgroups. Others have added a 6th category as shown in □ Table 31.6.

31.6.2 Brachial plexus injuries

Etiologies

- Etiologies include:
1. penetrating trauma
 2. traction (stretch injuries): more likely to affect the posterior and lateral cords than the medial cord and median nerve
 3. first rib fractures
 4. compression by hematoma

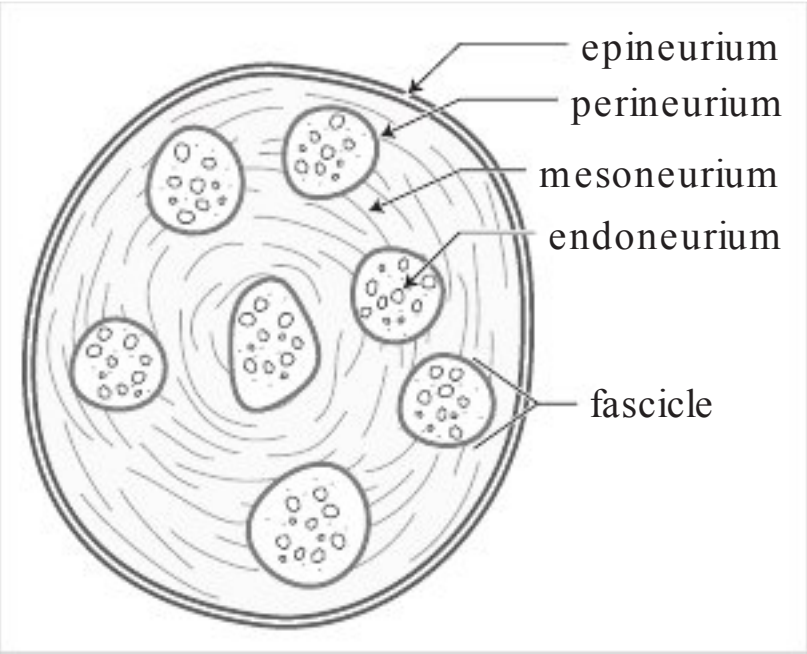


Fig. 31.1 Anatomy of a peripheral nerve

Table 31.6 Classification of peripheral nerve injury ^a	
Seddon system	Sunderland system
Neuropraxia	First-degree
Features common to both systems Physiologic transection (nerve in continuity). Basement membrane intact. Compression or ischemia → local conduction block (impaired axonal transport). □ No wallerian degeneration ^b . Motor involvement is typically>sensory. Autonomic function is preserved	
Recovers in hours to months; average is 6–8 weeks	Focal demyelination may occur. Recovery is usually complete in 2–3 weeks (not the “1 mm/day rule”)
Axonotmesis	Second-degree
Features common to both systems Complete interruption of axons and myelin sheaths. Supporting structures (including endoneurium) intact. □ Wallerian degeneration occurs	
	Recovers at 1 mm/day as axon follows “tubule.” Sometimes may only be diagnosed retrospectively. Recovery is poor in lesions requiring>18 months to reach target muscle
	Third-degree
	Endoneurium disrupted, epineurium & perineurium intact. Nerve may not appear seriously damaged on gross inspection. Recovery may range from poor to complete and depends on degree of intrafascicular fibrosis
	Fourth-degree
Neurotmesis	Interruption of all neural & supporting elements. Epineurium intact. Grossly: nerve is usually indurated & enlarged
	Fifth-degree
	Complete transection with loss of continuity
Nerve completely severed or disorganized by scar tissue. Spontaneous regeneration impossible	Sixth-degree ^c
	Mixed lesion. Combination of elements of first through fourth degree. There may be some preserved sensory fascicles (may produce a positive Tinel’s sign)
^a comparing and showing approximate equivalence of Seddon and Sunderland systems	
^b wallerian degeneration after British physiologist Augustus Volney Waller (1816–1870), AKA orthograde degeneration, AKA secondary degeneration: degeneration of the axon distal to a focal lesion	
^c not part of original Sunderland system	

Differentiating preganglionic from postganglionic injuries

Initial exam seeks to differentiate preganglionic injuries (proximal to dorsal root ganglion) which cannot be repaired surgically, from postganglionic injuries. Clues to a preganglionic injury include:

1. Horner's syndrome: pre-ganglionic injury interrupts white rami communicantes
2. paralysis of serratus anterior (long thoracic nerve): produces winging of scapula
3. paralysis of rhomboids (dorsal scapular nerve)
4. early neuropathic pain suggests nerve root avulsion. MRI or myelogram will show pseudomeningoceles at the avulsed levels
5. EMG: requires ≥ 3 weeks from injury for some findings. Look for:
 - a) denervation potentials in paraspinal muscles due to loss of neural input. The posterior ramus of the spinal nerve originates just distal to the dorsal root ganglion. Due to overlap, cannot localize to a specific segment
 - b) normal sensory nerve action potential (SNAP): preganglionic injuries leave the dorsal ganglion sensory cell body and the distal axon intact, so that normal SNAP can be recorded proximally even in an anesthetic region
6. pseudomeningocele on myelography or MRI: suggests nerve root avulsion (very proximal), however, 15% of pseudomeningoceles are not associated with avulsions, and 20% of avulsions do not have pseudomeningoceles^{36,37}

Types of brachial plexus injuries

(Duchene)-Erb's palsy

Upper brachial plexus injury (C5 & 6, some authors include C7) e.g. from forceful separation of humeral head from shoulder, commonly due to difficult parturition (see below) or motorcycle accident (downward force on shoulder can cause traumatic nerve root avulsion from the spinal cord). Paralysis of deltoid, biceps, rhomboids, brachioradialis, supra- & infra-spinatus, and occasionally supinator. C7 involvement produces weak wrist extension.

Motor: arm hangs at side internally rotated & extended at elbow and flexed at the wrist ("Bell-hop's tip position"). Hand motion is unaffected.

Klumpke's palsy

Injury to lower brachial plexus (C8 & T1, some authors include C7), from traction of abducted arm e.g. in catching oneself during a fall from a height, or by Pancoast tumor (lung apex tumor – check CXR with apical lordotic view). Characteristic claw deformity (also seen with ulnar nerve injury) with weakness and wasting of small hand muscles. Possible Horner's syndrome if T1 involved.

Birth brachial plexus injury (BBPI)

Incidence is 0.3–2.0 per 1000 live births (0.1% in infants with birthweight < 4000 gm³⁸). Rarely, a congenital case may be mistaken for BBPI.³⁹ Some contend that the plexus injury may occur when uterine contractions push the shoulder against the mother's pubic bone or with lowering of the shoulder with opposite inclination of the cervical spine.³⁹

Classification of BBPI injuries: Upper plexus injuries are most common, with about half having C5 & C6 injuries, and 25% involving C7 also.⁴⁰ Combined upper and lower lesions occur in $\approx 20\%$ Pure lower lesions (C7-T1) are rare, constituting only $\approx 2\%$ and seen most commonly in breech deliveries. Lesions are bilateral in $\approx 4\%$ A 4-level scale of intensity is shown $\square$ Table 31.7.⁴¹

Risk factors:

1. shoulder dystocia
2. high birth weight
3. primiparous mother
4. forceps⁴² or vacuum assisted delivery
5. breech presentation⁴³
6. prolonged labor
7. previous birth complicated by BBPI

Management of BBPI: Most surgeons observe all patients until age 3 months. Conservative surgeons may wait up to 9 months. More aggressive surgeons will explore the plexus at age 3 months if not antigravity in deltoid, biceps or triceps. In cases of proven avulsion (pseudomeningocele and EMG indicative of a preganglionic injury), nerve transfers are a valid option at 3 months.⁴⁴ EMG may show signs of reinnervation, but the recovery may not be robust enough.

Table 31.7 Birth brachial plexus injury			
Group	Lesion	Manifestation	Spontaneous recovery rate
1	C5 or C6 roots or superior trunk	paralysis of shoulder abduction, elbow flexion & forearm supination. Finger flexion is norma	90%
2	above + involvement of C7 or medial trunk	above + paralysis of finger extensors (but not flexors)	65%
3	above + finger flexors	essentially no hand movement. No Horner’s syndrome	≈ <50%
4	complete brachial plexus	flail arm + Horner’s syndrome	0%
	“dominant C7” paralysis variant	selective loss of shoulder abduction & elbow extension	

Table 31.8 Indications for neurosurgical intervention in GSW to the brachial plexus ⁴⁵	
1. complete loss in the distribution of at least one element a) no improvement clinically or on EMG in 2–5 months b) deficit in distribution that is responsive to surgery (e.g. C5, C6, C7, upper or middle trunk, lateral or posterior cords or their outflows) c) injuries with loss only in lower elements are not operated 2. incomplete loss with failure to control pain medically 3. pseudoaneurysm, clot or fistula involving plexus 4. true causalgia requiring sympathectomy	

Management of brachial plexus injuries

1. most injuries show maximal deficit at onset. Progressive deficit is usually due to vascular injuries (pseudoaneurysm, A-V fistula, or expansile clot), these should be explored immediately

2. clean, sharp, relatively fresh lacerating injuries (usually iatrogenic, scalpel induced) should be explored acutely and repaired with tension-free end-to-end anastamoses within 24–48 hours (after that then ends will more edematous and therefore more difficult to suture)

3. penetrating non-missile injuries with severe or complete deficit should be explored as soon as the primary wound heals

4. gunshot wounds (GSW) to the brachial plexus: deficit is usually due to axonotmesis or neurotmesis (see below). Sometimes nerves may be divided. Nerves showing partial function usually recover spontaneously; those with complete dysfunction rarely do so. Surgery is of little benefit for discrete injuries to the lower trunk, medial cord, or C8/T1 roots. Most are managed conservatively for 2–5 months. Indications for surgery are shown in □ Table 31.8

5. traction injuries: incomplete postganglionic injuries tend to improve spontaneously. If recovery is not satisfactory, perform EMG at 4–5 months and explore at 6 months

6. neuromas in continuity: those that do not conduct a SNAP have complete internal disruption and require resection and grafting. Methods of repair:

a) neurolysis:

• external neurolysis: most commonly performed in exploration. Value is questionable

• internal neurolysis: splitting the nerve into fascicles. Not recommended unless a clear neuroma in continuity is found eccentric in the nerve that conducts SNAP

b) nerve grafting. Sural nerve is the most commonly used interposition graft following resection of neuroma in continuity

c) nerve transfers. Donor nerve options:

• spinal accessory nerve

• intercostal nerves to musculocutaneous nerve

• fascicles of the ulnar nerve for the median nerve (Oberlin procedure)

• anterior interosseus nerve to median nerve

31.7 Missile injuries of peripheral nerves

This section deals primarily with gunshot wounds (GSW). Most injuries from a single bullet are due to shock and cavitation from the missile causing axonotmesis or neurotmesis, and are not from direct nerve transection. Approximately 70% will recover with expectant management.

However, if there is a lack of improvement on serial examinations including electrodiagnostic studies, intervention should be undertaken by about 5–6 months to avoid further difficulties due to nerve fibrosis and muscle atrophy.

See □ Table 31.8 for indications for surgery for missile injuries of the brachial plexus.

31.8 Thoracic outlet syndrome

31.8.1 General information

The thoracic outlet is a confined area at the apex of the lung bordered by the 1st rib below and the clavicle above through which passes the subclavian artery, vein, and brachial plexus.

Thoracic outlet syndrome (TOS) is a term implying compression of one or more of the enclosed structures producing a heterogeneous group of disorders. TOS tends to be diagnosed more often by general and vascular surgeons than by neurologists and neurosurgeons. Four unrelated conditions with different structures involved:

1. noncontroversial”, with characteristic symptom complex, reproducible clinical findings, confirmatory laboratory tests. Low incidence⁴⁶
 - arterial vascular: producing arm, hand and finger pallor and ischemia
 - venous vascular: producing arm swelling and edema
 - true neurologic: compressing the lower trunk or median cord of the brachial plexus (see below)
2. disputed neurologic: includes scalenus anticus syndrome (see below)

31.8.2 Differential diagnosis

1. herniated cervical disc
2. cervical arthrosis
3. lung cancer (pancoast tumor)
4. tardy ulnar nerve palsy
5. carpal tunnel syndrome
6. orthopedic shoulder problems
7. complex regional pain syndrome (reflex sympathetic dystrophy)

31.8.3 True neurologic TOS

General information

A rare condition primarily affecting adult women, usually unilateral.

Neurologic structures involved

1. most common: compression of the C8/T1 roots
2. or proximal lower trunk of the brachial plexus (BP)
3. less common: compression of the median cord of the BP

Etiologies

1. constricting band extending from the first rib to a rudimentary “cervical rib” or to an elongated C7 transverse process
2. scalenus (anticus) syndrome: controversial (below)
3. compression beneath the pectoralis minor tendon under the coracoid process: may result from repetitive movements of the arms above the head (shoulder elevation and hyperabduction)

Signs and symptoms

1. □ sensory changes in distribution of median cord (mainly along medial forearm), sparing median nerve sensory fibers (pass through upper and middle trunks)
2. hand clumsiness or weakness and wasting, especially abductor pollicis brevis and ulnar hand intrinsic (C8/T1 denervation/atrophy)
3. there may be tenderness over Erb’s point (2 to 3 cm above the clavicle in front of the C6 transverse process)
4. may be painless
5. usually unilateral

Confirmatory tests

1. EMG: unreliable (may be negative). Most common abnormality in neurogenic TOS is loss of medial antebrachial cutaneous SNAP
2. MRI does not show bony abnormalities well, but may occasionally demonstrate a kink in the lower BP. Can also rule-out conditions that may mimic TOS such as herniated cervical disc
3. cervical spine x-rays with obliques and apical lordotic CXR may demonstrate bony abnormalities. However, not every cervical rib produces symptoms (some patients with bilateral cervical ribs may have unilateral TOS).

Treatment

Controversial. Conservative treatment (usually including stretching and physical therapy) is about equally as effective as surgery and avoids attendant risks.

Decompression can be achieved by removing the muscles that surround the nerves (scaleneotomy), by transaxillary first rib resection, or both.

31.8.4 Scalenus (anticus) syndrome (disputed neurologic TOS)

Controversial. More commonly diagnosed in the 1940s and 1950s. There is a lack of consensus regarding the pathophysiology (including structures involved), clinical presentation, helpful tests, and optimal treatment. Removal of first thoracic rib is often advocated for treatment, frequently via a transaxillary approach. Unfortunately, injuries, especially to the lower trunk of the brachial plexus, may result from the surgery.

Other variations include an “upper plexus” type for which total anterior scalenectomy is advocated. Again, very controversial.

References

- [1] Poza JJ, Cobo AM, Marti-Masso JF. Peripheral neuropathy associated with polycythemia vera. *Neurologia*. 1996; 11:276–279
- [2] Ochoa JL, Verdugo RJ. Reflex Sympathetic Dystrophy: A Common Clinical Avenue for Somatoform Expression. *Neurol Clin*. 1995; 13:351–363
- [3] Denny-Brown D. Primary Sensory Neuropathy with Muscular Changes Associated with Carcinoma. *J Neurol Neurosurg Psychiatry*. 1948; 11:73–87
- [4] Camerlingo M, Nemni R, Ferraro B, et al. Malignancy and Sensory Neuropathy of Unexplained Cause: A Prospective Study of 51 Patients. *Arch Neurol*. 1998; 55:981–984
- [5] McLeod JG, Dyck PJ, Thomas PK. In: Paraneoplastic Neuropathies. *Peripheral Neuropathy*. Philadelphia: W.B. Saunders; 1993:1583–1590
- [6] Adams RD, Victor M. *Principles of Neurology*. 2nd ed. New York: McGraw-Hill; 1981
- [7] Turner JW, Parsonage MJ. Neuralgic amyotrophy (paralytic brachial neuritis); with special reference to prognosis. *Lancet*. 1957; 273
- [8] Tsairis P, Dyck PJ, Mulder DW. Natural History of Brachial Plexus Neuropathy: Report on 99 Patients. *Arch Neurol*. 1972; 27:109–117
- [9] Misamore GW, Lehman DE. Parsonage-Turner syndrome (acute brachial neuritis). *J Bone Joint Surg*. 1996; 78:1405–1408
- [10] Evans BA, Stevens JC, Dyck PJ. Lumbosacral Plexus Neuropathy. *Neurology*. 1981; 31:1327–1330
- [11] The Diabetes Control and Complications Trial Research Group. The Effect of Intensive Treatment of Diabetes on the Development and Progression of Long-Term Complications in Insulin-Dependent Diabetes Mellitus. *N Engl J Med*. 1993; 329:977–986
- [12] Asbury AK. Proximal Diabetic Neuropathy. *Ann Neurol*. 1977; 2:179–180
- [13] Dyck PJ, Thomas PK. *Peripheral Neuropathy*. 2nd ed. Philadelphia: W. B. Saunders; 1984
- [14] Garland H. Diabetic Amyotrophy. *BMJ*. 1955; 2:1287–1290
- [15] Barohn RJ, Sahenk Z, Warmolts JR, Mendell JR. The Bruns-Garland Syndrome (Diabetic Amyotrophy): Revisited 100 Years Later. *Arch Neurol*. 1991; 48:1130–1135
- [16] Pascoe MK, Low PA, Windebank AJ. Subacute Diabetic Proximal Neuropathy. *Mayo Clin Proc*. 1997; 72:1123–1132
- [17] Davis JL, Lewis SB, Gerich JE, et al. Peripheral Diabetic Neuropathy Treated with Amitriptyline and Fluphenazine. *JAMA*. 1977; 21:2291–2292
- [18] Max MB, Lynch SA, Muir J, Shoaf SE, et al. Effects of Desipramine, Amitriptyline, and Fluoxetine on Pain in Diabetic Neuropathy. *N Engl J Med*. 1992; 326:1250–1256
- [19] Mendel CM, Klein RF, Chappell DA, et al. A Trial of Amitriptyline and Fluphenazine in the Treatment of Painful Diabetic Neuropathy. *JAMA*. 1986; 255:637–639
- [20] Mendel CM, Grunfeld C. Amitriptyline and Fluphenazine for Painful Diabetic Neuropathy. *JAMA*. 1986; 256:712–714
- [21] Backonja M, Beydoun A, Edwards KR, Schwartz SL, Fonseca V, Hes M, LaMoreaux L, Garofalo E. Gabapentin for the symptomatic treatment of painful neuropathy in patients with diabetes mellitus: a randomized controlled trial. *JAMA*. 1998; 280:1831–1836
- [22] Morello CM, Leckband SG, Stoner CP, Moorhouse DF, Sahagian GA. Randomized double-blind study comparing the efficacy of gabapentin with amitriptyline on diabetic peripheral neuropathy pain. *Arch Intern Med*. 1999; 159:1931–1937
- [23] New Uses of Thalidomide. *Med Letter*. 1996; 38:15–16
- [24] Fuller GN, Jacobs JM, Guilloff RJ. Nature and Incidence of Peripheral Nerve Syndromes in HIV Infection. *J Neurol Neurosurg Psychiatry*. 1993; 56:372–381
- [25] Drugs for HIV Infection. *Med Letter*. 2000; 42:1–6
- [26] Kroll DA, Caplan RA, Posner K, et al. Nerve Injury Associated with Anesthesia. *Anesthesiology*. 1990; 73:202–207
- [27] Warner MA. Perioperative Neuropathies. *Mayo Clin Proc*. 1998; 73:567–574
- [28] Wadsworth TG, Williams JR. Cubital Tunnel External Compression Syndrome. *Br Med J*. 1973; 1:662–666
- [29] Warner MA, Marner ME, Martin JT. Ulnar Neuropathy: Incidence, Outcome, and Risk Factors in

- Sedated or Anesthetized Patients. *Anesthesiology*. 1994; 81:1332–1340
- [30] Alvine FG, Schurrer ME. Postoperative Ulnar-Nerve Palsy: Are There Predisposing Factors? *J Bone Joint Surg*. 1987; 69A:255–259
- [31] Stewart JD, Shantz SH. Perioperative ulnar neuropathies: A medicolegal review. *Can J Neurol Sci*. 2003; 30:15–19
- [32] Warner MA, Martin JT, Schroeder DR, et al. Lower-Extremity Motor Neuropathy Associated with Surgery Performed on Patients in a Lithotomy Position. *Anesthesiology*. 1994; 81:6–12
- [33] O'Donnell D, Rottman R, Kotelko D, et al. Incidence of Maternal Postpartum Neurologic Dysfunction. *Anesthesiology*. 1994; 81
- [34] Sanabria EAM, Nagashima T, Yamashita H, Ehara K, Kohmura E. Postoperative bilateral meralgia paresthetica after spine surgery: An overlooked entity? *Spinal Surgery*. 2003; 17:195–202
- [35] Kent CK, Moscucci M, Gallagher SG, et al. Neuropathies After Cardiac Catheterization: Incidence, Clinical Patterns, and Long-Term Outcome. *J Vasc Surg*. 1994; 19:1008–1014
- [36] Carvalho GA, Nikkhah G, Matthies C, Penkert G, Samii M. Diagnosis of root avulsions in traumatic brachial plexus injuries: value of computerized tomography myelography and magnetic resonance imaging. *J Neurosurg*. 1997; 86:69–76
- [37] Hashimoto T, Mitomo M, Hirabuki N, Miura T, Kawai R, Nakamura H, Kawai H, Ono K, Kozuka T. Nerve root avulsion of birth palsy: comparison of myelography with CT myelography and somatosensory evoked potential. *Radiology*. 1991; 178:841–845
- [38] Rouse DJ, Owen J, Goldenberg RL, Cliver SP. The effectiveness and costs of elective cesarean delivery for fetal macrosomia diagnosed by ultrasound. *JAMA*. 1996; 276:1480–1486
- [39] Gilbert A, Brockman R, Carlioz H. Surgical Treatment of Brachial Plexus Birth Palsy. *Clin Orthop*. 1991; 264:39–47
- [40] Boome RS, Kaye JC. Obstetric Traction Injuries of the Brachial Plexus: Natural History, Indications for Surgical Repair and Results. *J Bone Joint Surg*. 1988; 70B:571–576
- [41] van Ouwerkerk WJ, van der Sluijs JA, Nollet F, Barkhof F, Slooff AC. Management of obstetric brachial plexus lesions: state of the art and future developments. *Childs Nerv Syst*. 2000; 16:638–644
- [42] Piatt JH, Hudson AR, Hoffman HJ. Preliminary Experiences with Brachial Plexus Explorations in Children: Birth Injury and Vehicular Trauma. *Neurosurgery*. 1988; 22:715–723
- [43] Hunt D. Surgical Management of Brachial Plexus Birth Injuries. *Dev Med Child Neurol*. 1988; 30:821–828
- [44] Anand P, Birch R. Restoration of sensory function and lack of long-term chronic pain syndromes after brachial plexus injury in human neonates. *Brain*. 2002; 125:113–122
- [45] Kline DG, Hudson AR. *Nerve Injuries: Operative Results for Major Nerve Injuries, Entrapments, and Tumors*. Philadelphia: W. B. Saunders; 1995
- [46] Wilbourn AJ. The Thoracic Outlet Syndrome is Overdiagnosed. *Arch Neurol*. 1990; 47:328–330

Part XI
Neurophthalmology and
Neurotology

32	Neurophthalmology	558
33	Neurotology	572

XI

32 Neurophthalmology

32.1 Nystagmus

32.1.1 Definition

Involuntary rhythmic oscillation of the eyes, usually conjugate. Most common form is jerk nystagmus, in which the direction of the nystagmus is defined for the direction of the fast (cortical) component (which is not the abnormal component). Horizontal or upward gaze-provoked nystagmus may be due to sedatives or AEDs; otherwise vertical nystagmus is indicative of posterior fossa pathology.

32.1.2 Localizing lesion for various forms of nystagmus

1. seesaw nystagmus: intorting eye moves up, extorting eye moves down, pattern then reverses. Lesion in diencephalon. Also reported with chiasmal compression (occasionally accompanied with bitemporal hemianopia in parasellar masses)
2. convergence nystagmus: slow abduction of eyes followed by adducting (converging) jerks, usually associated with features of Parinaud's syndrome. May be associated with nystagmus retractorius (see below) with similar location of lesion
3. nystagmus retractorius: resulting from co-contraction of all EOM's. May accompany convergence nystagmus. Lesion in upper midbrain tegmentum (usually vascular disease or tumor, especially pinealoma)
4. downbeat nystagmus: nystagmus with the fast phase downward while in primary position. Most patients have a structural lesion in the posterior fossa, especially at the cervicomedullary junction (foramen magnum (FM)),¹ including Chiari I malformation, basilar impression, p-fossa tumors, syringobulbia.² Uncommonly occurs in multiple sclerosis (MS), spinocerebellar degeneration, and in some metabolic conditions (hypomagnesemia, thiamine deficiency, alcohol intoxication or withdrawal, or treatment with phenytoin, carbamazepine or lithium³)
5. upbeat nystagmus: lesion in medulla
6. abducting nystagmus occurs in INO. Lesion in pons (MLF)
7. Brun's nystagmus: lesion in pontomedullary junction (PMJ)
8. vestibular nystagmus: lesion in PMJ
9. ocular myoclonus: lesion in myoclonic triangle
10. periodic alternating nystagmus (PAN): lesion in FM and cerebellum
11. square wave jerks, macro square wave jerks, macro saccadic oscillations. Lesion in cerebellar pathways
12. "nystagmoid" eye movements (not true nystagmus)
 - a) ocular bobbing (p.570): lesion in pontine tegmentum
 - b) ocular dysmetria: overshoot of eye on attempted fixation followed by diminishing oscillations until eye "hones in" on target. Lesion in cerebellum or pathways (may be seen in Friedreich's ataxia)
 - c) ping-pong gaze (p.301)
 - d) "windshield wiper eyes" (p.301)

32.2 Papilledema

32.2.1 General information

AKA choked (optic) disk. Thought to be caused by axoplasmic stasis. One theory: elevated ICP is transmitted through the subarachnoid space of the optic nerve sheath to the region of the optic disc. Elevated ICP will usually obliterate retinal venous pulsation if the pressure is transmitted to the point where the central retinal vein passes through the subarachnoid space (≈ 1 cm posterior to the globe). Papilledema (PPD) may also be dependent on the ratio of retinal arterial to retinal venous pressure, with ratios $<1.5:1$ more commonly associated with papilledema than higher ratios.

Elevated ICP generally causes bilateral papilledema PPD (see below for unilateral papilledema). Papilledema may appear similar to optic neuritis on funduscopy, but the latter is usually associated with more severe visual loss and tenderness to eye pressure over the eye.

Pseudopapilledema may mimic papilledema in that the optic disc may appear swollen, but, unlike true papilledema, the peripapillary vessels are not obscured. It may be unilateral or bilateral. There

are a number of benign conditions that can cause pseudopapilledema (including a small optic cup, buried disc drusen...), and extensive workup is generally not indicated.

Papilledema typically takes 24–48 hours to develop following a sustained rise in ICP. It is rarely seen as early as ≈ 6 hours after onset, but not earlier. Papilledema does not cause visual blurring or reduction of visual fields unless very severe and prolonged.

32.2.2 Differential diagnosis of unilateral papilledema

1. compressive lesions
 - a) orbital tumors
 - b) tumors of optic nerve sheath (meningiomas)
 - c) optic nerve tumors (optic gliomas)
2. local inflammatory disorder
3. Foster Kennedy syndrome (p.99)
4. demyelinating disease (e.g. multiple sclerosis)
5. elevated ICP in the setting of something that prevents manifestation in one eye, including:
 - a) blockage which prevents transmission of elevated CSF pressure to that optic disc⁴
 - b) prosthetic eye (artificial eye)

32.3 Visual fields

32.3.1 General information

Normal human visual field: extends approximately from 35° nasally in each eye, to 90° temporally, and 50° above and below the horizontal meridian. The normal physiological blind spot (due to absence of light receptors in the optic disc due to penetration of the retina by the optic nerve and vessels) is located to the temporal side of the macular visual area in each eye.

32.3.2 Macular sparing/splitting

Macular splitting can occur both in lesions anterior or posterior to the lateral geniculate body (LGB). However, macular sparing tends to occur with lesions posterior to the LGB. Homonymous hemianopia with macular sparing tends to occur with lesions in the optic radiation or infarcts of primary visual cortex. There is more than 1 way for this to occur: input from the macula is spread over a large portion of the optic radiation and primary visual cortex, and the occipital pole (primary visual cortex) receives dual blood supply.

32.4 Visual field deficits

Can be tested by:

1. bedside confrontational testing: detects only gross peripheral field deficits. Stimulus is brought in from non-seeing to seeing field (towards macular vision area) along 8 meridians
2. formal perimetry
 - a) using a tangent screen
 - b) Goldmann perimetry
 - c) automated perimetry exam: Humphrey visual field (HVF)

□ Knee of Wilbrand. (Named for Hermann Wilbrand (1851–1935) German neurophthalmologist – the name of this physician is variously listed as: Hermann or Herman, and erroneously as von Willebrand, Willebrand, or Wildbrand...). A 1-2 mm anterior “bend” of the decussating fibers of the optic chiasm into the contralateral optic nerve before continuing to the optic tract.⁵ Initially identified histologically postmortem in subjects who had monocular enucleation. Optic nerve injury close to the chiasm produces a junctional scotoma involving an ipsilateral nerve fiber bundle defect and contralateral superior temporal quadrantanopia or contralateral temporal hemianopia stemming from injury to the proximal optic nerve and decussating “knee” fibers.^{5,6} Controversy as to the existence or significance of Wilbrand’s knee initially arose after further cadaveric studies suggested that Wilbrand’s knee is an anatomic artifact resulting from buckling of the decussating fibers into the contralateral optic nerve as the optic nerve and chiasm atrophy following enucleation.⁵ However, advanced optical imaging techniques have demonstrated a forward bend of the anterior inferior decussating fibers in chiasms with no pre-mortem pathology.⁷ However, case series with

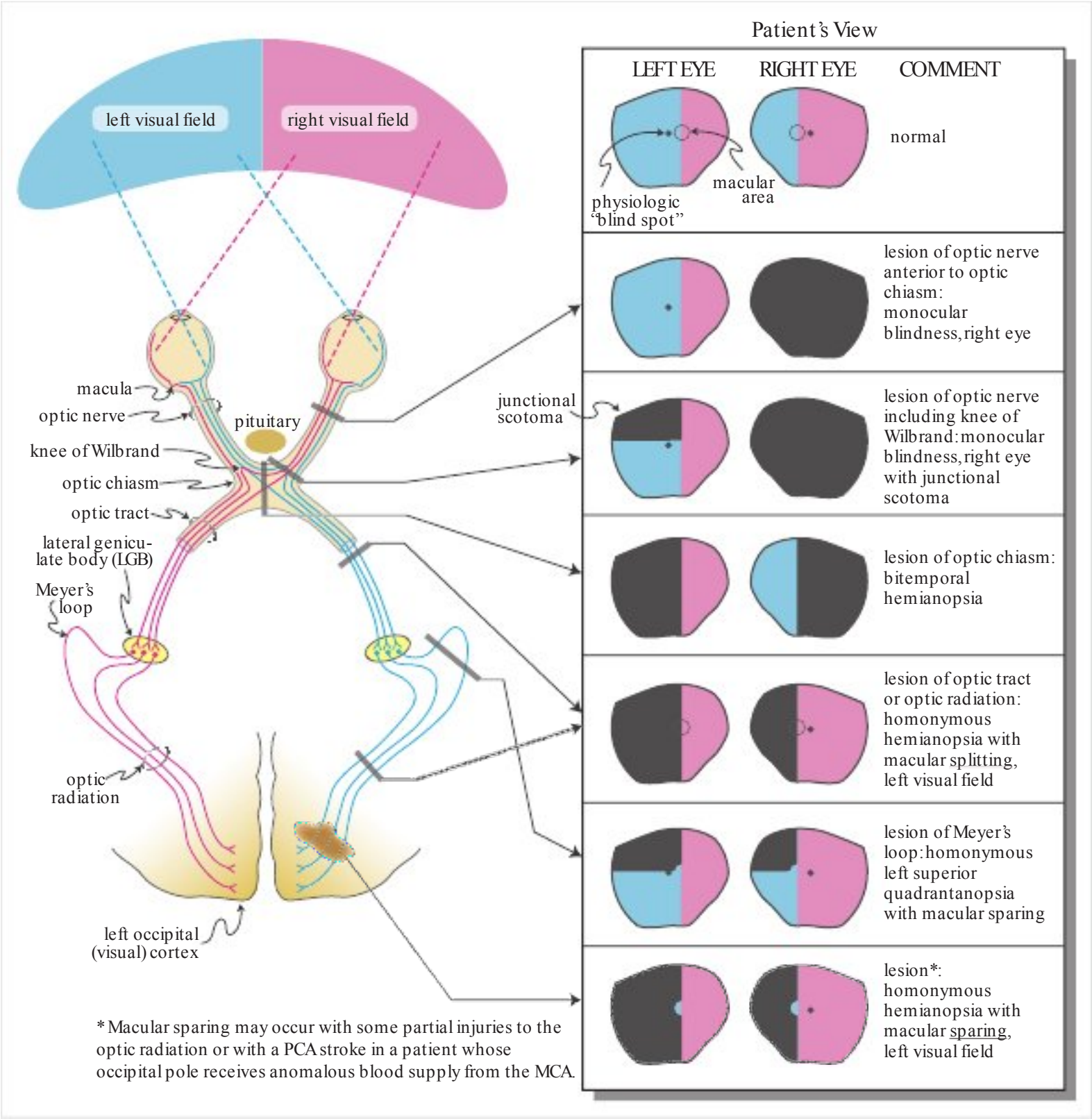


Fig. 32.1 Visual field deficits

intraoperative sectioning of the optic nerve at the level of the chiasm have not been shown to develop contralateral visual field deficits.^{8,9}

- Visual field deficit patterns. □ Fig. 32.1.
1. bitemporal hemianopsia
 2. posterior cerebral artery occlusion → infarction in the anterior visual cortex → contralateral homonymous hemianopia with macular sparing

32.5 Pupillary diameter

32.5.1 Pupilodilator (sympathetic)

Pupilodilator muscle fibers are sympathetic and are arranged radially in the iris.

First-order sympathetic nerve fibers arise in the posterolateral hypothalamus, and descend uncrossed in the lateral tegmentum of the midbrain, pons, medulla and cervical spinal cord to the intermediolateral cell column of the spinal cord from C8-T2 (ciliospinal center of Budge). Here they synapse with lateral horn cells (neurotransmitter: ACh) and give off 2nd order neurons (preganglionics).

Second-order neurons enter the sympathetic chain and ascend but do not synapse until they reach the superior cervical ganglion, where they give rise to 3rd order neurons.

Third-order neurons (postganglionics) course upward with the common carotid artery, those that mediate sweat in the face split off with the ECA. The rest travel with the ICA passing over the carotid sinus. Some fibers accompany V1 (ophthalmic division of trigeminal nerve), passing through (without synapsing) the ciliary ganglion, reaching the pupilodilator muscle of the eye as 2 long ciliary nerves (neurotransmitter: NE). Other fibers from the ICA travel with the ophthalmic artery to innervate the lacrimal gland and Müller's muscle (AKA the orbital muscle).

32.5.2 Pupilloconstrictor (parasympathetic)

Pupilloconstrictor muscle fibers are arranged as a sphincter in the iris.

Parasympathetic preganglionic fibers arise in the Edinger-Westphal nucleus (in high midbrain, superior colliculus level) and are situated peripherally on the intracranial portion of the oculomotor nerve (p.565).

32.5.3 Pupillary light reflex

Mediated by rods and cones of the retina which are stimulated by light, and transmit via their axons in the optic nerve. As with the visual path, temporal retinal fibers remain ipsilateral, whereas nasal retinal fibers decussate in the optic chiasm. Fibers subserving the light reflex bypass the lateral geniculate body (LGB) (unlike fibers for vision which enter the LGB) to synapse in the pretectal nuclear complex at the level of the superior colliculus. Intercalating neurons connect to both Edinger-Westphal parasympathetic motor nuclei. The preganglionic fibers travel within the third nerve to the ciliary ganglion as described above under Pupilloconstrictor (parasympathetic).

Monocular light normally stimulates bilaterally symmetric (i.e. equal) pupillary constriction (ipsilateral response is called direct, contralateral response is consensual).

32.5.4 Pupillary exam

To perform a complete bedside pupillary exam (see following sections for rationale for various aspects of the pupillary exam):

1. measure pupil size in a darkened room: anisocoria augmented in the dark indicates the smaller pupil is abnormal and suggests a sympathetic lesion
2. measure pupil size in a lighted room: anisocoria intensified in the light suggests the larger pupil is abnormal and that the defect is in the parasympathetics
3. note the reaction to bright light (direct and consensual)
4. near response (it is necessary to check this only if the light reaction is not good): the pupil normally constricts on convergence, and this response should be greater than the light reflex (accommodation is not necessary, and a visually handicapped patient can be instructed to follow their own finger as it is brought in)
 - a) light-near dissociation: pupillary constriction on convergence but absent light response (Argyll Robertson pupil). Etiologies:
 - classically described in syphilis
 - Parinaud's syndrome (p.99): dorsal midbrain lesion
 - oculomotor neuropathy (usually causes a tonic pupil as in oculomotor compression, see below): DM, EtOH
 - Adie's pupil: see below
5. swinging flashlight test: alternate the flashlight from one eye to the other with as little delay as possible; watch ≥ 5 seconds for the pupil to redilate (dilation after initial constriction is called pupillary escape and is normal – due to retinal adaptation). Normal: direct and consensual light reflexes are equal. Afferent pupillary defect (see below): consensual reflex is stronger than the direct (i.e. pupil is larger on direct illumination than contralateral illumination)

32.5.5 Alterations in pupillary diameter

Anisocoria

General information

□ Definition. Unequal pupil sizes (usually ≥ 1 mm difference).

□ Note. An afferent pupillary defect (APD) (even with total blindness in one eye) alone does not produce anisocoria (i.e. an APD together with anisocoria indicates two separate lesions).

Evaluation

1. history is critically important. Check for exposure to drugs that affect pupillary size, trauma. Look at old photos (e.g. driver's license) for physiologic anisocoria
2. exam: see Pupillary exam above
3. a non-contrast CT is usually not helpful and can provide a false sense of security

Differential diagnosis

1. physiologic anisocoria: occurs in $\approx 20\%$ of population (more common in people with a light iris). Familial and nonfamilial varieties exist. The difference in pupils is usually <0.4 mm. The inequality is the same in a light and dark room (or slightly worse in the dark)
2. pharmacologic pupil (see below): the most common cause of sudden onset of anisocoria
 - a) mydriatics (pupillary dilators):
 - sympathomimetics (stimulate the dilator pupillae): usually cause only 1–2 mm of dilation, may react slightly to light. Includes: phenylephrine, clonidine, naphazoline (an ingredient in OTC eye drops for allergies), eye contact with cocaine, certain plants (e.g. jimsonweed)
 - parasympatholytics (inhibit the sphincter pupillae): cause maximal dilation (up to 8 mm) that does not react to light. Includes: tropicamide, atropine, scopolamine (including patches for motion sickness), certain plants (e.g. deadly nightshade)
 - b) miotics (pupillary constrictors): pilocarpine, organophosphates (pesticides), flea powders containing anticholinesterase
3. Horner's syndrome: interruption of sympathetics to pupilodilator. The abnormal pupil is the smaller (miotic) pupil. If there is ptosis it will be on the side of the small pupil. See etiologies, etc. (p.564)
4. third nerve palsy (p.565). If there is ptosis, it will be on the side of the large pupil
 - a) oculomotor neuropathy ("peripheral" neuropathy of the third nerve): usually spares the pupil. Etiologies: DM (usually resolves in ≈ 8 weeks), EtOH...
 - b) third nerve compression: tends not to spare pupil (i.e. pupil is dilated). Produces loss of parasympathetic tone. Etiologies include:
 - aneurysm:
 - p-comm (the most common aneurysm to cause this)
 - basilar bifurcation (occasionally compresses the posterior III nerve)
 - uncal herniation: below
 - tumor
 - cavernous sinus lesions: including cavernous internal carotid aneurysm, carotid-cavernous fistula, cavernous sinus tumors
5. Adie's pupil (AKA tonic pupil): see below
6. local trauma to the eye: traumatic iridoplegia. Injury to the pupillary sphincter muscle may produce mydriasis or less often miosis, shape may be irregular
7. pontine lesions
8. eye prosthesis (artificial eye) AKA pseudoanisocoria
9. occasionally some patients have anisocoria that occurs only during migraine¹⁰
10. iritis
11. keratitis or corneal abrasion

Marcus Gunn pupil

AKA (relative) afferent pupillary defect (APD or RAPD), AKA amaurotic pupil. Finding: consensual pupillary reflex to light is stronger than the direct (normal responses are equal). Contrary to some textbooks, the amaurotic pupil is not larger than the other.¹¹ The presence of the consensual reflex is evidence of a preserved third nerve (with parasympathetics) on the side of the impaired direct reflex. Best detected with the swinging flashlight test (see above).

Etiologies

Lesion anterior to the chiasm ipsilateral to the side of the impaired direct reflex:

1. either in the retina (e.g. retinal detachment, retinal infarct e.g. from embolus)
2. or optic nerve, as may occur in:
 - a) optic or retrobulbar neuritis: commonly seen in MS, but may also occur after vaccinations or viral infections, and usually improves gradually
 - b) trauma to the optic nerve: indirect (p.836) or direct
 - c) compression by tumor anterior to the chiasm

Adie's pupil (tonic pupil)

An iris palsy resulting in a dilated pupil, due to impaired postganglionic parasympathetics. Thought to be due to a viral infection of the ciliary ganglion. When associated with loss of all muscle tendon reflexes it is called Holmes-Adie's (is not limited to knee jerks, as some texts indicate). Typically seen in a woman in her twenties.

Slit-lamp exam shows some parts of iris contract and others don't.

These patients exhibit light-near dissociation (see above): in checking near response it is necessary to wait a few seconds.

Denervation supersensitivity: usually occurs after several weeks (not in acute phase). Administer two drops of dilute pilocarpine (0.1–0.125%), a parasympathomimetic in each eye. Miosis (constriction) will occur in Adie's pupil within 30 minutes (normal pupils will react only to $\approx 1\%$ pilocarpine).

Pharmacologic pupil

General information

Follows administration of a mydriatic agent. The mydriatic agent may be "occult" when other care providers have not been alerted that this has been used on a patient (this should always be noted in the chart), or when health care personnel unwittingly inoculate agents, e.g. scopolamine, atropine¹²... into a patient's eye or into their own eye. May present with accompanying H/A, and if it is unknown that a mydriatic is involved, this may be misinterpreted e.g. as a warning of an expanding p-comm aneurysm.

A pharmacologically dilated pupil is very large (7–8 mm), and is larger than typical mydriasis due to third nerve compression (5–6 mm).

To differentiate pharmacologic pupil from a third nerve lesion: instill 1% pilocarpine (a parasympathomimetic) in both eyes (for comparison). A pharmacologic pupil does not constrict, whereas the normal side and a dilated pupil from a third nerve palsy will.

Agents

Drugs intentionally used by physicians to dilate the pupils (e.g. Mydracyl, see below). For other mydriatics, see above.

Management

Option: admit and observe overnight, pupil should normalize.

Using mydriatic agents to produce pupillary dilatation

Indications: to improve the ability to examine the retina. NB: ability to follow bedside examination of pupils will be lost for duration of drug effect. This could mask pupillary dilatation from third nerve compression due to herniation. Always alert other caregivers and place a note in the chart to document that the pupil has been pharmacologically dilated (see above), including the agent(s) used and the time administered.

□: 2 gtt of 0.5% or 1% tropicamide (Mydracyl®) blocks the parasympathetic supply to pupil, and produces a mydriasis that lasts a couple hrs to half a day. This can be augmented with 1 gtt 2.5% phenylephrine ophthalmic (Mydrin®, Neofrin®, Phenoptic® and others) which stimulates the sympathetics.

Oculomotor nerve compression

Third nerve compression may manifest initially with a mildly dilated pupil (5–6 mm). Possible etiologies include uncal herniation or expansion of a p-comm or basilar bifurcation aneurysm. However, within 24 hours, most of these cases will also develop an oculomotor palsy (with down and out deviation of the eye and ptosis). These pupils respond to mydriatics and to miotic agents (the latter helps differentiate this from a pharmacologic pupil, see above).

Although it is possible for a unilaterally dilated pupil alone to be the initial presentation in uncal herniation, in actuality almost all of these patients will have some other finding, e.g. alteration in mental status (confusion, agitation, etc.) before midbrain compression occurs (i.e. it would be rare for a person undergoing early uncal herniation with a dilating pupil to be awake, talking, appropriate and neurologically intact).

Table 32.1 Findings in Horner’s syndrome
• miosis (constricted pupil) • ptosis • enophthalmos • hyperemia of eye • anhidrosis of half of face

Neuromuscular blocking agents (NMBAs)

Due to the absence of nicotinic receptors on the iris, non-depolarizing muscle blocking agents, such as pancuronium (Pavulon®) normally do not alter pupillary reaction to light¹³ except in large doses where some of the first and second order neurons may be blocked.

Paradoxical pupillary reaction

- Pupils constrict when light is removed.
1. congenital stationary night blindness
 2. Best disease: autosomal dominant hereditary progressive macular dystrophy
 3. optic nerve hypoplasia
 4. retinitis pigmentosa

32.5.6 Horner’s syndrome

General information

Horner’s syndrome (HS) is caused by interruption of sympathetics to the eye and face anywhere along their path; see Pupilodilator (sympathetic) (p.560). Unilateral findings on the involved side in a fully developed Horner’s syndrome are shown in □ Table 32.1.

Miosis in HS

The miosis (pupillary constriction) in Horner’s syndrome is only ≈ 2–3 mm. This will be accentuated by darkening the room, which causes the normal pupil to dilate.

Ptosis and enophthalmos

Ptosis is due primarily to paralysis of the superior and inferior tarsal muscles (weakness of the inferior tarsal muscle is technically called “inverse ptosis”). Enophthalmos is due to Müller’s muscle paralysis, which also contributes a maximum of ≈ 2 mm to the ptosis. Ptosis in HS is partial, c.f. complete ptosis which is due to weakness of levator palpebra superioris which is not involved in Horner’s syndrome.

Possible sites of disruption of sympathetics

General information

See also anatomy of 1st, 2nd, and 3rd order sympathetic neurons (p.560).

1st order neuron (central neuron)

Interruption is often accompanied by other brainstem abnormalities. Etiologies of dysfunction: infarction from vascular occlusion (usually PICA), syringobulbia, intraparenchymal neoplasm.

2nd order neuron (preganglionic)

Etiologies of dysfunction: lateral sympathectomies, significant chest trauma, apical pulmonary neoplasms¹⁴ (Pancoast tumor), high thoracic or cervical neuroblastoma.

3rd order neuron (postganglionic)

The most common type. Etiologies of dysfunction: neck trauma, carotid vascular disease/studies – e.g. carotid dissections (p.1324) -, cervical bony abnormalities, migraine, skull base neoplasms, cavernous sinus lesions (e.g. meningioma). With involvement only of fibers on ICA, anhidrosis does not occur (i.e. sweating is preserved) on ipsilateral face since fibers to facial sweat glands travel with ECA.

Pharmacologic testing in Horner's syndrome

Establishing the diagnosis

If the diagnosis of Horner's syndrome is in doubt, the following may be used (not necessary when a pupil lag upon darkening the room can be demonstrated in the affected eye) (does not localize lesion as 1st order, etc.).

Cocaine. □: 1 gtt 4% cocaine OU (not the 10% solution that is commonly used in ENT procedures which will also anesthetize the sphincter pupillae, thus preventing miosis), repeat in 10 min. Observe pupils over 30 min. Cocaine blocks the NE re-uptake of postganglionics at the neuroeffector junction. In HS, no NE is released and cocaine will not dilate eye. If pupil dilates normally, no HS. Delayed dilatation occurs in partial HS.

Apraclonidine ophthalmic (Iopidine®) has essentially replaced cocaine for establishing diagnosis. It works like low-dose pilocarpine works with Adie's pupil. Iopidine causes the miotic pupil to dilate in Horner's syndrome due to denervation hypersensitivity in the pupilodilator muscle fibers.

Localizing the site of the lesion

First order HS usually is accompanied by other hypothalamic, brainstem, or medullary findings.

To differentiate a second from third-order: 1% hydroxyamphetamine (Paradrine®) releases NE from nerve endings at neuroeffector junction causing dilation of pupil except in 3rd order neuron lesions (injured postganglionics do not release NE).

32.6 Extraocular muscle (EOM) system

32.6.1 General information

Cr. N. III (oculomotor) innervates the ipsilateral medial rectus (MR), inferior rectus (IR), inferior oblique (IO), and superior rectus (SR). Cr. N IV (trochlear) innervates the ipsilateral superior oblique (SO), contralateral to the trochlear nucleus (p. 567). Cr. N. VI (abducens) innervates the ipsilateral lateral rectus (LR).

The frontal eye field is the cortical area that initiates voluntary (supranuclear) lateral saccadic eye movements ("pre-programmed", rapid, ballistic) to the opposite side, involved in suppressing reflexive saccades and generating voluntary, non-visual saccades. It is located in Brodmann's area 8 (in the frontal lobe, anterior to the primary motor cortex, □ Fig. 1.1). These corticobulbar fibers pass through the genu of the internal capsule to the paramedian pontine reticular formation (PPRF), which controls horizontal gaze, which sends fibers to the ipsilateral abducens/para-abducens (VI) nuclear complex, and via the medial longitudinal fasciculus (MLF) to the contralateral III nucleus to innervate the contralateral MR. Inhibitory fibers go to the ipsilateral third nerve to inhibit the antagonist MR muscle. Thus, the right PPRF controls lateral eye movements to right.

32.6.2 Internuclear ophthalmoplegia

Internuclear ophthalmoplegia (INO) is due to a lesion of the MLF (see above) rostral to the abducens nucleus. Findings in unilateral INO¹⁵ (see □ Fig. 32.2 for illustration):

1. on attempting to look to the side contralateral to the INO:
 - a) the eye ipsilateral to the lesion fails to ADDuct completely
 - b) abduction nystagmus in the contralateral eye (monocular nystagmus) often with some weakness of ABDuction
2. convergence is not impaired in isolated MLF lesions (INO is not an EOM palsy)

The most common causes of INO:

1. MS: the most common cause of bilateral INO in young adults
2. brainstem stroke: the most common cause of unilateral INO in the elderly

32.6.3 Oculomotor (Cr. N. III) nerve palsy (OMP)

General information

The oculomotor nerve exits the brainstem ventrally and has two components: motor neurons which originate in the oculomotor nucleus, and more peripherally situated parasympathetic fibers which arise from the Edinger-Westphal nucleus. The nerve passes through the cavernous sinus and enters the superior orbital fissure where it divides into a superior division (innervating the superior rectus and the levator palpebrae superioris) and an inferior division (supplying the medial rectus, inferior rectus and inferior oblique). The parasympathetic fibers travel with the inferior division and branch

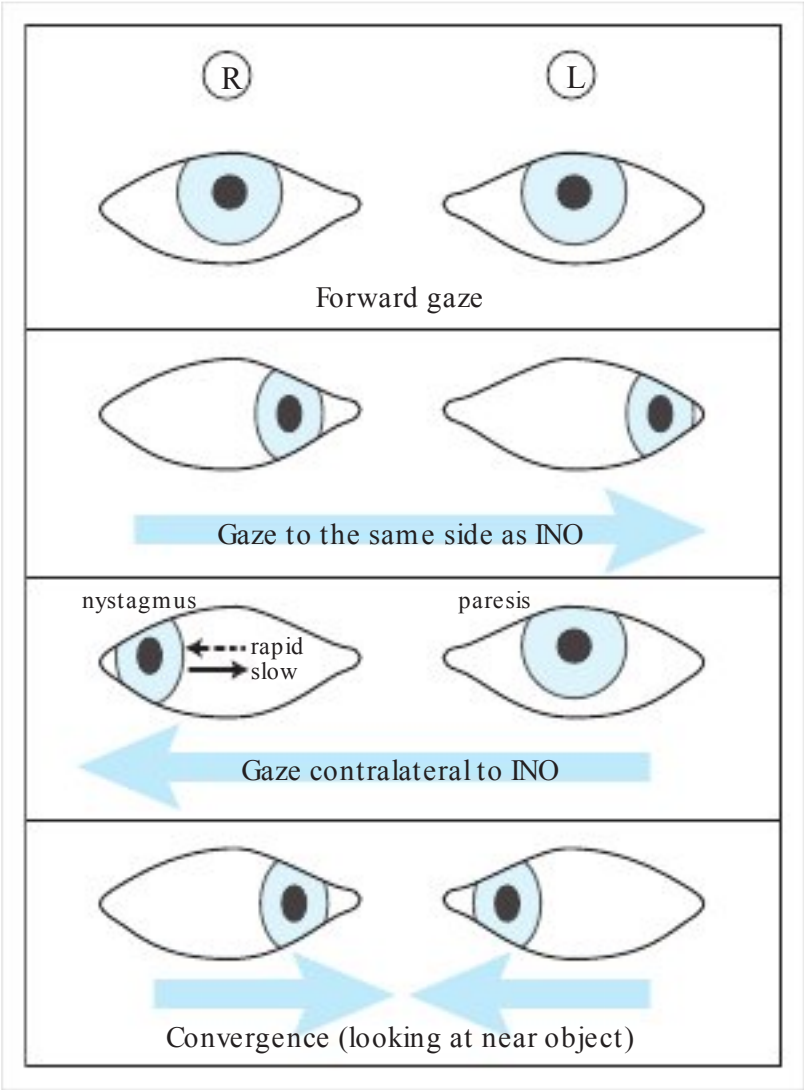


Fig. 32.2 Illustration of gaze findings with a left internuclear ophthalmoplegia

off to the ciliary ganglion where they synapse. Postganglionic fibers enter the posterior globe to innervate the ciliary muscle (relaxes the lens which “thickens” and accommodates for near vision) and the constrictor pupillae muscle.

Oculomotor nerve motor palsy causes ptosis with eye deviated “down & out.” Nuclear involvement of 3rd nerve is rare. NB: 3rd nerve palsy alone can cause up to 3 mm exophthalmos (proptosis) from relaxation of the rectus muscles.

Also see Painful ophthalmoplegia (p.568) and Painless ophthalmoplegia (p.569). For brainstem syndromes, see Benedikt’s syndrome (p.99), and Weber’s syndrome (p.99). Also, see Anisocoria (p.561).

Non pupil-sparing oculomotor palsy

The rule of the pupil in third nerve palsy

Elucidated in 1958 by Rucker. In effect, states “Third nerve palsy due to extrinsic compression of the nerve will be associated with impaired pupillary constriction.” However, it is often overlooked that in 3% the pupil is spared.¹⁶

Etiologies

Most cases are due to extrinsic compression of 3rd nerve. Etiologies include:

1. tumor: the most common tumors affecting 3rd nerve:
 - a) chordomas
 - b) clival meningiomas
2. vascular: the most common vascular lesions:
 - a) aneurysms of p-comm artery (pupil sparing with aneurysmal oculomotor palsy occurs in < 1%). □ Development of a new 3rd nerve palsy ipsilateral to a p-comm aneurysm may be a sign of expansion with the possibility of imminent rupture, and is traditionally considered an indication for urgent treatment
 - b) aneurysms of the distal basilar artery or bifurcation (basilar tip)
 - c) carotid-cavernous fistula (p.1256): look for pulsatile proptosis
3. uncal herniation
4. cavernous sinus lesions: usually cause additional cranial nerve findings (V1, V2, IV, VI); see Cavernous sinus syndrome (p.1401). Classically the third nerve palsy e.g. from enlarging cavernous aneurysm will not produce a dilated pupil because the sympathetics which dilate the pupil are also paralyzed¹ (p.1492)

Pupil sparing oculomotor palsy (pupil reacts to light)

General information

Usually from intrinsic vascular lesions occluding vaso-nervorum causing central ischemic infarction. Spares parasympathetic fibers located peripherally in 3rd nerve in 62–83% of cases.¹⁶

Etiologies

Etiologies include:

1. diabetic neuropathy
2. atherosclerosis (as seen in chronic HTN)
3. vasculopathies: including giant cell arteritis (p.195) (temporal arteritis)
4. chronic progressive ophthalmoplegia: usually bilateral
5. myasthenia gravis

Rarely, pupil-sparing OMP has been described following an intra-axial lesion, as in a midbrain infarction.¹⁷

Other causes of oculomotor palsy

Trauma, uncal herniation, laterally expanding pituitary adenomas, Lyme disease, cavernous sinus lesions: usually cause additional cranial nerve findings; see Multiple cranial nerve palsies (cranial neuropathies) (p.1399).

Lesions within the orbit tend to affect 3rd nerve branches unequally. Superior division lesion → ptosis and impaired elevation; inferior division lesion → impairment of depression, adduction and pupillary reaction.

32.6.4 Trochlear nerve (IV) palsy

Anatomy: the trochlear nucleus lies ventral to the cerebral aqueduct at the level of the inferior colliculi. Trochlear nerve axons pass dorsally around the aqueduct and decussate internally just caudal to the inferior colliculi. The nerve innervates the superior oblique muscle which primarily depresses the adducted eye, but in primary gaze it intorts and secondarily abducts and depresses the globe (i.e. it moves the eye down & out).

Some unique features of the trochlear nerve:

1. the only cranial nerve to decussate internally (i.e. the trochlear nucleus is on the contralateral side to the nerve that exits and to the superior oblique it goes to)
2. the only cranial nerve to exit posteriorly on the brain stem
3. the only cranial nerve passing through the superior orbital fissure that does not pass through the annulus of Zinn (AKA annulus tendineus or annular tendon)

Trochlear palsy results in eye deviation “up and in.” Patients tend to spontaneously tilt the head to the side opposite the IV palsy to “intort” the paretic eye and eliminate the diplopia. Diplopia is exacerbated when looking down (e.g. on descending stairs) especially when also looking inward, or when the examiner tilts the head towards the paretic side.

Isolated fourth nerve palsy is uncommon. It may occasionally occur with lesions of the cerebral peduncle or injury to the floor of the fourth ventricle near the aqueduct.

32.6.5 Abducens (VI) palsy

Produces a lateral rectus palsy. Clinically produces diplopia that is exaggerated with lateral gaze to the side of the palsy. Etiologies of isolated 6th nerve palsy include¹⁸:

1. vasculopathy: including diabetes and giant cell arteritis. Most cases resolve within 3 months (alternative cause should be sought in cases lasting longer)
2. increased intracranial pressure: palsy may occur with increased ICP even in the absence of direct compression of the nerve (a “false localizing” sign in this setting). Postulated to occur due to the fact that the VI nerve has a long intracranial course which may render it more sensitive to increased pressure. May be bilateral. Etiologies include:
 - a) traumatically increased ICP
 - b) increased ICP due to hydrocephalus (p.592), e.g. from p-fossa tumor
 - c) idiopathic intracranial hypertension (pseudotumor cerebri) (p.766)

3. cavernous sinus lesions: cavernous carotid aneurysm (p.1225), neoplasm (meningioma...), carotid cavernous fistula (p.1256)
4. inflammatory:
 - a) Gradenigo's syndrome (p.570), involvement at Dorello's canal
 - b) sphenoid sinusitis: (involvement at Dorello's canal)
5. intracranial neoplasm: e.g. clivus chordoma, chondrosarcoma
6. pseudoabducens palsy: may be due to
 - a) thyroid eye disease: the most common cause of chronic VI palsy. Will have positive forced duction test (eye cannot be moved by examiner)
 - b) myasthenia gravis: responds to edrophonium (Tensilon®) test
 - c) long-standing strabismus
 - d) Duane's syndrome (p.570)
 - e) fracture of the medial wall of the orbit with medial rectus entrapment
7. following lumbar puncture (p.1507): almost invariably unilateral
8. fracture through clivus (p.884)
9. idiopathic

32.6.6 Multiple extraocular motor nerve involvement

Lesions in cavernous sinus (see below) involve cranial nerves III, IV, VI and V1 & V2 (ophthalmic and maxillary divisions of trigeminal nerve), and spare II and V3.

Superior orbital fissure syndrome: dysfunction of nerves III, IV, VI and V1.

Orbital apex syndrome: involves II, III, IV, VI and partial V1.

4th nerve palsy may result from a contrecoup injury in frontal head trauma.

32.6.7 Painful ophthalmoplegia

Definition

Pain and dysfunction of ocular motility (may be due to involvement of one or more of cranial nerves III, IV, V & VI).

32

Etiologies

1. intraorbital
 - a) inflammatory pseudotumor (idiopathic orbital inflammation): see below
 - b) contiguous sinusitis
 - c) invasive fungal sinus infection producing orbital apex syndrome. Rhinocerebral mucormycosis (AKA zygomycosis): sinusitis with painless black palatal or nasal septal ulcer or eschar with hyphal invasion of blood vessels by fungi of the order Mucorales, especially rhizopus.¹⁹ Usually seen in diabetic or immunocompromised patients, occasionally in otherwise healthy patients.²⁰ Often involves dural sinuses and may cause cavernous sinus thrombosis
 - d) mets
 - e) lymphoma
2. superior orbital fissure/anterior cavernous sinus
 - a) Tolosa-Hunt syndrome: see below
 - b) mets
 - c) nasopharyngeal Ca
 - d) lymphoma
 - e) herpes zoster
 - f) carotid-cavernous fistula
 - g) cavernous sinus thrombosis
 - h) intracavernous aneurysm
3. parasellar region
 - a) pituitary adenoma
 - b) mets
 - c) nasopharyngeal Ca
 - d) sphenoid sinus mucocele
 - e) meningioma/chordoma
 - f) apical petrositis (Gradenigo's syndrome): see below
4. posterior fossa
 - a) p-comm aneurysm
 - b) basilar artery aneurysm (rare)

5. miscellaneous
 - a) diabetic ophthalmoplegia
 - b) migrainous ophthalmoplegia
 - c) cranial arteritis
 - d) tuberculous meningitis: may cause ophthalmoplegia, usually incomplete, most often primarily oculomotor nerve

32.6.8 Painless ophthalmoplegia

Differential diagnosis:

1. chronic progressive ophthalmoplegia: pupil sparing, usually bilateral, slowly progressive
2. myasthenia gravis: pupil sparing, responds to edrophonium (Tensilon®) test
3. myositis: usually also produces symptoms in other organ systems (heart, gonads...)

32.7 Neurophthalmologic syndromes

32.7.1 Pseudotumor (of the orbit)

General information

AKA “chronic granuloma” (a misnomer, since true epithelioid granulomas are rarely found). An idiopathic inflammatory disease confined to the orbit that may mimic a true neoplasm. Lymphocytic infiltration of extraocular muscles. Usually unilateral.

Typically presents with rapid onset of proptosis, pain, and EOM dysfunction (painful ophthalmoplegia with diplopia). Often follows URI, may be associated with scleral inflammation. Most commonly involves the superior orbital tissues.

Differential diagnosis

See Orbital lesions (p.1375) for list.

Key points for Graves’ disease (GD): the histologic appearance of GD (hyperthyroidism) may be indistinguishable from pseudotumor. Involvement with GD is usually bilateral.

Treatment

Surgery tends to cause a flare up, and is thus usually best avoided.

Steroids are the treatment of choice. □: 50–80 mg prednisone q d. Severe cases may necessitate treatment with 30–40 mg/d for several months.

Radiation treatment with 1000–2000 rads may be needed for cases of reactive lymphocytic hyperplasia.

32.7.2 Tolosa-Hunt syndrome

Nonspecific inflammation in the region of the superior orbital fissure, often with extension into the cavernous sinus, sometimes with granulomatous features. A diagnosis of exclusion. May be a topographical variant of orbital pseudotumor (see above). Clinical diagnostic criteria:

1. painful ophthalmoplegia
2. involvement of any nerve traversing the cavernous sinus. The pupil is usually spared (frequently not the case with aneurysms, specific inflammation, etc.)
3. symptoms last days to weeks
4. spontaneous remission, sometimes with residual deficit
5. recurrent attacks with remissions of months or years
6. no systemic involvement (occasional N/V, due to pain?)
7. dramatic improvement with systemic steroids: 60–80 mg prednisone PO q day (slow taper), relief within about 1 day
8. occasional inflammation of rectus muscle from contiguous inflammation

32.7.3 Raeder’s paratrigeminal neuralgia

Two essential components²¹:

1. unilateral oculosympathetic paresis (AKA partial Horner’s syndrome (HS), this usually lacks anhidrosis, and in this syndrome, possibly ptosis also)

2. homolateral trigeminal nerve involvement (usually tic-like pain, but may be analgesia or masseter weakness; pain, if present, must be tic-like and does not include e.g. unilateral head, face or vascular pain)

Localizing value of syndrome: region adjacent to trigeminal nerve in middle fossa. The cause is often not determined, but may rarely be due to aneurysm²² compressing V1 with sympathetics.

32.7.4 Gradenigo's syndrome

AKA apical petrositis. Mastoiditis with involvement of petrous apex (if pneumatized). Usually seen by ENT physicians. Classic triad:

1. abducens palsy: from inflammation of 6th nerve at Dorello's canal, which is where it enters the cavernous sinus just medial to the petrous apex
2. retro-orbital pain: due to inflammation of V1
3. draining ear

32.8 Miscellaneous neurophthalmologic signs

□ Corneal mandibular reflex. Eliciting the corneal reflex produces a jaw jerk or contralateral jaw movement (ipsilateral pterygoid contraction). A primitive pontine reflex, may be seen in a variety of insults to the brain (trauma, intracerebral hemorrhage...).

□ Duane syndrome. AKA retraction syndrome: paradoxical innervation causing co-contraction of the lateral and medial rectus muscles on attempted adduction with relaxation on abduction, produces mild enophthalmos with pseudoptosis. May be congenital (e.g. part of one of the following syndromes: acrorenal-ocular syndrome, Okihiro syndrome...).

□ Hippus. Rhythmic, irregular pupillary oscillations, changing by ≥ 2 mm. May confuse examination when checking pupillary responses; record the initial response. May be normal. No localizing value.

□ Marcus Gunn phenomenon. Not to be confused with Marcus Gunn pupil (p.562). Opening the mouth causes opening of a ptotic eye (abnormal reflex between proprioception of pterygoid muscles and third nerve). Reverse Marcus Gunn phenomenon: normal eye that closes with opening the mouth. Seen only in patients with peripheral facial nerve injuries, and probably results from aberrant regeneration.

□ Ocular bobbing²³. Abrupt, spontaneous, conjugate downward eye deviation with slow return to midposition, 2 to 12 times per min. It is associated with bilateral paralysis of horizontal gaze, including to doll's-eyes and calorics. Most commonly seen with destructive lesions of the pontine tegmentum (usually hemorrhage, but also infarction, glioma, trauma), but has also been described with compressive lesions.²⁴ Atypical bobbing is similar except that horizontal gaze is preserved, and can be seen with cerebellar hemorrhage, hydrocephalus, trauma, metabolic encephalopathy...

□ Opsoclonus²⁵. (Rare.) Rapid, conjugate, irregular, non-rhythmic (differentiates this from nystagmus) eye movements vertically or horizontally, persist (attenuated) during sleep (opsochoria if dysconjugate). Usually associated with diffuse myoclonus (fingers, chin, lips, eyelid, forehead, trunk and IEs); also, malaise, fatigability, vomiting and some cerebellar findings. Often resolves spontaneously within 4 mos.

□ Oscillopsia. Visual sensation that stationary objects are swaying side-to-side or vibrating.²⁶ Rarely the sole manifestation of Chiari I malformation²⁷ (often associated with downbeat nystagmus). Other causes include MS, or injury to both vestibular nerves; e.g. aminoglycoside ototoxicity,²⁸ bilateral vestibular neurectomies, see Dandy's syndrome (p.572).

□ Pseudo von Grafe sign. Lid retraction on downward gaze (true von Grafe sign is lid lag in hyperthyroidism) seen in aberrant nerve regeneration (inferior rectus innervation → activation of levator palpebrae).

□ Optic atrophy. Chronic, progressive optic atrophy is due to a compressive lesion (aneurysm, meningioma, osteopetrosis...) until proven otherwise.

References

- [1] Wilkins RH, Rengachary SS. Neurosurgery. New York 1985
- [2] Pinel JF, Larmande P, Guegan Y, et al. Down-Beat Nystagmus: Case Report with Magnetic Resonance Imaging and Surgical Treatment. Neurosurgery. 1987; 21:736–739
- [3] Williams DP, Troost BT, Rogers J. Lithium-Induced Downbeat Nystagmus. Arch Neurol. 1988; 45:1022–1023
- [4] Sher NA, Wirtschafter J, Shapiro SK, et al. Unilateral Papilledema in 'Benign' Intracranial Hypertension (Pseudotumor Cerebri). JAMA. 1983; 250:2346–2347
- [5] Horton JC. Wilbrand's knee of the primate optic chiasm is an artefact of monocular enucleation. Trans Am Ophthalmol Soc. 1997; 95:579–609
- [6] Grzybowski A. Harry Moss Traquair (1875-1954), Scottish ophthalmologist and perimetrist. Acta Ophthalmol. 2009; 87:455–459
- [7] Shin RK, Li TP. Visualization of Wilbrand's knee. Snowbird, UT 2013
- [8] Lee JH, Tobias S, Kwon JT, Sade B, Kosmorsky G. Wilbrand's knee: does it exist? Surg Neurol. 2006; 66:11–7; discussion 17
- [9] Zweckberger K, Unterberg AW, Schick U. Pre-chiasmatic transection of the optic nerve can save contralateral vision in patients with optic nerve sheath meningiomas. Clin Neurol Neurosurg. 2013; 115:2426–2431
- [10] Kawasaki A. Physiology, assessment, and disorders of the pupil. Curr Opin Ophthalmol. 1999; 10:394–400
- [11] Walsh FB, Hoyt WF. Clinical Neuro-Ophthalmology. Baltimore 1969
- [12] Nakagawa TA, Guerra L, Storgion SA. Aerosolized atropine as an unusual cause of anisocoria in a child with asthma. Pediatr Emerg Care. 1993; 9:153–154
- [13] Wijdicks EF. Determining Brain Death in Adults. Neurology. 1995; 45:1003–1011
- [14] Lepore FE. Diagnostic Pharmacology of the Pupil. Clin Neuropharmacol. 1985; 8:27–37
- [15] Zee DS. Internuclear ophthalmoplegia: pathophysiology and diagnosis. Baillieres Clin Neurol. 1992; 1:455–470
- [16] Trobe JD. Third nerve palsy and the pupil. Footnotes to the rule. Arch Ophthalmol. 1988; 106:601–602
- [17] Breen LA, Hopf HC, Farris BK, Gutmann L. Pupil-Sparing Oculomotor Nerve Palsy due to Midbrain Infarction. Arch Neurol. 1991; 48:105–106
- [18] Galetta SL, Smith JL. Chronic Isolated Sixth Nerve Palsies. Arch Neurol. 1989; 46:79–82
- [19] DeShazo RD, Chapin K, Swain RE. Fungal Sinusitis. N Engl J Med. 1997; 337:254–259
- [20] Radner AB, Witt MD, Edwards JE. Acute Invasive Rhinocerebral Zygomycosis in an Otherwise Healthy Patient: Case Report and Review. Clin Infect Dis. 1995; 20:163–166
- [21] Mokri B. Raeder's Paratrigeminal Syndrome. Arch Neurol. 1982; 39:395–399
- [22] Kashihara K, Ito H, Yamamoto S, et al. Raeder's Syndrome Associated with Intracranial Internal Carotid Artery Aneurysm. Neurosurgery. 1987; 20:49–51
- [23] Fisher CM. Ocular Bobbing. Arch Neurol. 1964; 11:543–546
- [24] Sherman DG, Salmon JH. Ocular Bobbing with Superior Cerebellar Artery Aneurysm: Case Report. J Neurosurg. 1977; 47:596–598
- [25] Smith JL, Walsh FB. Opsoclonus - Ataxic Conjugate Movements of the Eyes. Arch Ophthalmol. 1960; 64:244–250
- [26] Brickner R. Oscillopsia: A new symptom commonly occurring in multiple sclerosis. Arch Neurol Psychiatry. 1936; 36:586–589
- [27] Gingold SI, Winfield JA. Oscillopsia and Primary Cerebellar Ectopie: Case Report and Review of the Literature. Neurosurgery. 1991; 29:932–936
- [28] Marra TR, Reynolds NC, Stoddard JJ. Subjective Oscillopsia ("Jiggling Vision") Presumably Due to Aminoglycoside Ototoxicity: A Report of Two Cases. J Clin Neuro Ophthalmol. 1988; 8:35–38

33 Neurotology

33.1 Dizziness and vertigo

33.1.1 Differential diagnosis of dizziness

- Near syncope. Some overlap with syncope; see Syncope and apoplexy (p.1396)
 1. orthostatic hypotension
 2. cardiogenic hypotension
 - a) arrhythmia
 - b) valvular disease
 3. vasovagal episode
 4. hypersensitive carotid sinus; see Syncope and apoplexy (p.1396)
- Dysequilibrium
 1. multiple sensory deficits: e.g. peripheral neuropathy, visual impairment
 2. cerebellar degeneration
- Vertigo . Sensation of movement (usually spinning)
 1. inner ear dysfunction
 - a) labyrinthitis
 - b) Meniere's disease (see below)
 - c) trauma: endolymphatic leak
 - d) drugs: especially aminoglycosides
 - e) benign (paroxysmal) positional vertigo¹: AKA cupulolithiasis. Attacks of severe vertigo when the head is turned to certain positions (usually in bed). Due to calcium concretions in the semicircular canals. Self limited (most cases do not last > 1 year). No hearing loss
 - f) syphilis
 - g) vertebrobasilar insufficiency (p.1305)
 2. vestibular nerve dysfunction
 - a) vestibular neuritis: sudden onset of vertigo, gradual improvement
 - b) compression:
 - meningioma
 - vestibular schwannoma: usually slowly progressive ataxia instead of severe vertigo. BAER latencies usually abnormal. CT or MRI usually abnormal
 3. disabling positional vertigo : as described by Jannetta et al.,² constant disabling positional vertigo or dysequilibrium, causing $\approx$ constant nausea, no vestibular dysfunction nor hearing loss (tinnitus may be present). One possible cause is vascular compression of the vestibular nerve which may respond to microvascular decompression
 4. brainstem dysfunction
 - a) vascular disease; see Vertebrobasilar insufficiency (p.1305): less distinct vestibular symptoms, prominent nonvestibular symptoms
 - b) migraine: especially basilar artery migraine
 - c) demyelinating disease: e.g. multiple sclerosis
 - d) drugs: anticonvulsants, alcohol, sedatives/hypnotics, salicylates
 5. dysfunction of cervical proprioceptors: as in cervical osteoarthritis
- Poorly defined lightheadedness. Mostly psychiatric. May also include:
 1. hyperventilation
 2. hypoglycemia
 3. anxiety neurosis
 4. hysterical

33.1.2 Vestibular neurectomy

General information

Complete loss of vestibular function from one side is thought to produce transient vertigo due to the mismatch of vestibular input from the two ears. Theoretically, a central compensatory mechanism (the "cerebellar clamp") results in the amelioration of symptoms. In cases of unilateral fluctuating vestibular dysfunction, this compensatory mechanism may be impaired. Unilateral selective vestibular neurectomy (SVN) may convert the fluctuating or partial loss to a complete cessation of input

and facilitate compensation. Bilateral SVN is often complicated by oscillopsia (p.570) – AKA Dandy’s syndrome, with difficulty in maintaining balance in the dark due to loss of the vestibulo-ocular reflex – and is to be avoided.

Indications

The two conditions for which SVN is most commonly employed are Meniere’s disease (see below) and partial vestibular injury (viral or traumatic). SVN may be indicated in disabling cases refractory to medical or non-destructive surgical treatment when vestibular studies demonstrate continued or progressive uncompensated vestibular dysfunction.³

SVN preserves hearing and in Meniere’s disease is >90% effective in eliminating episodic vertiginous spells ($\approx$ 80% success rate in non-Meniere’s cases), but is unlikely to improve stability with rapid head movement.

Surgical approaches for SVN

1. retrolabyrinthine, AKA postauricular approach: anterior to sigmoid sinus. Primary choice in patients with Meniere’s disease who have not had previous endolymphatic sac (ELS) procedures since it permits simultaneous SVN and decompression of the endolymphatic sac. Requires mastoidectomy with skeletonization of the semicircular canals and ELS. The dural opening is bounded anteriorly by the posterior semicircular canal, posteriorly by the sigmoid sinus. Water-tight dural closure is difficult
2. retrosigmoid, AKA posterior fossa, AKA suboccipital approach: posterior to sigmoid sinus. The original approach plied by Dandy in pre-microsurgical era, usually sacrificed hearing, and occasionally facial nerve function. Better results today with microscopic techniques. Indicated for cases other than Meniere’s disease where there is no need for identification of the ELS. Also the best approach for positive identification of eighth nerve
3. middle fossa (extradural) approach: the fibers of the vestibular division may be more segregated from the cochlear fibers in the IAC than in the CPA, thus permitting more complete section of the vestibular nerve. May be appropriate for failed response to SVN by the above approaches. Disadvantages: requires temporal lobe retraction, does not allow exposure of ELS, and higher morbidity and risk of damage to facial nerve⁴ than retrolabyrinthine approach

Surgical considerations for selective vestibular neurectomy

(Also □ Fig. 1.7)

1. the vestibular nerve is in the superior half of the eighth nerve complex, and is slightly more gray in color than the cochlear division (due to less myelin⁵). They may be separated by a small vessel or by an indentation in the bundle
2. facial (VII) nerve:
 - a) whiter than the VIII nerve complex
 - b) lies anterior and superiorly to the VIII nerve
 - c) EMG monitoring of the facial nerve is recommended
 - d) direct stimulation confirms the identification
3. any vessels present on eighth nerve bundle must be preserved to save hearing (primarily, the artery of the auditory canal must be preserved)
4. if no plane of cleavage can be defined between vestibular & cochlear divisions, the superior half of the nerve bundle is divided
5. the endolymphatic sac lies $\approx$ midway between the posterior edge of the internal auditory meatus and the sigmoid sinus

33.2 Meniere’s disease

33.2.1 General information

Key concepts

- increased endolymphatic pressure
- clinical triad: vertigo, tinnitus & fluctuating hearing loss
- surgical options for failure of medical management include endolymphatic shunt or selective vestibular neurectomy

Probably due to a derangement of endolymphatic fluid regulation (a consistent finding is endolymphatic hydrops: increased endolymphatic volume and pressure with dilatation of endolymph spaces), with resultant fistulization into the perilymphatic spaces.

33.2.2 Epidemiology

Incidence $\approx$ 1 per 100,000 population.⁶ Most cases have onset between 30–60 years of age, rarely in youth or in the elderly. May become bilateral in 20%

33.2.3 Clinical

Clinical triad

1. attacks of violent vertigo (due to vestibular nerve dysfunction): usually the earliest and the most disabling symptom. Nausea, vomiting, and diaphoresis are frequent concomitants. Severe attacks may cause prostration. Vertigo may persist even after complete deafness. Balance is normal between attacks
2. tinnitus: often described as resembling the sound of escaping steam, not a true “ringing”
3. fluctuating low frequency hearing loss: may fluctuate for a periods of weeks to years, and may progress to permanent deafness if untreated (a sensation of fullness in the ear is commonly described,⁷ however, this is nonspecific and may occur with hearing loss for any reason)

Other clinical features

Drop attacks (“otolithic crises of Tumarkin”) occasionally occur.

Attack duration: $\approx$ 5–30 minutes (some say 2–6 hours), with a “post-ictal” period of fatigue lasting several hrs.

Frequency: varies from one or two attacks a year to several times per week.

Two subtypes differ from classical form: vestibular Meniere’s (episodic vertigo with normal hearing) and cochlear Meniere’s (few vestibular symptoms).

Natural course of syndrome is characterized by periods of remission. Eventually the vertiginous attacks either progress in severity, or “burn out” (being replaced by constant unsteadiness⁷).

Differential diagnosis

Also see Differential diagnosis: Dizziness and vertigo (p.572) for more details.

1. benign (paroxysmal) positional vertigo: AKA cupulolithiasis. Self limited (most cases last $<$ 1 year). No hearing loss
2. disabling positional vertigo: constant disabling positional vertigo or dysequilibrium, $\approx$ constant nausea, no vestibular dysfunction nor hearing loss (tinnitus may be present)
3. vestibular schwannoma: usually slowly progressive ataxia instead of episodic severe vertigo. BAER latencies usually abnormal. CT or MRI usually positive
4. vestibular neuronitis: sudden onset of vertigo with gradual improvement
5. vertebrobasilar insufficiency (VBI) (p.1305): less distinct vestibular symptoms, and prominence of nonvestibular symptoms

Diagnostic studies

1. electronystagmography (ENG) with bithermal caloric stimulation usually abnormal, may show blunted thermal responses
2. audiogram: low frequency hearing loss, fairly good preservation of discrimination and loudness recruitment, negative tone decay on impedance testing
3. BAER usually shows normal latencies
4. radiographic imaging (CT, MRI, etc.): no findings in Meniere’s disease
5. in bilateral cases, a VDRL should be checked to R/O luetic disease

Treatment

Medical treatment

1. reduced intake of salt (strict salt restriction is as effective as any medication) and caffeine
2. diuretics: taken daily until ear fullness abates, then PRN ear pressure (usually once or twice weekly suffices)

- a) acetazolamide: □ Diamox® sequels 500 mg p.o. q d × 1 week, increase to BID if symptoms persist. D/C if paresthesias develop. Do not use during 1st trimester of pregnancy
- 3. vestibular suppressants
 - a) diazepam (Valium®): probably the most effective
 - b) meclizine HCl (Antivert®): □ Adult dose for vertigo associated with the vestibular system (during attacks): 25–100 mg/day PO divided. Dose for motion sickness: 25–50 mg PO one hr prior to stimulus. Supplied: 12.5, 25 & 50 mg tabs. Side effects: drowsiness
- 4. vasodilators: postulated to be mediated by increased cochlear blood flow: inhalation of 5–10% CO₂ works well, but relief is short lived

Surgical treatment

Reserved for incapacitating cases refractory to medical management. When functional hearing exists, procedures that spare hearing are preferred because of high incidence of bilateral involvement. Procedures include:

- 1. endolymphatic shunting procedures: to mastoid cavity (Arenberg shunt) or to subarachnoid space. Reserved for cases with serviceable hearing. ≈ 65%success rate (see below). If symptoms are relieved ≥ 1 year, then a recurrence would be treated by shunt revision, if < 1 year then vestibular neurectomy
- 2. direct application of corticosteroids to the inner ear
- 3. nonselective vestibular ablation (in cases with nonserviceable hearing on the side of involvement)
 - a) surgical labyrinthectomy
 - b) middle ear perfusion with gentamicin
 - c) translabyrinthine section of the 8th nerve
- 4. selective vestibular neurectomy (p.572), in cases with serviceable hearing

Outcome

Endolymph shunting procedures

Outcomes from 112 endolymphatic shunting procedures are shown in □ Table 33.1.

Neurectomy procedures

Vestibulocochlear nerve section (based on early posterior fossa surgery by Dandy; entire eighth nerve bundle was sectioned in 587 patients; all were deaf post-op): 90%relieved of vertigo, 5% unchanged and 5%worse; 9%incidence of facial paralysis (3%incidence of permanent paralysis).

Selective vestibular nerve section (sparing cochlear portion, 95 patients from Dandy): 10%had improved hearing, 28%unchanged, 48%worse, 14%deaf.

Retrolabyrinthine approach: in 32 patients with Meniere’s syndrome (25 failed endolymph shunt) responding to survey, 85%had complete relief of vertigo, 6%improved, 9%no relief (one of whom responded to middle fossa neurectomy).⁵

Complications and untoward effects

Patients with little vestibular nerve function pre-op (determined by ENG) usually have little difficulty immediately following vestibular neurectomy; patients with more function may have a transient worsening post-op until they accommodate.

Among 42 patients undergoing retrolabyrinthine approach: none lost hearing as a result of surgery, no facial weakness, one CSF rhinorrhea requiring re-operation, and one meningitis with good outcome.⁵

Table 33.1 Outcome in 112 endolymphatic-subarachnoid shunts ⁷				
	Vertigo	Tinnitus	Hearing ^a	Ear pressure
improved	79 (70%) ^b	53 (47%)	19 (17%)	57 (51%)
stable	33 (29%)	49 (43%)	50 (45%)	24 (21%)
worse	(none)	10 (10%)	39 (35%)	31 (28%)
^a improved hearing considered serviceable (50 dBpure tone, 70%speech discrimination); additional 4 patients had improved but non-serviceable hearing				
^b 5 patients had recurrence of vertigo after 1–3 years				

In post-op failures, check ENG. If any vestibular nerve function is demonstrated on operated side, then the nerve section was incomplete; consider re-operating.

33.3 Facial nerve palsy

33.3.1 Severity grading

Severity of facial palsy is often graded with the House and Brackmann scale (see □ Table 41.3).

33.3.2 Localizing site of lesion

Central facial palsy (AKA supranuclear facial palsy)

The cortical representation for facial movement occurs in the motor strip along the lateral aspect (just above the most inferior opercular portion of the precentral gyrus). The keys to differentiating central paralysis (due to supranuclear lesions) from peripheral facial palsy are that central palsies:

1. are confined primarily to the lower face due to some bilateral cortical representation of upper facial movement
2. may spare emotional facial expression⁸ (e.g. smiling at a joke)

Nuclear facial palsy

The motor nucleus of the seventh nerve is located at the pontomedullary junction. Nuclear VII palsy results in paralysis of all VII nerve motor function. In nuclear facial palsies, other neurologic findings also often occur from involvement of adjacent neural structures by the underlying process (stroke, tumor...), e.g. in Millard-Gubler syndrome (p.99), there is ipsilateral abducens palsy + contralateral limb weakness. Tumors invading the floor of the 4th ventricle (e.g. medulloblastoma) may also cause nuclear facial palsy (from involvement of facial colliculus in the floor of 4th ventricle).

Facial nerve lesion

Motor fibers ascend within the pons and form a sharp bend (“internal genu”) around the sixth nerve (abducens) nucleus, forming a visible bump in the floor of the 4th ventricle (facial colliculus). The seventh nerve exits from the brain stem at the ponto-medullary junction (□ Fig. 2.6) where it may be involved in CPA tumors. It enters the supero-anterior portion of the internal auditory canal (□ Fig. 1.7). The geniculate ganglion (“external genu”) is located within the temporal bone. The first branch from the ganglion is the greater superficial petrosal nerve (GSPN) which passes to the pterygopalatine ganglion and innervates the nasal and palatine mucosa and the lacrimal gland of the eye; lesions proximal to this point produce a dry eye. The next branch is the branch to the stapedius muscle; lesions proximal to this point produce hyperacusis. Next, the chorda tympani joins the facial nerve bringing taste sensation from the anterior two-thirds of the tongue. Basal skull fractures may injure the nerve just proximal to this point. Travelling with the chorda tympani are fibers to the submandibular and sublingual glands. The facial nerve exits the skull at the stylomastoid foramen. It then enters the parotid gland, where it splits into the following branches to the facial muscles (cranial to caudal): temporal, zygomatic, buccal, mandibular, and cervical. Lesions within the parotid gland (e.g. parotid tumors) may involve some branches but spare others.

33.3.3 Etiologies

These etiologies produce primarily facial nerve palsy, also see Multiple cranial nerve palsies (cranial neuropathies) (p.1399). Note: 90–95% of all cases of facial palsy are accounted for by Bell’s palsy, herpes zoster oticus, and trauma (basal skull fractures).⁹

1. Bell’s palsy (p.577)
2. herpes zoster oticus (auris) (p.578)
3. trauma: basal skull fracture
4. birth:
 - a) congenital
 - *bilateral facial palsy (facial diplegia) of Möbius syndrome (p.1399): unique in that it affects upper face more than lower face
 - *congenital facial diplegia may be part of facioscapulohumeral or myotonic muscular dystrophy
 - b) traumatic

- 5. otitis media: with acute otitis media, facial palsy usually improves with antibiotics. With chronic suppurative otitis surgical intervention is required
- 6. central facial paralysis and nuclear facial paralysis: see Localizing site of lesion above
- 7. neoplasm: usually causes hearing loss, and (unlike Bell’s palsy) slowly progressive facial paralysis
 - a) most are either benign schwannomas of the facial or auditory nerve, or malignancies metastatic to the temporal bone. Facial neuromas account for ≈ 5% of peripheral facial nerve palsies¹⁰; the paralysis tends to be slowly progressive
 - b) parotid tumors may involve some branches but spare others
 - c) Masson’s vegetant intravascular hemangioendothelioma (p.770)
- 8. *neurosarcoidosis (p.189): VII is the most commonly affected cranial nerve
- 9. diabetes: 17% of patients >40 yrs old with peripheral facial palsy (PFP) have abnormal glucose tolerance tests. Diabetics have 4.5 times the relative risk of developing PFP than nondiabetics¹¹
- 10. *stage II Lyme disease (p.334) ¹²: facial diplegia is a hallmark
- 11. *Guillain-Barré syndrome: facial diplegia occurs in ≈ 50% of fatal cases
- 12. occasionally seen in Klippel-Feil syndrome
- 13. *isolated 4th ventricle (p.402): compression at the facial colliculus

* Items with an asterisk are often associated with facial diplegia (i.e. bilateral facial palsy), see also multiple cranial neuropathies (p.1399).

33.3.4 Bell’s palsy

General information

Bell’s palsy (BP), AKA idiopathic peripheral facial palsy (PFP), is the most common cause of facial paralysis (50–80% of PFPs). Incidence: 150–200/1-million/yr.

Etiology: by definition, PFP is called Bell’s palsy when it is not due to known causes of PFP (e.g. infection, tumor or trauma) and there are no other neurological (e.g. involvement of other cranial nerves) or systemic manifestations (e.g. fever, diabetes, possibly hypertension¹³).¹⁴ Thus, true BP is idiopathic, and is a diagnosis of exclusion. Most cases probably represent a viral inflammatory demyelinating polyneuritis¹⁵ usually due to the herpes simplex virus.¹⁶ Facial palsy due to Lyme disease can usually be recognized on clinical grounds.¹⁷ Severity may be graded on the House & Brackmann grading scale (see □ Table 41.3).

Presentation

A viral prodrome is frequent: URI, myalgia, hypesthesia or dysesthesia of the trigeminal nerve, N/V, diarrhea... Paralysis may be incomplete and remain so (Type I); it is complete at onset in 50%(Type II), the remainder progress to completion in 1 week. Usually exhibits distal to proximal progression: motor branches, then chorda tympani (loss of taste and decreased salivation), then stapedial branch (hyperacusis), then geniculate ganglion (decreased tearing). Associated symptoms are shown in □ Table 33.2, and are usually, but not always, ipsilateral. Herpes zoster vesicles develop in 4% of patients 2–4 days after onset of paralysis; and in 30% of patients 4–8 days after onset. During the recovery phase excessive lacrimation may occur (aberrant nerve regeneration).

Evaluation

Patients with PFP should be examined at an early stage to optimize outcome.

Electrodiagnostics: EMG may detect re-innervation potentials, aids prognostication. Nerve conduction study: electrical stimulation of the facial nerve near the stylomastoid foramen while recording EMG in facial muscles (a facial nerve may continue to conduct for up to ≈ 1 week even after complete transection).

Table 33.2 Associated symptoms with Bell’s palsy	
Symptom	%
facial & retroauricular pain	60%
dysgeusia	57%
hyperacusis	30%
reduced tearing	17%

Management

General measures

Eye protection: protection of the eye is critical. Artificial tears during the day, eye ointment at night, avoid bright light (using dark glasses during the day).

Medical management

Steroids: prednisolone 25 mg p.o. BID $\times$ 10 days, started within 72 hours of onset of symptoms, improves the chances of complete recovery at 3 & 9 months.

Acyclovir: does not help (alone or in combination with prednisolone).¹⁸

Surgical management

Surgical decompression: controversial. The definitive study has not been done. Rarely utilized. Indications may include:

1. complete facial nerve degeneration without response to nerve stimulation (although this absence is also used as an argument against surgery⁹)
2. progressively deteriorating response to nerve stimulation
3. no clinical nor objective (nerve testing) improvement after 8 wks (however, in cases where the diagnosis of Bell's palsy is felt to be certain, the active disease will have abated by $\approx$ 14 days after onset⁹)

Prognosis

Essentially all cases show some recovery (if none by 6 mos, other etiologies should be sought). Extent of recovery: 75–80% of cases recover completely, 10% partial, remainder poor. If recovery begins by 10–21 d, tends to be complete; if not until 3–8 wks $\rightarrow$ fair, if not until 2–4 mos $\rightarrow$ poor recovery. If paralysis is complete at onset, 50% will have incomplete recovery. Cases of incomplete paralysis at onset that do not progress to complete paralysis $\rightarrow$ complete recovery; incomplete paralysis at onset that progresses to complete $\rightarrow$ incomplete recovery in 75%. A worse prognosis is associated with: more proximal involvement, hyperacusis, decreased tearing, age $>$ 60 yrs, diabetes, HTN, psychoneuroses, and aural, facial or radicular pain.

33.3.5 Herpes zoster oticus facial paralysis

Symptoms are more severe than Bell's palsy, herpetic vesicles are usually present, and antibody titers to varicella-zoster virus rise. These patients have a higher risk of facial nerve degeneration.

33.3.6 Surgical treatment of facial palsy

General information

For cases with focal injury to the facial nerve (e.g. trauma, injury during surgery for CPA tumor...), dynamic reconstruction by nerve anastomoses are usually considered superior to static methods.¹⁹ For nonfocal causes, e.g. Bell's palsy, only "static" methods may be applicable. A functional neural repair is not possible if the facial muscles have atrophied or fibrosed.

Surgical options

Surgical treatment options include:

1. for intracranial injury to facial nerve (e.g. during CPA tumor surgery): intracranial reapproximation (with or without graft) offers the best hope for the most normal facial reanimation
 - a) timing
 - at time of tumor removal (for a divided facial nerve during removal of vestibular schwannoma^{20,21,22}): the best result that can be achieved with this is House-Brackmann Grade III. The operation fails to produce good results in $\approx$ 33% of cases²²
 - in delayed fashion, especially if the nerve was left in anatomic continuity
 - b) techniques
 - direct reanastomosis: difficult due to the frail nature of the VII nerve (especially when it has been stretched by a tumor)
 - cable graft: e.g. using greater auricular nerve²³ or sural nerve

2. extracranial facial nerve anastomosis
 - a) hypoglossal nerve (Cr. N. XII)-facial nerve anastomosis (see below)
 - b) spinal accessory nerve (Cr. N. XI)-facial nerve anastomosis (see below)
 - c) phrenic nerve-facial nerve anastomosis
 - d) glossopharyngeal (Cr. N. IX)-facial nerve anastomosis
 - e) crossface grafting (VII-VII): results have not been very good
3. “mechanical” or “static” means
 - a) facial suspension: e.g. with polypropylene (Marlex®) mesh²⁴
 - b) eye closure techniques (protects the eye from exposure and reduced tearing)
 - tarsorrhaphy: partial or complete
 - gold weights in eyelid
 - stainless-steel spring in eyelid

Timing of surgery

If the facial nerve is known to be interrupted (e.g. transected during removal of vestibular schwannoma) then early surgical treatment is indicated. When the status of the nerve is unknown or if in continuity but not functioning, then several months of observation and electrical testing should be allowed for spontaneous recovery. Very late attempts at anastomosis have less chance for recovery due to facial muscle atrophy.

Hypoglossal nerve-facial nerve (XII-VII) anastomosis

General information

Cannot be used bilaterally in patients with facial diplegia or in those with other lower cranial nerve deficits (or potential for same). In spite of some suggestions to the contrary, sacrificing the XII nerve does create some morbidity (tongue atrophy with difficulty speaking, mastication and swallowing in $\approx 25\%$ of cases, exacerbated when the facial muscles do not function on that side; aspiration may occur if vagus (Cr. N. X) dysfunction coexists with loss of XII).

Not as effective as it would theoretically seem possible. The resultant facial reanimation is often less than ideal (may permit mass movement). To avoid severe disappointment, the patient should thoroughly understand the likely side effects and that the facial movement will probably be much less than normal, often with poor voluntary control.

Usually performed in conjunction with anastomosis of the descendens hypoglossi to the distal hypoglossal nerve to try and reduce hemiatrophy of the tongue. Atrophy may also be reduced by using a “jump graft” without completely interrupting XII.²⁵

Technique

Position: supine, head turned slightly to the opposite side. Skin incision: 6–8 cm incision from just above the mastoid process obliquely downward across the neck to 2 cm below the angle of the jaw. The platysma is opened, and the tip of the mastoid is exposed by incising the insertion of the SCM and using a periosteal elevator. Incise the deep fascia; avoid the parotid gland, which is retracted superiorly. Rongeur the anterior third of the mastoid process (wax any exposed air cells) and identify the facial nerve as it exits the stylomastoid foramen between the mastoid process and the styloid process. Retract the posterior belly of the digastric inferiorly to aid the exposure.

The SCM is retracted laterally until the carotid sheath is identified, revealing the hypoglossal nerve. It loops around the occipital artery at this level (where it gives off the descendens hypoglossi) to pass between the carotid artery and jugular vein. The nerve is freed proximally to the point where it enters the carotid sheath and distally to the submandibular triangle where it is sharply divided.

The facial nerve is divided at the stylomastoid foramen and is approximated to the proximal hypoglossal nerve. The descendens hypoglossi is divided as far distally as possible and is then anastomosed to the distal stump of the hypoglossal nerve.

Variations

1. interposition jump grafts: spares function in the XII nerve (to minimize glottic denervation, the incision of XII should be distal to the descendens hypoglossi²⁵)
 - a) using cutaneous nerve jump graft²⁵
 - b) using muscle interposition jump graft²⁶
2. mobilizing the intratemporal portion of VII out of the fallopian canal (as previously described²⁷) and then anastomosing it using a bevelled cuts to a partially incised XII²⁸

Outcome

Results are better if performed early, although good results can occur up to 18 mos after injury. In 22 cases, 64% had good results, 14% fair, 18% poor, and 1 patient had no evidence of reinnervation. In 59% of cases, evidence of reinnervation was seen by 3–6 mos, in the remaining patients with reinnervation improvement was noted by 8 mos.²⁹ Recovery of forehead movement occurs in only ≈ 30% Return of tone precedes movement by ≈ 3 months.

Spinal accessory nerve-facial nerve (XI-VII) anastamosis

General information

First described in 1895 by Sir Charles Ballance.³⁰ Sacrifices some shoulder movement rather than use of tongue. Initial concerns about significant shoulder disability and pain resulted in the technique of using only the SCM branch of XI,³¹ however these problems have not occurred in the majority of patients even with use of the major division.³²

Technique

See reference.³²

Skin incision: curves across the mastoid tip along the anterior margin of the SCM. Strip and remove the anterior third of the mastoid process (wax any exposed air cells), identify the facial nerve and divide it as close to its exit from the stylomastoid foramen as possible. Locate the XI nerve 3–4 cm below the mastoid tip, and divide it distal to the SCM division. Mobilize the free end and anastamose it to the distal stump of VII. Results in loss of trapezius function, which may not cause deficit even if done bilaterally. Alternatively, the SCM branch of XI may be used, sparing the trapezius function, however the shorter length may be difficult to work with and in some individuals there may only be multiple small branches to the SCM.

33.4 Hearing loss

Two anatomic types: conductive and sensorineural.

33.4.1 Conductive hearing loss

- 1. Patients tend to speak with normal or low volume voice
- 2. etiologies: anything that interferes with ossicular movement. Included:
 - a) otitis media with middle ear effusion
 - b) otosclerosis
- 3. clinical findings with unilateral hearing loss (see □ Table 33.3):
 - a) Weber test will lateralize to side of hearing loss (Weber test: place a vibrating 256 or 512 Hz tuning fork on the center of the forehead; the sound will lateralize – sound louder – on the side of conductive hearing loss, or opposite to the side of SNHL)
 - b) Rinne test will be abnormal (BC>AC) on the side of hearing loss, called a negative Rinne (Rinne test: place a vibrating 256 or 512 Hz tuning fork on the mastoid process, when sound is no longer heard, move the fork to just outside the ear to see if air conduction [AC] is > bone conduction [BC])
- 4. middle ear impedance measurements are abnormal

Table 33.3 Interpretation of Weber and Rinne test results		
Weber	Rinne	Interpretation
nonlateralizing	AC>BC bilat	normal ^a
lateralizes to side A	normal bilaterally (AC>BC)	sensorineural hearing loss (SNHL) side B
lateralizes to side A	abnormal in side A (BC>AC)	conductive hearing loss side A
lateralizes to side A	abnormal in side B (BC>AC)	combined conductive + SNHL side B
^a normal, or symmetric hearing loss		

33.4.2 Sensorineural hearing loss (SNHL)

1. Patients tend to speak with loud voice
2. clinical findings with unilateral hearing loss (□ Table 33.3):
 - a) Weber test will lateralize to side of better hearing (Weber test: place a vibrating 256 or 512 Hz tuning fork on the center of the forehead; the sound will lateralize – sound louder – on the side of conductive hearing loss, or opposite to the side of SNHL)
 - b) Rinne test will be normal ($AC > BC$), called a positive Rinne (Rinne test: place a vibrating 256 or 512 Hz tuning fork on the mastoid process, when sound is no longer heard, move the fork to just outside the ear to see if air conduction [AC] is $>$ bone conduction [BC])
3. further divided into sensory or neural. Distinguished by otoacoustic emissions (only produced by a cochlea with functioning hair cells) or BSAERs
 - a) sensory: loss of outer hair cells in the cochlea. Etiologies: cochlear damage (usually causes high-frequency hearing loss) from noise exposure, ototoxic drugs (e.g. aminoglycosides), senile cochlear degeneration, viral labyrinthitis. Speech discrimination may be relatively preserved
 - b) neural: due to compression of the 8th cranial nerve. Etiologies: CP angle tumor (e.g. vestibular schwannoma). Typically much greater loss of word discrimination out of proportion to pure tone audiogram abnormalities

Sensory hearing loss may be distinguished from neural hearing loss by

1. otoacoustic emissions which are only produced by a cochlea with functioning hair cells
2. or BSAERs
3. an elevated stapedial reflex threshold out of proportion to PTA abnormalities is also highly diagnostic of a retrocochlear (neural) lesion

References

- [1] Brandt T, Daroff RB. The Multisensory Physiological and Pathological Vertigo Syndromes. *Ann Neurol*. 1980; 7:195–203
- [2] Jannetta PJ, Moller MB, Moller AR. Disabling Positional Vertigo. *N Engl J Med*. 1984; 310:1700–1705
- [3] Arriaga MA, Chen DA. Vestibular Nerve Section in the Treatment of Vertigo. *Contemp Neurosurg*. 1997; 19:1–6
- [4] McElveen JT, House JW, Hitselberger WE, et al. Retrolabyrinthine Vestibular Nerve Section: A Viable Alternative to the Middle Fossa Approach. *Otolaryngol Head Neck Surg*. 1984; 92:136–140
- [5] House JW, Hitselberger WE, McElveen J, et al. Retrolabyrinthine Section of the Vestibular Nerve. *Otolaryngol Head Neck Surg*. 1984; 92:212–215
- [6] Tarlov EC. Microsurgical Vestibular Nerve Section for Intractable Meniere's Disease. *Clin Neurosurg*. 1985; 33:667–684
- [7] Glasscock ME, Miller GW, Drake FD, et al. Surgical Management of Meniere's Disease with the Endolymphatic Subarachnoid Shunt. *Laryngoscope*. 1977; 87:1668–1675
- [8] Shambaugh GE. In: Facial Nerve Decompression and Repair. *Surgery of the Ear*. Philadelphia: W. B. Saunders; 1959:543–571
- [9] Adour KK. Diagnosis and Management of Facial Paralysis. *N Engl J Med*. 1982; 307:348–351
- [10] Shambaugh GE, Clemis JD, Paparella MM, Schumrick DA. In: Facial Nerve Paralysis. *Otolaryngology*. Philadelphia: W. B. Saunders; 1973
- [11] Adour KK, Wingerd J, Doty HE. Prevalence of Concurrent Diabetes Mellitus and Idiopathic Facial Paralysis (Bell's Palsy). *Diabetes*. 1975; 24:449–451
- [12] Treatment of Lyme Disease. *Med Letter*. 1988; 30:65–66
- [13] Abraham-Inpijn L, Devriese PP, Hart AAM. Predisposing Factors in Bell's Palsy: A Clinical Study with Reference to Diabetes Mellitus, Hypertension, Clotting Mechanism and Lipid Disturbance. *Clin Otolaryngol*. 1982; 7:99–105
- [14] Devriese PP, Schumacher T, Scheide A, et al. Incidence, Prognosis and Recovery of Bell's Palsy: A Survey of About 1000 Patients (1974-1983). *Clin Otolaryngol*. 1990; 15:15–27
- [15] Adour KK, Byl FM, Hilsinger RL, Kahn ZM, et al. The True Nature of Bell's Palsy: Analysis of 1000 Consecutive Patients. *Laryngoscope*. 1978; 88:787–801
- [16] Adour KK, Bell DN, Hilsinger RL. Herpes simplex virus in idiopathic facial paralysis (Bell palsy). *JAMA*. 1975; 233:527–530
- [17] Kuiper H, Devriese PP, de Jongh BM, Vos K, Dankert J. Absence of Lyme Borreliosis Among Patients With Presumed Bell's Palsy. *Arch Neurol*. 1992; 49:940–943
- [18] Sullivan FM, Swan IR, Donnan PT, Morrison JM, Smith BH, McKinstry B, Davenport RJ, Vale LD, Clarkson JE, Hammersley V, Hayavi S, McAteer A, Stewart K, Daly F. Early treatment with prednisolone or acyclovir in Bell's palsy. *N Engl J Med*. 2007; 357:1598–1607
- [19] Conley J, Baker DC. Hypoglossal-Facial Nerve Anastomosis for Reinnervation of the Paralyzed Face. *Plast Reconstr Surg*. 1979; 63:63–72
- [20] Pluchino F, Fornari M, Luccarelli G. Intracranial Repair of Interrupted Facial Nerve in Course of Operation for Acoustic Neuroma by Microsurgical Technique. *Acta Neurochir*. 1986; 79:87–93
- [21] Stephanian E, Sekhar LN, Janecka IP, Hirsch B. Facial Nerve Repair by Interposition Nerve Graft: Results in 22 Patients. *Neurosurgery*. 1992; 31:73–77
- [22] King TT, Sparrow OC, Arias JM, O'Connor AF. Repair of Facial Nerve After Removal of Cerebellopontine Angle Tumors: A Comparative Study. *J Neurosurg*. 1993; 78:720–725
- [23] Alberti PWRM. The Greater Auricular Nerve. Donor for Facial Nerve Grafts: A Note on its Topographical Anatomy. *Arch Otolaryngol*. 1962; 76:422–424
- [24] Strelzow VV, Friedman WH, Katsantonis GP. Reconstruction of the Paralyzed Face With the Polypropylene Mesh Template. *Arch Otolaryngol*. 1983; 109:140–144
- [25] May M, Sobol SM, Mester SJ. Hypoglossal-Facial Nerve Interpositional-Jump Graft for Facial Reanimation without Tongue Atrophy. *Otolaryngol Head Neck Surg*. 1991; 104:818–825
- [26] Drew SJ, Fullarton AC, Glasby MA, et al. Reinnervation of Facial Nerve Territory Using a Composite Hypoglossal Nerve-Muscle Autograft-Facial Nerve

- Bridge. An Experimental Model in Sheep. *Clin Otolaryngol.* 1995; 20:109–117
- [27] Hitselberger WE, House WF, Luetje CM. In: *Hypoglossal-Facial Anastomosis. Acoustic Tumors: Management.* Baltimore: University Park Press; 1979:97–103
- [28] Atlas MD, Lowinger DSG. A new technique for hypoglossal-facial nerve repair. *Laryngoscope.* 1997; 107:984–991
- [29] Pitty LF, Tator CH. Hypoglossal-Facial Nerve Anastomosis for Facial Nerve Palsy Following Surgery for Cerebellopontine Angle Tumors. *J Neurosurg.* 1992; 77:724–731
- [30] Duel AB. Advanced Methods in the Surgical Treatment of Facial Paralysis. *Ann Otol Rhinol Laryngol.* 1934; 43:76–88
- [31] Poe DS, Scher N, Panje WR. Facial Reanimation by XI-VII Anastomosis Without Shoulder Paralysis. *Laryngoscope.* 1989; 99:1040–1047
- [32] Ebersold MJ, Quast LM. Long-Term Results of Spinal Accessory Nerve-Facial Nerve Anastomosis. *J Neurosurg.* 1992; 77:51–54

Part XII

Primary Tumors of the
Nervous and Related
Systems: Tumors of
Neuroepithelial Tissue

34	General Information, Classification and Tumor Markers	584
35	Syndromes Involving Tumors	603
36	Astrocytomas	612
37	Other Astrocytic Tumors	629
38	Oligodendroglial Tumors and Tumors of the Ependyma, Choroid Plexus, and Other Neuroepithelial Tumors	638
39	Neuronal and Mixed Neuronal-Glial Tumors	651
40	Pineal Region and Embryonal Tumors	658
41	Tumors of Cranial, Spinal and Peripheral Nerves	670
42	Meningiomas	690
43	Other Tumors Related to the Meninges	701

XII

34 General Information, Classification and Tumor Markers

34.1 Classification of nervous system tumors

WHO classification of tumors of the nervous system

The 2007 WHO classification¹ identifies 7 categories of tumors of the nervous system (see □ Table 34.1) and a modified outline appears below^{1,2,3,4,5} (along with the unofficial category “intra-cranial and/or intraspinal embryonal remnants”, and pituitary adenomas (not part of the CNS)). Also to be considered: cysts (neurocysticercosis...), tumor-like masses (e.g. giant aneurysms), and local extension of regional tumors. Cytogenetic and molecular genetic information are playing an ever increasing role in the rapidly evolving classification of many brain tumors.

Table 34.1 Overview of WHO classification of nervous system tumors ³	
1.	tumors of neuroepithelial tissue
2.	tumors of cranial and paraspinal nerves
3.	tumors of the meninges
4.	lymphomas and hematopoietic neoplasms
5.	germ cell tumors
6.	tumors of the sellar region
7.	metastatic tumors

Table 34.2 2007 WHO classification of tumors of the nervous system ¹			
Tumor		ICD-O ^a	
A. TUMORS OF NEUROEPITHELIAL TISSUE ^b			
I. Astrocytes □ astrocytic tumors ^c (p.612)			
1.	astrocytomas that are typically infiltrating (p.612) (lower grade tumors in this category tend to progress in malignancy)		
	a.	diffuse astrocytoma (WHO II ^d) (p.612). Variants:	9400/3
		● fibrillary	9420/3
		● protoplasmic	9410/3
		● gemistocytic	9411/3
	b.	anaplastic (malignant) astrocytoma (WHO III) (p.616)	9401/3
	c.	glioblastoma (WHO IV) (p.616) (formerly glioblastoma multiforme (GBM)). Variants:	9440/3
		● giant cell glioblastoma	9441/3
		● gliosarcoma	9442/3
	d.	gliomatosis cerebri (p.619)	9381/3
	2.	more circumscribed lesions (these do not tend to progress to anaplastic astrocytoma and GBM)	
a.		pilocytic astrocytoma (p.629)	9421/1
		● pilomyxoid astrocytoma (WHO II) (p.632)	9425/3
b.		pleomorphic xanthoastrocytoma (PXA) (p.635)	9424/3
c.		subependymal giant cell astrocytoma (p.607): associated with tuberous sclerosis	9384/1

Table 34.2 continued			
Tumor		ICD-O ^a	
II. Oligodendrocytes □ oligodendroglial tumors			
1.	oligodendroglioma (WHO II) (p.638)		9450/3
2.	anaplastic oligodendroglioma (WHO III) (p.639)		9451/3
III. Oligoastrocytic tumors (nee: mixed gliomas)			
1.	oligoastrocytoma (WHO II) (p.641)		9382/3
2.	anaplastic (malignant) oligoastrocytoma (WHO III) (p.641)		9382/3
IV. Ependymocytes □ ependymal tumors			
1.	ependymoma (WHO II) (p.642). Variants:		9391/3
	a.	cellular (p.642)	9391/3
	b.	papillary (p.642)	9393/3
	c.	clear cell (p.642)	9391/3
	d.	tanycytic (p.642)	9391/3
2.	anaplastic (malignant) ependymoma (WHO III) (p.643)		9392/3
3.	myxopapillary ependymoma: filum terminale only (WHO I) (p.642)		9394/1
4.	subependymoma (WHO I) (p.643)		9383/1
V. Choroid plexus tumors (p.648)			
1.	choroid plexus papilloma		9390/0
2.	atypical choroid plexus papilloma		9390/1
3.	choroid plexus carcinoma		9390/3
VI. Other neuroepithelial tumors (p.649)			
1.	astroblastoma		9430/3
2.	chordoid glioma of the 3rd ventricle		9444/1
3.	angiocentric glioma		9431/1
VII. neuronal and mixed neuronal-glial tumors (p.651)			
1.	gangliocytoma		9492/0
2.	ganglioglioma (p.651)		9505/1
3.	dysembryoplastic neuroepithelial tumor (DNT) (p.646)		9413/0
4.	desmoplastic infantile astrocytoma/ganglioglioma (DIG) (p.645)		9412/1
5.	dysplastic gangliocytoma of cerebellum (Ihermitte-Duclos) (p.647)		9493/0
6.	anaplastic (malignant) ganglioglioma		9505/3
7.	central neurocytoma (p.645)		9506/1
8.	extraventricular neurocytoma		9506/1
9.	cerebellar liponeurocytoma		9506/1
10.	papillary glioneuronal tumor		9509/1
11.	rosette-forming glioneuronal tumor of the 4th ventricle		9509/1
12.	paraganglioma (of the filum terminale)		8680/1

Table 34.2 continued			
Tumor		ICD-O ^a	
VIII. Pinealocytes → pineal parenchymal tumors (p.659)			
1.	pineocytoma (pinealoma)	9361/1	
2.	pineoblastoma	9362/3	
3.	pineal parenchymal tumor of intermediate differentiation	9362/3	
4.	papillary tumor of the pineal region	9395/3	
IX. Embryonal tumors (p.663)			
1.	medulloblastoma (p.663). Variants:		9470/3
	a.	desmoplastic/nodular medulloblastoma (p.664)	9471/3
	b.	anaplastic medulloblastoma	9474/3
	c.	large-cell medulloblastoma (p.664)	9474/3
	d.	medulloblastoma with extensive nodularity	9471/3
2.	CNS primitive neuroectodermal tumors (PNET) (p.663)		9473/3
	a.	CNS neuroblastoma	9500/3
	b.	CNS ganglioneuroblastoma	9490/3
	c.	medulloepithelioma	9501/3
	d.	ependymoblastoma (p.666)	9392/3
3.	atypical teratoid/rhabdoid tumor (AT/RT) (p.666)		9508/3
B. TUMORS OF CRANIAL, SPINAL AND PERIPHERAL NERVES			
I. Schwannoma (neurilemmoma, neurinoma) (vestibular schwannoma AKA acoustic neuroma)		9560/0	
1.	cellular	9560/0	
2.	plexiform	9560/0	
3.	melanotic	9560/0	
II. Neurofibroma (p.604)		9540/0	
1.	plexiform (p.604)	9550/0	
III. Perineurioma (p.687)		9571/0	
1.	perineurioma, NOS	9571/0	
2.	malignant perineurioma	9571/3	
IV. Malignant peripheral nerve sheath tumor (MPNST) (neurogenic sarcoma, anaplastic neurofibroma, “malignant schwannoma”). Variants:		9540/3	
1.	epithelioid MPNST	9540/3	
2.	MPNST with mesenchymal differentiation	9540/3	
3.	melanotic MPNST	9540/3	
4.	MPNST with glandular differentiation	9540/3	

Table 34.2 continued		
Tumor		ICD-O ^a
C. TUMORS OF THE MENINGES		
I. Tumors of meningotheial cells		
1.	meningioma (p.690). Variants:	9530/0
a.	meningotheial (WHO I)	9531/0
b.	fibrous (fibroblastic) (WHO I)	9532/0
c.	transitional (mixed) (WHO I)	9537/0
d.	psammomatous (WHO I)	9533/0
e.	angiomatous (WHO I)	9534/0
f.	microcystic (WHO I)	9530/0
g.	secretory (WHO I)	9530/0
h.	lymphoplasmacyte-rich (WHO I)	9530/0
i.	metaplastic (WHO I)	9530/0
	the following meningiomas exhibit more malignant behavior	
	j.	clear cell (intracranial) (WHO II) 9538/1
	k.	chordoid (WHO II) 9538/1
	l.	atypical meningioma (WHO II) (p.693) 9539/1
	m.	papillary meningioma (WHO III) 9538/3
	n.	rhabdoid meningioma (WHO III) (p.694) 9538/3
	o.	anaplastic (malignant) meningioma (WHO III) (p.694) 9530/3
II. Mesenchymal, non-meningotheial tumors (p.701)		
1.	lipoma, e.g. of corpus callosum (p.260)	8850/0
2.	angiolipoma	8861/0
3.	hiberneuroma	8880/0
4.	liposarcoma (intracranial)	8850/3
5.	solitary fibrous tumor	8815/0
6.	fibrosarcoma	8810/3
7.	malignant fibrous histiocyoma	8830/3
8.	leiomyoma	8890/0
9.	leiomyosarcoma	8890/3
10.	rhabdomyoma	8900/0
11.	rhabdomyosarcoma	8900/3
12.	chondroma	9220/0
13.	chondrosarcoma	9220/3
14.	osteoma	9180/0

Table 34.2 continued		
Tumor		ICD-O ^a
15.	osteosarcoma	9180/3
16.	osteochondroma	9210/0
17.	hemangioma	9120/0
18.	epithelioid hemangioendothelioma	9133/1
19.	hemangiopericytoma (p.701)	9150/1
20.	anaplastic hemangiopericytoma	9150/3
21.	angiosarcoma	9120/3
22.	Kaposi sarcoma	9140/3
23.	Ewing sarcoma – PNET	9364/3
III. Primary melanocytic lesions		
1.	diffuse melanocytosis	8728/0
2.	melanocytoma	8728/1
3.	malignant melanoma (primary CNS) (p.701)	8720/3
4.	meningeal melanomatosis	8728/3
IV. Other neoplasms related to the meninges		
1.	hemangioblastoma (p.701)	9161/1
D. LYMPHOMAS AND HEMATOPOIETIC NEOPLASMS		
I. Malignant lymphoma (primary CNS lymphoma) (p.710)		9590/3
II. Plasmacytoma		9731/3
III. Granulocytic sarcoma		9930/3
E. GERM CELL TUMORS		
I. Germinoma (p.659)		9064/3
II. Embryonal carcinoma		9070/3
III. Endodermal sinus tumor (EST) (yolk sac tumor)		9071/3
IV. Choriocarcinoma		9100/3
V. Teratoma (from all 3 germ-cell layers) (p.660)		9080/1
1.	mature	9080/0
2.	immature	9080/3
3.	teratoma with malignant transformation	9084/3
VI. Mixed germ cell tumors		9085/3
F. TUMORS OF THE SELLAR REGION		
I. Craniopharyngioma (p.763). Variants:		9350/1
1.	adamantinomatous	9351/1
2.	papillary	9352/1

Table 34.2 continued		
Tumor		ICD-O ^a
II. Adenohypophyseal cells □ pituitary adenoma ^e (p.718)		8272/0
1.	prolactinoma ^e (p.722)	8271/0
2.	ACTH secreting adenoma	
3.	growth-hormone secreting adenoma	
4.	thyrotropin (TSH) secreting adenoma (p.726)	
5.	gonadotropin (LH and/or FSH) secreting adenoma	
III. Neurohypophysis and infundibulum (p.727)		
1.	granular cell tumor (p.727)	9582/0
2.	neurohypophyseal cells	9432/1
IV. Pituitary carcinoma (p.718)		
V. Spindle cell oncocytoma of the adenohypophysis		8291/0
G. METASTATIC TUMORS (p.800) Those commonly involving brain include:		
I. Lung cancer: especially small-cell (p.802)		
II. Breast		
III. Melanoma (p.803)		
IV. Renal cell (p.805)		
V. Lymphoma		
VI. GI		
H. LOCAL EXTENSIONS FROM REGIONAL TUMORS ^e		
I. Paraganglioma (chemodectoma)		
1.	glomus jugulare tumor (p.654)	
II. Notochord □ chordoma (p.778)		
III. Carcinoma		
I. CYSTS AND TUMOR-LIKE LESIONS ^e		
I. Rathke's cleft cyst (p.756)		
II. Ectodermal rests		
1.	epidermoid cyst (p.761)	
2.	cholesteatoma (p.761)	

Table 34.2 continued	
Tumor	ICD-O ^a
III. Dermoid cyst (p.760)	
IV. Colloid cyst of the third ventricle (p.756)	
V. Neurenteric/enterogenous cyst (p.290)	
VI. Neuroglial cyst	
VII. Hypothalamic neuronal hamartoma (p.261)	
VIII. Nasal glial heterotopia	
IX. Plasma cell granuloma	
J. UNCLASSIFIED TUMORS ^e	
^a ICD-O=morphology code of the International Classification of Diseases for Oncology (http://codes.iarc.fr). The extension after the slash is the “behavior code”: /0=benign, /1=low or uncertain malignant potential or borderline malignancy, /2=in situ lesions, /3=malignant tumors	
^b represents a significant portion of what is usually considered to be primary brain tumors	
^c the term “glioma” is occasionally used to refer to all glial tumors (e.g. “low-grade glioma” is often used when discussing low-grade tumors of any glial lineage), although in its usual sense glioma (especially “high grade gliomas”) refers only to astrocytic tumors	
^d “WHO II” means World Health Organization (WHO) grade II, “WHO III” means WHO grade III, etc.	
^e these tumors are not part of the 2007 WHO classification ^{5]}	

34.2 Brain tumors – general clinical aspects

34.2.1 General information

For details of presentation, see sections below for supratentorial and infratentorial tumors. Most brain tumors present with:

- Progressive neurologic deficit (68%): usually motor weakness (45%).
- Headache: was a presenting symptom in 54%(see below).
- Seizures in 26% Often focal in onset (due to cortical irritation in the area of the tumor), may generalize secondarily.

34.2.2 Focal neurologic deficits associated with brain tumors

In addition to nonfocal signs and symptoms (e.g. seizures, increased ICP...), as with any destructive brain lesion, tumors may produce progressive deficits related to the function of the involved brain. Some characteristic “syndromes”:

1. frontal lobe: abulia, dementia, personality changes. Often nonlateralizing, but apraxia, hemiparesis or dysphasia (with dominant hemisphere involvement) may occur
2. temporal lobe: auditory or olfactory hallucinations, déjà vu, memory impairment. Contralateral superior quadrantanopsia may be detected on visual field testing
3. parietal lobe: contralateral motor or sensory impairment, homonymous hemianopsia. Agnosias (with dominant hemisphere involvement) and apraxias may occur; see Clinical syndromes of parietal lobe disease (p.97)
4. occipital lobe: contralateral visual field deficits, alexia (especially with corpus callosum involvement with infiltrating tumors)
5. posterior fossa: (see above) cranial nerve deficits, ataxia (truncal or appendicular)

34.2.3 Headaches with brain tumors

General information

Headache (H/A) may occur with or without elevated ICP. H/A are present equally in patients with primary or metastatic tumor (≈ 50%of patients⁶). Classically described as being worse in the morning (possibly due to hypoventilation during sleep), this may actually be uncommon.⁶ Often exacerbated by coughing, straining, or (in 30%) bending forward (placing head in dependent position). Associated with nausea and vomiting in 40% may be temporarily relieved by vomiting (possibly due to hyperventilation during vomiting). These features along with the presence of a focal neurologic

deficit or seizure were thought to differentiate tumor H/A from others. However, H/A in 77% of brain tumor patients were similar to tension H/A, and in 9% were migraine-like.⁶ Only 8% showed the “classic” brain tumor H/A, two-thirds of these patients had increased ICP.

Etiologies of tumor headache

The brain itself is not pain sensitive. H/A in the presence of brain tumor may be due to any combination of the following:

1. increased intracranial pressure (ICP): which may be due to
 - a) tumor mass effect
 - b) hydrocephalus (obstructive or communicating)
 - c) mass effect from associated edema
 - d) mass effect from associated hemorrhage
2. invasion or compression of pain sensitive structures:
 - a) dura
 - b) blood vessels
 - c) periosteum
3. secondary to difficulty with vision
 - a) diplopia due to dysfunction of nerves controlling extra-ocular muscles
 - direct compression of III, IV, or VI
 - abducens palsy from increased ICP, see diplopia (p.592)
 - internuclear ophthalmoplegia due to brainstem invasion/compression
 - b) difficulty focusing: due to optic nerve dysfunction from invasion/compression
4. extreme hypertension resulting from increased ICP (part of Cushing’s triad)
5. psychogenic: due to stress from loss of functional capacity (e.g. deteriorating job performance)

34.2.4 Supratentorial tumors

See reference.⁷

Signs and symptoms include:

1. those due to increased ICP (see below):
 - a) from mass effect of tumor and/or edema
 - b) from blockage of CSF drainage (hydrocephalus): less common in supratentorial tumors (classically occurs with colloid cyst, may also occur with entrapped lateral ventricle)
2. progressive focal deficits: includes weakness, dysphasia (which occurs in 37–58% of patients with left-sided brain tumors⁸): see below
 - a) due to destruction of brain parenchyma by tumor invasion
 - b) due to compression of brain parenchyma by mass and/or peritumoral edema and/or hemorrhage
 - c) due to compression of cranial nerve(s)
3. headache: see below
4. seizures: not infrequently the first symptom of a brain tumor. Tumor should be aggressively sought in an idiopathic first time seizure in a patient >20 years (if negative, the patient should be followed with repeat studies at later dates). Rare with posterior fossa tumors or pituitary tumors
5. mental status changes: depression, lethargy, apathy, confusion
6. symptoms suggestive of a TIA (dubbed “tumor TIA”) or stroke, may be due to:
 - a) occlusion of a vessel by tumor cells
 - b) hemorrhage into the tumor: any tumor may hemorrhage, see Hemorrhagic brain tumors (p.1335)
 - c) focal seizure
7. in the special case of pituitary tumors (p.718):
 - a) symptoms due to endocrine disturbances
 - b) pituitary apoplexy (p.720)
 - c) CSF leak

Booking the case: Craniotomy for supratentorial tumor

Also see defaults & disclaimers (p.27). If awake craniotomy is required, see Booking the case: Awake craniotomy (p.1433).

1. position: (depends on location of tumor)
2. pre-op embolization (by neuroendovascular interventionalist) for some vascular tumors including some meningiomas

3. equipment:
 - a) microscope
 - b) ultrasonic aspirator
 - c) image guidance system
4. blood availability: type and cross 2 U PRBC
5. post-op: ICU
6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery through the skull to remove as much of the tumor as is safely possible
 - b) alternatives: nonsurgical management, radiation therapy for some tumors
 - c) complications: usual craniotomy complications (p.28) plus inability to remove all of the tumor

34.2.5 Infratentorial tumors

Signs and symptoms

Seizures are rare (unlike the situation with supratentorial tumors, (seizures arise from irritation of cerebral cortex).

1. most posterior fossa tumors present with signs and symptoms of increased intracranial pressure (ICP) due to hydrocephalus (HCP). These include:
 - a) headache: (see below)
 - b) nausea/vomiting: due either to increased ICP from HCP, or from direct pressure on the vagal nucleus or the area postrema (so-called “vomiting center”)
 - c) papilledema: estimated incidence is $\approx 50\text{--}90\%$ (more common when the tumor impairs CSF circulation)
 - d) gait disturbance/ataxia
 - e) vertigo
 - f) diplopia: may be due to VI nerve (abducens) palsy which may occur with increased ICP in the absence of direct compression of the nerve
2. S/S indicative of mass effect in various locations within the p-fossa
 - a) lesions in cerebellar hemisphere may cause: ataxia of the extremities, dysmetria, intention tremor
 - b) lesions of cerebellar vermis may cause: broad based gait, truncal ataxia, titubation
 - c) brainstem involvement usually results in multiple cranial nerve and long tract abnormalities, and should be suspected when nystagmus is present (especially rotatory or vertical)

Evaluation

Posterior fossa (infratentorial) tumors

See Posterior fossa lesions (p.905) for differential diagnosis (includes non-neoplastic lesions as well).

In pediatric patients with a posterior fossa tumor, an MRI of the lumbar spine should be done pre-op to rule-out drop mets (post-op there may be artifact from blood).

In adults, most intraparenchymal p-fossa tumors will be metastatic, and work-up for a primary should be undertaken.

Treatment of associated hydrocephalus

In cases with hydrocephalus at the time of presentation, some authors advocate initial placement of VP shunt or EVD prior to definitive surgery (waiting ≈ 2 wks before surgery) because of possibly lower operative mortality.⁹ Theoretical risks of using this approach include the following:

1. placing a shunt is generally a lifelong commitment, whereas not all patients with hydrocephalus from a p-fossa tumor will require a shunt
2. possible seeding of the peritoneum with malignant tumor cells e.g. with medulloblastoma. Consider placement of tumor filter (may not be justified given the high rate of filter occlusion and the low rate of “shunt metastases”¹⁰)
3. some shunts may become infected prior to the definitive surgery
4. definitive treatment is delayed, and the total number of hospital days may be increased
5. upward transtentorial herniation (p.303) may occur if there is excessively rapid CSF drainage

Either approach (shunting followed by elective p-fossa surgery, or semi-emergent definitive p-fossa surgery) is accepted. At Children’s Hospital of Philadelphia, dexamethasone is started and the surgery is performed on the next elective operating day, unless neurologic deterioration occurs necessitating emergency surgery.¹¹

Some surgeons place a ventriculostomy at the time of surgery. CSF is drained only after the dura is opened (to avoid upwards herniation) to help equilibrate the pressures between the infra- and supra-tentorial compartments. Post-op, the external ventricular drain EVD is usually set at a low height (≈ 10 cm above the EAM) for 24 hours, and is progressively raised over the next 48 hrs and should be D/C'd by ≈ 72 hrs post-op.

Booking the case: Craniotomy for infratentorial tumor

- Also see defaults & disclaimers (p.27) and retromastoid surgery for vestibular schwannomas (p.681).
- 1. position: (typically either prone or park bench, depending on tumor type/location and surgeon preference)
 - 2. pre-op embolization (by neuroendovascular interventionalist) for some vascular tumors such as hemangioblastoma
 - 3. equipment:
 - a) microscope
 - b) ultrasonic aspirator
 - c) image guidance system (optional)
 - 4. blood availability: type and cross 2 U PRBC
 - 5. post-op: ICU
 - 6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery through the skull remove as much of the tumor as is safely possible
 - b) alternatives: nonsurgical management, radiation therapy for some tumors
 - c) complications: usual craniotomy complications (p.28) plus inability to remove all of the tumor, hydrocephalus, CSF leak

34.3 Pediatric brain tumors

34.3.1 General information

Among all childhood cancers, brain tumors are the second only to leukemias in incidence (20%), and are the most common solid pediatric tumor,¹² comprising 40–50% of all tumors.¹³ Annual incidence: 2–5 cases per 100,000.

34.3.2 Types of tumors

The common pediatric brain tumors are gliomas (cerebellum, brain stem, and optic nerve), pineal tumors, craniopharyngiomas, teratomas, granulomas, and primitive neuroectodermal tumors (PNETs, primarily medulloblastoma).

Meningiomas: 1.5% of meningiomas occur in childhood and adolescence (usually between 10–20 years), comprising 0.4–4.6% of intracranial tumors¹⁴ (p.3263); see Meningiomas (p.690).

34.3.3 Infratentorial vs. supratentorial

It has traditionally been taught that most pediatric brain tumors ($\approx 60\%$) are infratentorial, and that these are $\approx$ equally divided among brain stem gliomas, cerebellar astrocytomas, and medulloblastomas. In reality, the ratio of supratentorial to infratentorial tumors is dependent on the specific age group studied, as illustrated in □ Table 34.3. □ Table 34.4 shows the breakdown for pooled data from 1350 pediatric brain tumors.

Astrocytomas are the most common supratentorial tumor in pediatrics as in adulthood.

34.3.4 Intracranial neoplasms during the first year of life

Brain tumors presenting during the first year of life is a different subset of tumors than those presenting later in childhood. In a busy neurosurgical unit in a children's hospital, they represented $\approx 8\%$ of children admitted with brain tumors, an average of only ≈ 3 admissions per year.¹⁷

90% of brain tumors in neonates are of neuroectodermal origin, teratoma being the most common. Some of these tumors may be congenital.¹⁸ Other supratentorial tumors include: astrocytoma,

Table 34.3 Location of pediatric brain tumors by age	
Age	% Infratentorial
0–6 mos	27%
6–12 mos	53%
12–24 mos	74%
2–16 yrs	42%

Table 34.4 Incidence of pediatric brain tumors ^a	
Tumor type	% of total
infratentorial tumors	54%
cerebellar astrocytomas (p. 630)	15%
medulloblastomas (p. 664)	14%
brain stem gliomas (p. 633)	12%
ependymomas (p. 642) ¹⁵	9%
supratentorial benign astrocytomas	13%
^a data pooled from 1350 pediatric brain tumors ¹⁶ (p 368)	

choroid plexus tumors, ependymomas, and craniopharyngiomas. Posterior fossa tumors include medulloblastoma and cerebellar astrocytoma.

Many of these tumors escape diagnosis until they are very large in size due to the elasticity of the infant skull, the adaptability of the developing nervous system to compensate for deficits, and the difficulty in examining a patient with limited neurologic repertoire and inability to cooperate. The most common presenting manifestations are vomiting, arrest or regression of psychomotor development, macrocrania, poor feeding/failure to thrive. They may also present with seizures.

34

34.4 Medications for brain tumors

34.4.1 Steroid use in brain tumors

The beneficial effect of steroids in metastatic tumors is often much more dramatic than with primary infiltrating gliomas.

Dexamethasone (Decadron®) dose for brain tumors:

- for patients not previously on steroids:
 - adult: 10 mg IVP loading, then 6 mg PO/IVP q 6 hrs.^{19,20} In cases with severe vasogenic edema, doses up to 10 mg q 4 hrs may be used
 - peds: 0.5–1 mg/kg IVP loading, then 0.25–0.5 mg/kg/d PO/IVP divided q 6 hrs. NB: avoid prolonged treatment because of growth suppressant effect in children
- for patients already on steroids:
 - for acute deterioration, a dose of approximately double the usual dose should be tried
 - see also “stress doses” (p. 146)

34.4.2 Prophylactic anticonvulsants with brain tumors

20–40% of patients with a brain tumor will have had a seizure by the time their tumor is diagnosed.²¹ Antiepileptic drugs (AEDs) are indicated in these patients. 20–45% more will ultimately develop a seizure.²¹ Prophylactic AEDs do not provide substantial benefit (reduction of risk >25% for seizure-free survival) and there are significant risks involved. Practice guidelines for AED use with brain tumors is shown below. Prophylactic AEDs are not indicated for isolated posterior fossa tumors.

Practice guideline: Prophylactic anticonvulsants with brain tumors

Level I¹: prophylactic AEDs should not be used routinely in patients with newly diagnosed brain tumors

Level II¹: in patients with brain tumors undergoing craniotomy, prophylactic AEDs may be used, and if there has been no seizure, it is appropriate to taper off AEDs starting 1 week post-op

34.5 Chemotherapy for brain tumors

34.5.1 General information

General information is presented here. Chemotherapy for some specific tumors is also included in the section devoted to that tumor where appropriate. Some agents used for CNS tumors: see □ Table 34.5.^{22,23}

34.5.2 Alkylating agents

Temozolomide (Temodar®) is a derivative of Dacarbazine (DTIC®) and is an oral alkylating agent. Temozolomide is a prodrug which undergoes rapid non-enzymatic conversion at physiologic pH to the active metabolite monomethyl triazenoimidazole carboxamide (MTIC). The cytotoxic effect of MTIC is associated with alkylation (adding an alkyl group, the smallest of which is a methyl group) of DNA at various sites primarily at the O6 and N7 positions on guanine. Some tumors can repair this damage via a protein (AGT) that is coded for by the O-6-methylguanine-DNA methyltransferase (MGMT) gene.

Table 34.5 Chemotherapeutic agents used for CNS tumors	
Agent	Mechanism
nitrosoureas: BCNU (carmustine), CCNU (AKA lomustine), ACNU (nimustine)	DNA crosslinks, carbamoylation of amino groups
alkylating (methylating) agents: procarbazine, temozolomide (Temodar®) (p. 595)	DNA alkylation, interferes with protein synthesis
carboplatin, cisplatin	chelation via intrastrand crosslinks
nitrogen mustards: cyclophosphamide, ifosfamide, cytoxan	DNA alkylation, carbonium ion formation
vinca alkaloids: vincristine, vinblastine, paclitaxel	microtubule function inhibitors
epidophyllotoxins (ETOP-oside, VP16, teniposide, VM26)	topoisomerase II inhibitors
topotecan, irinotecan (CPT-11)	topoisomerase I inhibitors
tamoxifen	protein kinase C inhibitor at high doses
bevacizumab (Avastin®)	anti-VEGF antibody may be useful in vestibular neuromas
hydroxyurea bleomycin taxol (paxlitaxol) methotrexate cytosine, arabinoside corticosteroids: dexamethasone, prednisone fluorouracil (FU)	

34.5.3 Nitrosoureas

Excellent BBB penetration (see below). Significant hematopoietic, pulmonary and renal toxicity.

34.5.4 Blood-brain barrier (BBB) and chemotherapy agents

Traditionally, the BBB has been considered to be a major hindrance to the use of chemotherapy for brain tumors. In theory, the BBB effectively excludes many chemotherapeutic agents from the CNS, thereby creating a “safe haven” for some tumors, e.g. metastases. This concept has been challenged.²⁴ Regardless of the etiology, the response of most brain tumors to systemic chemotherapy is usually very modest, with a notable exception being a favorable response of oligodendrogliomas (p.640) and gliomas lacking MGMT activity. Considerations regarding chemotherapeutic agents in relation to the BBB include:

1. some CNS tumors may partially disrupt the BBB, especially malignant gliomas²⁵
2. lipophilic agents (e.g. nitrosoureas) may cross the BBB more readily
3. selective intraarterial (e.g. intracarotid or intervertebral) injection²⁶: produces higher local concentration of agents which increases penetration of the BBB, with lower associated systemic toxicities than would otherwise occur
4. the BBB may be iatrogenically disrupted (e.g. with mannitol) prior to administration of the agent
5. the BBB may be bypassed by intrathecal administration of agents via LP or ventricular access device, e.g. methotrexate for CNS lymphoma (p.713)
6. biodegradable polymer wafers containing the agent may be directly implanted

34.5.5 Imaging studies following surgical removal of tumor

At many academic centers, it is common to get a noncontrast CT scan within 6–12 hours of surgery to assess for acute complications (primarily blood – within brain or epidural or subdural hematoma, amount of pneumocephalus, hydrocephalus...).

Then to assess the extent of tumor removal, a post-op brain CT or MRI without and with contrast should either be obtained within 2–3 days,²⁷ or should be delayed at least ≈ 30 days. The non-contrast scan is important to help differentiate blood from enhancement. The contrast images demonstrates areas of enhancement, which may represent residual tumor. After ≈ 48 hours, contrast enhancement due to post-operative inflammatory vascular changes ensues, which may be difficult to differentiate from tumor. This usually subsides by ≈ 30 days,²⁸ but may persist for 6–8 weeks.²⁹ This recommendation regarding the timing of post-op imaging does not apply to pituitary tumors (p.718). The effect of steroids on contrast enhancement is controversial,^{30,31} and may depend on many factors (including tumor type).

34.6 Intraoperative pathology consultations (“frozen section”)

34.6.1 Accuracy of intraoperative pathology consultations

Accuracy of intraoperative pathologic diagnosis can be increased by:

- Providing the pathologist with information regarding: patient demographics, clinical history, imaging results, relevant previous pathologic diagnoses, and clinical impression
- Larger specimen sizes when possible
- Avoiding artifact created by excessive crushing or coagulation

Frozen section diagnoses should be considered preliminary. The final diagnosis differs from the frozen section in approximately 3–10 % of cases.^{32,33,34} If the frozen section interpretation does not correlate with the clinical impression, direct discussion with the pathologist may be advisable.

34.6.2 Techniques for intraoperative tissue preparation

□ Touch preparation. The specimen is gently “touched” with a glass slide which is then rapidly fixed, stained and dehydrated for examination. This technique is particularly useful for tumors with discohesive cells (e.g. lymphoma, pituitary adenoma).

□ Smear or squash preparation. A small portion of specimen is smeared or compressed with moderate pressure between two glass slides, rapidly fixed, stained, and dehydrated for examination. This technique can be particularly useful for: multiple sclerosis (identifying histiocytes),

visualizing long cell processes in gliomas, and identifying cytoplasmic inclusions or intranuclear pseudoinclusions).^{35 (p 5–6)} The cohesive nature often seen in tumors such as metastases and meningiomas is apparent, as are areas of necrosis.

□ **Frozen section.** A portion of tissue is rapidly frozen in liquid nitrogen and cut into 4–6 micron sections, mounted on a slide, rapidly fixed, stained, and dehydrated for examination. Unlike touch and smear preparations, this allows more accurate assessment of lesion architecture, cellularity, and interface with adjacent brain tissue. Disadvantages include use of greater tissue with less available for permanent histology (important with small biopsies), as well as artifacts such as freezing ice crystal artifact (which, when present, suggests lesional tissue has been biopsied but limits interpretation of cellularity).^{35 (p 6)} When possible, some tissue should be preserved for processing without freezing to avoid artifacts.

At the time of frozen section interpretation, consideration for additional studies such as tissue cultures or flow cytometry should be entertained. With minute specimens, a discussion is warranted whether frozen section is required to preserve tissue for permanent studies.

34.6.3 Selected frozen section pitfalls or potential critical diagnoses

- Differentiating low grade gliomas from normal or reactive brain tissue can be challenging.^{32,33} Increased cellularity (best evaluated at low power), nuclear atypia when present, and increased perineuronal satellitosis can be helpful, though not always readily observed.^{35 (p 72–3, p 174–7)}
 Pearl: Secondary structures of Scherer may be useful in identifying gliomas in challenging cases and include increased perineuronal and perivascular satellitosis (limited perineuronal satellitosis is normal) and accumulation of neoplastic cells in the subpial molecular layer with subpial tumor spread.³⁶
- Metastasis vs Glioma: Usually not problematic on frozen section, except occasionally with markedly atypical gliomas with limited sampling.^{32,33,34} In such rare cases immunohistochemistry is helpful.
- Astrocytoma vs oligodendroglioma: Usually not a critical distinction at the time of frozen section diagnosis. However, in part due to frozen section artifact imparted to the nuclei, oligodendrogliomas can be interpreted as astrocytomas in the frozen section room.³² Smear preparations can at times be helpful due to decreased “frozen” artifact. Also, perineuronal satellitosis may in some cases be more pronounced in oligodendrogliomas.
 Pearl: The “fried egg” appearance in oligodendrogliomas is an artifact of formalin fixation for permanent sections and is NOT present on frozen sections, also hindering intraoperative interpretation.
- Glioma grading: Sampling bias, particularly in biopsy specimens, can lead to undergrading (i.e. mitotic figures underrepresented in the biopsy material, vascular proliferation or necrosis not present in the biopsy). However, overgrading of gliomas at the time of intraoperative consultation also can occur, including with low grade childhood gliomas such as pilocytic astrocytomas.³²
- Radionecrosis vs recurrent glioblastoma: Despite the fact that a history of prior radiation therapy is usually known at the time of intraoperative consultation, differentiation of the two entities can at times be difficult.³² Both lesions often are present simultaneously. Identifying obvious tumor cells and palisading necrosis suggests recurrent/residual glioblastoma. Radiation necrosis, which affects primarily white matter, is supported by large areas of geographic necrosis, sclerosis/hyalinization of vessels or fibrinoid necrosis of vessel walls, perivascular lymphocytes, calcifications, and the presence of macrophages.
- Ischemic infarct: May see ischemic red neuron change (if the biopsy includes gray matter), as well as histiocytes (similar to demyelinating lesions). Necrosis may mimic the center of ring enhancing lesions (such as glioblastoma or metastasis), though tumor necrosis will typically involve vessels and lack a macrophage response.^{35 (p 663–6)}
- Demyelinating lesions (e.g. tumefactive MS): Primarily affects white matter with discrete borders. Identifying histiocytes at the time of frozen section is critical for the diagnosis. The histiocytes can mimic astrocytes of gliomas on frozen section preparation; smear preparations are particularly helpful in the distinction.
- Lymphoma (PCNSL) and small cell carcinoma: Accurate frozen section diagnosis may be critical as both conditions typically require only biopsy unless there is significant mass effect. The two entities may resemble each other on frozen section as well as resembling astrocytomas, oligodendrogliomas, and other types of metastatic carcinomas.^{32,33,34,35 (p 395–7)} Touch preparations can be particularly useful in identifying PCNSL.
- Spindle cell tumor/meningioma: It has been demonstrated that at times distinguishing between meningioma and spindle cell lesions such as schwannomas can be difficult at the time of

intraoperative consultation.³² Classic features of meningiomas (whorls, psammoma bodies, intranuclear pseudoinclusions) may be absent, and freeze artifact can create Antoni B appearing areas in meningiomas.³² In addition, underdiagnosis of malignant meningiomas and sarcomas at the time of frozen section has been noted.^{32,33}

- Spinal cord astrocytoma vs. ependymoma: On occasion these entities can be difficult to distinguish during intraoperative consultation, especially as most spinal cord biopsies for frozen interpretation are minute. Due to the critical surgical implications of the accuracy of the frozen section diagnosis for intramedullary spinal gliomas, a careful discussion should be had between the surgeon and pathologist at the time of intraoperative consultation (refer to section on tumors of the spine and spinal cord).

34.6.4 Tissue preparation for permanent sections

Tissue is processed overnight through a variety of alcohol/xylene steps to remove water. This allows for embedding into paraffin so that thin sections can be cut for slide mounting. The specimens are then rehydrated by essentially reversing the alcohol/xylene steps in preparation for staining, and then dehydrated again for permanent slide coverslipping. This produces better histology with fewer artifacts, allows for processing of larger specimen volumes for evaluation, and allows for application of special stains as needed.

□ Fresh specimens. Tissue should be sent “fresh” (i.e. without preservatives such as formalin) when the following techniques are needed:

- electron microscopy
- flow cytometry: e.g. when lymphoma is suspected
- muscle
- cultures (typically: aerobic, anaerobic, acid fast and fungal) should be sent in situations where infection is a consideration

34.7 Select commonly utilized stains in neuropathology

34.7.1 Organism and special stains

1. Organism stains:

- tissue Gram stains (Brown & Brenn, Brown & Hopps)
- fungi (Periodic acid-Schiff: PAS, Gomori methenamine silver: GMS)
- acid fast bacilli (Ziehl-Neelsen, Kinyoun, FITE)

2. Special stains:

- luxol fast blue: stains myelin; absence of staining highlights demyelinating lesions. Presence within histiocytes demonstrates ingestion of myelin as seen in MS
- trichrome and reticulin stain: Both delineate the sarcomatous component of gliosarcomas. A reticulin stain demonstrates the connective tissue around acini of the normal pituitary gland, a feature that is lost with pituitary adenomas

34.7.2 Immunohistochemical stains

General information

Staining patterns: An individual tumor may lack a marker that is typically representative of its type.

□ Therefore, a positive stain is typically more significant than a negative stain.³⁷ General staining patterns³⁷ are shown in □ Table 34.6.

Glial fibrillary acidic protein (GFAP)

Polypeptide, MW = 49,000 Daltons. Stains intermediate filaments classically identified in astrocytes/astrocytic tumors. However, is also typically expressed in ependymomas, oligodendrogliomas (especially in minigemistocytes and gliofibrillary oligodendrocytes), and some choroid plexus papillomas.³⁸ (p 30-1),¹ (p 56, 76, 83) GFAP is only rarely found outside the CNS in such tissues as Schwann cells, epithelium of the lens, certain liver cells, chondrocytes, etc). □ It would be unusual for a metastatic lesion to demonstrate staining for GFAP. However, GFAP expression may be limited in certain subsets of GBM (e.g. small cell GBM).¹ (p 37)

Table 34.6 Immunohistochemical staining patterns for nervous system tumor masses of epithelioid cells ^a						
Neoplasm	Immunohistochemical stain response ^{b,c}					
	GFAP	CAM5.2	EMA	S-100	CgA	Syn
oligodendroglioma	+	–	–	+	–	0
ependymoma	0		0			–
choroid plexus papilloma	–	+			0	+
chordoma				–	–	
craniopharyngioma			–		–	
carcinoma			0	0		
pituitary adenoma		0	+	+	+	
paraganglioma				–	0	
meningioma			+		–	
melanoma		–		+	–	–
hemangioblastoma	0			0		0

^amodified from McKeever, P E, Immunohistochemistry of the Nervous System, in Dobbs, D J. Diagnostic Immunohistochemistry, Churchill Livingstone, NY, © 2002

^babbreviations: GFAP=glial fibrillary acidic protein, EMA=epithelial membrane antigen, CAM5.2 =cytokeratin CAM5.2, CgA=chromogranin A, syn=synaptophysin

^ca “+” or a “–” sign indicates presence or absence of the stain respectively; “0” entries indicate that the stain is not decisive for that particular tumor

S-100 protein

A low molecular weight (21,000 Daltons) calcium-binding protein that stains a variety of tissues including glia, neurons, chondrocytes, stellate cells of the adenohypophysis, myoepithelial cells, etc.^{37 (p 75)} Stains a variety of CNS neoplasms such as gliomas (though less specific than GFAP), PNET, ependymomas, chordomas, and craniopharyngiomas.^{1,37} Primary uses in neuropathology include to support the diagnosis of metastatic melanoma, and, in the peripheral nervous system, to confirm the diagnosis of schwannoma or neurofibroma (less intense staining in the latter).^{1 (p 156)} Clinically has been measured in serum (see below).

Cytokeratins (high and low molecular weight)

A variety of stains (low (e.g. CAM 5.2) & high molecular weight keratin, CK7, CK20, etc.) which stain epithelial cells. Useful to distinguish metastatic carcinoma (positive staining) from primary CNS tumors (note: cytokeratins can be expressed in choroid plexus papillomas, and GBM can express cytokeratins in rare cases).^{1 (p 39-40)} Different keratin staining combinations can be used to suggest possible sites of origin for metastatic tumors.

Epithelial membrane antigen (EMA)

Stains cell membranes in many carcinomas; useful to distinguish hemangioblastoma (negative) from metastatic renal cell carcinoma (positive). Also, meningiomas typically demonstrate positive staining, as do ependymomas.^{1 (p 169)}

MIB-1 (AKA monoclonal mouse anti-human Ki-67 antibody)

The Ki-67 antigen is expressed in all phases of the cell cycle except G0. Available since the early 1990s, it is a valuable marker of cell proliferation but can only be used with fresh-frozen specimens. MIB-1 is a monoclonal antibody developed using recombinant parts of the Ki-67 protein as an immunogen, and can be used on paraffin-embedded sections of fixed tissue. Cells leaving the G0/G1-phase and entering the S-phase (performing DNA synthesis) stain positive with MIB-1

immunohistochemical stain. This stain can be used to compute a semi-quantitative score. A high MIB-1 labeling index denotes high mitotic activity which often correlates with degree of malignancy. Often used in lymphomas, endocrine tumors, carcinoids, etc. See also use in astrocytomas (p.617) and in meningiomas (p.694).

Neuroendocrine stains

Includes:

In neuropathology, utilized in central neurocytoma, medulloblastoma, PNET, pineocytoma, ganglion cell tumors, paragangliomas, and choroid plexus tumors.³⁷ Metastases that are positive for neuroendocrine stains include small-cell carcinoma of the lung (most common), malignant pheochromocytoma, and Merkel cell tumor.

Includes:

1. chromogranin: stains synaptic vesicles. Perhaps less commonly utilized than synaptophysin in primary CNS tumor
2. synaptophysin: stains synaptic vesicles; has higher sensitivity but lower specificity than chromogranin.³⁷ (p 200) Often positive for central neurocytoma which typically lacks chromogranin staining.¹ (p 107)
3. CD56 (Neural cell adhesion molecule): a family of glycoproteins present in nervous tissue as well as other tissues such as thyroid, liver, etc.³⁷ (p 142-3, 264) Frequently used to confirm neuroendocrine differentiation.
4. neuron specific enolase (NSE): sensitive but not specific for neuronal or neuroendocrine differentiation despite its name (often referred to as “neuron non-specific enolase”).³⁷ (p 338) Due to this, it is less often utilized as a neuroendocrine marker.

Cluster of differentiation (CD) markers

A number of immunohistochemical stains that detect antigens on the surface of leukocytes, though many stain other cell types as well. Examples include:

- CD45: General leukocyte marker
- CD3 and CD5: T-cells
- CD20: B-cells
- CD38 and CD138: Plasma cells
- CD68: Histiocytes
- CD56 (Neural cell adhesion molecule): Classically stains natural killer cells, but also a neuroendocrine marker (see above)
- organism-specific immunohistochemical stains are available to detect certain organisms that infect the nervous system including HSV, CMV, and Toxoplasma gondii

34

Metastases that are positive for neuroendocrine stains include: small-cell carcinoma of the lung, malignant pheochromocytoma, Merkel cell tumor. Metastatic small-cell tumors to the brain staining positive for neuroendocrine stains are almost all due to lung primaries (all other primaries are a distant possibility).

34.7.3 Tumor markers used clinically

Human chorionic gonadotropin (hCG)

A glycoprotein, MW = 45,000. Secreted by placental trophoblastic epithelium. Beta chain (β -hCG) is normally present only in the fetus or in gravid or postpartum females, otherwise it indicates disease. Classically associated with choriocarcinoma (uterine or testicular), also found in patients with embryonal cell tumors, teratocarcinoma of testis, and others.

CSF β -hCG is 0.5–2% of serum β -hCG in non-CNS tumors. Higher levels are diagnostic of cerebral mets from uterine or testicular choriocarcinoma, or primary choriocarcinoma or embryonal cell carcinoma of pineal (p.658) or suprasellar region.

Alpha-fetoprotein

Alpha-fetoprotein (AFP) is a normal fetal glycoprotein (MW = 70,000) initially produced by the yolk sac, and later by the fetal liver. It is found in the fetal circulation throughout gestation, and drops rapidly during the first few weeks of life, reaching normal adult levels by age 1 yr. It is detectable only in trace amounts in normal adult males or nonpregnant females. It is present in amniotic fluid

in normal pregnancies, and is detectable in maternal serum starting at ≈ 12 –14 weeks gestation, increasing steadily throughout pregnancy until ≈ 32 weeks.³⁹

Abnormally elevated serum AFP may occur in Ca of ovary, stomach, lung, colon, pancreas, as well as in cirrhosis or hepatitis and in the majority of gravid women carrying a fetus with an open neural tube defect; see Prenatal detection of neural tube defects (p.290). Serum AFP > 500 ng/ml usually means primary hepatic tumor.

CSF-AFP is elevated in some pineal region germ-cell tumors (p.659). 16–25% of patients with testicular tumors get cerebral mets and elevated CSF AFP levels are reported in some.

Carcinoembryonic antigen (CEA)

A glycoprotein, MW = 200,000. Normally present in fetal endodermal cells. Originally described in the early 1960s in the serum in patients with colorectal adeno-Ca, now known to be elevated in many malignant and nonmalignant conditions (including cholecystitis, colitis, diverticulitis, hepatic involvement from any tumor, with 50–90% of terminal patients having elevation).

CSF CEA: levels > 1 ng/ml are reported with leptomeningeal spread of lung Ca (89%), breast Ca (60–67%), malignant melanoma (25–33%), and bladder Ca. May be normal even in CEA secreting cerebral mets if they don't communicate with the subarachnoid space. Only carcinomatous meningitis from lung or breast Ca consistently elevates CSF CEA in the majority of patients.

S-100 protein

Serum S-100 protein levels rise after head trauma, and possibly after other insults to the brain. Levels may also be elevated in Creutzfeldt-Jakob disease.

References

- [1] Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. WHO classification of tumors of the central nervous system. Lyon 2007
- [2] Kleihues P, Cavenee WK. World Health Organization classification of tumors: Pathology and genetics of tumors of the nervous system. Lyon 2000
- [3] Kleihues P, Louis DN, Scheithauer BW, Rorke LB, Reifenberger G, Burger PC, Cavenee WK. The WHO classification of tumors of the nervous system. J Neuropathol Exp Neurol. 2002; 61:215–25; discussion 226–9
- [4] Kleihues P, Burger PC, Scheithauer BW. The new WHO classification of brain tumors. Brain Pathol. 1993; 3:255–268
- [5] Escourolle R, Poirier J, Rubinstein LJ. Manual of Basic Neuropathology. 2nd ed. Philadelphia: W. B. Saunders; 1971
- [6] Forsyth PA, Posner JB. Headaches in Patients with Brain Tumors: A Study of 111 Patients. Neurology. 1993; 43:1678–1683
- [7] Mahaley MS, Mettlin C, Natarajan N, Laws ER, et al. National Survey of Patterns of Care for Brain-Tumor Patients. J Neurosurg. 1989; 71:826–836
- [8] Whittle IR, Pringle A-M, Taylor R. Effects of Resective Surgery for Left-Sided Intracranial Tumors on Language Function: A Prospective Study. Lancet. 1998; 351:1014–1018
- [9] Albright L, Reigel DH. Management of Hydrocephalus Secondary to Posterior Fossa Tumors. Preliminary Report. J Neurosurg. 1977; 46:52–55
- [10] Berger MS, Baumeister B, Geyer JR, Milstein J, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. J Neurosurg. 1991; 74:872–877
- [11] McLaurin RL, Venes JL. Pediatric Neurosurgery. Philadelphia 1989
- [12] Allen JC. Childhood Brain Tumors: Current Status of Clinical Trials in Newly Diagnosed and Recurrent Disease. Ped Clin N Am. 1985; 32:633–651
- [13] Laurent JP, Cheek WR. Brain Tumors in Children. J Pediatr Neurosci. 1985; 1:15–32
- [14] Youmans JR. Neurological Surgery. Philadelphia 1990
- [15] Duffner PK, Cohen ME, Freeman AI. Pediatric Brain Tumors: An Overview. Ca. 1985; 35:287–301
- [16] Section of Pediatric Neurosurgery of the American Association of Neurological Surgeons. Pediatric Neurosurgery. New York 1982
- [17] Jooma R, Hayward RD, Grant DN. Intracranial Neoplasms During the First Year of Life: Analysis of One Hundred Consecutive Cases. Neurosurgery. 1984; 14:31–41
- [18] Wakai S, Arai T, Nagai M. Congenital Brain Tumors. Surg Neurol. 1984; 21:597–609
- [19] Galicich JH, French LA. Use of Dexamethasone in the Treatment of Cerebral Edema Resulting from Brain Tumors and Brain Surgery. Am Pract Dig Treat. 1961; 12:169–174
- [20] French LA, Galicich JH. The Use of Steroids for Control of Cerebral Edema. Clin Neurosurg. 1964; 10:212–223
- [21] Glantz MJ, Cole BF, Forsyth PA, et al. Practice Parameter: Anticonvulsant Prophylaxis in Patients with Newly Diagnosed Brain Tumors. Report of the Quality Standards Subcommittee of the American Academy of Neurology. Neurology. 2000; 54:1886–1893
- [22] Chicoine MR, Silbergeld DL. Pharmacology for Neurosurgeons. Part I: Anticonvulsants, Chemotherapy, Antibiotics. Contemp Neurosurg. 1996; 18:1–6
- [23] Prados MD, Berger MS, Wilson CB. Primary Central Nervous System Tumors: Advances in Knowledge and Treatment. CA Cancer J Clin. 1998; 48:331–360
- [24] Stewart DJ. A Critique of the Role of the Blood-Brain Barrier in the Chemotherapy of Human Brain Tumors. J Neurooncol. 1994:121–139
- [25] Broadwell RD, Salzman M. In: The Blood Brain Barrier. Neurobiology of Brain Tumors. Baltimore: Williams and Wilkins; 1991:229–250
- [26] Madajewicz S, Chowhan N, Tfayli A, et al. Therapy for Patients with High Grade Astrocytoma Using Intraarterial Chemotherapy and Radiation Therapy. Cancer. 2000; 88:2350–2356
- [27] Barker FG, Prados MD, Chang SM, et al. Radiation Response and Survival Time in Patients with Glioblastoma Multiforme. J Neurosurg. 1996; 84:442–448
- [28] Laohaprasit V, Silbergeld DL, Ojemann GA, Eskridge JM, Winn HR. Postoperative CT Contrast Enhancement Following Lobectomy for Epilepsy. J Neurosurg. 1990; 73:392–395

- [29] Jeffries BF, Kishore PR, Singh KS, et al. Contrast Enhancement in the Postoperative Brain. *Radiology*. 1981; 139:409–413
- [30] Gerber AM, Savolaine ER. Modification of Tumor Enhancement and Brain Edema in Computerized Tomography by Corticosteroids: Case Report. *Neurosurgery*. 1980; 6:282–284
- [31] Hatam A, Bergström M, Yu ZY, et al. Effect of Dexamethasone Treatment in Volume and Contrast Enhancement of Intracranial Neoplasms. *J Comput Assist Tomogr*. 1983; 7:295–300
- [32] Plesec TP, Prayson RA. Frozen section discrepancy in the evaluation of central nervous system tumors. *Arch Pathol Lab Med*. 2007; 131:1532–1540
- [33] Shah AB, Muzumdar GA, Chitale AR, Bhagwati SN. Squash preparation and frozen section in intraoperative diagnosis of central nervous system tumors. *Acta Cytol*. 1998; 42:1149–1154
- [34] Uematsu Y, Owai Y, Okita R, Tanaka Y, Itakura T. The usefulness and problem of intraoperative rapid diagnosis in surgical neuropathology. *Brain Tumor Pathol*. 2007; 24:47–52
- [35] Burger PC. Smears and Frozen Sections in Surgical Neuropathology: A manual. PB Medical Publishing; 2009
- [36] Scherer HB. Structural Developments in Gliomas. *American Journal of Cancer*. 1938; 34:333–351
- [37] McKeever PE, Dabbs DJ. In: Immunohistochemistry of the Nervous System. Diagnostic Immunohistochemistry. New York: Churchill Livingstone; 2002:559–624
- [38] Russell DS, Rubenstein LJ. Pathology of Tumours of the Nervous System. 5th ed. Baltimore: Williams and Wilkins; 1989
- [39] Burton BK. Alpha-Fetoprotein Screening. *Adv Pediatr*. 1986; 33:181–196

35 Syndromes Involving Tumors

35.1 Neurocutaneous disorders

35.1.1 General information

Formerly called phakomatoses. Neurocutaneous disorders (NCD) are a group of conditions, each with unique neurologic findings and benign cutaneous lesions (NB: both skin and the CNS derive embryologically from ectoderm), usually with dysplasia of other organ systems (often including the eyes). With the exception of ataxia-telangiectasia (not discussed here) all exhibit autosomal dominant inheritance. There is also a high rate of spontaneous mutations. These syndromes should be kept in mind in a pediatric patient with a tumor, and other stigmata of these syndromes should be sought.

NCDs that are more likely to come to the attention of the neurosurgeon:

1. neurofibromatosis: see below
2. tuberous sclerosis (p.606)
3. von Hippel-Lindau disease (p.703)
4. Sturge Weber syndrome (p.608)
5. racemose angioma (Wyburn-Mason syndrome): midbrain and retinal AVMs

35.1.2 Neurofibromatosis

General information

Neurofibromatosis (NFT) is the most common of the NCDs. There are as many as 6 distinct types, the two most common of which (NF1 & NF2) are compared in □ Table 35.1 (variant forms also occur).

Table 35.1 Comparison of neurofibromatosis 1 & 2 ¹		
current designation →	Neurofibromatosis 1 (NF1) (p. 604)	Neurofibromatosis 2 (NF2) (p. 605)
alternate term	von Recklinghausen’s	bilateral acoustic NFT AKA MISME syndrome
obsolete term	peripheral NFT	central NFT
U.S. prevalence	100,000 people	≈ 3000 people
incidence	1/3000 births	1 in 40,000
inheritance	AD	AD
sporadic occurrence	30–50%	>50%
gene locus	17 (17q11.2)	22 (22q12.2)
gene product	neurofibromin	schwannomin (merlin)
vestibular schwannomas (VS)	almost never bilateral	bilateral VSs are the hall- mark
cutaneous schwannomas	no	70%
Lisch nodules	very common	not associated
cataracts	not associated	60–80%
skeletal anomalies	common	not associated
pheochromocytoma	occasional	not associated
MPNST ^a	≈ 2%	not associated
intellectual impairment	associated	not associated
associated medullary spinal cord tumors	astrocytoma	ependymoma
^a malignant peripheral nerve sheath tumor		

Schwannoma vs. neurofibroma

While similar in many ways, these tumors differ histologically. Schwannomas (nee: neurilemmomas) arise from schwann cells which produce myelin. Neurofibromas consist of neurites (axons or dendrites of immature or developing neurons), Schwann’s cells, and fibroblasts within a collagenous or myxoid matrix. In contrast to schwannomas which displace axons (centrifugal), neurofibromas are unencapsulated and engulf the nerve of origin (centripetal). Neurofibromas may occur as solitary lesions, or, may be multiple as part of NF1 in the setting of which there is potential for malignant transformation. Both tumors have Antoni A (compact) and Antoni B (loose) fibers, but neurofibromas tend to have more Antoni B fibers. A patient ≤ 30 years of age with a vestibular schwannoma is at increased risk of having NF2.

Neurofibromatosis 1
(NF1 AKA von Recklinghausen’s disease)

See reference.²

General information

More common than NF2, representing >90% of cases of neurofibromatosis.

Clinical

Diagnostic criteria: see □ Table 35.2.³

Associated conditions

1. Schwann-cell tumors on any nerve (but bilateral VSs are virtually nonexistent)
2. spinal and/or peripheral-nerve neurofibromas
3. multiple skin neurofibromas
4. aqueductal stenosis (p.258)
5. macrocephaly: secondary to aqueductal stenosis and hydrocephalus, increased cerebral white matter
6. intracranial tumors: hemispheric astrocytomas are the most common, solitary or multicentric meningiomas (usually in adults). Gliomas associated with NF1 are usually pilocytic astrocytomas. Brain stem astrocytomas include both contrast-enhancing pilocytic lesions and those that are non-enhancing and radiologically diffuse
7. unilateral defect in superior orbit → pulsatile exophthalmos
8. neurologic or cognitive impairment: 30–60% have mild learning disabilities
9. kyphoscoliosis (seen in 2–10% often progressive which then requires surgical stabilization)
10. visceral manifestations from involvement of autonomic nerves or ganglia within the organ. Up to 10% of patients have abnormal gastrointestinal motility/neuronal intestinal dysplasia related to neuronal hyperplasia within submucosal plexus
11. $\approx 20\%$ develop plexiform neurofibromas: tumors from multiple nerve fascicles that grow along the length of the nerve. Almost pathognomonic for NF1⁴
12. syringomyelia

Table 35.2 Diagnostic criteria for NF1 ³
Two or more of the following:
• ≥ 6 café au lait spots^a, each ≥ 5 mm in greatest diameter in prepubertal individuals, or ≥ 15 mm in greatest diameter in postpubertal patients • ≥ 2 neurofibromas of any type, or one plexiform neurofibroma (neurofibromas are usually not evident until age 10–15 yrs). May be painful • freckling (hyperpigmentation) in the axillary or intertriginous (inguinal) areas • optic glioma: see below • ≥ 2 Lisch nodules: pigmented iris hamartomas that appear as translucent yellow/brown elevations that tend to become more numerous with age • distinctive osseous abnormality, such as sphenoid dysplasia or thinning of long bone cortex with or without pseudarthrosis (e.g. of tibia or radius) • a first degree relative (parent, sibling or offspring) with NF1 by above criteria
^a café au lait spots: hyperpigmented oval light brown skin macules (flat). May be present at birth, increase in number and size during 1st decade. Are present in >99% of NF1 cases. Rare on face

- 13. malignant tumors that have increased frequency in NFT: neuroblastoma, ganglioglioma, sarcoma, leukemia, Wilm’s tumor, breast cancer⁵
- 14. pheochromocytoma: is occasionally present
- 15. “unidentified bright objects” (UBOs) on brain or spinal MRI in 53–79%of patients (bright on T2WI, isointense on T1WI) that may be hamartomas, heterotopias, foci of abnormal myelination or low grade tumors.⁶ Tend to resolve with age

Genetics

Simple autosomal dominant inheritance with variable expressivity but almost 100%penetrance after age 5 years. The NF1 gene is on chromosome 17q11.2 which codes for neurofibromin⁷ (neurofibromin is a negative regulator of the Ras oncogene). Loss of neurofibromin as in NF1 results in elevation of growth-promoting signals. The spontaneous mutation rate is high, with 30–50%of cases representing new somatic mutations.⁸

Counselling: prenatal diagnosis is possible by linkage analysis only if there are 2 or more affected family members.⁷ 70%of NF1 gene mutations can be detected using protein truncation analysis.

Management

- 1. optic gliomas
 - a) unlike optic gliomas in the absence of NFT, these are rarely chiasmal (usually involving the nerve), are often multiple, and have a better prognosis
 - b) most are non progressive, and should be followed ophthalmologically and with serial imaging (MRI or CT)
 - c) surgical intervention probably does not alter visual impairment. Therefore, surgery is reserved for special situations (large disfiguring tumors, pressure on adjacent structures...)
- 2. other neural tumors in patients with NF1 should be managed in the same manner as in the general population
 - a) focal, resectable, symptomatic lesions should be surgically removed
 - b) intracranial tumors in NF1 may often be unresectable, and in these cases chemotherapy and/or radiation therapy may be appropriate, with surgery reserved for cases with increasing ICP
 - c) when malignant degeneration is suspected (rare, but incidence of sarcomas and leukemias is increased), biopsy with or without internal decompression may be indicated

Neurofibromatosis 2 (NF2 AKA bilateral acoustic NFT)

See reference.⁹

General information

AKA MISME Syndrome (Multiple Inherited Schwannomas, Meningiomas, and Ependymomas).

Clinical

Diagnostic criteria: see □ Table 35.3.¹⁰

Other clinical features:

- 1. seizures or other focal deficits
- 2. skin nodules, dermal neurofibromas, café au lait spots (less common than in NF1)
- 3. multiple intradural spinal tumors are common (less common in NF1)¹¹: including intramedullary (especially ependymomas) and extramedullary (schwannomas, meningiomas...)
- 4. retinal hamartomas

Table 35.3 Diagnostic criteria for NF2 ¹⁰
Definite diagnosis if either: 1. bilateral vestibular schwannomas (VS) on imaging (MRI or CT) or 2. a first degree relative (parent, sibling or offspring) with NF2 and either: a) unilateral VS at age <30 years or b) any two of the following: meningioma, schwannoma (including spinal root), glioma (includes astrocytoma, ependymoma), posterior subcapsular lens opacity
Probable diagnosis if either: 1. unilateral VS at age <30 and any of the following: meningioma, schwannoma, glioma, posterior subcapsular lens opacity or 2. multiple meningiomas and either of the following: schwannoma, glioma, or posterior lens opacity

- 5. antigenic nerve growth factor is increased (does not occur with NF1)
- 6. despite its name, is not associated with neurofibromas

Two subtypes¹⁰:

- 1. the more common, severe form with younger age of onset (2nd to 3rd decade), with rapid progression of hearing loss and multiple associated tumors
- 2. milder form, presents later in life, with slower deterioration in hearing and fewer associated tumors

Genetics

Autosomal dominant inheritance. NF2 is due to a mutation at chromosome 22q12.2 Which results in the inactivation of schwannomin (AKA merlin, a semi-acronym for moesin-, ezrin, and radixin-like proteins), a tumor suppression peptide. NF2 patients with nonsense and frameshift mutations are more likely to have intramedullary tumors (but not any other type of tumor) compared with other mutation types.

Management considerations

- 1. bilateral vestibular schwannomas:
 - a) chance of preserving hearing is best when tumor is small. Thus, one should attempt to remove smaller tumor. If hearing is serviceable in that ear after surgery, then consider removing the second tumor, otherwise follow the second tumor as long as possible and perform a subtotal removal in an attempt to prevent total deafness
 - b) stereotactic radiosurgery therapy may be a treatment option
- 2. most NF2 patients will become deaf at some time during their life
- 3. prior to surgery, obtain MRI of cervical spine to R/O intraspinal tumors that may cause cord injuries during other operations
- 4. NB: pregnancy may accelerate the growth of eighth nerve tumors

35.1.3 Tuberous sclerosis complex

General information

Key concepts

- most cases are due to spontaneous mutation. Inherited cases are autosomal dominant. Incidence: 1 in 6K-10K live births
- clinical triad: seizures, mental retardation and sebaceous adenomas; the full clinical triad is seen in <1/3 of cases
- typical CNS finding: subependymal nodules (“tuber”) – a hamartoma
- common associated neoplasm: subependymal giant cell astrocytoma
- 2 tumor suppressor genes: TSC1 (on chromosome 9q34) codes for hamartin, and TSC2 (on chromosome 16p13) encodes tuberlin
- CT shows intracerebral calcifications (usually subependymal)

Tuberous sclerosis complex (TSC), AKA Bourneville’s disease, is a neurocutaneous disorder characterized by hamartomas of many organs including the skin, brain, eyes and kidneys. In the brain, the hamartomas may manifest as cortical tubers, glial nodules located subependymally or in deep white matter, or giant cell astrocytomas. Associated findings include pachygyria or microgyria.

Epidemiology/genetics/epigenetics

Incidence: 1 in 6,000–10,000 live births.¹² Point prevalence: 10.6 per 100,000 persons (from Rochester, MN¹³).

Autosomal dominant inheritance, however spontaneous mutation account for the majority of cases.¹⁴ Two distinct tumor suppressor genes have been identified: TSC1 (located on chromosome 9q34) codes for hamartin, and TSC2 (on chromosome 16p13) codes for tuberlin. Only 1 gene needs to be affected to develop TSC. These proteins work together to inhibit activation of rapamycin (mTOR).

Genetic counseling for unaffected parents with one affected child: 1–2% chance of recurrence.

Pathology

Subependymal nodules (“tubers”) are benign hamartomas that are almost always calcified, and protrude into the ventricles.

Subependymal giant cell astrocytoma (SEGA): a transformation lesion. Almost always located at the foramen of Monro. Occurs in 5–15%of patients with TSC.¹⁵ Histology shows fibrillary areas alternating with cells containing generous amounts of eosinophilic cytoplasm. Areas of necrosis and abundant mitotic figures may be seen, but are not associated with the typical malignant aggressiveness that these features usually denote.¹⁶

Clinical

Diagnostic criteria are shown in □ Table 35.4.

In the infant, the earliest finding is of “ash leaf” macules (hypomelanotic, leaf shaped) that are best seen with a Wood’s lamp. Infantile myoclonus may also occur.

In older children or adults, the myoclonus is often replaced by generalized tonic-clonic or partial complex seizures which occurs in 70–80% Facial adenomas are not present at birth, but appear in > 90%by age 4 yrs (these are not really adenomas of the sebaceous glands, but are small hamartomas of cutaneous nerve elements that are yellowish-brown and glistening and tend to arise in a butterfly malar distribution usually sparing the upper lip).

Retinal hamartomas occur in ≈ 50%(central calcified hamartoma near the optic disc or a more subtle peripheral flat salmon-colored lesion). A distinctive depigmented iris lesion may also occur.

Evaluation

Plain skull x-rays
May show calcified cerebral nodules.

CTscan
See reference.¹⁸

Intracerebral calcifications are the most common (97%of cases) and characteristic finding. Primarily located subependymally along the lateral walls of the lateral ventricles or near the foramina of Monro.

Low density lesions that do not enhance are seen in 61% Probably represent heterotopic tissue or defective myelination. Most common in occipital lobe.

Hydrocephalus (HCP) may occur even without obstruction. In the absence of tumor, HCP is usually mild. Moderate HCP usually occurs only in the presence of tumor.

Subependymal nodules are usually calcified, and protrude into the ventricle (“candle guttering” described the appearance on pneumoencephalography).

Table 35.4 Diagnostic criteria of tuberous sclerosis complex ¹⁷
• TSC: diagnosis requires 2 major criteria, or 1 major and 2 minor criteria • Probable TSC: 1 major+1 minor • Possible TSC: 1 major or 2 minor
Major criteria
• cutaneous manifestations: facial angiofibroma, ungual fibroma, >3 hypomelanotic macules, shagreen patch • brain and eye lesions: cortical tuber, subependymal nodules, subependymal giant cell astrocytoma, multiple retinal nodular hamartomas • tumors in other organs: cardiac, rhabdomyoma, lymphangioleiomyomatosis, renal angiomyolipoma
Minor criteria
• rectal polyps • pits in dental enamel • bone cysts • migration abnormalities of cerebral white matter • gingival fibromas • nonrenal hamartomas • achromic retinal patches • confetti skin lesions • multiple renal cysts

Paraventricular tumors (mostly giant cell astrocytomas) see pathology) are essentially the only enhancing lesion in TSC.

MRI

Subependymal tubers are high on T2 and low on T1 and only ≈ 10%enhance.

Low signal in subependymal lesions may represent calcification. SEGA enhance intensely (enhancing subependymal lesion are almost always SEGAs).

Radial bands sign: abnormal signal intensity extending in a radial manner, representing cells of varying degrees of neuronal and astrocytic differentiation as well as difficult to classify cells.¹⁹

Treatment

Paraventricular tumors should be followed. Tubers grow minimally, but SEGA progress should be removed if they are symptomatic. A transcallosal approach or ventriculoscopic removal are options.

Infantile myoclonus may respond to steroids. Seizures are treated with AEDs.

Surgery for intractable seizures may be considered when a particular lesion is identified as a seizure focus. Better seizure control, not cure, is the goal in TSC.

Patients ≥3 years of age with increasing size of SEGA lesions have had sustained reduction of SEGA volume on everolimus.²⁰

35.1.4 Sturge–Weber syndrome

General information

Key concepts

- cardinal signs: 1) localized cerebral cortical atrophy and calcifications, 2) ipsilateral port-wine facial nevus (usually in distribution of V1)
- contralateral seizures usually present
- plain skull films classically show “tram-tracking” (double parallel lines)

AKA encephalotrigeminal angiomas. A neurocutaneous disorder consisting of:

1. cardinal features:
- a) localized cerebral cortical atrophy and calcifications (especially cortical layers 2 and 3, with a predilection for the occipital lobes):

- calcifications appear as curvilinear double parallel lines (“tram-tracking”) on plain x-rays
 - cortical atrophy usually causes contralateral hemiparesis, hemiatrophy, and homonymous hemianopia (with occipital lobe involvement)
- b) ipsilateral port-wine facial nevus (nevus flammeus) usually in distribution of 1st division of trigeminal nerve (rarely bilateral)
2. other findings that may be present:
- a) ipsilateral exophthalmos and/or glaucoma, coloboma of the iris

b) oculomeningeal capillary hemangioma

c) convulsive seizures: contralateral to the facial nevus and cortical atrophy. Present in most patients starting in infancy

d) retinal angiomas

Genetics

Most cases are sporadic. Other cases are suggestive of recessive inheritance, with chromosome 3 being implicated.

Treatment

Treatment is supportive. Anticonvulsants are used for seizures. Lobectomy or hemispherectomy may be needed for refractory seizures. XRT: complications are common and benefits are lacking. Laser surgery for the cutaneous nevus is disappointing; better results obtain from masking the nevus with a skin colored tattoo.

35.1.5 Neurocutaneous melanosis (NCM)

Background

1. a rare, congenital, nonheritable phakomatosis in which large or numerous congenital melanocytic nevi are associated with benign and/or malignant melanocytic tumors of the leptomeninges²¹
2. pathogenesis: neuroectodermal defect during morphogenesis involving melanoblasts of skin and pia mater originating from neural crest cells²¹

Clinical features

1. two-thirds of patients with NCM have giant congenital melanocytic nevi²¹: pigmented nevi that are large, hairy, or both. (The chances that nevi represents NCM is higher when the nevi are located on head, posterior neck or paravertebral)
2. one-third have numerous lesions without a single giant lesion²¹
3. virtually all have large cutaneous melanocytic (pigmented) nevi located on the posterior torso²²
4. neurologic manifestations: usually before age 2 years. Signs of intracranial hypertension (lethargy, vomiting...), focal seizures, motor deficits or aphasia²¹
5. hydrocephalus: in almost 66% Usually due to obstruction of CSF flow or reduced absorption as a result of thickened leptomeninges²¹

Clinical diagnostic criteria

See reference.²³

1. large or multiple congenital melanocytic nevi with meningeal melanosis or melanoma
2. absence of cutaneous melanoma, except in patients with benign meningeal lesions (i.e. must rule-out meningeal metastases from cutaneous melanoma)
3. no evidence of meningeal melanoma, except in patients with benign cutaneous lesions

Associated conditions

NCM is sometimes associated with

1. neurocutaneous syndromes²¹
 - a) Sturge-Weber syndrome (p.608)
 - b) von Recklinghausen's neurofibromatosis (NF1) (p.604)
2. posterior fossa cystic malformations: e.g. Dandy Walker malformation (p.256); occurs in up to 10% These cases have worse prognosis due to malignant transformation²¹
3. intraspinal lipoma and syringomyelia²¹

Diagnostic testing

1. MRI: T1 and T2 signal shortening produced by melanin. IV gadolinium may demonstrate enhancement of tumor-infiltrated meninges²¹
2. histological exam of CNS lesions shows leptomeningeal melanosis (benign) which develops from the melanocytes of the pia matter. Melanoma (malignant) occurs in 40–62% of cases but distinction has little prognostic significance because of the poor outcome of the symptomatic NCM patient even in the absence of melanoma²¹

Management

The benefit of resecting skin lesions is questionable in the presence of leptomeningeal lesions.²⁴ NCM appears refractory to radiation therapy and chemotherapy²⁴

Neurosurgical involvement is usually limited to²³:

1. shunting for hydrocephalus
2. palliative operative decompression if early in the course
3. biopsy for tissue diagnosis in questionable cases

Prognosis

1. when neurological signs are present, prognosis is poor regardless of whether or not malignancy is present
2. >50% of patients die within 3 years after the first neurologic manifestation²¹

Table 35.5 Familial syndromes associated with CNS tumors	
Syndrome	CNS tumor
von Hippel-Lindau (p. 703)	hemangioblastoma
tuberous sclerosis (p. 606)	subependymal giant cell astrocytoma
neurofibromatosis type I (p. 604)	optic glioma, astrocytoma, neurofibroma
neurofibromatosis type II (p. 605)	vestibular schwannoma, meningioma, ependymoma, astrocytoma
Turcot syndrome (BTP syndrome) (p. 610) ²⁵	GBM, AA, & medulloblastoma, pineoblastoma
Li-Fraumeni (p. 610)	astrocytoma, PNET
Cowden (p. 647)	meningiomas
Ihermitte-Duclos (p. 647)	

35.2 Familial tumor syndromes

35.2.1 General information

Several familial syndromes are associated with CNS tumors as shown in □ Table 35.5.

35.2.2 Turcot syndrome

A rare inherited disorder characterized by multiple colorectal neoplasms (carcinomas or benign adenomatous polyps) together with neuroepithelial tumors of the CNS (GBM, AA, MB, pineoblastoma, ganglioglioma & ependymoma).¹² Type 1: GBM without familial polyposis (but often with non-polyposis colorectal cancer). Mean survival of Turcot patients with GBM is 27 months (longer than sporadic cases). Type 2: MB & familial adenomatous polyposis.

35.2.3 Li-Fraumeni syndrome

Rare (<400 families identified) inherited autosomal dominant mutation of the TP53 tumor suppressor gene. Patients have increased incidence of multiple types of tumors, including: sarcoma & osteosarcoma, breast cancer, astrocytoma and PNET, adrenocortical carcinoma, leukemia.

References

[1]

Burger PC, Scheithauer BW. AFIP Atlas of Tumor Pathology. Fourth series. Fascicle 7: Tumors of the Central Nervous System. Washington, D.C.: Armed Forces Institute of Pathology; 2007

[2]

Riccardi VM. von Recklinghausen Neurofibromatosis. N Engl J Med. 1981; 305:1617–1627

[3]

National Institutes of Health Consensus Development Conference. Neurofibromatosis: Conference Statement. Arch Neurol. 1988; 45:575–578

[4]

Packer RJ, Gutmann DH, Rubenstein A, Viskochil D, Zimmerman RA, Vezina G, Small J, Korf B. Plexiform neurofibromas in NF1: toward biologic-based therapy. Neurology. 2002; 58:1461–1470

[5]

Sharif S, Moran A, Huson SM, Iddenden R, Shenton A, Howard E, Evans DG. Women with neurofibromatosis 1 are at a moderately increased risk of developing breast cancer and should be considered for early screening. J Med Genet. 2007; 44:481–484

[6]

Sevick RJ, Barkovich AJ, Edwards MS, Koch T, Berg B, Lempert T. Evolution of white matter lesions in neurofibromatosis type 1: MR findings. AJR Am J Roentgenol. 1992; 159:171–175

[7]

Karnes PS. Neurofibromatosis: A Common Neurocutaneous Disorder. Mayo Clin Proc. 1998; 73:1071–1076

[8]

Walker L, Thompson D, Easton D, Ponder B, Ponder M, Frayling I, Baralle D. A prospective study of neurofibromatosis type 1 cancer incidence in the UK. Br J Cancer. 2006; 95:233–238

[9]

Martuza RL, Eldridge R. Neurofibromatosis 2: (Bilateral Acoustic Neurofibromatosis). N Engl J Med. 1988; 318:684–688

[10]

Parry DM, Eldridge R, Kaiser-Kupfer MI, Bouzas EA, Pikus A, Patronas N. Neurofibromatosis 2 (NF2): clinical characteristics of 63 affected individuals and clinical evidence for heterogeneity. Am J Med Genet. 1994; 52:450–461

[11]

Egelhoff JC, Bates DJ, Ross JS, Rothner AD, Cohen BH. Spinal MR Findings in Neurofibromatosis Types 1 and 2. AJNR. 1992; 13:1071–1077

[12]

Hottinger AF, Khakoo Y. Neurooncology of familial cancer syndromes. J Child Neurol. 2009; 24:1526–1535

[13]

Wiederholt WC, Gomez MR, Kurland LT. Incidence and Prevalence of Tuberous Sclerosis in Rochester, Minnesota, 1950 through 1982. Neurology. 1985; 35:600–603

[14]

Logue LG, Acker RE, Sienko AE. Best cases from the AFIP: angiomyolipomas in tuberous sclerosis. Radiographics. 2003; 23:241–246

[15]

Thiele EA. Managing epilepsy in tuberous sclerosis complex. J Child Neurol. 2004; 19:680–686

[16]

Chow CW, Klug GL, Lewis EA. Subependymal Giant-Cell Astrocytoma in Children: An Unusual

- Discrepancy Between Histological and Clinical Features. *J Neurosurg.* 1988; 68:880–883
- [17] Roach ES, Gomez MR, Northrup H. Tuberous sclerosis complex consensus conference: revised clinical diagnostic criteria. *J Child Neurol.* 1998; 13:624–628
- [18] McLaurin RL, Towbin RB. Tuberous Sclerosis: Diagnostic and Surgical Considerations. *Pediatr Neurosci.* 1985; 12:43–48
- [19] Bernauer TA. The radial bands sign. *Radiology.* 1999; 212:761–762
- [20] Franz DN, Agricola K, Mays M, Tudor C, Care MM, Holland-Bouley K, Berkowitz N, Miao S, Peyrard S, Krueger DA. Everolimus for subependymal giant cell astrocytoma: 5-year final analysis. *Ann Neurol.* 2015. DOI: 10.1002/ana.24523
- [21] Di Rocco F, Sabatino G, Koutzoglou M, Battaglia D, Caldarelli M, Tamburrini G. Neurocutaneous melanosis. *Childs Nerv Syst.* 2004; 20:23–28
- [22] DeDavid M, Orlov SJ, Provost N, Marghoob AA, Rao BK, Wasti Q, Huang CL, Kopf AW, Bart RS. Neurocutaneous melanosis: clinical features of large congenital melanocytic nevi in patients with manifest central nervous system melanosis. *J Am Acad Dermatol.* 1996; 35:529–538
- [23] McClelland S, III, Charnas LR, SantaCruz KS, Garner HP, Lam CH. Progressive brainstem compression in an infant with neurocutaneous melanosis and Dandy-Walker complex following ventriculoperitoneal shunt placement for hydrocephalus. Case report. *J Neurosurg.* 2007; 107:500–503
- [24] Mena-Cedillos CA, Valencia-Herrera AM, Arroyo-Pineda AI, Salgado-Jimenez MA, Espinoza-Montero R, Martinez-Avalos AB, Perales-Arroyo A. Neurocutaneous melanosis in association with the Dandy-Walker complex, complicated by melanoma: report of a case and literature review. *Pediatr Dermatol.* 2002; 19:237–242
- [25] Paraf F, Jothy S, Van Meir EG. Brain Tumor-Polyposis Syndrome: Two Genetic Diseases. *J Clin Oncol.* 1997; 15:2744–2758

36 Astrocytomas

36.1 Incidence, risk factors

Astrocytic tumors are the most common primary intra-axial brain tumor. The average annual age adjusted incidence from 2006 – 2010 was 5.17 per 100,000,¹ approximately 16,000 new cases/year in the U. S.

The best-established causes for brain tumors are syndromic (familial diseases...) and post-radiation therapy.

The controversy of cell phones as a risk factor: The substantial increase in the number of mobile phone subscriptions since the early 1990's has not been observed to be associated with increased incidence of brain tumor in the US.^{2,3,4} Nonetheless, primarily due to lack of data for extremely long exposure (>15 years), in May 2011, the WHO issued a warning of possible cancer risk in relation to the use of devices that emit nonionizing electromagnetic radiation, such as cellular phones.⁵

36.2 Classification and grading of astrocytic tumors

36.2.1 Classification by general morphology or behavior

In general

- Circumscribed vs. Diffuse
- Non-infiltrative vs. Infiltrative⁶
- Special vs. Ordinary

Detailed classification:

1. Infiltrating astrocytoma
 - a) Diffuse astrocytoma (AKA low grade astrocytoma): WHO II. Genetic hallmark: frequent TP53 mutation. Histological variants:
 - fibrillary: the most common subtype
 - gemistocytic
 - protoplasmic: rare
 - b) Mixed oligoastrocytoma WHO II
 - c) Anaplastic astrocytoma WHO III
 - d) Anaplastic oligoastrocytoma WHO III
 - e) Glioblastoma multiforme WHO IV
2. Non-infiltrative astrocytoma
 - a) Juvenile Pilocytic astrocytoma WHO I
 - b) Subependymal giant cell astrocytoma (Tuberous Sclerosis) WHO I
 - c) Desmoplastic infantile astrocytoma WHO I
 - d) Pilomyxoid astrocytoma WHO II
3. Unique astrocytoma (unique variant that the prognosis does not conform to the grading):
 - a) Pleomorphic xanthoastrocytoma WHO II
 - b) Gemistocytic astrocytoma WHO II

In diffuse/infiltrative tumors, since complete surgical excision is rarely attainable, there is higher recurrence rate and poorer prognosis in general.

Special vs Ordinary:

- dividing tumors according to favorable or poor behavior is not considered to be a contemporary classification system
- Special – favorable outcome, younger age group, includes pilocytic astrocytoma, microcytic cerebellar and subependymal giant cell astrocytoma
- Ordinary – poorer outcome, includes in WHO Grade II-IV or St. Anne/Mayo Grade I-IV group

36.2.2 Grading and neuropathology

General information

The grading of astrocytomas has been historically fraught with disagreement, and a number of grading systems have been proposed over the years. The first system of Bailey and Cushing was a 3-tiered system, the Kernohan system was 4-tiered, and since then a number of 3-tiered systems like the

Ringertz system,⁷ have been proposed. As a result, there is a lack of uniformity in, for example, what constitutes a glioblastoma from series to series over time. The current trend has been to use one of two different systems, the WHO definition or the St. Anne/Mayo system, both of which appear below.

Grading of astrocytomas remains controversial. Some special concerns:

- 1. sampling error: may have different degrees of malignancy in different areas
- 2. dedifferentiation (p.615): tumors tend to progress in malignancy over months or years
- 3. histological criteria that affect prognosis include: cellularity, presence of giant cells, anaplasia, mitosis, vascular proliferation with or without endothelial proliferation, necrosis, and pseudopalisading⁸
- 4. in addition to histology, issues that affect clinical behavior (many of which are not factored into most grading systems) include:
 - a) patient age
 - b) extent of tumor
 - c) topography: tumor location, especially in relation to critical structures

Obsolete grading systems

□ Ringertz’s system. The obsolete Ringertz’s system developed three grades for astrocytoma malignancy.⁷ Necrosis was used to divide the intermediate grade from the glioblastoma multi-forme. The Ringertz three-tiered grading system was more clinically correlated with the prognosis for astrocytomas.

□ Kernohan system. The obsolete Kernohan system,⁹ originally devised at the Mayo Clinic, divided these tumors into 4 grades (grade IV AKA glioblastoma multiforme) based on the degree of presence of a number of features such as anaplasia, nuclear pleomorphism, number of mitoses. The Kernohan system determined tumor grade according to the proportion of normal tissue remaining in the invading tumor and the type of invading edge of the tumor into normal tissue. Prognostically, this system distinguished only 2 clinically different groups (grades I/II, and grades III/IV) and is not used today. It is presented for completeness when reviewing older literature.

Current grading systems

The 2 main systems in use today are shown below, and differ primarily in the definition of Grade I.

□ WHO system. The World Health Organization (WHO) system is shown in □ Table 36.1.¹⁰ In the WHO system, grade I is reserved for special types of astrocytomas that are more circumscribed, including pilocytic astrocytomas, while the more typical astrocytic neoplasms are graded II through IV. The approximate equivalence to the Ringertz system, Kernohan grade, and St Anne-Mayo grading is also shown in □ Table 36.2.

Table 36.1 WHO classification of (“ordinary”) astrocytic tumors	
Designation	Criteria
II: diffuse astrocytoma	cytological atypia alone
III: anaplastic astrocytoma	anaplasia and mitotic activity
IV: glioblastoma (GBM)	also show microvascular proliferation and/or necrosis

Table 36.2 Approximate equivalence of Ringertz system, Kernohan grade (I-IV), and St Anne-Mayo grading to WHO system			
Modified Ringertz	Kernohan	St. Anne-Mayo	WHO 2007 ¹⁰
			Grade I (e.g. pilocytic as-trocytoma)
Astrocytoma (low grade)	I & II	Astrocytoma 1 & 2	Diffuse astrocytoma II
Anaplastic astrocytoma	III	Astrocytoma 3	Anaplastic astrocytoma III
Glioblastoma multiforme	IV	Astrocytoma 4	Glioblastoma multiforme IV

- WHO 2007 Grading of astrocytomas for specific types of astrocytomas:
- Grade I
 - Subependymal giant cell astrocytoma (SEGA)
 - Pilocytic astrocytoma
 - Grade II
 - Pilomyxoid astrocytoma
 - Diffuse astrocytoma
 - Pleomorphic xanthastrocytoma
 - Grade III – anaplastic astrocytoma
 - Grade IV
 - Glioblastoma
 - Giant cell glioblastoma
 - Gliosarcoma

□ St. Anne/Mayo grading system. Grading systems such as those described above are very dependent upon subjective impression of histological findings. The classification system known as the St. Anne/Mayo (SA/M) system¹¹ addresses histological considerations, and is reproducible and prognostically significant.¹² It is restricted to “ordinary” astrocytomas, as grade has not been shown to correlate with clinical behavior in pilocytic or microcystic cerebellar astrocytomas. It is similar to the WHO system except that SA/M grade I astrocytomas are a very rare diffuse astrocytoma without atypia.¹⁰

The SA/M system assesses the presence or absence of 4 criteria (see □ Table 36.3) and then assigns a grade based on the number of criteria present (□ Table 36.4). When the presence of any criteria is uncertain, it is considered to be absent.

The criteria tended to occur in a predictable sequence: nuclear atypia occurred in all grade 2 tumors, mitotic activity was seen in 92% of grade 3 tumors (and in none of the grade 2 tumors), necrosis and endothelial proliferation were restricted almost only to grade 4 tumors (they were seen in only 8% of grade 3 tumors).

The frequencies of occurrence among 287 “ordinary” astrocytomas were: grade 1 = 0.7% (a very rare tumor), grade 2 = 16%, grade 3 = 17.8% and grade 4 = 65.5%

Median survival was as follows¹¹: (there were only two grade 1 patients, one survived 11 years and the other was still alive after 15 years), grade 2 = 4 years, grade 3 = 1.6 years, and grade 4 = 0.7 years (8.5 months).

□ Relative frequency of astrocytoma grades. Glioblastoma comprise approximately 54% of all gliomas and about 6% of anaplastic astrocytoma. Less common ones like anaplastic oligodendroglioma and anaplastic oligoastrocytoma, together with the scarcer tumors such as anaplastic ependymoma and anaplastic ganglioglioma make up 7.2% of all gliomas.¹

Table 36.3 St. Anne/Mayo criteria	
• nuclear atypia: hyperchromatasia and/or obvious variation in size and shape	
• mitoses: regardless of normal or abnormal configuration	
• endothelial proliferation: vascular lumina are surrounded by “piled-up” endothelial cells (instead of the normal single layer). Does not include hypervascularity (which may occur in non-tumoral gliosis)	
• necrosis: only when obviously present. Does not include pseudopalisading when seen alone	

Table 36.4 St. Anne/Mayo grade	
Grade	No. of criteria
1	0
2	1
3	2
4	3 or 4

Future directions of astrocytoma grading

Guideline for future WHO classification¹³: In order to align with new developments in glioma molecular genetics, a consensus was reached amongst world leaders in neuropathology. Guidelines for the next WHO classification were issued:

- diagnostic entities should be defined as narrowly as possible to optimize interobserver reproducibility, clinicopathological predictions and therapeutic planning
 - diagnoses should be “layered” with histologic classification, WHO grade and molecular information listed below an “integrated diagnosis”
 - determinations should be made for each tumor entity as to whether molecular information is required, suggested or not needed for its definition
 - some pediatric entities should be separated from their adult counterparts
 - input for guiding decisions regarding tumor classification should be solicited from experts in complementary disciplines of neuro-oncology
 - entity-specific molecular testing and reporting formats should be followed in diagnostic reports.
- It is hoped that these guidelines will facilitate the forthcoming update of the fourth edition of the WHO classification of central nervous system tumors

Low-grade astrocytoma (WHO II) comments

AKA low-grade diffuse astrocytoma. Three histopathological types:

1. fibrillary astrocytoma: the most common histological subtype of Grade II
2. gemistocytic astrocytoma: particularly prone to progress to Grade II & IV
3. protoplasmic astrocytoma

These tumors tend to occur in children and young adults. Most present with seizures. There is a predilection for temporal, posterior frontal and anterior parietal lobes.¹⁴ They demonstrate low degrees of cellularity and preservation of normal brain elements within the tumor. Calcifications are rare. Anaplasia and mitoses are absent (a single mitosis is allowed). Blood vessels may be slightly increased in number. The ultimate behavior of these tumors is usually not benign. The most important favorable prognosticator is young age. Poor prognosis is associated with findings of increased ICP, altered consciousness, personality change, significant neurologic deficits,¹⁵ short duration of symptoms before diagnosis (suggesting rapid progression), and enhancement on imaging studies.

Dedifferentiation

The major cause of morbidity with low-grade astrocytomas is dedifferentiation to a more malignant grade. Low grade fibrillary astrocytomas tend to undergo malignant transformation more quickly (with six-fold increased rapidity) when diagnosed after age 45 years than when diagnosed earlier¹⁶ (see □ Table 36.5). Gemistocytic astrocytomas tend to dedifferentiate more rapidly than fibrillary astrocytomas. >60% of fibrillary astrocytomas have a mutation of the TP53 gene located on chromosome 17p; these tumors are more likely to dedifferentiate. Once dedifferentiation occurs, median survival is 2–3 years beyond that event. Genetic markers that correlate with a higher degree of malignant degeneration include^{17,18}:

1. loss of heterozygosity on chromosomes 10 & 17
2. alteration in tumor suppressor genes at 9p, 13q, 19q & 22q
3. changes in epidermal growth factor receptor (EGFR) and platelet-derived growth factor (PDGF)
4. transformation of the p53 suppressor gene
5. isocitrate dehydrogenase (IDH) mutations (p.616) – genetic aberrations that lead to epigenetic machinery dysfunction

Table 36.5 Dedifferentiation rate for low grade astrocytomas		
	Patients diagnosed @ age <45 yrs	Patients diagnosed @ age ≥45 yrs
mean time to de-differentiation	44.2 ± 17 mos	7.5 ± 5.7 mos
time to death	58 mos	14 mos

Malignant astrocytomas (WHO III and IV) comments

This category encompasses anaplastic astrocytoma (AA) and glioblastoma (GBM). Although both are “malignant”, AA and GBM have distinct differences. Among 1265 patients with malignant astrocytomas, the mean age was 46 yrs for AA, and 56 yrs for GBM. Mean duration of symptoms pre-op: 5.4 mos for GBM, and 15.7 mos for AA. Malignant astrocytomas may develop from low grade astrocytomas via dedifferentiation (see above), however they may also arise de novo.

Infratentorial glioblastoma (GBM) is rare, and often represents subarachnoid dissemination of a supratentorial GBM (used as an argument for irradiation in all patients with p-fossa GBM).¹⁹

Glioblastoma (multiforme) (WHO IV) comments

The most common primary brain tumor, it is also the most malignant astrocytoma. Current nomenclature omits “multiforme”.²⁰

Histological findings associated with GBM (not all may be present, and this list does not follow any of the standard grading systems above):

- gemistocytic astrocytes
- neovascularization with endothelial proliferation
- areas of necrosis
- pseudopalisading around areas of necrosis

36.3 Molecular genetics and epigenetics

36.3.1 Molecular pathways in the development of glioblastomas

Genome-wide profiling studies have shown remarkable genomic diversities among glioblastomas.^{21, 22} Molecular studies have helped identify at least 3 different pathways in the development of glioblastomas.²³

- 1st pathway: dysregulation of growth factor signaling through amplification and mutational activation of receptor tyrosine kinase (RTK) genes. RTKs are a diverse group of transmembrane proteins that act as receptors for growth factors like epithelial growth factor (EGF), vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF). They can also act as receptors for cytokines, hormones and other signaling pathways.
- 2nd pathway: the activation of the phosphatidylinositol-3-OH kinase (PI3K)/AKT/m TOR, which is an intracellular signaling pathway. It is essential in regulating cell survival.
- 3rd pathway: the inactivation of the p53 and retinoblastoma (Rb) tumor suppressor pathways.

36.3.2 Transcriptional silencing

O⁶-Methylguanine-DNA methyltransferase (MGMT) Methylation: MGMT is an independent prognosticator for response of malignant gliomas to chemotherapy with alkylating chemotherapeutic agents (e.g. nitrosourea or temozolomide)²⁴ which damage tumor DNA by attaching an alkyl group to the guanine base. The MGMT gene is located on chromosome 10q26 and codes for a DNA repair enzyme that specifically removes the mutagenic alkyl O⁶ methylguanine and restores guanine residues to their native state.^{25,26} The loss of MGMT expression is likely to be caused by transcriptional silencing through hypermethylation of the CpG islands on the genome^{27,28} and this hypermethylation phenomenon is frequently (45% to 75%) present in glioblastomas.

In short: Loss of MGMT expression makes alkylating agent (e.g. Temodar) more effective.

36.3.3 IDH 1/2 mutation

IDH1 Arg132 mutations and IDH2 Arg140 and Arg172 mutations accounting for >90% of aberrations.^{29,30} IDH1 and IDH2 mutations reduce the enzymatic capacity of these proteins to bind isocitrate, their substrate, and convert it into α -ketoglutarate (α -KG), generating carbon dioxide and replenishing NADH and NADPH as side products.³¹ This is one of the irreversible steps in the tricarboxylic acid cycle important for cellular respiration. Mutant IDH1 (cytoplasmic) and IDH2 (mitochondrial) enzymes also show a modified enzymatic capacity to convert α -KG into 2-hydroxyglutarate (2-HG), a small onco-metabolite. Equally important, IDH1 and IDH2 mutations stratify individuals into molecular subtypes with distinct clinical outcomes – the mutations are associated with lower-grade astrocytomas, oligodendrogliomas (grade II/III) and secondary gliomas with better overall survival, progression-free survival and chemosensitivity than glioblastomas that are wild type for both genes.^{29,30,31}

36.3.4 Primary vs. secondary glioblastoma

- Primary vs. secondary glioblastoma³²: First described in 1940 by German pathologist Scherer.
- primary glioblastoma: the majority of GBMs. Arise without evidence (clinical or histological) of a less malignant precursor. More common in older patients (mean age =55 years) after a short (<3 month) clinical history. Characterized by EGFR amplification (≈ 40%of cases) and/or overexpression (60%), PTEN mutations (30%), p16^{INK4a} deletion (30–40%), MDM2 amplification (<10%), and/or overexpression (50%), and in 50–80%of cases, loss of heterozygosity (LOH) on the entire chromosome 10.
 - secondary glioblastoma: develop by malignant degeneration of WHO grade II or III astrocytoma. Patients are younger (mean age =40 years) and have a slower clinical course. Secondary glioblastomas are less frequent than primary glioblastomas. 60%have TP53 mutations (present in >90%of the less malignant precursors). Malignant degeneration is characterized by allelic loss of chromosomes 19q and 10q. MGMT promoter methylation appears to occur with a higher frequency in secondary than in primary glioblastoma.^{33,34} Genetic sequencing of 200 glioblastomas uncovered IDH1 and IDH2 as recurrently mutated in 5%of primary gliomas and a majority of about 60–90% of secondary gliomas.^{29,30}

36.3.5 Subclassification of GBM

With the widespread use of molecular technology over the past two decades, considerable achievements have been made to identify the heterogeneity of glioblastoma, even though it is still under a single disease entity. Based on molecular genetic data from The Cancer Genome Atlas (TCGA) Research Network,³⁵ Verhaak et al used gene expression analysis to further sub-classify GBM into 4 subtypes: I. Classical, II. Mesenchymal, III. Proneural, and IV. Neural.³⁶

36.4 Miscellaneous pathological features

- Glial fibrillary acidic protein (GFAP). Most astrocytomas stain positive for GFAP (astrocytomas that may not stain GFAP positive: some poorly differentiated gliomas, purely gemistocytic astrocytomas since fibrillary astrocytes are required to stain positive).
- Cysts. Gliomas may have cystic central necrosis, but may also have an associated cyst even without necrosis. When fluid from these cysts is aspirated it can be differentiated from CSF by the fact that it is usually xanthochromic and often clots once removed from the body (unlike e.g. fluid from a chronic subdural). Although they may occur with malignant gliomas, cysts are more commonly associated with pilocytic astrocytomas (p.630).
- MIB-1 index. It has been suggested that a MIB-1 index ≥7–9%is indicative of an anaplastic tumor, while MIB-1 <5%favors a low-grade tumor. However, variability between observers and institutions precludes using the MIB-1 index as a sole discriminant between grade II & III astrocytomas.³⁷

36.5 Neuroradiological grading and findings

Astrocytomas typically arise in white matter (e.g. centrum semiovale) and traverse through white matter tracts (see below). See also MR-spectroscopy findings (p.233).

- CT scan & MRI grading. Grading gliomas by CT or MRI is imprecise,³⁸ but may be used as a preliminary assessment (see □ Table 36.6). Neuroradiologic grading is not applicable to pediatric patients or special astrocytomas (e.g. pilocytic astrocytomas).

Table 36.6 Grading gliomas by CT or MRI		
WHO grade	Typical radiographic findings	
II	CT: low density MRI: abnormal signal on T2WI	no enhancement with little or no mass effect
III	complex enhancement ^a	
IV	necrosis (ring enhancement)	
^a however, some may not enhance		

- Low-grade gliomas. Usually hypodense on CT. Most are hypointense on T1WI MRI, and show high intensity changes on T2WI that extend beyond the tumor volume. Most do not enhance on CT or MRI (although up to 40%do,³⁹ and these may have a worse prognosis). The UCSF preoperative grading system for low-grade infiltrating gliomas⁴⁰ assigns 1 point for the presence of each of 4 parameters shown in

Table 36.7

. The points are summed and the prognosis is shown in

Table 36.8

 (this scale needs to be validated at other institutions). Another study found poor prognosis associated with: age≥40 years, tumor≥6 cm dia, tumor crossing midline and the presence of neurologic deficit.⁴¹
- Malignant gliomas. Anaplastic astrocytomas (AA) may not enhance⁴² (31% of highly anaplastic and 59% of moderately anaplastic astrocytomas do not enhance on CT⁴³; MRI not studied). Calcifications and cysts occur in 10–20% of AA.⁴² Most glioblastomas enhance, but some rare ones do not.^{38,43}
- Ring-enhancement with glioblastoma (GBM). The nonenhancing center may represent necrosis or associated cyst (see above). The enhancing ring is cellular tumor, however, tumor cells also extend ≥15 mm beyond the ring.⁴⁴
- Positron emission tomography (PET) scan. Low grade fibrillary astrocytomas appear as hypometabolic “cold” spots with fluorodeoxyglucose PET scans. Hypermetabolic “hot” spots suggest high-grade astrocytomas and help distinguish high grade glioma that do not enhance on MRI from lower grade (II) astrocytomas.
- Angiographic appearance. AAs usually appear as an avascular mass. Tumor blush and AV-shunting with early draining veins are more characteristic of GBM.

36.6 Spread

Gliomas may spread by the following mechanisms⁴⁵ (note: <10% of recurrent gliomas recur away from the original site⁴⁶):

1. tracking through white matter
- a) corpus callosum (CC)

• through genu or body of CC → bilateral frontal lobe involvement (“butterfly glioma”)

• through splenium of CC → bilateral parietal or occipital lobes

b) cerebral peduncles → midbrain involvement

c) internal capsule → encroachment of basal ganglion tumors into centrum semiovale

d) uncinate fasciculus → simultaneous frontal and temporal lobe tumors

e) interthalamic adhesion → bilateral thalamic gliomas

Table 36.7 Preoperative grading of low-grade gliomas ⁴⁰	
Item	Yes/No
age >50 years	Yes = 1, No = 0
KPS ^a ≤ 80	Yes = 1, No = 0
located in eloquent brain ^b	Yes = 1, No = 0
maximal diameter >4 cm	Yes = 1, No = 0
^a KPS=Karnofsky performance score (p.1358) ^b for this study, eloquent brain is defined as any of: primary sensory or motor cortex, Wernicke’s or Broca’s area, basal ganglia/internal capsule, thalamus or primary visual cortex	

Table 36.8 Sum of points from Table 36.7		
Sum	5-year survival	5-year progression-free survival
0–1	97%	76%
2	81%	49%
3–4	56%	18%

- 2. CSF pathways (subarachnoid seeding): 10–25%frequency of meningeal and ventricular seeding by high grade gliomas⁴⁷
- 3. rarely, gliomas may spread systemically

36.7 Multiple gliomas

Discussion of multiple gliomatous masses has to acknowledge the concept that astrocytoma is a multifocal disease, not a focal one. Some terms are probably artificial, e.g. since gliomatosis cerebri probably represents a diffuse infiltrating glial tumor with areas that may dedifferentiate into higher grade and then is called multicentric glioma.

Settings in which multiple gliomatous masses are encountered:

- 1. conventional glioma that has spread by one of the mechanisms previously described (see above)
- 2. gliomatosis cerebri: a diffuse, infiltrating astrocytoma that invades almost all of the cerebral hemispheres and brainstem. Usually low-grade,³⁹ areas of anaplasia and glioblastoma may also occur⁴⁸ and may present as focal mass.⁴⁹ Occurs most frequently in 1st 2 decades
- 3. multiple primary gliomas: some of the following terms are inconsistently used interchangeably: “multicentric”, “multifocal”, and “multiple.” Reported range of occurrence is 2–20%of gliomas⁵⁰, ⁵¹ (lower end of range ≈ 2–4%is probably more accurate, the higher end of the range is probably accounted for by infiltrative extension⁵² (p 3117))
 - a) commonly associated with neurofibromatosis and tuberous sclerosis
 - b) rarely associated with multiple sclerosis and progressive multifocal leukoencephalopathy
- 4. meningeal gliomatosis: dissemination of glioma throughout the CSF, similar to carcinomatous meningitis (p.811). Occurs in up to 20%of autopsies on patients with high-grade gliomas. May present with cranial neuropathies, radiculopathies, myelopathy, dementia, and/or communicating hydrocephalus

In a series of 25 patients with multicentric glioma,⁵³ glioblastoma was the most common pathology (48%), followed by anaplastic astrocytoma (20%), and glioblastoma with simultaneous AA (20%).

□ Treatment considerations for multiple gliomas. There is little data available. In a nonrandomized study of 25 patients with multifocal glioma,⁵³ the 16 patients who underwent debulking did better than the 9 who did not. However, there was significant selection bias in choosing patients suitable for craniotomy.

Biopsy is generally required/recommended to confirm the diagnosis.

Σ

Once the diagnosis of multiple gliomatous masses has been ascertained, local therapies (e.g. surgery, interstitial radiation...) are impractical. Whole brain radiation and possibly chemotherapy are indicated. An exception would be to consider debulking tumor to prevent herniation in a patient deteriorating from mass effect.

36.8 Treatment

36.8.1 Low-grade astrocytomas (WHO grade II)

Treatment options

- 1. no treatment: follow serial neurologic exams and imaging studies, with intervention reserved for cases showing progression
- 2. radiation
- 3. chemotherapy
- 4. surgery
- 5. combinations of radiation and chemotherapy, with or without surgery

Analysis

No well-designed study has shown that any approach for supratentorial WHO grade II infiltrating astrocytomas in adults is clearly superior to any other. Some treatments may simply expose the

patient to the risk of treatment side effects. The argument holds that these tumors are slow growing, and that until progression on imaging or malignant degeneration is documented, that it may be no worse to not treat the patient.⁵⁴ Although this view has been challenged,⁵⁵ a definitive study has yet to be performed. The following are associated with more aggressive tumors and should prompt consideration for some form of treatment:

1. extremely young patients, or patients > 50 yrs age; increasing age at diagnosis is associated with more rapid dedifferentiation (p.615)
2. large tumors that enhance (tumor size is one of the most important prognosticators⁵⁶)
3. symptomatic patients, especially those with short clinical history
4. evidence of progression on imaging studies
5. Astrocytic/Astrocytic-dominant mixed glioma histology
6. Biopsy only without resection

Surgery for low grade gliomas

4 objectives in performing surgery for low grade gliomas⁵⁷:

1. To obtain histological confirmation/molecular genetic analysis
2. To improve neurological condition
3. To reduce risk of tumor growth
4. To prevent malignant transformation

Surgery is the principal treatment for low-grade glioma in most circumstances. Although there is no randomized control trial (RCT) comparing tumor resection to biopsy alone in low-grade glioma, current opinion advocates early resection. A recent Norway study showed a significant lower overall survival in a center favoring watchful waiting approach. More aggressive excision is associated with better outcome^{58,59,60} and later malignant transformation to anaplastic types.⁵⁹ Even in recurrent low-grade glioma, surgical resection is associated with a survival benefit.

Intraoperative Mapping and Awake Craniotomy

Complete resection is often not possible due to the infiltrative nature of low-grade gliomas and its frequent location near or at eloquent areas. Resection can be safely maximized by means of intraoperative mapping and awake-surgeries.⁶¹ A meta-analysis of 8091 patients showed the use of intraoperative stimulation brain mapping achieved more gross total removal with less late severe neurological deficits, and is recommended as a standard for glioma surgery especially if eloquent areas are involved.⁶² Multicentric gliomas, previously considered not resectable can also be resected with aid of awake intraoperative map-ping.⁶³ Despite this advances, the role of surgery remains limited for gliomatosis cerebri or very deep-seated lesions.

Surgery is the principal treatment in the following situations of low-grade astrocytomas:

1. surgical biopsy or partial resection is recommended in almost all cases to establish the diagnosis since clinical and radiographic data are not definitive¹⁴
2. pilocytic astrocytomas
 - a) cerebellar tumors occurring in children & young adults (p.630)
 - b) supratentorial pilocytic astrocytomas
3. when herniation threatens from large tumors or tumor cysts
4. tumors causing obstruction of CSF flow
5. may help in seizure control with refractory seizures
6. in an attempt to delay adjuvant therapy and its side-effects in children (especially XRT in those < 5 yrs old)¹⁴
7. for prevention of malignant transformation

The role of surgery is limited in the following situations of low-grade astrocytomas:

1. disseminated (poorly circumscribed) tumors
2. multifocal tumors
3. location in eloquent brain

Adjuvant therapy for low-grade gliomas

□ Radiation therapy (XRT). Early radiotherapy (54 Gy in fractions of 1.8 Gy) is recommended as adjuvant therapy and is shown to prolong median progression free survival from 3.4 to 5.3 years but does not affect overall survival.⁶⁴ In patients with the tumor radically resected, early RT did not prolong PFS and is recommended to be deferred until progression. Following incomplete resection, early RT significantly prolongs PFS and disease specific survival.⁶⁵ Two prospective trials found no

difference in OS or PFS between different XRT doses (EORTC trial⁵⁶: 45 Gy in 5 weeks vs. 59.4 Gy in 6.6 weeks; Intergroup study⁶⁶ 50.4 vs. 64.8 Gy). Side effects from WBXRT include: leukoencephalopathy and cognitive impairment; see Radiation injury and necrosis (p.1560). The frequency of side effects may⁶⁶ or may not⁶⁷ be higher at higher XRT doses.

□ Chemotherapy . Usually reserved for tumor progression. Temozolomide (Temodar®) may be effective in progressive WHO grade II astrocytomas (off label use).⁶⁸ Effectiveness of PCV (procarbazine, CCNU, and vincristine) was assessed by RTOG 9802. It showed no significant difference in 5-yr OS rates (RT+PCV versus RT: 72%versus 63% But on post hoc analysis of survival for patients surviving to 2 years, RT+PCV group had higher 5-yr OS than RT alone.⁶⁹

36.8.2 Malignant astrocytomas (WHO grades III and IV)

Surgery for newly diagnosed high-grade gliomas

The aim of surgery in high-grade gliomas is cytoreduction, to relieve mass effect, and to obtain adequate tissue for histological and molecular study. Cytoreductive surgery followed by external beam radiation and concurrent temozolomide has become the standard against which other treatments are compared.⁷⁰

□ Extent of resection. The extent of tumor removal and (in an inverse relationship) the volume of residual tumor on post-op imaging studies⁷¹ have a significant effect on time to tumor progression and median survival.⁷² It was demonstrated that excision for 97 percent or more was associated with increased survival time.⁷³ Whenever feasible, gross total resection of tumor with preservation of eloquent and critical structures should be the goal. Recent advances in tumor localization, intraoperative monitoring and mapping have allowed more effective and safer resection.

□ 5-aminolevulinic-acid (5-ALA) guided resection. In addition to techniques of stereotactic localization using preoperative imaging as well as intraoperative brain mapping, techniques to enhance visual identification of tumor intraoperatively may be used and include 5-aminolevulinic-acid (5-ALA). 5-ALA is metabolized into fluorescent porphyrins, which accumulate in malignant glioma cells. This property permits use of ultraviolet illumination during surgery as an adjunct to map out the tumor. This has been proven with RCT where use of 5-ALA leads to more complete resection (65%vs 36%, $p < 0.0001$), which translates into a higher 6-month progression free survival (41% vs 21.1% $p = 0.0003$) but no effect on OS.⁷⁴

Partial resection of a GBM carries significant risk of post-operative hemorrhage and/or edema (wounded glioma syndrome) with risk of herniation. Furthermore, the benefit of subtotal resection is dubious. Retrospective evidence suggested survival benefits in gross total resection but not with incomplete resection.⁷⁵ Therefore, surgical excision should only be considered when the goal of gross total removal is feasible.

As a result of the above, the following are usually not candidates for surgical debulking

1. extensive dominant lobe GBM
2. lesions with significant bilateral involvement (e.g. large butterfly gliomas)
3. elderly patients
4. Karnofsky score < 70 (in general, with infiltrating tumors, the neurologic condition on steroids is as good as it is going to get, and surgery rarely improves this)
5. multicentric gliomas

□ Stereotactic biopsy. Due to sampling error, stereotactic biopsy may underestimate the occurrence of GBM by as much as 25%

Indications for stereotactic biopsy (instead of initial resection) in suspected malignant astrocytomas⁷⁶:

1. tumors located in eloquent or inaccessible areas of brain
2. patients in poor medical condition precluding general anesthesia
3. to ascertain a diagnosis when one is not definitely established (including when considering a more definitive operation). Some CNS lymphomas mimic GBM radiographically (and without immunostaining, some have also been mistaken pathologically) biopsy should be given serious consideration (to avoid operating on a lymphoma which may be best treated with XRT & intrathecal chemotherapy)

Technique: Yield of biopsy is highest when targets within the low density (necrotic) center and enhancing rim are chosen.⁴⁴

Patients with left-sided tumors and dysphasia are at significant risk of worsening of language function following stereotactic biopsy (the risk of deterioration is low if there is no dysphasia before biopsy).⁷⁷

Adjuvant therapy after cytoreductive surgery for newly diagnosed GBM (Stupp regimen)

Temozolomide is an oral alkylating agent that is given as a prodrug which undergoes rapid non-enzymatic conversion at physiologic pH to the active metabolite monomethyl triazenoimidazole carboxamide (MTIC). The cytotoxic effect of MTIC is associated with alkylation (methylation) of DNA at various sites including the O⁶ and N⁷ positions on guanine.

Cytoreductive surgery followed by the Stupp regimen has become the standard for newly diagnosed GBM.⁷⁰ Stupp regimen included concomitant chemoradiotherapy and adjuvant chemotherapy. Chemoradiotherapy started within six weeks after histological diagnosis of GBM. Radiotherapy in the Stupp regimen consists of fractionated focal irradiation at a dose of 2 Gy per fraction once daily five days per week over a period of six week, for a total dose of 60 Gy, with a 2-3 cm margin of clinical target volume. The is compared to the usual XRT regime for malignant gliomas of 50-60 Gy (usually 50 Gy to a margin 2-3 cm greater than the enhanced volume on MRI with a boost to the enhancing volume to bring the total to 60 Gy⁴²). Concomitant chemotherapy consists of temozolomide 75 mg/m²/day, 7 days per week till the end of radiotherapy. Four weeks later, six cycles of adjuvant chemotherapy starts. Each cycle consists of 5-days of temozolomide repeated every 28 days. The dose was 150 mg/m²/day for first cycle and stepped up to 200 mg/m²/day. The median survival was 14.6 months with the Stupp regimen compared to 12.1 months with radiotherapy alone with median survival benefit of 2.1 months. The five-year survival rate was 9.8% for the Stupp regimen in contrast to 1.9%⁷⁸ Regardless of extent of resection and MGMT status, patients received the Stupp regimen had longer median survival. Patients with MGMT promoter region methylation had median survival time of 23.4 months compared to 12.6 months in the non-methylated group. In the MGMT unmethylated group, the Stupp regimen only improved median survival from 11.8 months to 12.6 months. Some groups do extend the adjuvant chemotherapy after the standard six-month regimen until tumor progression is observed, in one study this extended the median survival time from 16.5 months to 24.6 months.⁷⁹

Side effects: Temozolomide may cause myelosuppression. It should not be given unless the neutrophil count $\geq 1.5 \times 10^9/L$ and platelet count $\geq 100 \times 10^9/L$. For all patients with newly diagnosed GBM for concomitant temozolomide and radiotherapy, prophylaxis against *Pneumocystis carinii* pneumonia is required for the 42-day regimen.

□ Other treatment protocols for GBM. Gliadel® wafer carmustine (BCNU) 7.7 mg in a 200 mg prolipeprosan 20 hydrophobic polymer carrier (wafer) that can be applied to resection cavity after tumor excision. The wafers are degraded by hydrolysis and the drug is released over 2-3 wks. This exposes the tumor to 113 times the concentration of BCNU compared to IV administration. Following tumor removal, up to 8 of the 1.4 cm $\times$ 1 mm (dime-sized) wafers are applied to the tumor resection bed at the time of surgery.

It increases median survival to 13.8 months compared with 11.6 months in placebo groups for newly diagnosed GBM.⁸⁰ It showed no survival benefit for recurrent disease.⁸¹ Side effects: seizures, cerebral edema, healing abnormalities, intracranial infection.

Some other regimens that have not proven to be successful for GBM:

- AVAglio: a phase III trial with addition of bevacizumab to the Stupp regimen for newly diagnosed GBM⁸² and RTOG 0825 (another trial of similar design) showed improved PFS, but no significant improvement on OS.⁸³
- CENTRIC: a phase III trial studied another antiangiogenic (cilengitide, an integrin inhibitor) to the standard temozolomide and radiation therapy. There was no improvement in either PFS or OS.^{17,84}
- RTOG 93-05: stereotactic radiosurgery followed by conventional chemoradiation did not improve the median survival of GBM patient compared to those without stereotactic radiotherapy.⁸⁵
- Brachytherapy has shown no significant benefit as an adjunct to EBRT in the initial treatment of malignant astrocytomas.⁸⁶
- Whole brain XRT has not been shown to increase survival compared to focal XRT, and the risk of side effects is greater.⁸⁷
- Combination chemotherapy of procarbazine, lomustine and vincristine (PCV) was used before the proven benefit of the Stupp regimen. But randomized trial did not show any benefit.⁸⁸
- RTOG 0525: a phase 3 trial of dose-dense temozolomide at 100 mg/m² day 1-21 of a 28-day cycle showed no significant benefit over standard dosing for median OS or median PFS.⁸⁹

Treatment protocols for anaplastic astrocytoma (WHO Grade III)

For newly diagnosed anaplastic astrocytoma, surgery followed by radiotherapy was the standard treatment with median OS 5.7 yrs. In a retrospective review, radiotherapy followed by chemoradiation, concurrent chemoradiation using temozolomide did not achieve better median OS nor PFS compared with radiotherapy alone.⁹⁰ On the other hand, temozolomide was approved by FDA in 1999 for anaplastic astrocytoma refractory to both a nitrosourea and procarbazine. The median duration of all response was 50 weeks, median PFS was 4.4 months and median OS was 15.9 months.^{91,92}

36.8.3 Pseudoprogression

Since temozolomide has become the new standard of care in the treatment of GBM, there has been an increasing awareness of progressive contrast-enhancing areas on MRI that mimics tumor progression, typically seen ≤ 3 months after treatment. This phenomenon of pseudoprogression occurs in up to 28-60% of patients after XRT+temozolomide treatment. Histologically it resembles radiation necrosis and is believed to be associated with tumor kill by radiation. Increased tumor kill with chemotherapy results in more pseudoprogression. 91% in methylated MGMT patients vs. 41% in the unmethylated group.⁹³

Management: The MRI findings usually improve without treatment,⁹⁴ and steroids may help control symptoms.

Diagnosis: There is no definitive diagnostic test. MR perfusion has been tried to distinguish pseudoprogression from true progression but is not reliable. DWI with higher apparent diffusion coefficient (ADC), MR spectroscopy, and PET have also not achieved high sensitivity and specificity. Monitoring with serial MRIs and clinical exams seems to be an effective strategy.

36.8.4 Treatment for recurrent GBM

Less than 10% of recurrent gliomas recur away from the original tumor site.⁴⁶

- 1. Surgery: reoperation extends survival by an additional 36 weeks in patients with GBM, and 88 weeks in AA^{95,96} (duration of high quality survival was 10 weeks and 83 weeks respectively, and was lower with pre-op Karnofsky score <70). In addition to Karnofsky score, significant prognosticators for response to repeat surgery include: age and time from the first operation to reoperation (shorter times $\rightarrow$ worse prognosis).⁹⁷ Morbidity is higher with reoperation (5-18%); the infection rate is ≈ 3 x that for first operation, wound dehiscence is more likely.
- 2. Chemotherapy
 - a) temozolomide:
 - in a 2013 Cochrane review,⁹⁸ chemotherapy with temozolomide was not associated with extension of PFS nor OS

Table 36.9 Summary of systemic treatment for relapsed GBM				
Endpoint	Bevacizumab ¹⁰⁰	Bevacizumab + irinotecan ¹⁰⁰	Temozolomide ⁹⁹	PCV ¹⁰³
6-months PFS, %	43	50	7-36	38
Median PFS, months	4.2	5.6	1.8-3.7	N/A
ORR, %	28	38	3-11	3.5
Median DoR, months	5.6	4.3	N/A	N/A
Median OS, months	9.3	8.9	N/A	N/A
12-month OS, %	38	38	14.8-28.6	N/A
Decrease dose of corticosteroid ^a	46.5%	30.2%	N/A	N/A
Improved or stabilized neurocognitive function	59%-97% ¹⁰⁴	N/A	N/A	N/A
^a $\geq 50\%$ corticosteroid dose reduction, relative to baseline				

Table 36.10 Median survival for astrocytomas	
Grade	Median survival
I	8–10 yrs
II	7–8 yrs
III	≈ 2–3 yrs
IV	<1 yr

Table 36.11 MGMT Promotor Methylation Survival Outcome		
MGMT	Unmethylated	Methylated
Median OS (months)	12.2	18.2
2-year survival	7.8%	34.1%

- the RESCUE study looked at continuous dose-intense temozolomide at 50 mg/m²/day showed one year survival ranged 14.8–28.6%depended on the timing of progression and the start of dose-intense treatment⁹⁹
- b) bevacizumab (Avastin®) – a monoclonal antibody against VEGF. FDA approved in May, 2009 for progressive GBM following prior treatment based on two trials: the BRAIN study, AVF3708g¹⁰⁰ and NCI 06-C-0064E.¹⁰¹ Given as 10 mg/kg every 2 weeks until disease progres-sion. The reported 6-month PFS rate was 36.0% The median response durations were 3.9 months and 4.2 months from the two trials. The median OS was 9.3 months.¹⁰² Side effects: gastrointestinal perforations, wound healing complications, hemorrhage, fistula formation, arterial thromboembolic events, hypertension

Summary: For recurrent GBM, surgery is still the mainstay of treatment; in general, surgery is rec-ommended only for patients with KPS≥70.

36.9 Outcome

Survival with various grades of astrocytoma

In general, with “optimal treatment” the survival of the various grades of astrocytoma are approximately given in □ Table 36.10 (more details may be found in other sections – see also □ Table 36.13 and below for recursive partitioning analysis (RPA) for GBM).

Low-grade astrocytomas (WHO grade II)

For low grade infiltrating gliomas, see prognosis based on pre-op grading (□ Table 36.7).

Malignant astrocytomas (WHO grades III & IV)

Prognostic indicators:

1. patient age: consistently found to be the most significant prognosticator, with younger patients faring better.
2. histological features
3. performance status, e.g. Karnofsky score (KPS) at presentation (p.1358)
4. Mental Status changes and symptoms<3 months portends a worse prognosis
5. MGMT Methylation status

MGMT Promotor Methylation Survival Outcome (□ Table 36.11)²⁴

Analysis of 206 patients for MGMT methylation: Median overall survival for MGMT methylation positive patients treated with temozolomide and RT was 21.7 month, RT only 15.3 months; no sur-vival difference between treatments in patients with unmethylated MGMT.

Stupp Regime vs. radiotherapy alone, survival outcome^{70,78}

EORTC– NCIC Trial, 573 patients, 5-year Follow-up (□ Table 36.12)

Recursive partitioning analysis (RPA) with glioblastoma (□ Table 36.13)¹⁰⁵: RPA classification developed to compare survival categories and determine homogenous patient subsets. Useful for refining stratification and phase III study design. Can determine which patient subsets will benefit from specific treatments (and which may be spared unnecessary treatment).

Table 36.12 Stupp Regime vs. radiotherapy alone, survival outcome		
Survival	RT	RT+ TMZ
Median	12.1 months	14.6 months
2-year	10.9%	27.2%
3-year	4.4%	16.0%
4-year	3.0%	12.1%
5-year	1.9%	9.8%

Table 36.13 Recursive partitioning analysis (RPA) with glioblastoma				
Median survival			2-year survival	
RPA Class	Months	95% CI	%	95% CI
III	17	15-21	32	21-42
IV	15	13-16	19	15-24
V	10	9-12	11	7-16

The European Nomogram GBM Calculator¹⁰⁶: <http://www.eortc.be/tools/gbmcalculator>. Data from randomized trials by EORTC and NCIC analyzed for the prediction of survival of GBM patients.

References

[1] Ostrom QT, Gittleman H, Farah P, Ondracek A, Chen Y, Wolinsky Y, Stroup NE, Kruchko C, Barnholtz-Sloan JS. CBTUS statistical report: Primary brain and central nervous system tumors diagnosed in the United States in 2006-2010. *Neuro Oncol.* 2013; 15 Suppl 2:ii1–i56

[2] Inskip PD, Hoover RN, Devesa SS. Brain cancer incidence trends in relation to cellular telephone use in the United States. *Neuro Oncol.* 2010; 12:1147–1151

[3] Linet MS, Inskip PD. Cellular (mobile) telephone use and cancer risk. *Rev Environ Health.* 2010; 25:51–55

[4] Little MP, Rajaraman P, Curtis RE, Devesa SS, Inskip PD, Check DP, Linet MS. Mobile phone use and glioma risk: comparison of epidemiological study results with incidence trends in the United States. *BMJ.* 2012; 344

[5] World Health Organization. Electromagnetic fields and public health: mobile phones. 2011

[6] Berger MS, Leibel SA, Brunner JM, Finlay JL, Levin VA. In: *Primary Cerebral Tumours. Cancer of the Nervous System.* 2nd ed. Oxford: Oxford University Press; 2002:84–99

[7] Ringertz N. Grading of gliomas. *Acta Pathol Microbiol Scand.* 1950; 27:51–64

[8] Russell DS, Rubenstein LJ. *Pathology of Tumours of the Nervous System.* 5th ed. Baltimore: Williams and Wilkins; 1989:83–161

[9] Kernohan JW, Mabon RF, Svien HJ, et al. A Simplified Classification of the Gliomas. *Proc Staff Meet Mayo Clin.* 1949; 24:71–75

[10] Kleihues P, Louis DN, Wiestler OD, Burger PC, Scheithauer BW, Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. In: *WHO grading of tumors of the central nervous system. WHO classification of tumors of the central nervous system.* 4th ed. Lyon: International Agency for Research on Cancer; 2007:10–11

[11] Daumas-Duport C, Scheithauer B, O'Fallon J, Kelly P, et al. Grading of Astrocytomas: A Simple and Reproducible Method. *Cancer.* 1988; 62:2152–2165

[12] Kim TS, Halliday AL, Hedley-Whyte T, Convery K. Correlates of Survival and the Daumas-Duport Grading System for Astrocytomas. *J Neurosurg.* 1991; 74:27–37

[13] Louis DN, Perry A, Burger P, Ellison DW, Reifenberger G, von Deimling A, Aldape K, Brat D, Collins VP, Eberhart C, Figarella-Branger D, Fuller GN, Giangaspero F, Giannini C, Hawkins C, Kleihues P, Korshunov A, Kros JM, Beatriz Lopes M, Ng HK, Ohgaki H, Paulus W, Pietsch T, Rosenblum M, Rushing E, Soylemezoglu F, Wiestler O, Wesseling P. International Society Of Neuropathology–Haarlem consensus guidelines for nervous system tumor classification and grading. *Brain Pathol.* 2014; 24:429–435

[14] Berger MS, Apuzzo MLJ. In: *Role of Surgery in Diagnosis and Management. Benign Cerebral Glioma.* Park Ridge, Illinois: American Association of Neurological Surgeons; 1995:293–307

[15] Laws ER, Taylor WF, Clifton MB, et al. Neurosurgical Management of Low-Grade Astrocytoma of the Cerebral Hemispheres. *J Neurosurg.* 1984; 61:665–673

[16] Shafqat S, Hedley-Whyte ET, Henson JW. Age-Dependent Rate of Anaplastic Transformation in Low-Grade Astrocytoma. *Neurology.* 1999; 52:867–869

[17] Nutt CL, Stemmer-Rachamimov AO, Cairncross JG, Louis DN, Ali-Osman F. In: *Molecular Pathology of Nervous System Tumors. Brain Tumors.* Humana Press; 2005:33–54

[18] James CD, Ali-Osman F. In: *Molecular Genetics of Tumors of the Central Nervous System. Brain Tumors.* Humana Press; 2005:19–32.

[19] Kopelson G, Linggood R. Infratentorial Glioblastoma: The Role of Neuraxis Irradiation. *Int J Radiation Oncology Biol Phys.* 1982; 8:999–1003

[20] Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. *WHO classification of tumors of the central nervous system.* Lyon 2007

- [21] Maher EA, Brennan C, Wen PY, Durso L, Ligon KL, Richardson A, Khatry D, Feng B, Sinha R, Louis DN, Quackenbush J, Black PM, Chin L, DePinho RA. Marked genomic differences characterize primary and secondary glioblastoma subtypes and identify two distinct molecular and clinical secondary glioblastoma entities. *Cancer Res.* 2006; 66:11502–11513
- [22] Liang Y, Diehn M, Watson N, Bollen AW, Aldape KD, Nicholas MK, Lamborn KR, Berger MS, Botstein D, Brown PO, Israel MA. Gene expression profiling reveals molecularly and clinically distinct subtypes of glioblastoma multiforme. *Proc Natl Acad Sci U S A.* 2005; 102:5814–5819
- [23] Furnari FB, Fenton T, Bachoo RM, Mukasa A, Stommel JM, Stegh A, Hahn WC, Ligon KL, Louis DN, Brennan C, Chin L, DePinho RA, Cavenee WK. Malignant astrocytic glioma: genetics, biology, and paths to treatment. *Genes Dev.* 2007; 21:2683–2710
- [24] Hegi ME, Diserens AC, Gorlia T, Hamou MF, de Tribolet N, Weller M, Kros JM, Hainfellner JA, Mason W, Mariani L, Bromberg JE, Hau P, Mirimanoff RO, Cairncross JG, Janzer RC, Stupp R. MGMT gene silencing and benefit from temozolomide in glioblastoma. *N Engl J Med.* 2005; 352:997–1003
- [25] Margison GP, Kleihues P. Chemical carcinogenesis in the nervous system. Preferential accumulation of O6-methylguanine in rat brain deoxyribonucleic acid during repetitive administration of N-methyl-N-nitrosourea. *Biochem J.* 1975; 148:521–525
- [26] Goth R, Rajewsky MF. Persistence of O6-ethylguanine in rat-brain DNA: correlation with nervous system-specific carcinogenesis by ethylnitrosourea. *Proc Natl Acad Sci U S A.* 1974; 71:639–643
- [27] Qian XC, Brent TP. Methylation hot spots in the 5' flanking region denote silencing of the O6-methylguanine-DNA methyltransferase gene. *Cancer Res.* 1997; 57:3672–3677
- [28] Esteller M, Hamilton SR, Burger PC, Baylin SB, Herman JG. Inactivation of the DNA repair gene O6-methylguanine-DNA methyltransferase by promoter hypermethylation is a common event in primary human neoplasia. *Cancer Res.* 1999; 59:793–797
- [29] Yan H, Parsons DW, Jin G, McLendon R, Rasheed BA, Yuan W, Kos I, Batinic-Haberle I, Jones S, Riggins GJ, Friedman H, Friedman A, Reardon D, Hernon J, Kinzler KW, Velculescu VE, Vogelstein B, Bigner DD. IDH1 and IDH2 mutations in gliomas. *N Engl J Med.* 2009; 360:765–773
- [30] Parsons DW, Jones S, Zhang X, Lin JC, Leary RJ, Angenendt P, Mankoo P, Carter H, Siu IM, Gallia GL, Olivi A, McLendon R, Rasheed BA, Keir S, Nikolskaya T, Nikolsky Y, Busam DA, Tekleab H, Diaz LA, Jr, Hartigan J, Smith DR, Strausberg RL, Marie SK, Shinjo SM, Yan H, Riggins GJ, Bigner DD, Karchin R, Papadopoulos N, Parmigiani G, Vogelstein B, Velculescu VE, Kinzler KW. An integrated genomic analysis of human glioblastoma multiforme. *Science.* 2008; 321:1807–1812
- [31] Yen KE, Bittinger MA, Su SM, Fantin VR. Cancer-associated IDH mutations: biomarker and therapeutic opportunities. *Oncogene.* 2010; 29:6409–6417
- [32] Ohgaki H, Kleihues P. The definition of primary and secondary glioblastoma. *Clin Cancer Res.* 2013; 19:764–772
- [33] Bello MJ, Alonso ME, Aminosos C, Anselmo NP, Arjona D, Gonzalez-Gomez P, Lopez-Marin I, de Campos JM, Gutierrez M, Isla A, Kusak ME, Lassalletta L, Sarasa JL, Vaquero J, Casartelli C, Rey JA. Hypermethylation of the DNA repair gene MGMT: association with TP53 G:C to A:T transitions in a series of 469 nervous system tumors. *Mutat Res.* 2004; 554:23–32
- [34] Nakamura M, Watanabe T, Yonekawa Y, Kleihues P, Ohgaki H. Promoter methylation of the DNA repair gene MGMT in astrocytomas is frequently associated with G:C→A:T mutations of the TP53 tumor suppressor gene. *Carcinogenesis.* 2001; 22:1715–1719
- [35] Cancer Genome Atlas Research Network. Comprehensive genomic characterization defines human glioblastoma genes and core pathways. *Nature.* 2008; 455:1061–1068
- [36] Verhaak RG, Hoadley KA, Purdom E, Wang V, Qi Y, Wilkerson MD, Miller CR, Ding L, Golub T, Mesirov JP, Alexe G, Lawrence M, O'Kelly M, Tamayo P, Weir BA, Gabriel S, Winckler W, Gupta S, Jakkula L, Feiler HS, Hodgson JG, James CD, Sarkaria JN, Brennan C, Kahn A, Spellman PT, Wilson RK, Speed TP, Gray JW, Meyerson M, Getz G, Perou CM, Hayes DN, Cancer Genome Atlas Research Network. Integrated genomic analysis identifies clinically relevant subtypes of glioblastoma characterized by abnormalities in PDGFRA, IDH1, EGFR, and NF1. *Cancer Cell.* 2010; 17:98–110
- [37] Kleihues P, Louis DN, Scheithauer BW, Rorke LB, Reifenberger G, Burger PC, Cavenee WK. The WHO classification of tumors of the nervous system. *J Neuropathol Exp Neurol.* 2002; 61:215–25; discussion 226–9
- [38] Kondziolka D, Lunsford LD, Martinez AJ. Unreliability of Contemporary Neurodiagnostic Imaging in Evaluating Suspected Adult Supratentorial (Low-Grade) Astrocytoma. *J Neurosurg.* 1993; 79:533–536
- [39] Zee CS, Conti P, Destian S, et al. Apuzzo MLJ. In: *Imaging Features of Benign Gliomas. Benign Cerebral Glioma.* Park Ridge, Illinois: American Association of Neurological Surgeons; 1995:247–274
- [40] Chang EF, Smith JS, Chang SM, Lamborn KR, Prados MD, Butowski N, Barbaro NM, Parsa AT, Berger MS, McDermott MM. Preoperative prognostic classification system for hemispheric low-grade gliomas in adults. *J Neurosurg.* 2008; 109:817–824
- [41] Pignatti F, van den Bent M, Curran D, Debruyne C, Sylvester R, Therasse P, Afra D, Cornu P, Bolla M, Vecht C, Karim AB. Prognostic factors for survival in adult patients with cerebral low-grade glioma. *J Clin Oncol.* 2002; 20:2076–2084
- [42] Narayan P, Olson JJ. Management of anaplastic astrocytoma. *Contemp Neurosurg.* 2001; 23:1–6
- [43] Chamberlain MC, Murovic J, Levin VA. Absence of Contrast Enhancement on CT Brain Scans of Patients with Supratentorial Malignant Gliomas. *Neurology.* 1988; 38:1371–1373
- [44] Greene GM, Hitchon PW, Schelper RL, et al. Diagnostic Yield in CT-Guided Stereotactic Biopsy of Gliomas. *J Neurosurg.* 1989; 71:494–497
- [45] Scherer HJ. The Forms of Growth in Gliomas and their Practical Significance. *Brain.* 1940; 63:1–35
- [46] Choucair AK, Levin VA, Gutin PH, et al. Development of Multiple Lesions During Radiation Therapy and Chemotherapy. *J Neurosurg.* 1986; 65:654–658
- [47] Erlich SS, Davis RL. Spinal Subarachnoid Metastasis from Primary Intracranial Glioblastoma Multiforme. *Cancer.* 1978; 42:2854–2864
- [48] Artigas J, Cervos-Navarro J, Iglesias JR, et al. Gliomatosis Cerebri: Clinical and Histological Findings. *Clin Neuropathol.* 1985; 4:135–148
- [49] Wilson NW, Symon L, Lantos PL. Gliomatosis Cerebri: Report of a Case Presenting as a Focal Cerebral Mass. *J Neurol.* 1987; 234:445–447
- [50] Barnard RO, Geddes JF. The Incidence of Multifocal Cerebral Gliomas: A Histological Study of Large Hemisphere Sections. *Cancer.* 1987; 60:1519–1531
- [51] van Tassel P, Lee Y-Y, Bruner JM. Synchronous and Metachronous Malignant Gliomas: CT Findings. *AJNR.* 1988; 9:725–732
- [52] Harsh GR, Wilson CB, Youmans JR. In: *Neuroepithelial Tumors of the Adult Brain. Neurological Surgery.* 3rd ed. Philadelphia: W. B. Saunders; 1990:3040–3136
- [53] Salvati M, Caroli E, Orlando ER, Frati A, et al. Multicentric glioma: our experience in 25 patients and critical review of the literature. *Neurosurg Rev.* 2003; 26:275–279
- [54] Cairncross JG, Laperriere NJ. Low-Grade Glioma: To Treat or Not to Treat? *Arch Neurol.* 1989; 46:1238–1239

- [55] Shaw EG. Low-Grade Gliomas: To Treat or Not to Treat? A Radiation Oncologist's Viewpoint. *Arch Neurol*. 1990; 47:1138–1139
- [56] Karim ABMF, Maat B, Hatlevoll R, et al. A randomized trial on dose-response in radiation therapy of low-grade cerebral glioma: European Organization for Research and Treatment of Cancer (EORTC) Study 22844. *Int J Radiation Oncology Biol Phys*. 1996; 36:549–556
- [57] van den Bent MJ, Snijders TJ, Bromberg JE. Current treatment of low grade gliomas. *Memo*. 2012; 5:223–227
- [58] McGirt MJ, Goldstein IM, Chaichana KL, Tobias ME, Kothbauer KF, Jallo GI. Extent of surgical resection of malignant astrocytomas of the spinal cord: outcome analysis of 35 patients. *Neurosurgery*. 2008; 63:55–60; discussion 60–1
- [59] Capelle L, Fontaine D, Mandonnet E, Taillandier L, Golmard JL, Bauchet L, Pallud J, Peruzzi P, Baron MH, Kujas M, Guyotat J, Guillemin R, Frenay M, Tailibert S, Colin P, Rigau V, Vandenbos F, Pinelli C, Duffau H. Spontaneous and therapeutic prognostic factors in adult hemispheric World Health Organization Grade II gliomas: a series of 1097 cases: clinical article. *J Neurosurg*. 2013; 118:1157–1168
- [60] Shaw EG, Berkey B, Coons SW, Bullard D, Brachman D, Buckner JC, Stelzer KJ, Barger GR, Brown PD, Gilbert MR, Mehta M. Recurrence following neurosurgeon-determined gross-total resection of adult supratentorial low-grade glioma: results of a prospective clinical trial. *J Neurosurg*. 2008; 109:835–841
- [61] De Benedictis A, Moritz-Gasser S, Duffau H. Awake mapping optimizes the extent of resection for low-grade gliomas in eloquent areas. *Neurosurgery*. 2010; 66:1074–84; discussion 1084
- [62] De Witt Hamer PC, Robles SG, Zwinderman AH, Duffau H, Berger MS. Impact of intraoperative stimulation brain mapping on glioma surgery outcome: a meta-analysis. *J Clin Oncol*. 2012; 30:2559–2565
- [63] Terakawa Y, Yordanova YN, Tate MC, Duffau H. Surgical management of multicentric diffuse low-grade gliomas: functional and oncological outcomes: clinical article. *J Neurosurg*. 2013; 118:1169–1175
- [64] van den Bent MJ, Afra D, de Witte O, Ben Hassel M, Schraub S, Hoang-Xuan K, Malmstrom PO, Collette L, Pierart M, Mirimanoff R, Karim AB. Long-term efficacy of early versus delayed radiotherapy for low-grade astrocytoma and oligodendroglioma in adults: the EORTC 22845 randomised trial. *Lancet*. 2005; 366:985–990
- [65] Hanzely Z, Polgar C, Fodor J, Brucher JM, Vitanovics D, Mangel LC, Afra D. Role of early radiotherapy in the treatment of supratentorial WHO Grade II astrocytomas: long-term results of 97 patients. *J Neurooncol*. 2003; 63:305–312
- [66] Shaw E, Arusell R, Scheithauer B, O'Fallon J, O'Neill B, Dinapoli R, Nelson D, Earle J, Jones C, Cascino T, Nichols D, Ivnik R, Hellman R, Curran W, Abrams R. Prospective randomized trial of low- versus high-dose radiation therapy in adults with supratentorial low-grade glioma: initial report of a North Central Cancer Treatment Group/Radiation Therapy Oncology Group/Eastern Cooperative Oncology Group study. *J Clin Oncol*. 2002; 20:2267–2276
- [67] Laack NN, Brown PD, Ivnik RJ, Furth AF, Ballman KV, Hammack JE, Arusell RM, Shaw EG, Buckner JC. Cognitive function after radiotherapy for supratentorial low-grade glioma: a North Central Cancer Treatment Group prospective study. *Int J Radiat Oncol Biol Phys*. 2005; 63:1175–1183
- [68] Quinn JA, Reardon DA, Friedman AH, Rich JN, Sampson JH, Provenzale JM, McLendon RE, Gururangan S, Bigner DD, Herndon JE, II, Avgeropoulos N, Finlay J, Tourt-Uhlig S, Affronti ML, Evans B, Stafford-Fox V, Zaknoen S, Friedman HS. Phase II trial of temozolomide in patients with progressive low-grade glioma. *J Clin Oncol*. 2003; 21:646–651
- [69] Shaw EG, Wang M, Coons SW, Brachman DG, Buckner JC, Stelzer KJ, Barger GR, Brown PD, Gilbert MR, Mehta MP. Randomized trial of radiation therapy plus procarbazine, lomustine, and vincristine chemotherapy for supratentorial adult low-grade glioma: initial results of RTOG 9802. *J Clin Oncol*. 2012; 30:3065–3070
- [70] Stupp R, Mason WP, van den Bent MJ, Weller M, Fisher B, Taphoorn MJ, Belanger K, Brandes AA, Marosi C, Bogdahn U, Curschmann J, Janzer RC, Ludwin SK, Gorlia T, Allgeier A, Lacombe D, Cairncross JG, Eisenhauer E, Mirimanoff RO. Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma. *N Engl J Med*. 2005; 352:987–996
- [71] Grabowski MM, Recinos PF, Nowacki AS, Schroeder JL, Angelov L, Barnett GH, Vogelbaum MA. Residual tumor volume versus extent of resection: predictors of survival after surgery for glioblastoma. *J Neurosurg*. 2014; 121:1115–1123
- [72] Keles GE, Anderson B, Berger MS. The Effect of Extent of Resection on Time to Tumor Progression and Survival in Patients with Glioblastoma Multiforme of the Cerebral Hemisphere. *Surg Neurol*. 1999; 52:371–379
- [73] Lacroix M, Abi-Said D, Fourney DR, Gokaslan ZL, Shi W, DeMonte F, Lang FF, McCutcheon IE, Hassenbusch SJ, Holland E, Hess K, Michael C, Miller D, Sawaya R. A multivariate analysis of 416 patients with glioblastoma multiforme: prognosis, extent of resection, and survival. *J Neurosurg*. 2001; 95:190–198
- [74] Stummer W, Pichlmeier U, Meinel T, Wiestler OD, Zanella F, Reulen HJ. Fluorescence-guided surgery with 5-aminolevulinic acid for resection of malignant glioma: a randomised controlled multicentre phase III trial. *Lancet Oncol*. 2006; 7:392–401
- [75] Kreth FW, Thon N, Simon M, Westphal M, Schackert G, Nikkhah G, Hentschel B, Reifenberger G, Pietsch T, Weller M, Tonn JC. Gross total but not incomplete resection of glioblastoma prolongs survival in the era of radiochemotherapy. *Ann Oncol*. 2013; 24:3117–3123
- [76] Coffey RJ, Lunsford LD, Taylor FH. Survival After Stereotactic Biopsy of Malignant Gliomas. *Neurosurgery*. 1988; 22:465–473
- [77] Thomson A-M, Taylor R, Fraser D, Whittle IR. Stereotactic Biopsy of Nonpolar Tumors in the Dominant Hemisphere: A Prospective Study of Effects on Language Functions. *J Neurosurg*. 1997; 89:923–926
- [78] Stupp R, Hegi ME, Mason WP, van den Bent MJ, Taphoorn MJ, Janzer RC, Ludwin SK, Allgeier A, Fisher B, Belanger K, Hau P, Brandes AA, Gijtenbeek J, Marosi C, Vecht CJ, Mokhtari K, Wesseling P, Villa S, Eisenhauer E, Gorlia T, Weller M, Lacombe D, Cairncross JG, Mirimanoff RO. Effects of radiotherapy with concomitant and adjuvant temozolomide versus radiotherapy alone on survival in glioblastoma in a randomised phase III study: 5-year analysis of the EORTC-NCIC trial. *Lancet Oncol*. 2009; 10:459–466
- [79] Roldan Urgoiti GB, Singh AD, Easaw JC. Extended adjuvant temozolomide for treatment of newly diagnosed glioblastoma multiforme. *J Neurooncol*. 2012; 108:173–177
- [80] Westphal M, Ram Z, Riddle V, Hilt D, Bortey E. Gliadel wafer in initial surgery for malignant glioma: long-term follow-up of a multicenter controlled trial. *Acta Neurochir (Wien)*. 2006; 148:269–75; discussion 275
- [81] Hart MG, Grant R, Garside R, Rogers G, Somerville M, Stein K. Chemotherapy wafers for high grade glioma. *Cochrane Database Syst Rev*. 2011. DOI: 10.1002/14651858.CD007294.pub2
- [82] Genentech Study Showed That Avastin Helped People with Newly Diagnosed Glioblastoma Live Longer without Their Disease Worsening When Added to Radiation and Chemotherapy. 2012
- [83] Gilbert MR, Dignam J, Won M, Blumenthal DT, et al. RTOG 0825: Phase III double-blind placebo-controlled trial evaluating bevacizumab in patients with newly diagnosed glioblastoma. *J Clin Oncol*. 2013; 31

- [84] Merck: Phase III Trial of Cilengitide Did Not Meet Primary Endpoint in Patients With Newly Diagnosed Glioblastoma. 2013
- [85] Souhami L, Seiferheld W, Brachman D, Podgorsak EB, Werner-Wasik M, Lustig R, Schultz CJ, Sause W, Okunieff P, Buckner J, Zamorano L, Mehta MP, Curran WJ, Jr. Randomized comparison of stereotactic radiosurgery followed by conventional radiotherapy with carmustine to conventional radiotherapy with carmustine for patients with glioblastoma multiforme: report of Radiation Therapy Oncology Group 93-05 protocol. *Int J Radiat Oncol Biol Phys.* 2004; 60:853–860
- [86] Sneed PK, McDermott MW, Gutin PH. Interstitial brachytherapy procedures for brain tumors. *Semin Surg Oncol.* 1997; 13:157–166
- [87] Shapiro WR, Green SB, Burger PC, et al. Randomized Trial of Three Chemotherapy Regimens and Two Radiotherapy Regimens in Postoperative Treatment of Malignant Glioma: Brain Tumor Cooperative Group Trial 8001. *J Neurosurg.* 1989; 71:1–9
- [88] Randomized trial of procarbazine, lomustine, and vincristine in the adjuvant treatment of high-grade astrocytoma: a Medical Research Council trial. *J Clin Oncol.* 2001; 19:509–518
- [89] Gilbert MR, Wang M, Aldape KD, Stupp R, Hegi ME, Jaeckle KA, Armstrong TS, Wefel JS, Won M, Blumenthal DT, Mahajan A, Schultz CJ, Erridge S, Baumert B, Hopkins KI, Tzuk-Shina T, Brown PD, Chakravarti A, Curran WJ, Jr, Mehta MP. Dose-dense temozolomide for newly diagnosed glioblastoma: a randomized phase III clinical trial. *J Clin Oncol.* 2013; 31:4085–4091
- [90] Shonka NA, Theeler B, Cahill D, Yung A, Smith L, Lei X, Gilbert MR. Outcomes for patients with anaplastic astrocytoma treated with chemoradiation, radiation therapy alone or radiation therapy followed by chemotherapy: a retrospective review within the era of temozolomide. *J Neurooncol.* 2013; 113:305–311
- [91] Yung WK, Prados MD, Yaya-Tur R, et al. Multicenter phase II trial of temozolomide in patients with anaplastic astrocytoma or anaplastic oligoastrocytoma at first relapse. *Temodal Brain Tumor Group. J Clin Oncol.* 1999; 17:2762–2771
- [92] Food and Drug Administration (FDA). Briefing book for the March 13, 2003 ODAC meeting regarding accelerated approval clinical phase 4 commitments NDA 21-029 Temodar® (temozolomide). 2003
- [93] Brandes AA, Tosoni A, Spagnolli F, Frezza G, Leonardi M, Calbucci F, Franceschi E. Disease progression or pseudoprogression after concomitant radiochemotherapy treatment: pitfalls in neuro-oncology. *Neuro Oncol.* 2008; 10:361–367
- [94] Brandsma D, Stalpers L, Taal W, Sminia P, van den Bent MJ. Clinical features, mechanisms, and management of pseudoprogression in malignant gliomas. *Lancet Oncol.* 2008; 9:453–461
- [95] Harsh GR, Levin VA, Gutin PH, Wilson CB, et al. Reoperation for Recurrent Glioblastoma and Anaplastic Astrocytoma. *Neurosurgery.* 1987; 21:615–621
- [96] Ammirati M, Galicich JH, Arbit E, et al. Reoperation in the Treatment of Recurrent Intracranial Malignant Gliomas. *Neurosurgery.* 1987; 21:607–614
- [97] Brem H, Piantadosi S, Burger PC, et al. Placebo-Controlled Trial of Safety and Efficacy of Intraoperative Controlled Delivery by Biodegradable Polymers of Chemotherapy for Recurrent Gliomas. *Lancet.* 1995; 345:1008–1012
- [98] Hart MG, Garside R, Rogers G, Stein K, Grant R. Temozolomide for high grade glioma. *Cochrane Database Syst Rev.* 2013; 4. DOI: 10.1002/14651858.CD007415.pub2
- [99] Perry JR, Belanger K, Mason WP, Fulton D, Kavan P, Easaw J, Shields C, Kirby S, Macdonald DR, Eisenstat DD, Thiessen B, Forsyth P, Pouliot JF. Phase II trial of continuous dose-intense temozolomide in recurrent malignant glioma: RESCUE study. *J Clin Oncol.* 2010; 28:2051–2057
- [100] Friedman HS, Prados MD, Wen PY, Mikkelsen T, Schiff D, Abrey LE, Yung WK, Paleologos N, Nicholas MK, Jensen R, Vredenburgh J, Huang J, Zheng M, Cloughesy T. Bevacizumab alone and in combination with irinotecan in recurrent glioblastoma. *J Clin Oncol.* 2009; 27:4733–4740
- [101] Kreisl TN, Kim L, Moore K, Duic P, Royce C, Stroud I, Garren N, Mackey M, Butman JA, Camphausen K, Park J, Albert PS, Fine HA. Phase II trial of single-agent bevacizumab followed by bevacizumab plus irinotecan at tumor progression in recurrent glioblastoma. *J Clin Oncol.* 2009; 27:740–745
- [102] Cohen MH, Shen YL, Keegan P, Pazdur R. FDA drug approval summary: bevacizumab (Avastin) as treatment of recurrent glioblastoma multiforme. *Oncologist.* 2009; 14:1131–1138
- [103] Schmidt F, Fischer J, Herrlinger U, Dietz K, Dichgans J, Weller M. PCV chemotherapy for recurrent glioblastoma. *Neurology.* 2006; 66:587–589
- [104] Henriksson R, Asklund T, Poulsen HS. Impact of therapy on quality of life, neurocognitive function and their correlates in glioblastoma multiforme: a review. *J Neurooncol.* 2011; 104:639–646
- [105] Mirimanoff RO, Gorlia T, Mason W, Van den Bent MJ, Kortmann RD, Fisher B, Reni M, Brandes AA, Curschmann J, Villa S, Cairncross G, Allgeier A, Lacombe D, Stupp R. Radiotherapy and temozolomide for newly diagnosed glioblastoma: recursive partitioning analysis of the EORTC 26981/22981-NCIC CE3 phase III randomized trial. *J Clin Oncol.* 2006; 24:2563–2569
- [106] Gorlia T, van den Bent MJ, Hegi ME, Mirimanoff RO, Weller M, Cairncross JG, Eisenhauer E, Belanger K, Brandes AA, Allgeier A, Lacombe D, Stupp R. Nomograms for predicting survival of patients with newly diagnosed glioblastoma: prognostic factor analysis of EORTC and NCIC trial 26981-22981/CE3. *Lancet Oncol.* 2008; 9:29–38

37 Other Astrocytic Tumors

37.1 Pilocytic astrocytomas

37.1.1 General information

Key concepts

- a subgroup of astrocytomas with better prognosis (10-year survival: 94%) than infiltrating fibrillary or diffuse astrocytomas
- age ≤ 20 yrs in 75% which is lower than for typical astrocytomas
- common locations: cerebellar hemisphere, optic nerve/chiasm, hypothalamus
- radiographic appearance: discrete appearing, contrast enhancing lesion, often cystic with mural nodule
- pathology: compacted and loose textured astrocytes with Rosenthal fibers and/or eosinophilic granular bodies
- danger of overgrading and overtreating if not recognized. Histology alone may be inadequate for diagnosis; knowledge of radiographic appearance is critical

Background and terminology

Pilocytic astrocytoma (PCA) is the currently recommended nomenclature for these tumors that have been previously referred to variously as cystic cerebellar astrocytomas, juvenile pilocytic astrocytomas, optic gliomas, and hypothalamic gliomas.¹ (p 77–96) However, since treatment decisions vary based on location and neural involvement, it is still helpful to discuss differences in management of these subtypes.

PCAs differ markedly from infiltrating fibrillary astrocytomas in terms of their ability to invade tissue and for malignant degeneration.

37.1.2 Location

PCAs arise throughout the neuraxis and are more common in children and young adults:

1. optic gliomas & hypothalamic gliomas:
 - a) PCAs arising in the optic nerve are called optic gliomas (p.631)
 - b) when they occur in the region of the chiasm they cannot always be distinguished clinically or radiographically from so-called hypothalamic gliomas (p.632) or gliomas of the third ventricular region
2. cerebral hemispheres: tends to occur in older patients (i.e. young adults) than optic nerve/hypothalamic lesions. These PCAs are potentially confused with fibrillary astrocytomas possessing more malignant potential. PCAs are often distinguished by a cystic component with an enhancing mural nodule (would be atypical for a fibrillary astrocytoma), & some PCAs have dense calcifications¹
3. brainstem gliomas (p.633): usually are fibrillary infiltrating type and only a small proportion are pilocytic. Those that are PCAs may comprise the majority of the prognostically favorable group described as “dorsally exophytic”²
4. cerebellum: formerly referred to as cystic cerebellar astrocytoma (p.630)
5. spinal cord: PCAs may also occur here, but little information is available on these. Again, patients tend to be younger than with spinal cord fibrillary astrocytomas

37.1.3 Pathology

PCAs are composed of loosely knit tissue comprised of stellate astrocytes in microcystic regions containing eosinophilic granular bodies intermixed with regions of compact tissue consisting of elongated and fibrillated cells often associated with Rosenthal fiber formation¹ (Rosenthal fibers: sausage or corkscrew shaped cytoplasmic eosinophilic inclusion bodies consisting of glial filament aggregates resembling hyaline; stain bright red on Masson trichrome smears).

These latter two distinctive features facilitate the diagnosis. Another characteristic finding is that the tumors easily break through the pia to fill the overlying subarachnoid space. PCAs may also

infiltrate into the perivascular spaces. Vascular proliferation is common. Multinucleated giant cells with peripherally located nuclei are common, especially in PCAs of the cerebellum or cerebrum. Mitotic figures may be seen, but are not as ominous as with fibrillary astrocytomas. Areas of necrosis may also be seen. In spite of well-demarcated margins grossly and on MRI, at least 64%of PCAs infiltrate the surrounding parenchyma, especially the white matter³ (the clinical significance of this is uncertain, one study found no statistically significant decrease in survival⁴).

Differentiating from a diffuse or infiltrating fibrillary astrocytoma: Unless some of the distinctive findings described above are seen, pathology alone may not be able to differentiate. This may be especially problematic with small specimens obtained e.g. with stereotactic biopsy. Factors that suggest the diagnosis include young age, and knowledge of the radiographic appearance is often critical (see below).

Malignant degeneration: Malignant degeneration has been reported, often after many years. This may occur without radiation therapy (XRT),⁵ although in most cases XRT had been administered.⁶

37.1.4 Radiographic appearance

On CT or MRI, PCAs are usually well circumscribed, 94%enhance with contrast³ (unlike most low-grade fibrillary astrocytomas), frequently have a cystic component with a mural nodule, and have little or no surrounding edema. Although they may occur anywhere in the CNS, 82%are periventricular.³ Calcifications are only occasionally present.³ 4 main imaging patterns of cerebellar or cerebral PCAs are shown in □ Table 37.1.

37.1.5 Epidemiology

Usually presents during second decade of life (ages 10–20). 75%occur in age <20 years.⁷ No evidence of gender predilection.

37.1.6 Pilocytic astrocytoma of the cerebellum

General information

Key concepts

- often cystic, half of these have mural nodule
- usually presents during the second decade of life (ages 10–20 yrs)
- also, see Key concepts for pilocytic astrocytomas in general (p.629)

Formerly referred to by the nonspecific and confusing term cystic cerebellar astrocytoma. One of the more common pediatric brain tumors ($\approx 10\%$), comprising 27–40%of pediatric p-fossa tumors.⁹(p 367–74),¹⁰(p 3032) They may also occur in adults, where the mean age is lower and the post-operative survival is longer than for fibrillary astrocytomas.¹¹

Presentation

Signs and symptoms of pilocytic astrocytoma (PCA) of the cerebellum are usually those of any p-fossa mass, i.e. those of hydrocephalus or cerebellar dysfunction; see Posterior fossa (infratentorial) tumors (p.592).

Table 37.1 Common imaging characteristics of cerebellar or cerebral PCAs		
%	Description	
21%	nonenhancing cyst with enhancing mural nodule	over 66%are cystic with enhancing mural nodule
46%	enhancing cyst with enhancing mural nodule	
16%	mass with nonenhancing central area (necrosis)	
17%	solid mass with minimal or no cyst	

Pathology

The classic “juvenile pilocytic astrocytoma” of the cerebellum is a distinctive entity with its macroscopic cystic architecture and microscopic spongy appearance.¹ For other microscopic findings, see above.

These tumors may be solid, but are more often cystic (hence the older term “cystic cerebellar astrocytoma”), and tend to be large at the time of diagnosis (cystic tumors: 4–5.6 cm dia; solid tumors: 2–4.8 cm dia). Cysts contain highly proteinaceous fluid (averaging ≈ 4 Hounsfield units higher density than CSF on CT⁸).

50% of cystic tumors have a mural nodule and a cyst lining of reactive, non-neoplastic cerebellar tissue or ependymal lining (non-enhancing on CT), whereas the remaining 50% lack a nodule and have a cyst wall of poorly cellular tumor¹² (enhances on CT).

Histological classification of Winston

The Winston classification system¹³ is shown in □ Table 37.2. 72% of cerebellar PCAs tended to cluster with either Type A or B characteristics, 18% in his series had both, and 10% had neither.

Treatment guidelines

The natural history of these tumors is slow growth. Treatment of choice is surgical excision of the maximal amount of the tumor that can be removed without producing deficit. In some, invasion of brainstem or involvement of cranial nerves or blood vessels may limit resection. In tumors composed of a nodule with a true cyst, excision of the nodule is sufficient; the cyst wall is non-neoplastic and need not be removed. In tumors with a so-called “false cyst” where the cyst wall is thick and enhances (on CT or MRI), this portion must be removed also. Because of the high 5 and 10 year survival rates together with the high complication rate of radiation therapy over this time interval – see Radiation injury and necrosis (p.1560) –, and the fact that many incompletely resected tumors enlarge minimally if at all over periods of 5, 10 or even 20 years, it is recommended to not radiate these patients post-op. Rather, they should be followed with serial CT or MRI and be re-operated if there is recurrence.¹⁴ Radiation therapy is indicated for nonresectable recurrence (i.e. reoperation is preferred if possible) or for recurrence with malignant histology. Chemotherapy is preferable to XRT in younger patients.¹⁵

Also, see Posterior fossa (infratentorial) tumors (p.592) for guidelines regarding hydrocephalus, etc.

Prognosis

Children with Winston Type A cerebellar PCAs had 94% 10-yr survival, whereas those with Type B had only 29% 10-yr survival.

Tumor recurrence is relatively common, and although it has been said that they generally occur within ≈ 3 yrs of surgery,¹⁶ this is controversial and very late recurrences (violating Collins’ law, which says that a tumor may be considered cured if it does not recur within a time period equal to the patient’s age at diagnosis + 9 months) are well known.¹⁴ Also, some tumors excised partially fail to show further growth, representing a form of cure.

About 20% of cases develop hydrocephalus requiring treatment following surgery.¹⁷ So-called “drop metastases” are rare with PCAs.

37.1.7 Optic glioma

General information

Accounts for ≈ 2% of gliomas in adults, and 7% in children. The incidence is higher (≈ 25%) in neurofibromatosis (NFT) (p.603).

Table 37.2 Classification of cerebellar astrocytoma
• Type A: microcysts, leptomeningeal deposits, Rosenthal fibers, foci of oligodendroglioma
• Type B: perivascular pseudorosettes, high cell density, mitosis, calcification
• common features of types A & B: hypervascularity, endothelial proliferation, parenchymal desmoplasia, pleomorphism

May arise in any of the following patterns:

1. one optic nerve (without chiasmal involvement)
2. optic chiasm: less commonly involved in patients with NFT than in sporadic cases
3. multicentric in both optic nerves sparing the chiasm: almost only seen in NFT
4. may occur in conjunction with or be part of a hypothalamic glioma (see below)

Pathology

Most are composed of low-grade (pilocytic) astrocytes. Rarely, a malignant chiasmal glioma occurs.

Presentation

Painless proptosis is an early sign in lesions involving one optic nerve. Chiasmal lesions produce variable and nonspecific visual defects (usually monocular) without proptosis. Large chiasmal tumors may cause hypothalamic and pituitary dysfunction, and may produce hydrocephalus by obstruction at the foramen of Monro. Gliosis of the optic nerve head may be seen on fundoscopy.

Evaluation

Plain x-rays: not usually helpful, although in some cases dilatation of the optic canal can be seen in optic canal views.

CT/MRI: CT scan is excellent for imaging structures within the orbit. MRI is helpful for demonstrating chiasmal or hypothalamic involvement. On CT or MRI, involvement of the optic nerve produces contrast enhancing fusiform enlargement of the nerve usually extending >1 cm in length.

Treatment

Tumor involving a single optic nerve, sparing the chiasm, producing proptosis and visual loss should be treated with a transcranial approach with excision of the nerve from the globe all the way back to the chiasm (a transorbital (Kronlein) approach is not appropriate since tumor may be left in the nerve stump). In addition to the anticipated blindness in the involved eye, this may produce a junctional scotoma (p.730).

Chiasmal tumors are generally not treated surgically except for biopsy (especially when it is difficult to distinguish an optic nerve glioma from a hypothalamic glioma), CSF shunting, or to remove the rare exophytic component to try and improve vision.

Further treatment: Chemotherapy¹⁵ (especially in younger patients) or XRT is used for chiasmal tumors, for multicentric tumors, post-op if tumor is found in the chiasmal stump end of the resected nerve, and for the rare malignant tumor. Typical XRT treatment planning is for 45 Gy given in 25 fractions of 1.8 Gy.

37.1.8 Hypothalamic glioma

Pilocytic astrocytomas of the hypothalamus and third ventricular region occur primarily in children. Radiographically, the lesion may have an intraventricular appearance. Many of these tumors have some chiasmal involvement and the distinction from optic nerve glioma cannot be made (see above).

May present with so-called “diencephalic syndrome”, a rare syndrome seen in peds, usually caused by an infiltrating glioma of the anterior hypothalamus. Classically: cachexia (loss of subcutaneous fat) associated with hyperactivity, over-alertness and an almost euphoric affect. May also see: hypoglycemia, failure to thrive, macrocephaly.

When complete resection is not possible, further treatment may be needed as outlined under optic gliomas (above).

37.1.9 Pilomyxoid astrocytoma (PMA)

WHO grade II. Related to pilocytic astrocytomas (PCA) but more aggressive with greater tendency to recur and spread in CSF.¹⁸ May be an infantile form of PCA with a case report of “maturation” to a typical PCA.¹⁹ Typical onset in infancy (10 months).

Histologically: dominant mucoid matrix, monomorphic bipolar cells, and angiocentric cell arrangement. By definition, does not contain Rosenthal fibers.

May also occur in spinal cord, with a case report of extraneural peritoneal mets spread through a VP shunt.²⁰

37.1.10 Brainstem glioma

General information

Key concepts

- not a homogeneous group. MRI can differentiate malignant from benign lesions
- trend: lower grade tumors tend to occur in the upper brainstem, and higher grade tumors in the lower brainstem/medulla
- usually presents with multiple cranial nerve palsies and long tract findings
- most are malignant, have poor prognosis, and are not surgical candidates
- role of surgery primarily limited to dorsally exophytic lesions and shunting

Brainstem gliomas (BSG) tend to occur during childhood and adolescence (77%are <20 yrs old, they comprise 1%of adult tumors²¹). BSG are one of the 3 most common brain tumors in pediatrics – see Pediatric brain tumors (p.593) – comprising ≈ 10–20%of pediatric CNS tumors.²

Presentation

See reference.²²

Upper brainstem tumors tend to present with cerebellar findings and hydrocephalus, whereas lower brainstem tumors tend to present with multiple lower cranial nerve deficits and long tract findings. Due to their invasive nature, signs and symptoms usually do not occur until the tumor is fairly extensive in size.

Signs and symptoms:

1. gait disturbance
2. headache (p.590)
3. nausea/vomiting
4. cranial nerve deficits: diplopia, facial asymmetry
5. distal motor weakness in 30%
6. papilledema in 50%
7. hydrocephalus in 60% usually due to aqueductal obstruction (often late, except with periaque-ductal tumors, e.g. below)
8. failure to thrive (especially in age ≤2 yrs)

Pathology

BSG is a heterogeneous group. There may be a tendency towards lower grade tumors in the upper brainstem (76%were low-grade) versus the lower brainstem (100%of the glioblastomas were in the medulla).²³ A cystic component is seen rarely. Calcifications are also rare. 4 growth patterns that can be identified by MRI²⁴ that may correlate with prognosis²⁵:

1. diffuse: all are malignant (most are anaplastic astrocytomas, the rest are glioblastomas). On MRI these tumors extend into the adjacent region in vertical axis (e.g. medullary tumors extend into pons and/or cervical cord) with very little growth towards obex, remaining intraaxial
2. cervicomedullary: most (72%) are low-grade astrocytomas. The rostral extent of these tumors is limited to the spinomedullary junction. Most bulge into the obex of the 4th ventricle (some may have an actual exophytic component)
3. focal: extent limited to medulla (does not extend up into pons nor down into spinal cord). Most (66%) are low-grade astrocytomas
4. dorsally exophytic: may be an extension of “focal” tumors (see above). Many of these may actually be low grade gliomas including:
 - a) pilocytic astrocytomas (p.629)
 - b) gangliogliomas (p.651): very rare, only 13 cases reported as of 1984. Compared to other BSGs, these patients tend to be slightly older and the medulla is involved more frequently²⁶

Evaluation

MRI

The diagnostic test of choice. MRI evaluates status of ventricles, gives optimal assessment of tumor (CT is poor in the posterior fossa) and detects exophytic component. T1WI: almost all are

hypointense, homogeneous (excluding cysts). T2WI: increased signal, homogeneous (excluding cysts). Gadolinium enhancement is highly variable.²⁴

CT

Most do not enhance on CT, except possibly an exophytic component. If there is marked enhancement, consider other diagnoses (e.g. high grade vermian astrocytoma).

Treatment

Surgery

Biopsy: should not be performed when the MRI shows a diffuse infiltrating brainstem lesion²⁷ (does not change treatment or outcome).

Treatment is usually non-surgical. Exceptions where surgery may be indicated:

1. tumors with a dorsally exophytic component²: see below these may protrude into 4th ventricle or CP angle, tend to enhance with IV contrast, tend to be lower grade
2. some success has been achieved with non-exophytic tumors that are not malignant astrocytomas (surgery in malignant astrocytomas is without benefit)²⁵ (detailed follow-up is lacking)
3. shunting for hydrocephalus

Dorsally exophytic tumors

These tumors are generally histologically benign (e.g. gangliogliomas) and are amenable to radical subtotal resection. Prolonged survival is possible, with a low incidence of disease progression at short-term follow-up.²

Surgical goals in exophytic tumors include:

1. enhanced survival by subtotal removal of exophytic component²⁸: broad attachment to the floor of 4th ventricle is typical and usually precludes complete excision (although some “safe entry” zones have been described²⁹). An ultrasonic aspirator facilitates debulking
2. establishing diagnosis: radiographic differentiation of exophytic brainstem gliomas tumors from other lesions (e.g. medulloblastoma, ependymoma and dermoids) may be difficult
3. tumors that demonstrate recurrent growth after resection remained histologically benign and were amenable to re-resection²

Complications of surgery generally consisted of exacerbation of pre-operative symptoms (ataxia, cranial nerve palsies...) which usually resolved with time.

Medical

No proven chemotherapeutic regimen. Steroids are usually administered. In pediatrics, there is some indication of response to Temodar® (temozolomide) (p.595).

Radiation

Traditionally given as 45–55 Gy over a six week period, five days per week. When combined with steroids, symptomatic improvement occurs in 80% of patients.

Possible improved survival with so called “hyperfractionation” where multiple smaller doses per day are used.

Prognosis

Most children with malignant BSG will die within 6–12 months of diagnosis. XRT may not prolong survival in patients with grade III or IV tumors. A subgroup of children have a more slowly growing tumor and may have up to 50% five-year survival. Dorsally exophytic tumors comprised of pilocytic astrocytomas may have a better prognosis.

37.1.11 Tectal gliomas

General information

A topically defined diagnosis generally consisting of low-grade astrocytomas. Considered a benign subgroup of brainstem glioma. Because of location, tends to present with hydrocephalus. Was dramatically referred to as “the smallest tumor in the body that can lead to the death of the patient”.³⁰ Focal neurologic findings are rare – diplopia, visual field deficits, nystagmus, Parinaud’s syndrome (p.99), ataxia, seizures... – and are often reversible after the hydrocephalus is corrected.

Epidemiology

Comprises ≈ 6% of surgically treated pediatric brain tumors.³¹ Presents primarily in childhood. Median age of patients becoming symptomatic = 6–14 years.³¹

Pathology

Since many of these are not biopsied, meaningful statistical analysis is not possible. Pathologies identified include: WHO II diffuse astrocytoma, pilocytic astrocytomas, WHO II ependymoma, anaplastic astrocytoma, oligodendroglioma & oligoastrocytoma.

Radiographic evaluation

CT scan detects hydrocephalus, but may miss the tumor in ≈ 50%³² Calcification on CT has been described in 9–25%^{32,33}

MRI is the study of choice for diagnosis and follow-up. Typically appears as a mass projecting dorsally from the quadrigeminal plate. Isointense on T1WI, iso- or hyperintense on T2WI.^{31,34} Enhancement with gadolinium occurs in 18% and is of uncertain prognostic significance.

Treatment

General information

Due to the indolent course, open surgery is not recommended. Options include:

- 1. VP shunt: the standard treatment for years. Long-term results are good with a functioning shunt
- 2. endoscopic third ventriculostomy: may avoid the need for a shunt. Endoscopic biopsy³⁵ may be done at the same time through the same burr hole if it is technically feasible (requires a dilated foramen of Monro, which is often present). Long-term results unknown
- 3. endoscopic aqueductoplasty (with or without stenting): an option for some. Long-term results unknown

Stereotactic radiosurgery: May be offered for tumor progression (criteria are not defined: radiographic progression may not be associated with clinical deterioration³⁴). Dosing should be limited to ≤14 Gray at the 50–70% isodose line to avoid radiation-induced side effects.³⁶

Prognosis

Tumor progression: described in 15–25%

Follow-up: no accepted guidelines. Serial neurologic exams and MRIs every 6–12 months has been suggested.³¹

37.2 Pleomorphic xanthoastrocytoma (PXA)

37.2.1 General information

Key concepts

- low-grade glioma, possibly from subpial astrocytes → superficial location, >90% supratentorial, most common in children or young adults
- mural nodule with cystic component in 25%, meninges involved in >67%
- pathology: pleomorphic cells (xanthomatous (lipid laden) cells, fibrillary and giant multinucleated astrocytes). Usually circumscribed, occasionally invasive
- WHO grade II unless high mitotic index or necrosis, which is WHO grade III
- treatment: maximal safe resection. XRT or chemo ≈ only for grade III

A low-grade glioma thought to arise from subpial astrocytes which may explain their superficial location and abundance of reticulin fibers. Over 90% are supratentorial. Predilection for temporal lobes (50%), followed by parietal, occipital & frontal lobes. Most have a cystic component (may be multiloculated, but >90% have a large, single cyst).

37.2.2 Epidemiology

≈ 1% of astrocytomas. Usually occurs in children or young adults (most are <18 years age). No gender difference.

37.2.3 Clinical

Usual presentation: seizures. May also produce focal deficit or increased ICP.

37.2.4 Differential diagnosis

1. imaging: meningioma is also superficial with dural tail, may also resemble low grade fibrillary astrocytoma
2. pathology: may be confused with anaplastic astrocytoma

37.2.5 Pathology

WHO grade II (MIB is usually <1%) unless there is a high mitotic index or necrosis which qualifies as WHO grade III “PXA with anaplastic features”.³⁷ Compact, superficial tumor with marked cellular pleomorphism (fibrillary and giant multinucleated astrocytes, large xanthomatous (lipid laden) GFAP staining cells (bespeaking glial origin)), abundant reticulin and frequent perivascular chronic inflammatory cells. The reticulin fibers surround two cell types:

1. spindle cells: fusiform cell shape with elongate nuclei
2. pleomorphic cells: round cells with heterochromic, pleomorphic nuclei that may be mononucleated or multinucleated. Variable intracellular lipid content

Usually circumscribed, occasionally infiltrates cortex. Marked cellular pleomorphism may cause these tumors to be mistaken for anaplastic astrocytoma. Vascular proliferation and necrosis are absent,³⁸ most but not all lack mitotic figures. Some PXAs undergo anaplastic change.³⁹ There have also been several reported cases of malignant transformation to anaplastic astrocytoma or glioblastoma.⁴⁰

37.2.6 Imaging

The cyst, when present, may partially enhance on CT or MRI. A mural nodule is present in 25%. May have “dural tail” (67% show leptomeningeal involvement, 13% show involvement of all 3 meningeal layers). Peritumoral edema may be mild to moderate, calcifications are rare.⁴¹

CT: solid portion of tumor is ill-defined and may be isodense to grey matter.

MRI: T1WI: hypointense cystic component with ill-defined isointense solid component that strongly enhances with gadolinium. T2WI: hyperintense cystic component with ill-defined isointense solid component.

37.2.7 Treatment

1. surgery: primary treatment
 - a) gross total resection if it can be accomplished without unacceptable neurologic deficit, otherwise subtotal resection
 - b) extent of resection: most strongly associated with recurrence free survival⁴²
 - c) incomplete resections should be followed since these tumors may grow very slowly over many years before retreatment is necessary, and repeat excision should be considered
2. radiation therapy: controversial
 - a) literature suggests either no difference in overall survival or possibly a trend toward prolonged survival³⁸
 - b) considered with: residual disease, high mitotic index, or necrosis
3. chemotherapy: role not defined

37.2.8 Prognosis

Overall survival with gross total resection or subtotal resection, with or without radiation and chemotherapy: 5 years = 80% 10 years = 71%³⁷

Extent of resection, mitotic index, and necrosis appear to be the best predictors of outcome.^{41,42}

References

- [1] Burger PC, Scheithauer BW. Atlas of Tumor Pathology. Tumors of the Central Nervous System. Washington, D.C.: Armed Forces Institute of Pathology; 1994
- [2] Pollack IF, Hoffman HJ, Humphreys RP, Becker L. The Long-Term Outcome After Surgical Treatment of Dorsally Exophytic Brain-Stem Gliomas. *J Neurosurg.* 1993; 78:859–863
- [3] Coakley KJ, Huston J, Scheithauer BW, Forbes G, Kelly PJ. Pilocytic Astrocytomas: Well-Demarcated Magnetic Resonance Appearance Despite Frequent Infiltration Histologically. *Mayo Clin Proc.* 1995; 70:747–751
- [4] Hayostek CJ, Shaw EG, Scheithauer B, et al. Astrocytomas of the Cerebellum: A Comparative Clinicopathologic Study of Pilocytic and Diffuse Astrocytomas. *Cancer.* 1993; 72:856–869
- [5] Bernell WR, Kepes JJ, Seitz EP. Late Malignant Recurrence of Childhood Cerebellar Astrocytoma. *J Neurosurg.* 1972; 37:470–474
- [6] Schwartz AM, Ghatak NR. Malignant Transformation of Benign Cerebellar Astrocytoma. *Cancer.* 1990; 65:333–336
- [7] Wallner KE, Gonzales MF, Edwards MSB, Wara WM, Sheline GE. Treatment of juvenile pilocytic astrocytoma. *J Neurosurg.* 1988; 69:171–176
- [8] Zimmerman RA, Bilaniuk CT, Bruno LA, et al. CT of Cerebellar Astrocytoma. *Am J Roentgenol.* 1978; 130:929–933
- [9] Section of Pediatric Neurosurgery of the American Association of Neurological Surgeons. *Pediatric Neurosurgery.* New York 1982
- [10] Youmans JR. *Neurological Surgery.* Philadelphia 1990
- [11] Ringertz N, Nordenstam H. Cerebellar Astrocytoma. *J Neuropathol Exp Neurol.* 1951; 10:343–367
- [12] Gol A. Cerebellar Astrocytomas in Children. *Am J Dis Child.* 1963; 106:21–24
- [13] Winston K, Gilles FH, Leviton A, et al. Cerebellar Gliomas in Children. *J Natl Cancer Inst.* 1977; 58:833–838
- [14] Austin EJ, Alvord EC. Recurrences of Cerebellar Astrocytomas: A Violation of Collins' Law. *J Neurosurg.* 1988; 68:41–47
- [15] Packer RJ, Lange B, Ater J, et al. Carboplatin and Vincristine for Recurrent and Newly Diagnosed Low-Grade Gliomas of Childhood. *J Clin Oncol.* 1993; 11:850–856
- [16] Bucy PC, Thieman PW. Astrocytomas of the Cerebellum. A Study of Patients Operated Upon Over 28 Years Ago. *Arch Neurol.* 1968; 18:14–19
- [17] Stein BM, Tenner MS, Fraser RAR. Hydrocephalus Following Removal of Cerebellar Astrocytomas in Children. *J Neurosurg.* 1972; 36:763–768
- [18] Tihan T, Fisher PG, Kepner JL, Godfraind C, McComb RD, Goldthwaite PT, Burger PC. Pediatric astrocytomas with monomorphous pilomyxoid features and a less favorable outcome. *J Neuropathol Exp Neurol.* 1999; 58:1061–1068
- [19] Ceppa EP, Bouffet E, Griebel R, Robinson C, Tihan T. The pilomyxoid astrocytoma and its relationship to pilocytic astrocytoma: report of a case and a critical review of the entity. *J Neurooncol.* 2007; 81:191–196
- [20] Arulrajah S, Huisman TA. Pilomyxoid astrocytoma of the spinal cord with cerebrospinal fluid and peritoneal metastasis. *Neuropediatrics.* 2008; 39:243–245
- [21] Packer RJ, Nicholson HS, Vezina LG, Johnson DL. Brainstem gliomas. *Neurosurg Clin N Am.* 1992; 3:863–879
- [22] Laurent JP, Cheek WR. Brain Tumors in Children. *J Pediatr Neurosci.* 1985; 1:15–32
- [23] Reigel DH, Scarff TB, Woodford JE. Biopsy of Pediatric Brain Stem Tumors. *Childs Brain.* 1979; 5:329–340
- [24] Epstein FJ, Farmaer J-P. Brain-Stem Glioma Growth Patterns. *J Neurosurg.* 1993; 78:408–412
- [25] Epstein F, McCleary EL. Intrinsic Brain-Stem Tumors of Childhood: Surgical Indications. *J Neurosurg.* 1986; 64:11–15
- [26] Garcia CA, McGarry PA, Collada M. Ganglioglioma of the Brain Stem. Case Report. *J Neurosurg.* 1984; 60:431–434
- [27] Albright AL, Packer RJ, Zimmerman R, et al. Magnetic Resonance Scans Should Replace Biopsies for the Diagnosis of Diffuse Brain Stem Gliomas: A Report from the Children's Cancer Group. *Neurosurgery.* 1993; 33:1026–1030
- [28] Hoffman HJ, Becker L, Craven MA. A Clinically and Pathologically Distinct Group of Benign Brainstem Gliomas. *Neurosurgery.* 1980; 7:243–248
- [29] Kyoshima K, Kobayashi S, Gibo H, Kuroyanagi T. A Study of Safe Entry Zones via the Floor of the Fourth Ventricle for Brain-Stem Lesions. *J Neurosurg.* 1993; 78:987–993
- [30] Kernohan WJ, Armed Forces Institute of Pathology. In: Tumors of the central nervous system. Atlas of Tumor Pathology. Washington, DC 1952:19–42
- [31] Stark AM, Fritsch MJ, Claviez A, Dorner L, Mehdorn HM. Management of tectal glioma in childhood. *Pediatr Neurol.* 2005
- [32] Bogner L, Turjman F, Villanyi E, Mottolese C, Guyotat J, Fischer C, Juvet A, Lapras C. Tectal plate gliomas. Part II: CT scans and MR imaging of tectal gliomas. *Acta Neurochir (Wien).* 1994; 127:48–54
- [33] Pollack IF, Pang D, Albright AL. The long-term outcome in children with late-onset aqueductal stenosis resulting from benign intrinsic tectal tumors. *J Neurosurg.* 1994; 80:681–688
- [34] Grant GA, Avellino AM, Loeser JD, Ellenbogen RG, Berger MS, Roberts TS. Management of intrinsic gliomas of the tectal plate in children. A ten-year review. *Pediatr Neurosurg.* 1999; 31:170–176
- [35] Oka K, Kin Y, Go Y, Ueno Y, Hirakawa K, Tomonaga M, Inoue T, Yoshioka S. Neuroendoscopic approach to tectal tumors: a consecutive series. *J Neurosurg.* 1999; 91:964–970
- [36] Kihlstrom L, Lindquist C, Lindquist M, Karlsson B. Stereotactic radiosurgery for tectal low-grade gliomas. *Acta Neurochir Suppl.* 1994; 62:55–57
- [37] Fouladi M, Jenkins J, Burger P, Langston J, Merchant T, Heideman R, Thompson S, Sanford A, Kun L, Gajjar A. Pleomorphic xanthoastrocytoma: favorable outcome after complete surgical resection. *Neurooncol.* 2001; 3:184–192
- [38] Kepes JJ, Rubinstein LJ, Eng LF. Pleomorphic Xanthoastrocytoma: A Distinctive Meningeal Glioma of Young Subjects with Relatively Favorable Prognosis. A Study of 12 Cases. *Cancer.* 1979; 44:1839–1852
- [39] Weldon-Linne CM, Victor TA, Groothuis DR, Vick NA. Pleomorphic Xanthoastrocytoma: Ultrastructural and Immunohistochemical Study of a Case with a Rapidly Fatal Outcome Following Surgery. *Cancer.* 1983; 52:2055–2063
- [40] Kumar S, Retnam TM, Menon G, Nair S, Bhattacharya RN, Radhakrishnan VV. Cerebellar hemisphere, an uncommon location for pleomorphic xanthoastrocytoma and lipidized glioblastoma multiformis. *Neurol India.* 2003; 51:246–247
- [41] Pahapill PA, Ramsay DA, Del Maestro RF. Pleomorphic xanthoastrocytoma: case report and analysis of the literature concerning the efficacy of resection and the significance of necrosis. *Neurosurgery.* 1996; 38:822–8; discussion 828–9
- [42] Giannini C, Scheithauer BW, Burger PC, Brat DJ, Wollan PC, Lach B, O'Neill BP. Pleomorphic xanthoastrocytoma: what do we really know about it? *Cancer.* 1999; 85:2033–2045

38 Oligodendroglial Tumors and Tumors of the Ependyma, Choroid Plexus, and Other Neuroepithelial Tumors

38.1 Oligodendroglial tumors

38.1.1 General information

Key concepts

- frequently presents with seizures
- predilection for the frontal lobes
- histology: classic features of “fried egg” cytoplasm (on permanent pathology) & “chicken wire” vasculature are unreliable. Calcifications are common
- grading: controversial. Recommendation: low grade and high grade
- recommended treatment: surgery for mass effect or low grade lesions (high-grade lesions are controversial). Chemotherapy for all (with or without surgery), XRT only for anaplastic transformation

38.1.2 Epidemiology

Oligodendroglioma (ODG) have long been thought to comprise only $\approx 2\text{--}4\%$ of primary brain tumors^{1,2} or $4\text{--}8\%$ of cerebral gliomas²; but recent evidence indicates these tumors have been underdiagnosed (many are misinterpreted as fibrillary astrocytomas, especially the infiltrative portion of these tumors) and ODGs may represent up to $25\text{--}33\%$ of glial tumors.^{3,4} Ratio of male:female = 3:2. Primarily a tumor of adults: average age ≈ 40 years (peak between 26–46 years), but with a smaller earlier peak in childhood between 6–12 years.⁵ CSF metastases reportedly occur in up to 10% but 1% may be a more realistic estimate.¹ Spinal ODGs comprise only $\approx 2.6\%$ of intramedullary tumors of the cord and filum.

38.1.3 Clinical

Classic presentation of ODG: a patient with seizures for many years prior to the diagnosis being made when they would present with an apoplectic event due to peritumoral intracerebral hemorrhage. This scenario is less common in the CT/MRI era.

Seizures are the presenting symptoms in $\approx 50\text{--}80\%$ of cases.^{1,5} The remainder of presenting symptoms are nonspecific for ODG, and are more often related to local mass effect and less commonly to $\uparrow$ ICP. Presenting symptoms are shown in $\square$ Table 38.2.

38.1.4 Evaluation

Calcifications are seen in $28\text{--}60\%$ of ODGs on plain radiographs,¹ and on 90% of CTs.

Table 38.1 Location of oligodendrogliomas	
Location	%
supratentorial	>90%
frontal lobes	45%
hemisphere (outside frontal lobes)	40%
within third or lateral ventricle	15%
infratentorial+ spinal cord	<10%

Table 38.2 Presenting symptoms in 208 oligodendrogliomas ¹	
Symptom	%
seizures	57%
headache	22%
mental status changes	10%
vertigo/nausea	9%

Table 38.3 Features associated with low-grade and high-grade oligodendrogliomas		
Feature	WHO II (low grade)	WHO III (high grade)
contrast enhancement on CT or MRI	absent	present
endothelial proliferation on histology	absent	present
pleomorphism (large variability in nuclear and cytoplasmic size and shape)	absent	present
tumor proliferation (evidenced by mitotic figures or high MIB-1 index ^a)	absent	present
astrocytic component	absent	present
^a for information on the MIB-1 index (p.599)		

38.1.5 Pathology

General information

73% of tumors have microscopic calcifications.⁶ Isolated tumor cells consistently penetrate largely intact parenchyma, an associated solid tumor component may or may not be present.⁴ The solid portion, when present, classically demonstrates lucent perinuclear halos giving a “fried egg” appearance (actually an artifact of formalin fixation, which is not present on frozen section and may make diagnosis difficult on frozen). A “chicken-wire” vascular pattern has also been described.⁷ These features are felt to be unreliable, and cells with monotonous round nuclei (often in cellular sheets) with an eccentric rim of eosinophilic cytoplasm lacking obvious cell processes are more consistent features.⁸

16% of hemispheric ODGs are cystic⁶ (cysts form from coalescence of microcysts from micro-hemorrhages, unlike astrocytomas which actively secrete fluid).

33–41% have a component of ependymal or neoplastic astrocytic cells (so called oligoastrocytomas or mixed gliomas⁹ or collision tumors (p.645)).

GFAP staining: Since most ODGs contain microtubules instead of glial filaments,¹⁰ ODGs usually do not stain for GFAP (p.598) although some do.¹¹ In mixed gliomas, the astrocytic component may stain for GFAP.

Grading

A work-in-progress. Historically, a number of attempts at grading ODGs have been proposed and then abandoned because of lack of prognostic significance (for a review, see reference⁸). For example, the system of Smith et al.¹² (see below) was based on 5 histopathologic features which have been shown not to be independent determinants of tumor progression (only pleomorphism has been shown to be statistically correlated with survival⁸). Necrosis does not appear to reliably predict a poor prognosis.⁸

For prognostic purposes, it is suggested that ODGs be stratified into two groups:

- oligodendroglioma (WHO grade II) or low grade
- anaplastic oligodendroglioma (WHO grade III) or high grade.^{2,8}

Although there is not uniform agreement on the means for differentiating the two, the factors shown in □ Table 38.3 should be taken into account as they have been demonstrated to have prognostic significance. Using the spatial grading system for low grade gliomas, no ODGs are of the Type 1 tumor (solid tumor without infiltrative component).

Table 38.4 Smith grading of oligodendrogliomas					
Grade	Nuclear to cyto-plasm ratio	Maximal cell density	Pleomorphism	Endothelial pro-liferation	Necrosis
A	↓	↓	–	–	–
B	↑ ^a	↑ ^a	+ ^a	–	–
C	↑	↑	+	+	–
D	↑	↑	+	+	+
↑ means high; ↓ means low; + means feature is present; – means feature is absent					
^a considered grade B if one or more of these are present					

Smith grading system

- See reference.¹²
- Tumors composed of at least 51%oligodendroglial elements are graded on 5 histologic features:
1. maximal nuclear/cytoplasmic ratio: the normal oligodendroglial cell ratio is considered low (↓), anything above that is coded as high (↑)
 2. maximal cell density: coded according to predominant appearance of sample. Cell density similar to white matter is low (↓), sheets of cells with little or no intervening extracellular neuropil is high (↑)
 3. pleomorphism: coded according to predominant appearance of sample. Coded as present (+) if there is large variability in nuclear and cytoplasmic size and shape
 4. endothelial proliferation: coded present (+) if 1 or more instance noted
 5. necrosis: coded present (+) if 1 or more instance of coagulation necrosis and/or areas of debris filled with macrophages noted

Tumor grade is then determined from □ Table 38.4.

38.1.6 Treatment

General information

Σ
Recommendation (see text for details). Following an appropriate surgical procedure (if indicated), chemotherapy is the primary treatment modality. XRT is reserved for anaplastic transformation, if it should occur. ⁸

Chemotherapy

Most ODGs respond to some form of chemotherapy, usually in <3 mos, often with a reduction in size. The response is variable in degree and duration.¹³ No pathological or clinical feature of high-grade ODGs has been identified that reliably predicts response to chemotherapy. However, allelic loss of chromosome 1p, and combined loss of chromosome arms 1p and 19q, are associated with response to chemo; and losses of both 1p and 19q were associated with longer tumor-free survival after chemo.¹⁴

The most experience is with PCV (procarbazine 60 mg/m² IV, CCNU AKA lomustine (CeeNU ®) 110 mg/m² PO, and vincristine 1.4 mg/m² IV, all given on a 29 day cycle repeated every 6 weeks).^{15,16} Also studied: temozolomide for recurrent anaplastic oligoastrocytoma showed some efficacy.¹⁷

Surgery

- Indications for surgery:
1. ODGs with significant mass effect regardless of grade: surgery decreases the need for corticoste-roids, reduces symptoms and prolongs survival⁸

2. tumors without significant mass effect:

- a) low-grade ODGs and oligoastrocytoma: surgery is recommended for resectable lesions. Gross total removal should be attempted when possible (survival is improved even more than with astrocytomas¹⁸), but not at the expense of neurologic function
- b) high-grade ODGs: data for improved survival is less convincing, and some studies show no advantage of gross total removal over partially resected or biopsied-only high-grade lesions.⁸ Older dogma was that aggressive removal leads to longer survival,¹⁹ and results in fewer side effects than “partial debulking” operations²⁰

Grossly, the tumor appears as a pink to red, friable mass. There may be a false plane of demarcation between tumor and what appears to be normal brain.

Postoperative radiation

Benefits of postoperative irradiation is controversial.⁵ In a retrospective analysis with no set selection criteria, survival was better in patients receiving >45 Gy.²¹ In another series, no difference in 5 year survival following surgery was seen with or without XRT (amount of radiation not specified).²² Radiation side effects of memory loss, dementia and personality changes are more common with the longer survival seen in many of these cases.²³

38.1.7 Prognosis

Pure ODGs have a better prognosis than mixed oligoastrocytomas which are better than pure astrocytomas (an oligodendroglial component, no matter how small, confers a better prognosis).

With the Smith grading system (see above), median survival in 323 cases was as follows: grade A was 94 months, grades B & C were not statistically different and were 51 and 45 months respectively, and grade D was 17 months.

10 year survival of 10–30% has been quoted for tumors that are completely or predominantly ODGs.²¹ As a group, median survival for surgically treated lesions is given as 35 months post-op (mean 52 months).¹

The presence of calcifications is debated as a prognosticator; in one series, calcified ODG on plain films had a longer median survival of 108 months (vs. 58 months for non-calcified).¹

Frontal lobe ODGs survived longer than those in temporal lobes (37 months vs. 28 months post-operative survival),¹ possibly due to increased ease of radical resection with the former.

Chromosomal 1p loss (or combined 1p and 19q loss) is also associated with longer survival.^{14,24}

38.2 Oligoastrocytic tumors

38.2.1 Molecular biology

May show changes typical for diffuse astrocytoma (TP53 mutation & LOH on 17p) or for ODG (LOH on 1p and 19q). No molecular genetic markers have been identified to distinguish oligoastrocytoma from either astrocytoma or ODG. Unlike ODG, the prognostic/therapeutic value of LOH on 1p is less clear.²⁴

Oligoastrocytoma (WHO grade II)

Two distinct neoplastic cell types, 1 type resembles oligodendroglioma cells, and the other resembles cells in diffuse astrocytomas. Some cells may have features of both. The 2 cell types may be segregated or diffusely admixed.

Anaplastic oligoastrocytoma (WHO grade III)

Increased cellularity, nuclear atypia, pleomorphism, and high mitotic activity. Necrosis and microvascular proliferation may be present. Differentiating from GBM may be difficult since GMS may have areas resembling anaplastic ODG (the term “glioblastoma with oligodendroglioma component” is a disputed term suggested for these – unproven suggestion that survival may be better than for ordinary GBM²⁵).

38.3 Ependymal tumors

38.3.1 Ependymoma

General information regarding intracranial and spinal ependymomas

Ependymomas arise from ependymal cells lining the cerebral ventricles and the central canal of the spinal cord. They may occur anywhere along the neuraxis, in pediatrics they are most common in the posterior fossa (see below), in adults they tend to be intraspinal (p.788).

Epidemiology

- intracranial: comprises only $\approx 5\text{--}6\%$ of intracranial gliomas, 69% occur in children,²⁶ comprise 9% of pediatric brain tumors.²⁷ Incidence of pediatric intracranial ependymomas: ≈ 200 cases/yr in the U.S.
- spinal: $\approx 60\%$ of spinal cord gliomas (the most common primary intramedullary spinal cord glioma below the mid-thoracic region – see Intramedullary spinal cord tumors (p.787) – 96% occur in adults,²⁶ especially those of filum terminale; see myxopapillary ependymoma below).

The mean age at diagnosis is shown in □ Table 38.5.

Ependymomas have the potential to spread via the CSF through the neuraxis (including to the spinal cord), a process known as “seeding”, resulting in so-called “drop mets” in the spinal cord in 11% The incidence is higher with higher grade.²⁷ Systemic spread occurs on rare occasion.

Pathology

Although they are usually circumscribed with a covering layer of ependyma, ependymomas may be invasive.

Classification is a work in progress. Ependymomas from different locations (p-fossa, supratentorial, spinal cord) are genetically distinct.²⁸ The World Health Organization (WHO) classification of ependymal tumors:

1. ependymoma (WHO II) – variants:
 - a) cellular
 - b) papillary: “classic lesion” occurring in brain or spinal cord. Can metastasize in up to 30% of cases. Dark, small nuclei. 2 cytoplasmic patterns:
 - differentiation along glial line: these form perivascular pseudorosettes (areas of radiating processes lacking nuclei surrounding blood vessels) which, when they occur, are diagnostic
 - cuboidal cells: these can form true rosettes (where the cytoplasmic processes extend to form a lumen-like structure with no central blood vessel – as if trying to recreate an ependymal lining e.g. of a ventricle). True rosettes are classically associated with ependymomas, however, perivascular pseudorosettes are more common
 - c) clear cell
 - d) tanycytic: rare. Tumor cells appear similar to “ependymoglia” or “tanocytes” (stretched cells present to a limited degree in the normal CNS). True rosettes are absent. No preference for age, sex or location within CNS.²⁹ Treatment of choice: gross total resection²⁹
2. myxopapillary ependymoma: (WHO I) distinctive, occurs only in filum terminale. Papillary, with microcystic vacuoles and mucosubstance

Table 38.5 Mean age at diagnosis of ependymoma ²⁶		
Location (in 101 patients)	All patients (yrs)	Children (yrs) (age <15 yrs)
intracranial	17.5	5
infratentorial	14.5	4.5
supratentorial	22	6.5
intraspinal	40	
intramedullary	47	
cauda region	32	

- 3. subependymomas: (WHO I) typically occur in anterior lateral ventricles or posterior fourth ventricle, with prominent role of subependymal glial cells. Not uncommon at autopsy, rarely surgical. Classically do not enhance; see Ependymal and subependymal enhancement (p.1385)
- 4. anaplastic ependymomas: (WHO III) pleomorphism, multinucleation, giant cells, mitotic figures, vascular changes and areas of necrosis; the term ependymblastoma has occasionally been used for more anaplastic lesions, but this term is best reserved for a distinct, rare childhood primitive neuroectodermal tumor (p.666). It is unclear if the degree of anaplasia has any effect on outcome

General information for intracranial ependymomas

Key concepts

- usually benign tumors, often fibrillary with epithelial appearance. Perivascular pseudorosettes or true rosettes may be seen in classic (papillary) form
- most often occur in the floor of the 4th ventricle, presenting with hydrocephalus (increased ICP) and cranial nerve VI & VII palsies
- evaluation: includes imaging the entire neuraxis (usually with enhanced MRI: cervical, thoracic, lumbar & brain) because of potential for seeding through CSF
- worse prognosis the younger the patient (especially age <24 months)
- treatment: the best outcomes are associated with gross total removal (no enhancing tumor on post-op MRI) followed by XRT. XRT may be withheld for age <3
- do LP ≈ 2 weeks post-op to send ≈ 10 cc of CSF for cytology for prognostication

Usually well circumscribed and benign (although anaplastic (malignant) ependymomas do occur), commonly arises in the floor of the fourth ventricle (60–70%are infratentorial, all of these occur near 4th ventricle,²⁶ they comprise 25%of tumors in region of 4th ventricle^{30 (p 2792)}). Children with p-fossa ependymomas often have anaplastic tumors with a higher risk of spread through the neuraxis. Supratentorial ependymomas are often cystic. Rarely occur outside the CNS in: mediastinum, lung or ovaries. Although not as malignant histologically as medulloblastomas, ependymomas have a worse prognosis due to their propensity to invade the obex which precludes complete removal.

Clinical

Symptoms

Mostly those of posterior fossa mass with increased ICP^{30 (p 2795)} (from hydrocephalus) and cranial nerve involvement.

Symptoms of increased ICP:

- 1. headache: 80%
- 2. N/V: 75%
- 3. ataxia or vertigo: 60%
- 4. seizures: only in ≈ 30%of supratentorial lesions; comprise only 1%of patients with intracranial tumors presenting with seizures

Signs

Cranial nerve involvement: invasion of the floor of the 4th ventricle may involve the facial colliculus producing facial nerve palsy (p.576) (involvement of internal genu of VII) and abducens palsy (from VI nucleus).

Evaluation

MRI: imaging study of choice. Image the entire craniospinal axis with and without contrast because of possibility of drop mets. Usually appears as a mass in the floor of fourth ventricle, often with obstructive hydrocephalus. May be difficult to distinguish from medulloblastoma (MBS) radiographically, see differentiating features (p.1365).

CT: not as detailed for evaluation of posterior fossa.

Myelogram: water-soluble contrast myelography is about as sensitive as gadolinium enhanced MRI in detecting “drop mets.” Myelography also provides CSF for cytology for staging purposes.

Treatment

Surgical resection

Goal of surgery: maximal possible resection of intracranial portion without causing neurological deficits (since extent of resection is an important prognosticator). Gross total resection may not be possible when invasion of the floor is extensive, or when tumor extends through the foramen of Luschka (bradycardia may prevent GTR).

2 weeks postoperatively, perform LP to look for “drop mets”: 10 cc of CSF is sent for cytology to quantitate (if any) number of malignant cells (may be used to follow treatment). If LP is positive, then by definition there are drop mets. If negative, it is not as helpful (sensitivity is not high). CSF from an EVD is not as sensitive as LP.

Lesions in fourth ventricle region are approached via midline suboccipital craniectomy.

Radiation therapy (XRT)

Ependymomas rank 2nd only to medulloblastomas in radiosensitivity. XRT is administered after surgical excision (survival is improved with post-op XRT^{26,31}: 50% survival time was 2 yrs longer with XRT than without,²⁶ and 5-year survival increased from 20–40% without XRT to 40–80% with XRT³¹), however, for patients age <3 years, see below.

1. cranial XRT
 - a) traditional therapy: 45–48 Gy to tumor bed³¹ (recurrence treated with additional 15–20 Gy)^{30 (p 2797)}
 - b) recent recommendations: 3-D conformal XRT with higher doses (59.4 Gy delivered to tumor bed + 1 cm margins)³²
 - c) intensity modulated proton beam therapy appears equivalent in terms of local control, but may be better at sparing normal tissue³³
2. spinal XRT: most radiate only if drop mets or if positive CSF cytology (however, prophylactic spinal is controversial³⁴)
 - a) low dose XRT to entire spinal axis (median dose = 30 Gy in one series³¹)
 - b) boost to any regions showing drop mets
3. XRT is undesirable in age <3 due to side effects. XRT was avoided in ≈ 30% of patients <3 years age with comparable survival when XRT was reserved for treatment failures.^{35,36} This concept of selective XRT may be applicable to older children as well³⁷

Chemotherapy

Role is very limited.

1. has little impact on newly diagnosed cases. Adjuvant chemo after XRT in patients >3 years showed no benefit
2. may reduce vascularity of ependymomas which may facilitate GTR (sometimes in a second stage operation)
3. may be considered for infants <≈ 3 years age to delay use of XRT (see above)
4. chemo at the time of recurrence may arrest tumor progression for short periods

Outcome

Operative mortality^{30 (p 2797)}: 20–50% in early series; more recently: 5–8%

Operative morbidity: advise patients/families pre-op of the likelihood need for post-op gastric feeding tube (G-tube) and tracheostomy (these may be temporary).

Age: peds vs. adults: 5-year survival is 20–30% in the pediatric group,^{27,38} compared with up to 80% in adults. Patients 24–35 months old did better (5-year survival = 73%) than those younger than 24 months (26% 5-YS) or those older than 36 months (36% 5-YS).³⁹

Pathology: prognosis is worse with anaplastic ependymoma (WHO III) than with “standard” grade (WHO II).^{40,41} However, excluding WHO III tumors, malignant features in an ependymoma do not necessarily portend a worse prognosis.⁴²

Extent of resection: the risk of recurrence is highest following subtotal resection. Gross total resection (GTR) (surgical) of primary intracranial tumor followed by craniospinal XRT as outlined above yields 41% 5-year survival.

Treatment failure: WHO Grade II tumors tend to recur initially at the site of origin.⁴⁰ However, primary failure in 9–25% of patients is via drop mets.^{39,43}

38.4 Neuronal and mixed glial tumors

38.4.1 Desmoplastic infantile astrocytoma/ganglioglioma

The former entities “desmoplastic cerebral astrocytoma of infancy” and “desmoplastic infantile ganglioglioma” have been combined to “desmoplastic infantile astrocytoma and ganglioglioma” (DIG).⁴⁴ A lesion with either astrocytic or dual glioneuronal differentiation. Prognosis is usually favorable.

38.4.2 Central neurocytoma

General information

Key concepts

- Rare, WHO Grade II neuronal tumor
- Primarily seen in young adults
- Gross total resection can be curative
- If MIB-1 labeling index >2-4% there is an increased risk of recurrence after subtotal resection
- If MIB-1 labeling is elevated, radiation therapy after subtotal resection can reduce the risk of recurrence

Central neurocytoma is a WHO Grade II neuronal tumor that constitutes 0.1-0.5% of brain tumors. These are generally attached to the septum pellucidum within the lateral ventricles or they are within the third ventricle. Incidence peaks in the 3rd decade, but they can also present in children and the elderly. No gender predominates. The most common presentation is with increased intracranial pressure and ventriculomegaly.^{45,46}

Pathology

Central neurocytoma cells have small, round nuclei. Cells often have a “fried egg” appearance on H&E staining, which can mimic oligodendroglioma. There are two primary architectures: honeycomb (also mimicking oligodendroglioma) and fibrillary. Rosettes can also be seen. Immunohistochemistry is often positive for neuronal markers synaptophysin and Neu-N. When the diagnosis is not clear, electron microscopy is used to demonstrate neuronal features including: a prominent golgi apparatus, parallel microtubules, and dense core neurosecretory granules.^{45,46} 1p/19q deletion has not been reported in central neurocytoma, but it can be seen in extraventricular neurocytoma.⁴⁷

Variants

1. Extraventricular neurocytoma is a more rare subtype that may be located in the cerebral parenchyma, cerebellum, thalamus, brainstem, pineal region, and spinal cord. Tumor cells can be focally infiltrative into surrounding tissue. Pathology findings include vessel wall hyalinization and ganglionic differentiation. Immunohistochemistry is positive for synaptophysin in the cytoplasm and neuropil. Neu-N staining is often positive. Electron microscopy findings include round nuclei and neurosecretory granules, as in central neurocytoma. 1p and 19q deletions, in isolation or in combination, can be seen in extraventricular neurocytoma.⁴⁷
2. Liponeurocytoma is a lipid-laden neurocytoma that most often appears in the cerebellum. Rarely, similar tumors can also appear in the lateral ventricles. Some authors suggest the more inclusive name “Liponeurocytoma” should replace the separate WHO classification of “Cerebellar liponeurocytoma.” Others contend that the supratentorial exceptions are central neurocytomas which contain neurocytes that underwent lipidization, instead of having actual adipose metaplasia.⁴⁸

Imaging

CT scan: 25-50% of these tumors show calcifications. Tumors are usually iso- to hyperdense, with hypodense areas that represent cystic degeneration.⁴⁵

MRI: Tumors appear heterogeneously isointense on T1 and hyperintense on T2. MR spectroscopy often shows a high glycine peak.⁴⁹

Moderate to strong contrast enhancement shows on both CT and MR images.⁴⁶

Treatment

1. Total resection is often curative. After gross total resection, radiation therapy is generally not necessary.⁴⁵
2. Subtotal resection can be followed by stereotactic radiosurgery, especially if MIB-1 labeling is >2-4%.^{45,46,49,50}
3. Chemotherapy for recurrent and inoperable tumors has been reported with many agents, including: alkylating agents (carmustine, cyclophosphamide, ifosfamide, lomustine), platinum-based agents (carboplatin and cisplatin), etoposide, topotecan, and vincristine.^{49,50}
4. After treatment, patients should have long-term follow-up imaging as surveillance for tumor recurrence.⁴⁶

Prognosis

Total resection can be curative.^{45,49} Recurrence risk after subtotal resection varies with the MIB-1 labeling index. In a 2004 study of neurocytomas with MIB-1 labeling >2% the 5-year local control rate after subtotal resection alone was 7% but if subtotal resection was followed with radiation therapy it was 70%.⁵⁰ In a 2013 study, patients with MIB-1 labeling <4% had no recurrence at 4 years after subtotal resection. If MIB-1 labeling was >4% at subtotal resection, recurrence was found in 50% of patients at 2 years and 75% at 4 years.⁴⁹ Most local recurrences occur within 3-6 years. Recurrence is more common in extraventricular neurocytoma.⁵⁰

38.4.3 Cerebellar liponeurocytoma

Née lipomatous medulloblastoma. Occurs exclusively in cerebellum of adults (mean age: 50 years). No gender preference.

Histology: clusters of neoplastic neurocytes with lipidization (resembling adipocytes) with background of small neoplastic cells with morphological features more suggestive of neurocytes. Synaptophysin (p.600) and MAP-2 immunostaining is consistent and diffuse, focal GFAP staining is common. Usually no mitotic figures. MIB-1 index 1–3%

38.4.4 Dysembryoplastic neuroepithelial tumors (DNT) or (DNET)

See references.^{51,52}

Epidemiology

Incidence: not accurately known because the diagnosis may be missed. Estimated range: 0.8–5% of all primary brain tumors. Typically occurs in children and young adults.

Most common locations: temporal or frontal. Parietal and especially occipital lobe involvement is rare. DNTs have been reported in the cerebellum, pons & basal ganglia.

Pathology

A WHO Grade I glioma. Thought to arise embryologically from the secondary germinal layer (which includes subependymal layer, cerebellar external granular layer, hippocampal dentate fascia & subpial granular layer).

Multinodularity at low-power is a key feature, and the primary constituent cells are oligodendrocytes and to a lesser extent, astrocytes that are often pilocytic. Occasionally difficult to differentiate from oligodendroglioma.

Two distinct forms⁵³ (do not appear to have different prognoses):

1. simple form: glioneural elements consisting of axon bundles perpendicular to the cortical surface, lined with oligodendroglial-like cells that are S-100 positive and GFAP negative. Normal appearing neurons floating in a pale eosinophilic matrix are scattered between these columns (no resemblance to ganglion cells, unlike gangliogliomas)
2. complex form: glioneural elements as described above in the simple form, with glial nodules scattered throughout. The glial component may mimic a low-grade fibrillary astrocytoma. Foci of cortical dysplasia occur

Clinical

Typically associated with longstanding medically intractable seizures, usually complex partial. Symptoms usually begin before age 20.

Imaging

Cortical lesions with no surrounding edema and no midline mass effect.

CT: hypodense with distinct margins. Deformity of overlying calvaria is common.

MRI: T1WI: hypointense. T2WI: hyperintense, septations may be seen. If there is enhancement, it is usually nodular.

PET scan: hypometabolic with [18F]-fluorodeoxyglucose. Negative [11C]-methionine uptake (unlike all other gliomas).

Outcome

Seizure control: usually improves after surgery. Degree of control seems to correlate with completeness of removal. Improvement in seizures correlates inversely with the duration of intractable seizures.

Recurrence/continued growth: recurrence after complete removal, or tumor growth after partial resection is rare. Adjuvant treatment (XRT, chemotherapy...) is of no benefit in these benign tumors. Mitoses or endothelial proliferation, seen on occasion, do not affect outcome. Malignant transformation is very rare.

38.4.5 Dysplastic gangliocytoma of cerebellum (Lhermitte-Duclos disease)

General information

AKA: ganglioneuroma of the cerebellum, purkinjoma, granular cell hypertrophy of the cerebellum, gangliocytoma dysplasticum, hamartoma of the cerebellum.

Rare (200 case reports⁵⁴) cerebellar lesion with features of both a malformation and a low grade (WHO I) neoplasm that has the propensity to progress (enlarge) and recur after surgery. May be focal or diffuse. Diffuse enlargement of cerebellar folia.

Strongly associated with Cowden syndrome: AKA multiple hamartoma syndrome. Autosomal dominant. Incidence: 1 in 250,000 live births.⁵⁵ Associated with thyroid, breast & uterine Ca, mucosal neuromas & meningiomas.

Pathology

Derangement of normal laminar cellular architecture of the cerebellum with:

1. thickening of the outer molecular cell layer
2. loss of middle Purkinje cell layer
3. infiltration of inner granular cell layer with dysplastic ganglion cells

Clinical

Typically a middle aged adult with signs and symptoms of a cerebellar mass. May also present with hydrocephalus or may be an incidental finding.

Imaging

CT: hypo- to iso-dense, nonenhancing lesion with mass effect.

MRI: T1WI: hypo- to iso-intense. T2WI: hyperintense, heterogeneous. Nonenhancing. Characteristic striated appearance⁵⁶ (tiger stripes) due to widened cerebellar folia. May contain calcifications. DWI: hyperintense. ADC map: hypointense.

NB: in a child with MRI findings of Lhermitte-Duclos disease (LDD) (even if classic), a medulloblastoma is statistically more likely^{57,58} (especially medulloblastoma with extensive nodularity⁵⁹ (MBEN)).

Treatment

Controversial. A few cases with a benign course have been described.⁶⁰ Shunting for hydrocephalus. Biopsy is recommended⁵⁸ particularly for pediatric cases to rule-out medulloblastoma. Surgical excision may be considered when there is significant mass effect.⁶¹ Efficacy of XRT is unknown.

38.5 Choroid plexus tumors

38.5.1 General information

Most are histologically benign (choroid plexus papilloma (CPP), WHO I), although intermediate (atypical choroid plexus papilloma, WHO II) and malignant tumors (choroid plexus carcinoma (CPC), WHO III) may occur. Malignant degeneration from WHO I or II to grade III was seen in 2 out of 124 patients with 59 months mean follow-up.⁶² All may produce drop mets in the CSF, but WHO III do so more commonly. Although usually slow growing, they sometimes grow rapidly.

Atypical CPP have more mitotic figures than CPP without frank signs of malignancy seen in CPC,⁶³ and up to 2 of the following 4 features may be observed: increased cellularity, nuclear pleomorphism, blurring of the papillary pattern, areas of necrosis.

38.5.2 Epidemiology

Prevalence: 0.4–1% of all intracranial tumors. 1.5–6% of tumors in peds.

Although they may occur at any age, 70% of patients are <2 yrs old.⁶⁴ Some tumors occur in neonates, supporting the hypothesis that some of these are congenital.⁶⁵

Location: in adults these tumors are usually infratentorial, whereas in children they tend to occur supratentorially (a reverse from the situation for most other tumors) in the lateral ventricle⁶⁵ with a predilection for the left side. See Intraventricular lesions (p.1381) for differential diagnosis. They can be located anywhere there is choroid plexus, with the most frequent locations: the lateral or fourth ventricles, the CPA (from extension of choroid plexus through the foramen of Luschka).

38.5.3 Presentation

Most present with symptoms of increased ICP from hydrocephalus (H/A, N/V, craniomegaly), others may present with seizures, subarachnoid hemorrhage (with meningismus), or focal neurologic deficit (hemiparesis, sensory deficits, cerebellar signs, or cranial nerve palsies of III, IV and VI).

Hydrocephalus, which may result from: overproduction of CSF (although total removal of these tumors does not always cure the hydrocephalus – especially in patients with high CSF protein, hemorrhage from tumor or surgery, or ependymitis), obstruction of CSF outflow, or communicating hydrocephalus from CSF borne particulates.

38.5.4 Imaging

Brain MRI or CT without and with contrast usually demonstrates a densely enhancing multilobulated intraventricular mass classically with projecting “fronds.” Hydrocephalus is common.

38.5.5 Treatment

General information

There is no role for chemotherapy or radiation for WHO I lesions. For choroid plexus carcinoma, chemotherapy benefits a subset of patients.⁶⁶

Surgical treatment

Benign lesions may be cured surgically with total removal, and even the malignant tumors respond well to surgery. The operation may be difficult due to fragility of the tumor and bleeding from the choroidal arteries. However, persistence with a second and sometimes even third operation is recommended as 5-year survival rate of 84% can be achieved.⁶⁵ Post-operative subdural collections after transcortical tumor excision may occur, and may result from a persistent ventriculosubdural fistula, which may require subdural-peritoneal shunting.⁶⁴

38.5.6 Recurrence

12 recurrences (6% of WHO I and 29% of WHO II patients) requiring neurosurgical intervention occurred in 124 complete resections with 59 months mean follow-up.⁶²

38.6 Other neuroepithelial tumors

1. astroblastoma
2. chordoid glioma of the 3rd ventricle⁶⁷: rare, benign tumor of adulthood. Solid, enhancing mass of the 3rd ventricle. Female:male ratio = 3:1. Mitotic activity is absent in most tumors. GFAP immunostaining is common, S100 reactivity is variable. Histologically similar appearing to chordoid meningioma, which lacks GFAP staining. Attachment to wall of 3rd ventricle (hypothalamus) may prevent total removal
3. angiocentric glioma

References

- [1] Mork SJ, Lindegaard KF, Halvorsen TB, et al. Oligodendroglioma: Incidence and Biological Behavior in a Defined Population. *J Neurosurg.* 1985; 63:881–889
- [2] Daumas-Duport C, Tucker M-L, Kolles H, et al. Oligodendrogliomas: Part II - A new grading system based on morphological and imaging criteria. *J Neurooncol.* 1997; 34:61–78
- [3] Coons SW, Johnson PC, Scheithauer BW, et al. Improving Diagnostic Accuracy and Interobserver Concordance in the Classification and Grading of Primary Gliomas. *Cancer.* 1997; 79:1381–1393
- [4] Daumas-Duport C, Varlet P, Tucker M-L, et al. Oligodendrogliomas: Part I - Patterns of Growth, Histological Diagnosis, Clinical and Imaging Correlations: A Study of 153 Cases. *J Neurooncol.* 1997; 34:37–59
- [5] Chin HW, Hazel JJ, Kim TH, et al. Oligodendrogliomas. I. A Clinical Study of Cerebral Oligodendrogliomas. *Cancer.* 1980; 45:1458–1466
- [6] Roberts M, German W. A Long Term Study of Patients with Oligodendrogliomas. *J Neurosurg.* 1966; 24:697–700
- [7] Coons SW, Johnson PC, Pearl DK, Olafsen AG. The Prognostic Significance of Ki-67 Labeling Indices for Oligodendrogliomas. *Neurosurgery.* 1997; 41:878–885
- [8] Fortin D, Cairncross GJ, Hammond RR. Oligodendroglioma: An Appraisal of Recent Data Pertaining to Diagnosis and Treatment. *Neurosurgery.* 1999; 45:1279–1291
- [9] Hart MN, Petito CK, Earle KM. Mixed Gliomas. *Cancer.* 1974; 33:134–140
- [10] Rutka JT, Murakami M, Dirks PB, et al. Role of Glial Filaments in Cells and Tumors of Glial Origin: A Review. *J Neurosurg.* 1997; 87:420–430
- [11] Kros JM, Schouten WCD, Janssen PJA, et al. Proliferation of Gemistocytic Cells and Glial Fibrillary Acidic Protein (GFAP)-Positive Oligodendroglial Cells in Gliomas: A MIB-1/GFAP Double Labeling Study. *Acta Neuropathol (Berl).* 1996; 91:99–103
- [12] Smith MT, Ludwig CL, Godfrey AD, et al. Grading of Oligodendrogliomas. *Cancer.* 1983; 52:2107–2114
- [13] Cairncross JG, Macdonald D, Ludwin S, et al. Chemotherapy for Anaplastic Oligodendroglioma. *J Clin Oncol.* 1994; 12:2013–2021
- [14] Cairncross JG, Ueki K, Zlatescu MC, et al. Specific Genetic Predictors of Chemotherapeutic Response and Survival in Patients with Anaplastic Oligodendrogliomas. *J Natl Cancer Inst.* 1998; 90:1473–1479
- [15] Levin VA, Edwards MS, Wright DC, et al. Modified Procarbazine, CCNU and Vincristine (PCV-3) Combination Chemotherapy in the Treatment of Malignant Brain Tumors. *Cancer Treat Rep.* 1980; 64:237–244
- [16] Glass J, Hochberg FH, Gruber ML, Louis DN, et al. The Treatment of Oligodendrogliomas and Mixed Oligodendroglioma-Astrocytomas with PCV Chemotherapy. *J Neurosurg.* 1992; 76:741–745
- [17] Yung WK, Prados MD, Yaya-Tur R, et al. Multicenter phase II trial of temozolomide in patients with anaplastic astrocytoma or anaplastic oligoastrocytoma at first relapse. *Temodal Brain Tumor Group. J Clin Oncol.* 1999; 17:2762–2771
- [18] Berger MS, Rostomily RC. Low Grade Gliomas: Functional Mapping Resection Strategies, Extent of Resection, and Outcome. *J Neurooncol.* 1997; 34:85–101
- [19] Earnest F, Kernohan JW, Craig WM. Oligodendrogliomas: A Review of 200 Cases. *Arch Neurol Psychiat.* 1950; 63:964–976
- [20] Ciric I, Ammirati M, Vick N, et al. Supratentorial Gliomas: Surgical Considerations and Immediate Postoperative Results. *Neurosurgery.* 1987; 21:21–26
- [21] Gonzales M, Sheline GE. Treatment of Oligodendrogliomas With or Without Postoperative Radiation. *J Neurosurg.* 1988; 68:684–688
- [22] Reedy DP, Bay JW, Hahn JF. Role of Radiation Therapy in the Treatment of Cerebral Oligodendroglioma. *Neurosurgery.* 1983; 13:499–503
- [23] Taphoorn MJ, Heimans JJ, Snoek FJ, et al. Assessment of Quality of Life in Patients Treated for Low-Grade Glioma: A Preliminary Report. *J Neurol Neurosurg Psychiatry.* 1992; 55:372–376
- [24] Smith JS, Perry A, Borell TJ, Lee HK, O'Fallon J, Hosek SM, Kimmel D, Yates A, Burger PC, Scheithauer BW, Jenkins RB. Alterations of chromosome arms 1p and 19q as predictors of survival in oligodendrogliomas, astrocytomas, and mixed oligoastrocytomas. *J Clin Oncol.* 2000; 18:636–645
- [25] Kraus JA, Lamszus K, Glesmann N, Beck M, Wolter M, Sabel M, Krex D, Klockgether T, Reifenberger G, Schlegel U. Molecular genetic alterations in glioblastomas with oligodendroglial component. *Acta Neuropathol (Berl).* 2001; 101:311–320
- [26] Mork SJ, Loken AC. Ependymoma: A Follow-Up Study of 101 Cases. *Cancer.* 1977; 40:907–915
- [27] Duffner PK, Cohen ME, Freeman AI. Pediatric Brain Tumors: An Overview. *Ca.* 1985; 35:287–301
- [28] Taylor MD, Poppleton H, Fuller C, Su X, Liu Y, Jensen P, Magdaleno S, Dalton J, Calabrese C, Board J, Macdonald T, Rutka J, Guha A, Gajjar A, Curran T, Gilbertson RJ. Radial glia cells are candidate stem cells of ependymoma. *Cancer Cell.* 2005; 8:323–335
- [29] Kleihues P, Louis DN, Scheithauer BW, Rorke LB, Reifenberger G, Burger PC, Cavenee WK. The WHO classification of tumors of the nervous system. *J Neuropathol Exp Neurol.* 2002; 61:215–25; discussion 226–9
- [30] Youmans JR. *Neurological Surgery.* Philadelphia 1982
- [31] Shaw EG, Evans RG, Scheithauer BW, et al. Postoperative Radiotherapy of Intracranial Ependymoma in Pediatric and Adult Patients. *Int J Radiation Oncology Biol Phys.* 1987; 13:1457–1462
- [32] Merchant TE, Mulhern RK, Krasin MJ, Kun IE, Williams T, Li C, Xiong X, Khan RB, Lustig RH, Boop FA, Sanford RA. Preliminary results from a phase II trial of conformal radiation therapy and evaluation of radiation-related CNS effects for pediatric patients with localized ependymoma. *J Clin Oncol.* 2004; 22:3156–3162
- [33] MacDonald SM, Safai S, Trofimov A, Wolfgang J, Fullerton B, Yeap BY, Bortfeld T, Tarbell NJ, Yock T. Proton radiotherapy for childhood ependymoma: initial clinical outcomes and dose comparisons. *Int J Radiat Oncol Biol Phys.* 2008; 71:979–986
- [34] Vanuytsel L, Brada M. The Role of Prophylactic Apical Irradiation in Localized Intracranial Ependymoma. *Int J Radiation Oncology Biol Phys.* 1991; 21:825–830
- [35] van Veelen-Vincent ML, Pierre-Kahn A, Kalifa C, Sainte-Rose C, Zerah M, Thorne J, Renier D. Ependymoma in childhood: prognostic factors, extent of

- surgery, and adjuvant therapy. *J Neurosurg.* 2002; 97:827–835
- [36] Grundy RG, Wilne SA, Weston CL, Robinson K, Lashford LS, Ironside J, Cox T, Chong WK, Campbell RH, Bailey CC, Gattamaneni R, Picton S, Thorpe N, Mallicci C, English MW, Punt JA, Walker DA, Ellison DW, Machin D. Primary postoperative chemotherapy without radiotherapy for intracranial ependymoma in children: the UKCCSG/SIOP prospective study. *Lancet Oncol.* 2007; 8:696–705
- [37] Little AS, Sheean T, Manoharan R, Darbar A, Teo C. The management of completely resected childhood intracranial ependymoma: the argument for observation only. *Childs Nerv Syst.* 2009; 25:281–284
- [38] Sutton LN, Goldwein J, Perilongo G, et al. Prognostic Factors in Childhood Ependymomas. *Pediatr Neurosurg.* 1990; 16:57–65
- [39] Zacharoulis S, Ji L, Pollack IF, Duffner P, Geyer R, Grill J, Schild S, Jaing TH, Massimino M, Finlay J, Spoto R. Metastatic ependymoma: a multi-institutional retrospective analysis of prognostic factors. *Pediatr Blood Cancer.* 2008; 50:231–235
- [40] Kawabata Y, Takahashi JA, Arakawa Y, Hashimoto N. Long-term outcome in patients harboring intracranial ependymoma. *J Neurosurg.* 2005; 103:31–37
- [41] Tihan T, Zhou T, Holmes E, Burger PC, Ozuysal S, Rushing EJ. The prognostic value of histological grading of posterior fossa ependymomas in children: a Children's Oncology Group study and a review of prognostic factors. *Mod Pathol.* 2008; 21:165–177
- [42] Ross GW, Rubinstein LJ. Lack of Histopathological Correlation of Malignant Ependymomas with Postoperative Survival. *J Neurosurg.* 1989; 70:31–36
- [43] Foreman NK, Love S, Thorne R. Intracranial ependymomas: analysis of prognostic factors in a population-based series. *Pediatr Neurosurg.* 1996; 24:119–125
- [44] Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. WHO classification of tumors of the central nervous system. Lyon 2007
- [45] Patel DM, Schmidt RF, Liu JK. Update on the diagnosis, pathogenesis, and treatment strategies for central neurocytoma. *J Clin Neurosci.* 2013; 20:1193–1199
- [46] Sharma MC, Deb P, Sharma S, Sarkar C. Neurocytoma: a comprehensive review. *Neurosurg Rev.* 2006; 29:270–85; discussion 285
- [47] Agarwal S, Sharma MC, Sarkar C, Suri V, Jain A, Sharma MS, Ailawadhi P, Garg A, Mallick S. Extraventricular neurocytomas: a morphological and histogenetic consideration. A study of six cases. *Pathology (Phila).* 2011; 43:327–334
- [48] Chakraborti S, Mahadevan A, Govindan A, Yasha TC, Santosh V, Kovoor JM, Ramamurthi R, Alapatt JP, Hedge T, Shankar SK. Supratentorial and cerebellar liponeurocytomas: report of four cases with review of literature. *J Neurooncol.* 2011; 103:121–127
- [49] Kaur G, Kane AJ, Sughrue ME, Oh M, Safaei M, Sun M, Tihan T, McDermott MW, Berger MS, Parsa AT. MIB-1 labeling index predicts recurrence in intraventricular central neurocytomas. *J Clin Neurosci.* 2013; 20:89–93
- [50] Rades D, Fehlaue F, Schild SE. Treatment of atypical neurocytomas. *Cancer.* 2004; 100:814–817
- [51] Daumas-Duport C, Scheithauer BW, Chodkiewicz J-P, Laws ER, Vedrenne C. Dysembryoplastic Neuroepithelial Tumor: A Surgically Curable Tumor of Young Patients with Intractable Seizures. *Neurosurgery.* 1988; 23:545–556
- [52] Daumas-Duport C, Varlet P, Bacha S, Beuvon F, Cervera-Pierot P, Chodkiewicz JP. Dysembryoplastic neuroepithelial tumors: nonspecific histological forms – a study of 40 cases. *J Neurooncol.* 1999; 41:267–280
- [53] Adada B, Sayed K. Dysembryoplastic neuroepithelial tumors. *Contemp Neurosurg.* 2004; 26:1–5
- [54] Robinson S, Cohen AR. Cowden disease and Lhermitte-Duclos disease: an update. Case report and review of the literature. *Neurosurg Focus.* 2006; 20
- [55] Nelen MR, van Staveren WC, Peeters EA, Hassel MB, Gorlin RJ, Hamm H, Lindboe CF, Fryns JP, Sijmons RH, Woods DG, Mariman EC, Padberg GW, Kremer H. Germline mutations in the PTEN/MMAC1 gene in patients with Cowden disease. *Hum Mol Genet.* 1997; 6:1383–1387
- [56] Meltzer CC, Smirniotopoulos JG, Jones RV. The striated cerebellum: an MR imaging sign in Lhermitte-Duclos disease (dysplastic gangliocytoma). *Radiology.* 1995; 194:699–703
- [57] Chen KS, Hung PC, Wang HS, Jung SM, Ng SH. Medulloblastoma or cerebellar dysplastic gangliocytoma (Lhermitte-Duclos disease)? *Pediatr Neurol.* 2002; 27:404–406
- [58] Someshwar S, Hogg JP, Nield LS. Lhermitte-Duclos disease or neoplasm? *Applied Neurology.* 2007; 3:37–39
- [59] Suresh TN, Santosh V, Yasha TC, Anandh B, Mohanty A, Indiradevi B, Sampath S, Shankar SK. Medulloblastoma with extensive nodularity: a variant occurring in the very young-clinicopathological and immunohistochemical study of four cases. *Childs Nerv Syst.* 2004; 20:55–60
- [60] Capone Mori A, Hoeltzenbein M, Poetsch M, Schneider JF, Brandner S, Boltshauser E. Lhermitte-Duclos disease in 3 children: a clinical long-term observation. *Neuropediatrics.* 2003; 34:30–35
- [61] Carlson JJ, Milburn JM, Barre GM. Lhermitte-Duclos disease: case report. *J Neuroimaging.* 2006; 16:157–162
- [62] Jeibmann A, Wrede B, Peters O, Wolff J, Paulus W, Hasselblatt M. Malignant progression in choroid plexus papillomas. *J Neurosurg.* 2007; 107:199–202
- [63] Jeibmann A, Hasselblatt M, Gerss J, Wrede B, Egensperger R, Beschoner R, Hans VH, Rickert CH, Wolff JE, Paulus W. Prognostic implications of atypical histologic features in choroid plexus papilloma. *J Neuropathol Exp Neurol.* 2006; 65:1069–1073
- [64] Boyd MC, Steinbok P. Choroid Plexus Tumors: Problems in Diagnosis and Management. *J Neurosurg.* 1987; 66:800–805
- [65] Ellenbogen RG, Winston KR, Kupsky WJ. Tumors of the Choroid Plexus in Children. *Neurosurgery.* 1989; 25:327–335
- [66] Wrede B, Liu P, Wolff JE. Chemotherapy improves the survival of patients with choroid plexus carcinoma: a metaanalysis of individual cases with choroid plexus tumors. *J Neurooncol.* 2007; 85:345–351
- [67] Brat DJ, Scheithauer BW, Staugaitis SM, Cortez SC, Brecher K, Burger PC. Third ventricular chordoid glioma: a distinct clinicopathologic entity. *J Neuropathol Exp Neurol.* 1998; 57:283–290

39 Neuronal and Mixed Neuronal-Glial Tumors

39.1 Ganglioglioma

39.1.1 General information

Key concepts

- composed of two cell types: ganglion cells (neurons) and glial cells
- extremely rare (<2% of intracranial neoplasms)
- seen primarily in the first 3 decades of life
- characterized by slow growth and a tendency to calcify

The term “ganglioglioma” was introduced in 1930 by Courville.¹ A tumor composed of two types of cells: ganglion cells (neurons) which may arise from primitive neuroblasts, and glial cells, usually astrocytic in any phase of differentiation.²

39.1.2 Epidemiology

Incidence

Typically quoted³ as 0.3–0.6%. One series⁴ found gangliogliomas in 1.3% of all brain tumors (including mets), or 3% of primary brain tumors. Considering only children and young adults, incidence ranges from 1.2–7.6% of brain tumors.³

Demographics

Occurs primarily in children and young adults (peak age of occurrence: 11 yrs).

39.1.3 Location

May occur in various parts of the nervous system (cerebral hemispheres, spinal cord, brainstem, cerebellum, pineal region, thalamus, intrasellar, optic nerve, and peripheral nerve have been reported³). Most occur above the tentorium, primarily in or near the 3rd ventricle, in the hypothalamus or in the temporal or frontal lobes.⁵ Brainstem gangliogliomas (p.633) occur rarely.

39.1.4 Pathology

Mixture of 2 types of neoplastic cells: neuronal (ganglion) and astrocytic (glial). Very slow growing.

Two major classifications: ganglioneuromas (less common, more benign; predominance of neuronal component) and gangliogliomas (preponderance of glial cells).

Grossly: white matter mass; well-circumscribed, firm, with occasional cystic areas and calcified regions. Most dissect easily from brain, but the solid portion may show an infiltrative tendency.³

Microscopically: ganglion cells must demonstrate nerve cell differentiation, e.g. Nissl substance and axons or dendrites. Pitfall: differentiating neoplastic neurons from neurons entrapped by an invading astrocytoma may be difficult. Also, neoplastic astrocytes may resemble neurons on light microscopy. 2 of 10 patients had areas of oligodendroglioma. One series found necrotic areas in 7 of 14 patients, minimal calcification, and Rosenthal bodies.⁶ Suggested criteria for diagnosis⁷:

1. clusters of large cells potentially representing neurons (required for diagnosis)
2. no perineural clustering of glial cells around the suspected neoplastic neurons
3. fibrosis (desmoplasia)
4. calcification

Aggressive malignant changes in the glial component may dictate a poor outcome, although an “aggressive” background is not unusual and may not indicate malignancy.

39.1.5 Presentation

Most common presenting symptom was seizure, or a change in a pre-existing seizure pattern. Often, the seizures are difficult to control medically.

39.1.6 Radiologic Evaluation

Neuroradiologic findings are not specific for this tumor.

Plain skull x-ray: calcification was noted in 2 of 6 patients.³

CT: all of 10 patients had a low density lesion on non-contrast CT; 8 enhanced slightly with contrast; 5 of the 10 had calcification on CT.⁴ 6 of the 10 were in temporal lobe (this predilection has been noted in many but not all series), and 4 were in frontal lobe. Frequently appears cystic on CT, but still may be found to be solid at operation. Mass effect rare (suggests slow growth).

MRI: high signal on T1WI, low signal on T2WI. Calcifications appear as low signal on both.³

Angiography: shows either an avascular or a minimally vascular mass.

39.1.7 Treatment

Recommendation is wide radical excision when possible (may be more limited in spinal cord and brainstem tumors). Close follow-up is recommended, and re-resection should be considered for recurrence. The role of XRT is unknown, and due to the deleterious effects together with the good long-term prognosis, it is not recommended initially but may be considered for recurrence.⁸

39.1.8 Prognosis

Russell and Rubinstein⁹ first proposed that the grade of the astrocytic component of the tumor determines the prognosis. This has been supported by some case reports, but clinical series have not been able to correlate histology with outcome.⁸ Thus, anaplasia is not significantly associated with a worse prognosis.⁸

The majority of patients did well and were asymptomatic after resection. 1 patient in a series of 10 died 3 days post-op from cerebral edema.

In 58 patients, 5-year survival was 89%and 10-year survival was 84%⁸ In 9 brainstem gangliogliomas, 5-year survival was 78%

The value of radiation therapy is not known. Consider radiation when growth is evident on follow-up CT, or when infiltration is felt to occur at time of surgery.

1 patient had degeneration to glioblastoma when a recurrence was discovered 5 years after removal (this patient received radiation therapy).

The prognosis with following subtotal resection of brainstem gangliogliomas is better than for brainstem gliomas as a group.⁵

39.2 Paraganglioma

39.2.1 General information

AKA chemodectoma, AKA glomus tumors. □ Table 39.1 shows the designation of these tumors in various sites.

Table 39.1 Designation based on site of origin	
Site	Designation
carotid bifurcation (most common)	carotid body tumors
auricular branch of vagus (middle ear)	glomus tympanicum
superior vagal ganglion (jugular foramen)	glomus jugulare
inferior vagal (nodose) ganglion (nasopharynx at skull base) (least common)	glomus intravagale (AKA glomus vagale)
adrenal medulla & sympathetic chain	pheochromocytoma

These tumors arise from paraganglion cells (not chemoreceptor cells as previously thought, therefore the term chemodectoma is losing favor). Slow growing tumors (<2 cm in 5 years). Histologically benign (<10% associated with lymph node involvement or distant spread). Most contain secretory granules on EM (mostly epinephrine & nor-epinephrine, and these tumors may occasionally secrete these catecholamines with risk of life-threatening HTN and/or cardiac arrhythmias).

Glomus tumors may occur in 2 patterns:

1. familial: non multicentric. Up to 50%
2. nonfamilial. may be multicentric (metachronous) 5%

39.2.2 Pheochromocytoma

General information

Located in the adrenal gland. May be sporadic, or as part of familial syndrome (von Hippel-Lindau disease (p.703), MEN 2A & 2B, & neurofibromatosis). Consider genetic testing if age at diagnosis is < 50 years for mutations of VHL and other genetic abnormalities (RET, SDHS, SDHB, SDHC¹⁰).

Laboratory studies

1. fractionated plasma metanephrines: 96% sensitivity, 85% specificity.¹¹ More sensitive than serum catecholamines with sporadic elevations. Pheochromocytoma is ruled out if plasma normetanephrine (NMN) < 112 pg/ml and metanephrine (MN) < 61 pg/ml. Highly suspicious if NMN > 400 pg/ml or MN > 236 pg/ml
2. 24 hr urine collection for: total catecholamines (epinephrine and nor-epinephrine) and metanephrines (88% sensitivity, 99.7% specificity¹²). Note: testing for vanillylmandelic acid (VMA) is no longer done as it does not measure fractionated metanephrines.
3. where elevation is found, a clonidine suppression test can be done. Normal response consists of a fall in plasma catecholamines to ≤ 50% of baseline and below 500 pg/ml (there will be a reduction in essential hypertension, but no change with pheochromocytoma or other tumor production)

Imaging

Indicated when laboratory tests confirm pheochromocytoma.

MRI with contrast is preferred over CT.

CT may be used when MRI is contraindicated, but is less sensitive, especially for lesions < 1 cm diameter.

¹²³I MIBG (iodine-123-meta-iodobenzylguanidine) scintigraphy detects extra-adrenal pheochromocytomas with 83–100% sensitivity, 95–100% specificity. If not available ¹³¹I MIBG may be used with 77–90% sensitivity, 95–100% specificity.

39.2.3 Carotid body tumors

General information

Possibly the most common paraganglioma (pheochromocytoma may be more common). Approximately 5% are bilateral; the incidence of bilaterality increases to 26% in familial cases (these are probably autosomal dominant).

Clinical

Usually present as painless, slow growing mass in upper neck. Large tumors may → cranial nerve involvement (especially vagus and hypoglossal). May also cause stenosis of ICA → TIAs or stroke.

Evaluation

1. carotid angiogram: demonstrates predominant blood supply (usually external carotid, with possible contributions from vertebral and thyrocervical trunk). May also detect bilateral lesions. Characteristic finding: splaying of bifurcation
2. MRI (or CT): evaluates extent, and assesses for intracranial extension

Treatment

Resection reported to carry a high complication rate, including stroke (8–20%) and cranial nerve injury (33–44%). Mortality rate is 5–13%

39.2.4 Glomus tumors

General information

Glomus tumors may be subdivided into glomus jugulare and glomus tympanicum tumors. Glomus jugulare tumors arise from the jugular bulb (in the jugular foramen at the junction of the sigmoid sinus and jugular vein). Glomus tympanicum tumors are centered higher than glomus jugulare. Glomus tumors are rare (0.6% of all head and neck tumors), yet the glomus tympanicum is the most common neoplasm of the middle ear. Glomus jugulare tumors (GJT) arise from glomus bodies, usually in the area of the jugular bulb, and track along vessels. May have finger-like extension into the jugular vein (which may embolize during resection).¹³ Most are slow growing, although rapidly growing tumors do occur.

Vascular supply: very vascular. Main feeders of GJT are from the external carotid (especially inferior tympanic branch of ascending pharyngeal artery, and branches of posterior auricular, occipital, and internal maxillary), with additional feeders from petrous portion of the ICA. Glomus tympanicum tumors feed from the auricular artery.

Epidemiology

Female:male ratio is 6:1. Bilateral occurrence is almost nonexistent.

Pathology

General information

Histologically indistinguishable from carotid body tumors. May invade locally, both through temporal bone destruction and especially along pre-existing pathways (along vessels, eustachian tube, jugular vein, carotid artery). Intradural extension is rare. Malignancy may occur, but is rare. These tumors rarely metastasize.

Secretory properties

These tumors usually possess secretory granules (even the functionally inactive tumors) and may actively secrete catecholamines (similar to pheochromocytomas, occurs in only 1–4% of GJT¹⁴). Nor-epinephrine will be elevated in functionally active tumors since glomus tumors lack the methyl-transferase needed to convert this to epinephrine. Alternatively, serotonin and kallikrein may be released, and may produce a carcinoid-like syndrome (bronchoconstriction, abdominal pain and explosive diarrhea, violent H/A, cutaneous flushing, hypertension, hepatomegaly and hyperglycemia).¹⁵ During surgical manipulation, these tumors may also release histamine and bradykinin, causing hypotension and bronchoconstriction.¹⁶

Clinical

Symptoms

Patients commonly present with hearing loss and pulsatile tinnitus. Dizziness is the third most common symptoms. Ear pain may also occur.

Signs

Hearing loss may be conductive (e.g. due to obstruction of the ear canal) or sensorineural due to invasion of the labyrinth often with accompanying vertigo (the eighth nerve is the most common cranial nerve involved). Various combinations of palsies of cranial nerves IX, X, XI & XII occur – see Jugular foramen syndromes (p.100) – with occasional VII palsy (usually from involvement within the temporal bone). Ataxia and/or hydrocephalus can occur with massive lesions that cause brain-stem compression. Occasionally patients may present with symptoms due to secretory products (see below).

Otoscopic exam → pulsatile reddish-blue mass behind eardrum (occasionally, lamentably biopsied by ENT physician with possible ensuing massive blood loss).

Differential diagnosis

See Cerebellopontine angle (CPA) lesions (p.1365). The major differential is neurilemmomas (vestibular schwannomas), both enhance on CT. A cystic component and extrinsic compression of the jugular bulb are characteristic of neurilemmomas. Angiography will differentiate difficult cases.

Evaluation

Neurophysiologic testing
Audiometric and vestibular testing should be performed.

Imaging
1. CT or MRI used to delineate location and extent of tumor; CT is better for assessing bony involvement of the skull base
2. angiography: confirms diagnosis (helping to rule out vestibular schwannoma), and ascertains patency of contralateral jugular vein in event that jugular on side of tumor must be sacrificed; jugular bulb and/or vein are usually partially or completely occluded

Endocrine/laboratory studies
See also details (p.653).

Classification

A number of classification schemes have been proposed. The modified Jackson classification is shown in □ Table 39.2.

Treatment

Surgical resection is usually simple and effective for small tumors confined to the middle ear. For larger tumors that invade and destroy bone, the relative role of surgery and/or radiation is not fully determined. With large tumors, surgery carries the risk of significant cranial nerve palsies.

Medical management
General information
For tumors that actively secrete catecholamines, medical therapy is useful for palliation or as adjunctive treatment before embolization or surgery. Alpha and beta blockers given before embolization or surgery blocks possibly lethal blood pressure lability and arrhythmias. Adequate blockade takes ≈ 2–3 weeks of alpha blocker and at least 24 hours of beta blocker therapy; in emergency, 3 days of treatment may suffice.

Alpha blockers
Reduce BP by preventing peripheral vasoconstriction.
1. phenoxybenzamine (Dibenzylin®): long acting; peak effect 1–2 hrs. Start with 10 mg PO BID and gradually increase to 40–100 mg per day divided BID
2. phentolamine (Regitine®): short acting. Usually used IV for hypertensive crisis during surgery or embolization.
□: 5 mg IV/IM (peds: 1 mg) 1–2 hrs pre-op, repeat PRN before and during surgery

Table 39.2 Modified Jackson classification ¹⁷		
Type	Description	Intracranial extension
I	small; involves jugular bulb, middle ear & mastoid	none
II	extends under IAC	possible
III	extends into petrous apex	possible
IV	extends beyond petrous apex into clivus or infratemporal fossa	possible

Beta blockers

Reduces catecholamine induced tachycardia and arrhythmias (may also prevent hypotension that might occur if only alpha blockade is used). These drugs are not always needed, but when used 6 NB: these drugs must not be started before starting alpha-blockers (to prevent hypertensive crisis and myocardial ischemia).

1. propranolol (Inderal®): □: oral dose is 5–10 mg q 6 hrs. IV dose for use during surgery is 0.5–2 mg slow IVP
2. labetalol (Normodyne®) (p.126): may have some efficacy in blocking $\alpha 1$ selective and β non-selective (potency <propranolol)

Serotonin, bradykinin, histamine release blockers

These agents may provoke bronchoconstriction that does not respond to steroids, but may respond to inhaled β -agonists or inhaled anticholinergics. Somatostatin may be used to inhibit release of serotonin, bradykinin, or histamines. Since this drug has a short half-life, it is preferable to give octreotide (p.742) 100 mcg sub-Q q 8 hrs.

Radiation therapy

XRT may relieve symptoms and stop growth in spite of persistence of tumor mass. 40–45 Gy in fractions of 2 Gy has been recommended.¹⁸ Lower doses of ≈ 35 Gy in 15 fractions of 2.35 Gy appear as effective and have fewer side effects.¹⁹ Generally used as primary treatment only for large tumors or in patients too elderly or infirmed to undergo surgery. Some surgeons pretreat 4–6 mos pre-operatively with XRT to decrease vascularity²⁰ (controversial).

Embolization

1. generally reserved for large tumors with favorable blood supply (i.e. vessels that can be selectively embolized with no danger of particles passing thru to normal brain)
2. post-embolization tumor swelling may compress brainstem or cerebellum
3. may be used preoperatively to reduce vascularity. Performed 24–48 hours pre-op (not used prior to that, because of post-embolization edema)
4. caution with actively secreting tumors which may release vasoactive substances (e.g. epinephrine) upon infarction from the embolization
5. may also be used as primary treatment ($\pm$ radiation) in patients who are not surgical candidates. In this case, is only palliative, as tumor will develop new blood supply
6. absorbable (Gelfoam®) and non-absorbable (Ivalon®) materials have been used

Surgical treatment

General information

The tumor is primarily extradural, with extremely vascular surrounding dura.

Suboccipital approach may cause dangerous bleeding and usually results in incomplete resection. Team approach by a neurosurgeon in conjunction with a neuro-otologist and possibly head and neck surgeon has been advocated.²¹ This approach utilizes an approach to the skull base through the neck.

ECA feeders are ligated early, followed rapidly by draining veins (to prevent systemic release of catecholamines).

Sacrifice of the jugular vein (JV) is tolerated if the contralateral JV is patent (often, the ipsilateral JV will already be occluded).

Surgical complications and outcome

The most common complications are CSF fistula, facial nerve palsy, and varying degrees of dysphagia (from dysfunction of lower cranial nerves). Dysfunction of any of the cranial nerves VII thru XII can occur, and a tracheostomy should be performed if there is any doubt of lower nerve function, and a gastrostomy feeding tube may be needed temporarily or permanently. Lower cranial nerve dysfunction also predisposes to aspiration, the risk of which is also increased by impaired gastric emptying and ileus that may occur due to reduced cholecystokinin (CCK) levels post-op. Excessive blood loss can also occur.

Even after gross total tumor removal, recurrence rate may be as high as one third.^{20,22}

39.3 Neuroblastomas

39.3.1 General information

Tumors arising from sympathetic ganglion.²³ May occur anywhere in the sympathetic nervous system, most commonly from adrenal gland (40%), followed by sympathetic ganglia of thoracic (15%), cervical (5%) and pelvic regions (5%). Neoplasms under this rubric include:

1. neuroblastomas: the most undifferentiated and aggressive in this group
2. ganglioneuroblastomas
3. ganglioneuromas

Note: Olfactory neuroblastomas are called esthesioneuroblastomas (p.1388).

39.3.2 Presentation

May present with abdominal mass, local or radicular pain, or (with high thoracic or cervical tumors) Horner's syndrome. Spinal cord compression may occur from invasion through the neural foramen, and scoliosis may occur. Catecholamine precursors (homovanillic acid (HVA), vanillylmandelic acid (VMA) and dopamine) may be excreted and cause HTN (can be assayed in urine). Periorbital tumor metastases may produce raccoon's eyes (usually unilateral ecchymosis and proptosis). Many of the low-grade tumors regress spontaneously and never present.

References

- [1] Courville CB. Ganglioglioma. Tumor of the Central Nervous System: Review of the Literature and Report of Two Cases. *Arch Neurol Psychiatry*. 1930; 24:439–491
- [2] Rubinstein LJ. Tumors of the Central Nervous System. *Atlas of Tumor Pathology, Second Series, Fascicle 6*. Washington, DC: Armed Forces Institute of Pathology; 1972
- [3] Demierre B, Stichnoth FA, Hori A, et al. Intracerebral Ganglioglioma. *J Neurosurg*. 1986; 65:177–182
- [4] Kalyan-Raman UP, Olivero WC. Ganglioglioma: A Correlative Clinicopathological and Radiological Study of Ten Surgically Treated Cases with Follow-Up. *Neurosurgery*. 1987; 20:428–433
- [5] Garcia CA, McGarry PA, Collada M. Ganglioglioma of the Brain Stem. Case Report. *J Neurosurg*. 1984; 60:431–434
- [6] Sutton LN, Packer RJ, Rorke LB, et al. Cerebral Gangliogliomas During Childhood. *Neurosurgery*. 1983; 13:124–128
- [7] Miller DC, Lang FF, Epstein FJ. Central Nervous System Gangliogliomas. Part 1: Pathology. *J Neurosurg*. 1993; 79:859–866
- [8] Lang FF, Epstein FJ, Ransohoff J, et al. Central Nervous System Gangliogliomas. Part 2: Clinical Outcome. *J Neurosurg*. 1993; 79:867–873
- [9] Russell DS, Rubenstein LJ. Ganglioglioma: A Case with a Long History and Malignant Evolution. *J Neuropathol Exp Neurol*. 1962; 21:185–193
- [10] van Nederveen FH, Gaal J, Favier J, Korpershoek E, Oldenburg RA, de Bruyn EM, Sleddens HF, Derckx P, Riviere J, Dannenberg H, Petri BJ, Komminoth P, Pacak K, Hop WC, Pollard PJ, Mannelli M, Bayley JP, Perren A, Niemann S, Verhofstad AA, de Bruine AP, Maher ER, Tissier F, Meatchi T, Badoual C, Bertherat J, Amar L, Alataki D, Van Marck E, Ferrau F, Francois J, de Herder WW, Peeters MP, van Linge A, Lenders JW, Gimenez-Roqueplo AP, de Krijger RR, Dinjens WN. An immunohistochemical procedure to detect patients with paraganglioma and pheochromocytoma with germline SDHB, SDHC, or SDHD gene mutations: a retrospective and prospective analysis. *Lancet Oncol*. 2009; 10:764–771
- [23] Brodeur GM, Pritchard J, Berthold F, et al. Revisions of the international criteria for neuroblastoma diagnosis, staging, and response to treatment. *J Clin Oncol*. 1993; 11:1466–1477
- [11] Kudva YC, Sawka AM, Young WF, Jr. Clinical review 164: The laboratory diagnosis of adrenal pheochromocytoma: the Mayo Clinic experience. *J Clin Endocrinol Metab*. 2003; 88:4533–4539
- [12] de Jong WH, Eisenhofer G, Post WJ, Muskiet FA, de Vries EG, Kema IP. Dietary influences on plasma and urinary metanephrines: implications for diagnosis of catecholamine-producing tumors. *J Clin Endocrinol Metab*. 2009; 94:2841–2849
- [13] Chretien PB, Engelman K, Hoyer RC, et al. Surgical Management of Intravascular Glomus Jugulare Tumor. *Am J Surg*. 1971; 122:740–743
- [14] Jackson CG, Harris PF, Glasscock MEI, et al. Diagnosis and Management of Paragangliomas of the Skull Base. *Am J Surg*. 1990; 159:389–393
- [15] Farrior JB, Hyams VJ, Benke RH, Farrior JB. Carcinoid Apudoma Arising in a Glomus Jugulare Tumor: Review of Endocrine Activity in Glomus Jugulare Tumors. *Laryngoscope*. 1980; 90:110–119
- [16] Jensen NF. Glomus Tumors of the Head and Neck: Anesthetic Considerations. *Anesth Analg*. 1994; 78:112–119
- [17] Jackson CG, Glasscock ME, Nissen AJ, et al. Glomus Tumor Surgery: The Approach, Results, and Problems. *Otolaryngol Clin North Am*. 1982; 15:897–916
- [18] Kim J-A, Elkon D, Lim M-L, Constable WC. Optimum Dose of Radiotherapy for Chemodectomas of the Middle Ear. *Int J Radiation Oncology Biol Phys*. 1980; 6:815–819
- [19] Cummings BJ, Beale FA, Garrett PG, Harwood AR, et al. The Treatment of Glomus Tumors in the Temporal Bone by Megavoltage Radiation. *Cancer*. 1984; 53:2635–2640
- [20] Spector GJ, Fierstein J, Ogura JH. A Comparison of Therapeutic Modalities of Glomus Tumors in the Temporal Bone. *Laryngoscope*. 1976; 86:690–696
- [21] Schmidek HH, Sweet WH. *Operative Neurosurgical Techniques*. New York 1982
- [22] Hatfield PM, James AE, Schulz MD. Chemodectomas of the Glomus Jugulare. *Cancer*. 1972; 30:1164–1168

40 Pineal Region and Embryonal Tumors

40.1 Pineal region tumors

40.1.1 General information

Key concepts

- wide variety of pathology: germ cell tumors (mostly germinomas, teratomas), astrocytomas,& pineal tumors (mostly pineoblastomas) account for most tumors
- since tumors may be of mixed cell types, CSF tumor markers (β -hCG, AFP...) are not as useful for diagnosis as they are for following response to treatment
- traditionally a test dose of XRT was employed, but there is a growing trend to obtain tissue diagnosis in all cases if possible before instituting treatment

Pineal region¹: the area of the brain bounded dorsally by the splenium of the corpus callosum and the tela choroidea, ventrally by the quadrigeminal plate and midbrain tectum, rostrally by the posterior aspect of the 3rd ventricle, and caudally by the cerebellar vermis.

A striking feature is the diversity of lesions (neoplastic and nonneoplastic) that may occur in this location due to the variety of tissues and conditions normally present, as shown in □ Table 40.1.

40.1.2 Pineal cysts (PCs)

General information

Usually an incidental finding (i.e. not symptomatic), seen on $\approx$ 4% of MRIs² or on 25–40% of autopsies³ (many are microscopic). The most common ones are intra-pineal glial-lined cysts with diameter < 1 cm. Etiology is obscure, PCs are nonneoplastic, and may be due to ischemic glial degeneration or due to sequestration of the pineal diverticulum. They have been regarded as benign, but

Table 40.1 Conditions giving rise to pineal region tumors	
Substrate in pineal region	Tumor that may arise
pineal glandular tissue	pineocytomas and pineoblastomas
glial cells	astrocytomas (including pilocytic), oligodendrogliomas, glial cysts (AKA pineal cyst)
arachnoid cells	meningiomas, arachnoid cysts (non-neoplastic). Meningiomas characteristically displace the internal cerebral vein inferiorly
ependymal lining	ependymomas
sympathetic nerves	chemodectomas
rests of germ cells	germ cell tumors: choriocarcinoma, germinoma, embryonal carcinoma, endodermal sinus tumor (yolk sac tumor), and teratoma
absence of blood-brain barrier (BBB) in pineal gland	makes it a susceptible site for hematogenous metastases
remnants of ectoderm	epidermoid or dermoid cysts
Non neoplastic lesions that may mimic tumors	
vascular	vein of Galen aneurysm (p. 1255), AVM
infectious	cysticercosis (p. 371)

the natural history is not known with certainty.⁴ PCs may contain clear, slightly xanthochromic, or hemorrhagic fluid. Rarely, they may enlarge, and like other pineal region masses, may become symptomatic by causing hydrocephalus by aqueductal compression,⁵ gaze paresis⁶ including Parinaud's syndrome (p.99), or hypothalamic symptoms.

Positional H/As have been attributed to PCs, the theory is that the cyst could intermittently compress the vein of Galen and/or sylvian aqueduct.⁷ This remains unproven since asymptomatic compression of the vein of Galen and the quadrigeminal plate has been demonstrated on MRI.⁸

Imaging

May escape detection on CT because the cyst fluid density is often similar to CSF. MRI T1WI shows round or ovoid abnormality in region of pineal recess, signal varies with protein content (isointense or slightly hyperintense). T2WI occasionally show increased intensity.⁴ Gadolinium occasionally enhances the cyst wall with a maximum thickness of 2 mm; irregularities of the wall with nodular enhancement suggests the lesion is not benign.

Epidermoid-dermoid cysts may also occur in the pineal region, and are larger and have different signal characteristics on MRI.

Management

Asymptomatic PCs <2 cm diameter with typical appearance should be followed clinically and with annual imaging studies. Surgery to relieve symptoms or to obtain a diagnosis is suggested for symptomatic lesions or for ones that show changes on MRI.

Surgery options for patients with hydrocephalus:

1. CSF shunt: may not relieve gaze disturbance (from pressure on tectal plate)
2. cyst excision: relieves symptoms and establishes diagnosis. Low morbidity
3. stereotactic or endoscopic aspiration: may not get enough tissue for diagnosis
4. endoscopic third ventriculostomy (ETV) (p.1517): useful only for typical PC as it does not obtain tissue for pathology. A few cases of regression of PCs after ETV have been reported⁹

40.1.3 Pineal region neoplasms

General information

Tumors in this region are more common in children (3–8% of pediatric brain tumors) than in adults ($\leq 1\%$).¹⁰ Over 17 tumor types occur in this region.¹¹ Germinoma is the most common tumor (21–44% in American/European population, 43–70% in Japan), followed by astrocytoma, teratoma and pineoblastoma.¹² Many tumors are of mixed cell type.

Germ cell tumors (GCT), ependymomas and pineal cell tumors metastasize easily through the CSF (“drop metastases”).

Pineal gland tumors

Pineal cell tumors

A pineocytoma (AKA pinealcytoma) is a well differentiated neoplasm arising from pineal epithelium. Pineoblastoma (AKA pinealblastoma) is a malignant tumor that is considered a primitive neuroectodermal tumor (PNET) (p.666). Both can metastasize through the CSF, and both are radiosensitive.

Germ cell tumors (GCT)

When they arise in the CNS, GCTs occur in the midline in the suprasellar and/or pineal region (simultaneous suprasellar and pineal region lesions is diagnostic of a GCT, so-called synchronous germ cell tumors, comprise 13% of GCTs, and are highly sensitive to XRT¹³). In the pineal region, these tumors occur predominantly in males. In females, GCTs are more common in the suprasellar region.¹⁴ Aside from benign teratomas, all intracranial GCTs are malignant and may metastasize via CSF and systemically. Types of GCTs:

1. germinomas: malignant tumors of primitive germ cells that occur in the gonads (called testicular seminomas in males, dysgerminomas in females) or in the CNS. Survival with these is much better than with nongerminomatous tumors
2. non-germinomatous germ cell tumors (NGGCT) include:
 - a) embryonal carcinoma
 - b) choriocarcinoma

Table 40.2 Occurrence of CSF tumor markers with pineal germ cell tumors ^a			
Tumor	β-hCG ^b	AFP	PLAP ^c
choriocarcinoma	≈ 100%	–	–
germinoma	10–50%	–	+
embryonal carcinoma	–	+	–
yolk sac carcinoma	–	+	–
immature teratoma	–	+	–
mature teratoma	–	–	–
^a adapted with permission from personal communication, Ashraf Samy Youssef, M.D., Ph.D. ^b abbreviations: β-hCG=beta human chorionic gonadotropin, AFP=alpha-fetoprotein, PLAP=placental alkaline phosphatase ^c elevated PLAP may also occur in serum			

- c) endodermal sinus tumor (EST) AKA yolk sac carcinoma: usually malignant
- d) teratoma
 - mature
 - immature

Tumor markers

GCTs characteristically (but not always) give rise to tumor markers in the CSF; see Tumor markers used clinically (p.600). Elevated CSF beta-human chorionic gonadotropin (β-hCG) is classically associated with choriocarcinomas, but also occurs with up to 50%of germinomas (which are more common). Alpha-fetoprotein (AFP) is elevated with endodermal sinus tumors, embryonal carcinoma and occasionally with teratomas. Elevated placental alkaline phosphatase (PLAP) in serum or CSF occurs with intracranial germinomas.¹⁵ □ Table 40.2 summarizes these findings. When positive, tumor markers can be followed serially to assess treatment and to look for recurrence (they should be checked in serum and CSF). NB: tumor markers alone are not usually sufficient for making a definitive diagnosis of a pineal region tumor since many of these tumors are mixed cell type.

Pediatric

A breakdown of pediatric pineal region tumors in one series is shown in □ Table 40.3 (series A). In 36 patients <18 yrs age, 17 distinct histological tumor types were identified: 11 germinomas (the most common tumor), 7 astrocytomas, and the remaining 18 had 15 different tumors.¹⁷

Adult

GCTs and pineal cell tumors occur primarily in childhood and young adults. Thus, over the age of 40, a pineal region tumor is more likely to be a meningioma or a glioma. Series B in □ Table 40.3 includes both adult and pediatric patients.

Clinical

Almost all patients have hydrocephalus by the time of presentation, causing typical signs and symptoms of headache, vomiting, lethargy, memory disturbance, abnormally increasing head circumference in infants, and seizures. Parinaud’s syndrome (p.99), or the syndrome of the sylvian aqueduct, may be present. Precocious puberty may occur only in boys with choriocarcinomas or germinomas with syncytiotrophoblastic cells, due to luteinizing hormone-like effects of β-hCG secreted in the CSF. Suprasellar GCT: triad of diabetes insipidus, visual deficit and panhypopituitarism.¹⁴
Drop metastases from CSF seeding can produce radiculopathy and/or myelopathy.

Management

The optimal management strategy for pineal region tumors has yet to be determined.
“ Test dose” radiation: Controversial (p.662). This is giving way to the doctrine of obtaining histology in most cases (e.g. by stereotactic biopsy) because of the harmful effects of XRT and because 36–50% of pineal tumors are benign or radioresistant.¹⁸ The concept was that if a pineal region

Table 40.3 Pineal region tumors		
Tumor	Series A ^a (%)	Series B ^b (%)
germinoma	30	27
astrocytoma	19	26
pineocytoma	6	12
malignant teratoma	6	
unidentified germ-cell tumor	6	
choriocarcinoma	3	1.1
malignant teratoma/embryonal cell tumor	3	1.6
glioblastoma	3	
teratoma	3	4.3
germinoma/ectodermal sinus tumor	3	
dermoid	3	
embryonal cell tumor	3	
pineoblastoma	3	12
pineocytoma/pineoblastoma	3	
endodermal sinus tumor	3	
glial cyst (pineal cyst) ¹⁶	3	2.7
arachnoid cyst	3	
metastases		2.7
meningioma		2.7
ependymoma		4.3
oligodendrogliomas		0.54
ganglioglioneuroma		2.7
lymphoma		2.7
^a 36 children ≤ 18 yrs ¹⁷		
^b 370 tumors in patients 3–73 yrs old ¹⁰		

tumor enhanced uniformly and had the classic appearance of a germinoma on MRI, a test dose of 5 Gy was given, and if the tumor would shrink then the diagnosis of germinoma was virtually certain and XRT was continued without surgery. This may needlessly expose a patient with benign or radio-resistant tumors to XRT.¹⁰ “Trial XRT” should be avoided in tumors suspected of being teratomas or epidermoid cyst on MRI, and the response may be misleading in the relatively common situation of tumors with mixed cell types.

- Management suggestions
1. get MRI of cervical, thoracic and lumbar spine to assess for drop mets
 2. send for GCT markers (β-hCG, AFP, PLAP) (p.660). Somewhat helpful, but not adequate for diagnosis: if negative for GCT markers, it may be a pineal cell tumor, or it may be a GCT without markers, see Tumor markers (p.660); if positive, it can still be a mixed cell-type tumor:
 - a) serum
 - b) CSF (if able to safely obtain; LP is contraindicated with large intracranial mass and/or obstructive hydrocephalus; CSF may be obtained from EVD if placed)

3. obtain histology in most cases. Most often this involves a biopsy, which should be generous (to avoid missing other histologies in mixed cell tumors)
 - a) if hydrocephalus: transventricular biopsy
 - b) if no hydrocephalus:
 - open biopsy or
 - stereotactic biopsy or
 - ? by CACE (see below)
4. based on markers and histology:
 - a) germinoma: XRT+chemo
 - b) all other tumors: one option is resection followed by adjuvant therapy (usually not very helpful) – see Indications (p.662) for controversies

Hydrocephalus

Patients presenting acutely due to hydrocephalus may be best treated with external ventricular drainage (EVD). This permits control over the amount of CSF drained, prevents peritoneal seeding with tumor (a rare event¹⁹), and may avoid having a permanent shunt placed in the significant number of patients who will not need one after tumor removal (although $\approx 90\%$ of patients with a pineal GCT require a shunt). Ventricular access, via EVD or Frazier burr hole (p.1450), in the post-op period is important in the event of acute hydrocephalus.

Stereotactic procedures

May be used to ascertain diagnosis (biopsy), or to treat symptomatic pineal region cysts.^{20,21} Caution is advised since the pineal region has numerous vessels (vein of Galen, basal veins of Rosenthal, internal cerebral veins, posterior medial choroidal artery)²² which may be displaced from their normal position. The complication rate of stereotactic biopsy is: $\approx 1.3\%$ mortality, $\approx 7\%$ morbidity, and 1 case of seeding in 370 patients, and the diagnostic rate is $\approx 94\%$ ¹⁰ A shortcoming of stereotactic biopsy is that it may fail to disclose the histologic heterogeneity of some tumors.

Two main stereotactic trajectories: 1) anterolateral (low frontal) approach below the internal cerebral veins, and 2) posterolateral trans-parieto-occipital.¹¹ One study found that the trajectory correlated with complications, and they recommended the anterolateral approach.²³ However, the correlation of trajectory and complications was not born out in another study,¹⁰ and they found that the complication rate was higher in firm tumors (pineocytomas, teratomas, and astrocytomas) and they recommend an open approach when the tumor appears difficult to penetrate on the first attempt at biopsy.

Stereotactic radiosurgery may be appropriate for treatment of some lesions.

Computer-assisted cisternal endoscopic approach (CACE)

Employs a supracerebellar infratentorial approach that permits visualization of neurovascular structures and avoids traversing brain parenchyma.¹¹

Radiation treatment

For controversies regarding “test dose” XRT, see Management (p.660). Germinomas are very sensitive to radiation (and chemotherapy), and are probably best treated with these modalities and followed.

XRT is also utilized post-op for other malignant tumors. For highly malignant tumors or if there is evidence of CSF seeding, craniospinal XRT with a boost to the tumor bed is appropriate.

If possible, XRT is best avoided in the young child. Chemotherapy may be used for age < 3 yrs until the child is older when XRT is better tolerated.¹⁴

Surgical treatment of the tumor

Indications

Controversial. Some authors feel that most tumors (except germinomas, which are best treated with XRT) are amenable to open resection.²⁴ Others feel that resection should be limited to $\approx 25\%$ of tumors which are¹⁰:

1. radioresistant (e.g. malignant nongerminoma GCTs): 35–50% of pineal region tumors (larger numbers occur in series not limited to pediatric patients)
2. benign (e.g. meningioma, teratomas...)
3. well encapsulated
4. NB: malignant germ cell tumors should be without evidence of metastases (those with metastases do not benefit from surgery on the primary tumor)
5. pineocytoma: recommendation is for surgical excision + SRS for any residual

Options

1. direct surgery: obtains generous tissue for biopsy. Curative for benign lesions. Not the optimum treatment for malignant tumors and germinomas without complications

2. biopsy followed by adjuvant therapy: the preferred management (p.660) for malignancies and germinomatous germ cell tumors

Surgical approaches

Choice is aided by the pre-op MRI and includes:

1. most common approach: midline infratentorial-supracerebellar approach of Horsley and Krause as refined by Stein.²⁵ Cannot be used if the angle of the tentorium is too steep (best assessed on MRI). May be done in the sitting position – risk of air embolism (p.1445) – or in the Concorde position (p.1445)
2. occipital transtentorial: wide view. Risk of injury to occipital visual cortex or splenium of corpus callosum. Recommended for lesions centered at or superior to the tentorial edge or located above the vein of Galen or for rare cysts with superior extension. The occipital lobe is retracted laterally, and the tentorium is incised 1 cm lateral to the straight sinus
3. transventricular: indicated for large, eccentric lesions with ventricular dilatation. Usually via a cortical incision in the posterior portion of the superior temporal gyrus. Risks: visual defect, seizures, and on dominant side language dysfunction
4. lateral paramedian infratentorial
5. transcallosal: largely abandoned except for tumors extending into corpus callosum and third ventricle
6. paramedian infratentorial-supracerebellar approach may be used for cysts that do not extend superiorly or contralaterally⁴: avoids midline venous structures

Important surgical considerations:

The base of the pineal gland is the posterior wall of the 3rd ventricle. The splenium of the corpus callosum lies above, and the thalamus surrounds both sides. The pineal projects posteriorly and inferiorly into the quadrigeminal cistern. The deep cerebral veins are a major obstacle to operations in this region. Venous drainage of the pineal region must be preserved.

Surgical outcome

Mortality rate: 5–10%¹⁰ Postoperative complications include: new visual field deficits, epidural fluid collection, infection, and cerebellar ataxia.

40.2 Embryonal tumors

40.2.1 General information

A few words about PNETs

Initially, the term primitive neuroectodermal tumor (PNET) encompassed a wide variety of previously individually named tumors which all seemed to share certain pathologic features suggesting origin from a common progenitor cell in the subependymal matrix (primitive neuroectodermal cells) (although the actual cell of origin is unknown). They are histologically indistinguishable but genetically distinct.²⁶ Now, the recommendation is to call these “embryonal tumors”,²⁷ but the term PNET is entrenched. These tumors include: retinoblastoma, pineoblastoma, neuroblastoma, esthesioneuroblastoma. Medulloblastoma (MB) is more than just a PNET of the posterior fossa (see below), as alterations involved in evolution of MBs such as beta-catenin and APC mutations are absent in pineoblastomas and supratentorial PNETs (sPNETs). At least some MBs originate from the external granular layer (EGL) of the cerebellum.

Embryonal tumors

Location: Embryonal tumors most commonly arise in the cerebellar vermis (medulloblastoma), but also occur in cerebrum, pineal, brainstem or spinal cord. Primary spinal cord PNETs are extremely rare (approximately 30 cases reported by 2007²⁸). sPNETs have a worse prognosis than MB (see below).

Dissemination: Embryonal tumors (ETs) may disseminate via the CSF spontaneously,²⁹ or iatrogenically (following surgery or shunting, the latter is a rare cause of tumor dissemination¹⁹). Thus, all patients with ETs require spinal axis evaluation (gadolinium enhanced MRI is about as sensitive as water-soluble myelography) and cytologic examination of CSF. Prophylactic craniospinal XRT is indicated following surgical removal, but cranial XRT is avoided if at all possible before 3 years of age to avoid intellectual impairment and growth retardation; see Radiation injury and necrosis (p.1560). Extraneural metastases can also occur.

Collin’s law: AKA period of risk of recurrence (PRR) is often applied to children who have been treated for embryonal tumors (especially medulloblastoma) but may also be used with any tumor thought to arise from a gestational event. It states that PRR is equal to the age at diagnosis plus 9 months.³⁰ Patients that remain free of recurrence beyond the PRR have a much lower risk of recurrence, however recurrence beyond this time has been reported in a small number ($\approx 1.4\%$) of cases,³¹ and other tumors may occur e.g. as a result of induction by XRT used to treat the initial tumor.

40.2.2 Medulloblastoma (MB)

General information

Key concepts

- a small-cell embryonal tumor of the cerebellum found predominantly in children (peak: 1st decade). The most common pediatric brain malignancy
- usually arises in the cerebellar vermis in the region of the apex of the roof of the 4th ventricle (fastigium), often producing hydrocephalus
- brainstem invasion usually limits complete surgical excision
- all patients must be evaluated for “drop mets”

Epidemiology

In children: MBs comprise 15–20% of intracranial tumors,³² 30–55% of p-fossa tumors. MB is the most common malignant pediatric brain tumor.³³ MBs comprise $<1\%$ of adult brain neoplasms. Peak incidence: during 1st decade. Median age at diagnosis: 5–7 years (75% are diagnosed by age 15). Male:female ratio is 2:1. Familial cancer syndromes that include MB: Gorlin syndrome, Turcot syndrome (p.610).

Pathology

All MB are WHO grade IV.³⁴
Histologic subtypes³⁴:

1. classic (90%): small, densely packed undifferentiated cells with hyperchromatic nuclei, scant cytoplasm (and inconstant cell clusters in Homer-Wright rosettes)³⁵ (sometimes called “blue tumor”) (monotonous appearance)
2. desmoplastic (6%): similar to classic type with “glomeruli” AKA pale islands (collagen bundles and scattered, less cellular areas). Marked tendency for neuronal differentiation. More common in adults. Prognosis controversial: may be the same³⁶ or less aggressive³⁷ than classic MB
3. large cell (4%³⁸): large, round, and/or pleomorphic nucleoli, higher mitotic activity. In the few case reports, all were male. More aggressive than classic. Resembles atypical teratoid/rhabdoid tumors of cerebellum, but has different phenotype and cytogenic features

Molecular biology

The molecular genetic alterations in MBs can be divided into 3 groups:

1. non-random chromosomal abnormalities: (e.g. consistent deletion of 17p markers) has been shown in 35–40%
2. information from gene profiling:
 - a) ZIC and NSCL1 were the genes most closely correlated with MBs
 - b) certain genes were associated with more favorable outcome (using 8 genes a pattern associated with 80% 5-year survival compared to 17% when the pattern was lacking)²⁶
3. abnormalities in signal transduction pathways: e.g. neurotrophin signaling pathway (important in cerebellar development) or Sonic hedgehog (Shh)³⁹

Seeding and metastases

$\approx 10\text{--}35\%$ have seeded the craniospinal axis at the time of diagnosis,³² and extraneural mets occur in 5% of patients,³³ sometimes promoted by shunting⁴⁰ (although this is uncommon¹⁹).

Clinical

Clinical history is typically brief (6–12 weeks). MBs usually arise in the cerebellar vermis, at the apex of the roof of the 4th ventricle (fastigium in the region of the posterior medullary velum), which predisposes to early obstructive hydrocephalus. Usual presenting symptoms: H/A, N/V, and truncal & appendicular ataxia. Infants with hydrocephalus may present with irritability, lethargy, or progressive macrocrania.⁴¹ Spinal drop mets may produce back pain, urinary retention or leg weakness. Common signs: papilledema, ataxia, nystagmus, EOM palsies.

Evaluation

Usually appears as a solid, IV-contrast-enhancing lesion on CT or MRI (however, a rare diffuse variant in children <3 yrs, medulloblastoma with extensive nodularity⁴² (MBEN), has been described). Most are located in the midline in the region of the 4th ventricle (laterally situated tumors are more common in adults). Most have hydrocephalus. Ependymoma (p.1365) is the main entity to differentiate from on imaging.

CT: noncontrast → typically hyperdense (due to high cellularity), contrast → most enhance. 20% have calcifications.

MRI: T1WI → hypo- to isointense. T2WI → heterogeneous due to tumor cysts, vessels and calcifications.³⁵ Most enhance (including MBEN)

Spinal imaging: MRI with IV gadolinium or CT/myelography with water-soluble contrast should be done to rule-out “drop mets.” Staging is done either pre-op or within 2–3 weeks of surgery.

Treatment

Stratification of patients into risk groups guides therapy (□ Table 40.4).

MB are highly radiosensitive and moderately chemosensitive.

Treatment of choice: surgical debulking of as much tumor as possible (without causing neurological injury) followed by craniospinal XRT (radiation is necessary because of propensity to recur and to seed). Invasion of or attachment to the floor of the fourth ventricle (brainstem in the region of the facial colliculus) often limits excision. It is better to leave a small residual on the brain-stem (these patients do fairly well) than it is to chase every last remnant into the brain-stem (neurologic deficit is more likely with this).

Surgical exposure of midline cerebellar medulloblastomas requires opening of the foramen magnum, usually removal of the posterior arch of C1, and occasionally the arch of C2. Tumor spread with arachnoidal thickening (“sugar coating”) may occur.

XRT: optimal irradiation dose: 35–40 Gy to whole craniospinal axis + 10–15 Gy boost to tumor bed (usually posterior-fossa) and to any spinal mets seen, all fractionated over 6–7 wks.^{43,44} Reduce dosages by 20–25% for age <3 yrs, or use chemotherapy instead. Lower dose radiation (25 Gy) to the neuraxis may provide acceptable control when confirmed gross total excision is achieved.⁴⁵

Chemotherapy: there is no standardized chemotherapy regimen. Lomustine (CCNU), cisplatin and vincristine (VCR) are primarily used, but are usually reserved for recurrence, for poor risk patients (below), or for children <3 yrs age. Significant survival advantage was shown in poor-risk children with adjuvant chemotherapy (5-year actuarial disease-free survival rate = 87%) compared to those without (33%). No difference was observed among standard-risk patients.⁴⁶

Shunts: 30–40% of children require permanent VP shunts following p-fossa resection. The risk of shunt-related seeding has been quoted as high as 10–20%³² but this is probably overestimated.¹⁹ In the past, tumor filters were frequently used. They are less commonly used today because of the high incidence of obstruction.

Table 40.4 Risk stratification in medulloblastoma
Standard-risk patients
No residual tumor on post-op MRI and negative CSF results. 5-year survival is >5% and progression-free survival = 50% ^{48,49}
Poor-risk patients
Bulky residual tumor >1.5 cm ² post-op and dissemination in the brain, spine or CSF. Worse prognosis. 5-year disease-free survival is 35–50% ⁵⁰
Intermediate risk patients
An intermediate risk group probably exists, but has been poorly characterized

Prognosis

Poor prognosticators⁴⁷

- younger age (especially if <3 yrs)
- disseminated (metastatic) disease
- inability to perform gross-total removal (especially if residual > 1.5 cm² in patient with localized disease)
- histological differentiation along glial, ependymal, or neuronal lines

One stratification scheme is shown in □ Table 40.4.

The sex of the child is an important predictor for survival of MB; girls had a much better outcome.⁵¹ Gene expression profiling is highly predictive of response to therapy, predicting outcome with much greater accuracy than current staging criteria.²⁶ The ability of multiple biological and clinical markers to predict outcomes for patients with MB is currently under investigation.^{52,53}

Long-term survivors of MB are at significant risk for permanent endocrinologic, cognitive, and psychological sequelae of treatments. Infants and very young children with MB remain a difficult therapeutic challenge because they have the most virulent form of the disease and are at highest risk for treatment-related sequelae.

Most common site of recurrence is p-fossa.

Collins' law (p.664) has also been used to define the period of risk of recurrence (PRR) but exceptions to the law have been reported.³¹

40.2.3 Supratentorial primitive neuroectodermal tumors

General information

Supratentorial primitive neuroectodermal tumor (sPNETs) are highly malignant lesions primarily affecting young children (65% occur in age <5 years) and account for 2.5–6% of childhood brain tumors. Occur rarely in adults. No gender predilection. Histologically indistinguishable from medulloblastoma (MB), they have a distinct genetic profile, are more aggressive, and often respond poorly to MB-specific therapies (especially pineoblastomas). Overall survival rate for sPNETs is substantially lower than that for MBs, with an expected 3- year progression-free survival of approximately 50% for localized supratentorial PNETs.^{54,55}

Ependymoblastoma

A highly cellular embryonal form of ependymal tumor.⁵⁶ Occurs most often in age <5 yrs. Prognosis is poor, with median post-op survival ranging from 12–20 months, and almost 100% mortality rate at 3 yrs. As with other tumors in this category, there is a tendency for subarachnoid seeding.

40.2.4 Atypical teratoid/rhabdoid tumors (AT/RT)

A unique embryonal tumor of the CNS. Many of these tumors were probably previously misdiagnosed as MBs. Occurs primarily in infants and children (>90% are <5 years of age, with most age <2 years). A minority are associated with primary renal rhabdoid tumor. 50% of AT/RTs occur in posterior fossa with a predilection for the cerebellopontine angle (CPA).

33% have CSF spread at presentation. Most patients die within 1 year of diagnosis.

Histopathology: some tumors are composed entirely of rhabdoid cells, others have a combination of rhabdoid and areas resembling PNET/MB. Other cell types include: malignant mesenchymal cells (usually spindle cells), malignant epithelial cells (glandular or squamous).

Molecular biology: AT/RT and the rhabdoid renal tumors have a deletion or monosomy of chromosome 22.

40.2.5 Esthesioneuroblastoma

General information

Esthesioneuroblastoma (ENB), originally described in 1924, AKA olfactory neuroblastoma, AKA olfactory esthesioneuroblastoma, AKA esthesioneurocytoma, AKA olfactory placode tumor.⁵⁷ A rare nasal neoplasm with an incidence 0.4 per 1,000,000 people.⁵⁸ Believed to arise from the olfactory neural crest cells in the upper nares, it is considered to be malignant. These tumors occur over a wide age range (3 to 90 years), with a bimodal peak between the second and third decade and a second peak in the sixth and seventh decades.

Imaging

MRI: isointense with brain on T1-weighted imaging and intermediate to high signal intensity on T2-weighted imaging and enhance heterogeneously with gadolinium. Signal characteristics may mimic meningioma. For higher stage lesions, the cribiform plate may be eroded, better seen on thin cut CT. The most important factor determining resectability is intracranial extension. Magnetic resonance aids in the distinction between extradural tumor, dural invasion, or parenchymal brain invasion. None of these are specific to this tumor.

Differential diagnosis

Includes SNUC, nasal melanoma, nasal squamous cell carcinoma, and meningioma.

Diagnosis

Endoscopic biopsy is typically performed in the otolaryngology office prior to surgery. A clinical oncology exam should be performed, and if there is suspicion for metastatic disease a PET scan should be ordered which is sensitive for metastatic disease.

Clinical classification systems

The modified Kadish system⁵⁹ (which added category D to the original Kadish system⁶⁰) is shown in □ Table 40.5. This classification appears to correlate with survival.⁵⁹ Alternative systems by Biller et al.⁶¹ and Dulguerov and Calcaterra,⁶² (see □ Table 40.5) attempt to subdivide the Kadish C classification, however the more popular modified Kadish system is more frequently used.

Pathologic grading

Hyams grading, a system used to define all upper respiratory tract carcinomas is utilized which assess nuclear pleomorphism, mitotic activity, rosette presence, necrosis, and summates these to produce Hyams 1–4 classification.⁶³ It has been shown in meta-analysis as well as large series that Hyams grade 1 and 2 predict benign disease course, as compared to Hyams 3 and 4 which predict poor disease course. It is recommended grading be performed in all cases.^{64,65}

Treatment

Primary Treatment is controversial. Some institutions believe in upfront combined radiation therapy and chemotherapy prior to craniofacial resection. However most practice upfront surgery, which classically consisted of endoscopic resection with negative margins for Kadish A and B lesions, and for Kadish C and D lesions craniofacial resection which was bifrontal craniotomy with associated lateral rhinotomy. However, with the advent of endoscopic techniques the lateral rhinotomy is often replaced with a purely endoscopic approach unless there is inferior lateral orbital or maxillary involvement, in which case the lateral rhinotomy is frequently used. Finally, some institutions are now managing Kadish stages purely endoscopically unless they are unable to get negative margins at the time of surgery, then conversion to an open approach is performed or SRS is performed, however this is controversial.

Outcome

Median overall survival is typically 7.2 ± 0.7 years.⁶⁴

Table 40.5 Clinical classification systems for esthesioneuroblastoma		
Modified Kadish ⁵⁹	Biller et al. ⁶¹	Dulguerov and Calcaterra ⁶²
A: Confined to Nasal Cavity	T1: Nasal/Paranasal Sinuses	T1: Nasal/Paranasal Sinuses
B: Extends to Paranasal Sinus	T2: Periorbital/Anterior Fossa Extension	T2: Erosion of Cribiform Plate
C: Local Extension (orbit or cribriforms plate)	T3: Brain Involvement, Resectable Margins	T3: Periorbital/Anterior Fossa Extension
D: Distant Metastasis	T4: Unable to Obtain Negative Margins- Unresectable	T4: Brain Involvement

Mean progression free survival is 4.8 ± 0.7 years. The 5 and 10 year survivals are 63% and 40%⁶⁴

Population based analysis of the Surveillance Epidemiology and End Results (SEER) database confirm that Kadish staging, lymph node involvement, and age at diagnosis have significant prognostic value.⁶⁶ These findings have been confirmed in a large meta-analysis recently published by Kane et al., in 2010⁶⁵ Further, higher Hyams grading (grades 3 and 4) correlate with a poorer prognosis.^{64,65}

Salvage treatment: For patients with recurrent disease, this typically occurs in 2 patterns: that of intracranial recurrence or those with distant metastasis.^{67,68} Intracranial recurrence is typically treated with repeat transcranial resection, however stereotactic radiosurgery is a viable option.^{67,68,69} In patients with distant metastasis, those with cervical lymph node metastasis should undergo modified radical neck dissection to understand the extent of disease. This typically leads to chemotherapy, of which platinum based therapies remain the standard of therapy at this time.^{67,70,71}

References

[1] Ringertz N, Nordenstam H, Flyger G. Tumors of the Pineal Region. *J Neuropathol Exp Neurol.* 1954; 13:540–561

[2] Di Costanzo A, Tedeschi G, Di Salle F, Golia F, Morrone R, Bonavita V. Pineal Cysts: An Incidental MRI Finding? *J Neurol Neurosurg Psychiatry.* 1993; 56:207–208

[3] Hasegawa A, Ohtsubo K, Mori W. Pineal Gland in Old Age: Quantitative and Qualitative Morphological Study of 168 Human Autopsy Cases. *Brain Res.* 1987; 409:343–349

[4] Torres A, Krisht AF, Akouri S. Current Management of Pineal Cysts. *Contemp Neurosurg.* 2005; 27:1–5

[5] Maurer PK, Ecklund J, Parisi JE, Ondra S. Symptomatic Pineal Cysts: Case Report. *Neurosurgery.* 1990; 27:451–454

[6] Wisoff JH, Epstein F. Surgical Management of Symptomatic Pineal Cysts. *J Neurosurg.* 1992; 77:896–900

[7] Klein P, Rubinstein LJ. Benign Symptomatic Glial Cysts of The Pineal Gland: A Report of Seven Cases and Review of the Literature. *J Neurol Neurosurg Psychiatry.* 1989; 52:991–995

[8] Mamourian AC, Towfighi J. Pineal Cysts: MR Imaging. *AJNR.* 1986; 7:1081–1086

[9] Di Chirico A, Di Rocco F, Velardi F. Spontaneous regression of a symptomatic pineal cyst after endoscopic third-ventriculostomy. *Childs Nerv Syst.* 2001; 17:42–46

[10] Regis J, Bouillot P, Rouby-Volot F, et al. Pineal Region Tumors and the Role of Stereotactic Biopsy: Review of the Mortality, Morbidity, and Diagnostic Rates in 370 Cases. *Neurosurgery.* 1996; 39:907–914

[11] Youssef AS, Keller JT, van Loveren HR. Novel application of computer-assisted cisternal endoscopy for the biopsy of pineal region tumors: cadaveric study. *Acta Neurochir (Wien).* 2007; 149:399–406

[12] Oi S, Matsumoto S. Controversy pertaining to therapeutic modalities for tumors of the pineal region: a worldwide survey of different patient populations. *Childs Nerv Syst.* 1992; 8:332–336

[13] Sugiyama K, Uozumi T, Kiya K, Mukada K, Arita K, Kurisu K, Hotta T, Ogasawara H, Sumida M. Intracranial germ-cell tumor with synchronous lesions in the pineal and suprasellar regions: report of six cases and review of the literature. *Surg Neurol.* 1992; 38:114–120

[14] Hoffman HJ, Ostubo H, Hendrick EB, et al. Intracranial Germ-Cell Tumors in Children. *J Neurosurg.* 1991; 74:545–551

[15] Shinoda J, Yamada H, Sakai N, Ando T, Hirata T, Miwa Y. Placental alkaline phosphatase as a tumor marker for primary intracranial germinoma. *J Neurosurg.* 1988; 68:710–720

[16] Todo T, Kondo T, Shinoura N, Yamada R. Large Cysts of the Pineal Gland: Report of Two Cases. *Neurosurgery.* 1991; 29:101–106

[17] Edwards MSB, Hudgins RJ, Wilson CB, et al. Pineal Region Tumors in Children. *J Neurosurg.* 1988; 68:689–697

[18] Oi S, Matsuzawa K, Choi JU, Kim DS, Kang JK, Cho BK. Identical characteristics of the patient populations with pineal region tumors in Japan and in Korea and therapeutic modalities. *Childs Nerv Syst.* 1998; 14:36–40

[19] Berger MS, Baumeister B, Geyer JR, Milstein J, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. *J Neurosurg.* 1991; 74:872–877

[20] Stern JD, Ross DA. Stereotactic Management of Benign Pineal Region Cysts: Report of Two Cases. *Neurosurgery.* 1993; 32:310–314

[21] Musolino A, Cambria S, Rizzo G, Cambria M. Symptomatic Cysts of the Pineal Gland: Stereotactic Diagnosis and Treatment of Two Cases and Review of the Literature. *Neurosurgery.* 1993; 32:315–321

[22] Kelly PJ. Comment on Musolino A, et al.: Symptomatic Cysts of the Pineal Gland: Stereotactic Diagnosis and Treatment of Two Cases and Review of the Literature. *Neurosurgery.* 1993; 32:320–321

[23] Dempsey PK, Kondziolka D, Lunsford LD. Stereotactic Diagnosis and Treatment of Pineal Region Tumors and Vascular Malformations. *Acta Neurochir.* 1992; 116:14–22

[24] Kelly PJ. Comment on Regis J, et al.: Pineal Region Tumors and the Role of Stereotactic Biopsy: Review of the Mortality, Morbidity, and Diagnostic Rates in 370 Cases. *Neurosurgery.* 1996; 39:912–913

[25] Stein BM. The infratentorial supracerebellar approach to pineal lesions. *J Neurosurg.* 1971; 35:197–202

[26] Pomeroy SL, Tamayo P, Gaasenbeek M, Sturla LM, Angelo M, McLaughlin ME, Kim JY, Goumnerova LC, Black PM, Lau C, Allen JC, Zagzag D, Olson JM, Curran T, Wetmore C, Biegel JA, Poggio T, Mukherjee S, Rifkin R, Califano A, Stolovitzky G, Louis DN, Mesirov JP, Lander ES, Golub TR. Prediction of central nervous system embryonal tumour outcome based on gene expression. *Nature.* 2002; 415:436–442

[27] Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. WHO classification of tumors of the central nervous system. Lyon 2007

[28] Kumar R, Reddy SJ, Wani AA, Pal L. Primary spinal primitive neuroectodermal tumor: case series and review of the literature. *Pediatr Neurosurg.* 2007; 43:1–6

[29] Tomita T, McLone DG. Spontaneous Seeding of Medulloblastoma: Results of Cerebrospinal Fluid Cytology and Arachnoid Biopsy from the Cisterna Magna. *Neurosurgery.* 1983; 12:265–267

[30] Collins VP, Loeffler RK, Tivey H. Observations on growth rates of human tumors. *Am J Roentgenol Radium Ther Nucl Med.* 1956; 76:988–1000

[31] Sure U, Berghorn WJ, Bertalanffy H. Collins' law. Prediction of recurrence or cure in childhood medulloblastoma? *Clin Neurol Neurosurg.* 1997; 99:113–116

[32] Laurent JP, Cheek WR. Brain Tumors in Children. *J Pediatr Neurosci.* 1985; 1:15–32

[33] Allen JC. Childhood Brain Tumors: Current Status of Clinical Trials in Newly Diagnosed and Recurrent Disease. *Ped Clin N Am.* 1985; 32:633–651

[34] Eberhart CG, Kepner JL, Goldthwaite PT, Kun IE, Duffner PK, Friedman HS, Strother DR, Burger PC. Histopathologic grading of medulloblastomas: a

- Pediatric Oncology Group study. *Cancer*. 2002; 94:552–560
- [35] Blaser SI, Harwood-Nash DC. Neuroradiology of pediatric posterior fossa medulloblastoma. *J Neurooncol*. 1996; 29:23–34
- [36] Pramanik P, Sharma MC, Mukhopadhyay P, Singh VP, Sarkar C. A comparative study of classical vs. desmoplastic medulloblastomas. *Neurol India*. 2003; 51:27–34
- [37] Kleihues P, Louis DN, Scheithauer BW, Rorke LB, Reifenberger G, Burger PC, Cavenee WK. The WHO classification of tumors of the nervous system. *J Neuropathol Exp Neurol*. 2002; 61:215–25; discussion 226–9
- [38] Giangaspero F, Rigobello L, Badiali M, et al. Large-cell medulloblastoma. *Am J Surg Pathol*. 1992; 16:687–693
- [39] Corcoran RB, Scott MP. Oxysterols stimulate Sonic hedgehog signal transduction and proliferation of medulloblastoma cells. *Proc Natl Acad Sci U S A*. 2006; 103:8408–8413
- [40] Kessler LA, Dugan P, Concannon JP. Systemic Metastases of Medulloblastoma Promoted by Shunting. *Surg Neurol*. 1975; 3:147–152
- [41] Park TS, Hoffman HJ, Hendrick EB, et al. Medulloblastoma: Clinical Presentation and Management. *J Neurosurg*. 1983; 58:543–552
- [42] Suresh TN, Santosh V, Yasha TC, Anandh B, Mohanty A, Indiradevi B, Sampath S, Shankar SK. Medulloblastoma with extensive nodularity: a variant occurring in the very young-clinicopathological and immunohistochemical study of four cases. *Childs Nerv Syst*. 2004; 20:55–60
- [43] Merchant TE, Wang MH, Haida T, Lindsley KL, Finlay J, Dunkel IJ, Rosenblum MK, Leibel SA. Medulloblastoma: long-term results for patients treated with definitive radiation therapy during the computed tomography era. *Int J Radiat Oncol Biol Phys*. 1996; 36:29–35
- [44] Packer RJ, Gajjar A, Vezina G, Rorke-Adams L, Burger PC, Robertson PL, Bayer L, LaFond D, Donahue BR, Marymont MH, Muraszko K, Langston J, Spoto R. Phase III study of craniospinal radiation therapy followed by adjuvant chemotherapy for newly diagnosed average-risk medulloblastoma. *J Clin Oncol*. 2006; 24:4202–4208
- [45] Tomita T, McLone DG. Medulloblastoma in Childhood: Results of Radical Resection and Low-Dose Radiation Therapy. *J Neurosurg*. 1986; 64:238–242
- [46] Packer RJ, Sutton LN, Goldwein JW, Perilongo G, Bunin G, Ryan J, Cohen BH, D'Angio G, Kramer ED, Zimmerman RA, et al. Improved survival with the use of adjuvant chemotherapy in the treatment of medulloblastoma. *J Neurosurg*. 1991; 74:433–440
- [47] Gilbertson RJ. Medulloblastoma: signalling a change in treatment. *Lancet Oncol*. 2004; 5:209–218
- [48] David KM, Casey AT, Hayward RD, Harkness WF, Phipps K, Wade AM. Medulloblastoma: is the 5-year survival rate improving? A review of 80 cases from a single institution. *J Neurosurg*. 1997; 86:13–21
- [49] Albright AL, Wisoff JH, Zeltzer PM, Boyett JM, Rorke LB, Stanley P. Effects of medulloblastoma resections on outcome in children: a report from the Children's Cancer Group. *Neurosurgery*. 1996; 38:265–271
- [50] Evans AE, Jenkins RD, Spoto R, et al. The Treatment of Medulloblastoma: Results of a Prospective Randomized Trial of Radiation Therapy With and Without CCNU, Vincristine, and Prednisone. *J Neurosurg*. 1990; 72:572–582
- [51] Weil MD, Lamborn K, Edwards MS, Wara WM. Influence of a child's sex on medulloblastoma outcome. *JAMA*. 1998; 279:1474–1476
- [52] Gajjar A, Hernan R, Kocak M, Fuller C, Lee Y, McKinnon PJ, Wallace D, Lau C, Chintagumpala M, Ashley DM, Kellie SJ, Kun L, Gilbertson RJ. Clinical, histopathologic, and molecular markers of prognosis: toward a new disease risk stratification system for medulloblastoma. *J Clin Oncol*. 2004; 22:984–993
- [53] Ray A, Ho M, Ma J, Parkes RK, Mainprize TG, Ueda S, McLaughlin J, Bouffet E, Rutka JT, Hawkins CE. A clinicobiological model predicting survival in medulloblastoma. *Clin Cancer Res*. 2004; 10:7613–7620
- [54] Reddy AT, Janss AJ, Phillips PC, Weiss HL, Packer RJ. Outcome for children with supratentorial primitive neuroectodermal tumors treated with surgery, radiation, and chemotherapy. *Cancer*. 2000; 88:2189–2193
- [55] Hong TS, Mehta MP, Boyett JM, Donahue B, Rorke LB, Yao MS, Zeltzer PM. Patterns of failure in supratentorial primitive neuroectodermal tumors treated in Children's Cancer Group Study 921, a phase III combined modality study. *Int J Radiat Oncol Biol Phys*. 2004; 60:204–213
- [56] Mork SJ, Rubinstein LJ. Ependymblastoma. A Reappraisal of a Rare Embryonal Tumor. *Cancer*. 1985; 55:1536–1542
- [57] Berger L, Luc G, Richard D. L'Esthesioneuroepitheliome Olfactif. *Bull Assoc Franc Etude Cancer*. 1924; 13:410–421
- [58] Theilgaard SA, Buchwald C, Ingeholm P, Kornum Larsen S, Eriksen JG, Sand Hansen H. Esthesioneuroblastoma: a Danish demographic study of 40 patients registered between 1978 and 2000. *Acta Otolaryngol*. 2003; 123:433–439
- [59] Chao KS, Kaplan C, Simpson JR, Haughey B, Spector GJ, Sessions DG, Arquette M. Esthesioneuroblastoma: the impact of treatment modality. *Head Neck*. 2001; 23:749–757
- [60] Kadish S, Goodman M, Wang CC. Olfactory neuroblastoma. A clinical analysis of 17 cases. *Cancer*. 1976; 37:1571–1576
- [61] Biller HF, Lawson W, Sachdev VP, Som P. Esthesioneuroblastoma: surgical treatment without radiation. *Laryngoscope*. 1990; 100:1199–1201
- [62] Dulguerov P, Calcaterra T. Esthesioneuroblastoma: the UCLA experience 1970–1990. *Laryngoscope*. 1992; 102:843–849
- [63] Hyams V. Tumors of the upper respiratory tract and ear. Washington, D.C.: Armed Forces Institute of Pathology; 1988
- [64] Van Gompel JJ, Giannini C, Olsen KD, Moore E, Piccirilli M, Foote RL, Buckner JC, Link MJ. Long-term outcome of esthesioneuroblastoma: hyams grade predicts patient survival. *J Neurol Surg B Skull Base*. 2012; 73:331–336
- [65] Kane AJ, Sughrue ME, Rutkowski MJ, Aranda D, Mills SA, Buencamino R, Fang S, Barani IJ, Parsa AT. Post-treatment prognosis of patients with esthesioneuroblastoma. *J Neurosurg*. 2010; 113:340–351
- [66] Gardner G, Robertson JH. Hearing Preservation in Unilateral Acoustic Neuroma Surgery. *Ann Otol Rhinol Laryngol*. 1988; 97:55–66
- [67] Dias FL, Sa GM, Lima RA, Kligerman J, Leoncio MP, Freitas EQ, Soares JR, Arcuri RA. Patterns of failure and outcome in esthesioneuroblastoma. *Arch Otolaryngol Head Neck Surg*. 2003; 129:1186–1192
- [68] Gore MR, Zanation AM. Salvage Treatment of Local Recurrence in Esthesioneuroblastoma: A Meta-analysis. *Skull Base*. 2011; 21:1–6
- [69] Van Gompel JJ, Carlson ML, Pollock BE, Moore EJ, Foote RL, Link MJ. Stereotactic radiosurgical salvage treatment for locally recurrent esthesioneuroblastoma. *Neurosurgery*. 2013; 72:332–9; discussion 339–40
- [70] Foote RL, Morita A, Ebersold MJ, Olsen KD, Lewis JE, Quast LM, Ferguson JA, O'Fallon WM. Esthesioneuroblastoma: the role of adjuvant radiation therapy. *Int J Radiat Oncol Biol Phys*. 1993; 27:835–842
- [71] Kim HJ, Cho HJ, Kim KS, Lee HS, Kim HJ, Jung E, Yoon JH. Results of salvage therapy after failure of initial treatment for advanced olfactory neuroblastoma. *J Craniomaxillofac Surg*. 2008; 36:47–52

41 Tumors of Cranial, Spinal and Peripheral Nerves

41.1 Vestibular schwannoma

41.1.1 General information

Key concepts

- Histologically benign tumor of cranial nerve VIII located in the cerebellopontine angle (CPA)
- Usually arises from inferior division (controversial) of the vestibular portion of the VIII nerve
- 3 most common early symptoms (clinical triad): hearing loss (insidious and progressive), tinnitus (high pitched) and dysequilibrium (true vertigo is uncommon)
- W/U: All patients: ☐ MRI (without & with contrast), ☐ audiometrics (pure tone audiogram and speech discrimination). In addition for small VSs (≤ 15 mm dia): ☐ ENG, ☐ VEMP, ☐ ABR
- Histology: comprised of Antoni A (narrow elongated bipolar cells) and Antoni B fibers (loose reticulated)
- Choice of management option (observation, surgery, XRT or chemotherapy (Avastin®)) depends heavily on tumor size, growth, hearing status, VII function and presence of NF2

Vestibular schwannoma (VS) is a histologically benign schwann-cell sheath tumor that usually arises from of the inferior division of the vestibular nerve (not the cochlear portion). VSs arise as a result of the loss of a tumor-suppressor gene on the long arm of chromosome 22 (in sporadic cases this is a somatic mutation; in neurofibromatosis Type 2 (NF2) this is either inherited or represents a new mutation that may then be transmitted to offspring).

Older terms included for reference, that should be avoided^{1,2}: acoustic neuroma, acoustic neurinomas (neurinoma being obsolete term for schwannoma), neurilemoma or neurilemmoma.

41.1.2 Epidemiology

One of the most common intracranial tumors, comprising 8–10% of tumors in most series.³ Annual incidence is probably about 1.5 cases per 100,000 population – over the past couple decades this estimate has increased and the typical size at diagnosis has decreased as a result of the proliferation of MRI scans.⁴ In the US, annual incidence has been found to vary between 1.1 and 1.3 per 100,000 population between 2004 and 2007.⁵ VSs typically become symptomatic after age 30. At least 95% are unilateral.

Neurofibromatosis Type 2
The incidence of vestibular schwannomas (VS) is increased in neurofibromatosis (NFT), with bilateral VS being pathognomonic of neurofibromatosis Type 2 (NFT2), see central NFT (p.605). Any patient <40 yrs old with unilateral VS should also be evaluated for NFT2. Cytologically, the VSs of NFT2 are identical to sporadic cases, however in NFT2 the tumors form grape-like clusters that may infiltrate the nerve fibers (unlike most sporadic VSs which displace the eighth nerve).

41.1.3 Pathology

Tumors are composed of Antoni A fibers (narrow elongated bipolar cells) and Antoni B fibers (loose reticulated). Verocay bodies are also seen, and consist of acellular eosinophilic areas surrounded by parallel arrangement of spindle shaped schwann cells (they are not a cell type).

41.1.4 Clinical

Symptoms

General information

Symptoms are shown in ☐ Table 41.1. The type of symptoms are closely correlated with tumor size. Most initially cause the triad of ipsilateral sensorineural hearing loss, tinnitus and balance difficulties. Larger tumors can cause facial numbness, weakness or twitching, and possibly brainstem symptoms. Rarely, large tumor may produce hydrocephalus. With current imaging modalities (CT and especially MRI), increasing numbers of smaller lesions are being detected.

Table 41.1 Symptoms in vestibular schwannoma (131 patients ³)	
Symptom	%
hearing loss	98%
tinnitus	70%
dysequilibrium ^a	67%
H/A	32%
facial numbness	29%
facial weakness	10%
diplopia	10%
N/V	9%
otalgia	9%
change of taste	6%
^a or vertigo	

Symptoms from 8th nerve compression

Unilateral sensorineural hearing loss, tinnitus and dysequilibrium are related to pressure on the eighth nerve complex in the IAC. These are the earliest symptoms, and by the time of diagnosis, virtually all tumors have caused otologic symptoms.

Hearing loss is insidious and progressive in most (c.f. the hearing loss in Meniere’s disease which fluctuates), however 10%report sudden hearing loss (see below). 70%have a high frequency loss pattern, and word discrimination is usually affected (especially noticeable in telephone conversation).

The tinnitus is usually high pitched.

Unsteadiness manifests primarily as difficulty with balance; true vertigo occurs in <20%

Sudden hearing loss: The differential diagnosis for sudden hearing loss (SHL) is extensive.⁶ Idiopathic SHL (i.e. no identified etiology: must rule out neoplasm, infection, autoimmune, vascular and toxic causes) occurs in an estimated 10 per 100,000 population.⁷ 1%of patients with SHL will be found to have a VS, and SHL may be the presenting symptom in 1–14%of patients with VS.⁸ SHL with VS is presumably due to an infarction of the acoustic nerve, or acute occlusion of the cochlear artery. Treatment options for SHLinclude:

- 1. steroids: e.g. prednisone 60 mg PO q d × 10 d then tapered⁸
- 2. famciclovir (Famvir®) 500 mg po TID × 10 d
- 3. □ heparin has been shown not to be of help
- 4. conservative treatment: rest, restriction of salt, alcohol and tobacco⁹
- 5. experimental: consideration may be given to thrombolytic therapy (p.1286), e.g. rt-PA

Symptoms from 5th and 7th nerve compression

Otalgia, facial numbness and weakness, and taste changes occur as the tumor enlarges and compresses the fifth and seventh nerves. These symptoms usually do not occur until the tumor is >2 cm. This highlights an interesting paradox: facial weakness is a rare or late occurrence, even though the 7th nerve is almost always distorted early; whereas facial numbness occurs sooner once trigeminal compression occurs (often in the presence of normal facial movement), despite the fact that the 5th nerve is farther away.¹⁰ This may be due to the resiliency of motor nerves relative to sensory nerves.

Symptoms from compression of brainstem and other cranial nerves

Larger tumors cause brainstem compression (with ataxia, H/A, N/V, diplopia, cerebellar signs, and if unchecked, coma, respiratory depression and death) and lower cranial nerve (IX, X, XII) palsies (hoarseness, dysphagia...). Obstruction of CSF circulation by larger tumors (usually >4 cm) may produce hydrocephalus with increased ICP.

Rarely, 6th nerve involvement may cause diplopia.

Signs

Hearing loss due to VIII involvement is the earliest cranial nerve finding. 66% of patients have no abnormal physical finding except for hearing loss (for other findings, see □ Table 41.2).

Since hearing loss is sensorineural, Weber test (p.580) will lateralize to the uninvolved side, and if there is enough preserved hearing, Rinne test (p.580) will be positive (i.e. normal; air conduction > bone conduction) on both sides.

Facial nerve (VII) dysfunction is uncommon before treatment. When present, it is usually graded clinically on the House and Brackmann scale (see □ Table 41.3).

Vestibular involvement causes nystagmus (may be central or peripheral) and abnormal electro-nystagmography (ENG) with caloric stimulation.

Table 41.2 Signs in 131 vestibular schwannomas (excluding hearing loss) ³	
Sign	%
abnormal corneal reflex	33
nystagmus	26
facial hypoesthesia	26
facial weakness (palsy)	12
abnormal eye movement	11
papilledema	10
Babinski sign	5

Table 41.3 Clinical grading of facial nerve function(House and Brackmann ¹¹)		
Grade	Function	Description
1	normal	normal facial function in all areas
2	mild dysfunction	1. gross: slight weakness noticeable on close inspection; may have very slight synkinesis 2. at rest: normal symmetry and tone 3. motion: a) forehead: slight to moderate movement b) eye: complete closure with effort c) mouth: slight asymmetry
3	moderate dysfunction	1. gross: obvious but not disfiguring asymmetry: noticeable but not severe synkinesis 2. motion: a) forehead: slight to moderate movement b) eye: complete closure with effort c) mouth: slightly weak with maximal effort
4	moderate to severe dysfunction	1. gross: obvious weakness and/or disfiguring asymmetry 2. motion: a) forehead: none b) eye: incomplete closure c) mouth: asymmetry with maximum effort
5	severe dysfunction	1. gross: only barely perceptible motion 2. at rest: asymmetry 3. motion: a) forehead: none b) eye: incomplete closure
6	total paralysis	no movement

Differential diagnosis

See Cerebellopontine angle (CPA) lesions (p.1365). The major differentials are: meningioma, or neuroma of an adjacent cranial nerve (e.g. trigeminal).

41.1.5 Evaluation

General information

- 1. Brain MRI without and with contrast. FIESTA MRI if available. If MRI is contraindicated, then a CT scan without and with contrast
- 2. Temporal bone CT for detailed bony anatomy if surgery is contemplated
- 3. audiometric evaluation:
 - a) pure tone audiogram (see below)
 - b) speech discrimination evaluation (see below)
 - c) patients with small VSs (≤ 15 mm dia) also get:
 - ENG: (p.674) assesses superior vestibular nerve
 - VEMP: (p.675) assesses inferior vestibular nerve
 - ABR: (p.675) prognosticates chance of hearing preservation

Audiometric and audiologic studies

General information

Baseline studies are helpful for management treatment decisions and for later comparison and to assess the contralateral ear.

Pure tone audiogram (PTA)

May be useful as first-step screening test. Air conduction assesses the entire system, bone conduction assesses from the cochlea and proximally. PTA assesses the functionality of hearing (to help in treatment decision making) and acts as a baseline for future comparison. The pure tone average (also abbreviated PTA) is a single numerical score that is an average of the thresholds for frequencies across the audio spectrum (at 500, 1000 & 2000 Hz). On a standard audiogram, X's denote the left ear (AS) and O's denote the right ear (AD).

Progressive unilateral or asymmetric sensorineural hearing loss of high tones occurs in >95% of VSs.¹² High-frequency hearing loss also happens to be the most common type of hearing loss with age or with noise induced sensorineural hearing loss, but is usually symmetrical. Only ≈ 1 in 1000 patients with asymmetric hearing have a VS.¹ Other causes of asymmetrical sensorineural hearing loss¹³: other CPA lesions (e.g. meningioma), inner ear lesions, intraaxial lesions (including 9 infarctions), multiple sclerosis. On hearing screening tests, an unexplained PTA difference from one ear to the other >10–15 dB is suspicious and should be investigated further.

Speech discrimination evaluation

Speech discrimination is maintained in conductive hearing loss, moderately impaired in cochlear hearing loss, and worst with retrocochlear lesions. No longer used for diagnostic purposes; a score of 4% suggests a retrocochlear lesion, as does a score that is worse than would be predicted based on PTA testing (the speech recognition threshold should be similar to PTA thresholds below 4 kHz). Has found usefulness in determining serviceability of hearing and prognosticating for hearing preservation surgery. Open-set word recognition score (WRS □ Table 41.4) is a more sensitive measure of communication ability than PTA.

Table 41.4 Open-set word recognition score	
Class	WRS%
I	70–100%
II	50–69%
III	1–49%
IV	0

Table 41.5 Gardener and Robertson modified hearing classification ^a				
Class	Clinical utility	Description	Pure tone audiogram ^b (dB)	Speech discrimination ^b
I	serviceable	good-excellent	0–30	70–100%
II	serviceable	serviceable	31–50	50–59%
III	non-serviceable	non-serviceable	51–90	5–49%
IV	non-serviceable	poor	91-max	1–4%
V	non-serviceable	none	not testable	0
^a modification ¹⁵ of the Silverstein and Norrell system ¹⁶				
^b if PTA and speech discrimination score do not qualify in the same class, use the lower class				

Table 41.6 American Academy of Otolaryngology-Head and Neck Surgery Foundation hearing classification system				
Class	Clinical utility	Pure tone threshold (dB) ^a		Speech discrimination score ^b (%)
A	“useful”	≤30	AND	≥70
B	“useful”	>30 AND ≤50	AND	≥50
C	“aidable”	>50	AND	≥50
D	“nonfunctional”	any level		<50
^a average of pure tone hearing thresholds by air conduction at 0.5, 1, 2 & 3 kHz				
^b speech discrimination at 40 dB or maximum comfortable loudness				

Definition of serviceable hearing

There are many definitions of what constitutes serviceable hearing. Also, even nonserviceable hearing can offer some benefit. If WRS is good (≥70%) but PTA is poor, a hearing aid may provide significant benefit.

Two commonly used scoring systems for hearing are shown here:

1. Modified Gardener-Robertson system for grading hearing: shown in □ Table 41.5. Class I patients may use a phone on that side, class II patients can localize sounds.
2. The American Academy of Otolaryngology – Head and Neck Surgery Foundation (AAO-HNS) hearing classification system¹⁴: shown in □ Table 41.6.

Some definitions of serviceable hearing (see text that follows for details):

1. AAO-HNS class A or B
2. “50/50 rule”: Gardner-Robertson class I or II (pure tone audiogram threshold ≤50 dB and speech discrimination score ≥50%)
3. some prefer a 70/30 rule (70%WRS, 30 dB PTA)
4. in a patient with good hearing in the contralateral ear, a speech discrimination score (SDS) of < 70%in the affected ear is not considered good hearing; whereas if the contralateral ear is totally deaf, a SDS of ≥50%can be useful¹⁷

Additional tests that are helpful with small VSs (≤15 mm diameter)

The ENG and VEMP evaluate the superior and inferior division of the vestibular nerve (VN) respectively. The inferior VN is closer to the cochlear nerve than the superior VN (□ Fig. 1.7), and small tumors (≤4 mm) of the inferior VN tend to be deeper and closer to the cochlear nerve than similar sized tumors of the superior division which tend to be more superficial and more easily removed.

Electronystagmography (ENG): Only tests the horizontal semicircular canal □ assesses the superior vestibular nerve which innervates it. Normally, each ear contributes an equal portion of the response. The ENG is considered abnormal if there is >20%difference between the two sides. Response

may be normal with a small tumor arising from the inferior division of vestibular nerve. NB: the vestibular nerve may continue to function until almost all of the nerve fibers are affected. Mnemonic for the directionality of the nystagmus (direction is classified based on the fast phase of the nystagmus). COWS (Cold Opposite, Warm Same). NB: this is different than caloric testing for brain death and is a source of confusion.

Vestibular evoked myogenic potential (VEMP): Assesses inferior vestibular nerve by imparting acoustic energy to the saccule.¹⁸ Independent of hearing (can be done even with deafness from profound sensorineural hearing loss). Electrodes are placed on sternocleidomastoid muscle (SCM).

Auditory brainstem responses (ABR): AKA BAER (p.240). The most common findings are prolonged I-III and I-V interpeak latencies. No longer used for diagnostic purposes (sensitivity is only $\approx 88\text{--}90\%$ (i.e. will miss 10–12% of VSs) and specificity is only 85% ABR is useful for prognostication – poor wave morphology correlates with lower chance of preserving hearing (even with good hearing).

Radiographic evaluation

MRI

Thin slice axial plane gadolinium enhanced MRI is the diagnostic procedure of choice with sensitivity close to 98% and almost 0% false positive rate. Characteristic findings: round or oval enhancing tumor centered on IAC. Large VSs (>3 cm dia) may show cystic appearing areas on CT or MRI; in actuality these areas are usually solid. Adjacent trapped CSF cisterns may also give cystic appearance. In a pilot study, hyperintensity on T2 has been associated with soft and suckable tumors at the time of surgery¹⁹ and a trend towards better VII function preservation.

FIESTA MRI (fast imaging employing steady state acquisition): uses CSF as the contrast agent (□ does not use gadolinium), may improve visualization of the tumor and nerves.

CTscan

CT with IV contrast is second choice for imaging modality. If MRI is contraindicated and clinical suspicion of VS is strong but the CT is negative, small lesions may be visualized by introducing 3–4 ml of subarachnoid air via lumbar puncture, and scanning the patient with the affected side up (to trap air in region of IAC), non-filling of the IAC is indicative of an intracanalicular mass. Even with air contrast, CT was normal in 6% in Mayo series.³

Many VSs enlarge the ostium of the IAC (called trumpeting). Normal diameter of the IAC: 5–8 mm. 3–5% of VSs do not enlarge the IAC on CT (percentage is likely higher in small VSs vs. large ones).

Thin-cut temporal bone CT should be obtained for operative planning. Important features to identify:

- For middle fossa approach: bony coverage of Geniculate ganglion to identify dehiscence
- For translab approach:
 - Extent of pneumatization of the mastoid and position of sigmoid sinus. An anterior sinus with poorly pneumatized mastoids can indicate a tight space for this approach
 - Position of the jugular bulb. If high riding, can indicate a tight space in translab approach
- For retrosigmoid transmeatal approach: location and thickness of bone coverage over the posterior semicircular canal and vestibular aqueduct. The extent of peritubular air cells and retro facial air cells needs to be assessed in planning the approach and preventing CSF leaks.

41.1.6 Management

Management options

Management options include:

1. expectant management: follow symptoms, hearing (audiometrics) and tumor growth on serial imaging (MRI or CT). Intervention is performed for progression. Growth patterns observed:
 - a) little or no growth: applies to most (83%) VSs confined within the IAC and 30% extending into CPA (see natural history of growth below)
 - b) slow growth $\approx 2\text{ mm/yr}$
 - c) rapid growth: $\geq 10\text{ mm/yr}$
 - d) a few actually shrink⁴
2. radiation therapy (alone, or in conjunction with surgery)
 - a) external beam radiation therapy (EBRT)
 - stereotactic radiation
 - stereotactic radiosurgery (SRS) (p.1564): single dose
 - stereotactic radiotherapy (SRT) (p.1564): fractionated

3. surgery: approaches include the following (see below for details)
 - a) retrosigmoid (AKA suboccipital): may be able to spare hearing
 - b) translabyrinthine (and its several variations): sacrifices hearing, may be slightly better for sparing VII
 - c) middle fossa approach (extradural subtemporal): only for small lateral VSs
4. chemotherapy: some preliminary promise for progressive NF2-related vestibular schwannomas with bevacizumab (Avastin®), an anti-VEGF (vascular endothelial growth factor) monoclonal antibody (see below). Side effects: hemorrhage occurs in $\approx 7\%$ due to vessel necrosis

Patient/tumor factors influencing management decisions

In addition to the usual factors involved in the decision process with brain tumors, e.g. the patient's general medical condition, age, natural history, etc., elements unique to VSs include: chances of preserving VII & V nerve function and hearing (in those with serviceable hearing) (all of which are related to tumor size), and the presence of NF2.

Specifics:

1. natural history of growth
 - a) usual quoted range: $\approx 1\text{--}10\text{ mm/yr}$. However this can be quite variable
 - b) strictly intracanalicular tumors: only 17% grew outside the meatus (in 552 VSs over 3.6 years mean follow-up (230 were intrameatal at time of diagnosis, 322 had extrameatal extension)⁴)
 - c) extrameatal tumors (with extension into CP angle): 30% grew $>2\text{ mm}$ (in 522 VSs over 3.6 years mean follow-up⁴)
 - d) VSs that did not grow in the 5 years after diagnosis did not grow after that
 - e) 6% actually decrease in size²⁰
2. natural history of hearing function in untreated intracanalicular VSs in AAO-HNS Group A ($\square$ Table 41.6) patients²¹
 - a) 50% deteriorated to a lower class over 4.6 years (loss of $\geq 10\text{ dB PTA}$ or $\geq 10\%$ SDS)
 - b) after 4.6 years of observation, the proportion of patients eligible for hearing preservation treatment (as determined by a word recognition score class I (70–100% SDS)) was reduced to 28% (a 44% reduction) and by AAO-HNS class A to 9% (a 53% reduction)
 - c) the risk of losing hearing was not related to: age, gender, acoustic tumor size (all tumors were intracanalicular) or tumor sublocalization (fundus, central, porus)
 - d) hearing loss was positively correlated to the absolute volumetric tumor growth rate (tumors that eventually expand out of the IAC have a faster rate and degree of hearing loss compared to tumors remaining in the IAC)
 - e) the risk of losing hearing was significantly lower for patients with 100% word recognition score. Over 4.6 years observation, 89% remained in WRS class I ($\square$ Table 41.4) compared to only 43% for patients with only a small (1–10%) loss of WRS at diagnosis
3. size: as tumors exceed 15 mm diameter, treatment complications increase
 - a) significantly lower chance for hearing preservation
 - b) increased incidence of VII injury
4. presence of cysts: cystic tumors may display sudden and dramatic growth⁴
5. serviceable hearing: see Definition of serviceable hearing (p. 674)
6. hearing in contralateral ear

Management algorithm

- Small tumors ($<15\text{ mm}$ diameter) with perfect hearing (WRS 100%):
 - Observe both radiographically (CT or MRI scan) plus serial hearing tests:
 - Scan: recommend treatment for growth $>2\text{ mm}$ between studies. Scan schedule guide:
 - Every 6 months $\times 2$ years after diagnosis, then (if stable)
 - Annually until 5 years after diagnosis, then (if stable)
 - At years 7, 9 & 14 after diagnosis⁴
 - Annual audiology evaluations
 - Hearing deterioration (WRS $<100\%$) but no growth: see below
 - Rationale: in patients with small tumors and normal WRS, comparing the results of hearing preservation following surgery or SRS to the natural history, the conclusion is that established tumor growth should be the main determinant for treatment¹⁷
- Small tumors with serviceable hearing: management is very controversial
 - In general, patients with serviceable hearing but WRS $<100\%$ have a 50% chance of preserving their serviceable hearing (50/50 rule or AAO-HNS class A or B) with either observation, SRS or microsurgical resection

- Hearing preservation rates better than 50% have been reported for both microsurgery and radiosurgery in very selected patients (smaller tumors, located medially in the IAC, intact pre-op ABR amplitudes and latencies)
- Decisions to try to beat the natural history (50% chance of hearing loss at 10 years) should be very individualized based on tumor factors (size, location) and patient's factors (ABR, age, co-morbidities, preferences)
- Final management decisions are often dictated by non medical reasons (patient's perception, social situation, financial considerations, support system etc.)
- Medium size tumors (15-25 mm diameter)
 - Tumors > 15-20 mm should be treated.^{4,17} This is mostly true for young patients
 - Close observation to establish growth is a valid option in older patients or patients with medical co-morbidities
 - Complication rate increases and facial outcome worsens with increasing tumor size
 - NF2 patients present a challenge and should be evaluated individually. In general the success rate in the management of their tumors is lower (higher cranial nerve deficit and higher recurrence rate).^{22,23} Early management is considered more favorable for good outcome.²⁴ A retrospective study found significant hearing improvement and tumor shrinkage in >50% of NF2 patients with progressive VS using bevacizumab (Avastin®) (see above)
- Large tumors (>25 mm diameter): treatment is recommended
 - Microsurgical resection is favored to reduce mass effect and decompress the brainstem
 - SRS is also useful in larger tumors for older patients or those with significant co-morbidities

Selection options for intervention

Once treatment is elected (see algorithm above), the type of treatment must be selected.

Comparison of microsurgery vs. radiosurgery (SRS)

1. hearing preservation
 - a) for patients with testable pre-operative hearing
 - summary: radiosurgery or stereotactic radiation appears to be better at preserving hearing than microsurgery. The difference is minor for tumors < 10 mm and very good pre-operative hearing (70% SDS, and 30 dB PTA). The advantage of radiation is more pronounced for larger tumors and greater pre-operative hearing loss. Details:
 - SRS: overall, at 3, 5, and 10 years, 81%, 77% and 66% of the patients maintained their GR hearing class (□ Table 41.5). For patients receiving a tumor margin dose of 13 Gy or less, those same percentages were 93%, 87% and 87%.²⁵ Hearing preservation appears to be related to the radiation dose to the cochlea rather than to the tumor itself.²⁶
 - microsurgery: hearing preservation is significantly related to the tumor size and to the experience of the surgical team. Hearing preservation in Samii's series of 1000 VS²³ improved from 24% in the first 200 cases to 49% in later cases. Hearing preservation in microsurgery has improved with the use of direct cochlear nerve monitoring²⁷ compared to auditory brainstem responses monitoring only. Hearing preservation in patients with class A, small tumors and direct cochlear nerve monitoring (compound nerve action potential) was 91%.²⁸ With microsurgery, the durability of the hearing preservation is also excellent with only 15% of patients with class A hearing after surgery, and 33% of patients with class B hearing after surgery slipping one class at 5 years follow-up.²⁹
2. facial nerve preservation
 - a) preservation has been excellent with both microsurgery and radiosurgery
 - b) microsurgery: 98.5% overall³⁰ and 100% in tumors not touching the brainstem. Staged resection has been advocated by some to improve facial nerve preservation in giant VS (>4–4.5 cm)³¹
 - c) radiosurgery: 98% of patients.²⁵ The incidence of facial neuropathy has significantly decreased since the SRS dose was decreased to 12–13 Gy. Facial neuropathy in the recent series occurred in patients having received 18–20 Gy
3. trigeminal neuropathy (TGN)
 - a) a complication classically feared in large tumors especially following SRS
 - b) SRS: 7% incidence of TGN (mainly in patients receiving higher doses, i.e. 18 Gy). No patients who received a dose < 13 Gy developed TGN.²⁵
 - c) microsurgery: post-op TGN is not reported in most series
4. tumor control (local control rates, LCR):
 - a) tumor control has been a concern with radiosurgery and with the more recent decrease in dose from 18–20 Gy to 12–14 Gy, long term data are lacking
 - b) microsurgery: tumor recurrence has been poorly studied. Quoted rates in the literature vary between 0.5% at 6 years³⁰ to 9.2%.³²

- c) SRS: tumor recurrence requiring re-treatment at 5 years was 4%²⁵ but 18% of patients presented with transient increase in the size of the tumor (“pseudogrowth”) at a mean of 8 months, with later regression in half and stabilization to the new size in the other half

Vertigo and dizziness

For patients with episodic vertigo or balance difficulties as the predominant symptom; also, see points under Selection options for intervention (p.677):

1. remember: patients with VS are susceptible to other causes of vertigo as well, and patients should undergo ENG and functional balance assessment
2. vertigo that is due to the VS is often self-limited, and improves in 6–8 weeks to a reasonably tolerable level with no treatment (patients may do better with so-called “vestibular rehab”)
3. residual dizziness and balance disturbances are common whether stereotactic radiosurgery (SRS) or microsurgery (MS) is used, but are typically less after MS
4. a recent study based on patient’s reported vertigo intensity has shown that symptoms of vertigo are improved by any treatment versus observation (van Gompel et al³³)
5. after SRS: a minimum of 5-6 mos, and sometimes up to eighteen months may be required to produce beneficial effects. Symptoms are improved more rapidly after MS than with SRS
6. following MS: The severity of dizziness after MS depends on the pre-operative vestibular function on the affected side. If the ipsilateral vestibular function is absent pre-operatively, then patients will not suffer from dizziness or nausea post-op. if ipsilateral vestibular function is intact pre-op, then patients may be very dizzy and nauseated in particular for the first 24 hrs
7. conclusion:
 - observation may be the best choice for $\approx 20\%$ of patients
 - when treatment is desired:
 - surgery is the best choice for most VSs producing vertigo
 - SRS may be the right choice for some, especially: elderly patients (> 70 yrs) with other health problems, for recurrence of VS, and for individual preference

Hydrocephalus

When hydrocephalus is present, it may require separate treatment with a CSF shunt – see Surgical considerations (p.679) – and may possibly be done at the same time as surgery for the VS (if surgery for the VS is indicated).

41.1.7 Surgical treatment

Approaches

General information

Three basic surgical approaches:

1. those with possibility of hearing preservation
 - a) middle fossa (MF): poor access to posterior fossa (see below)
 - b) retrosigmoid (RS) (see below) AKA retrosigmoid-transmeatal approach
2. translabyrinthine (TL): non hearing preserving (see below)

Excellent results have been reported with each of these approaches. These guidelines assume that the surgical team is comfortable with all three approaches.

Decision algorithm for approach

The choice of approach is dictated by hearing salvageability and tumor size as follows:

1. salvageable hearing (□ Table 41.7 for definition and guidelines)
 - a) if tumor is intracanalicular (no extension beyond a few mm into the posterior fossa (CPA); note: differences of opinion exist regarding how much tumor in the CPA can be removed via MF): use the middle fossa approach. Note: some authors exclusively use the retrosigmoid approach for hearing preservation even for these tumors with excellent results
 - b) if tumor extends > few mm into the posterior fossa: use the retrosigmoid approach (it is generally accepted that the cisternal part of the tumor is not well exposed by the middle fossa approach, especially regarding the ability to dissect tumor off the nerves)
2. non salvageable hearing (see □ Table 41.7 for definition and guidelines)
 - a) Use translabyrinthine approach or retrosigmoid approach

Table 41.7 Hearing salvageability
Serviceable and unsalvageable hearing
Definition of serviceable hearing
A generous definition of serviceable hearing: PTA<50 dB and SDS>50% ^a
Unsalvageable hearing
Serviceable hearing is unlikely to be preserved post-op when 1. pre-op SDS<75% 2. or pre-op PTA loss>25 dB 3. or pre-op BAER has abnormal wave morphology 4. or tumor>2–2.5 cm diameter
^a see also other definitions of serviceable hearing (p.674)

- b) Either can be used irrespective of tumor size. Surgical team preference is the main deciding factor. Some influencing aspects:
 - a young patient with no cerebellar atrophy might favor a translab approach
 - an anteriorly situated sigmoid sinus and/or a high riding jugular bulb restrict the working space in the translab approach, and might favor a retrosigmoid approach

Surgical considerations

General information

The first surgical removal of a vestibular schwannoma was performed over a century ago in 1894.³⁴ The facial nerve is pushed forward by the tumor in $\approx 75\%$ of cases (range: 50–80%), but may occasionally be pushed rostrally, less often inferiorly, and rarely posteriorly. It may even continue to function while it is flattened to a mere ribbon on the tumor capsule surface.

Anesthesia with minimal muscle relaxants allows intra-op seventh nerve monitoring. In only $\approx 10\%$ of large tumors is the cochlear nerve a separate band on the tumor capsule, in the remainder it is incorporated into the tumor.

Although total excision of tumor is usually the goal of surgery, facial nerve preservation must take precedence over degree of resection. Near total resection (very small sliver of tumor left on the facial nerve) or subtotal resection are all excellent options if the tumor is tightly adherent to the facial nerve or the brainstem. Both have excellent long term tumor control rate with either observation or post-op radiation.

If hydrocephalus is present, it used to be standard practice to place a CSF shunt and wait ≈ 2 weeks before the definitive operation.³⁵ While still acceptable, this is less commonly done at present, and shunting or EVD is often performed under the same anesthesia.

Large tumors may be approached by a staged surgical approach to debulk tumor and preserve facial nerve, or a planned subtotal resection followed by radiation. For tumors >3 cm, such approach seems to lead to improved results of facial nerve function³⁶

The extra anesthesia time involved in translab approach may be detrimental in the elderly.

Middle fossa approach

- indications:
 - a) hearing preservation
 - b) laterally placed tumors
 - c) small tumors (usually <2.5 cm)
- pros:
 - a) allows drilling and exposure of the IAC all the way to the geniculate ganglion (good for laterally placed tumors)
 - b) basically an extradural subtemporal operation
- cons:
 - a) potential damage to temporal lobe with risk of seizures
 - b) facial nerve is the most superficial nerve in this exposure and therefore the surgeon works “around” the facial nerve (possibility of injury)
- technique summary
 - a) lumbar drain

Table 41.8 Pros & cons of translab approach

Disadvantages	Advantages
• sacrifices hearing (acceptable when hearing is already non-functional or unlikely to be spared by other approach) • may take longer than retrosigmoid approach • possibly higher rate of post-op CSF leak	• early identification of VII may result in higher preservation rate • less risk to cerebellum and lower cranial nerves • patients do not get as “ill” from blood in cisterna magna, etc. (essentially an extracranial approach)

- b) usually straight incision, starting in front of the tragus, extending cephalad for 6 cm, held open with a self retaining retractor
- c) the temporalis muscle is incised vertically (along the muscle fibers) along the most posterior aspect of the exposure, as well and reflected anteriorly
- d) craniotomy: 4 cm × 3 cm
- e) elevate the middle fossa dura, section the middle meningeal artery. Identify and preserve the greater superficial petrosal nerve (GSPN), arcuate eminence, V3, and true edge of the petrous bone (the false edge is the groove occupied by the superior petrosal sinus)
- f) drill and expose the internal auditory canal all the way to Bill’s bar (for tumors extending laterally)
- g) localize the facial nerve with the nerve stimulator
- h) open the IAC dura along the main axis of the IAC, avoiding VII
- i) identify the vestibular, cochlear and facial nerves
- j) dissect the tumor off the nerves

Translabyrinthine approach

Often preferred by neurotologists.

- 1. pros & cons: See □ Table 41.8
- 2. technique summary
 - a) Position: supine head rotated to contralateral side, can be done either with pins or doughnut if it is anticipated that no retractors will be used
 - b) Prep the abdomen for fat graft (almost always used)
 - c) skin incision should be tailored to the location of the sigmoid sinus (observe location of the sigmoid sinus and pinna of the ear on the pre-op MRI). Usually smaller opening than retrosigmoid approach
 - does not require a craniotomy. For large tumors requiring an “extended translab”, 1–2 cm of retrosigmoid dura should be exposed during the mastoidectomy to allow for retraction of the sigmoid sinus
 - dural opening along the IAC after identification of VII with stimulator
 - for large tumor: section the superior petrosal sinus and section the tentorium to gain better intradural exposure
 - closure requires fat graft

Booking the case: Translabyrinthine approach for vestibular schwannoma

Also see defaults & disclaimers (p. 27).

- 1. position: supine with shoulder roll
- 2. equipment:
 - a) microscope
 - b) high speed drill
 - c) ultrasonic aspirator
- 3. some surgeons work with neurotologist to assist with the IAC and for follow-up
- 4. neuromonitoring: facial EMG (does not require EEG tech), SSEPs for tumors involving the brain-stem (requires EEG tech)
- 5. post-op: ICU
- 6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery through an incision behind the ear to remove a tumor growing inside the skull on the nerve to the ear. Possible need for post-op lumbar drain. Fat graft (≈ always used)

- b) alternatives: nonsurgical management with follow-up MRIs, other surgical approaches, radiation (stereotactic radiosurgery)
- c) complications: CSF leak with possible meningitis, loss of hearing in ipsilateral ear (if not already lost), paralysis of facial muscles on the side of surgery with possible need for surgical procedures to help correct (correction is often far from perfect), facial numbness, post-op balance difficulties/vertigo, brainstem injury with stroke

Retrosigmoid approach

AKA posterior fossa, AKA suboccipital approach.^{37,38}

- pros:
 - a) familiar to most neurosurgeons □ often preferred by neurosurgeons
 - b) quick access to the tumor
 - c) hearing preservation possible
 - d) NOTE: this approach is very versatile. Samii²³ resected all his acoustic tumors via a retrosigmoid approach; he achieved a significant amount of brain relaxation and improved exposure by using in the sitting position, which is generally not used in the USA because of associated complications (p.1445)
- cons:
 - a) cerebellar retraction: not a problem for tumors <4 cm, provided the craniotomy is sufficiently lateral and the cisterna magna and the CP angle cistern has been opened
 - b) headaches: it has been suggested that headaches are more common following retrosigmoid craniotomy than after translabyrinthine craniotomy. Postulated mechanisms: purely extradural drilling in translab with no bone dust in subarachnoid space. More anterior skin incision in translab and less disruption of the suboccipital musculature and greater occipital nerve

Booking the case: Retrosigmoid craniotomy for vestibular schwannoma

Also see defaults & disclaimers (p. 27).

1. position: lateral decubitus with tumor side up
2. equipment:
 - a) microscope
 - b) ultrasonic aspirator
 - c) image guided navigation system (if used) (may be helpful for placing skin incision and craniotomy more than for tumor localization)
3. some surgeons work with neurotologist to assist with the IAC and for follow-up
4. neuromonitoring: facial EMG (does not require EEG tech), BAERS, near field monitoring (CNAP: compound nerve action potential)
5. post-op: ICU
6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery through an incision behind the ear to remove a tumor growing inside the skull on the nerve to the ear. Possible need for post-op lumbar drain. Possible fat graft (optional)
 - b) alternatives: nonsurgical management with follow-up MRIs, other surgical approaches, radiation (stereotactic radiosurgery)
 - c) complications: CSF leak with possible meningitis, loss of hearing in ipsilateral ear (if not already lost), paralysis of facial muscles on the side of surgery with possible need for surgical procedures to help correct (correction is often far from perfect), facial numbness post-op balance difficulties/vertigo, brainstem injury with stroke. Facial numbness (infrequent)

Technique summary

1. position: lateral decubitus with tumor side up, head in pins rotated (might need shoulder roll), zygomatic arch horizontal. 30° elevation of the head is paramount; see Posterior fossa (suboccipital) craniectomy, Lateral oblique position (p. 1446)
2. percutaneous lumbar drain (optional)

Table 41.9 Aids in localizing VII nerve origin⁴¹

- VII nerve originates in the pontomedullary sulcus near the lateral end of the sulcus, 1–2 mm anterior to the VIII nerve
- the pontomedullary sulcus ends just medial to the foramen of Luschka (extending from the lateral recess of the IV ventricle, □ Fig. 1.9)
- a tuft of choroid plexus usually extends out of the foramen of Luschka on the posterior surface of IX and X nerve, just inferior to the origin of VII
- the flocculus of the cerebellum projects from the lateral recess into the CPA just posterior to the origin of VII and VIII
- VII origin is 4 mm cephalad and 2 mm anterior to that of the IX nerve

3. incision is shaped like the pinna of the ear, 3 finger breaths behind the external auditory canal
4. the craniotomy has to be lateral enough to expose part of the sigmoid and part of the transverse sinuses and to allow a straight line of sight to the lateral end of the IAC
5. to prevent CSF leak, seal all bone edges with bone wax
6. dural opening along the lines of the craniotomy
7. exposure is enhanced by opening the cerebello-pontine angle cistern and the cisterna magna under the microscope and draining CSF (20–40 ml of CSF can also be drained via a lumbar sub-arachnoid catheter)
8. the petrosal vein is often sacrificed at the beginning of the procedure to allow the cerebellum to relax and fall back and to avoid tearing off the transverse sinus. Be careful not to coagulate the SCA that often runs with the petrosal vein
9. using the facial nerve stimulator, the posterior aspect of the tumor is inspected to make sure the facial nerve has not been pushed posteriorly
10. the thin layer of arachnoid that covers most tumors is identified. Vessels within the arachnoid may contribute to cochlear function and may be preserved by keeping them with the arachnoid
11. the plane between tumor and cerebellum may be followed to the brainstem, and occasionally to the VII nerve (this plane is harder to follow once bleeding from tumor debulking occurs)
12. to help locate the origin of the VII nerve at the brainstem see □ Table 41.9 and CPA anatomy in □ Fig. 1.9
13. the posterolateral tumor capsule is opened, and internal decompression is performed. The tumor is collapsed inward and the capsule is kept intact and is rolled laterally off of VII and is eventually removed. The most difficult area to separate VII from tumor is just proximal to the entrance to the porus acusticus. A general recommendation is to accept a subtotal or near total resection to preserve anatomic continuity of the facial nerve in cases where it is identified by stimulation but because it is so flattened it cannot be seen as a separate structure on the surface of the tumor.
14. after the extracanalicular portion of tumor is removed, the dura over the IAC is incised, and the IAC is drilled open and tumor is removed from this portion. To preserve hearing, the bony labyrinth must not be violated. The posterior semicircular canal (SCC) is the most vulnerable structure (□ Fig. 41.1). The vestibule of the SCCs is also at risk but is less likely to be entered. The maximal amount of temporal bone drilling that can be accomplished without entering the posterior SCC can be determined from the pre-op CT. The operculum of the temporal bone, is a small step-off palpable with a nerve hook posteriorly from the porus acusticus. It marks the location of the vestibular aqueduct and is a good landmark for the posterior extent of the drilling in the retrosigmoid exposure of the IAC. Measuring the distance from the IAC to the posterior semicircular canal on a pre-op CT and measuring the thickness of the bone overlying the posterior semicircular canal are recommended for safe exposure of the IAC, in particular for hearing preservation. However, opening the labyrinth cannot always be avoided; and any opening should be plugged with bone-wax or muscle.³⁹ If the facial nerve is not intact and is not going to be grafted, then the IAC should be plugged, e.g. by bone wax covered with a small piece of hammered muscle (hammering makes the muscle sticky by activating extrinsic clotting factors) and Gelfoam®.

NB: large tumors: in some large tumors, the capsule may be adherent to the brainstem and so portions of tumor must be left; recurrence rate among these is $\approx$ 10–20%⁴⁰ Large tumors may also involve V superiorly (sometimes VII is pushed up against V), and inferiorly may involve IX, X, and XI. The lower cranial nerves can usually be spared by dissecting them off of the tumor capsule, and protecting them with cottonoids.

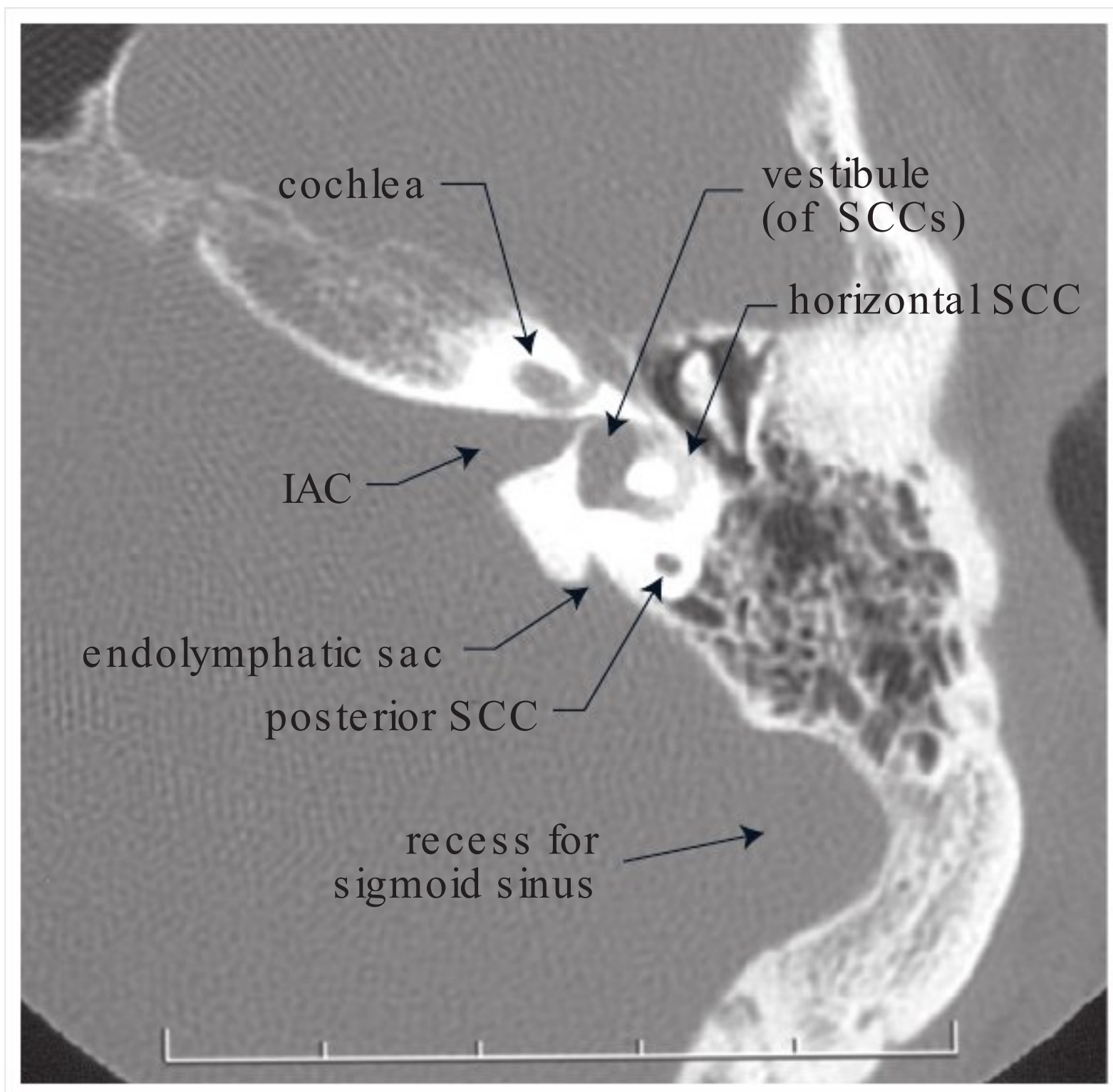


Fig. 41.1 Structures of the left temporal bone. CT scan (left petrous bone, axial slice) provided courtesy of Chris Danner, M.D.

Post-op care and care for complications

Cranial nerve and brainstem dysfunction

Facial nerve (VII)

If eye closure is impaired due to VII dysfunction: □ natural tears 2 gtts to affected eye q 2 hrs and PRN. Apply Lacrilube® to affected eye and tape it shut q hs. If there is complete VII palsy with little chance of early recovery, or if facial sensation (Vth nerve) is also impaired, tarsorrhaphy is performed within a few days.

Facial re-animation (e.g. hypoglossal-facial anastomosis) is performed after 1–2 months if VII was divided, or if no function returns after 1 year with an anatomically intact nerve.

Vestibular nerve (VIII)

Vestibular dysfunction is common post-op, nausea and vomiting due to this (and also intracranial air) is common. Balance difficulties due to this clear rapidly, however, ataxia from brainstem dysfunction may have a permanent component.

Lower cranial nerves

The combination of IX, X and XII dysfunction creates swallowing difficulties and creates a risk of aspiration.

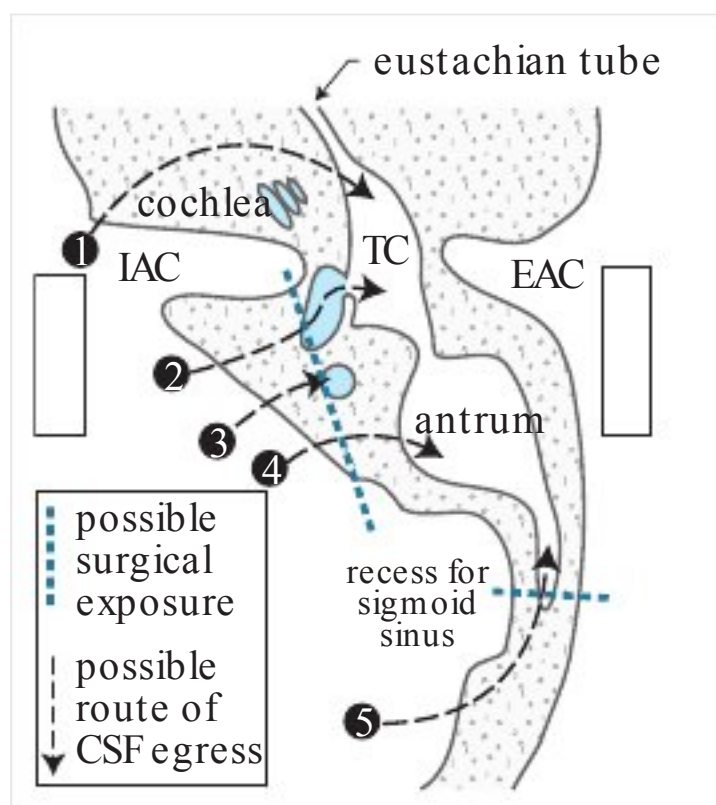


Fig. 41.2 Possible routes for CSF rhinorrhea following vestibular schwannoma surgery (see text) (right petrous bone, axial slice). Adapted from *Surgical Neurology*, Vol. 43, Nutik S L, Korol H W, *Cerebrospinal Fluid Leak After Acoustic Neuroma Surgery*, 553–7, 1995, with permission from Elsevier Science

Brainstem dysfunction

Brainstem dysfunction may occur from dissection of tumor off of the brainstem. This may produce ataxia, contralateral paresthesias in the body. Although there may be improvement, once present, there is often some permanent residual.

CSF fistula

Also, see CSF fistula (cranial) (p.384) for general information. CSF fistula may develop through the skin incision, the ear (CSF otorrhea) through a ruptured tympanic membrane, or via the eustachian tube and then through the nose (rhinorrhea) or down the back of the throat.

Rhinorrhea may occur through any of the following routes (circled numbers in □ Fig. 41.2):

- □ via the apical cells to the tympanic cavity (TC) or eustachian tube (the most common path)
- entry into the bony labyrinth – in order to reach the middle ear would require rupture e.g. of the oval window by overpacking bone wax into the labyrinth)
 - □ through the vestibule of the horizontal semicircular canal (SCC)
 - □ through the posterior SCC (the posterior SCC is the most common area that is entered by drilling)
- □ follows the perilabyrinthine cells and tracts to the mastoid antrum
- □ through the mastoid air cells surgically exposed at the craniotomy site

Most leaks are diagnosed within 1 week of surgery, although 1 presented 4 years post-op.⁴² They appear to be more common with more lateral unroofing of the IAC.⁴² Meningitis complicates a CSF leak in 5–25% of cases, and usually develops within days of the onset of leak.⁴² Hydrocephalus may promote the development of a CSF fistula.

Treatment: 25–35% of leaks stop spontaneously (one series reported 80%).⁴² Treatment options include:

1. non-surgical:
 - a) elevate HOB
 - b) a percutaneous lumbar subarachnoid drain may be tried,^{43,44} although some debate its efficacy,³⁸ and there is a theoretical risk of drawing bacteria into the CNS
2. surgical treatment for persistent leaks: In general, post-operative CSF leak (including rhinorrhea) is best addressed with immediate surgical re-exploration
 - a) in the case of a translabyrinthine approach with absent ipsilateral hearing: to treat rhinorrhea, pack and permanently close the Eustachian tube via a trans tympanic membrane approach. This is very effective and avoids re-opening the surgical incision and removing the previously placed fat graft.
 - b) if hearing is preserved (which excludes translab), every effort should be made to preserve the Eustachian tube function to preserve middle ear function. Re-explore the surgical field, re-wax the aircells and place additional fat graft, fascia, pericranium or other sealant over the exposed aircells. This aggressive management is the most definitive and rapid treatment, and avoids prolonged bed rest required by placing a lumbar drain and trying to control the leak in a conservative way
3. a CSF leak may be an indication of altered CSF hydrodynamics. Most of these patients demonstrate frank ventriculomegaly (hydrocephalus). In some patients the leak may function as a

Table 41.10 Cranial nerve preservation in retrosigmoid removal of VSs ^a		
Size of tumor	Preserved function	
	VII nerve	VIII nerve
<1 cm	95–100%	57%
1–2 cm	80–92%	33%
>2 cm	50–76%	6%
^a series of 135 VSs ⁴⁸ (p 729) and other sources ⁴⁰ (p 3337), ⁴⁵		

pressure relief valve and thereby ameliorate the ventriculomegaly (i.e. there would be hydrocephalus if there wasn't a leak). Adjunctive CSF shunting is usually also necessary or the repair will be more likely to fail

Outcome and follow-up

Complete surgical removal was reported in 97–99%of cases.⁴⁵

Surgical morbidity and mortality

Also see Post-op considerations for p-fossa cranis (p.1451). Estimated frequency of some complications⁴⁶: the most common complication is CSF leakage in 4–27%⁴² (see above), meningitis in 5.7% stroke in 0.7% subsequent requirement for CSF shunt (for hydrocephalus or to treat leak) in 6.5% The mortality rate is ≈ 1%at specialized centers.^{23,45,47}

Cranial nerve dysfunction

□ Table 41.10 shows statistics of VII and VIII cranial nerve preservation following suboccipital removal of VSs in several combined groups of patients. For more details, see below.

Post-radiation cranial neuropathies generally appear 6–18 months following stereotactic radiosurgery (SRS),⁴⁹ and since more than half of these resolve within 3–6 months after the onset the recommendation is treat these with a course of corticosteroids.

Facial nerve (VII)

See □ Table 41.3 for the House and Brackmann grading scale. Grades 1–3 are associated with acceptable function. Facial nerve preservation is related to tumor size.

Surgery: With the use of modern facial nerve monitoring techniques, anatomical integrity of the facial nerve can be achieved in >90%even for very large tumors and in close to 99%in medium size tumors.⁵⁰ In cases where the nerve is so flattened that tumor has to be left over the nerve to preserve its anatomical integrity, functional outcome of the facial nerve is nevertheless lower, especially with larger tumors. Excellent outcome (HB grade I-III) was only achieved in 75%of the patients with large tumors and 91%in medium size tumors.

SRS for tumors ≤3 cm diameter: With modern SRS dosimetry (12-13 Gy for patients with serviceable hearing and 13-14 Gy for patients with non serviceable hearing, the incidence of new facial nerve weakness was 4%⁵¹

Vestibulo-acoustic nerve (VIII)

Patients with unilateral VS and Class I or II hearing (□ Table 41.5) comprised ≈ 12%of cases in a large series.⁵² Preservation of hearing is critically dependent on tumor size, with little chance of preservation with tumors >1–1.5 cm diameter. Chances of preserving hearing may possibly be improved by intra-operative brainstem auditory evoked potential monitoring.⁵³ In centers treating large numbers of VSs, hearing preservation rates of 35–71%can be achieved with tumors <1.5 cm^{52, 54} (although a range of 14–48%may be more realistic⁵⁵). Hearing may rarely be improved post-op.⁵⁶

SRS: for tumors ≤3 cm diameter,⁵⁷ hearing was preserved in 26%of 65 cases with pre-op pure tone threshold <90 dB. Hearing loss has been correlated with increase in tumor size.⁵⁸ NB: there is a high rate of hearing loss at 1 year. SRT: useful hearing was preserved in 93%⁵⁹

Vestibular nerve function is rarely normal post-op. Attempts at “vestibular” sparing surgery have shown no better results than surgery not specifically addressing this issue. Most patients with unilateral loss of vestibular nerve function will learn to compensate to a significant degree with input from the contralateral side, if normal. Patients with ataxia as a result of brainstem injury from the tumor or the surgery will have more difficulties post-op. Some patients will seem to do well initially post-op with respect to vestibular nerve function, only to undergo a delayed deterioration several

months post-op. These cases likely represent aberrant regeneration of the vestibular nerve fibers and may be extremely difficult to manage. Some experts advocate cutting the vestibular nerve, as for Meniere’s disease (p.573).

Trigeminal nerve (V)

Postoperative trigeminal nerve symptoms occur transiently in 22%and permanently in 11%following microsurgery, similar to the results of SRS.⁶⁰ New facial numbness occurred in 2%with SRT.⁵⁹

Lower cranial nerves

Injuries to IX, X and XI occur infrequently following surgery on large tumors that distort the nerves and displace them inferiorly against the occipital bone.

Recurrence

Following microsurgery (MS)

Recurrence is highly dependent on extent of removal. However, recurrence can develop in tumors that were apparently totally removed, or when subtotal resection was performed. This can occur many years after treatment. Tumor progression rate following subtotal resection is ≈ 20%⁵⁵ All patients should be followed with imaging (CT or MRI). In older series with up to 15 yrs follow-up, local control rate (LCR) after “total resection” is ≈ 94% More recent series with MRI follow-up indicate recurrence rates of 7–11%(3–16 yrs follow-up).⁵⁵

Use of EBRT

EBRT may improve LCR in incompletely resected tumors as shown in □ Table 41.11 (note: with the long survival expected with benign tumors, post XRT complications may occur).

Microsurgery vs. SRS

The long-term results for SRS using the current recommended dose of 14 Gy are still not known³⁴ (in spite of journal article titles suggesting otherwise⁶²). In a non-randomized retrospective study⁶⁰ of VSs<3 cm dia, the short-term LCR (median 24 mos follow-up) was 97%for microsurgery vs. 94%for stereotactic radiosurgery (SRS). However, for benign tumors, long-term follow-up is critical (possibly 5–10 years⁴⁹), and this study suggests that the long-term LCR will be better for MS than SRS. SRS studies with long-term follow-up⁶³ are not directly comparable because in the cases with longest follow-up, higher radiation doses were used with a resultant higher incidence of radiation complications, and an anticipated better LCR.

Initially there may be temporary enlargement of the tumor accompanied by loss of central contrast enhancement following SRS in ≈ 5%of patients⁶⁴ (with up to 2%of patients showing actual initial tumor growth), and so the need for further treatment after SRS should be postponed until there is evidence of sustained growth.⁶⁵ Surgery should therefore be avoided during the interval from 6 to 18 months after SRS because this is time of maximum damage from the radiation.⁶⁵

Although the numbers are small, there have been indications that the rate of VII nerve injury may be higher in patients undergoing microsurgery following SRS failure to achieve LCR compared to cases where microsurgery was the initial procedure,^{66,67} however, this has been disputed.⁶⁵ Lastly, there is a potential for malignant transformation of VSs following SRS including triton tumors^{68,69} (malignant neoplasms with rhabdoid features) or the induction of skull base tumors (which has been reported with external beam radiation⁷⁰), as well as the risk of late arterial occlusion (the AICA lies near the surface of VSs), any of which may occur many years later.

Treatment for recurrence following microsurgery

Repeat surgery for recurrent VS is an option. One series of 23 patients⁷¹ showed that 6 of 10 patients with moderate or normal VII function maintained at least moderate function after reoperation,

Table 41.11 Local control rates of surgery vs. surgery+ EBRT for VSs ⁶¹		
Extent of surgical removal	Local control rate (LCR)	
	Surgery	Surgery + EBRT ^a
gross total	60/62 (97%)	no data
near total (90–99%)	14/15 (93%)	2/2 (100%)
subtotal (<90%)	7/13 (54%)	17/20 (85%) ^a
biopsy only	no data	3/3 (100%)
^a with doses <45 Gy, LCR was 33% with >45 Gy LCR was 94%		

3 patients had increased ataxia, and 1 patient had a cerebellar hematoma. The use of SRS has been endorsed by some for recurrence of VS following one or more MS procedures.⁵⁵ Using SRS for recurrent VSs resulted in worsening of facial nerve function in 23% of patients with Grade I-III function before SRS (median follow-up=43 mos), and 14% developed new trigeminal symptoms.⁵⁵ 6% of patients developed tumor progression after SRS.

Hydrocephalus

May occur following treatment (MS or SRS) for VS, and may even occur years later. The increased CSF pressure may also predispose to development of a CSF fistula.

41.2 Tumors of peripheral nerves: Perineurioma

A nerve sheath tumor. Variants:

1. intraneural perineurioma: usually solitary lesion of adolescence or young adulthood, affecting primarily peripheral nerves (cranial nerve involvement is rare). Pseudo-onion bulb formation with cylindrical enlargement of the nerve over 2–10 cm. Mitotic activity is rare, MIB-1 labeling index is low. Chromosome 22 loss is characteristic,⁷² no NF1 association. Treatment: conservative sampling of lesion, not resection
2. soft tissue perineurioma: uncommon. Only rarely can an associated nerve be identified. Almost exclusively benign, but malignant variety does occur. Female:male ratio=4:1. In males, hands are often affected. Discrete, but not encapsulated, diameter=1.5–20 cm. Treatment: gross total excision is curative

References

- [1] National Institutes of Health Consensus Development Conference. Acoustic Neuroma: Consensus Statement. Bethesda, MD 1991
- [2] Eldridge R, Parry D. Summary: Vestibular Schwannoma (Acoustic Neuroma) Consensus Development Conference. *Neurosurgery*. 1992; 30:962–964
- [3] Harner SG, Laws ER. Clinical Findings in Patients with Acoustic Neuromas. *Mayo Clin Proc*. 1983; 58:721–728
- [4] Stangerup SE, Caye-Thomasen P, Tos M, Thomsen J. The natural history of vestibular schwannoma. *Otol Neurotol*. 2006; 27:547–552
- [5] Lau T, Olivera R, Miller T, Jr, Downes K, Danner C, van Loveren HR, Agazzi S. Paradoxical trends in the management of vestibular schwannoma in the United States. *J Neurosurg*. 2012; 117:514–519
- [6] Jaffe B. Clinical Studies in Sudden Deafness. *Adv Otorhinolaryngol*. 1973; 20:221–228
- [7] Byl F. Seventy-Six Cases of Presumed Sudden Hearing Loss Occurring in 1973: Prognosis and Incidence. *Laryngoscope*. 1977; 87:817–824
- [8] Berenholz LP, Eriksen C, Hirsh FA. Recovery From Repeated Sudden Hearing Loss With Corticosteroid Use in the Presence of an Acoustic Neuroma. *Ann Otol Rhinol Laryngol*. 1992; 101:827–831
- [9] Moskowitz D, Lee KJ, Smith HW. Steroid Use in Idiopathic Sudden Sensorineural Hearing Loss. *Laryngoscope*. 1984; 94:664–666
- [10] Tarlov EC. Microsurgical Vestibular Nerve Section for Intractable Meniere's Disease. *Clin Neurosurg*. 1985; 33:667–684
- [11] House WF, Brackmann DE. Facial Nerve Grading System. *Otolaryngol Head Neck Surg*. 1985; 93:184–193
- [12] Hardy DG, Macfarlane R, Baguley D, et al. Surgery for Acoustic Neurinoma: An Analysis of 100 Translabrynthine Operations. *J Neurosurg*. 1989; 71:799–804
- [13] Daniels RL, Swallow C, Shelton C, Davidson HC, Krejci CS, Harnsberger HR. Causes of unilateral sensorineural hearing loss screened by high-resolution fast spin echo magnetic resonance imaging: review of 1,070 consecutive cases. *Am J Otol*. 2000; 21:173–180
- [14] Committee on Hearing and Equilibrium of the American Academy of Otolaryngology-Head and Neck Surgery Foundation. Guidelines for the evaluation of hearing preservation in acoustic neuroma (vestibular schwannoma). *Otolaryngol Head Neck Surg*. 1995; 113:179–180
- [15] Gardner G, Robertson JH. Hearing Preservation in Unilateral Acoustic Neuroma Surgery. *Ann Otol Rhinol Laryngol*. 1988; 97:55–66
- [16] Silverstein H, McDaniel A, Norrell H, Haberkamp T. Hearing Preservation After Acoustic Neuroma Surgery with Intraoperative Direct Eighth Cranial Nerve Monitoring: Part II. A Classification of Results. *Otolaryngol Head Neck Surg*. 1986; 95
- [17] Stangerup SE, Caye-Thomasen P, Tos M, Thomsen J. Change in hearing during 'wait and scan' management of patients with vestibular schwannoma. *J Laryngol Otol*. 2008; 122:673–681
- [18] Murofushi T, Matsuzaki M, Mizuno M. Vestibular evoked myogenic potentials in patients with acoustic neuromas. *Arch Otolaryngol Head Neck Surg*. 1998; 124:509–512
- [19] Copeland WR, Hoover JM, Morris JM, Driscoll CL, Link MJ. Use of preoperative MRI to predict vestibular schwannoma intraoperative consistency and facial nerve outcome. *J Neurol Surg B Skull Base*. 2013; 74:347–350
- [20] Bederson JB, von Ammon K, Wichmann WW, Yasargil MG. Conservative Treatment of Patients with Acoustic Tumors. *Neurosurgery*. 1991; 28:646–651
- [21] Caye-Thomasen P, Dethloff T, Hansen S, Stangerup SE, Thomsen J. Hearing in patients with intracanalicular vestibular schwannomas. *Audiol Neurotol*. 2007; 12:1–12
- [22] Asthagiri AR, Parry DM, Butman JA, Kim HJ, Tsilou ET, Zhuang Z, Lonser RR. Neurofibromatosis type 2. *Lancet*. 2009; 373:1974–1986
- [23] Samii M, Matthies C. Management of 1000 Vestibular Schwannomas (Acoustic Neuromas): Surgical Management with an Emphasis on Complications and How to Avoid Them. *Neurosurgery*. 1997; 40:11–23
- [24] Brackmann DE, Fayad JN, Slaterry WH, III, Friedman RA, Day JD, Hitselberger WE, Owens RM. Early proactive management of vestibular schwannomas in neurofibromatosis type 2. *Neurosurgery*. 2001; 49:274–80; discussion 280-3
- [25] Lobato-Polo J, Kondziolka D, Zorro O, Kano H, Flickinger JC, Lunsford LD. Gamma knife radiosurgery in younger patients with vestibular schwannomas. *Neurosurgery*. 2009; 65:294–300; discussion 300-1

- [26] Timmer FC, Hanssens PE, van Haren AE, Mulder JJ, Cremers CW, Beynon AJ, van Overbeeke JJ, Graa-mans K. Gamma knife radiosurgery for vestibular schwannomas: results of hearing preservation in relation to the cochlear radiation dose. *Laryngo-scope*. 2009; 119:1076–1081
- [27] Danner C, Mastrodimos B, Cueva RA. A comparison of direct eighth nerve monitoring and auditory brainstem response in hearing preservation surgery for vestibular schwannoma. *Otol Neurotol*. 2004; 25:826–832
- [28] Yamakami I, Yoshinori H, Saeki N, Wada M, Oka N. Hearing preservation and intraoperative auditory brainstem response and cochlear nerve compound action potential monitoring in the removal of small acoustic neurinoma via the retrosigmoid approach. *J Neurol Neurosurg Psychiatry*. 2009; 80:218–227
- [29] Wang AC, Chinn SB, Than KD, Arts HA, Telian SA, El-Kashlan HK, Thompson BG. Durability of hearing preservation after microsurgical treatment of ves-tibular schwannoma using the middle cranial fossa approach. *J Neurosurg*. 2013; 119:131–138
- [30] Samii M, Gerganov V, Samii A. Improved preserva-tion of hearing and facial nerve function in vestibul-ar schwannoma surgery via the retrosigmoid approach in a series of 200 patients. *J Neurosurg*. 2006; 105:527–535
- [31] Patni AH, Kartush JM. Staged resection of large acoustic neuromas. *Otolaryngol Head Neck Surg*. 2005; 132:11–19
- [32] Roche PH, Ribeiro T, Khalil M, Soumare O, Thomas-sin JM, Pellet W. Recurrence of vestibular schwan-nomas after surgery. *Prog Neurol Surg*. 2008; 21:89–92
- [33] Van Gompel JJ, Patel J, Danner C, Zhang AN, Samy Youssef AA, van Loveren HR, Agazzi S. Acoustic neu-roma observation associated with an increase in symptomatic tinnitus: results of the 2007-2008 Acoustic Neuroma Association survey. *J Neurosurg*. 2013; 119:864–868
- [34] Pitts LH, Jackler RK. Treatment of Acoustic Neuro-mas. *N Engl J Med*. 1998; 339:1471–1473
- [35] Ojemann RG. Microsurgical Suboccipital Approach to Cerebellopontine Angle Tumors. *Clin Neurosurg*. 1978; 25:461–479
- [36] Porter RG, LaRouere MJ, Kartush JM, Bojrab DI, Pieper DR. Improved facial nerve outcomes using an evolving treatment method for large acoustic neu-romas. *Otol Neurotol*. 2013; 34:304–310
- [37] Rhoton AL, Jr. The cerebellopontine angle and pos-terior fossa cranial nerves by the retrosigmoid approach. *Neurosurgery*. 2000; 47:S93–129
- [38] Ebersold MJ, Harner SG, Beatty CW, Harper CM, et al. Current Results of the Retrosigmoid Approach to Acoustic Neurinoma. *J Neurosurg*. 1992; 76:901–909
- [39] Tatagiba M, Samii M, Matthies C, El Azm M, Schön-mayr R. The Significance for Postoperative Hearing of Preserving the Labyrinth in Acoustic Neurinoma Surgery. *J Neurosurg*. 1992; 77:677–684
- [40] Youmans JR. *Neurological Surgery*. Philadelphia 1990
- [41] Rhoton AL. Microsurgical Anatomy of the Brainstem Surface Facing an Acoustic Neuroma. *Surg Neurol*. 1986; 25:326–339
- [42] Nutik SL, Korol HW. Cerebrospinal Fluid Leak After Acoustic Neuroma Surgery. *Surg Neurol*. 1995; 43:553–557
- [43] Symon L, Pell MF. Cerebrospinal Fluid Rhinorrhea Following Acoustic Neurinoma Surgery: Technical Note. *J Neurosurg*. 1991; 74:152–153
- [44] Ojemann RG. Management of Acoustic Neuromas (Vestibular Schwannomas). *Clin Neurosurg*. 1993; 40:498–539
- [45] Sekhar LN, Gormely WB, Wright DC. The Best Treat-ment for Vestibular Schwannoma (Acoustic Neuro-ma): Microsurgery or Radiosurgery? *Am J Otol*. 1996; 17:676–689
- [46] Wiegand DA, Fickel V. Acoustic Neuromas. The Patient's Perspective. Subjective Assessment of Symptoms, Diagnosis, Therapy, and Outcome in 541 Patients. *Laryngoscope*. 1989; 99:179–187
- [47] Gormley WB, Sekhar LN, Wright DC, et al. Acoustic Neuroma: Results of Current Surgical Management. *Neurosurgery*. 1997; 41:50–60
- [48] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
- [49] Flickinger JC, Kondziolka D, Pollock BE, Lunsford LD. Evolution in Technique for Vestibular Schwannoma Radiosurgery and Effect on Outcome. *Int J Radiation Oncology Biol Phys*. 1996; 36:275–280
- [50] Samii M, Gerganov VM, Samii A. Functional out-come after complete surgical removal of giant ves-tibular schwannomas. *J Neurosurg*. 2010; 112:860–867
- [51] Pollock BE, Driscoll CL, Foote RL, Link MJ, Gorman DA, Bauch CD, Mandrekar JN, Krecke KN, Johnson CH. Patient outcomes after vestibular schwannoma management: a prospective comparison of micro-surgical resection and stereotactic radiosurgery. *Neurosurgery*. 2006; 59:77–85; discussion 77-85
- [52] Glasscock ME, Hays JW, Minor LB, Haynes DS, Carra-sco VN. Preservation of Hearing in Surgery for Acoustic Neuromas. *J Neurosurg*. 1993; 78:864–870
- [53] Ojemann RG, Levine RA, Montgomery WM, et al. Use of Intraoperative Auditory Evoked Potentials to Preserve Hearing in Unilateral Acoustic Neuroma Removal. *J Neurosurg*. 1984; 61:938–948
- [54] Brackmann DE, House JR III, Hitselberger WE. Tech-nical Modifications to the Middle Cranial Fossa Approach in Removal of Acoustic Neuromas. Los Angeles, CA 1993
- [55] Pollock BE, Lunsford LD, Flickinger JC, Clyde BL, Kondziolka D. Vestibular Schwannoma Manage-ment. Part I. Failed Microsurgery and the Role of Delayed Stereotactic Radiosurgery. *J Neurosurg*. 1998; 89:944–948
- [56] Shelton C, House WF. Hearing Improvement After Acoustic Tumor Removal. *Otolaryngol Head Neck Surg*. 1990; 103:963–965
- [57] Hirsch A, Norén G. Audiological Findings After Ster-eotactic Radiosurgery in Acoustic Neuromas. *Acta Otolaryngol (Stockh)*. 1988; 106:244–251
- [58] Flickinger JC, Lunsford LD, Coffey RJ, Linskey ME, et al. Radiosurgery of Acoustic Neurinomas. *Cancer*. 1991; 67:345–353
- [59] Selch MT, Pedroso A, Lee SP, Solberg TD, Agazaryan N, Cabatan-Awang C, DeSalles AA. Stereotactic radiotherapy for the treatment of acoustic neuro-mas. *J Neurosurg*. 2004; 101:362–372
- [60] Pollock BE, Lunsford LD, Kondziolka D, et al. Out-come Analysis of Acoustic Neuroma Management: A Comparison of Microsurgery and Stereotactic Radiosurgery. *Neurosurgery*. 1995; 36:215–229
- [61] Wallner KE, Sheline GE, Pitts LH, Wara WM, et al. Efficacy of Irradiation for Incompletely Excised Acoustic Neurilemmas. *J Neurosurg*. 1987; 67:858–863
- [62] Kondziolka D, Lunsford LD, McLaughlin MR, Flick-inger JC. Long-Term Outcomes After Radiosurgery for Acoustic Neuromas. *N Engl J Med*. 1998; 339:1426–1433
- [63] Noren G, Hirsch A, Mosskin M. Long-Term Efficacy of Gamma Knife Radiosurgery in Vestibular Schwannomas. *Acta Neurochir*. 1993; 122
- [64] Linskey ME, Lunsford LD, Flickinger JC. Neuroimag-ing of Acoustic Nerve Sheath Tumors After Stereo-tactic Radiosurgery. *AJNR*. 1991; 12:1165–1175
- [65] Pollock BE, Lunsford LD, Kondziolka D, et al. Vestib-ular Schwannoma Management. Part II. Failed Radiosurgery and the Role of Delayed Microsurgery. *J Neurosurg*. 1998; 89:949–955
- [66] Slattery WH, Brackmann DE. Results of Surgery Fol-lowing Stereotactic Irradiation for Acoustic Neuro-mas. *Am J Otol*. 1995; 16:315–321

- [67] Wiet RJ, Micco AG, Bauer GP. Complications of the Gamma Knife. *Arch Otolaryngol Head Neck Surg.* 1996; 122:414–416
- [68] Yakulis R, Manack L, Murphy AI. Postradiation Malignant Triton Tumor: A Case Report and Review of the Literature. *Arch Pathol Lab Med.* 1996; 120:541–548
- [69] Comey CH, McLaughlin MR, Jho HD, Martinez AJ, Lunsford LD. Death From a Malignant Cerebellopontine Angle Triton Tumor Despite Stereotactic Radiosurgery. *JNeurosurg.* 1998; 89:653–658
- [70] Lustig LR, Jackler RK, Lanser MJ. Radiation-Induced Tumors of the Temporal Bone. *Am J Otol.* 1997; 18:230–235
- [71] Beatty CW, Ebersold MJ, Harner SG. Residual and Recurrent Acoustic Neuromas. *Laryngoscope.* 1987; 97:1168–1171
- [72] Emory TS, Scheithauer BW, Hirose T, Wood M, Onofrio BM, Jenkins RB. Intraneural perineurioma. A clonal neoplasm associated with abnormalities of chromosome 22. *Am J Clin Pathol.* 1995; 103:696–704

42 Meningiomas

42

42.1 General information

Key concepts

- slow growing, extra-axial tumor, usually benign, arise from arachnoid (not dura)
- imaging (MRI or CT): classically broad based attachment on dura often with dural tail, typically enhance densely, may cause hyperostosis of adjacent bone
- MRI: isointense on T1WI, hypodense on T2WI
- 32%of incidentally discovered meningiomas do not grow over 3 years follow-up
- surgical indications: documented growth on serial imaging and/or symptoms referable to the lesion that are not satisfactorily controlled medically
- most (but not all) are cured if completely removed, which is not always possible
- most commonly located along falx, convexity, or sphenoid bone
- frequently calcified. Classic histological finding: psammoma bodies

Meningiomas are the most common primary intracranial tumors. They are usually slow growing, circumscribed (non-infiltrating), benign lesions. Histologically malignant (incidence: $\approx$ 1.7%of meningiomas¹) and/or rapidly growing varieties are also described; a rapidly growing lesion that looks like a meningioma may be a hemangiopericytoma (p.701). Actually arise from arachnoid cap cells (not dura). May be multiple in up to 8%of cases,² this finding is more common in neurofibromatosis. Occasionally forms a diffuse sheet of tumor (meningioma en plaque). This section considers intracranial meningiomas.

May occur anywhere that arachnoid cells are found (between brain and skull, within ventricles, and along spinal cord). Ectopic meningiomas may arise within the bone of the skull (primary intraosseous meningiomas)³ and others occur in the subcutaneous tissue with no attachment to the skull. Most are asymptomatic (see below).

42.2 Epidemiology

As many as 3%of autopsies on patients >60 yrs age reveals a meningioma.⁴ Meningiomas account for 14.3–19%of primary intracranial neoplasms.⁵ Incidence peaks at 45 years age. Female:male ratio is 1.8:1.

1.5%occur in childhood and adolescence, usually between 10–20 years age.⁶(p 3263) 19–24%of adolescent meningiomas occur in patients with neurofibromatosis type I (von Recklinghausen’s).

42.3 Common locations

42.3.1 General information

□ Table 42.1 lists common locations. Other locations include: CP-angle, clivus, planum sphenoidale and foramen magnum. $\approx$ 60–70%occur along the falx (including parasagittal), along sphenoid bone (including tuberculum sellae), or over the convexity. Childhood meningiomas are rare, 28%are intraventricular, and the posterior fossa is also a common site.

42.3.2 Sphenoid wing (or ridge) meningiomas

Three basic categories⁸:

1. lateral sphenoid wing (or pterional): behavior and treatment are usually similar to convexity meningioma
2. middle third (or alar)
3. medial (clinoidal): tend to encase the ICA and the MCA as well as cranial nerves in the region of the superior orbital fissure and the optic nerve. May compress brainstem. Total removal is often not possible

Table 42.1 Location of adult meningiomas (series of 336 cases ⁷)	
Location	%
parasagittal	20.8
convexity	15.2
tuberculum sellae	12.8
sphenoidal ridge	11.9
olfactory groove	9.8
falx	8
lateral ventricle	4.2
tentorial	3.6
middle fossa	3
orbital	1.2
spinal	1.2
intrasylvian	0.3
extracalvarial	0.3
multiple	0.9

42.3.3 Parasagittal and falx meningiomas

Up to 50%invade the superior sagittal sinus (SSS). Grouped based on location along AP direction of SSS as:

- 1. anterior (ethmoidal plate to coronal suture): 33% Most often present with H/A and mental status changes
- 2. middle (between coronal and lambdoidal sutures): 50% Most often present as Jacksonian seizure and progressive monoplegia
- 3. posterior (lambdoidal suture to torcular Herophili): 20% Most often present with H/A, visual symptoms, focal seizures, or mental status changes

Classification systems for the extent of SSS invasion include one by Bonnal and Brotchi,⁹ and a more recent one by Sindou et al.¹⁰ shown in □ Fig. 42.1.

Parasagittal meningiomas may originate at the level of the motor strip, and a common initial manifestation of these is a contralateral foot drop.¹¹

42.3.4 Olfactory groove meningiomas

Presentation (usually asymptomatic until they are large) may include:

- 1. Foster Kennedy syndrome (p.99):anosmia (patient is usually unaware of this), ipsilateral optic atrophy, contralateral papilledema
- 2. mental status changes: often with frontal lobe findings (apathy, abulia...)
- 3. urinary incontinence
- 4. posteriorly located lesions may compress the optic apparatus causing visual impairment
- 5. large lesions may compress the fornix and cause short-term memory loss
- 6. seizure

The morbidity, mortality and difficulty in achieving total removal increase significantly for tumors > 3 cm in size.¹²

Pre-op MRA, CTA or angiogram may be helpful to assess location of anterior cerebral arteries relative to the tumor. 70–80%of these get the majority of their blood supply from the anterior ethmoidal artery, which is usually not embolized due to risk to ophthalmic artery (and blindness). If there are substantial middle meningeal feeders, these may be embolized, but the benefit tends to be small.

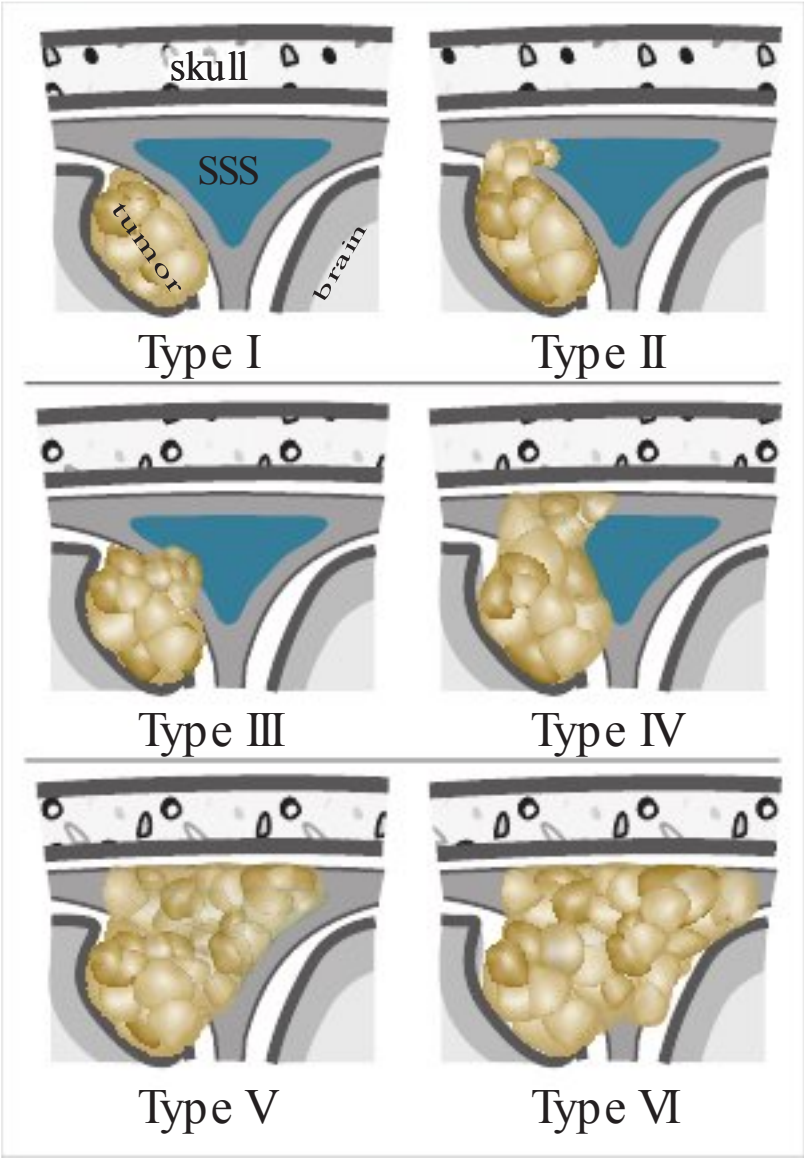


Fig. 42.1 Grading system for meningioma invasion of the superior sagittal sinus Modified from Sindou MP et al., J Neurosurg, 105: pp 514–25, 2006. Shown: schematic coronal section through superior sagittal sinus (SSS).
Type I=attachment to lateral wall of sinus
Type II=invasion of lateral recess
Type III=invasion of lateral wall
Type IV=invasion of lateral wall and roof
Type V=total sinus occlusion, contralateral wall spared
Type VI=total sinus occlusion, invasion of all walls

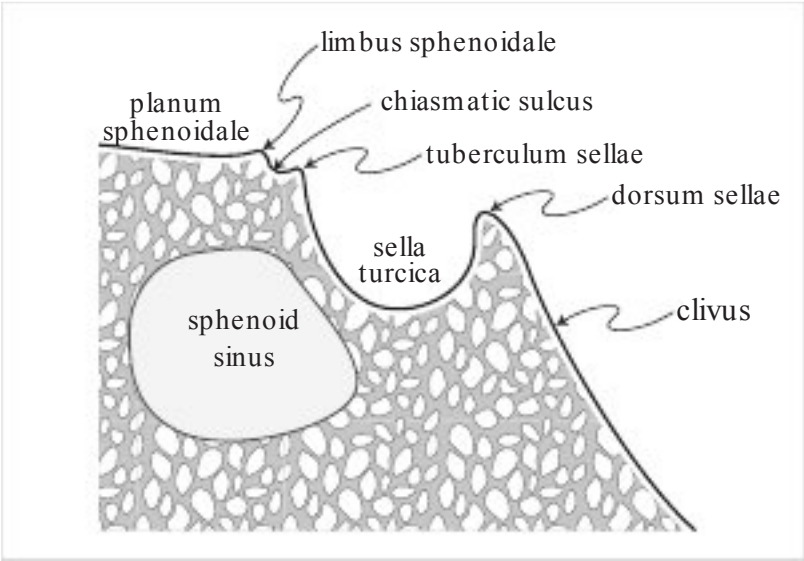


Fig. 42.2 Anatomic locations of planum sphenoidale and tuberculum sellae

42.3.5 Planum sphenoidale meningiomas

Arise from the flat part of the sphenoid bone (see □ Fig. 42.2) anterior to the chiasmatic sulcus in the posterior part of the anterior cranial fossa.

42.3.6 Tuberculum sellae meningiomas (TSM)

The site of origin of these tumors is only about 2 cm posterior to that of olfactory groove meningiomas.¹² The tuberculum sellae is the bony elevation between the chiasmatic sulcus and the sella turcica (see □ Fig. 42.2). By definition, the anterior margin of the chiasmatic sulcus (the limbus sphenoidale) is the demarcation between the anterior and middle cranial fossa. Therefore these tumors originate in the middle fossa (unlike planum sphenoidale meningiomas which are in the anterior fossa).

TSMs are notorious for producing visual loss (chiasmal syndrome = primary optic atrophy + bitemporal hemianopsia). When a TSM grows posteriorly into the sella turcica it may be mistaken for a pituitary macroadenoma (see □ Fig. 89.3 for MRI and differentiating features).

42.3.7 Foramen magnum meningiomas

As with any foramen magnum (FM) lesion (p.1367); the neurologic symptoms and signs can be very confusing and often do not initially suggest a tumor in this location.

In the French Cooperative Study, there were 106 FM meningiomas,¹³ 31%arose from the anterior lip, 56%were lateral, and 13%arose from the posterior lip of the FM. Most are intradural, but they can be extradural or a combination (the latter 2 have a lateral origin and are often invasive, which makes total removal more difficult).¹⁴ They may be above, below, or on both sides of the vertebral artery.¹⁴

42.4 Pathology

Four critical histopathological variables:

- 1. grade see □ Table 42.2
- 2. histological subtype see □ Table 42.2
- 3. proliferation indices (p.694)
- 4. brain invasion (p.694)

There are a number of pathologic classification systems,^{15,16 (p 465),17} and transitional forms between the major types exist. More than one histological pattern may be seen in a given tumor. The WHO 2000 classification is shown in □ Table 42.2.

- 1. meningiomas with low risk of both recurrence and/or aggressive growth (WHO grade III)
 - a) meningothelial or meningotheliomatous, AKA syncytial: the most common. Sheets of polygonal cells. Some use the term angiomatous for meningotheliomatous variety with closely packed blood vessels
 - b) fibrous or fibroblastic: cells separated by connective tissue stroma. Consistency is more rubbery than meningotheliomatous or transitional
 - c) transitional: intermediate between meningotheliomatous and fibrous. Cells tend to be spindle shaped, but areas of typical meningotheliomatous cells occur. Whorls, some of which are calcified (psammoma bodies)
 - d) psammomatous: calcified meningothelial whorls
 - e) angiomatous
 - f) microcystic: AKA “humid” or vacuolated meningioma. The characteristic dilated extracellular spaces are usually empty, but occasionally contain substance that stains positive for PAS (? glycoprotein) or contain fat.¹⁸ The cysts may coalesce and form grossly or radiologically visible cysts and may resemble astrocytomas
 - g) secretory
 - h) lymphoplasmacyte-rich
- 2. meningiomas with greater recurrence risk and/or aggressive growth include
 - a) atypical meningioma: increased mitotic activity (1–2 mitotic figure/high-powered field), increased cellularity, focal areas of necrosis, giant cells. Cellular pleomorphism is not unusual

Table 42.2 WHO classification of meningiomas	
WHO Grade	Meningiomas
WHO I	meningothelial fibrous (fibroblastic) transitional (mixed) psammomatous angiomatous microcystic secretory lymphoplasmacyte-rich metaplastic
WHO II	chordoid clear cell (intracranial) atypical
WHO III	papillary rhabdoid (see text) anaplastic

but is not significant in and of itself. Increasing atypia appears to correlate with increasing aggressiveness

- b) rhabdoid meningiomas: usually have malignant features and behave aggressively. Behavior in the absence of malignant features is undetermined¹⁹
- c) malignant meningiomas: AKA anaplastic, papillary or sarcomatous. Characterized by frequent mitotic figures, cortical invasion, rapid recurrence even after apparent total removal,²⁰ and, rarely, metastases (see below). Frequent mitotic figures (≥ 4 mitoses per high-power field) or the presence of papillary features are strong predictors of malignancy. May be more common in younger patients

Obsolete terms (in the current WHO classification) presented for context in older literature: metaplastic, myxomatous, xanthomatous (abundant cytoplasmic lipids; appear vacuolated), lipomatous, granular, chondroblastic, osteoblastic, melanotic. Angioblastic or (meningeal) hemangiopericytomas; true hemangiopericytomas are sarcomas (p.701). (Others use the term “angioblastic” for tumors histologically similar to hemangioblastoma. Angioblastic meningiomas were felt to have more malignant clinical characteristics than other forms.¹⁶ (p 479–83))

Proliferation indices

Due to variation between institutions and observers, it is advised that proliferation indices (e.g. Ki-67 or MIB-1) not be used as the sole discriminant for grading. However, these indices do correlate with prognosis (see □ Table 42.3). Adding the phrase “with high proliferative activity” is suggested for tumors with a very high index.¹⁹

42.4.1 Brain invasion

The presence of brain invasion increases the likelihood of recurrence to levels similar to atypical meningiomas (not anaplastic),²² but is not an indicator of malignant grade. Brain invasion in atypical meningiomas does not dictate malignant behavior. Adding the phrase “with brain invasion” is suggested to denote higher risk of recurrence.¹⁹

42.4.2 Metastases

Very rarely a meningioma may metastasize outside the CNS. Most of these are angioblastic or malignant. Lung, liver, lymph nodes and heart are the most common sites.

42.4.3 Differential diagnosis/diagnostic considerations of meningioma

- 1. multiple meningiomas: suggests neurofibromatosis 2 (NF2)
- 2. pleomorphic xanthoastrocytoma (PXA) (p.635): may mimic meningiomas since they tend to be peripherally located and may have a dural tail
- 3. Rosai-Dorfman disease: especially if extracranial lesions are also identified. A connective tissue disorder with sinus histiocytosis and massive painless lymphadenopathy (most have cervical lymphadenopathy). Usually in young adults. Isolated intracranial involvement is rare. MRI: dural-based enhancing mass with signal characteristics similar to meningioma, may have dural tail. Most common intracranial locations: cerebral convexities, parasagittal, suprasellar, cavernous sinus. Pathology: dense fibrocollagenous connective tissue with spindle cells and lymphocytic infiltration, stains for CD68 & S-100. Histiocytic proliferation without malignancy. Foamy histiocytes are characteristic. Surgery and immunosuppressive therapy not effective. Low-dose XRT may be the best option

Table 42.3 Ki-67 proliferation index in meningiomas ^{21a}		
Description & WHO grade	Mean Ki-67 index ^a	Recurrence rate
Common meningioma (WHO grade I)	0.7%	9%
Atypical meningioma (WHO grade II)	2.1%	29%
Anaplastic meningioma (WHO grade III)	11%	50%
^a not recommended for grading (see text)		

42.5 Presentation

Symptoms depend on the location of the tumor, and some specific locations are associated with well-described symptom complexes.

Seizures may occur with supratentorial meningiomas as a result of irritation of the cerebral cortex.

Asymptomatic meningiomas

Meningiomas are the most common primary intracranial tumors, and most remain asymptomatic throughout the patient's life.²³ The routine use of CT & MRI for numerous indications inevitably results in the discovery of incidental (asymptomatic) meningiomas. In a population based study (the study population was middle class Caucasians and result may not be generalizable to other groups),²³ incidental meningiomas were seen in 0.9% of MRIs. In another series, 32% of primary brain tumors seen on imaging studies were meningiomas, and 39% of these were asymptomatic.²⁴ Of 63 cases followed for >1 year with nonsurgical management, 68% showed no increase in size over an average follow-up of 36.6 mos, whereas 32% increased in size over 28 mos average follow-up.²⁴ Asymptomatic meningiomas with calcification seen on CT and/or hypointensity on T2WI MRI appeared to have a slower growth rate.²⁴

Data is lacking to make evidence-based management guidelines. A suggestion is to obtain a follow-up imaging study 3–4 months after the initial study to rule-out rapid progression, and then repeat annually for 2–3 years. The development of symptoms would prompt performing a study at that time.

Treatment is indicated for lesions that produce symptoms that cannot be satisfactorily controlled medically, or for those that demonstrate significant continued growth on serial imaging studies. When surgery was performed, the perioperative morbidity rate was statistically significantly higher in patients >70 years old (23%) than in those <70 (3.5%).²⁴

42.6 Evaluation

42.6.1 MRI

Occasionally may be isointense with brain on T1WI and T2WI, but most enhance with gadolinium. Brain edema may or may not be present. Calcifications appear as signal voids on MRI. Gives information regarding patency of dural venous sinuses (accuracy in predicting sinus involvement is $\approx 90\%$ ⁵). “Dural tail” is a common finding.²⁶

42.6.2 CT

Appear as homogeneous, densely enhancing mass with broad base of attachment along dural border. Non-contrast Hounsfield numbers of 60–70 in a meningioma usually correlates with presence of psammomatous calcifications. There may be little cerebral edema, or it may be marked and may extend throughout the white matter of the entire hemisphere.

Intraventricular meningiomas: 50% produce extraventricular edema. On angio, these may falsely appear malignant.

Prostate cancer may mimic meningioma (prostate mets to brain are rare, but prostate frequently goes to bone, and may go to skull and can cause hyperostosis).

42.6.3 Angiography

Classic pattern: “comes early, stays late” (appears early in arterial phase, blush persists beyond venous phase). Meningiomas characteristically have external carotid artery feeders. Exceptions: low frontal median (e.g. olfactory groove) meningiomas which feed from the ICA (ethmoidal branches of the ophthalmic artery). Suprasellar meningiomas may also be fed by large branches of the ophthalmic arteries. Parasellar meningiomas tend to feed from the ICA. Secondary vascular supply may be derived from pial branches of the anterior, middle, and posterior cerebral arteries.

Artery of Bernasconi & Cassinari AKA artery of tentorium (a branch of the meningohypophyseal trunk) AKA the “Italian” artery: enlarged in lesions involving tentorium (e.g. tentorial meningiomas).

Angiography also gives information about occlusion of dural venous sinuses, especially for parasagittal/falx meningiomas. Oblique views are often best for evaluating patency of the superior sagittal sinus (SSS). Angiography can also help confirm diagnosis by the distinctive prolonged homogeneous tumor blush. Angiography also provides an opportunity for pre-op embolization (see below).

Pre-op embolization: Reduces the vascularity of these often bloody tumors, facilitating surgical removal. Timing of subsequent surgery is controversial. Some advocate waiting 7–10 days to permit tumor necrosis which simplifies resection.^{27,28} Complications include: hemorrhage (intratumoral and SAH), cranial nerve deficits (usually transient), stroke from embolization through ICA or VA anastomoses, scalp necrosis, retinal embolus, and potentially dangerous tumor swelling. Some meningiomas (e.g. olfactory groove) are less amenable to embolization.

42.6.4 Plain x-rays

May show: calcifications within the tumor (in $\approx 10\%$), hyperostosis or blistering of the skull (including floor of frontal fossa with olfactory groove meningiomas), enlargement of vascular grooves (especially middle meningeal artery).

42.7 Treatment

42.7.1 General information

Surgery is the treatment of choice for symptomatic meningiomas. Incidental meningiomas with no brain edema or those presenting only with seizures that are easily controlled medically may be managed expectantly with serial imaging as meningiomas tend to grow slowly, and some may “burn out” and cease growing (p.695).

Radiation therapy is considered for patients who are not surgical candidates, for some deep inaccessible tumors, for multiply recurrent meningiomas or for atypical or malignant meningiomas either after initial subtotal resection or after first recurrence.

42.7.2 Surgical technique

General information

These tumors are often very bloody. Preoperative embolization and autologous blood donation may be helpful for specific tumors. General principles of meningioma surgery²⁹:

1. early interruption of the blood supply to the tumor
2. internal decompression (using ultrasonic aspirator, cautery loops...)
3. dissection of the tumor capsule from the brain by cutting and coagulating vascular and arachnoid attachments while infolding the tumor into the area of decompression with minimal retraction on adjacent brain
4. removal of attached bone and dura when possible

Position

As usual, the head should be elevated $\approx 30^\circ$ above the right atrium.

For meningiomas involving the superior sagittal sinus (SSS)³⁰:

- for tumors involving the anterior third of the SSS: supine semi-sitting position
- for tumors of the middle third of the SSS: lateral position with the side of the tumor down, the neck tilted 45° toward the upward shoulder
- for tumors of the posterior third of the SSS: prone position

Sinus involvement

Greenberg IMHO

Attempting to occlude or bypass the middle third of the superior sagittal sinus involved with meningioma is treacherous. Even in expert hands, there is significant risk of venous infarction/sinus occlusion with 8% morbidity and 3% mortality,¹⁰ and complete removal is still not assured.³¹ Venous drainage may occur through the dura adjacent to the sinus, in the skin, bone of the skull and even the tumor itself may participate. It is almost always preferable to leave residual tumor and consider treating it with radiation therapy than to cause a venous infarction.

Alternatives for treatment of dural sinus involvement include:

□ Superior sagittal sinus (SSS). If the tumor occludes the SSS, it has been suggested that the sinus can be resected carefully preserving veins draining into the patent portions of the sinus. □ However, this should be undertaken with great trepidation since patients still not infrequently develop venous infarcts, probably as a result of loss of minimal sinus flow and venous channels in the dura. Before ligating the sinus, the lumen should be inspected for a tail of tumor within

partial occlusion of superior sagittal sinus:

1. anterior to the coronal suture, the sinus may usually be divided safely
2. posterior to the coronal suture (or, perhaps more accurately – posterior to the vein of Trolard), it must not be divided or else severe venous infarction will occur
 - a) with superficial involvement (Type I, □ Fig. 42.1), tumor may be dissected off the sinus with care to preserve patency
 - b) with extensive involvement:
 - sinus reconstruction: hazardous. Thrombosis rate using venous graft approaches 50% and is close to 100% with artificial grafts (e.g. Gore-Tex) which should not be used
 - it may be best to leave residual tumor, and follow with CT or MRI. If the residual tumor grows, or if the Ki-67 score is high (p.694), SRS may be used; SRS may also be used as initial treatment for tumors that are <2.3–3 cm (p.1564).

□ Transverse sinus (TS). A patent dominant TS must not be suddenly occluded

Sphenoid wing, parasagittal or falx meningiomas (general principles)

Once tumor is exposed a partial internal debulking is performed. Then the point of attachment (to the falx or sphenoid bone) is peeled away using bipolar cautery to divide feeding vessels. Then the main portion of the tumor may be separated from brain, with the tumor being avascular once the vascular pedicle has been transected.

Parasagittal and falx meningiomas

The inferior portion of the tumor may adhere to branches of the anterior cerebral artery. Middle or posterior third tumors are exposed using a horseshoe incision based in the direction of the major scalp feeding vessels. The patient may be placed in a lateral position, or the sitting position may be used with doppler monitoring for air embolism (p.1445). Anterior third tumors are approached using a bicoronal skin incision with the patient supine. For tumors that cross the midline, burr holes are placed to straddle the SSS. For managing superior sagittal sinus involvement, see above.

Since these tumors are often debulked from the inside, removal tends to be bloodier than meningiomas that can be removed in 1 piece. The ability to embolize these tumors pre-op is somewhat limited, but may be an adjunct. Technique: Cut through tumor leaving a thin layer on dura. Then remove the now relatively avascular part that impresses the brain. Then make an incision through the dura near the tumor; it tends to be bloody, but once you have control of both sides of the dura you can begin to excise the dura around the tumor (you may need to leave a cuff on SSS if it is involved).

Sphenoid wing meningiomas

Apterional craniotomy (p.1453) is utilized. The neck is extended to allow gravity to retract the brain off of the floor of the skull.

Lateral sphenoid wing meningiomas: The approach to these tumors is often similar to convexity meningiomas. The height of the skin incision and bone opening should be high enough to encompass the tumor.

Medial sphenoid wing meningiomas: A lumbar drain is used. The head is turned 30° off the vertical. Aggressive extradural removal of sphenoid wing is performed. An FTOZ approach may provide additional exposure. The sylvian fissure is split widely. The ICA and MCA are often encased by tumor (look for the appearance of “grooves” on the surface of the tumor on MRI, which indicates vessels, e. g. MCA). To locate the ICA, identify MCA branches and follow them proximally into the tumor. The optic nerve is best identified at the optic canal. Avoid excessive retraction of the optic apparatus. The deep portion of the tumor often has numerous small parasitic vessels from the ICA (which makes this part very bloody), and may also invade the lateral wall of the cavernous sinus (which creates risk of cranial nerve deficits with attempted removal). Therefore, the recommendation is to leave some tumor behind and use radiosurgery to deal with it.

Olfactory groove meningiomas

Approached via a bifrontal craniotomy (preserving the periosteum to cover the frontal air sinus and floor of frontal fossa at the end of the case). Small tumors may be approached via unilateral craniotomy on the side with the most tumor).^{6 (p 3284)} For large tumors, a lumbar CSF drain will help with brain relaxation.¹² The head is rotated 20° to one side to facilitate dissection of the anterior cerebral arteries and optic nerve while preserving visualization of both sides of the tumor involvement.³² The neck is slightly extended. The dura is opened low, and the superior sagittal sinus is ligated and divided at this location. Amputation of the frontal pole should be done if necessary to avoid excessive retraction. Vascular feeding arteries come through the floor of the frontal fossa in the midline. Initially, the anterior tumor capsule is opened and the tumor debulked from within heading towards the floor of the frontal fossa to interrupt the blood supply. The posterior capsule of the tumor is dissected carefully as this portion of the tumor may encase branches of the anterior cerebral artery, and/or optic nerves and chiasm. A large tumor with suprasellar extension usually displaces the optic nerve and chiasm inferiorly.¹² If necessary, the frontopolar branch and other small branches may be sacrificed without problem.³³ Post-op risks include CSF leak through the ethmoid sinuses.

Tuberculum sellae meningiomas

These tumors typically displace both optic nerves posteriorly and laterally.¹² Occasionally, the nerves are completely engulfed by tumor.

Cerebellopontine angle meningiomas

Usually arise from the meninges covering the petrous bone. May be divided into those that occur anterior to, and those that occur posterior to the IAC.

Foramen magnum meningiomas

Tumors arising from the posterior or posterolateral lip of the foramen magnum (FM) are removed relatively easily. Anterior and lateral FM tumors may be operated by the posterolateral approach, and for anterior tumors,¹⁴ a transcondylar approach may alternatively be used.³⁴

With meningiomas below the vertebral artery (VA), the lower cranial nerves are displaced superiorly with the VA. However, when the tumor is above the VA, the position of the lower cranial nerves cannot be predicted.¹⁴

Large tumors may adhere to or encase neurovascular structures, and these should be internally debulked and then dissected free.

Posterior suboccipital approach: Used for meningiomas arising from the posterior lip of the FM or slightly posterolateral.

The patient is positioned prone or three-quarter prone. Neck flexion should be kept to a minimum to avoid brainstem compression by the tumor.³⁵ The surgeon must remain vigilant for the PICA and vertebral arteries, which may be encased.

42.7.3 Radiation therapy (XRT)

Generally regarded as ineffective as primary modality of treatment. Many prefer not to use XRT for “benign” lesions. Efficacy of XRT in preventing recurrence is controversial (see below under Recurrence); some surgeons reserve XRT for malignant (invasive), vascular, rapidly recurring (“aggressive”), or non-resectable meningiomas.

For recurrent atypical or anaplastic meningioma with residual disease post-op, XRT with 55–60 Gy is recommended.

42.8 Outcome

5 year survival for patients with meningioma³⁶: 91.3%

Recurrence

The extent of surgical tumor removal is the most important factor in the prevention of recurrence. The Simpson grading system for the extent of meningioma removal is shown in □ Table 42.4. An often overlooked aspect of the Simpson grading system is that it refers exclusively to removal of intradural tumor, and thus leaving tumor e.g. in the sagittal sinus could still be compatible with complete removal. Recurrence after gross total tumor removal occurred in 11–15% of cases, but was 29% when removal is incomplete (length of follow-up not specified)⁷; 5-year recurrence rates of 37%

Table 42.4 Simpson grading system for removal of meningiomas ³⁷	
Grade	Degree of removal
I	macroscopically complete removal with excision of dural attachment and abnormal bone (including sinus resection when involved)
II	macroscopically complete with endothermy coagulation (Bovie, or laser) of dural attachment
III	macroscopically complete without resection or coagulation of dural attachment or of its extradural extensions (e.g. hyperostotic bone)
IV	partial removal leaving tumor in situ
V	simple decompression (± biopsy)

³⁸-85%⁹ after partial resection are also quoted. The overall recurrence rate at 20 years was 19% in one series,⁴⁰ and 50% in another.³⁹ Malignant meningiomas have a higher recurrence rate than benign ones.

Value of XRT

A retrospective series of 135 non-malignant meningiomas followed 5–15 years post-op at UCSF revealed a recurrence rate of 4% with total resection, 60% for partial resection without XRT, and 32% for partial resection with XRT.⁴¹ Mean time to recurrence was longer in the XRT group (125 mos) than in the non-XRT group (66 mos). These results suggest that XRT may be beneficial in partially resected meningiomas. Alternatively, one can follow these patients with CT or MRI and use XRT for documented progression.

In addition to the usual side effects of XRT – see Radiation injury and necrosis (p.1560) – there is also a case report of a malignant astrocytoma developing after XRT was used to treat a meningioma.⁴²

References

[1] Mahmood A, Caccamo DV, Tomecek FJ, Malik GM. Atypical and Malignant Meningiomas: A Clinicopathological Review. *Neurosurgery*. 1993; 33:955–963

[2] Sheehy JP, Crockard HA. Multiple Meningiomas: A Long-Term Review. *J Neurosurg*. 1983; 59:1–5

[3] Kulali A, Ilcayto R, Rahmanli O. Primary calvarial ectopic meningiomas. *Neurochirurgia (Stuttg)*. 1991; 34:174–177

[4] Nakasu S, Hirano A, Shimura T, et al. Incidental Meningiomas in Autopsy Studies. *Surg Neurol*. 1987; 27:319–322

[5] Wara WM, Sheline GE, Newman H, et al. Radiation Therapy of Meningiomas. *AJR*. 1975; 123:453–458

[6] Youmans JR. *Neurological Surgery*. Philadelphia 1990

[7] Yamashita J, Handa H, Iwaki K, et al. Recurrence of Intracranial Meningiomas, with Special Reference to Radiotherapy. *Surg Neurol*. 1980; 14:33–40

[8] Cushing H, Eisenhardt L. In: *Meningiomas of the Sphenoidal Ridge. A. Those of the Deep or Clinoidal Third. Meningiomas: Their Classification, Regional Behaviour, Life History, and Surgical End Results*. Springfield, Illinois: Charles C Thomas; 1938:298–319

[9] Bonnal J, Brotchi J. Surgery of the superior sagittal sinus in parasagittal meningiomas. *J Neurosurg*. 1978; 48:935–945

[10] Sindou MP, Alvernia JE. Results of attempted radical tumor removal and venous repair in 100 consecutive meningiomas involving the major dural sinuses. *J Neurosurg*. 2006; 105:514–525

[11] Eskandary H, Hamzel A, Yasamy MT. Foot Drop Following Brain Lesion. *Surg Neurol*. 1995; 43:89–90

[12] Al-Mefty O, Sekhar LN, Janecka IP. In: *Tuberculum Sella and Olfactory Groove Meningiomas. Surgery of Cranial Base Tumors*. New York: Raven Press; 1993:507–519

[13] George B, Lot G, Velut S. Tumors of the Foramen Magnum. *Neurochirurgie*. 1993; 39:1–89

[14] George B, Lot G, Boissonnet H. Meningioma of the Foramen Magnum: A Series of 40 Cases. *Surg Neurol*. 1997; 47:371–379

[15] Zulch KJ. Histologic Typing of Tumors of the Central Nervous System. *International Histological Classification of Tumors*, no. 21. Geneva: World Health Organization; 1979

[16] Russell DS, Rubenstein LJ. *Pathology of Tumours of the Nervous System*. 5th ed. Baltimore: Williams and Wilkins; 1989

[17] Kleihues P, Burger PC, Scheithauer BW. The new WHO classification of brain tumors. *Brain Pathol*. 1993; 3:255–268

[18] Michaud J, Gagné F. Microcystic meningioma. *Clinicopathologic Report of Eight Cases*. *Arch Pathol Lab Med*. 1983; 107:75–80

[19] Kleihues P, Louis DN, Scheithauer BW, Rorke LB, Reifenberger G, Burger PC, Cavenee WK. The WHO classification of tumors of the nervous system. *J Neuropathol Exp Neurol*. 2002; 61:215–25; discussion 226-9

[20] Thomas HG, Dolman CL, Berry K. Malignant Meningioma: Clinical and Pathological Features. *J Neurosurg*. 1981; 55:929–934

[21] Kolles H, Niedermayer I, Schmitt C, Henn W, Feld R, Steudel WI, Zang KD, Feiden W. Triple approach for diagnosis and grading of meningiomas: histology, morphometry of Ki-67/Feulgen stainings, and cytogenetics. *Acta Neurochir (Wien)*. 1995; 137:174–181

[22] Perry A, Scheithauer BW, Stafford SL, Lohse CM, Wollan PC. "Malignancy" in meningiomas: A clinicopathologic study of 116 patients with grading implications. *Cancer*. 1999; 85:2046–2056

[23] Vernooji MW, Ikram A, Tanghe HL, et al. Incidental findings on brain MRI in the general population. *N Engl J Med*. 2007; 357:1821–1828

[24] Kuratsu J-I, Kochi M, Ushio Y. Incidence and Clinical Features of Asymptomatic Meningiomas. *J Neurosurg*. 2000; 92:766–770

- [25] Zimmerman RD, Fleming CA, Saint-Louis LA, et al. Magnetic Resonance of Meningiomas. *AJNR*. 1985; 6:149–157
- [26] Taylor SL, Barakos JA, Harsh GR, Wilson CB. Magnetic Resonance Imaging of Tuberculum Sellae Meningiomas: Preventing Preoperative Misdiagnosis as Pituitary Macroadenoma. *Neurosurgery*. 1992; 31:621–627
- [27] Chun JY, McDermott MW, Lamborn KR, Wilson CB, Higashida R, Berger MS. Delayed surgical resection reduces intraoperative blood loss for embolized meningiomas. *Neurosurgery*. 2002; 50:1231–5; discussion 1235–7
- [28] Kai Y, Hamada J, Morioka M, Yano S, Todaka T, Ushio Y. Appropriate interval between embolization and surgery in patients with meningioma. *AJNR Am J Neuroradiol*. 2002; 23:139–142
- [29] Ojemann RG. Management of Cranial and Spinal Meningiomas. *Clin Neurosurg*. 1992; 40:321–383
- [30] Colli BO, Carlotti CG. Parasagittal meningiomas. *Contemp Neurosurg*. 2007; 29:1–8
- [31] Heros RC. Meningiomas involving the sinus. *J Neurosurg*. 2006; 105:511–513
- [32] Bogaev CA, Sekhar LN, Sekhar LN, Fessler RG. In: Olfactory groove and planum sphenoidale meningiomas. *Atlas of Neurosurgical Techniques*. New York: Thieme Medical Publishers, Inc.; 2006:608–617
- [33] Ojemann RG, Schmidek HH, Sweet WH. In: *Surgical Management of Olfactory Groove Meningiomas. Operative Neurosurgical Techniques*. 3rd ed. Philadelphia: W.B. Saunders; 1995:393–401
- [34] Hakuba A, Tsujimoto T, Sekhar LN, Janecka IP. In: *Transcondyle Approach for Foramen Magnum Meningiomas. Surgery of Cranial Base Tumors*. New York: Raven Press; 1993:671–678
- [35] David CA, Spetzler R. Foramen Magnum Meningiomas. *Clin Neurosurg*. 1997; 44:467–489
- [36] Mahaley MS, Mettlin C, Natarajan N, Laws ER, et al. National Survey of Patterns of Care for Brain-Tumor Patients. *J Neurosurg*. 1989; 71:826–836
- [37] Simpson D. The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry*. 1957; 20:22–39
- [38] Mirimanoff RO, Dosoretz DE, Lingood RM, et al. Meningioma: Analysis of Recurrence and Progression Following Neurosurgical Resection. *J Neurosurg*. 1985; 62:18–24
- [39] Adegbite AV, Khan MI, Paine KWE, et al. The Recurrence of Intracranial Meningiomas After Surgical Treatment. *J Neurosurg*. 1983; 58:51–56
- [40] Jaaskelainen J. Seemingly complete removal of histologically benign intracranial meningioma: late recurrence rate and factors predicting recurrence in 657 patients. A multivariate analysis. *Surg Neurol*. 1986; 26:461–469
- [41] Barbaro NM, Gutin PH, Wilson CB, et al. Radiation Therapy in the Treatment of Partially Resected Meningiomas. *Neurosurgery*. 1987; 20:525–528
- [42] Zuccarello M, Sawaya R, deCourten-Myers. Glioblastoma Occurring After Radiation Therapy for Meningioma: Case Report and Review of Literature. *Neurosurgery*. 1986; 19:114–119

43 Other Tumors Related to the Meninges

43.1 Mesenchymal, non-meningothelial tumors

43.1.1 Hemangiopericytoma

A sarcoma arising from pericytes (surrounding blood vessels). May metastasize (usually to bone, lung or liver). Occur $\approx$ anywhere (soft tissues, muscles, thoracic aorta, kidney, omentum...). May mimic meningioma on CT or MRI (MRS may help distinguish¹). Recurrence is common, sometimes late. Neurosurgically relevant sites:

1. intracranial: includes intraventricular
2. spinal

Treatment: Surgery is primary treatment. XRT may reduce recurrence rate. Chemotherapy is used for metastases or for tumors failing local control measures.

43.1.2 Primary cerebral sarcoma

Rare. May result from sarcomatous change in preexisting tumor such as meningioma, glioblastoma, or oligodendroglioma.

43.2 Primary melanocytic lesions

Primary CNS melanoma

Probably arises from melanocytes in the leptomeninges. May spread through CSF pathways. May occasionally metastasize outside the CNS to produce systemic metastases.²

The peak age for this tumor is in the 4th decade (compared to the 7th decade for primary cutaneous melanoma).³

43.3 Hemangioblastoma

43.3.1 General information

Key concepts

- highly vascular well-circumscribed solid or cystic neoplasm of CNS or retina
- the most common primary intra-axial tumor in the adult posterior fossa
- may occur sporadically or as part of von Hippel-Lindau disease
- on imaging, may be solid, or cystic with enhancing mural nodule
- ☐ CBC: may be associated with erythrocytosis (polycythemia)

Hemangioblastomas⁴ (p. 772–82) (HGB) are histologically benign tumors. Intracranially, they occur almost exclusively in the p-fossa (hemangioblastomas are the most common primary intra-axial p-fossa tumor in adults). May occur in cerebellar hemisphere, vermis or brainstem. Less than 100 supratentorial cases have been reported. May also occur in spinal cord (p. 789) (1.5–2.5% of spinal cord tumors). Relationship and/or identity with angioblastic meningiomas is controversial. Also difficult to distinguish histologically from a renal cell carcinoma.

HGB may occur sporadically, but 20% occur as part of von Hippel-Lindau (VHL) disease (see below). Retinal HGB and/or angiomas occur in 6% of patients with cerebellar HGBs.

43.3.2 Hemangioblastomas (in general)

Epidemiology

HGB represent 1–2.5% of intracranial tumors. They comprise 7–12% of primary p-fossa tumors.⁵ 5–30% of cases of cerebellar HGB and 80% of spinal HGB are associated with VHL (see above).

Sporadic cases tend to present in the 4th decade, whereas VHL cases present earlier (peak in 3rd decade). In sporadic cases, the HGB are solitary and originate in the cerebellum (83–95%), spinal cord (3–13%), medulla oblongata (2%)⁶ or cerebrum (1.5%).⁵ $\approx$ 30% of patients with cerebellar HGB have VHL.⁷

Presentation

S/S of cerebellar HGB are usually those of any p-fossa mass – H/A, N/V, cerebellar findings...; see Posterior fossa (infratentorial) tumors (p.592) – and obstructive hydrocephalus may occur. HGB is rarely documented as a cause of apoplexy due to intracerebral hemorrhage (ICH) (lobar or cerebellar), however, some studies indicate that if cases of ICH are carefully examined, abnormal vessels consistent with HGB (and occasionally misidentified as AVM) may be found with surprising frequency (in spite of negative CT and/or angiography).⁸

Retinal HGBs tend to be located peripherally, and may hemorrhage and cause retinal detachment. Erythrocytosis may be due to erythropoietin liberated by the tumor.

Pathology

No report of malignant change. May spread thru CSF after surgery, but remain benign. No true capsule, but usually well circumscribed (narrow zone of infiltration). May be solid, or cystic with a mural nodule (70% of cerebellar lesions are cystic; nodules are very vascular, appear red, are often located near pial surface, and may be as small as 2 mm; cyst fluid is clear yellow with high protein). In cystic lesions, the cyst wall is lined with non-neoplastic compressed cerebellum. The cyst develops because the vessel walls are so thin that they leak water, proteins don't cross as readily.

Cardinal feature: numerous capillary channels, lined by a single layer of endothelium, surrounded by reticulin fibers (stains positive with reticulin stain). Macrophages stain PAS positive.

Three types of cells:

1. endothelial
2. pericytes: surrounded by basement membrane
3. stromal: polygonal. Foamy clear cytoplasm, often lipid laden. Origin controversial

Three types of HGB recognized⁹:

1. juvenile: thin walled capillaries & dilated vessels tightly packed
2. transitional: thin walled capillaries & dilated vessels intermingled with stromal cells, some of which are lipid laden (sudanophilic)
3. clear cell: neoplasm made up almost entirely of sheets of xanthoma cells with a rich vascular stroma

Cyst patterns¹⁰:

1. no associated cysts: 28%
2. peritumoral cyst alone: 51%
3. intratumoral cyst: 17%
4. peritumoral AND intratumoral cysts: 4%

Evaluation

Patients with a p-fossa HGB (radiologically suspected or histologically proven) should undergo MRI of entire neuraxis because of possibility of spinal HGBs (may be distant from p-fossa lesion; may suggest possibility of VHL).

CT: solid lesions are usually isodense with intense contrast enhancement. Cystic HGBs remain low density with contrast, with the nodule enhancing.

MRI: preferable to CT due to the tumor's predilection for the p-fossa. May show serpentine vascular signal voids, especially in the periphery of the lesion. Also, peripheral hemosiderin deposits may occur from previous hemorrhages.⁵

Vertebral angiography: usually demonstrates intense vascularity (most other tumors of the p-fossa are relatively avascular). May be required in HGBs where nodule is too small to be imaged on CT/MRI. 4 patterns: 1) vascular mural nodule on side of avascular cyst, 2) vascular lesion surrounding avascular cyst, 3) solid vascular mass, & 4) multiple, separate vascular nodules.

Labs: often discloses polycythemia (no hematopoietic foci within tumor). In cases with suggestive history, labwork to rule-out catecholamine production from pheochromocytoma may be indicated; see Endocrine/laboratory studies (p.655).

Treatment

Surgery

Surgical treatment may be curative in cases of sporadic HGB, not in VHL.

Pre-operative embolization may help reduce the vascularity.

Cystic HGBs require removal of mural nodule (otherwise, cyst will recur). The cyst wall is not removed unless there is evidence of tumor within the cyst wall on MRI (typically thick-walled cysts) or visually at the time of surgery.¹⁰ 5-ALA fluorescence may aid in visual localization of small hemangioblastomas within the cyst wall.¹¹

Solid HGBs tend to be more difficult to remove. They are treated like AVMs (avoid piecemeal removal), working along margin and devascularizing blood supply. A helpful technique is to shrink the tumor by laying a length of bipolar forceps along tumor surface and coagulating. HGBs with attachment to floor of 4th ventricle may be hazardous to remove (cardio-respiratory complications).

Multiple lesions: if $\geq 0.8\text{--}1$ cm diameter: may treat as in solitary lesion. Smaller and deeper lesions may be difficult to locate at time of surgery.

Cystic brainstem HGB: the solid nodule of the tumor is removed under the microscope by bipolarizing and cutting the gliotic adhesions to the parenchyma. Removal of the cyst wall is not necessary. There is often a cleavage between the tumor and the floor of the fourth ventricle which facilitates tumor removal. To reduce bleeding, avoid piecemeal removal. Preserve large draining veins until the arterial feeders to the mural nodule have been isolated and resected.¹²

Radiation treatment

Effectiveness is dubious. May be useful to reduce tumor size or to retard growth, e.g. in patients who are not surgical candidates, for multiple small deep lesions, or for inoperable brainstem HGB. Does not prevent regrowth following subtotal excision.

43.3.3 von Hippel-Lindau disease (VHL)

General information

Key concepts

- disorder with hemangioblastomas (HGB) 1° of cerebellum, retina, brainstem & spinal cord, as well as renal cysts/tumors, pheochromocytomas (among others)
- autosomal dominant, due to inactivation of tumor suppressor gene on 3p25
- expression and age of onset are variable, but $\approx$ always manifests by age 60
- mean age of developing HGBs is at least 10 years younger than sporadic HGBs

A multisystem neoplastic disorder characterized by a tendency to develop hemangioblastomas (HGB) of the retina, brain and spinal cord, renal clear cell carcinoma (RCC), pheochromocytomas, endolymphatic sac tumors, and others^{6,13} (retinal location is 2nd most common after cerebellar, □ Table 43.1). The variability of von Hippel-Lindau disease (VHL) has lead some to suggest the use of the term hemangioblastomatosis.

Epidemiology

Incidence: 1 in 31,000 to 36,000 live births. $\approx$ 30% of patients with cerebellar HGB have VHL⁷

Genetics

Autosomal dominant inheritance with $\approx$ 95% penetrance at age 60 yrs.^{6,16} 4% of VHL are asymptomatic carriers. The VHL gene is a tumor suppressor gene on chromosome 3p25, and biallelic inactivation is required for tumor development.⁷ Most patients inherit a VHL gene (allele) with the germline mutation from the affected parent and a normal somatic (wild-type) VHL gene from the unaffected parent.

Subtypes of VHL

See reference.¹⁷

- Type I. May have any manifestation of VHL except pheochromocytoma
- Type II. Pheochromocytoma is characteristic
- Type IIA. Have low risk of renal cell Ca and neuroendocrine pancreatic tumor

Table 43.1 Associations with von Hippel-Lindau disease ^a	
Common lesions	Frequency in VHL
hemangioblastomas	
• cerebellum (solid or cystic)	80%
• retina	41–59%
• brainstem	10–25%
• spinal cord	10–50%
pancreatic tumors or cysts	22–80%
renal clear cell Ca & cysts	14–60%
polycythemia	9–20% of intracranial HGBs
Rare lesions (pertinent to nervous system)	Frequency in VHL
supratentorial hemangioblastoma	3–6%
cystadenomas of the broad ligament	10% of ♀
papillary cystadenomas of epididymis	25–60% ♂
endolymphatic sac tumors	10–15%
adrenal medullary pheochromocytoma (tends to be bilateral)	7–24%
^a see references ^{6,14,15} for more	

- Type IIB. Higher risk of renal cell Ca and neuroendocrine pancreatic tumor
- Type IIC. Risk of pheochromocytoma only (without risk of HGB or RCC)

Diagnostic criteria

Suggested diagnostic criteria for VHL:

1. in 80% of patients with VHL there is a multigenerational family history, and only 1 manifestation (CNS HGB or visceral lesion) is necessary to make the diagnosis in these patients
2. if no family history (20% of VHL, many of these represent a de novo mutation): 2 manifestations including 1 CNS or retinal HGB are required¹⁸
3. genetic testing in uncertain cases (see below)

Tumors associated with VHL

1. cerebellar hemangioblastomas (HGB):
 - a) prevalence: 44–72% of VHL patients
 - b) mean age of diagnosis in VHL patients with cerebellar hemangioblastomas is at least 10 years younger than sporadic cerebellar hemangioblastomas
 - c) cysts are commonly associated with cerebellar, brainstem and spinal HGBs
 - d) cysts grow at a faster rate than the HGBs, □ symptoms related to mass effect are frequently secondary to the cysts
 - e) cerebellar HGBs were located in the superficial, posterior and superior half of the cerebellar hemispheres¹⁰
 - f) 93% of the cerebellar HGBs were located in the cerebellar hemispheres and 7% in the vermis
 - g) the HGBs are also more frequently found in the superficial posterior half of the brainstem and the spinal cord
 - h) the HGBs have multiple sequential growth and quiescent phases
2. spinal cord hemangioblastomas
 - a) occur in 13–44% of VHL patients
 - b) 90% are located rostrally within the cervical and thoracic cord. Almost all (96%) of the tumors are located in the posterior half of the spinal cord, 4% are located in the ventral half of the spinal cord. 1–3% are found in the lumbosacral nerve roots

- c) by way of comparison, 80% of spinal cord HGB are associated with VHL, whereas only 5–31% of cerebellar HGB are associated with VHL
- d) 95% of symptom-producing spinal HGBs are associated with syringomyelia
- 3. brainstem hemangioblastomas
 - a) usually located in the posterior medulla oblongata usually around the obex and the region within the area postrema
- 4. pheochromocytomas (PCC): 20% of PCC are associated with VHL. PCC occur in 7–20% of families with VHL
- 5. endolymphatic sac tumors (ELST):
 - a) locally invasive benign tumors that occur in 10–15% of VHL patients (30% of these will develop bilateral ELSTs – VHL is the only disease with bilateral ELSTs). Rarely metastasize
 - b) presents with hearing loss in 95% (may be acute (86%) or insidious (14%), tinnitus (90%), vertigo or imbalance (66%), aural fullness (30%), and facial paresthesias (8%)
 - c) mean age of onset of hearing loss: 22 years (range: 12–50)¹⁹
- 6. retinal hemangioblastomas²⁰
 - a) occur in >50% of VHL patients. Mean age at presentation: 25 years
 - b) frequently bilateral, multifocal and recurrent
 - c) often asymptomatic. Visual symptoms occur with progressive growth, edema, retinal detachments and hard exudates
 - d) typically located in the periphery and near or on the optic disc
 - e) microangiomas measuring a few hundred microns without dilated feeding vessels may be located in the periphery
 - f) retrobulbar HGB are rare (5.3% in NIH cohort)²¹
 - g) severity of optic disease correlates with CNS and renal involvement
 - h) early diagnosis and treatment with laser photocoagulation, and cryotherapy can prevent visual loss. Low dose external XRT may be an option for refractory cases
- 7. renal-cell carcinoma (RCC)^{15,22,23,24,25,26,27,28}
 - a) the most common malignant tumor in VHL. Usually a clear cell carcinoma
 - b) lifetime risk for RCC in VHL: $\approx 70\%$
 - c) the growth rate of RCC is highly variable
 - d) RCC is the cause of death in 15–50% of patients
 - e) metastases respond poorly to chemotherapy and radiation
 - f) bilateral and multiple lesions are common
 - g) partial nephrectomy or tumor enucleation is preferred to avoid/delay dialysis and transplantation
 - h) nephron- or renal-sparing surgery recommended for tumors < 3 cm
 - i) promising techniques: cryo- and radiofrequency ablation of tumors < 3 cm
- 8. renal cysts^{15,24,27,28,29}
 - a) 50–70% of VHL patients have bilateral and multiple renal cysts
 - b) rarely cause profound renal impairment
 - c) chronic renal failure or renal hypertension not as common as with polycystic kidney disease
- 9. epididymal cystadenomas
 - a) benign lesions that arise from the epididymal duct
 - b) found in 10–60% of male VHL patients
 - c) typically appear in the teenage years
 - d) may cause infertility if bilateral
 - e) may be multiple
- 10. broad ligament cystadenomas
 - a) arise from the embryonic mesonephric duct
 - b) true incidence unknown
 - c) rarely reported and usually not recognized in women with VHL
- 11. pancreatic neuroendocrine tumors and cysts
 - a) 35 to 70% of patients with VHL develop an endocrine tumor or cyst
 - b) pancreatic cysts are generally asymptomatic and often multiple
 - c) pancreatic neuroendocrine tumors are usually non-functional and 8% of them are malignant
 - d) differential diagnosis: pancreatic islet cell tumors, MEN2

Treatment

Resection of individual CNS tumors is usually reserved until symptomatic to decrease the number of operations over a lifetime since the tumors in VHL are usually multiple, tend to recur, and the growth pattern is saltatory. Surgery is the treatment of choice for accessible cystic HGBs. For details, see Treatment, under Hemangioblastoma (p. 702).

Stereotactic radiosurgery (SRS)³⁰: May provide local control rates of >50% over 5 years. SRS has been recommended for asymptomatic HBG >5 mm diameter if they are cystic or progressing in size during surveillance.³¹ Cranial treatment plan: using a median dose of 22 Gy (range: 12–40 Gy) prescribed to the median 82%isodose line in 1–4 sessions. In cystic lesions, treatment is confined to the contrast enhancing mural nodule (the cyst wall is not treated). Spinal treatment plan: median dose of 21 Gy (range 20–25 Gy) prescribed to the median 77%isodose line in 1–3 sessions. Radiosurgery is usually contraindicated in hemangioblastomas with a cyst.

Surveillance

Because of the lifetime risk of developing tumors, regular surveillance is needed. Various protocols have been proposed,^{33,34} including those by the NIH¹⁵ and the Danish clinical recommendations.³⁵ The algorithm recommended by the VHL Family Alliance for patients with VHL and at-risk relatives is shown in □ Table 43.2. (Screening at-risk relatives can be stopped at age 60 years if no abnormalities have been detected.)

Individuals who do not carry the altered gene on DNA testing do not require surveillance.

Table 43.2 Health-care provider’s surveillance guidelines for patients with or at risk for VHL ^a	
Age	Surveillance
Any age	DNA testing for VHL marker is available to identify family members at risk
From birth	check for neurologic deficit, nystagmus, strabismus, white pupil...& refer to retinologist for abnormal findings. Newborn hearing screening
1 year	retina exam ^b (especially if positive for VHL mutation)
2–10 years	Annual: • PE^c including orthostatic blood pressure measurement, neurologic exam, retina exam^b • blood test or 24° urine for catecholamines & metanephrines (p. 653). If elevated: abdominal MRI or MIBG scan (p. 653) • abdominal U/S starting at age 8 Every 2–3 years: complete audiology exam. Annually if hearing loss, tinnitus or vertigo
11–19 years	Every 6 months: retina exam ^b Annual: • PE (including scrotal exam in males), neuro exam • 24° urine for catecholamines & metanephrines (p. 653). If elevated: abdominal MRI or MIBG scan (p. 653) • abdominal U/S (kidneys, pancreas & adrenals). If abnormal: abdominal MRI or CT (except in pregnancy) Every 1–2 years or if symptoms develop: • gadolinium MRI of brain & spine. Annually at onset of puberty or before and after pregnancy (only for emergencies during pregnancy) • complete audiology exam. If abnormal, or if tinnitus or vertigo at any time: MRI of IAC to look for ELST
≥20 years	Annual: • dilated retina exam^b • PE (including scrotal exam in males), neuro exam • blood test or 24° urine for catecholamines & metanephrines (p. 653). If elevated: abdominal MRI or MIBG scan (p. 653) • check kidneys, pancreas & adrenals with abdominal U/S and at least every other year unenhanced/enhanced abdominal CT (not during pregnancy) Every 2 years: • (or before and after pregnancy, except for emergencies) gadolinium MRI of brain & spine • complete audiology exam. If abnormal, or if tinnitus or vertigo at any time: MRI of IAC to look for ELST
Prior to surgery or childbirth	• blood test or 24° urine for catecholamines & metanephrines (p. 653) to rule out pheochromocytoma
^a adapted ³² ^b indirect ophthalmoscope exam by retinologist familiar with VHL ^c abbreviations: PE=physical exam by physician familiar with VHL, ELST=endolymphatic sac tumor	

Prognosis

The lifespan of patients with VHL is decreased. 30–50% die of renal cell Ca (RCC). Metastases from RCC and neurologic complications from cerebellar HGB are the primary causes of death.

Metastases respond poorly to chemotherapy and XRT.

Resources

Genetic screening for VHL can be done at a few centers. Information for patients and families can be found at www.vhl.org/.

References

- [1] Barba I, Moreno A, Martinez-Perez I, et al. Magnetic resonance spectroscopy of brain hemangiopericytomas: high myoinositol concentrations and discrimination from meningiomas. *J Neurosurg.* 2001; 94:55–60
- [2] Savitz MH, Anderson PJ. Primary Melanoma of the Leptomeninges: A Review. *Mt Sinai J Med.* 1974; 41:774–791
- [3] Gibson JB, Burrows D, Weir WP. Primary Melanoma of the Meninges. *J Pathol Bacteriol.* 1957; 74:419–438
- [4] Wilkins RH, Rengachary SS. *Neurosurgery.* New York 1985
- [5] Ho VB, Smirniotopoulos JG, Murphy FM, Rushing EJ. Radiologic-Pathologic Correlation: Hemangioblastoma. *AJNR.* 1992; 13:1343–1352
- [6] Catapano D, Muscarella LA, Guarnieri V, Zelante L, D'Angelo VA, D'Agruma L. Hemangioblastomas of central nervous system: molecular genetic analysis and clinical management. *Neurosurgery.* 2005; 56:1215–21; discussion 1221
- [7] Hottinger AF, Khakoo Y. Neurooncology of familial cancer syndromes. *J Child Neurol.* 2009; 24:1526–1535
- [8] Wakai S, Inoh S, Ueda Y, et al. Hemangioblastoma Presenting with Intraparenchymatous Hemorrhage. *J Neurosurg.* 1984; 61:956–960
- [9] Silver ML, Hennigar G. Cerebellar Hemangioma (Hemangioblastoma). A Clinicopathological Review of 40 Cases. *J Neurosurg.* 1952; 9:484–494
- [10] Jagannathan J, Lonser RR, Smith R, DeVroom HL, Oldfield EH. Surgical management of cerebellar hemangioblastomas in patients with von Hippel-Lindau disease. *J Neurosurg.* 2008; 108:210–222
- [11] Utsuki S, Oka H, Sato K, Shimizu S, Suzuki S, Fujii K. Fluorescence diagnosis of tumor cells in hemangioblastoma cysts with 5-aminolevulinic acid. *J Neurosurg.* 2009. DOI: 10.3171/2009.5.JNS08442
- [12] Agrawal A, Kakani A, Vagh SJ, Hiwale KM, Kolte G. Cystic hemangioblastoma of the brainstem. *J Neurosci Rural Pract.* 2010; 1:20–22
- [13] Glenn GM, Linehan WM, Hosoe S, Latif F, et al. Screening for von Hippel-Lindau Disease by DNA Polymorphism Analysis. *JAMA.* 1992; 267:1226–1231
- [14] Wanebo JE, Lonser RR, Glenn GM, Oldfield EH. The natural history of hemangioblastomas of the central nervous system in patients with von Hippel-Lindau disease. *J Neurosurg.* 2003; 98:82–94
- [15] Butman JA, Linehan WM, Lonser RR. Neurologic manifestations of von Hippel-Lindau disease. *JAMA.* 2008; 300:1334–1342
- [16] Go RCP, Lamiell JM, Hsia YE, et al. Segregation and Linkage Analysis of von Hippel-Lindau Disease Among 220 Descendants from one Kindred. *Am J Human Genet.* 1984; 36:131–142
- [17] Friedrich CA. Genotype-phenotype correlation in von Hippel-Lindau syndrome. *Hum Mol Genet.* 2001; 10:763–767
- [18] Melmon KL, Rosen SW. Lindau's Disease. Review of the Literature and Study of a Large Kindred. *Am J Med.* 1964; 36:595–617
- [19] Manski TJ, Heffner DK, Glenn GM, Patronas NJ, Pikus AT, Katz D, Lebovics R, Sledjeski K, Choyke PL, Zbar B, Linehan WM, Oldfield EH. Endolymphatic sac tumors. A source of morbid hearing loss in von Hippel-Lindau disease. *JAMA.* 1997; 277:1461–1466
- [20] Chew EY. Ocular manifestations of von Hippel-Lindau disease: clinical and genetic investigations. *Trans Am Ophthalmol Soc.* 2005; 103:495–511
- [21] Meyerle CB, Dahr SS, Wetjen NM, Jirawuthiworavong GV, Butman JA, Lonser RR, Oldfield E, Rodriguez-Coleman H, Wong WT, Chew EY. Clinical course of retrobulbar hemangioblastomas in von Hippel-Lindau disease. *Ophthalmology.* 2008; 115:1382–1389
- [22] Niemela M, Lemeta S, Summanen P, Bohling T, Sainio M, Kere J, Poussa K, Sankila R, Haapasalo H, Kaariainen H, Pukkala E, Jaaskelainen J. Long-term prognosis of haemangioblastoma of the CNS: impact of von Hippel-Lindau disease. *Acta Neurochir (Wien).* 1999; 141:1147–1156
- [23] Choyke PL, Glenn GM, Walther MM, Zbar B, Linehan WM. Hereditary renal cancers. *Radiology.* 2003; 226:33–46
- [24] Meister M, Choyke P, Anderson C, Patel U. Radiological evaluation, management, and surveillance of renal masses in Von Hippel-Lindau disease. *Clin Radiol.* 2009; 64:589–600
- [25] Maher ER, Kaelin WG, Jr. von Hippel-Lindau disease. *Medicine (Baltimore).* 1997; 76:381–391
- [26] Hes FJ, Feldberg MA. Von Hippel-Lindau disease: strategies in early detection (renal-, adrenal-, pancreatic masses). *Eur Radiol.* 1999; 9:598–610
- [27] Bisceglia M, Galliani CA, Senger C, Stallone C, Sessa A. Renal cystic diseases: a review. *Adv Anat Pathol.* 2006; 13:26–56
- [28] Truong LD, Choi YJ, Shen SS, Ayala G, Amato R, Krishnan B. Renal cystic neoplasms and renal neoplasms associated with cystic renal diseases: pathogenetic and molecular links. *Adv Anat Pathol.* 2003; 10:135–159
- [29] Bradley S, Dumas N, Ludman M, Wood L. Hereditary renal cell carcinoma associated with von Hippel-Lindau disease: a description of a Nova Scotia cohort. *Can Urol Assoc J.* 2009; 3:32–36
- [30] Moss JM, Choi CY, Adler JR, Jr, Soltys SG, Gibbs IC, Chang SD. Stereotactic radiosurgical treatment of cranial and spinal hemangioblastomas. *Neurosurgery.* 2009; 65:79–85; discussion 85
- [31] Chang SD, Meisel JA, Hancock SL, Martin DP, McManus M, Adler JR, Jr. Treatment of hemangioblastomas in von Hippel-Lindau disease with linear accelerator-based radiosurgery. *Neurosurgery.* 1998; 43:28–34; discussion 34–5
- [32] VHL Family Alliance. VHL Handbook. Section 5: Suggested screening guidelines. 2009
- [33] Constans JP, Meder F, Maiuri F, Donzelli R, Spaziante R, de Divitiis E. Posterior fossa hemangioblastomas. *Surg Neurol.* 1986; 25:269–275
- [34] Hes FJ, van der Luijt RB. [Von Hippel-Lindau disease: protocols for diagnosis and periodical clinical monitoring. National Von Hippel-Lindau Disease Working Group]. *Ned Tijdschr Geneesk.* 2000; 144:505–509
- [35] Poulsen ML, Budtz-Jorgensen E, Bisgaard ML. Surveillance in von Hippel-Lindau disease (vHL). *Clin Genet.* 2009. DOI: 10.1111/j.1399-0004.2009.01281.x

Part XIII

Tumors Involving
Non-Neural Origin:
Metastases, Lymphomas,
Chordomas

44	Lymphomas and Hematopoietic Neoplasms	710
45	Pituitary Tumors – General Information and Classification	718
46	Pituitary Adenomas – Evaluation and Nonsurgical Management	730
47	Pituitary Adenomas – Surgical Management, Outcome, and Recurrence Management	747
48	Cysts and Tumor-Like Lesions	756
49	Pseudotumor Cerebri and Empty Sella Syndrome	766
50	Tumors and Tumor-Like Lesions of the Skull	775
51	Tumors of the Spine and Spinal Cord	783
52	Cerebral Metastases	800
53	Spinal Epidural Metastases	814

44 Lymphomas and Hematopoietic Neoplasms

44.1 CNS lymphoma

44.1.1 General information

44

Key concepts

- may be primary or secondary (pathologically identical)
- suspected with homogeneously enhancing lesion(s) in the central gray matter or corpus callosum (on MRI or CT) especially in AIDS patients
- may present with multiple cranial-nerve palsies
- diagnosis highly likely if tumor seen in conjunction with uveitis
- very responsive initially to steroids → short-lived disappearance (may produce “ghost tumors”)
- treatment: usually XRT ± chemotherapy. Role of neurosurgery usually limited to biopsy and/or placement of ventricular access reservoir for chemotherapy
- risk factors: immunosuppression (AIDS, transplants), Epstein-Barr virus, collagen vascular diseases

CNS involvement with lymphoma may occur secondarily from a “systemic” lymphoma, or may arise primarily in the CNS. It is controversial whether most intracranial malignant lymphomas are primary¹ or secondary.²

44.1.2 Primary vs. secondary lymphoma

Secondary CNS lymphoma

Non CNS lymphoma is the fifth most common cause of cancer deaths in the U.S., 63% of new cases are non-Hodgkin's. Secondary CNS involvement usually occurs late in the course. Metastatic spread of systemic lymphoma to the cerebral parenchyma occurs in 1–7% of cases at autopsy.³

Primary CNS lymphoma

Older names include: reticulum cell sarcoma and microglioma⁴ since they were thought to possibly arise from microglia which were considered part of the reticuloendothelial system.

A rare, malignant primary CNS neoplasm comprising 0.85–2% of all primary brain tumors and 0.2–2% of malignant lymphomas.⁵ May occasionally metastasize outside the CNS.

44.1.3 Epidemiology

The incidence of primary CNS lymphoma (PCNSL) is rising relative to other brain lesions, and will likely exceed that of low-grade astrocytomas and approach meningiomas. This is in part due to the occurrence of PCNSL in AIDS and transplant patients, but the incidence has also increased in the general population over the past 20 years.⁶

Male:female ratio = 1.5:1 (based on literature review⁷).

Median age at diagnosis: 52 yrs⁷ (younger among immunocompromised patients: ≈ 34 yrs).

Most common supratentorial locations: frontal lobes, then deep nuclei; periventricular also common. Infratentorially: cerebellum is the most common location.

44.1.4 Conditions with increased risk of primary CNS lymphomas (PCNSL)

1. collagen vascular disease
 - a) systemic lupus erythematosus
 - b) Sjögren's syndrome: an autoimmune connective tissue disorder
 - c) rheumatoid arthritis
2. immunosuppression
 - a) organ transplant recipients: related to chronic immunosuppression. Falls under category of post transplant lymphoproliferative disease (PTLD)⁸

- b) severe-congenital immunodeficiency syndrome (“SCIDS”)
- c) AIDS^{9,10}: CNS lymphoma occurs in $\approx 10\%$ of AIDS patients, and is the first presentation in 0.6%
- d) possibly increased incidence in the elderly due to reduced competency of immune system
- 3. Epstein-Barr virus¹¹ is associated with a broad spectrum of lymphoproliferative disorders, and is detectable in $\approx 30\text{--}50\%$ of systemic lymphomas, however, it has been associated with almost 100% of PCNSL,¹² especially AIDS-related cases^{13 (p 317)}

44.1.5 Pathology

Characteristic sites: corpus callosum, basal ganglia, periventricular.

The neoplastic cells are identical to those of systemic lymphomas. Most are bulky tumors that are contiguous with the ventricles or meninges.

Histologic distinguishing features: tumor cells form cuffs around blood vessels which demonstrate multiplication of basement membranes (best demonstrated with silver reticulum stain).

Frozen section distorts the cells and may lead to a misdiagnosis of malignant glioma.^{13 (p 320)}

Immunohistochemical stains differentiates B-cell lymphomas from T-cell lymphomas (B-cell types are more common, especially in PCNSL and in AIDS).

EM shows absence of junctional complexes (desmosomes) that are usually present in epithelial derived tumors.

Intravascular lymphomatosis¹⁴: Formerly: (malignant) angioendotheliomatosis. A rare lymphoma with no solid mass in which malignant lymphoid cells are found in the lumen of small blood vessels in affected organs. CNS involvement is reported in most cases. Presentation is non specific: patients are often febrile, and may present with progressive multifocal cerebrovascular events (including stroke or hemorrhage), spinal cord or nerve root symptoms including cauda equina syndrome (p.1050), encephalopathy or peripheral or cranial neuropathies.¹⁵ Initial transient cerebral symptoms may mimic TIAs or seizures. The ESR is often elevated prior to initiation of steroids. Lymphoma cells may be seen in the CSF.

Painful skin nodules or plaques occur in $\approx 10\%$ of cases, generally involving the abdomen or lower extremities, and these cases may be diagnosed with skin biopsy (differential diagnosis here includes angioendotheliomatosis, a benign capillary and endothelial cell disorder). Otherwise, diagnosis often requires brain biopsy (open or stereotactic), in which involved areas on imaging studies are targeted. Pathology: malignant lymphoid cells distend and occlude small arteries, veins and capillaries with little or no parenchymal extension.^{13 (p 324)} Treatment with combination chemotherapy can result in long-term remission in some patients, but early diagnosis before permanent damage occurs is critical (diagnosis is rarely made pre-mortem).

44.1.6 Presentation

General information

Presentation is similar with primary or secondary CNS lymphoma: the two most common manifestations are those due to epidural spinal cord compression and those of carcinomatous meningitis (multiple cranial nerve deficits) (p.811). Seizures occur in up to 30% of patients.¹

Symptoms

1. presents with non-focal non-specific symptoms in over 50% of patients; at time of presentation most commonly includes:
 - a) mental status changes in one-third
 - b) symptoms of increased ICP (H/A, N/V)
 - c) generalized seizures in 9%
2. focal symptoms in 30–42% of cases:
 - a) hemimotor or hemisensory symptoms
 - b) partial seizures
 - c) multiple cranial-nerve palsies (due to carcinomatous meningitis)
3. combination of focal and non-focal symptoms

Signs

1. non-focal in 16%
 - a) papilledema
 - b) encephalopathy
 - c) dementia

2. focal findings in 45% of cases:
 - a) hemimotor or hemisensory deficits
 - b) aphasia
 - c) visual field deficits
3. combination of focal and non-focal signs

Uncommon but characteristic syndromes

1. uveocyclitis, coincident with (in 6% of cases) or preceding the diagnosis of (in 11% of cases) lymphoma
2. subacute encephalitis with subependymal infiltration
3. MS-like illness with steroid-induced remission

44

44.1.7 Evaluation

All patients should be assessed (history, physical, and if appropriate, laboratory tests) for any of the conditions associated with lymphoma (p. 710). Since primary CNS lymphoma is less common than secondary involvement, any patient with CNS lymphoma should have work-up for occult systemic lymphoma including:

1. careful physical exam of all lymph nodes (LN)
2. evaluation of perihilar and pelvic LN (CXR, CT of chest & abdomen)
3. routine blood and urine testing
4. bone marrow biopsy
5. MRI of the entire spine
6. testicular ultrasound in males
7. ophthalmologic examination (including slit-lamp evaluation of both eyes) in all
 - a) for possible uveitis
 - b) $\approx 28\%$ of patients with PCNSL will also have intraocular lymphoma. Often resistant to methotrexate, but responds to low dose ocular XRT (7–8 Gy)

44.1.8 Diagnostic tests

General information

On imaging (CT or MRI) 50–60% occur in one or more cerebral lobes (in grey or white matter). 25% occur in deep midline structures (septum pellucidum, basal ganglion, corpus callosum). 25% are infratentorial. 10–30% of patients have multiple lesions at the time of presentation. In contrast, systemic lymphomas that spread to the CNS tend to present with leptomeningeal involvement instead of parenchymal tumors.¹⁶

CT

Non-AIDS-related cases tend to enhance homogeneously, whereas AIDS-related cases often have a necrotic center and appear as multifocal ring-enhancing lesions¹⁷ (the wall is thicker than with an abscess).

Non-AIDS related cases: CNS lymphomas should be suspected with homogeneously enhancing lesion(s) in the central gray or corpus callosum. 75% are in contact with ependymal or meningeal surfaces (this together with dense enhancement may produce a “pseudomeningioma pattern”, however lymphomas lack calcifications and tend to be multiple).

60% are hyperdense to brain, only 10% are hypodense. Characteristically, $>90\%$ of these tumors enhance; this is densely homogeneous in over 70%. As a result, when rare non-enhancing cases occur it often leads to a delay in diagnosis.¹⁸ The appearance of enhanced PCNSL on CT has been likened to “fluffy cotton balls.” There may be surrounding edema¹⁹ and there is usually mass effect.

There is an almost diagnostic tendency of rapid partial to complete resolution on CT (and even at the time of surgery) following the administration of steroids, earning the nickname of “ghost-cell tumor”^{20,21} or disappearing tumor.

MRI

No pathognomonic feature. May be difficult to discern if tumor is located subependymally (signal characteristics similar to CSF); proton-weighted image may avoid this pitfall. Nonenhancing lymphoma (on MRI or CT) is rare²² (some of these may enhance after XRT) but may be underreported. Bright on DWI (restricted diffusion), isointense to hypointense on ADC map.

CSF

Should only be obtained if no mass effect. Usually abnormal, but non-specific. Most common abnormalities are elevated protein (in >80%), and increased cell count (in 40%). Cytology is positive for lymphoma cells (pre-operatively) in only 10% (sensitivity may be higher with leptomeningeal involvement as in non-AIDS patients than with parenchymal involvement commonly seen in AIDS). Repeating up to 3 LPs may increase yield. Although the diagnosis of lymphoma can be made from CSF, cells obtained in this manner are not adequate for detailed tissue typing which is possible with solid tissue from biopsy.

Angiography

Rarely helpful. 60% of cases show only an avascular mass. 30–40% show diffuse homogeneous staining or blush.

44

44.1.9 Treatment

Surgery

Surgical decompression with partial or gross total removal does not alter patient's prognosis. The main indications for surgery:

- Biopsy: obtain solid tissue to ascertain that the tumor is a lymphoma, and to determine the type of lymphoma. Stereotactic techniques are often well-suited for these often deep tumors.²³

Radiation therapy

The standard treatment after tissue biopsy is whole-brain radiation therapy. Doses used tend to be lower than for other primary brain tumors. ≈ 40 –50 Gy total are usually given in 1.8–3 Gy daily fractions.

Chemotherapy

General information

In non-AIDS cases, chemotherapy combined with XRT prolongs survival compared to XRT alone.²⁴

Methotrexate (MTX)

The addition of intraventricular MTX (rather than just intrathecal via LP) delivered through a ventricular access device (6 doses of 12 mg twice a week, with IV leucovorin rescue) may result in even better survival.²⁵ In the event of an intrathecal MTX overdose (OD), interventions recommended²⁶: ODs of up to 85 mg can be well tolerated with little sequelae; immediate LP with drainage of CSF can remove a substantial portion of the drug (removing 15 ml of CSF can eliminate ≈ 20 –30% of the MTX within 2 hrs of OD). This can be followed by ventriculolumbar perfusion over several hours using 240 ml of warmed isotonic preservative-free saline entering through the ventricular reservoir and exiting through a lumbar subarachnoid catheter. For major OD of >500 mg, add intrathecal administration of 2,000 U of carboxypeptidase G2 (an enzyme that inactivates MTX). In cases of MTX OD, systemic toxicity should be prevented by treating with IV dexamethasone and IV (not IT) leucovorin.

Rituximab

Available since 1997 for treatment of refractory systemic B-cell non-Hodgkins lymphoma. Intrathecally, may be more effective for CD33+ lymphomas.

44.1.10 Prognosis

With no treatment, median survival is 1.8–3.3 months following diagnosis.

With radiation therapy,¹ median survival is 10 months, with 47% 1-year median survival, and 16% 2-year median survival. 3-year survival is 8% and 5-year survival is 3–4%. With intraventricular MTX, median time to recurrence was 41 mos.²⁵ Occasionally, prolonged survival may be seen.²⁷

About 78% of cases recur, usually ≈ 15 months after treatment (late recurrences also are seen). Of these recurrences, 93% are confined to the CNS (often at another site if the original site responded well), and 7% are elsewhere.

In AIDS-related cases, the prognosis appears worse. Although complete remission occurs in 20–50% following XRT, the median survival is only 3–5 months,^{28,29} usually related to AIDS-related opportunistic infection. However, neurologic function and quality of life improve in $\approx 75\%$ ²⁸

Although there are individual studies that show trends, there are no prognostic features that consistently correlate with survival.

44.2 Multiple myeloma

44.2.1 General information

Multiple myeloma (MM) (sometimes referred to simply as myeloma) is a neoplasm of a single clone of plasma cells characterized by proliferation of plasma cells in bone marrow, infiltration of adjacent tissues with mature and immature plasma cells, and the production of an immunoglobulin, usually monoclonal IgG or IgA (referred to collectively as M-protein³⁰). Circulating pre-myeloma cells lodge in appropriate microenvironments (e.g. in bone marrow) where they differentiate and expand. Although MM is often referred to in the context of “metastatic lesions” to bone, it is also sometimes considered a primary bone tumor. If only a single lesion is identified, then it is referred to as a plasmacytoma (see below).

44.2.2 Epidemiology

In the U.S., incidence is $\approx 1\text{--}2$ per 100,000 in caucasians, and is $\approx$ twice that in blacks. MM accounts for 1% of malignancies, and 10% of hematologic cancers. The peak age of occurrence is 60–70 yrs of age, with $<2\%$ of patients being <40 yrs old. Slightly more common in males. Monoclonal gammopathy without MM occurs in $\approx 0.15\%$ of the population, and in long-term follow-up 16% of these develop MM with an annual rate of 0.18%³¹

44.2.3 Presentation

General information

MM presents as a result of the following (underscored items are characteristic for MM):

1. proliferation of plasma cells: interferes with normal immune system function $\rightarrow$ increased susceptibility to infection
2. bone involvement
 - a) bone marrow involvement $\rightarrow$ destruction of hematopoietic capacity $\rightarrow$ normocytic normochromic anemia, leukopenia, thrombocytopenia
 - b) bone resorption
 - $\rightarrow$ weakening of the bone $\rightarrow$ pathologic fractures (see below)
 - $\rightarrow$ hypercalcemia (present initially in 25% of MM patients, see below)
 - c) swelling or local tenderness of bone
 - d) bone pain: characteristically induced by movement, and absent at rest
 - e) spinal involvement
 - invasion of spinal canal in $\approx 10\%$ of cases $\rightarrow$ spinal cord compression $\rightarrow$ myelopathy
 - nerve root compression (radiculopathy)
3. overproduction of certain proteins by plasma cells. May lead to:
 - a) hyperviscosity syndrome
 - b) cryoglobulinemia
 - c) amyloidosis
 - d) renal failure: multifactorial, but monoclonal light chains play a role

Skeletal disease

MM involvement is by definition multiple, and is usually restricted to sites of red marrow: ribs, sternum, spine, clavicles, skull, or proximal extremities. Lesions of the spine and/or skull are the usual reasons for presentation to the neurosurgeon.

Bone resorption in MM is not due simply to mechanical erosion by plasma cells. Increased osteoclastic activity has been observed.

Plasma cell tumors of the skull involving the cranial vault usually do not produce neurologic symptoms. Cranial nerve palsies can arise from skull base involvement. Orbital involvement may produce proptosis (exophthalmos).

Neurologic involvement

- Neurologic manifestations can occur as a result of:
- 1. tumor involvement of bone causing compression (see above)
 - a) tumor in spine with compression of spinal cord or nerve roots
 - b) tumor in skull with compression of brain or cranial nerves
 - 2. deposition of amyloid within the flexor retinaculum of the wrist →carpal tunnel syndrome; the median nerve itself does not contain amyloid, and therefore responds well to surgical division of the transverse carpal ligament (p.524)
 - 3. diffuse progressive sensorimotor polyneuropathy: occurs in 3–5%of patients with MM
 - a) about half are due to amyloidosis (p.549)
 - b) polyneuropathy can also occur without amyloidosis, especially in the rare osteosclerotic variant of MM
 - 4. multifocal leukoencephalopathy has been described in MM³²
 - 5. hypercalcemia: may produce a dramatic encephalopathy with confusion, delirium or coma. Neurologic symptoms of hypercalcemia associated with MM are more common than in hypercalcemia of other etiologies
 - 6. very rare: intraparenchymal metastases³³

44.2.4 Evaluation

The diagnostic criteria for MM is shown in □ Table 44.1. Tests that may be used in evaluating patients with MM or suspected MM include:

- 1. 24 hour urine for kappa Bence-Jones protein present in 75% (Note: Small proteins (comprised of light chains of immunoglobulins) found in the urine of ≈ 80%of patients with MM (may also occur in other conditions). Monoclonal proteins cannot be detected in the urine or serum of ≈ 1% of MM patients; two or more monoclonal bands are produced in ≈ 0.5–2.5%of patients with MM.³⁴)
- 2. bloodwork: serum protein electrophoresis (SPEP) and immune electrophoresis (IEP) (looking for IgG kappa band)
- 3. skeletal radiologic survey. Characteristic x-ray finding: multiple, round, “punched-out” (sharply demarcated) lytic lesions in the bones typically involved (see above). Osteosclerotic lesions are seen in <3%of patients with MM. Diffuse osteoporosis may also be seen
- 4. CBC: anemia eventually develops in most patients with MM. It is usually of moderate severity (Hgb ≈ 7–10 gm%) with a low reticulocyte count
- 5. technetium-99 m nuclear bone scan is usually negative in untreated MM (due to rarity of spontaneous new bone formation) and is less sensitive than conventional radiographs. Therefore it is not usually helpful except perhaps to implicate etiologies other than MM to explain the observed findings. After treatment, bone scan may become positive as osteoblastic activity ensues (“flare” response)
- 6. serum creatinine: for prognostication
- 7. bone marrow biopsy: virtually all MM patients have “myeloma cells” (although sensitive, this is not specific and other diagnostic criteria should be sought)

44.2.5 Treatment

Many aspects of treatment fall into the purvey of the oncologist (see review³¹). Some aspects pertinent to neurosurgical care include:

- 1. XRT (p.808): MM is very radiosensitive. Focal XRT for pain due to readily identifiable bone lesions, may allow pathologic fractures to heal and is effective in spinal cord compression

Table 44.1 Criteria for diagnosis of MM ^a
1. Cytologic criteria a) marrow morphology: plasma cells and/or myeloma cells ≥ 10% of 1000 or more cells b) biopsy proven plasmacytoma
2. Clinical and laboratory criteria a) myeloma protein (M-component) in serum (usually >3 gm/dl) or urine IEP b) osteolytic lesions on x-ray (generalized osteoporosis qualifies if marrow contains >30% plasma or myeloma cells) c) myeloma cells in ≥ 2 peripheral blood smears
^a diagnosis requires ³⁵ : 1A & 1B, or 1A or 1B and 2A, 2B, or 2C

2. mobilization: immobilization due to pain and fear of pathologic compression fractures leads to further detrimental increases in serum calcium and weakness
3. pain control: mild pain often responds well to salicylates (contraindicated in thrombocytopenia). Local XRT is also effective (see below)
4. percutaneous kyphoplasty (p.1011) may be used for some spine lesions (preferred over vertebroplasty because of reduced potential of spreading neoplasm)
5. therapy for hypercalcemia usually improves symptoms related to that derangement
6. bisphosphonates (p.1122) inhibit bone resorption and rapidly reduces hypercalcemia. Pamidronate is currently preferred over older agents
7. bortezomib (Velcade®): the first proteasome inhibitor, indicated for treatment of refractory MM

44.2.6 Prognosis

Untreated MM has a 6 month median survival. Solitary plasmacytoma has a 50%10-year survival. If there is an solitary site of involvement but M-protein is present (i.e. essentially a plasmacytoma except for the M-protein), elimination of the M-protein following XRT indicates a 50–60%chance of remaining free of MM, if the M-protein doesn't resolve, there is a high chance of developing MM.

44.3 Plasmacytoma

44.3.1 General information

A neoplasm of a single clone of plasma cells similar to multiple myeloma (see above) but meeting the following criteria:

1. there must be no other lesions on complete skeletal survey (not bone scan)
2. bone marrow aspirate must show no evidence of myeloma
3. and serum and urine electrophoresis should show no M-protein

MM will develop in 55–60%of patients with a solitary plasmacytoma in 5 years, and in 70–80%by 10 yrs.

44.3.2 Treatment

1. local XRT provides good local control rates
2. percutaneous kyphoplasty (p.1011): preferred over vertebroplasty because of reduced potential of spreading neoplastic cells

References

- [1] O'Neill BP, Illig JJ. Primary Central Nervous System Lymphoma. *Mayo Clin Proc.* 1989; 64:1005–1020
- [2] Kawakami Y, Tabuchi K, Ohnishi R, et al. Primary Central Nervous System Lymphoma. *J Neurosurg.* 1985; 62:522–527
- [3] Jellinger K, Radaszkiewicz T. Involvement of the Central Nervous System in Malignant Lymphomas. *Virchows Arch (Pathol Anat).* 1976; 370:345–362
- [4] Helle TL, Britt RH, Colby TV. Primary Lymphoma of the Central Nervous System. *J Neurosurg.* 1984; 60:94–103
- [5] Alic L, Haid M. Primary Lymphoma of the Brain: A Case Report and Review of the Literature. *J Surg Oncol.* 1984; 26:115–121
- [6] Eby NL, Grufferman S, Flannelly CM, Scholf SC, Vogel FS, Burger PC. Increasing Incidence of Primary Brain Lymphoma in the U.S. *Cancer.* 1988; 62:2461–2465
- [7] Murray K, Kun L, Cox J. Primary Malignant Lymphoma of the Central Nervous System: Results of Treatment of 11 Cases and Review of the Literature. *J Neurosurg.* 1986; 65:600–607
- [8] Penn I. Development of Cancer as a Complication of Clinical Transplantation. *Transplant Proc.* 1977; 9:1121–1127
- [9] Levy RM, Bredesen DE, Rosenblum ML. Neurological manifestations of the acquired immunodeficiency syndrome (AIDS): Experience at UCSF and review of the literature. *J Neurosurg.* 1985; 62:475–495
- [10] Jean WC, Hall WA. Management of Cranial and Spinal Infections. *Contemp Neurosurg.* 1998; 20:1–10
- [11] Hochberg FH, Miller G, Schooley RT, et al. Central-Nervous-System Lymphoma Related to Epstein-Barr Virus. *N Engl J Med.* 1983; 309:745–748
- [12] MacMahon EME, Glass JD, Hayward SD, et al. Epstein-Barr Virus in AIDS-Related Primary Central Nervous System Lymphoma. *Lancet.* 1991; 338:969–973
- [13] Burger PC, Scheithauer BW, Vogel FS. *Surgical Pathology of the Nervous System and Its Coverings.* 4th ed. New York: Churchill Livingstone; 2002
- [14] Calamia KT, Miller A, Shuster EA, et al. Intravascular Lymphomatosis: A Report of Ten Patients with Central Nervous System Involvement and a Review of the Disease Process. *Adv Exp Med Biol.* 1999; 455:249–265
- [15] Glass J, Hochberg FH, Miller DC. Intravascular Lymphomatosis. A Systemic Disease with Neurologic Manifestations. *Cancer.* 1993; 71:3156–3164
- [16] So YT, Beckstead JH, Davis RL. Primary central nervous system lymphoma in acquired immune deficiency syndrome: A clinical and pathological study. *Ann Neurol.* 1986; 20:566–572
- [17] Poon T, Matoso I, Tchertkoff V, et al. CT features of primary cerebral lymphoma in AIDS and non-AIDS patients. *J Comput Assist Tomogr.* 1989; 13:6–9
- [18] DeAngelis LM. Cerebral Lymphoma Presenting as a Nonenhancing Lesion of Computed Tomographic/

- Magnetic Resonance Scan. *Ann Neurol.* 1993; 33:308–311
- [19] Enzmann DR, Krikorian J, Norman D, et al. Computed Tomography in Primary Reticulum Cell Sarcoma of the Brain. *Radiology.* 1979; 130:165–170
- [20] Vaquero J, Martinez R, Rossi E, et al. Primary Cerebral Lymphoma: the 'Ghost Tumor'. *J Neurosurg.* 1984; 60:174–176
- [21] Gray RS, Abrahams JJ, Hufnagel TJ, et al. Ghost-cell tumor of the optic chiasm; primary CNS lymphoma. *J Clin Neuroophthalmol.* 1989; 9:98–104
- [22] DeAngelis LM. Cerebral lymphoma presenting as a nonenhancing lesion on computed tomographic/magnetic resonance scan. *Ann Neurol.* 1993; 33:308–311
- [23] O'Neill BP, Kelly PJ, Earle JD, et al. Computer-Assisted Stereotactic Biopsy for the Diagnosis of Primary Central Nervous System Lymphoma. *Neurology.* 1987; 37:1160–1164
- [24] DeAngelis LM, Yahalom J, Heinemann M-H, et al. Primary Central Nervous System Lymphomas: Combined Treatment with Chemotherapy and Radiotherapy. *Neurology.* 1990; 40:80–86
- [25] DeAngelis LM, Yahalom J, Thaler HT, Kher U. Combined Modality Therapy for Primary CNS Lymphomas. *J Clin Oncol.* 1992; 10:635–643
- [26] O'Marcaigh AS, Johnson CM, Smithson WA, et al. Successful Treatment of Intrathecal Methotrexate Overdose by Using Ventriculolumbar Perfusion and Intrathecal Instillation of Carboxypeptidase G2. *Mayo Clin Proc.* 1996; 71:161–165
- [27] Hochberg FH, Miller DC. Primary Central Nervous System Lymphoma. *J Neurosurg.* 1988; 68:835–853
- [28] Baumgartner JE, Rachlin JR, Beckstead JH, Levy RM, et al. Primary Central Nervous System Lymphomas: Natural History and Response to Radiation Therapy in 55 Patients with Acquired Immunodeficiency Syndrome. *J Neurosurg.* 1990; 73:206–211
- [29] Formenti SC, Gill PS, Lean E, et al. Primary Central Nervous System Lymphoma in AIDS: Results of Radiation Therapy. *Cancer.* 1989; 63:1101–1107
- [30] Keren DF, Alexanian R, Goeken JA, Gorevic PD, Kyle RA, Tomar RH. Guidelines for Clinical and Laboratory Evaluation of Patients with Monoclonal Gammopathies. *Arch Pathol Lab Med.* 1999; 123:106–107
- [31] Bataille R, Harousseau J-L. Multiple Myeloma. *N Engl J Med.* 1997; 336:1657–1664
- [32] McCarthy J, Proctor SJ. Cerebral Involvement in Multiple Myeloma. Case Report. *J Clin Pathol.* 1978; 31:259–264
- [33] Norum J, Wist E, Dahil IM. Cerebral Metastases from Multiple Myeloma. *Acta Oncol.* 1991; 30:868–869
- [34] Foerster J, Lee GR, Bithell TC, Foerster J, Athens JW, Lukens JN. In: Multiple Myeloma. Wintrobe's Clinical Hematology. 9th ed. Philadelphia: Lea and Febiger; 1993:2219–2249
- [35] Costa G, Engle RL, Schilling A, et al. Melphalan and Prednisone: An Effective Combination for the Treatment of Multiple Myeloma. *Am J Med.* 1973; 54:589–599

45 Pituitary Tumors – General Information and Classification

45.1 General information

Key concepts

- most are benign adenomas arising from the anterior pituitary (adenohypophysis)
- presentation (see below): most commonly present due to hormonal effects (includes: hyperprolactinemia, Cushing’s syndrome, acromegaly...), mass effect (most commonly: bitemporal hemianopsia from compression of optic chiasm), as an incidental finding, or infrequently with pituitary apoplexy (p. 720)
- work-up for a newly diagnosed intrasellar lesion: see □ Table 46.1
- prolactinoma is the only type for which medical therapy (DA agonists) may be the primary treatment in certain cases (p. 737). For other tumor types, options primarily consist of surgery (trans-sphenoidal or transcranial), or XRT
- post-op concerns include: diabetes insipidus, adrenal insufficiency, CSF leak

See also review of pituitary embryology & neuroendocrinology (p. 149).

45.2 General tumor types

45.2.1 Pituitary adenomas

Most primary pituitary tumors are benign adenomas which arise from the anterior pituitary gland (adenohypophysis). Adenomas may be classified by a number of schemes, including: by endocrine function (aided by immunostaining), by light microscopy (p.727) with routine histological staining and by electron microscopic appearance.

Microadenoma: A pituitary tumor <1 cm diameter. Currently, 50%of pituitary tumors are <5 mm at time of diagnosis. These may be difficult to find at the time of surgery.

Macroadenomas: Tumors >1 cm diameter.

45.2.2 Pituitary carcinoma

See reference.¹

Rare (<140 reports). Usually invasive and secretory (most common hormones: ACTH, PRL). Can metastasize, at which point prognosis is poor (66% 1-year mortality). Little improvement with further surgery, XRT, or chemotherapy.

45.2.3 Neurohypophyseal tumors

Neurohypophyseal tumors (tumors of the neurohypophysis i.e. the posterior pituitary) are rare; see Pituicytoma (p.728).

45.3 Epidemiology

Pituitary tumors represent ≈ 10%of intracranial tumors (incidence is higher in autopsy series). They are most common in the 3rd and 4th decades of life, and affect both sexes equally. The incidence is increased in multiple endocrine adenomatosis or neoplasia (MEA or MEN) (especially type I: autosomal dominant inheritance with high penetrance, also involves pancreatic islet cell tumors (which may produce gastrin and hence Zollinger-Ellison syndrome) and parathyroids (hyperparathyroidism), and in which the pituitary tumors are usually nonsecretory).

45.4 Differential diagnosis of pituitary tumors

See differential diagnosis (p.1371), which also includes non-neoplastic etiologies.

45.5 Clinical presentation of pituitary tumors

45.5.1 General information

Classically, pituitary tumors are divided into functional (or secreting), and non-functional (AKA endocrine-inactive, which are either nonsecretory, or else secrete products such as gonadotropin that do not cause endocrinologic symptoms).

In general, secreting tumors tend to present earlier as a result of symptoms caused by physiologic effects of excess hormones that they secrete² (this applies less e.g. to prolactinomas in males since the symptoms may be mild or unrecognized). Nonsecreting tumors usually do not present until they are sufficiently large enough to cause neurologic deficits by mass effect.

45.5.2 Presentation

General information

Presentation may be the result of: endocrine syndromes, mass effect, incidental finding (essentially only with macroadenomas), pituitary apoplexy.

Endocrinologic disturbance

Hormone oversecretion (secretory tumor)

≈ 65% of adenomas secrete an active hormone (48% prolactin, 10% GH, 6% ACTH, 1% TSH)³:

1. prolactin (PRL) (p. 722): can cause amenorrhea-galactorrhea syndrome in females), impotence in males Etiologies:
 - a) prolactinoma (p. 722): neoplasia of pituitary lactotrophs
 - b) stalk effect (p. 732): pressure on the pituitary stalk may reduce the inhibitory control over PRL secretion
2. growth hormone (GH): elevated GH is due to a pituitary adenoma > 95% of the time
 - a) in adults: causes acromegaly (p. 725)
 - b) in prepubertal children (before epiphyseal plate closure): produces pituitary gigantism (very rare)
3. corticotropin AKA adrenocorticotrophic hormone (ACTH):
 - a) Cushing's disease (endogenous hypercortisolism): see below
 - b) Nelson syndrome (p. 724): can develop only in patients who have had an adrenalectomy
4. thyrotropin (TSH) (p. 726): secondary (central) hyperthyroidism
5. gonadotropins (luteinizing hormone (LH) and/or follicle stimulating hormone (FSH)): usually does not produce a clinical syndrome

Underproduction of pituitary hormones

May be caused by compression of the normal pituitary by large tumors. More common with non-secretory tumors than with secretory tumors. In order of sensitivity to compression (i.e. the order in which pituitary hormones become depressed from mass effect): GH, gonadotropins (LH & FSH), TSH, ACTH (mnemonic: Go Look For The Adenoma). Chronic deficiency of all pituitary hormones (panhypopituitarism) may produce pituitary cachexia (AKA Simmonds' cachexia).

□ NB: selective reduction of a single pituitary hormone is very atypical with pituitary adenomas. May occur with autoimmune hypophysitis (p. 1373), which most commonly involves ACTH or ADH (causing DI⁴ – see below)

1. Deficiency of specific hormones
2. growth hormone deficiency (note: growth hormone stimulation test (p. 736) is more sensitive and specific for GH deficiency than measuring basal GH levels):
 - a) in children: produces growth delay
 - b) in adults: produces vague symptoms with metabolic syndrome (decreased lean body mass, centripetal obesity, reduced exercise tolerance, impaired sense of well-being)
 - c) hypogonadism: amenorrhea (women), loss of libido, infertility
3. hypothyroidism: cold intolerance, myxedema, entrapment neuropathies (e.g. carpal tunnel syndrome), weight gain, memory disturbance, integumentary changes (dry skin, coarse hair, brittle nails), constipation, increased sleep demand
4. hypoadrenalism: orthostatic hypotension, easy fatigability
5. diabetes insipidus: almost never seen pre-operatively with pituitary tumors (except possibly with pituitary apoplexy, see below). If DI is present, other etiologies should be sought, including:
 - a) autoimmune hypophysitis (p. 1373)
 - b) hypothalamic glioma
 - c) suprasellar germ cell tumor

6. gonadotropin deficiency (hypogonadotrophic hypogonadism) with anosmia is part of Kallmann syndrome⁵

Mass effect (other than compression of the pituitary)

Because they tend to get to a larger size before detection, this is more common with nonfunctioning tumors. Of functional tumors, prolactinoma is the most likely to become large enough to cause mass effect (especially in males or non-menstruating females); ACTH tumor is least likely. Nonspecific symptoms include headaches. □ seizures are rarely attributable to pituitary adenomas and other etiologies should be sought. Mass effect may occur suddenly as a result of expansion with pituitary apoplexy (see below).

Structures commonly compressed and their manifestations include:

1. optic chiasm: classically produces bitemporal hemianopsia (non-congruous). May also cause decreasing visual acuity
2. involvement of third ventricle may produce obstructive hydrocephalus
3. cavernous sinus

a) pressure on cranial nerves contained within (III, IV, V1, V2, VI): ptosis, facial pain, diplopia (below)

b) occlusion of the cavernous sinus: proptosis, chemosis

c) encasement of the carotid artery by tumor: may cause slight narrowing, but complete occlusion is rare
4. invasive adenomas (p.721) infrequently present with CSF rhinorrhea⁶ with invasive prolactinomas this may be precipitated by shrinkage resulting from medical treatment
5. macroadenomas may produce H/A possibly via increased intrasellar pressure

Pituitary apoplexy

General information

Key concepts

- due to expansion of a pituitary adenoma from hemorrhage or necrosis
- typical presentation: paroxysmal H/A with endocrinologic and/or neurologic deficit (usually ophthalmoplegia or visual loss)
- management: immediate administration of glucocorticoids, and transsphenoidal decompression within 7 days in most cases

Definition

Neurologic and/or endocrinologic deterioration due to sudden expansion of a mass within the sella turcica.

Etiology

Sudden intrasellar expansion may occur as a result of hemorrhage, necrosis^{7,8} and/or infarction within a pituitary tumor and adjacent pituitary gland. Occasionally, hemorrhage occurs into a normal pituitary gland or Rathke’s cleft cyst.⁹

Epidemiology

In Wilson’s series, 3%of his patients with macroadenomas had an episode of pituitary apoplexy. In another series of 560 pituitary tumors, a high incidence of 17%was found (major attack in 7% minor in 2% asymptomatic in 8%).¹⁰ It is common for apoplexy to be the initial presentation of a pituitary tumor.¹¹

Clinical features of pituitary apoplexy

Patients often present with abrupt onset of H/A, visual disturbance, and loss of consciousness. Neurologic involvement includes:

1. visual disturbances: one of the most common findings. Includes:

a) ophthalmoplegia (unilateral or bilateral): opposite the situation with a pituitary tumor, ophthalmoplegia occurs more often (78%) than visual pathway deficits (52–64%)¹²

b) one of the typical field cuts (p.730) seen in pituitary tumors

2. reduced mental status: due to ↑ ICP or hypothalamic involvement
3. cavernous sinus compression can cause venous stasis and/or pressure on any of the structures within the cavernous sinus
 - a) trigeminal nerve symptoms
 - b) proptosis
 - c) ophthalmoplegia (Cr. N. III palsy is more common than VI)
 - d) ptosis may be an early symptom^{13,14}
 - e) pressure on carotid artery
 - f) compression of sympathetics within the cavernous sinus may produce a form of Horner's syndrome with unilateral ptosis, miosis, & anhidrosis limited to the forehead
 - g) carotid artery compression may cause stroke or vasospasm
4. when hemorrhage breaks through the tumor capsule and the arachnoid membrane into the chiasmatic cistern, signs and symptoms of SAH may be seen
 - a) N/V
 - b) meningismus
 - c) photophobia
5. increased ICP may produce lethargy, stupor or coma
6. hypothalamic involvement may produce
 - a) hypotension
 - b) thermal dysautoregulation
 - c) cardiac dysrhythmias
 - d) respiratory pattern disturbances
 - e) diabetes insipidus
 - f) altered mental status: lethargy, stupor or coma
7. suprasellar expansion can produce acute hydrocephalus

Evaluation

CT or MRI shows hemorrhagic mass in sella turcica and/or suprasellar region, often distorting the anterior third ventricle.

Cerebral angiography should be considered in cases where differentiating pituitary apoplexy from aneurysmal SAH is difficult.

Management of pituitary apoplexy

Pituitary function is consistently compromised, necessitating rapid administration of corticosteroids and endocrine evaluation.

In the absence of visual deficits, prolactinomas may be treated with bromocriptine.

Rapid decompression is required for: sudden constriction of visual fields, severe and/or rapid deterioration of acuity, or neurologic deterioration due to hydrocephalus. Surgery in ≤7 days of pituitary apoplexy resulted in better improvement in ophthalmoplegia (100%), visual acuity (88%) and field cuts (95%) than surgery after 7 days, based on a retrospective study of 37 patients.¹⁵ Decompression is usually via a transsphenoidal route (transcranial approach may be advantageous in some cases). Goals of surgery:

1. to decompress the following structures if under pressure: optic apparatus, pituitary gland, cavernous sinus, third ventricle (relieving hydrocephalus)
2. obtain tissue for pathology
3. complete removal of tumor is usually not necessary
4. for hydrocephalus: ventricular drainage is generally required

45.6 Specific types of pituitary tumors

45.6.1 Invasive pituitary adenomas

About 5% of pituitary adenomas become locally invasive. The genetic make-up of these tumors may differ from more benign adenomas,¹⁶ even though the histology is similar. Numerous classifications systems have been devised for invasive adenomas, Wilson's system¹⁷ (modified from Hardy^{18,19}) is shown in □ Table 45.1.

The clinical course is variable, with some tumors being more aggressive than others. Occasionally, these tumors grow to gigantic sizes (>4 cm dia), and these are often very aggressive and follow a malignant course.²⁰

At times, an adenoma may push the medial wall of the cavernous sinus ahead of it without actually perforating this dural structure (i.e. not actually invading the sinus).²¹ This is difficult to

Table 45.1 Anatomic classification of pituitary adenoma (modified Hardy system) ¹⁷	
Extension	
• Suprasellar extension 0: none - A: expanding into suprasellar cistern - B: anterior recesses of 3rd ventricle obliterated - C: floor of 3rd ventricle grossly displaced • Parasellar extension - D^a: intracranial (intradural) - E: into or beneath cavernous sinus (extradural)	
Invasion/Spread	
• Floor of sella intact - I: sella normal or focally expanded; tumor < 10 mm - II: sella enlarged; tumor ≥ 10 mm • Sphenoid extension - III: localized perforation of sellar floor - IV: diffuse destruction of sellar floor • Distant spread - V: spread via CSF or blood-borne	
^a specify: 1) anterior, 2) middle, or 3) posterior fossa	

reliably identify on MRI, and the most definitive sign of cavernous sinus invasion is carotid artery encasement.²²

Presentation

1. visual system
 - a) most present due to compression of the optic apparatus, usually producing gradual visual deficit (however, sudden blindness is not unheard of)
 - b) extraocular muscle deficits may occur with cavernous sinus invasion, and usually develop after visual loss
 - c) exophthalmos may occur with orbital invasion due to compromise of orbital venous drainage
2. hydrocephalus: suprasellar extension may obstruct one or both foramen of Monro
3. invasion of the skull base may lead to nasal obstruction. CSF rhinorrhea may occasionally be precipitated by tumor shrinkage in response to dopamine agonists (e.g. bromocriptine) as a result of uncovering areas of bone erosion. This carries the risk of ascending meningitis.²³
4. tumors that secrete prolactin (p.719) often present with findings of hyperprolactinemia and with these, the prolactin levels are usually > 1000 ng/ml (caution: giant invasive adenomas with very high PRL production may have a falsely low PRL level due to “hook effect” (p.733)

45.6.2 Hormonally active pituitary tumors

Prolactinomas

The most common secretory adenoma. Arise from neoplastic transformation of anterior pituitary lactotrophs. See □ Table 46.4 for DDx of hyperprolactinemia.

Manifestations of prolonged hyperprolactinemia:

1. females: amenorrhea-galactorrhea syndrome (AKA Forbes-Albright syndrome, AKA Ahumada-del Castillo syndrome). Variants: oligomenorrhea, irregular menstrual cycles. 5% of women with primary amenorrhea will be found to have a PRL-secreting pituitary tumor.²⁴ Remember: pregnancy is the most common cause of secondary amenorrhea in females of reproductive potential. The galactorrhea may be spontaneous or expressive (only on squeezing the nipples)
2. males: impotence, decreased libido. Galactorrhea is rare (estrogen is also usually required). Gynecomastia is rare. Prepubertal prolactinomas may result in small testicles and feminine body habitus
3. either sex:
 - a) infertility is common
 - b) bone loss (osteoporosis in women, and both cortical and trabecular osteopenia in men) due to a relative estrogen deficiency, not due to the elevated prolactin itself

At the time of diagnosis, 90% of prolactinomas in women are microadenomas, vs. 60% for males (probably due to gender specific differences in symptoms resulting in earlier presentation in females). Some tumors secrete both PRL and GH.

Cushing’s disease

General information and Cushing’s syndrome

Cushing’s syndrome (CS) is a constellation of findings caused by hypercortisolism. Cushing’s disease (p.723) – endogenous hypercortisolism due to hypersecretion of ACTH by an ACTH secreting pituitary adenoma – is just one cause of CS. The most common cause of CS is iatrogenic (administration of exogenous steroids). Possible etiologies of endogenous hypercortisolism are shown in □ Table 45.2. To determine the etiology of CS, see Dexamethasone suppression test (p.734).

Conversion factors²⁵ for ACTH and cortisol between U.S. units and SI units are shown in Eq (45.1) and (45.2).

ACTH : 1pg=ml ¼ 1ng/ liter

ø45:1p

Cortisol : 1µg=dl ¼ 27:59 nmol/ liter

ø45:2p

Ectopic ACTH secretion

Hypercortisolism may also be due to ectopic secretion of ACTH usually by tumors, most commonly small-cell carcinoma of the lung, thymoma, carcinoid tumors, pheochromocytomas, and medullary thyroid carcinoma. In addition to findings of Cushing’s syndrome, patients are typically cachectic due to the malignancy which is usually rapidly fatal.

Prevalence of Cushing’s disease

40 cases/million population. ACTH-producing adenomas comprise 10–12% of pituitary adenomas.²⁶ Cushing’s disease is 9 times more common in women, whereas ectopic ACTH production is 10 times more common in males. Non-iatrogenic CS is 25% as common as acromegaly.

At the time of presentation, over 50% of patients with Cushing’s disease have pituitary tumors <5 mm in diameter, which are very difficult to image with CT or MRI. Most are basophilic, some (especially the larger ones) may be chromophobic. Only ≈ 10% are large enough to produce some mass effect, which may cause enlargement of the sella turcica, visual field deficit, cranial nerve involvement and/or hypopituitarism.

Clinical findings in Cushing’s disease

Findings are those of Cushing’s syndrome (hypercortisolism from any cause) include:

1. weight gain
- a) generalized in 50% of cases

Table 45.2 Causes of endogenous hypercortisolism

Site of pathology	Secretion product	Percent of cases	ACTH levels
pituitary corticotroph adenoma: Cushing’s disease (p.723)	ACTH	60–80%	slightly elevated ^a
ectopic ACTH production (p.723); most are lung tumors, others: pancreas...		1–10%	very elevated
adrenal (adenoma or carcinoma)	cortisol	10–20%	low
hypothalamic or ectopic secretion of corticotropin-releasing hormone (CRH) producing hyperplasia of pituitary corticotrophs; pseudo-Cushing’s state (p.723)	CRH	rare	elevated

^aACTH may be normal or slightly elevated; normal ACTH levels in the presence of hypercortisolism are considered inappropriately elevated

- b) centripetal fat deposition in 50% trunk, upper thoracic spine (“buffalo hump”), supraclavicular fat pad, neck, “dewlap tumor” (episternal fat), with round plethoric face (“moon facies”) and slender extremities
- 2. hypertension
- 3. ecchymoses and purple striae, especially on flanks, breasts and lower abdomen
- 4. amenorrhea in women, impotence in men, reduced libido in both
- 5. hyperpigmentation of skin and mucous membranes: due to MSH cross-reactivity of ACTH. Occurs only with elevated ACTH, i.e. Cushing’s disease (not Cushing’s syndrome) or ectopic ACTH production (also below)
- 6. atrophic, tissue-paper thin skin with easy bruising and poor wound healing
- 7. psychiatric: depression, emotional lability, dementia
- 8. osteoporosis
- 9. generalized muscle wasting with complaints of easy fatigability
- 10. elevation of other adrenal hormones: androgens may produce hirsutism and acne
- 11. sepsis: associated with advanced Cushing’s syndrome

Laboratory findings in Cushing’s disease

- 1. hyperglycemia: diabetes or glucose intolerance
- 2. hypokalemic alkalosis
- 3. loss of diurnal variation in cortisol levels
- 4. normal or elevated ACTH levels
- 5. failure to suppress cortisol with low-dose (1 mg) dexamethasone test (p.734)
- 6. elevated 24-hour urine free-cortisol
- 7. CRH levels will be low (not commonly measured)

Nelson’s syndrome (or Nelson syndrome) (NS)

General information

Key concepts

- a rare condition that follows 10–30%of total bilateral adrenalectomies (TBA) performed for Cushing’s disease; see indications for TBA (p. 743)
- classic triad: hyperpigmentation (skin & mucus membranes), abnormal ↑ ACTH, and progression of pituitary tumor (the last criteria is now controversial)
- treatment options: surgery (transsphenoidal or transcranial), XRT, medication

A rare condition that follows 10-30%of total bilateral adrenalectomies (TBA) performed for Cushing’s disease; see indications for TBA (p.743). NS is due to continued growth of corticotroph (ACTH-secreting) adenoma cells. Usually occurs 1–4 years after TBA (range: 2 mos-24 years).²⁶ Theoretical explanation (unproven)²⁷: following TBA, hypercortisolism resolves, and CRH levels increase back to normal from the (reduced) suppressed state; corticotroph adenomas in patients with NS have an increased & prolonged response to CRH resulting in increased growth. Also, corticotrophs in NS and CD show reduced inhibition by glucocorticoids. It is controversial if some cases may be related to insufficient glucocorticoid replacement after TBA.²⁶

Manifestations

See reference.²⁷

- 1. hyperpigmentation (due to melanin stimulating hormone (MSH) cross reactivity of ACTH and actual increased levels of MSH due to increased proopiomelanocortin production). Often the earliest sign that Nelson’s syndrome is developing. Look for linea nigra (midline pigmentation from pubis to umbilicus) and hyperpigmentation of scars, gingivae, and areolae. Ddx of hyperpigmentation includes: primary adrenal insufficiency (high levels of ACTH), ectopic ACTH secretion, hemochromatosis (more bronze color), jaundice (yellowish)
- 2. tumor growth →increased mass effect (p.720) or invasion: the most serious consequence. These corticotroph tumors are among the most aggressive of pituitary tumors.^{28 (p 545)} May produce any of the problems associated with macroadenomas (optic nerve compression, cavernous sinus invasion, pituitary insufficiency, H/A, bony invasion...) as well as necrosis with precipitous intracranial hypertension²⁹; see pituitary apoplexy (p.720)

- 3. malignant transformation of the corticotroph tumor (very rare)
- 4. hypertrophy of adrenal tissue rests: may be located in the testes → painful testicular enlargement and oligospermia. Rarely the rests can secrete enough cortisol to normalize cortisol levels or even cause a recurrence of Cushing's disease despite the adrenalectomy

Evaluation

- 1. Laboratories
 - a) ACTH > 200 ng/L (usually thousands of ng/L) (normal: usually < 54 ng/L)
 - b) exaggerated ACTH response to CRH (not required for diagnosis)
 - c) other pituitary hormones may be affected as with any macroadenoma causing mass effect (p. 720) and endocrine screening (p. 731) should be done
- 2. Formal visual field testing (p. 730): should be done in patients with suprasellar extension or in those being considered for surgery (as a baseline for comparison)

Treatment

See treatment (p. 739).

Acromegaly

General information

Key concepts

- abnormally high levels of growth hormone in an adult. >95% of cases are due to a benign pituitary somatotroph adenoma, >75% are > 10 mm at time of diagnosis
- effects include soft tissue and skeletal changes, cardiomyopathy, colon Ca
- work-up: endocrine tests (p. 730), cardiology consult, colonoscopy
- treatment (p. 741): surgery for most, and then if necessary, medical therapy (p. 741) and/or XRT (p. 745)
- suggested criteria for biochemical cure (p. 753): normal IGF-1, growth hormone level < 5 ng/ml, AND GH nadir of < 1 ng/ml after OGST (p. 736)

Incidence: 3 cases/1-million persons/year. >95% of cases of excess GH result from a pituitary somatotroph adenoma. Growth hormone carcinoma is extremely rare. Ectopic GH secretion may occur uncommonly with: carcinoid tumor, lymphoma, pancreatic islet-cell tumor. By the time of diagnosis, >75% of pituitary GH tumors are macroadenomas (> 10 mm dia) with cavernous sinus invasion and/or suprasellar extension.

25% of acromegalics have thyromegaly with normal thyroid studies. 25% of GH adenomas also secrete prolactin. Acromegaly occurs rarely as part of a genetic syndrome, including: multiple endocrine neoplasia type 1 (MEN 1), McCune-Albright syndrome, familial acromegaly, and Carney complex.³⁰

Clinical

Elevated levels of GH in children before closure of the epiphyseal plates in the long bones produces gigantism. Usually presents in the teen years.

In adults, elevated GH levels produces acromegaly (age: usually > 50 yrs) with findings that may include^{31,32} (also see □ Table 45.3):

- 1. skeletal overgrowth deformities
 - a) increasing hand and foot size
 - b) thickened heel pad
 - c) frontal bossing
 - d) prognathism
- 2. cardiovascular
 - a) cardiac findings (structural and functional): arrhythmias, valvular disease, concentric myocardial hypertrophy
 - b) hypertension (30%)
- 3. soft tissue swelling (includes macroglossia)
- 4. glucose intolerance
- 5. peripheral nerve entrapment syndromes (including carpal tunnel syndrome)

Table 45.3 Risks of long-term exposure to excess growth hormone (GH) ³⁴	
Arthropathy	
1. unrelated to age of onset or GH levels 2. usually with longstanding acromegaly 3. reversibility^a: a) rapid symptomatic improvement b) bone & cartilage lesions irreversible	
Peripheral neuropathy	
1. intermittent anesthetics, paresthesias 2. sensorimotor polyneuropathy 3. impaired sensation 4. reversibility^a: a) symptoms may improve b) onion bulbs (whorls) do not regress	
Cardiovascular disease	
1. cardiomyopathy a) reduced LV diastolic function b) increased LV mass and arrhythmias c) fibrous hyperplasia of connective tissue 2. HTN: exacerbates cardiomyopathic changes 3. reversibility^a: may progress even with normal GH	
Respiratory disease	
1. upper airway obstruction: caused by soft tissue overgrowth and decreased pharyngeal muscle tone with sleep apnea in ≈ 50% 2. reversibility^a: generally improves	
Neoplasia	
1. increased risk of malignancies (especially colon-Ca) & soft-tissue polyps 2. reversibility^a: unknown	
Glucose intolerance	
1. occurs in 25% of acromegalics (more common with family history of DM) 2. reversibility^a: improves	
^a reversibility with normalization of GH levels	

6. debilitating headache

7. excessive perspiration (especially palmar hyperhidrosis)

8. oily skin

9. joint pain

10. sleep apnea

11. fatigue

12. colon cancer: risk is ≈ 2 × risk of general population³³

Patients with elevated levels of GH (including partially treated cases) have 2–3 times the expected mortality rate,³⁴ primarily due to hypertension, diabetes, pulmonary infections, cancer, and cardiovascular disease (□ Table 45.3). Soft-tissue swelling and nerve entrapment may be reversible with normalization of GH levels, however many disfiguring changes and health risks are permanent (□ Table 45.3 for specifics).

Thyrotropin (TSH)-secreting adenomas

General information

Rare: comprise ≈ 0.5–1% of pituitary tumors.^{3,35} Produces central (secondary) hyperthyroidism (note: central hyperthyroidism may also occur with pituitary resistance to thyroid hormones³⁶): elevated circulating T3 and T4 levels, with elevated or inappropriately normal TSH³⁶ (TSH should be

undetectable in primary hyperthyroidism). Up to 33% of tumors positive for TSH immunostaining are nonsecretory.³⁶ Many of these tumors are plurihormonal, but the secondary hormone is usually clinically silent. Most of these tumors are aggressive and invasive and are large enough at presentation to also produce mass effect (especially if prior thyroid ablative procedures have been done, which occurs in up to 60% of cases due to lack of recognition of pituitary abnormality^{36,37}).

Clinical

Symptoms of hyperthyroidism: anxiety, palpitations (due to a-fib), heat intolerance, hyperhidrosis, and weight loss despite normal or increased intake. Signs: hyperactivity, lid lag, tachycardia, irregular rhythm when a-fib is present, hyperreflexia, tremor. Exophthalmos and infiltrative dermopathy (e.g. pretibial myxedema) are present only in Grave's disease.

45.6.3 Pathological classification of pituitary tumors

Light microscopic appearance of adenomas

Older classification system. Of limited usefulness. With newer techniques (EM, immunohistochemistry, radio-immuno assay...) many tumors previously considered nonsecretory have been found to have all the components necessary to secrete hormones.

In order of decreasing frequency:

1. chromophobe: most common (ratio of chromophobe to acidophil is 4-20:1). Originally considered "non-secretory", in actuality may produce prolactin, GH, or TSH
2. acidophil (eosinophilic): produce prolactin, TSH, or usually GH
3. basophil → gonadotropins, β -lipotropin, or usually ACTH → Cushing's disease

Classification of adenomas based on secretory products

1. endocrine-active tumors: $\approx 70\%$ of pituitary tumors produce 1 or 2 hormones that are measurable in the serum and cause defined clinical syndromes, these are classified based on their secretory product(s)
2. endocrine-inactive (nonfunctional) tumors³⁸ (note: a and b constitute the bulk of endocrine-inactive adenomas):
 - a) null-cell adenoma
 - b) oncocyoma
 - c) gonadotropin-secreting adenoma
 - d) silent corticotropin-secreting adenoma
 - e) glycoprotein-secreting adenoma

Tumors of the neurohypophysis and infundibulum

General information

The most common tumors encountered in the posterior pituitary are metastases (owing to the rich blood supply).

Granular cell tumors

AKA (infundibular) granular cell tumor (GCT). WHO grade I. Obsolete terms: choristoma,³⁹ granular cell myoblastoma, pituicytoma (this term is now reserved for a circumscribed glial neoplasm – see below). Tumors with nests of large cells having granular, eosinophilic cytoplasm.

While rare, GCTs are the most common primary tumor of the neurohypophysis and pituitary stalk/infundibulum⁴⁰ with a predilection for the stalk (these result in suprasellar extension). GCTs have been identified in the gastrointestinal tract, genitourinary tract, orbital region as well as in other locations of the central nervous system with no connection to the pituitary gland or hypothalamus (e.g. spinal meninges⁴¹). Female:male ratio $\geq 2:1$. Asymptomatic microscopic clusters of granular cells (tumorettes) are more common, with an incidence up to 17%⁴²

The most common presentation is with visual field deficits due to optic chiasm compression.³⁹ However, any symptom typical of a hormonally inactive sellar mass may occur.

Imaging: may appear radiographically identical to adenomas. Rarely considered in the differential diagnosis pre-op. Isodense on CT and isointense on T1WI MRI, dense homogeneous enhancement on CT & MRI.

Treatment: if GTC is suspected pre-op, a transcranial approach is preferred over transsphenoidal because of the vascularity which has prevented total resection in 60–70% of reported cases.⁴³ XRT may be considered for subtotal resection.⁴⁰

Pituicytoma

Less favored alternate terms include posterior pituitary astrocytoma. Rare (mostly case reports). Circumscribed tumor with spindle cells, arising from the neurohypophysis or infundibulum.⁴⁴ WHO grade I. Reported only in adults.

Treatment: surgical excision. Subtotal removal may be followed by recurrence over several years.

References

[1] Ragel BT, Couldwell WT. Pituitary carcinoma: a review of the literature. *Neurosurg Focus*. 2004; 16

[2] Ebersold MJ, Quast LM, Laws ER, et al. Long-Term Results in Transsphenoidal Removal of Nonfunctioning Pituitary Adenomas. *J Neurosurg*. 1986; 64:713–719

[3] Biller BM, Swearingen B, Zervas NT. A decade of the Massachusetts General Hospital Neuroendocrine Clinical Center. *J Clin Endocrinol Metab*. 1997; 82:1668–1674

[4] Abe T, Matsumoto K, Sanno N, Osamura Y. Lymphocytic Hypophysitis: Case Report. *Neurosurgery*. 1995; 36:1016–1019

[5] Lieblich JM, Rogol AD, White BJ, Rosen SW. Syndrome of anosmia with hypogonadotropic hypogonadism (Kallmann syndrome): clinical and laboratory studies in 23 cases. *Am J Med*. 1982; 73:506–519

[6] Nutkiewicz A, DeFeo DR, Kohout RI, et al. Cerebrospinal Fluid Rhinorrhea as a Presentation of Pituitary Adenoma. *Neurosurgery*. 1980; 6:195–197

[7] Reid RL, Quigley ME, Yen SC. Pituitary Apoplexy: A Review. *Arch Neurol*. 1985; 42:712–719

[8] Cardoso ER, Peterson EW. Pituitary Apoplexy: A Review. *Neurosurgery*. 1984; 14:363–373

[9] Onesti ST, Wisniewski T, Post KD. Pituitary Hemorrhage into a Rathke's Cleft Cyst. *Neurosurgery*. 1990; 27:644–646

[10] Wakai S, Fukushima T, Teramoto A, Sano K. Pituitary Apoplexy: Its Incidence and Clinical Significance. *J Neurosurg*. 1981; 55:187–193

[11] Rovit RL, Fein JM. Pituitary Apoplexy, A Review and Reappraisal. *J Neurosurg*. 1972; 37:280–288

[12] Liu JK, Couldwell W. Pituitary apoplexy: Diagnosis and management. *Contemp Neurosurg*. 2003; 25:1–5

[13] Yen MY, Liu JH, Jaw SJ. Ptosis as the early manifestation of pituitary tumour. *Br J Ophthalmol*. 1990; 74:188–191

[14] Telesca M, Santini F, Mazzucco A. Adenoma related pituitary apoplexy disclosed by ptosis after routine cardiac surgery: occasional reappearance of a dismal complication. *Intensive Care Med*. 2009; 35:185–186

[15] Bills DC, Meyer FB, Laws ER, Jr, Davis DH, Ebersold MJ, Scheithauer BW, Ilstrup DM, Abboud CF. A retrospective analysis of pituitary apoplexy. *Neurosurgery*. 1993; 33:602–8; discussion 608-9

[16] Pei L, Melmed S, Scheithauer B, et al. Frequent Loss of Heterozygosity at the Retinoblastoma Susceptibility Gene (RB) Locus in Aggressive Pituitary Tumors: Evidence for a Chromosome 13 Tumor Suppressor Gene Other Than RB. *Cancer Res*. 1995; 55:1613–1616

[17] Wilson CB, Tindall GT, Collins WF. In: *Neurosurgical Management of Large and Invasive Pituitary Tumors*. Clinical Management of Pituitary Disorders. New York: Raven Press; 1979:335–342

[18] Hardy J, Kohler PO, Ross GT. In: *Transsphenoidal Surgery of Hypersecreting Pituitary Tumors*. Diagnosis and Treatment of Pituitary Tumors. New York: Excerpta Medica/American Elsevier; 1973:179–194

[19] Hardy J, Thompson RA, Green R. In: *Transsphenoidal Surgery of Intracranial Neoplasm*. Adv Neurol. New York: Raven Press; 1976:261–274

[20] Krisht AF. Giant Invasive Pituitary Adenomas. *Contemp Neurosurg*. 1999; 21:1–6

[21] Laws ER. Comment on Knosp E, et al.: Pituitary Adenomas with Invasion of the Cavernous Sinus Space: A Magnetic Resonance Imaging Classification Compared with Surgical Findings. *Neurosurgery*. 1993; 33

[22] Scotti G, Yu CY, Dillon WP, et al. MR Imaging of Cavernous Sinus Involvement by Pituitary Adenomas. *AJR*. 1988; 151:799–806

[23] Barlas O, Bayindir C, Hepgul K, Can M, Kiris T, Sencer E, Unal F, Aral F. Bromocriptine-induced cerebrospinal fluid fistula in patients with macroprolactinomas: report of three cases and a review of the literature. *Surg Neurol*. 1994; 41:486–489

[24] Amar AP, Couldwell WT, Weiss MH. Prolactinomas: Focus on Indications, Outcomes, and Management of Recurrences. *Contemp Neurosurg*. 1989; 21:1–6

[25] Esposito F, Dusick JR, Cohan P, et al. Early morning cortisol levels as a predictor of remission after transsphenoidal surgery for Cushing's disease. *J Clin Endocrinol Metab*. 2006; 91:7–13

[26] Banasiak MJ, Malek AR. Nelson syndrome: comprehensive review of pathophysiology, diagnosis, and management. *Neurosurg Focus*. 2007; 23

[27] Assie G, Bahurel H, Coste J, Silvera S, Kujas M, Dugue MA, et al. Corticotroph tumor progression after adrenalectomy in Cushing's Disease: a reappraisal of Nelson's syndrome. *J Clin Endocrinol Metab*. 2007; 49:381–386

[28] Bertagna X, Raux-Demay M-C, Guilhaume B, et al., Melmed S. In: *Cushing's Disease. The Pituitary*. 2nd ed. Malden, MA: Blackwell Scientific; 2002:496–560

[29] Kasperlik-Zaluska AA, Bonicki W, Jeske W, Janik J, et al. Nelson's syndrome - 46 years later: clinical experience with 37 patients. *Zentralbl Neurochir*. 2006; 67:14–20

[30] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. *Endocr Pract*. 2004; 10:213–225

[31] Melmed S. Acromegaly. *N Engl J Med*. 1990; 322:966–977

[32] Melmed S. Medical progress: Acromegaly. *N Engl J Med*. 2006; 355:2558–2573

[33] Renehan AG, Shalet SM. Acromegaly and colorectal cancer: risk assessment should be based on population-based studies. *J Clin Endocrinol Metab*. 2002; 87:1909–1909

[34] Acromegaly Therapy Consensus Development Panel. Consensus Statement: Benefits Versus Risks of Medical Therapy for Acromegaly. *Am J Med*. 1994; 97:468–473

[35] Beck-Peccoz P, Brucker-Davis F, Persani L, Smallridge RC, Weintraub BD. Thyrotropin-secreting pituitary tumors. *Endocr Rev*. 1996; 17:610–638

[36] Clarke MJ, Erickson D, Castro MR, Atkinson JL. Thyroid-stimulating hormone pituitary adenomas. *J Neurosurg*. 2008; 109:17–22

[37] Beck-Peccoz P, Persani L. Medical management of thyrotropin-secreting pituitary adenomas. *Pituitary*. 2002; 5:83–88

[38] Wilson CB. Endocrine-Inactive Pituitary Adenomas. *Clin Neurosurg*. 1992; 38:10–31

[39] Cohen-Gadol AA, Pichelmann MA, Link MJ, Scheithauer BW, Krecke KN, Young WF, Jr, Hardy J, Giannini C. Granular cell tumor of the sellar and suprasellar region: clinicopathologic study of 11 cases and literature review. *Mayo Clin Proc*. 2003; 78:567–573

[40] Schaller B, Kirsch E, Tolnay M, Mindermann T. Symptomatic granular cell tumor of the pituitary gland: case report and review of the literature. *Neurosurgery*. 1998; 42:166–70; discussion 170-1

[41] Markesbery WR, Duffy PE, Cowen D. Granular cell tumors of the central nervous system. *J Neuropathol Exp Neurol*. 1973; 32:92–109

- [42] Fuller GN, Wesseling P, Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. In: Granular cell tumors of the neurohypophysis. WHO classification of tumors of the central nervous system. 4th ed. Lyon: International Agency for Research on Cancer; 2007:241–242
- [43] Gueguen B, Merland JJ, Riche MC, Rey A. Vascular Malformations of the Spinal Cord: Intrathecal Perimedullary Arteriovenous Fistulas Fed by Medullary Arteries. *Neurology*. 1987; 37:969–979
- [44] Wesseling P, Brat DJ, Fuller GN, Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Bosman FT, Jaffe ES, Lakhani SR, Ohgaki H. In: Pituicytoma. WHO classification of tumors of the central nervous system. 4th ed. Lyon: International Agency for Research on Cancer; 2007:243–244

46 Pituitary Adenomas – Evaluation and Nonsurgical Management

46.1 Evaluation

46.1.1 History and physical

Directed to look for signs and symptoms of:

1. endocrine hyperfunction (see Functional pituitary tumors above), including:
 - a) prolactin: amenorrhea (women), nipple discharge (primarily in women since estrogen is also required), impotence (males)
 - b) thyroid: heat intolerance
 - c) growth hormone: change in ring size or shoe size or coarsening of facial features, gigantism (children)
 - d) cortisol: hyperpigmentation, Cushingoid features
2. endocrine deficits due to mass effect (p. 720) on pituitary
3. visual field deficit: bedside confrontational testing to rule-out visual field deficit (classically bitemporal hemianopsia, see below)
4. deficits of cranial nerves within cavernous sinus
 - a) III, IV, VI: disorder of pupil and extraocular muscles
 - b) VI, V2: reduced sensation in forehead, nose, upper lip and cheek

46.1.2 Diagnostic tests

Overview

Initial (screening) tests to work-up a patient presenting with a known or suspected pituitary mass are shown in □ Table 46.1. Further testing is indicated for abnormal results or for strong suspicion of specific syndromes (see indicated page for details).

Visual fields

Formal visual field testing: by perimetry with a tangent screen (using the small red stimulus since desaturation of color is an early sign of chiasmal compression) or by Goldman or automated Humphrey perimeter (the latter requires good cooperation from the patient to be valid).

Visual field deficit patterns

Depends in part on location of chiasm with respect to sella turcica: the chiasm is located above the sella in 79% posterior to the sella turcica (postfixed chiasm) in 4% in front of the sella (pre-fixed) in 5%^{2 (p 2135)}

1. compression of the optic chiasm:
 - a) bitemporal hemianopsia (p. 560) that obeys the vertical meridian: classic visual field deficit associated with a pituitary tumor. Due to impingement on crossing nasal fibers in the chiasm
 - b) other reported patterns that occur rarely: monocular temporal hemianopsia
2. optic nerve compression: more likely in patients with a postfixed chiasm
 - a) loss of vision in the ipsilateral eye. If carefully sought, there is usually a superior outer (temporal) quadrantanopsia in the contralateral eye^{2 (p 2135)} (so-called junctional scotoma AKA “pie in the sky” defect) from compression of the anterior knee of Wilbrand (p. 1214); may also be an early finding even without a post-fixed chiasm
 - b) may produce central scotoma or monocular reduction in visual acuity
3. compression of the optic tract: may occur with a pre-fixed chiasm. Produces homonymous hemianopsia

Initial endocrinologic evaluation (screening) modified)

See reference.³

Also, see □ Table 46.1. May give indication of tumor type, determines whether any hormones need to be replaced, and serves as a baseline for comparison following treatment. Includes clinical assessment for signs and symptoms, as well as laboratory tests. Screening tests should be checked in

Table 46.1 Summary of initial (screening) work-up for pituitary tumors		
Evaluation		Rationale
☐ Formal visual fields (usually Humphrey visual fields (HVF))		• compression of optic chiasm → visual field deficit (usually bitemporal hemianopsia)
Endocrine screening	☐ 8 A.M. cortisol ^a & 24-hour urine free cortisol ^a	• cortisol ↑ in hypercortisolism (Cushing’s syndrome) (p. 731) • cortisol ↓ in hypoadrenalism (primary or secondary)
	☐ free T4 ^b , TSH	Hypothyroidism • T4 ↓ & TSH ↑ in primary hypothyroidism (this may cause thyrotroph hyperplasia in pituitary gland) • T4 ↓ & TSH nl or ↓ in secondary hypothyroidism (as in hypopituitarism)
	(alternatively, total T4 may be used, if preferred ^b)	Hyperthyroidism (thyrotoxicosis) • T4 ↑ & TSH ↓ in primary hyperthyroidism • T4 ↑ & TSH ↑ in TSH-secreting pituitary adenomas
	☐ prolactin	• ↑ or ↑ ↑ with prolactinoma
		• slight ↑ with stalk effect (usually <90 ng/ml)
	☐ gonadotropins (FSH, LH) and sex steroids (♀: estradiol, ♂: testosterone)	• ↓ in hypogonadotrophic hypogonadism (from mass effect causing compression of the pituitary gland) • ↑ with gonadotropin secreting adenoma
	☐ insulin-like growth factor-1 (IGF-1) AKA somatomedin-C ^b	• ↑ in acromegaly • ↓ in hypopituitarism (one of the most sensitive markers)
	☐ fasting blood glucose	↓ in hypoadrenalism (primary or secondary)
☐ Radiographic studies. Either: • Brain MRI and pituitary MRI without & with contrast (test of choice), usually with protocol for navigation system. Some surgeons also get noncontrast brain CT to look at bony anatomy (esp. sphenoid sinus) • If MRI contraindicated: CT without & with contrast (with coronal reconstruction) + cerebral angiogram		
^a 8 A.M. cortisol is the best test for hypocortisolism (e.g. to look for pituitary insufficiency), 24-hour urine free cortisol is the best test for hypercortisolism (p. 731) ¹ (e.g. to look for Cushing’s syndrome) ^b IGF-1 is the primary test for excess growth hormone (GH); direct measurement of GH is unreliable		

all patients with pituitary tumors. Note: selective loss of a single pituitary hormone together with thickening of the pituitary stalk is strongly suggestive of autoimmune hypophysitis (p. 1373).

1. adrenal axis screening; see tests to assess cortisol reserve (p. 735)

a) Cortisol levels normally peak between 7-8 AM. AM cortisol may normally be elevated slightly above the reference range. 8 AM cortisol level: better for detecting hypocortisolism.¹ Normal: 6-18 mcg/100 ml. Interpretation:

- 8 AM cortisol < 6 mcg/100 ml: suggestive of adrenal insufficiency
- 8 AM cortisol 6-14 mcg/100 ml: nondiagnostic
- 8 AM cortisol > 14 mcg/100 ml: adrenal insufficiency is unlikely

b) in questionable cases, including to distinguish pseudo-Cushing states from Cushing’s syndrome (p. 734)

c) 24-hour urine free cortisol: more accurate for hypercortisolism¹ (almost 100% sensitive and specific, false negative rare except in stress or chronic alcoholism). If not elevated several times above normal, at least 2 additional determinations should be made⁴

2. thyroid axis: the basis for thyroid screening is shown in ☐ Table 46.2

a) screening: T4 level (total or free), thyroid-stimulating hormone (TSH) (AKA thyrotropin). Normal values: free T4 index is 0.8–1.5, TSH 0.4–5.5 mcU/ml, total T4 4–12 mcg/100 ml (NB: be sure to check both T4 AND TSH)

b) further testing: thyrotropin-releasing hormone (TRH) stimulation test (indicated if T4 is low or borderline): check baseline TSH, give 500 mcg TRH IV, check TSH at 30 & 60 mins. Normal response: peak TSH twice baseline value at 30 mins. Impaired response with a low T4 indicates pituitary deficiency. Exaggerated response suggests primary hypothyroidism

Table 46.2 Basis for thyroid screening		
Rationale	T ₄	TSH
Primary hypothyroidism ^a (problem with thyroid gland itself) • chronic primary hypothyroidism may produce secondary pituitary hyperplasia (pituitary pseudotumor) indistinguishable from adenoma on CT or MRI. Must be considered in any patient with a pituitary mass^{7,8} • pathophysiology: loss of negative feedback from thyroid hormones causes increased TRH release from the hypothalamus producing secondary hyperplasia of thyrotrophic cells in the adenohypophysis (thyrotroph hyperplasia). The patient may present due to pituitary enlargement (visual symptoms, elevated PRL from stalk effect, enlarged sella turcica on x-rays...) • chronic stimulation from elevated TRH may rarely produce thyrotroph adenomas • labs: T₄ low or normal, TSH elevated (>90–100 in patients presenting with thyrotroph hyperplasia), prolonged and elevated TSH response to TRH stimulation test (see text)	↓	↑
Secondary hypothyroidism ^a (insufficient TSH stimulation of thyroid) • pituitary hypothyroidism accounts for only ≈ 2–4% of all hypothyroid cases⁹ • ≈ 23% of patients with chromophobe adenomas develop secondary hypothyroidism if untreated (pituitary compression causes reduced TSH) • labs: T₄ low, TSH low or normal, reduced response to TRH stimulation test (see text)	↓	↓ or nl
Primary hyperthyroidism (problem with thyroid gland itself) • etiologies: localized hyperactive thyroid nodule, circulating antibody that stimulates the thyroid, or diffuse thyroid hyperplasia (Graves' disease, AKA ophthalmic hyperthyroidism) • labs: T₄ elevated, TSH subnormal (usually undetectable)	↑	↓
Secondary hyperthyroidism (central hyperthyroidism) • etiologies ◦ TSH-secreting pituitary adenoma (rare) ◦ pituitary resistance to thyroid hormones (disrupts negative feedback loop) • labs: T₄ elevated, TSH elevated or inappropriately normal	↑	↑ or nl
^a □ Caution: replacing thyroid hormone with inadequate cortisol reserves (as may occur in panhypopituitarism) can precipitate adrenal crisis; see management (p. 738)		

3. gonadal axis
- a) screening:
- serum gonadotropins: FSH & LH

• sex steroids: estradiol in women, testosterone in men (measure total testosterone)
- b) further testing: none dependable in differentiating pituitary from hypothalamic disorders
4. prolactin levels (PRL): for prolactin neurophysiology (p. 151)
- a) interpretation is shown in □ Table 46.3. See □ Table 46.4 for differential diagnosis of hyperprolactinemia. Prolactin level correlates with size of prolactinomas⁵: if PRL is <200 ng/ml, ≈ 80% of tumors are microadenomas, and 76% of these will have normal PRL after surgery; if PRL >200, only ≈ 20% are microadenomas
- b) blood samples should be obtained midmorning (i.e. not soon after awakening) and not after stress, breast stimulation, or physical examination, which may increase PRL levels
- c) be aware of the following when interpreting PRL levels:
- because of variations in secretion (daily fluctuations can be as high as 30%) and intrinsic inaccuracies of radioimmunoassay, PRL levels should be rechecked if there is a reason to question a specific result

• heterophilic antibodies (seen in individuals routinely exposed to animal serum products) can cause anomalous results

• stalk effect: PRL is the only pituitary hormone primarily under inhibitory regulation (p. 151). Injury to or compression of the hypothalamus or pituitary stalk from surgery or compression by any type of tumor can cause modest elevation of PRL due to decrease in prolactin inhibitory factor (PRIF). Rule of thumb: the percent chance of an elevated PRL being due to a prolactinoma is equal to one half the PRL level. Persistent post-op PRL elevation may occur even with total tumor removal as a result of injury to stalk (usually ≤90 ng/ml; stalk effect doubtful if PRL >150). For stalk effect, follow these patients, do not use bromocriptine

Table 46.3 Significance of prolactin levels ^a		
PRL (ng/ml)	Interpretation	Situations observed in
3–30 ^b	normal	non-pregnant female
10–400		pregnancy (□ Table 46.4)
2–20		postmenopausal female
25 ^b -150	moderate elevation	• prolactinoma • “stalk effect” (see text) • other causes^d
> 150 ^c	significant elevation	prolactinoma ^d
^a Note: ectopic sites of prolactin secretion have rarely been reported (e.g. in a teratoma ¹⁰)		
^b normal values vary, use your lab’s reference range		
^c some authors recommend 200 ng/ml as the cutoff for probable prolactinomas ¹¹		
^d for DDx of hyperprolactinemia see □ Table 46.4		

Table 46.4 Differential diagnosis of elevated prolactin (PRL) level (hyperprolactinemia) ^a	
- pregnancy-related - during pregnancy^b: 10–400 ng/ml - postpartum: PRL decreases ≈ 50%(to ≈ 100 ng/ml) in the first week postpartum, and is usually back to normal in 3 weeks - in the lactating female: suckling increases PRL, which is critical for lactogenesis (once initiated, nonpregnant PRL levels can maintain lactation). First 2–3 months postpartum: basal PRL=40–50 ng/ml, suckling → increases × 10–20. 3–6 months postpartum: basal PRL levels become normal or slightly elevated, and double with suckling. PRL should normalize by 6 months after weaning - pituitary adenoma - prolactinoma: larger prolactin microadenomas and macroadenomas usually produce PRL>100 ng/ml - stalk effect (p. 732): rule of thumb, the percent chance of an elevated PRL being due to a prolactinoma is equal to one half the PRL level - some tumors secrete both PRL and GH - drugs: dopamine receptor antagonists (e.g. phenothiazines, metoclopramide), oral contraceptives (estrogens), tricyclic antidepressants, verapamil, H2 antagonists (e.g. ranitidine), some SSRIs in particular paroxetine (Paxil®)¹²... - primary hypothyroidism: TRH, a prolactin releasing factor (PRF), (p. 151) will be elevated - empty sella syndrome (p. 766) - post-ictal (p. 467): PRL usually normalizes within 1–2 hrs after a seizure - breast or chest-wall trauma/surgery: usually ≤ 50 ng/ml - excessive exercise: usually ≤ 50 ng/ml - stress: in some cases the stress of having the blood test is enough to elevate PRL, anorexia nervosa - ectopic secretion: reported in renal cell or hepatocellular tumors, uterine fibroids, lymphomas - infiltrating hypothalamic tumors - renal failure - cirrhosis - macroprolactinemia: see text	
^a hyperprolactinemia from causes other than prolactinomas rarely exceeds 200 ng/ml	
^b always R/O pregnancy as a cause of amenorrhea & hyperprolactinemia in a female with reproductive potential	

- “prolactin level >200 ng/ml”: if the lab reports the prolactin level as “>200” (or some other high value) instead of an actual number, it usually indicates a very high prolactin level that exceeds the upper limits of the assay. Call the lab and ask them to determine the actual value. This usually requires the lab to run serial dilutions until the PRL is in a range that their assay can quantify (they may be able to do it with the specimen they have, or else the patient will need to have another blood draw). The reasons this is important: (1) treatment decisions: PRL>500 usually indicates that surgery alone will not be able to normalize the PRL (p. 739); (2) to assess response to treatment: it is essential to know what value you are starting with to determine response to medication, surgery, XRT...
- hook effect: extremely high PRL levels may overwhelm the assay (the large numbers of PRL molecules prevent the formation of the necessary PRL-antibody-signal complexes for

- radioimmunoassay) and produce falsely low results. Therefore, for large adenomas with a normal PRL level, have the lab perform several dilutions of the serum sample and re-run the PRL, especially in patients with clinical hyperprolactinemia
 - macroprolactinemia: a situation where prolactin molecules polymerize and bind to immunoglobulins. Prolactin in this form has reduced biological activity but produces a laboratory finding of hyperprolactinemia. Clinical significance is controversial,⁶ asymptomatic patients usually do not require treatment
5. growth hormone:
 - a) IGF-1 (somatomedin-C) level (p.736) is the recommended initial test (testing for elevated IGF-1 is extremely sensitive for acromegaly)
 - b) checking a single random GH level may not be a reliable indicator (p.736) and is therefore not recommended
 6. neurohypophysis (posterior pituitary): deficits are rare with pituitary tumors
 - a) screening: check adequacy of ADH by demonstrating concentration of urine with water deprivation
 - b) further testing: measurement of serum ADH in response to infusion of hypertonic saline

Specialized endocrinologic tests

Cushing’s syndrome

Tests for hypercortisolism

These tests are used to determine if hypercortisolism (Cushing’s syndrome, CS) is present or not, regardless of etiology. Usually only needed if the screening 24-hr urine free cortisol (p.731) is equivocal (the basis of these tests is shown in □ Table 46.5).

1. overnight low-dose dexamethasone (DMZ) suppression tests¹³:
 - a) overnight low dose test: give DMZ 1 mg PO @11 P.M. and draw serum cortisol the next day at 8 A.M. Results:
 - cortisol < 1.8 mcg/dl (note: this is the currently accepted normal value; previously it was 5 mcg/dl): Cushing’s syndrome is ruled out (except for a few patients with CS who suppress at low DMZ doses, possibly due to low DMZ clearance¹⁴)
 - cortisol 1.8 -10 mcg/dl: indeterminate, retesting is necessary
 - cortisol > 10 mcg/dl: CS is probably present. False positives can occur in the so-called pseudo-Cushing’s state where ectopic CRH secretion produces hyperplasia of pituitary corticotrophs that is clinically indistinguishable from pituitary ACTH producing tumors (requires further testing¹⁴). Seen in: 15%of obese patients, in 25%of hospitalized and chronically ill patients, in high estrogen states, in uremia, and in depression. The combined DMZ-CRH test can be used to identify this (see reference¹⁴). False positives also may occur in alcoholics or patients on phenobarbital or phenytoin due increased metabolism of DMZ caused by induced hepatic microsomal degradation
 - b) 2 day low dose test (used when overnight test is equivocal): give DMZ 0.5 mg PO q 6 hrs for 2 days starting at 6 A.M.; 24 hr urine collections are obtained prior to test and on the 2nd day of DMZ administration. Normal patients suppress urinary 17-hydroxycorticosteroids (OHCS) to less than 4 mg/24 hrs, whereas ≈ 95%of patients with CS have abnormal response (higher amounts in urine)¹⁴
2. 11 PM salivary cortisol: this is the time of the usual cortisol nadir. Test must be run at NIH approved lab. Accuracy is as good as low-dose DMZ suppression test

Distinguishing Cushing’s disease from ectopic ACTH secretion

These tests are used to distinguish primary Cushing’s disease (CD) (pituitary ACTH hypersecretion) from ectopic ACTH production and adrenal tumors (may be required since 40%of CD patients have a normal MRI¹)

Table 46.5 Basis for biochemical tests in Cushing’s syndrome (CS)
• normally, low DMZ doses suppress ACTH release through negative feedback on hypothalamic-pituitary axis, reducing urine and serum corticosteroids • in ≥ 98% of cases of Cushing’s syndrome, suppression occurs, but at a much higher threshold • adrenal tumors and most (85–90%) cases of ectopic ACTH production (especially bronchial Ca) will not suppress even with high dose DMZ • ACTH response to CRH is exaggerated in CS • DMZ does not interfere with measurement of urinary and plasma cortisol and 17-hydroxycorticosteroids

1. random serum ACTH: if <5 ng/L indicates ACTH independent CS (e.g. adrenal tumor). Not sensitive or specific due to variability of ACTH levels
2. abdominal CT: usually shows unilateral adrenal mass with adrenal tumors, or normal or bilateral adrenal enlargement in ACTH-dependent cases
3. high-dose dexamethasone (DMZ) suppression test: (NB: up to 20% of patients with CD do not suppress with high-dose DMZ. Phenytoin may also interfere with high-dose DMZ suppression¹⁵)
 - a) overnight high-dose test: obtain a baseline 8 A.M. plasma cortisol level
 - b) then give DMZ 8 mg PO @ 11 P.M. and measure plasma cortisol level the next morning at 8 A.M.
 - c) in 95% of CD cases plasma cortisol levels are reduced to $<50\%$ of baseline, whereas in ectopic ACTH or adrenal tumors it will usually be unchanged
4. metyrapone (Metopirone®) test: performed on an inpatient basis. Give 750 mg metyrapone (suppresses cortisol synthesis) PO q 4 hrs for 6 doses. Most patients with CD will have a rise in 17-OHCS in urine of 70% above baseline, or an increase in serum 11-deoxycortisol 400-fold above baseline
5. corticotropin-releasing hormone (CRH) stimulation test: CD responds to exogenous CRH 0.1 mcg/kg IV bolus with even further increased plasma ACTH and cortisol levels; ectopic ACTH and adrenal tumors do not¹⁶
6. inferior petrosal sinus (IPS) sampling (or, cavernous sinus sampling is preferred by some): done by interventional neuroradiologist. Uses a microcatheter to measure ACTH levels on each side at baseline, and then at 2, 5 & 10 minutes after stimulation with IV CRH (with simultaneous peripheral ACTH levels at each interval). General information:
 - a) IPS sampling is not needed when the following criteria of CD are met¹⁷:
 - ACTH-dependent Cushing's disease
 - suppression with high-dose dexamethasone test (see above)
 - visible pituitary adenoma on MRI
 - b) may also determine likely side of a microadenoma within the pituitary (thus may be able to avoid bilateral adrenalectomy which requires lifelong gluco- and mineralo-corticoid replacement and risks Nelson's syndrome (p. 724) in 10–30% 15–30% of the time this test falsely lateralizes the tumor due to the communication through the circular sinus
 - c) a baseline IPS ACTH to peripheral ACTH ratio $>1.4:1$ is consistent with primary Cushing's disease
 - d) a post CRH ratio >3 is also consistent with primary Cushing's disease
 - e) complication rate: 1–2% includes puncture of the sinus wall

Assessing cortisol reserve

1. cosyntropin stimulation test¹⁸:
 - a) draw a baseline cortisol level (fasting is not required; test can be performed at any time of day)
 - b) give cosyntropin (Cortrosyn®) (a potent ACTH analogue) 1 ampoule (250 mcg) IM or IV
 - c) then check cortisol levels at 30 mins (optional) and at 60 mins
 - d) normal response: peak cortisol level >18 mcg/dl AND an increment >7 mcg/dl, or a peak >20 mcg/dl regardless of the increment
 - e) subnormal response: indicates adrenal insufficiency. In primary adrenal insufficiency, pituitary ACTH secretion will be elevated. In secondary adrenal insufficiency, chronically reduced ACTH causes adrenal atrophy and unresponsiveness to acute stimulation with this exogenous ACTH analogue
 - f) normal response: rules out primary and overt secondary adrenal insufficiency, but may be normal in mild cases of reduced pituitary ACTH or early after pituitary surgery where adrenal atrophy has not occurred. In these cases further testing may be positive: see metyrapone test (p. 753) or ITT (see below)
2. insulin tolerance test (ITT): “gold standard” for assessing integrity of the hypothalamic-pituitary-adrenal axis. Cumbersome to do. Abnormal in 80% of CS. Assesses ACTH, cortisol & GH reserve
 - a) rationale: an appropriate cortisol increment in response to insulin-induced hypoglycemia suggests patient will also be able to respond to other stresses (acute illness, surgery...)
 - b) contraindications: seizure disorder, ischemic cardiac disease, untreated hypothyroidism
 - c) pre-test preparation: D/C estrogen replacement for 6 weeks prior to test. Have 50 ml of D50 and 100 mg IV hydrocortisone available during test
 - d) protocol: give regular insulin 0.1 U/kg IV push, and draw blood for glucose, cortisol and GH at 0, 10, 20, 30, 45, 60, 90 and 120 mins (monitor blood sugar by fingerstick during test, and give IV glucose if patient becomes symptomatic). If fingerstick blood sugar is not <50 mg/dl by 30 minutes and patient is asymptomatic, give additional regular insulin 5 U IVP. There must be 2 specimens after adequate hypoglycemia

- e) results:
- ifadequate hypoglycemia (<40 mg/dl) was not accomplished: cortisol or GH deficiency cannot be diagnosed
 - normal: cortisol increment >6 mcg/dl to a peak >20
 - peak cortisol=16–20: steroids needed only for stress
 - peak cortisol<16: glucocorticoid replacement needed
 - Cushing syndrome: increment <6

Acromegaly

For suspected acromegaly, the most useful test is an IGF-1 level.

1. insulin-like growth factor-1 (IGF-1) (formerly somatomedin-C) level: an excellent integrative marker of average GH secretion. Normal levels depend on age (peaking during puberty), gender, pubertal stage and lab. Typical fasting levels by age are shown in □ Table 46.6. Estrogen may suppress IGF-1 levels
2. growth hormone (GH): normal basal fasting level is <5 ng/ml. In patients with acromegaly, GH is usually >10 ng/ml but can be normal. Normal basal levels do not reliably distinguish normal patient from GH deficiency.¹⁹ Furthermore, due to pulsatile secretion of GH, normal patients may have sporadic peaks up to 50 ng/ml.²⁰ Occasionally acromegaly may be present even with GH levels as low as 37 pg/ml.²¹ □random GH levels are not generally useful for diagnosing acromegaly (see above for IGF-1)
3. other tests used uncommonly
 - a) oral glucose suppression test (OGST): less precise and more expensive than measuring IGF-1, however may be more useful than IGF-1 for monitoring initial response to therapy. GH levels are measured at 0, 30, 60, 90 & 120 minutes after a 75 gm oral glucose load. If the GH nadir is not <1 ng/ml, the patient is acromegalic.^{22,23} GH suppression may also be absent with liver disease, uncontrolled DM & renal failure. □ Relatively contraindicated in patients with DM and high glucose levels
 - b) growth-hormone releasing hormone (GHRH) levels: may help diagnose ectopic GH secretion in a patient with proven acromegaly with no evidence of pituitary tumor on imaging. If an extrapituitary source is suspected, chest and abdominal CT and/or MRI should also be obtained²⁴
 - c) GHRH stimulation test: results may be discordant in up to 50%of patients with acromegaly²² and is thus rarely used (as of this writing, pharmaceutical production of GHRF has been discontinued)
4. octreotide scan: SPECT imaging 4 and 24 hours after injection with 6.5 mCi of indium-111 OctreoScan, a somatostatin receptor imaging agent

Radiographic evaluation

General information

≈ 50%of pituitary tumors causing Cushing’s syndrome are too small to be imaged on CT or MRI (therefore endocrinologic testing is required to prove the pituitary origin). See differential diagnosis of intrasellar lesions (p.1371); some are indistinguishable radiographically.

Table 46.6 Normal IGF-1 by age	
Age (yrs)	Level (ng/ml)
1–5	49–327
6–8	52–345
9–11	74–551
12–15	143–996
16–20	141–903
21–39	109–358
40–54	87–267
>54	55–225

Normal AP diameter of pituitary gland: female of childbearing age ($\approx 13\text{--}35$ yrs): ≤ 11 mm, for all others normal is ≤ 9 mm. (Note: pituitary glands in adolescent girls may be physiologically enlarged (mean height: 8.2 ± 1.4 mm) as a result of hormonal stimulation of puberty.²⁵)

Skull x-ray

A lateral skull x-ray may help define the bony anatomy of the sphenoid sinus in cases where trans-sphenoidal surgery is contemplated (currently, CT scan is usually employed).

MRI

Imaging test of choice for pituitary tumors.

Ordering MRI:

- typical order on initial workup: brain MRI and pituitary MRI without and with contrast (pituitary protocol includes thin coronal cuts through sella showing cavernous sinus and optic chiasm) with image navigation protocol (e.g. BrainLab™, or Stealth™...)
- if you are hunting for a microadenoma, then a dynamic MRI increases the chances that you might catch the tumor at a time when it enhances differentially from the gland.
- for follow-up of macroadenomas: routine coronal & sagittal pituitary MRI without & with contrast suffices

Findings: Information about invasion of cavernous sinus, and about location and/or involvement of para-sellar carotids. MRI may fail to demonstrate tumor in 25–45% of cases of Cushing's disease.²⁶ 3 T vs. 1.5 T MRI: based on 5 cases of Cushing's disease, a 3 T MRI showed the adenoma more clearly in 2 cases, in 1 case it showed the tumor on the correct side opposite to where the 1.5 T MRI showed it, and in 2 cases neither 1.5 T nor 3 T MRI could show the microadenoma).²⁷

Microadenoma: 75% are low signal on T1WI, and high signal on T2WI (but 25% can behave in any way, including completely opposite to above). Enhancement is very time-dependent. Imaging must be done with 5 minutes of contrast administration to see a discrete microadenoma. Initially, gadolinium enhances the normal pituitary (no blood brain barrier) but not the pituitary tumor. After ≈ 30 minutes, the tumor enhances about the same. Dynamic MRI scans have been used to increase the sensitivity (contrast is injected while the MRI scanner is running).

Neurohypophysis: normally is high signal on T1WI²⁸ (possibly due to phospholipids). Absence of this "bright spot" often correlates with diabetes insipidus as may occur with autoimmune hypophysitis (p. 1373), however, failure to image the bright spot is not uniformly abnormal.

Deviation of the pituitary stalk may also indicate the presence of a microadenoma. Normal thickness of the pituitary stalk is approximately equal to basilar artery diameter. Thickening of stalk is usually NOT adenoma, differential diagnosis for a thickened stalk: lymphoma, autoimmune hypophysitis (p. 1373), granulomatous disease, hypothalamic glioma.

CT

Generally superseded by MRI. May be appropriate when MRI is contraindicated (e.g. pacemaker). When done, should include direct coronal imaging, or coronal reconstructions from thin-cut axial CT. If MRI cannot be done, consider also cerebral angiography to demonstrate parasellar carotid arteries and to R/O aneurysm as a possibility.

Calcium in pituitary usually signifies hemorrhage or infarction within tumor.

Angiography

Sometimes used in cases considered for transsphenoidal surgery (e.g. as a complement to CT) to localize the parasellar carotids (note: MRI provides this information, and evaluates invasion of cavernous sinuses, usually obviating the need for angiography).

46.2 Management/treatment recommendations

46.2.1 General information

See treatment of pituitary apoplexy (p. 720). For large invasive adenomas, see below.

Note: Prolactinoma is the only pituitary tumor for which medical therapy (dopamine agonist) is the primary treatment modality (in certain cases).

Hormone replacement therapy (HRT)

HRT may be required for patients with documented endocrine deficits pre-op or post-op in cases where pituitary function is not normal. Critical issues:

1. corticosteroids
 - a) indications: inadequate cortisol reserve as demonstrate by failing a cosyntropin stimulation test; failure to achieve a peak cortisol level >18 mcg/dl in response to cosyntropin (p.735)
 - b) may start cortisol immediately after bloodwork for cosyntropin test is drawn (do not need to wait for test results) – then, when test results available, continue or stop therapy based on test results
 - c) ☐ physiologic replacement dose: cortisol 20 mg po q AM and 10 mg po q 4 PM. Stress doses may be needed (p.146) in some situations
2. thyroid hormone replacement
 - a) ☐ thyroid replacement can precipitate adrenal crisis if started before cortisol in a patient with adrenal insufficiency (as may occur in panhypopituitarism)
 - ☐ do a cosyntropin stimulation test (p.735) and start cortisol
 - thyroid replacement may be initiated after 1 full day of cortisol.
 - ☐: start with synthroid 125 mcg/d
 - b) although there are warnings not to do surgery on a hypothyroid patient, the reality is that it takes 3–4 weeks for adequate replacement and hypothyroid patients frequently undergo surgery before then with no untoward effect
3. testosterone replacement: may increase intratumoral levels of estradiol which may promote tumor growth. ☐ wait for stabilization of tumor before starting

46.2.2 Management of large, invasive adenomas

See reference.²⁹

1. prolactinomas
 - a) dopamine agonists (DA) (p.739) unless there is unstable deficit
 - b) for unstable deficit, or if the tumor does not respond to DAs: debulk the tumor transsphenoidally and then rechallenge with DA therapy
2. tumors secreting growth hormone or ACTH: an aggressive surgical approach is indicated with these tumors since the secretion product is harmful and effective medical adjuvants are lacking
 - a) pre-treat invasive GH-secreting tumors with somatostatin analogue therapy before surgery to reduce surgical risks (general and cardiac)
 - b) elderly patients or tumors >4 cm diameter: debulk tumor transsphenoidally and/or adjuvant therapy (XRT and/or medications)
 - c) young age and size <4 cm: radical surgery (may utilize a cranio-orbito-zygomatic skull base approach; may be curative)
3. nonfunctional adenomas
 - a) elderly patient: expectant management is an option, with intervention for signs of progression (radiographic or neurologic)
 - b) central tumor or elderly patient with progression: transsphenoidal tumor debulking and/or XRT (residual tumor in the region of the cavernous sinus may show little or no change over several years, and with these nonfunctional tumors, there is less harm in following them than if there is a harmful secretion product)
 - c) parasellar tumor and/or young age: radical surgery A (often not curative)

46.2.3 Hormonally inactive macroadenomas – management

General information

Due to poor response rates to medication, when treatment is indicated, surgery and/or XRT are usually the initial treatment of choice (see below for XRT).

Medical management of hormonally inactive macroadenomas

Bromocriptine has been tried with mild reductions in tumor size in only $\approx 20\%$ of patients. The poor results are probably due to the paucity of dopaminergic receptors on cell membranes in these tumors. Octreotide reduces tumor volume in $\approx 10\%$ of cases. These agents have been used pre-op in some cases to try to decrease tumor size for surgery.

Follow-up recommendations for medically managed hormonally inactive macroadenomas

For asymptomatic microadenomas (<1 cm dia), recommend: F/U pituitary MRI at years 1, 2, 5 and ± 10 (can stop F/U after 10 and possibly 5 years if no growth).

For tumors >1 cm, recommend: check visual fields, pituitary bloodwork (to R/O pituitary insufficiency) and pituitary MRI at years 0.5, 1, 2 & 5, and any time symptoms develop.

Gonadotropin-secreting tumors

Rarely, a non-functional tumor may secrete gonadotropins (FSH, LH). This does not produce a clinical syndrome. Normal and neoplastic pituitary gonadotrophs have gonadotropin-releasing hormone (GnRH) receptors, and may respond to long-acting GnRH agonists (by down-regulating receptors) or GnRH antagonists, but significant reductions in tumor size does not occur.

Surgical indications for hormonally inactive pituitary macroadenomas

1. tumors causing symptoms by mass effect: visual field deficit (classically: bitemporal hemianopsia, panhypopituitarism)
2. some surgeons recommend surgery for macroadenomas that elevate the chiasm even in the absence of endocrine abnormalities or visual field deficit because of the possibility of injury to the optic apparatus (see below for invasive pituitary macroadenomas)
3. acute and rapid visual or other neurologic deterioration. May represent ischemia of the chiasm, or tumor hemorrhage/infarction causing expansion (pituitary apoplexy). The major danger is blindness (hypopituitarism, while concerning, can be treated with replacement therapy). Visual loss usually requires emergent decompression. Some surgeons feel that a transcranial approach is necessary, but transsphenoidal decompression is usually satisfactory^{29,30}
4. to obtain tissue for pathological diagnosis in questionable cases
5. Nelson's syndrome (p.724)
 - a) surgery (transsphenoidal or transcranial): the primary treatment. The aggressiveness of the tumor sometimes requires total hypophysectomy
 - b) XRT (possibly SRS) is used following subtotal excision
 - c) medical therapy is usually ineffective. Agents that could be considered include³¹: dopamine agonists, valproic acid, somatostatin analogues, rosiglitazone, and serotonin agonists

46

46.2.4 Prolactinomas – management

General information

1. prolactin level (PRL) <500 ng/ml in tumors that are not extensively invasive (see below for invasive tumors): PRL may be normalized with surgery
2. PRL >500 ng/ml: the chances of normalizing PRL surgically are very low.³² Algorithm:
 - a) if no acute progression (worsening vision...), an initial attempt at purely medical control should be made as the chances of normalizing PRL surgically with pre-op levels >500 ng/ml are very low³² (these tumors may shrink dramatically with bromocriptine)
 - b) response should be evident by 4–6 weeks (significant decrease in PRL, improvement of visual deficits, or shrinkage on MRI)
 - c) if tumor not controlled medically ($\approx 18\%$ will not respond to bromocriptine): surgery followed by reinstitution of medical therapy may normalize PRL

Medical management with Dopamine agonists

Dopamine Agonists

Side effects³³: (may vary with different preparations) nausea, H/A, fatigue, orthostatic hypotension with dizziness, cold induced peripheral vasodilatation, depression, nightmares and nasal congestion. Side effects are more troublesome during the first few weeks of treatment. Tolerance may be improved by bedtime dosing with food, slow dose escalation, sympathomimetics for nasal congestion, and acetaminophen 1–2 hrs before dosing to reduce H/A. Psychosis and vasospasm are rare side effects that usually necessitates discontinuation of the drug.

Drug info: Bromocriptine (Parlodel®)

A semi-synthetic ergot alkaloid that binds to dopamine receptors (dopamine agonist) on normal and tumor lactotrophs, inhibiting synthesis and secretion of PRL and other cell processes resulting in decreased cell division and growth. Bromocriptine lowers prolactin level regardless of whether the source is an adenoma or normal pituitary (e.g. as a result of stalk effect) to <10% of pretreatment values in most patients. It also frequently reduces the tumor size in 6–8 weeks in 75% of patients with macroadenomas, but only as long as therapy is maintained and only for tumors that actually produce prolactin. Only $\approx 1\%$ of prolactinomas continue to grow while the patient is on bromocriptine. Prolactinomas may enlarge rapidly upon discontinuation of the drug. However, permanent normoprolactinemia can occur (see below).

Pregnancy issues: bromocriptine can restore fertility. Continued therapy during pregnancy is associated with a 3.3% incidence of congenital anomalies and 11% spontaneous abortion rate which is the same as for the general population. Estrogen elevation during pregnancy stimulates hyperplasia of lactotrophs and some prolactinomas, but the risk of symptomatic enlargement of microadenomas and totally intrasellar macroadenomas is <3% vs. 30% risk for macroadenomas.³⁴

Prolonged treatment with bromocriptine may reduce the chances of surgical cure if this should be chosen at a later date. With a microadenoma, one year of bromocriptine may reduce the surgical cure rate by as much as 50%, possibly due to induced fibrosis.³⁵ Thus, it is suggested that if surgery is to be done that it be done in the first 6 months of bromocriptine therapy. Shrinkage of large tumors due to bromocriptine may cause CSF rhinorrhea.³⁶ Side effects : see above.

□: start with 1.25 mg (half of a 2.5 mg tablet) PO q hs (nighttime dosing reduces some side effects) (vaginal administration is an alternative). Add additional 2.5 mg per day as necessary (based on PRL levels), making a dosage change every 2–4 weeks for microadenomas, or every 3–4 days for macroadenomas causing mass effect. Initial recheck of prolactin level after about 4 weeks at a reasonable dose to verify response. □ To shrink large tumors or for extremely high PRL levels, higher doses are usually needed initially (e.g. 7.5 mg TID for ≈ 6 mos), and then lower doses may be able to maintain normal levels (typical maintenance dosage: 5–7.5 mg daily (range: 2.5–15 mg) which may be given as a single dose or divided TID). Supplied: 2.5 mg scored tabs; 5 mg capsules.

Drug info: Cabergoline (Dostinex®)

An ergot alkaline derivative that is a selective D2 dopamine agonist (bromocriptine (see above) affects both D2 and D1 receptors).³⁷ The elimination half-life is 60–100 hrs which usually permits dosing 1–2 times weekly. Control of PRL and resumption of ovulatory cycles may be better than with bromocriptine.³⁸ Side effects : (see above) H/A and GI symptoms are reportedly less problematic than with bromocriptine. □ Cardiac valve disease³⁹ (incidence-rate ratio, 4.9; 95% CI) affecting the mitral, aortic, and tricuspid valves possibly leading to regurgitation (the drug activates 5-HT_{2B} receptors which induces prolonged mitogenic effects in fibrocytoblasts, which may lead to valvular fibroplasia) which has not been observed at doses used for prolactinomas (is associated with doses used for Parkinson's disease which are $>10 \times$ pituitary doses): recommendation: do not discontinue cabergoline for this reason if dose is <2 mg/wk. □ Contraindications: eclampsia or pre-eclampsia, uncontrolled HTN. Dosage should be reduced with severe hepatic dysfunction.

□: Start with 0.25 mg PO twice weekly, and increase each dose by 0.25 mg every 4 weeks as needed to control PRL (up to a maximum of 3 mg per week). Typical dose is 0.5–1 mg twice weekly. Some combine the total dose and give it once weekly. Initial recheck of prolactin level after about 4 weeks to verify response. Supplied: 0.5 mg scored tablets.

Drug info: Pergolide (Permax®)

A long-acting ergot alkaloid dopamine agonist that reduces PRL levels for >24 hrs. Not available in the U.S. for use in humans. Once daily dosing improves compliance. Side effects : □ Withdrawn due to risk of cardiac valve disease (see cabergoline (Dostinex®) above).

□ Start with 0.05 mg PO q hs, and increase by 0.025–0.05 increments (up to a maximum of ≈ 0.25 mg/d) until desired PRL levels are achieved.

Response to medical treatment

Treatment response to DA is assessed with serial prolactin levels as shown in □ Table 46.7. It is uncommon for a prolactinoma to enlarge without an increase in prolactin level.⁵

Discontinuation of dopamine agonists: Long-term therapy with DA agonists has some cytocidal effect on pituitary tissue. In an early report, discontinuation of treatment after 24 months was associated with >95%recurrence rate.⁴⁰ Recent literature suggests a 20–30%chance of normoprolactinemia off medication in select patients.⁴¹

Recommendations⁴¹: if response to DA agonist is satisfactory, treat for 1–4 years (microadenomas: check prolactin yearly, macroadenomas are more likely to grow and should be checked more often). Microadenomas or macroadenomas that are no longer visible on MRI are candidates for DA agonist withdrawal. For microadenomas: discontinue the drug; for macroadenomas: slowly taper the drug then discontinue. Recurrence rate is highest during 1st year, □ check prolactin levels and clinical symptoms every 3 months during the 1st year. Long-term follow up is required, especially for macroadenomas.

46.2.5 Acromegaly – management

See references.^{23,42,43}

Surgery

Surgery is the primary treatment modality for acromegaly when treatment is indicated.

1. asymptomatic elderly patients do not require treatment since there is little evidence that intervention alters life expectancy in this group
2. if no contraindications, surgery (usually transsphenoidal) is currently the best initial therapy (worse prognosis with macroadenomas) providing more rapid reduction in GH levels and decompression of neural structures (e.g. optic chiasm) and improves the efficacy of subsequent somatostatin analogues.⁴³ Surgery is not recommended for elderly patients
3. medical therapy (p.741); reserved for:
 - a) patients not cured by surgery (reoperation doesn’t work very often for acromegaly). Note: the definition of “biochemical cure” with acromegaly is not standardized (p.754), surgery is still helpful for those “not cured” and improves efficacy of other therapies; IGF-1 may take months to normalize after surgery
 - b) or for those who cannot tolerate surgery (e.g. due to cardiomyopathy, severe hypertension, airway obstruction..., these contraindications may improve with medical therapy and then surgery can be reconsidered)
 - c) or for recurrence after surgery or XRT
4. XRT (p.745): for failure of medical therapy. Not recommended as initial treatment. NB: some practitioners use XRT for surgical failure, and employ medical therapy while waiting for XRT to have an effect. GH levels decline very slowly after XRT, see details & side effects (p.739)

Medical therapy

Overview

1. dopamine agonists (DAs): although not mentioned in the AACE guidelines,²² it may be worth trying a DA to see if the tumor responds (≈ 20%respond). If responsive, DAs are especially well suited for GH tumors that cosecrete PRL
 - a) bromocriptine: (see below) although it benefits only a minority, a first line drug since it is cheaper than pegvisomant or octreotide and is given PO
 - b) cabergoline (see above)
 - c) pergolide (see above)
 - d) others: lisuride, depo-bromocriptine (bromocriptine-LAR)

Table 46.7 Prolactin level with DA agonist treatment	
PRL level (ng/ml)	Recommendation
<20	maintain
20–50	reassess dose
>50	consider surgery

2. somatostatin analogues: indications: as initial medical therapy, or if no response to DAs, some also use this pre-op to improve surgical success rate
 - a) octreotide & octreotide-LAR (see below)
 - b) lanreotide, lanreotide SR & long-acting aqueous gel lanreotide (Autogel)
3. GH antagonists: pegvisomant (see below) considered for failures to above (not a primary therapy)
4. combination therapy: may be more effective than individual drugs. Pegvisomant or octreotide + dopamine agonist if no response to 1 drug alone

Specific agents

Drug info: Bromocriptine (Parlodel®)

Neoplastic somatotrophs may respond fortuitously to dopamine agonists and reduce growth hormone (GH) secretion. Bromocriptine lowers GH levels to <10 ng/ml in 54% of cases, to <5 ng/ml in only $\approx 12\%$. Tumor shrinkage occurs in only $<20\%$. Higher doses are usually required than for prolactinomas. If effective, the drug may be continued but should be periodically withdrawn to assess the GH level; see Side effects (p. 739). Estimated annual cost: \$3,200 in the U.S.

□ For growth hormone tumors that respond to bromocriptine, the usual dosage is 20–60 mg/d in divided doses (higher doses are unwarranted). The maximal daily dose is 100 mg.

Drug info: Octreotide (Sandostatin®)

A somatostatin analogue that is 45 times more potent than somatostatin in suppressing GH secretion but is only twice as potent in suppressing insulin secretion, has a longer half-life (≈ 2 hrs after SQ injection, compared to $\approx$ minutes for somatostatin), and does not result in rebound GH hypersecretion. GH levels are reduced in 71%, IGF-1 levels are reduced in 93%. 50–66% have normal GH levels, 66% achieve normal IGF-1 levels. Tumor volume reduces significantly in about 30% of patients. Many symptoms including H/A usually improve within the first few weeks of treatment. Annual cost to the patient: at least $\approx$ \$7,800 in the U.S. Usually given in combination with bromocriptine.

After 50 mcg SQ injection, GH secretion is suppressed within 1 hr, nadirs at 3 hrs, and remains reduced for 6–8 hrs (occasionally up to 12 hrs). Side effects: reduced GI motility and secretion, diarrhea, steatorrhea, flatulence, nausea, abdominal discomfort (all of these usually remit in 10 days), clinically insignificant bradycardia in 15%, cholesterol cholelithiasis (in 10–25%) or bile sludge. Asymptomatic stones require no treatment and routine ultrasonography is not required. Mild hypothyroidism or worsening of glucose intolerance may occur.

□: Start with 50–100 mcg SQ q 8 hrs. Increase up to a maximum of 1500 mcg/d (doses >750 mcg/d are rarely needed). Average dose required is 100–200 mcg SQ q 8 hrs.

Sandostatin LAR Depot: long acting release (LAR) form given by IM injection. □: give a test dose of short acting octreotide SQ in the office, and if no reaction (e.g. N/V...) begin LAR injections with 20 mg IM q 4 weeks, increase to 30 mg if GH >5 mU/L just before 4th dose. Control can be achieved in some patients with dosing q 8–12 weeks.⁴⁴

Drug info: Pegvisomant (Somavert®)

A genetically engineered competitive GH-receptor antagonist. Treatment for ≥ 12 mos results in normal IGF-1 levels in 97% of patients.⁴⁵ No change in pituitary tumor size has been observed.⁴⁶ Indications: failure of somatostatin in patient with GH secreting adenoma (patient is switched to pegvisomant, it is not added to regimen). Side effects: significant but reversible liver function abnormalities occur in $<1\%$. Serum GH increases, probably as a result of loss of negative feedback on IGF-1 production. Antibodies to GH occurred in 17% but tachyphylaxis was not observed.

□: 5–40 mg/d SQ (dose must be titrated to keep IGF-1 in the normal range, to avoid GH deficiency conditions).

46.2.6 Cushing's disease – management

Management algorithm

1. if pituitary MRI shows a mass: transsphenoidal surgery
2. if pituitary MR is negative (up to 40% of patients with Cushing's disease have negative MRI): perform inferior petrosal sinus (IPS) sampling (p.735)
 - a) if IPS sampling is positive: surgery
 - b) if IPS sampling is negative: look for extra-pituitary source of ACTH (abdominal CT)
3. if pituitary surgery is performed but biochemical cure – see criteria (p.753) – is not obtained with surgery:
 - a) unlike acromegaly, a partial reduction is not helpful to the patient
 - b) consider re-exploration if pituitary source is still suspected
 - c) stereotactic radiosurgery or medical therapy (see below)
 - d) adrenalectomy in appropriate patients (see below)

Transsphenoidal surgery

Transsphenoidal surgery is the treatment of choice for most (medical therapy is inadequate as initial therapy since there is no effective pituitary suppressive medication). Cure rates are $\approx 85\%$ for microadenomas (i.e. tumors ≤ 1 cm dia), but are lower for larger tumors. Even with microadenomas, hemihypophysectomy on the side of the tumor is usually required for cure (the tumor is difficult to completely extirpate) with attendant increased risk of CSF leak. If this fails, consideration should then be for total hypophysectomy. Failure of total hypophysectomy prompts consideration for bilateral adrenalectomy (total hypophysectomy virtually eliminates risk of Nelson's syndrome following adrenalectomy – see below).

Stereotactic radiosurgery

Often normalizes serum cortisol levels. Useful for: recurrence after surgery, inaccessible tumors (e.g. cavernous sinus)⁴⁷...

Adrenalectomy

Total bilateral adrenalectomy (TBA) corrects hypercortisolism in 96–100%³¹ (unless there is an extra-adrenal remnant), but lifelong gluco- and mineralo-corticoid replacement are required and up to 30% develop Nelson's syndrome (p.724); incidence reduced by total hypophysectomy or possibly by pituitary XRT.

Indications: continued hypercortisolism with:

1. non-resectable pituitary adenoma
2. failure of medical therapy to control symptoms after transsphenoidal surgery
3. life-threatening Cushing's disease (CD)
4. CD with no evidence of pituitary tumor; testing should include high-dose DMZ suppression test (p.735) and/or inferior petrosal sinus sampling (p.735)

Follow-up after TBA to rule-out Nelson's syndrome: there is no standardized regimen. Suggestion: check serum ACTH levels q 3–6 months $\times$ 1 year, q 6 months $\times$ 2 years, q year thereafter. A pituitary MRI is done if an ACTH level is > 100 ng/L, otherwise, annual MRIs are sufficient⁴⁸ $\times$ 3 years and then if ACTH levels remain low, get an MRI every other year.

Medical therapy

For patients who fail surgical therapy or for whom surgery cannot be tolerated, medical therapy and/or radiation are utilized. Occasionally may be used for several weeks prior to planned surgery to control significant manifestations of hypercortisolism, e.g. diabetes, HTN, psychiatric disturbances... (p.723).

Ketoconazole (Nizoral®)³³: an antifungal agent that blocks adrenal steroid synthesis. The initial drug of choice. Over 75% of patients have normalization of urinary free cortisol and 17-hydroxycorticosteroid levels. Side effects: reversible elevations of serum hepatic transaminase (in 15%), GI discomfort, edema, skin rash. Significant hepatotoxicity occurs in 1 of 15,000 patients. Watch $\square$ Table 8.2.

$\square$ Start with 200 mg PO BID. Adjust dosage based on 24-hr urine free cortisol and 17-hydroxycorticosteroid levels. Usual maintenance doses 400–1200 mg daily in divided doses (maximum of 1600 mg daily).

Aminoglutethimide (Cytadren®)³³: inhibits the initial enzyme in the synthesis of steroids from cholesterol. Normalizes urinary free cortisol in $\approx 50\%$ of cases. Side effects: dose-dependent reversible effects include sedation, anorexia, nausea, rash and hypothyroidism (due to interference with thyroid hormone synthesis).

□ Start with 125–250 mg PO BID. Effectiveness may diminish after several months and dose escalation may be needed. Generally do not exceed 1000 mg/d.

Metyrapone (Metopirone®): inhibits 11- β -hydroxylase (involved in one of the final steps of cortisol synthesis) may be used alone or in combination with other drugs. Normalizes mean daily plasma cortisol in $\approx 75\%$. Side effects: lethargy, dizziness, ataxia, N/V, primary adrenal insufficiency, hirsutism and acne.

□ Usual dose range is 750–6000 mg/d usually divided TID with meals. Initial effectiveness may diminish with time.

Mitotane (Lysodren®): related to the insecticide DDT. Inhibits several steps in glucocorticoid synthesis, and is cytotoxic to adrenocortical cells (adrenolytic agent). 75% of patients enter remission after 6–12 months of treatment, and the medication may sometimes be discontinued (however hypercortisolism may recur). Side effects: may be limiting, and include anorexia, lethargy, dizziness, impaired cognition, GI distress, hypercholesterolemia, adrenal insufficiency (which may necessitate supernormal doses of glucocorticoids for replacement due to induced glucocorticoid degradation).

□ Start with 250–500 mg PO q hs, and escalate dose slowly. Usual dose range is 4–12 gm/d usually divided TID-QID. Initial effectiveness may diminish with time.

Cyproheptadine (Periactin®): a serotonin receptor antagonist that corrects the abnormalities of Cushing's disease in a small minority of patients, suggesting that some cases of "pituitary" Cushing's disease are really due to a hypothalamic disorder. Combined therapy with bromocriptine may be more effective in some patients. Side effects: sedation & hyperphagia with weight gain usually limit usefulness.

□ Usual dosage range: 8–36 mg/d divided TID.

46.2.7 Thyrotropin (TSH)-secreting adenomas – management

General information

1. transsphenoidal surgery has been the traditional first-line treatment.⁴⁹ These tumors may be fibrous and difficult to remove⁵⁰
2. for incomplete resection: post-op XRT is employed
3. if hyperthyroidism persists: medical therapy is added with agents including octreotide, bromocriptine (more effective for tumors that co-secrete PRL), and oral cholecystographic agents (which inhibit conversion of T4 to T3) e.g. iopanoic acid

Medical therapy

Normal and neoplastic anterior hypophyseal thyrotroph cells possess somatostatin receptors and most respond to octreotide (see below). Occasionally, beta-blockers or low-dose antithyroid drugs (e.g. Tapazole® (methimazole) ≈ 5 mg PO TID for adults) may additionally be required.

Octreotide (Sandostatin®)

Doses required are usually < than with acromegaly. TSH levels decline by >50% in 88% of patients, and become normal in $\approx 75\%$. T4 and T3 levels decrease in almost all, with 75% becoming normal. Tumor shrinkage occurs in $\approx 33\%$

□ Start with 50–100 mcg SQ q 8 hrs. Titrate to TSH, T4 and T3 levels.

46.3 Radiation therapy for pituitary adenomas

46.3.1 General information

Conventional EBXRT usually consists of 40–50 Gy administered over 4–6 weeks.

46.3.2 Side effects

Radiation injury to the remaining normal pituitary results in hypocortisolism, hypogonadism, or hypothyroidism in 40–50% of patients after 10 years. It may also injure the optic nerve and chiasm

Table 46.8 Recurrence rate of pituitary tumors removed transsphenoidally ^a		
Extent of removal	Post-op XRT?	Recurrence rate
subtotal	no	50%
gross total		21%
subtotal	yes	10%
gross total		0
^a 108 macroadenomas, 6 mos to 14 years follow-up ⁵²		

(possibly causing blindness), cause lethargy, memory disturbances, cranial nerve palsies, and tumor necrosis with hemorrhage and apoplexy. Cure rates but also complications are higher after proton beam therapy.

46.3.3 Recommendation

Radiation therapy should not be routinely used following surgical removal. Follow patient with yearly MRI. Treat recurrence with repeat operation. Consider radiation if recurrence cannot be removed and tumor continues to grow.

46.3.4 Sellar radiation therapy for nonfunctional pituitary tumors

In one series of 89 nonfunctioning pituitary tumors ranging 0.5–5 cm diameter (mean = 2 cm) not totally resected because of involvement of cavernous sinus (or other inaccessible sites), half were treated with radiation therapy (XRT). The recurrence rate was neither lower (and was actually higher) nor later in the XRT group.⁵¹ However, another series of 108 pituitary macroadenomas found the recurrence rates shown in □ Table 46.8 which tend to favor radiation therapy.

When used, doses of 40 or 45 Gy in 20 or 25 fractions, respectively, is recommended.⁵³ The onco-cytic variant of null cell pituitary tumors appears to be more radioresistant than the nononcocytic undifferentiated cell adenoma.⁵³

46.3.5 Sellar radiation therapy for acromegaly

Not the preferred initial treatment. Works better with lower initial GH levels. In most patients, GH levels begin to fall during the first year after XRT, dropping by ≈ 50% after 2 years, and decrease gradually thereafter, reaching ≤ 10 ng/ml in 70% of patients after 10 years. It takes up to 20 years for 90% of patients to achieve GH levels < 5 ng/ml. During this latency period, patients are exposed to unacceptably high levels of GH (octreotide may be used while waiting). Patients are also still at risk for radiation side effects mentioned above. Options include: EBRT, stereotactic radiosurgery (about equally effective). Estimated cost: \$20,000.

46.3.6 Sellar radiation therapy for Cushing’s disease

XRT corrects hypercortisolism in 20–40% and produces some improvement in another 40%. Improvement may not be seen for 1–2 yrs post treatment.

References

[1] Chandler WF. Treatment of disorders of the pituitary gland: pearls and pitfalls from 30 years of experience. Clin Neurosurg. 2008; 56:18–22

[2] Walsh FB, Hoyt WF. Clinical Neuro-Ophthalmology. Baltimore 1969

[3] Tindall GT, Barrow DL. Current Management of Pituitary Tumors: Part I. Contemp Neurosurg. 1988; 10:1–6

[4] Watts NB. Cushing’s Syndrome: An Update. Contemp Neurosurg. 1995; 17:1–7

[5] Gillam MP, Molitch ME, Lombardi G, Colao A. Advances in the treatment of prolactinomas. Endocr Rev. 2006; 27:485–534

[6] Olukoga AO. Macroprolactinemia is clinically important. J Clin Endocrinol Metab. 2002; 87:4833–4834

[7] Bilaniuk LT, Moshang T, Cara J, et al. Pituitary Enlargement Mimicking Pituitary Tumor. J Neurosurg. 1985; 63:39–42

[8] Atchison JA, Lee PA, Albright L. Reversible Suprasellar Pituitary Mass Secondary to Hypothyroidism. JAMA. 1989; 262:3175–3177

[9] Watanakunakorn C, Hodges RE, Evans TC. Myxedema. A Study of 400 Cases. Arch Intern Med. 1965; 116:183–190

- [10] Kallenberg GA, Pesce CM, Norman B, et al. Ectopic Hyperprolactinemia Resulting From an Ovarian Teratoma. *JAMA*. 1990; 263:2472–2474
- [11] Randall RV, Scheithauer BW, Laws ER, et al. Pituitary Adenomas Associated with Hyperprolactinemia. *Mayo Clin Proc*. 1985; 60:753–762
- [12] Cowen PJ, Sargent PA. Changes in plasma prolactin during SSRI treatment: evidence for a delayed increase in 5-HT neurotransmission. *J Psychopharmacol*. 1997; 11:345–348
- [13] Tyrell JB, Aron DC, Forsham PH, Greenspan FS. In: *Glucocorticoids and Adrenal Androgens. Basic and Clinical Endocrinology*. 3rd ed. Norwalk: Appleton and Lange; 1991:323–362
- [14] Yanovski JA, Cutler GB, Chrousos GP, Nieman IK. Corticotropin-releasing hormone stimulation following low-dose dexamethasone administration: A new test to distinguish Cushing's syndrome from pseudo-Cushing's states. *JAMA*. 1993; 269:2232–2238
- [15] McCutcheon IE, Oldfield EH, Barrow DL, Selman W. In: *Cortisol: Regulation, Disorders, and Clinical Evaluation. Neuroendocrinology*. Baltimore: Williams and Wilkins; 1992:117–173
- [16] Chrousos GP, Schulte HM, Oldfield EH, et al. The Corticotropin-Releasing Factor Stimulation Test: An Aid in the Evaluation of Patients with Cushing's Syndrome. *N Engl J Med*. 1984; 310:622–626
- [17] Esposito F, Dusick JR, Cohan P, et al. Early morning cortisol levels as a predictor of remission after transsphenoidal surgery for Cushing's disease. *J Clin Endocrinol Metab*. 2006; 91:7–13
- [18] Watts NB, Tindall GT. Rapid assessment of corticotropin reserve after pituitary surgery. *JAMA*. 1988; 259:708–711
- [19] Abboud CF. Laboratory Diagnosis of Hypopituitarism. *Mayo Clin Proc*. 1986; 61:35–48
- [20] Melmed S. Acromegaly. *N Engl J Med*. 1990; 322:966–977
- [21] Dimaraki EV, Jaffe CA, DeMott-Friberg R, Chandler WF, Barkan AL. Acromegaly with apparently normal GH secretion: implications for diagnosis and follow-up. *J Clin Endocrinol Metab*. 2002; 87:3537–3542
- [22] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. *Endocr Pract*. 2004; 10:213–225
- [23] Melmed S. Medical progress: Acromegaly. *N Engl J Med*. 2006; 355:2558–2573
- [24] Frohman LA. Ectopic hormone production by tumors: growth hormone-releasing factor. *Neuroendocrine Perspect*. 1984; 3:201–224
- [25] Peyster RG, Hoover ED, Viscarello RR, et al. CT Appearance of the Adolescent and Preadolescent Pituitary Gland. *AJNR*. 1983; 4:411–414
- [26] Watson JC, Shawker TH, Nieman IK, et al. Localization of Pituitary Adenomas by Using Intraoperative Ultrasound in Patients with Cushing's Disease and No Demonstrable Pituitary Tumor on Magnetic Resonance Imaging. *J Neurosurg*. 1998; 89:927–932
- [27] Kim LJ, Lekovic GP, White WL, Karis J. Preliminary Experience with 3-Tesla MRI and Cushing's Disease. *Skull Base*. 2007; 17:273–277
- [28] Kucharczyk W, Davis DO, Kelly WM, et al. Pituitary adenomas: high-resolution MR imaging at 1.5 T. *Radiology*. 1986; 161:761–765
- [29] Krisht AF. Giant Invasive Pituitary Adenomas. *Contemp Neurosurg*. 1999; 21:1–6
- [30] Wilson CB. Endocrine-Inactive Pituitary Adenomas. *Clin Neurosurg*. 1992; 38:10–31
- [31] Banasiak MJ, Malek AR. Nelson syndrome: comprehensive review of pathophysiology, diagnosis, and management. *Neurosurg Focus*. 2007; 23
- [32] Barrow DL, Mizuno J, Tindall GT. Management of Prolactinomas Associated with Very High Serum Prolactin Levels. *J Neurosurg*. 1988; 68:554–558
- [33] Blevins LS. Medical Management of Pituitary Adenomas. *Contemp Neurosurg*. 1997; 19:1–6
- [34] Molitch ME. Pregnancy and the hyperprolactinemic woman. *N Engl J Med*. 1985; 312:1364–1370
- [35] Landolt AM, Osterwalder V. Perivascular Fibrosis in Prolactinomas: Is it Increased by Bromocriptine? *J Clin Endocrinol Metab*. 1984; 58:1179–1183
- [36] Barlas O, Bayindir C, Hepgul K, Can M, Kiris T, Sencer E, Unal F, Aral F. Bromocriptine-induced cerebrospinal fluid fistula in patients with macroprolactinomas: report of three cases and a review of the literature. *Surg Neurol*. 1994; 41:486–489
- [37] Cabergoline for Hyperprolactinemia. *Med Letter*. 1997; 39:58–59
- [38] Webster J, Piscitelli G, Polli A, et al. A Comparison of Cabergoline and Bromocriptine in the Treatment of Hyperprolactinemic Amenorrhea. *N Engl J Med*. 1994; 331:904–909
- [39] Schade R, Andersohn F, Suissa S, Haverkamp W, Garbe E. Dopamine agonists and the risk of cardiac-valve regurgitation. *N Engl J Med*. 2007; 356:29–38
- [40] Johnston DG, Hall K, Kendall-Taylor P, Patrick D, Watson M, Cook DB. Effect of dopamine agonist withdrawal after long-term therapy in prolactinomas. Studies with high-definition computerised tomography. *Lancet*. 1984; 2:187–192
- [41] Schlechte JA. Long-term management of prolactinomas. *J Clin Endocrinol Metab*. 2007; 92:2861–2865
- [42] Acromegaly Therapy Consensus Development Panel. Consensus Statement: Benefits Versus Risks of Medical Therapy for Acromegaly. *Am J Med*. 1994; 97:468–473
- [43] Colao A, Attanasio R, Pivonello R, et al. Partial surgical removal of growth hormone-secreting pituitary tumors enhances the response to somatostatin analogs in acromegaly. *J Clin Endocrinol Metab*. 2006; 91:85–92
- [44] Turner HE, Thornton-Jones VA, Wass JA. Systematic dose-extension of octreotide LAR: the importance of individual tailoring of treatment in patients with acromegaly. *Clin Endocrinol (Oxf)*. 2004; 61:224–231
- [45] van der Lely AJ, Hutson RK, Trainer PJ, Besser GM, Barkan AL, Katznelson L, Klibanski A, Herman-Bonert V, Melmed S, Vance ML, Freda PU, Stewart PM, Friend KE, Clemmons DR, Johannsson G, Stavrou S, Cook DM, Phillips LS, Strasburger CJ, Hackett S, Zib KA, Davis RJ, Scarlett JA, Thorner MO. Long-term treatment of acromegaly with pegvisomant, a growth hormone receptor antagonist. *Lancet*. 2001; 358:1754–1759
- [46] Pegvisomant (Somavert) for acromegaly. *Med Letter*. 2003; 45:55–56
- [47] Sheehan JM, Vance ML, Sheehan JP, Ellegala DB, Laws ER, Jr. Radiosurgery for Cushing's disease after failed transsphenoidal surgery. *J Neurosurg*. 2000; 93:738–742
- [48] Assie G, Bahurel H, Coste J, Silvera S, Kujas M, Dugue MA, et al. Corticotroph tumor progression after adrenalectomy in Cushing's Disease: a reappraisal of Nelson's syndrome. *J Clin Endocrinol Metab*. 2007; 49:381–386
- [49] Clarke MJ, Erickson D, Castro MR, Atkinson JL. Thyroid-stimulating hormone pituitary adenomas. *J Neurosurg*. 2008; 109:17–22
- [50] Sanno N, Teramoto A, Osamura RY. Long-term surgical outcome in 16 patients with thyrotropin pituitary adenoma. *J Neurosurg*. 2000; 93:194–200
- [51] Ebersold MJ, Quast LM, Laws ER, et al. Long-Term Results in Transsphenoidal Removal of Nonfunctioning Pituitary Adenomas. *J Neurosurg*. 1986; 64:713–719
- [52] Ciric I, Mikhael M, Stafford T, et al. Transsphenoidal Microsurgery of Pituitary Macroadenomas with Long-Term Follow-Up Results. *J Neurosurg*. 1983; 59:395–401
- [53] Breen P, Flickinger JC, Kondziolka D, Martinez AJ. Radiotherapy for Nonfunctional Pituitary Adenoma: Analysis of Long-Term Tumor Control. *J Neurosurg*. 1998; 89:933–938

47 Pituitary Adenomas – Surgical Management, Outcome, and Recurrence Management

47.1 Surgical treatment for pituitary adenomas

47.1.1 Medical preparation for surgery

1. stress dose steroids: given to all patients during and immediately after surgery
2. hypothyroidism: ideally, hypothyroid patients should have >4 weeks of replacement to reverse hypothyroidism, however:
 - a) ☐ do not replace thyroid hormone until the adrenal axis is assessed; giving thyroid replacement to a patient with hypoadrenalism can precipitate adrenal crisis. If hypoadrenal, begin cortisol replacement first, may begin thyroid hormone replacement after 24 hours of cortisol
 - b) surgery is done frequently on patients with hypothyroidism and appears to be tolerated well in the vast majority of cases

47.1.2 Surgical approaches – overview

1. transsphenoidal: an extra-arachnoid approach, requires no brain retraction, no external scar (aside from where a fat graft is procured, if used). Usually the procedure of choice. Indicated for microadenomas, macroadenomas without significant extension laterally beyond the confines of the sella turcica, patients with CSF rhinorrhea, and tumors with extension into sphenoid air sinus
 - a) sublabial
 - b) trans-nares: an alotomy may be used to enlarge the exposure through the nares if necessary
2. transthemoidal approach¹ (p 343–50)
3. transcranial approaches:
 - a) indications: most pituitary tumors are operated by the transsphenoidal technique (see above), even if there is significant suprasellar extension. However, a craniotomy may be indicated for the following²:
 - minimal enlargement of the sella with a large suprasellar mass, especially if the diaphragma sellae is tightly constricting the tumor (producing a “cottage loaf” tumor) and the suprasellar component is causing chiasmal compression³ (p 124)
 - extrasellar extension into the middle fossa that is larger than the intrasellar component
 - unrelated pathology may complicate a transsphenoidal approach: rare, e.g. a parasellar aneurysm
 - unusually fibrous tumor that could not be completely removed on a previous transsphenoidal approach
 - recurrent tumor following a previous transsphenoidal resection
 - b) choices of approach
 - subfrontal: provides access to both optic nerves. May be more difficult in patients with pre-fixed chiasm
 - frontotemporal (pterional): places optic nerve and sometimes carotid artery in line of vision of tumor. There is also incomplete access to intrasellar contents. Good access for tumors with significant lateral extrasellar extension
 - ☐ subtemporal: usually not a viable choice. Poor visualization of optic nerve/chiasm and carotid. Does not allow total removal of intrasellar component

47.1.3 Transsphenoidal surgery

Booking the case: Transsphenoidal surgery

Also see defaults & disclaimers (p. 27) and pre-op orders (p. 751).

1. position: supine, horseshoe head rest or (especially if image guided navigation is used) pin headholder
2. equipment:
 - a) microscope
 - b) C-arm (if used)
 - c) image guided navigation system (if used)

- d) endoscopy cart for cases performed endoscopically (surgeon preference)
- 3. instrumentation: transsphenoidal instrument set (usually includes speculum, curettes, long instruments including bipolars)
- 4. some surgeons use ENT to perform the approach and closure and for follow-up
- 5. post-op: ICU
- 6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: removal of pituitary tumor through the nose, possible placement of fat graft from abdomen
 - b) alternatives: surgery through the skull (trans-cranial), radiation
 - c) complications: CSF leak with possible meningitis, problems with pituitary hormones which may sometimes be permanent (which would require lifetime replacement therapy), injury to optic nerve with visual loss, injury to carotid artery with possible bleeding and/or stroke

Technique

General information

For pre- and post-op orders, see below.

Details of the surgery are beyond the scope of this text, see references.^{3,4,5,6}

Intraoperative disasters

Usually related to loss of landmarks.³ Can be minimized using intraoperative navigation or fluoroscopy to verify location.

- injury of carotid artery:
 - typically injured in lateral aspect of opening. Bone may be dehiscent over the ICA
 - signaled by profuse arterial bleeding
 - can usually be packed off, if fat/fascia graft from thigh or abdomen is available it may be used; otherwise use e.g. woven surgical)
 - the operation is halted, and a STAT post-op arteriogram must be done
 - if a pseudoaneurysm or site of injury is identified angiographically, it must be eliminated before a potentially lethal hemorrhage; accomplished either by endovascular techniques or by surgical trapping with clips above and below
- opening through the clivus and erroneous biopsy of the pons
- opening through the floor of the frontal fossa with injury to the olfactory nerves and entry into inferior frontal lobes

Overview of the procedure

1. lumbar drain: may be used with some macroadenomas to inject fluid in order to help bring the tumor down (see below), also may be used for post-op CSF drainage following transsphenoidal repair of CSF fistula
2. medications (in addition to pre-op meds, see below): intraoperatively 100 mg hydrocortisone IV q 8 hrs
3. positioning
 - a) elevate thorax 10–15°: reduces venous pressure
 - b) head stabilization: if image-guided navigation is to be used, the head is placed in a Mayfield headholder or a headband with registration array is attached and the head is placed on a horseshoe head rest. If no image guidance, a horseshoe headrest may be used
 - c) position option 1: surgeon standing to right of patient
 - shoulder-roll
 - top of head canted slightly to left
 - neck position: For microscope: extend neck slightly with the head in either a Mayfield head-holder or on a horseshoe headrest. For endoscope: do not extend neck (more comfortable for holding instruments)
 - ET tube positioned down and to patient's left (to get it out of the way)
 - microscope: observer's eyepiece on the left
 - d) position option 2: surgeon standing above patient's head: head pointing straight up towards ceiling, neck slightly extended
 - e) abdomen or right thigh is prepped for fat graft
4. C-arm fluoro: image-guided navigation can eliminate the need for fluoro. Orient the C-arm for a true lateral by aligning the mandibular rami and/or by superimposing the floor of the left and

- right frontal fossae. If this proves difficult, lay a Penfield 4 on the nasion oriented from lateral canthus to lateral canthus, then aim the fluoro to shoot “down the barrel” of the Penfield 4
5. after approach to floor of sella is complete (see below), outline the upper and lower boundaries of the sella using image navigation or for fluoro using an instrument (e.g. suction tip) – obtain hard-copy of images for documentation purposes
 6. opening the sellar floor:
 - a) starting the opening: open exactly in the midline using the nasal septum as a landmark (NB: the septum of the sphenoid sinus is unreliable as a midline indicator, and often curves inferiorly towards one of the carotid arteries).
 - macroadenomas may have thinned the bone to the point that it just flakes off
 - otherwise, use a bayoneted chisel or high-speed diamond burr to start the opening
 - b) use a Kerrison rongeur to expand the opening. □ CAUTION: stay away from the extreme lateral sella to avoid entering the cavernous sinus or injuring the carotid artery
 7. coagulate the dura centrally in an “X” pattern (NOT “+” pattern) with bipolar cautery. Macroadenomas may cause yellowish discoloration of the dura directly over the tumor
 8. consider aspirating through dura with a 20 gauge spinal needle to R/O large venous sinus (dura often has bluish discoloration), aneurysm, or empty sella
 9. incise the dura in the “X” pattern in the midline with a #11 scalpel on a bayoneted handle
 10. tumor removal for macroadenoma:
 - a) gently bring tumor into the field with ring curettes, and remove with pituitary rongeurs or aspirate with suction. Some tumors are very fibrous and may be difficult to remove
 - b) do not pull on the lateral component of the tumor with pituitary rongeur due to risk of injuring carotid artery
 - c) if the suprasellar component will not come down, it may be brought down by having the anesthesiologist inject 5 ml aliquots of saline into a lumbar drain while monitoring blood pressure and pulse^{3 (p 135),7}
 - d) once the tumor is debulked internally, try to develop a plane between the tumor capsule and the pituitary. A good place to start looking is inferiorly where the dura can be separated from the tumor capsule and then followed on the surface. Sometimes the tumor capsule cannot be removed due to severe bleeding
 - e) complete tumor removal is often not possible, and the goal of the surgery then is “containment”
 - f) endoscopic techniques and image guided navigation may be employed to assist in removal of macroadenomas
 11. tumor removal for microadenoma
 - a) if the side of the tumor is known, begin exploration of the gland on that side by making incision with #11 blade and using a dissector to try and locate the tumor (like a “grain of rice in a blueberry”)
 - b) for Cushing’s disease, if no tumor is identified on pre-op MRI⁸:
 - intraoperative ultrasound may help localize tumor in $\approx 70\%$ of cases⁹ but a specialized U/S probe is required
 - if IPS sampling showed a lateralizing ACTH gradient: start with a paramedian incision on the side of the higher ACTH gradient; if no adenoma is encountered, the contralateral paramedian and then midline incisions are used to explore the gland
 - if IPS sampling and MRI do not suggest tumor location: the gland is explored sequentially with 2 paramedian incisions and then a midline incision
 - if the adenoma cannot be found, a hemihypophysectomy is performed on the side of higher ACTH levels if IPS sampling shows a lateralizing gradient, or on the side with more suspicious tissue on frozen section. Total hypophysectomy is not routinely performed⁸
 - c) most adenomas are purplish-grey and easily aspirated, however some may be more fibrous. The normal pituitary gland is firm and rubbery (the adeno-hypophysis is orange-pink, the neurohypophysis is a whitish-grey), and normally does not curette very easily
 - d) use image guidance or fluoro to determine approximate location of diaphragma sellae. Do not go cephalad to this to avoid a CSF leak, to avoid entering the circular venous sinus in the dura here and to avoid trauma to the optic chiasm
 12. after removal of macroadenoma, check depth of tumor bed on fluoro or image guidance, and make sure it correlates with approximate tumor volume on MRI
 13. the sella may be packed in a number of ways,⁶ one method:
 - a) if CSF leak occurs: place muscle or fat in defect within sella. Some recommend against the use of muscle because it always putrefies.^{3 (p 129)} Do not overpack to avoid recreating mass effect with the graft

- b) recreate the floor of the sella using nasal cartilage placed within the sella. Alternatively, a nonporous Medpor® polyethylene transsphenoidal sellar implant (Porex Surgical Products <http://www.porexurgical.com>) may be used
- c) if CSF leak occurs pack sphenoid sinus with fat from abdomen (option: fat with fascia on surface)
- d) fibrin glue may optionally be used to help hold any of these components in place

Approach to sphenoid sinus for microscopic removal

Often done by ENT co-surgeon. One method:

1. insert temporary speculum into nose. For this discussion, the right nares will be described
2. use endoscope to locate middle concha. Follow this posteriorly to identify os into sphenoid sinus which is usually located posterior to and slightly above the posterior extent of the middle concha
3. inject local anesthetic with epinephrine to blanch mucosa
4. insert sickle knife into os with sharp side facing the septum (medially) and incise the mucosa as the knife is drawn outward
5. use a Freer to dissect the thusly created mucosal flaps off the medial septum (pull one up, the other down)
6. break through the posterior part of the septum so that both sides of the floor of the sphenoid sinus are exposed. Cartilage or bone from this step is saved to use later in reconstructing the floor of the sella if desired
7. open the floor of the sphenoid and take it all the way to the right os (you will probably not see the left os)
8. place the Hardy speculum or equivalent
9. strip mucosa off the walls of the sphenoid sinus using a Blakely and slow pulling motion

Endoscopic tumor removal

There is no proof that removal is superior to microscopic approach through a speculum.

Advantages over microscopic removal: Visualization is better, especially within tumor bed.

Disadvantages: less familiarity with use of endoscopes by most neurosurgeons compared to ENT surgeons. Lack of 3D visualization (may be overcome using 3D endoscopes). Need for single handed technique (may be overcome by having assistant hold endoscope, or using an endoscope holder e.g. Mitaka), need for a binasal approach (if it is desired to use two hands for the actual surgery).

47.1.4 Peri-operative complications

1. hormonal imbalance:
 - a) acute post-op concerns:
 - alterations in ADH: transient abnormalities are common – see typical post-op patterns (p. 752) – including DI. DI lasting >3 mos is uncommon
 - cortisol deficiency → hypocortisolism → Addisonian crisis if severe
 - b) long-term: hypopituitarism in ≈ 5% (retrospective series¹⁰)
 - TSH deficiency → hypothyroidism → (rarely) myxedema coma if severe
 - adrenal insufficiency
 - deficiency of sex hormones → hypogonadotropic hypogonadism
2. secondary empty sella syndrome (chiasm retracts into evacuated sella → visual impairment)
3. hydrocephalus with coma¹¹: may follow removal of tumors with suprasellar extension (trans-sphenoidally or transcranially). Consider ventriculostomy placement if hydrocephalus is present (even if not symptomatic). Possible etiologies:
 - a) traction on the attached 3rd ventricle
 - b) cerebral edema due to vasopressin release from manipulation of the pituitary and/or stalk
 - c) tumor edema following resection
4. infection
 - a) pituitary abscess^{12,13}
 - b) meningitis
5. CSF rhinorrhea (fistula): 3.5% incidence¹⁴
6. carotid artery rupture: rare. May occur intraoperatively (see above) or in delayed fashion after surgery, often ≈ day 10 post-op (due to breakdown of fibrin around carotid, or possibly due to rupture of a pseudoaneurysm created at surgery)
7. entry into cavernous sinus with possible injury of any structure within
8. nasal septal perforation

47.1.5 Frontotemporal (pterional) approach

A right sided approach is usually employed (less risk to dominant hemisphere). Exceptions: when the left eye is the side of worse vision; if there is predominant left sided tumor extension; if there is other pathology on the left (e.g. aneurysm).

Positioning is that same as for ACoA aneurysm with the □ Fig. 94.5. The frontal lobe is elevated, and the temporal tip is gently retracted posteriorly. Bridging veins to the temporal tip must be coagulated to avoid rupture, as for any pterional approach. The approach is similar to that for an ACoA aneurysm (i.e. more emphasis is placed on frontal lobe elevation than on temporal tip retraction), except that unlike ACoA aneurysm, exposure of the ICA is not needed because proximal control is not necessary.

The tumor capsule can usually be seen between the two optic nerves. The capsule is coagulated with bipolar cautery, and is incised. The tumor is then debulked from within. By staying within the capsule, risk of injury to the pituitary stalk and optic chiasm is minimized. Significant amounts of tumor can be removed by aspiration if it is soft and suckable.

□ Caution: the blood supply to the optic chiasm is from the inferior aspect. Skeletonizing the chiasm or attempting to tease away tumor adherent to it may worsen vision.

47.1.6 Peri-operative management

47

Pre-op orders

1. Polysporin® ointment (PSO) applied in both nostrils the night before surgery
2. antibiotics, one of the following regimens may be used:
 - a) chloramphenicol 500 mg IVPB at 11 PM & 6 AM
OR
 - b) chloramphenicol 500 mg PO at MN & IV at 6 AM; ampicillin 1 gm PO at MN & IV at 6 AM
OR
 - c) Unasyn® 1.5 gm (1 gm ampicillin + 0.5 gm sulbactam) IVPB at MN & 6 AM
3. steroids, either:
 - a) hydrocortisone sodium succinate (Solu-Cortef®) 50 mg IM at 11 PM & 6 AM. On call to OR: hang 1 LD5LR + 20 mEq KCl/l + 50 mg Solu-Cortef at 75 ml/hr
OR
 - b) hydrocortisone 100 mg PO at MN & IV at 6 AM
4. intra-op: continue 100 mg hydrocortisone IV q 8 hrs

Post-op orders

1. intake & output (I's & O's) q 1 hr; urine specific gravity (SG) q 4° and anytime urine output (UO) > 250 ml/hr
2. activity: BR with HOB @ 30°.
3. diet: ice chips PRN. Patient is not to drink through a straw (to avoid negative pressure on sphenoid sinus with risk of aggravating CSF fistula)
4. no incentive spirometry (to avoid negative pressure on sphenoid sinus with risk of aggravating CSF fistula)
5. IVF: base IV D5 1/2 NS + 20 mEq KCl/L at appropriate rate (75–100 ml/hr)
PLUS: replace UO > base IV rate ml for ml with 1/2 NS.
NB: if patient receives significant fluids intra-operatively, then they may have an appropriate post-op diuresis, in which case consider replacing only $\approx 2/3$ of UO > base IV rate with 1/2 NS
6. meds
 - a) antibiotics: continue chloramphenicol 500 mg IVPB q 6 hr (also continue ampicillin if used pre-op), change to PO when tolerated, D/C when nasal packing removed
 - b) steroids (some surgeons routinely use post-op steroids until the adequacy of endogenous steroids is established, especially with Cushing's disease, see below). Either:
 - hydrocortisone 50 mg IM/IV q 6 hrs, on POD #2 change to prednisone 5 mg PO q 6 hrs $\times$ 1 day, then 5 mg PO BID, D/C on POD #6
OR
 - hydrocortisone 50 mg IM/IV/PO BID, taper 10 mg/dose/day to physiologic dose of 20 mg q AM and 10 mg q PM until adrenal axis assessed
 - c) diabetes insipidus (DI): patients are placed on "DI watch" (monitoring U.O. and serum and urine labs) see below for typical patterns.
Diagnostic criteria: U.O. > 250 ml/hr $\times$ 1–2 hrs, and SG < 1.005 (usually < 1.003) (dilute urine), often with elevation of serum Na⁺

If DI develops, attempt to keep up with fluid loss with IVF (see above); if rate is too high for IV or PO replacement ($>300\text{ cc/hr} \times 4\text{ hrs}$ or $>500\text{ cc/hr} \times 2\text{ hrs}$), check urine S.G. and if <1.005 then give a vasopressin preparation (see below, or see □ Table 5.7). □ Caution: danger of overtreating in case of triphasic response (see below), therefore use EITHER:

- 5 U aqueous vasopressin (Pitressin®) IVP/IM/SQ q 6 hrs PRN
OR
- desmopressin (DDAVP®) injection SQ/IV titrated to UO. Usual adult dose: 0.5–1 ml (2–4 mcg) daily in 2 divided doses
AVOID
- □ avoid tannate oil suspension, because it is a long acting preparation (and may overtreat) and has erratic absorption
THEN: when nasal packs out, EITHER
- intranasal DDAVP (100 mcg/ml): range 0.1–0.4 ml (10–40 mcg) intranasally BID (typically 0.2 ml BID) PRN
OR
- clofibrate (Atromid S®) 500 mg PO QID (does not always work)

- 7. labs: renal profile with osmolarity q 6 hrs, 8 A.M. serum cortisol
- 8. nasal packs: remove on post-op day 3–6

Urinary output: patterns of postoperative diabetes insipidus

Manage diabetes insipidus (DI) as described above in post-op orders. Post-op DI generally follows one of three patterns¹⁵; see Diabetes insipidus for details (p. 120):

- 1. transient DI: lasts until $\approx 12\text{--}36\text{ hrs}$ post-op then normalizes
- 2. “prolonged” DI: lasts months, or rarely may be permanent
- 3. “triphasic response” (least common). 3 stages:
 - a) DI (short duration): due to injury to posterior pituitary
 - b) normalization or SIADH-like picture: due to release of ADH from neuron endings form hypothalamus. It is during this phase that there is a risk of severe iatrogenic hyponatremia from overtreatment initiated during the initial DI phase
 - c) DI (long-term)

Discontinuation of steroids post-op

Simple management schemes

Some surgeons do not routinely assess ACTH reserve post-op for patients that were not hypocortisolemic pre-op:

- taper and stop hydrocortisone 24–48 hrs post-op. Then, check 6 AM serum cortisol level 24 hrs after discontinuing hydrocortisone and interpret the results as shown in □ Table 47.1.¹⁶
- If there is any question about reserve, the patient can be discharged on hydrocortisone 50 mg PO q AM and 25 mg PO q 4:00 PM until adrenal reserve can be formally assessed (see below)

Assessment of postoperative ACTH (corticotropin) reserve

Simple assessment protocol for patients who go home on hydrocortisone and were not on it pre-op.

- taper hydrocortisone over 2–3 weeks down to 20 mg po q AM and 10 mg q 4 PM (a little higher than maintenance to provide for some stress coverage) for several days
- then hold the PM dose and check an 8 AM serum cortisol the next day

Table 47.1 Interpretation of 6 AM cortisol levels		
6 AM cortisol	Interpretation	Management
≥ 9 mcg/dl	normal	no further tests or treatment
3–9 mcg/dl	possible ACTH deficiency	place patient on hydrocortisone ^a (p. 144)
≤ 3 mcg/dl	ACTH deficient	
^a perform cosyntropin stimulation test (p. 735) 1 month post-op; D/C steroids if normal; if subnormal, then permanent replacement required		

- to avoid adrenal insufficiency in patients with incompetent reserve: as soon as the blood is drawn have the patient take their morning dose and resume regular dosing until the test results are available
- If this 8 AM cortisol shows any significant adrenal function, then taper the patient off hydrocortisone

Metirapone (Metopirone®) test

This test more accurately assesses the hypothalamic-pituitary-adrenal axis and is useful if there is suspicion of reduced reserve of pituitary ACTH production. Metirapone inhibits 11-β-hydroxylation in the adrenal cortex, reducing production of cortisol and corticosterone with concomitant increase of serum 11-deoxycortisol precursors and its 17-OHCS metabolites which appear in the urine. In response, a normal pituitary increases ACTH production.

1. All patients should have a cosyntropin stimulation test (p.735) first to rule-out primary adrenal insufficiency .
2. ☐ Do not do this test if there is known primary adrenal insufficiency.
3. ☐ Do not do this test as an outpatient.
4. Test protocol
 - a) give 2–3 grams metirapone PO at midnight
 - b) check serum 11-deoxycortisol level the next morning
 - c) a normal response is a 11-deoxycortisol level >7 mcg/dl
 - d) CAUTION: in patients with very little reserve, the reduced cortisol may provoke adrenal insufficiency (this test is safer than the higher doses used for urinary 17-OHCS testing)

Postoperative CT/MRI scan

A study using CT in 12 patients with macroadenomas following transsphenoidal surgery without radiation therapy demonstrated that the maximal height of the pituitary “mass” did not return to normal immediately post-op (even with total tumor removal), rather a period of 3–4 months was required.¹⁷

Σ

The optimal timing of the initial post-op CT or MRI to function as a baseline to rule-out future recurrence after transsphenoidal surgery is ≈ 3–4 months post-op.

47.2 Outcome following transsphenoidal surgery

47.2.1 General information

In a series of 108 macroadenomas, gross total removal was unusual in tumors with >2 cm suprasellar extension.¹⁴

47.2.2 Visual deficit

In cases with compression of the optic apparatus, there can be significant improvements in vision following surgery.^{14,18}

47.2.3 Biochemical outcome

Prolactinomas

In a series of 108 macroadenomas, endocrinologic cure was attained in 25% of prolactin-secreting tumors.¹⁴

Acromegaly

Criteria of biochemical cure

The criteria for biochemical cure of acromegaly is not standardized. There may be a discord between IGF-1 levels and mean GH levels.¹⁹ Many use a GH cutoff level; range of levels

described: <2.5–5 ng/ml. Others feel that an elevated IGF-1 represents lack of cure even if GH <5. However, normal IGF-1 levels may not be mandatory.²⁰ Still others require a normal IGF-1 AND a normal response to an oral glucose suppression test (OGST) (p.736).

Low GH levels that do not also suppress to <1 ng/ml after an OGST are considered controlled but not cured (even with normal IGF-1 levels).²¹ If asymptomatic, expectant management with close follow-up is recommended.²¹

Σ
Biochemical cure criteria for acromegaly is not standardized. Recommendation ²¹ : 1) IGF-1 levels within age-matched reference range 2) basal (morning) serum GH level <5 ng/ml, AND GH nadir <1 ng/ml in OGST

47 Outcome with acromegaly

Transsphenoidal surgery results in biochemical cure in 85% of cases with adenomas <10 mm diameter, no evidence of local invasion, and random GH levels <40 ng/ml pre-op. Overall, ≈ 50% of all acromegalics undergoing transsphenoidal surgery had a biochemical cure.²² Only 30% of macroadenomas and very few with marked suprasellar extension have surgical cure. Patients not cured with surgery require lifelong medical suppression. These tumors may also recur years later after apparent cure. Patients should be monitored every 6–12 months for recurrence.²¹

Cushing’s disease

There are numerous methodologies for determining biochemical cure for Cushing’s disease. One difficulty is that exogenous steroids are often given post-op to avoid potential hypoadrenalism or Addisonian crisis or for nausea. Some options:

1. immediate post-op early morning cortisol levels⁸:
 - a) all steroids are withheld post-op (including dexamethasone as an antiemetic) unless biochemical and/or clinical evidence of hypocortisolism (clinical signs: nausea, anorexia, H/A, arthralgias). □ Requires close monitoring and administration of steroids if symptoms develop
 - b) serum ACTH and cortisol levels are drawn between 6–9:00 AM on post-op days 1 & 2
 - c) early remission defined as a lowest cortisol level ≤140 nmol/L (≤5 mcg/dl)
 - 97%(31/32) patients with early remission had sustained remission with mean follow-up of 32 months
 - only 12.5%(1/8) without early remission showed evidence of sustained remission
 - this has been used to select patients for possible early re-exploration
 - early ACTH levels usually drop, but do not consistently become subnormal and are not reliable in predicting sustained remission⁸
2. provocative tests
 - a) overnight low-dose dexamethasone suppression test: an AM cortisol level on post-op day 3 that is ≤8 mcg/dl after an overnight 1 mg dexamethasone suppression test is predictive of sustained remission in 97%³
 - b) CRH stimulation test²⁴
3. measurements usually conducted 3 days to 2 weeks post-op following 24 hours of steroid cessation after initial post-op coverage with glucocorticoids
 - a) 24-hour urinary free cortisol
 - b) serum cortisol: the criteria of a cortisol level <50 nmol/l (<1.8 mcg/dl)^{25,26,27} is probably too stringent^{28,29,8}
 - c) serum ACTH

The overall remission rate since 1980 is 64–93% with the highest rates (86–98%) in patients with noninvasive microadenomas identifiable on MRI.⁸

Following effective treatment, all of the following usually improve but may not normalize:

1. HTN and hyperglycemia: within ≈ 1 year
2. osteoporosis related to CD: over ≈ 2 years
3. psychiatric symptoms

Thyrotropin (TSH)-secreting adenomas

Following debulking, small amounts of residual tumor may continue to produce sufficient TSH for hyperthyroidism to persist.³⁰ Following surgery+XRT, only $\approx 40\%$ achieve a cure (defined as no residual tumor at surgery or on imaging, and normal free T3 with TSH levels at or below normal).

47.3 Management of recurrent pituitary adenomas

Recurrence incidence: $\approx 12\%$ with most recurring 4–8 years post-op (in the same series¹⁴).

For tumors demonstrating significant regrowth or symptoms following initial resection, consideration for re-resection may be given. Once the tumor is debulked, consideration should be given to XRT, either immediately following the second operation, or, if recurrence after a second operation then almost certainly after a third debulking.

References

- [1] Schmidek HH, Sweet WH. Operative Neurosurgical Techniques. New York 1982
- [2] Wilson CB. Endocrine-Inactive Pituitary Adenomas. Clin Neurosurg. 1992; 38:10–31
- [3] Powell M, Lightman SL. Management of Pituitary Tumours: A Handbook. New York 1996
- [4] Hardy J. Transsphenoidal Hypophysectomy. J Neurosurg. 1971; 34:582–594
- [5] Kern EB, Pearson BW, McDonald TJ, et al. The Transseptal Approach to Lesions of the Pituitary and Parasellar Region. Laryngoscope. 1979; 89S:1–34
- [6] Spaziante R, de Divitiis E, Cappabianca P. Reconstruction of the Pituitary Fossa in Transsphenoidal Surgery: An Experience of 140 Cases. Neurosurgery. 1985; 17:453–458
- [7] Zhang X, Fei Z, Zhang J, et al. Management of Non-functioning Pituitary Adenomas with Suprasellar Extension by Transsphenoidal Microsurgery. Surg Neurol. 1999; 52:380–385
- [8] Esposito F, Dusick JR, Cohan P, et al. Early morning cortisol levels as a predictor of remission after transsphenoidal surgery for Cushing's disease. J Clin Endocrinol Metab. 2006; 91:7–13
- [9] Watson JC, Shawker TH, Nieman IK, et al. Localization of Pituitary Adenomas by Using Intraoperative Ultrasound in Patients with Cushing's Disease and No Demonstrable Pituitary Tumor on Magnetic Resonance Imaging. J Neurosurg. 1998; 89:927–932
- [10] Fatemi N, Dusick JR, Mattozo C, McArthur DL, Cohan P, Boscardin J, Wang C, Swerdloff RS, Kelly DF. Pituitary hormonal loss and recovery after transsphenoidal adenoma removal. Neurosurgery. 2008; 63:709–18; discussion 718–9
- [11] Decker RE, Chalif DJ. Progressive Coma After the Transsphenoidal Decompression of a Pituitary Adenoma with Marked Suprasellar Extension: Report of Two Cases. Neurosurgery. 1991; 28:154–158
- [12] Domingue JN, Wilson CB. Pituitary Abscesses. J Neurosurg. 1977; 46:601–608
- [13] Robinson B. Intracellar Abscess After Transsphenoidal Pituitary Adenectomy. Neurosurgery. 1983; 12:684–686
- [14] Ciric I, Mikhael M, Stafford T, et al. Transsphenoidal Microsurgery of Pituitary Macroadenomas with Long-Term Follow-Up Results. J Neurosurg. 1983; 59:395–401
- [15] Verbalis JG, Robinson AG, Moses AM. Postoperative and Post-Traumatic Diabetes Insipidus. Front Horm Res. 1985; 13:247–265
- [16] Watts NB, Tindall GT. Rapid assessment of corticotropin reserve after pituitary surgery. JAMA. 1988; 259:708–711
- [17] Teng MMH, Huang CI, Chang T. The Pituitary Mass After Transsphenoidal Hypophysectomy. AJNR. 1988; 9:23–26
- [18] Cohen AR, Cooper PR, Kupersmith MJ, et al. Visual Recovery After Transsphenoidal Removal of Pituitary Adenoma. Neurosurgery. 1985; 17:446–452
- [19] Turner HE, Thornton-Jones VA, Wass JA. Systematic dose-extension of octreotide LAR: the importance of individual tailoring of treatment in patients with acromegaly. Clin Endocrinol (Oxf). 2004; 61:224–231
- [20] Ayuk J, Clayton RN, Holder G, Sheppard MC, Stewart PM, Bates AS. Growth hormone and pituitary radiotherapy, but not serum insulin-like growth factor-I concentrations, predict excess mortality in patients with acromegaly. J Clin Endocrinol Metab. 2004; 89:1613–1617
- [21] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. Endocr Pract. 2004; 10:213–225
- [22] Davis DH, Laws ER, Ilstrup DM, et al. Results of Surgical Treatment for Growth Hormone-Secreting Pituitary Adenomas. J Neurosurg. 1993; 79:70–75
- [23] Chen JC, Amar AP, Choi S, Singer P, Couldwell WT, Weiss MH. Transsphenoidal microsurgical treatment of Cushing's disease: postoperative assessment of surgical efficacy by application of an overnight low-dose dexamethasone suppression test. J Neurosurg. 2003; 98:967–973
- [24] Nishizawa S, Oki Y, Ohta S, Yokota N, et al. What can predict postoperative "endocrinological cure" in Cushing's disease? Neurosurgery. 1999; 45:239–244
- [25] Trainer PJ, Lawrie HS, Verhelst J, et al. Transsphenoidal resection in Cushing's disease: undetectable serum cortisol as the definition of successful treatment. Clinical Endocrinology (Oxf). 1993; 38:73–78
- [26] Rees DA, Hanna FW, Davies JS, Mills RG, Vafidis J, Scanlon MF. Long-term follow-up results of transsphenoidal surgery for Cushing's disease in a single centre using strict criteria for remission. Clinical Endocrinology (Oxf). 2002; 56:541–551
- [27] Yap LB, Turner HE, Adams CB, Wass JA. Undetectable postoperative cortisol does not always predict long-term remission in Cushing's disease: A single centre audit. Clinical Endocrinology (Oxf). 2002; 56:25–31
- [28] Simmons NE, Alden TD, Thorner MO, Laws ER. Serum cortisol response to transsphenoidal surgery for Cushing disease. J Neurosurg. 2001; 95:1–8
- [29] Rollin GA, Ferreira NP, Junges M, Gross JL, Czepielewski MA. Dynamics of serum cortisol levels after transsphenoidal surgery in a cohort of patients with Cushing's disease. J Clin Endocrinol Metab. 2004; 89:1131–1139
- [30] Sanno N, Teramoto A, Osamura RY. Long-term surgical outcome in 16 patients with thyrotropin pituitary adenoma. J Neurosurg. 2000; 93:194–200

Cysts and Tumor-Like Lesions

Rathke’s cleft cyst

Rathke’s cleft cyst (RCC) are nonneoplastic lesions that are thought to be remnants of Rathke’s pouch. They are primarily intrasellar, and are found incidentally in 13–23% of necropsies.¹ The adenohypophysis arises from proliferation of the anterior wall of Rathke’s pouch, and so RCC have a similar lineage to pituitary adenomas and are rarely found together.² RCC are often discussed in contrast to craniopharyngiomas (CP) (see above). Some features are compared in □ Table 48.1.

RCC usually appear as low-density cystic lesions on CT. One half show capsular enhancement. MRI appearance is variable.³ Rule of thumb: a lesion with a nodule in the sella is usually a RCC.

Colloid cyst

General information

Key concepts

- slow-growing benign tumor comprising <1% of intracranial tumors
- classically occurs in the anterior 3rd ventricle, blocking foramina of Monro → obstructive hydrocephalus involving only the lateral ventricles (≈ pathognomonic)
- enhances minimally or not at all on CT/MRI
- natural history: risk of sudden death has been described, but is controversial
- treatment, when indicated, is surgical. Main options: transcallosal, transcortical/transventricular (only if hydrocephalus), ventriculoscopic

AKA neuroepithelial cysts. Comprise 2% of gliomas, and about 0.5–1% of all intracranial tumors.⁴ Usual age of diagnosis: 20–50 yrs.

Pathogenesis

Origin: unknown. Implicated structures include: paraphysis (evagination in roof of third ventricle, rudimentary in humans), diencephalic ependyma in the recess of the postvelar arch, ventricular neuroepithelium.

Comprised of a fibrous epithelial-lined wall filled with either mucoid or dense hyloid substance. A slow growing, benign tumor.

Most commonly found in the third ventricle in the region of the foramina of Monro, but may be seen elsewhere, e.g. in septum pellucidum.⁵

Clinical signs and symptoms

Symptoms are shown in □ Table 48.2. Signs are shown in □ Table 48.3, most commonly presents either with signs of intermittent acute intracranial hypertension (classically attributed to movement of the cyst on its pedicle causing episodic obstruction of the foramina of Monro, rarely born out at

Table 48.1 Comparison of craniopharyngioma to Rathke’s cleft cyst		
Feature	Craniopharyngioma	Rathke’s cleft cyst
site of origin	anterior superior margin of pituitary	pars intermedia of pituitary
cell lining	stratified squamous epithelium	single layer cuboidal epithelium
cyst contents	cholesterol crystals	resembles motor oil
surgical treatment	total removal is the goal	partial excision and drainage ³
cyst wall	thick	thin

Table 48.2 Symptoms of colloid cyst at presentation ^a		
Symptom	No.	%
headache	26	68%
gait disturbance	18	47%
disturbed mentation	14	37%
vomiting (± nausea)	14	37%
blurred vision	9	24%
incontinence	5	13%
dizziness	5	13%
tinnitus	5	13%
seizures	4	10%
acute deterioration	4	10%
diplopia	3	8%
“drop attacks”	1	
diabetes insipidus	1	
asymptomatic	1	
^a 38 patients, pre-CT era ⁴		

Table 48.3 Signs at presentation ^a		
Sign	No.	%
papilledema	18	47%
gait disturbance	12	32%
normal exam	10	26%
hyperreflexia	9	24%
Babinski reflex	8	21%
incoordination	5	13%
nystagmus	5	13%
tremor	4	10%
hyporeflexia	3	8%
6th nerve palsy	2	5%
^a 38 patients with colloid cysts, pre-CT era ⁴		

operation) or with chronic hydrocephalus (from chronic obstruction). Most cysts <1 cm diameter do not produce hydrocephalus and are asymptomatic.

□ Sudden death. A high rate of sudden death has been reported with colloid cysts (20%in pre-CT era⁶) but is probably overestimated. The obsolete theory was that these tumors are mobile and thus could shift position and acutely block CSF flow with resultant herniation. Progressive obstruction from tumor growth does often produce chronic hydrocephalus, and it is possible that at some point

the brain may decompensate in some cases. Changes in CSF dynamics resulting from procedures (LP, ventriculography...) may have also contributed.⁷ Another proposed mechanism is disturbance of hypothalamic-mediated cardiovascular reflex control.⁷

48.2.4 Evaluation

Imaging (MRI or CT) demonstrates the tumor usually located in the anterior 3rd ventricle. Here, it often blocks both foramina of Monro causing almost pathognomonic hydrocephalus involving only the lateral ventricles (sparing the 3rd and 4th). Differential diagnosis includes basilar artery aneurysms, hamartomas, primary or secondary neoplasm, and xanthogranulomas.⁸

MRI: usually the optimal imaging technique. However, there are cases where cysts are iso-intense on MRI and CT is superior⁹ (scrutinize the midline T1WI MRI images). When the lesion is identifiable, MRI clearly demonstrates the location of the cyst and relation to nearby structures, usually obviating an angiogram. MRI appearance: variable. Usually hyperintense on T1WI, hypointense on T2WI. Some data suggests that symptomatic patients are more likely to display T2 hyperintense cysts on MRI, indicating higher water content which may reflect a propensity for continued cyst expansion.¹⁰ Enhancement: minimal, sometimes involving only capsule.

CT scan: findings are variable. Most are hyperdense (however, iso- and hypo-dense colloid cysts occur), and about half enhance slightly. Density may correlate with viscosity of contents, hyperdense cysts were harder to drain percutaneously.¹¹ CT is usually not quite as good as MRI, especially with isodense cysts. These tumors calcify only rarely.

- LP: contraindicated prior to placement of shunt due to risk of herniation.

48.2.5 Treatment

General information

Optimal treatment remains controversial. Initially, shunting without treating the cyst was advocated.¹² The nature of the obstruction (both foramina of Monro) requires bilateral ventricular shunts (or, unilateral shunt with fenestration of the septum pellucidum). Presently, one form or another of direct surgical treatment is usually recommended for some or all of the following reasons:

1. to prevent shunt dependency
2. to reduce the possibility of tumor progression
3. since the mechanism of sudden neurologic deterioration may be due to factors such as cardiovascular instability from hypothalamic compression and not due to hydrocephalus

Surgical treatment

Management options

Also see Approaches to the third ventricle (p.1461).

1. transcallosal approaches: not dependent on dilated ventricles. Higher incidence of venous infarction or forniceal injury (see below)
2. transcortical approach (p.1466): higher incidence of post-op seizures ($\approx 5\%$). Not feasible with normal sized ventricles (e.g. in patient with VP shunt)
3. stereotactic drainage: see below
4. ventriculoscopic removal: see below

A meta-analysis of 1278 patients comparing endoscopic and various microsurgical techniques found that the microsurgical group had a significantly greater extent of resection (96.8% vs 58.2%), lower rates of recurrence (1.48% vs 3.91%), and lower rates of reoperation than the endoscopic group (0.38% vs 3.0%). Both groups had similar rates of mortality (1.4% vs 0.6%) and shunt dependency (6.2% vs 3.9%). Overall, the complication rate was lower in the endoscopic group than in the microsurgical group (10.5% vs 16.3%). Within the microsurgery group, the transcallosal approach had a lower overall morbidity rate (14.4%) than the transcortical approach (24.5%).¹³

Using natural history for treatment decisions

A review of 58 asymptomatic patients (average age 57 yr) with incidentally discovered colloid cysts of the third ventricle with mean follow up of 79 months demonstrated the incidence of symptomatic worsening at 2, 5, and 10 year follow up to be 0%0% and 8% respectively. Of the 34 patients who obtained follow up imaging, 32 demonstrated no change in cyst size or ventricular caliber. The average age of these patients was significantly higher than that of the patients undergoing surgery for symptomatic lesions (57 vs 41) and thus may reflect a patient cohort with differing natural histories.¹⁴

NB: Many “asymptomatic” patients may have headaches at diagnosis. Careful evaluation of headache etiology (i.e. posttraumatic, migrainous, tension, etc) should be undertaken to determine whether the headaches are due to the colloid cyst or if the cyst is asymptomatic.

Pollock et al.¹⁰ reviewed 155 patients with newly diagnosed colloid cysts, performing a recursive partitioning analysis, and divided patients into three classes as shown in □ Table 48.4.

Additionally, a significantly greater number Class III patients had hyperintense cyst contents on T2 weighted imaging (44% vs 13%) and that symptomatic patients were more likely to have increased T2 signal (44% vs 8%) compared to asymptomatic counterparts. The authors suggest that asymptomatic patients with low T2 signal containing cysts may reflect a group with low potential for cyst expansion and development of cyst related symptoms (even in the presence of ventriculomegaly) – thus they may represent a population that may be safely managed in a nonoperative fashion.¹⁰ Nevertheless, most surgeons would recommend surgery for a patient with ventriculomegaly and symptoms such as headache even if they may not be definitely related.

Transcallosal approach

Access to the 3rd ventricle via either the foramen of Monroe or by interforncial approach. Since colloid cysts tend to occur exactly at the foramen of Monroe, it is rarely necessary to enlarge the foramen to locate the tumor. See Transcallosal approach to lateral or third ventricle (p.1462).

Stereotactic drainage of colloid cysts

May be useful,¹⁵ especially in patients with normal ventricles from shunting, but the contents may be too viscous,¹⁶ and the tough capsule may make blind penetration difficult. Total or even subtotal aspiration may not require further treatment in some patients; however, recurrence rate is higher than with surgical removal.¹⁷

Early morbidity was relatively high from this procedure (not widely reported in literature) possibly from vascular injury or mechanical trauma; this has improved. May be more feasible with intra-operative ventriculography¹⁸ or with a ventriculoscope¹⁹ (some say this is the initial procedure of choice,²⁰ with craniotomy reserved for treatment failures).

- Two features that correlate with unsuccessful stereotactic aspiration²¹:
- 1. high viscosity: correlates with hyperdensity on CT (low viscosity correlated with hypo- or iso-dense CT appearance; no MRI finding correlated with viscosity)
 - 2. deflection of the cyst from tip of aspirating needle due to small size

Table 48.4 Classes of colloid cyst based on recursive partitioning analysis					
Class	Age	Cyst diameter	Ventricles	Symptomatic patients /% of total)	Treatment options
I	>50 yrs	<10 mm	Normal	12%	May be monitored clinically and with serial imaging (CT or MRI)
II	>50 yrs	<10 mm	Ventriculomegaly	50%	If asymptomatic, may be monitored clinically and with serial imaging (CT or MRI)
III	<50 yrs	>10 mm	Ventriculomegaly	85%	Surgical removal is recommended

- Stereotactic technique²²:
1. insertion point of stereotactic needle is just anterior to right coronal suture
 2. start with sharp-tipped 1.8 mm probe, and advance to 3–5 mm beyond target site (to accommodate for displacement of cyst wall)
 3. use a 10 ml syringe and apply 6–8 ml of negative aspiration pressure
 4. if this does not yield any material, repeat with a 2.1 mm probe
 5. although complete cyst evacuation is desirable, if this cannot be accomplished an acceptable goal of aspiration is re-establishment of patency of the ventricular pathways (may be verified by injecting 1–2 cc of iohexol)

48.3 Epidermoid and dermoid tumors

48.3.1 General information

AKA epidermoid or dermoid cysts.

Both are usually developmental, benign tumors that may arise when retained ectodermal implants are trapped by two fusing ectodermal surfaces. The growth rate of these tumors is linear, like skin (rather than exponential, as with neoplastic tumors).

- They may occur in the following locations:
1. calvaria: skull involvement (p.776) occurs when ectodermal rests are included in the developing cranium, epidural extension may occur with growth
 2. intracranial: the most common sites include
 - a) suprasellar: commonly produce bitemporal hemianopsia and optic atrophy, and only occasionally pituitary (endocrine) symptoms (including DI)
 - b) sylvian fissure: may present with seizures
 - c) cerebellopontine angle (CPA): may produce trigeminal neuralgia, especially in young patient
 - d) basilar-posterior fossa: may produce lower cranial nerve findings, cerebellar dysfunction, and/or corticospinal tract abnormalities
 - e) within the ventricular system: occur within the 4th ventricle more commonly than any other
 3. scalp
 4. within the spinal canal:
 - a) most arise in the thoracic or upper lumbar spine
 - b) epidermoids of the lower lumbar spine may occur iatrogenically following LP; see Lumbar puncture (p.1504)
 - c) dermoids of the spinal canal are usually associated with a dermal sinus tract (p.270) and may produce recurrent bouts of spinal meningitis.

48.3.2 Comparison of dermoids and epidermoids

Distinguishing features between the two tumors are shown in □ Table 48.5.

Table 48.5 Comparison of epidermoids and dermoid		
Feature	Epidermoid	Dermoid
Frequency	0.5–1.5% of brain tumors	0.3% of brain tumors
Lining	stratified squamous epithelium	also include dermal appendage organs (hair follicles and sebaceous glands)
Contents	keratin, cellular debris, and cholesterol, occasional hair	same as epidermoids, plus hair and sebum
Location	more common laterally (e.g. CP angle)	more commonly near midline
Associated anomalies	tend to be isolated lesions	associated with other congenital anomalies in up to 50% of cases
Meningitis	may have recurrent aseptic meningitis, including Mollaret's meningitis (p.762)	may have repeated bouts of bacterial meningitis

48.3.3 Epidermoid cysts

General information

Key concepts

- usually arise from ectoderm trapped within or displaced into the CNS
- predilection for: CP angle, 4th ventricle, suprasellar region, spinal cord
- sometimes AKA cholesteatoma (not to be confused with cholesterol granuloma)
- grow at linear rate (unlike exponential rate of true neoplasms)
- imaging: CSF-like mass (hi-signal on DWMRI is the best test to differentiate)
- may produce aseptic meningitis (Mollaret’s meningitis is one form)
- treatment: surgical excision. XRT has no role

AKA cholesteatoma (not cholesterol granuloma, (see below), AKA pearly tumor, AKA ectodermal inclusion cyst (see □ Table 48.5 for comparison to dermoids). Although epidermoids and cholesteatomas are histologically identical (both arise from epithelium entrapped in an abnormal location, epidermoids are intradural, cholesteatomas are extradural), the term cholesteatoma is most often used to describe the lesion in the middle ear where the entrapped epithelium usually arises from chronic middle ear infections which lead to a retraction pocket (rarely, may instead be congenital).

- May arise from any of the following²³:
1. displaced dorsal midline ectodermal cell rests trapped during neural tube closure between gestational weeks 3–5
 2. multipotential embryonic cell rests
 3. epithelial cell rests carried to the CPA with the developing otic vesicle
 4. epidermal cells displaced into CNS, e.g. by LP – see Lumbar puncture (p.1504) – or repeated percutaneous cranial subdural taps²⁴

Epidemiology

Epidermoids comprise 1% of intracranial tumors⁵⁵⁶, and ≈ 7% of CPA tumors. Peak age of occurrence: 40 years. No gender difference.

Histology

Epidermoids are lined by stratified squamous epithelium, and contain keratin (from desquamated epithelium), cellular debris, and cholesterol.²⁵ Growth occurs at a linear rate like normal skin, unlike the exponential growth of true neoplasms.²⁶ The cyst contents may be liquid or may have a flaky consistency. They tend to spread along normal cleavage planes and surround vital structures (cranial nerves, ICA...). Bony destruction occurs in a minority, usually with larger tumors. Rare degeneration to squamous cell cancer²⁷ primarily in cases of repeated recurrences after multiple surgeries.

Distinction from cholesterol granuloma

Epidermoid cysts are sometimes mistakenly equated with cholesterol granulomas,²⁸ possibly because of the similarity between the terms cholesteatoma and cholesterol granuloma. However, these are distinct lesions.²⁹ Cholesterol granulomas usually occur following chronic inflammation (usually in pneumatized portions of the temporal bone: petrous apex, mastoid air cells, middle ear space). Some differences are delineated in □ Table 48.6.

Presentation

1. may present as any mass lesion in the same location
2. CPA lesions can produce V, VII or VIII neuropathies
3. recurrent episodes of aseptic meningitis caused by rupture of the cyst contents, which may also lead to hydrocephalus.
 - a) Symptoms include fever and meningeal irritation.

Table 48.6 Characteristics of epidermoid &cholesteatoma vs. cholesterol granuloma			
Feature	Epidermoid	Cholesteatoma	Cholesterol granuloma
origin	ectodermal cells in abnormal location (within CNS, intradural)	(within ear, extradural)	chronic inflammatory cells surrounding cholesterol crystals (? from breakdown of RBC membranes)
precursor	usually congenital, occasionally acquired, e.g. after IP (p. 1507)	usually acquired (following chronic infection ? due to epithelial cells from tympanic membrane), occasionally congenital	chronic middle ear infection or idiopathic hemotympanum
symptoms	vary depending on location	chronic hearing loss, ear drainage, pain or numbness around ear	usually involve vestibular or cochlear dysfunction
Imaging (may not reliably distinguish among these)	CT: low density; no enhancement; bone erosion in only 33% MRI: T1 WI: intensity slightly> CSF; T2 WI: tumor & CSF similar hi intensity		CT: homogeneous & isodense; rim enhancement; extensive destruction of petrous bone MRI: increased signal on both T1 WI and T2 WI
gross appearance	pearly white		brown (from hemosiderin)
Microscopic pathology ³²	hyperkeratotic cyst lined with stratified squamous epithelium		fibroblastic proliferation, hemosiderin-laden macrophages, cholesterol clefts, giant cell reaction
ideal treatment	aggressive near-total excision		subtotal resection followed by drainage & restoration of pneumatization ³³

- b) CSF shows pleocytosis, hypoglycorrhachia, elevated protein, and negative cultures. Cholesterol crystals may be seen and can be recognized by their amorphous birefringent appearance.
- c) Mollaret’s meningitis is a rare variant of aseptic meningitis which includes the finding of large cells in the CSF that resemble endothelial cells (which may be macrophages⁵⁶⁴) that may be seen in some patients with epidermoid cysts^{30,31}

Imaging

MRI □ Fig. 48.1: mimics CSF on T1 WI (low signal, may be slightly>CSF) and T2 WI (high signal). Tumors are usually also high signal on T2 WI, but most enhance with contrast on T1 WI (epidermoids do not enhance). An epidermoid may pass from the posterior fossa through the incisura to the middle fossa.

Diffusion weighted imaging (DWI) is the best test to differentiate epidermoids from CSF (e.g. as in similar appearing arachnoid cyst). Epidermoids show intense signal on DWI as a result of restriction of water movement.

Treatment

Caution when removing epidermoid cysts to minimize spilling contents as they are quite irritating and may cause severe chemical meningitis (Mollaret’s meningitis, see above). Berger²³ advocates intraoperative irrigation with hydrocortisone (100 mg/L of LR) to reduce the risk of post-op communicating hydrocephalus. Peri-operative IV steroids and copious saline irrigation during surgery may provide similar results. The tumor is not in the cyst wall, and the surgical plan is generally to remove as much as possible but to leave capsule adherent to critical structures such as brainstem and blood vessels as the morbidity of removal is high and a small residual does not preclude satisfactory outcome.

In spite of adequate removal, it is not unusual to see persistent brainstem distortion on post-op imaging.²⁹ Post-op radiation is not indicated as the tumor is benign and XRT does not prevent recurrence.³⁴

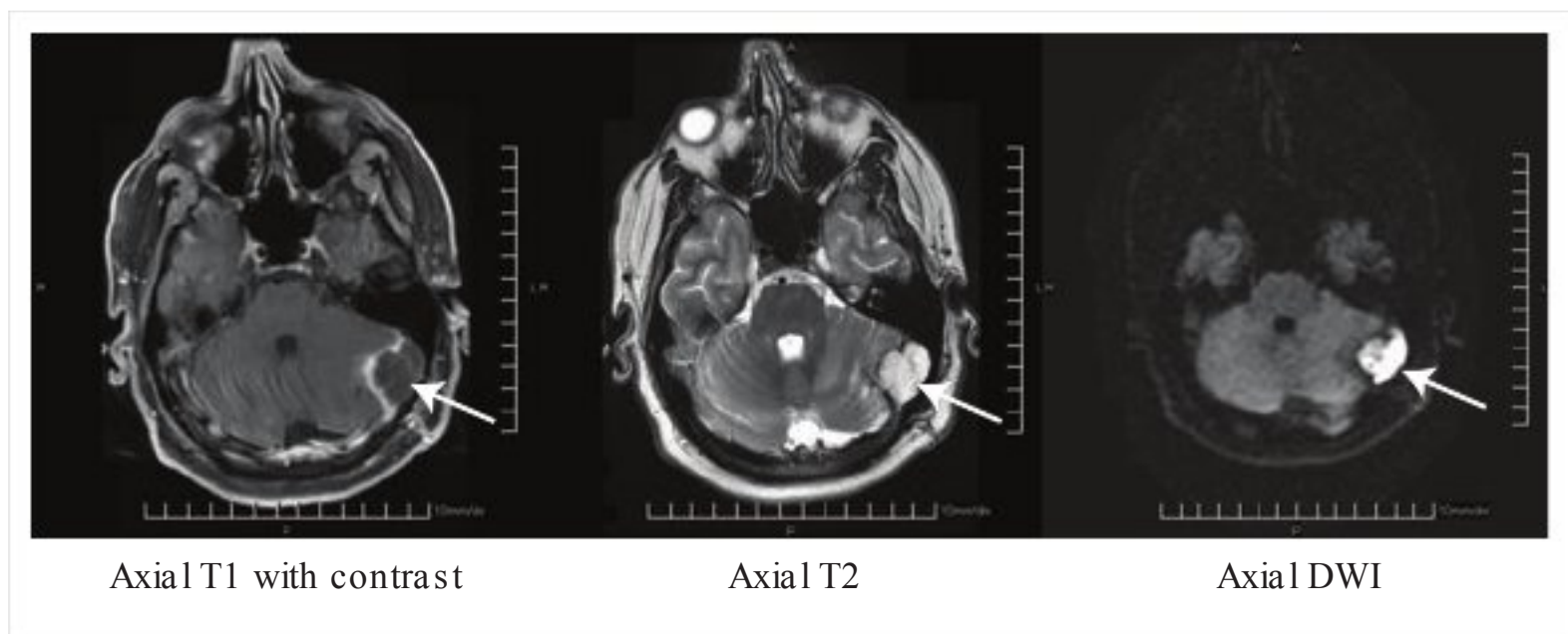


Fig. 48.1 MRI demonstrating left cerebellopontine angle epidermoid. Note that the CSF is dark on the DWI

48.4 Craniopharyngioma

48.4.1 General information

Craniopharyngiomas (CP) are tumors that develop from residual cells of Rathke's pouch (p. 149), and tend to arise from the anterior superior margin of the pituitary. They are lined with stratified squamous epithelium. Some CP may arise primarily within the third ventricle.³⁵ Almost all CP have solid and cystic components; fluid in the cysts varies, but usually contains cholesterol crystals. CP do not undergo malignant degeneration; but difficulty in cure makes them malignant in behavior.³⁶ (p 905–15) CP are distinct from Rathke's cleft cyst, but share some similarities (see below).

Calcification: microscopically 50% Plain x-ray: 85% in childhood, 40% in adults.

48.4.2 Epidemiology

Incidence: 2.5–4% of all brain tumors; about 50% occur in childhood (9% of Matson's series). Peak incidence: age 5–10 yrs.

48.4.3 Anatomy

Arterial supply: usually small feeders from ACA and A-comm, or from ICA and P-comm (do not receive blood from PCA or BA-bifurcation unless blood supply of floor of third ventricle is parasitized).

48.4.4 Surgical treatment

Pre-op endocrinologic evaluation

As for pituitary tumor (p. 730). Hypoadrenalism may be corrected rapidly, but hypothyroidism takes longer; either condition can increase surgical mortality.

Approach

Usually via large right frontotemporal flap as low as possible along base of frontal fossa (lateral sphenoid wing rongeuired/drilled). Approach to tumor is extra-axial, whether subfrontal or frontotemporal. All tumors should be aspirated (even if they appear solid radiographically). Then, with microscope, possible approaches include:

1. subchiasmatic: through space between optic nerves and anterior to chiasm. It was thought that a "prefixed chiasm" (i.e. congenitally short optic nerves with chiasm unusually close to the planum sphenoidale) was more common in patients with CP, making this approach more difficult. However, in reality the chiasm is probably bowed anteriorly by the tumor within the third ventricle giving the illusion of a prefixed chiasm in most cases
2. opticocarotid (between right ICA and right optic nerve/tract)
3. lamina terminalis (tumor often needs to be brought down and removed subchiasmatically)^{35,37}

4. lateral to carotid artery
5. transfrontal-transsphenoidal: drill off tuberculum sellae

Alternative approaches to frontotemporal

1. pure transsphenoidal: if dark fluid is aspirated with no CSF evident, it is possible to leave a stent from the tumor cavity to the sphenoid air-sinus to permit continued drainage
2. transcallosal: strictly for tumors limited to the third ventricle
3. a combined subfrontal/pterional approach capitalizes on the advantages of each (head is positioned with slight lateral rotation)

Spare the following structures: small arterial feeders to undersurface of the chiasm (major supply) and tract; at least a remnant of pituitary stalk (recognized by unique pattern of longitudinal striations which are the long portal veins). If the tumor easily pulls down from above then this is permissible, however do not pull too hard or else hypothalamic injury may result.

Post-op

1. steroids: these patients are all considered hypo-adrenal. Give hydrocortisone in physiologic doses (for mineralocorticoid activity) in addition to dexamethasone (glucocorticoid that treats edema) taper. Taper steroids slowly to avoid aseptic (chemical) meningitis
2. diabetes insipidus (DI) (p.752): often shows up early. May be part of a “triphasic response.” Best managed initially with fluid replacement. If necessary, use short acting vasopressin (prevents iatrogenic renal shutdown if a SIADH-like phase develops during vasopressin therapy)

48.4.5 Radiation

Controversial. Side effects: include endocrine dysfunction, optic neuritis, dementia. Post-op XRT probably helps prevent regrowth when residual tumor is left behind,³⁸ however, in pediatric cases it may be best to postpone XRT (to minimize deleterious effect on IQ), recognizing that reoperation may be necessary for recurrence.

48.4.6 Outcome

5–10% mortality in most series, most from hypothalamic injury (unilateral hypothalamic lesions are rarely clinically evident; bilateral injuries may produce hyperthermia and somnolence; damage to anterior osmoreceptors may → loss of thirst sensation). Five year survival is ≈ 55–85% (range from 30–93% has been reported).

48.4.7 Recurrence

Most recurrences are in <1 year, few >3 yrs (very delayed recurrence usually follow what was thought to be “total” removal). Morbidity/mortality is higher with re-operation.

References

- [1] Maggio WW, Cail WS, Brookeman JR, et al. Rathke's Cleft Cyst: Computed Tomographic and Magnetic Resonance Imaging Appearances. *Neurosurgery*. 1987; 21:60–62
- [2] Nishio S, Mizuno J, Barrow DL, Takei Y, Tindall GT. Pituitary Tumors Composed of Adenohypophysial Adenoma and Rathke's Cleft Cyst Elements: A Clinicopathological Study. *Neurosurgery*. 1987; 21:371–377
- [3] Voelker JL, Campbell RL, Muller J. Clinical, Radiographic, and Pathological Features of Symptomatic Rathke's Cleft Cysts. *J Neurosurg*. 1991; 74:535–544
- [4] Little JR, MacCarty CS. Colloid Cysts of the Third Ventricle. *J Neurosurg*. 1974; 39:230–235
- [5] Ciric I, Zivin I. Neuroepithelial (Colloid) Cysts of the Septum Pellucidum. *J Neurosurg*. 1975; 43:69–73
- [6] Guner M, Shaw MDM, Turner JW, et al. Computed Tomography in the Diagnosis of Colloid Cyst. *Surg Neurol*. 1976; 6:345–348
- [7] Ryder JW, Kleinschmidt BK, Keller TS. Sudden Deterioration and Death in Patients with Benign Tumors of the Third Ventricle Area. *J Neurosurg*. 1986; 64:216–223
- [8] Tatter SB, Ogilvy CS, Golden JA, Ojemann RG, Louis DN. Third ventricular xanthogranulomas clinically and radiologically mimicking colloid cysts. Report of two cases. *J Neurosurg*. 1994; 81:605–609
- [9] Mamourian AC, Cromwell LD, Harbaugh RE. Colloid Cyst of the Third Ventricle: Sometimes More Conspicuous on CT Than MR. *AJNR*. 1998; 19:875–878
- [10] Pollock BE, Schreiner SA, Huston J, III. A theory on the natural history of colloid cysts of the third ventricle. *Neurosurgery*. 2000; 46:1077–81; discussion 1081–3
- [11] El Khoury C, Brugieres P, Decq P, Cosson-Stanescu R, Combes C, Ricolfi F, Gaston A. Colloid cysts of the third ventricle: are MR imaging patterns predictive of difficulty with percutaneous treatment? *AJNR Am J Neuroradiol*. 2000; 21:489–492
- [12] Torkildsen A. Should Extirpation be Attempted in Cases of Neoplasm in or Near the Third Ventricle of the Brain? Experiences with a Palliative Method. *J Neurosurg*. 1948; 5:249–275
- [13] Sheikh AB, Mendelson ZS, Liu JK. Endoscopic versus microsurgical resection of colloid cysts: a systematic review and meta-analysis of 1,278 patients. *World Neurosurg*. 2014; 82:1187–1197

- [14] Pollock BE, Huston J, III. Natural history of asymptomatic colloid cysts of the third ventricle. *J Neurosurg.* 1999; 91:364–369
- [15] Bosch DA, Rahn T, Backlund EO. Treatment of Colloid Cyst of the Third Ventricle by Stereotactic Aspiration. *Surg Neurol.* 1978; 9:15–18
- [16] Rivas JJ, Lobato RD. CT-Assisted Stereotaxic Aspiration of Colloid Cysts of the Third Ventricle. *J Neurosurg.* 1985; 62:238–242
- [17] Mathiesen T, Grane P, Lindquist C, von Holst H. High Recurrence Rate Following Aspiration of Colloid Cysts in the Third Ventricle. *J Neurosurg.* 1993; 78:748–752
- [18] Musolino A, Fosse S, Munari C, et al. Diagnosis and Treatment of Colloid Cysts of the Third Ventricle by Stereotactic Drainage. Report on Eleven Cases. *Surg Neurol.* 1989; 32:294–299
- [19] Apuzzo MLJ, Chandrasoma PT, Zelman V, Giannotta SL, et al. Computed Tomographic Guidance Stereotaxis in the Management of Lesions of the Third Ventricular Region. *Neurosurgery.* 1984; 15:502–508
- [20] Apuzzo MLJ. Comment on Garrido E, et al.: Cerebral Venous and Sagittal Sinus Thrombosis After Transcallosal Removal of a Colloid Cyst of the Third Ventricle: Case Report. *Neurosurgery.* 1990; 26
- [21] Kondziolka D, Lunsford LD. Stereotactic Management of Colloid Cysts: Factors Predicting Success. *J Neurosurg.* 1991; 75:45–51
- [22] Hall WA, Lunsford LD. Changing Concepts in the Treatment of Colloid Cysts. An 11-Year Experience in the CT Era. *J Neurosurg.* 1987; 66:186–191
- [23] Berger MS, Wilson CB. Epidermoid Cysts of the Posterior Fossa. *J Neurosurg.* 1985; 62:214–219
- [24] Gutin PH, Boehm J, Bank WO, Edwards MS, Rosegay H. Cerebral convexity epidermoid tumor subsequent to multiple percutaneous subdural aspirations. Case report. *J Neurosurg.* 1980; 52:574–577
- [25] Fleming JFR, Botterell EH. Cranial Dermoid and Epidermoid Tumors. *Surg Gynecol Obstet.* 1959; 109:57–79
- [26] Alvord EC. Growth Rates of Epidermoid Tumors. *Ann Neurol.* 1977; 2:367–370
- [27] Link MJ, Cohen PL, Breneman JC, Tew JM, Jr. Malignant squamous degeneration of a cerebellopontine angle epidermoid tumor. Case report. *J Neurosurg.* 2002; 97:1237–1243
- [28] Sabin HI, Bardi LT, Symon L. Epidermoid Cysts and Cholesterol Granulomas Centered on the Posterior Fossa: Twenty Years of Diagnosis and Management. *Neurosurgery.* 1987; 21:798–803
- [29] Altschuler EM, Jungreis CA, Sekhar LN, Jannetta PJ, et al. Operative Treatment of Intracranial Epidermoid Cysts and Cholesterol Granulomas: Report of 21 Cases. *Neurosurgery.* 1990; 26:606–614
- [30] Abramson RC, Morawetz RB, Schlitt M. Multiple Complications from an Intracranial Epidermoid Cyst: Case Report and Literature Review. *Neurosurgery.* 1989; 24:574–578
- [31] Szabo M, Majtenyi C, Gusea A. Contribution to the Background of Mollaret's Meningitis. *Acta Neuropathol.* 1983; 59:115–118
- [32] Friedman I. Epidermoid Cholesteatoma and Cholesterol Granuloma: Experimental and Human. *Ann Otol Rhinol Laryngol.* 1959; 68:57–79
- [33] Chang P, Fagan PA, Atlas MD, Roche J. Imaging destructive lesions of the petrous apex. *Laryngoscope.* 1998; 108:599–604
- [34] Keville FJ, Wise BL. Intracranial Epidermoid and Dermoid Tumors. *J Neurosurg.* 1959; 16:564–569
- [35] Klein HJ, Rath SA. Removal of Tumors of the III Ventricle Using Lamina Terminalis Approach: Three Cases of Isolated Growth of Craniopharyngiomas in the III Ventricle. *Childs Nerv Syst.* 1989; 5:144–147
- [36] Wilkins RH, Rengachary SS. *Neurosurgery.* New York 1985
- [37] Patterson RH, Denylevich A. Surgical Removal of Craniopharyngiomas by a Transcranial Approach through the Lamina Terminalis and Sphenoid Sinus. *Neurosurgery.* 1980; 7:111–117
- [38] Manaka S, Teramoto A, Takakura K. The Efficacy of Radiotherapy for Craniopharyngioma. *J Neurosurg.* 1985; 62:648–656

49 Pseudotumor Cerebri and Empty Sella Syndrome

49.1 Pseudotumor cerebri

49.1.1 General information

Key concepts

- papilledema and symptomatic ICP elevation >20 cm H₂O in the absence of intracranial mass or infection. Often associated with dural sinus thrombosis
- a preventable cause of (often permanent) blindness from optic atrophy
- more common in obese females of childbearing age than general population
- recommended work-up:
 - preferred imaging studies: □ brain MRI (without & with contrast) and MRV. Imaging should be normal (allowed exception: slit-like ventricles)
 - □ LP. Findings: opening pressure (>25 cm H₂O) & normal CSF analysis
 - □ ophthalmologic eval: test visual fields, acuity, and check for papilledema
- usually self-limited, recurrence is common, chronic in some patients
- risk of blindness is not reliably correlated to duration of symptoms, papilledema, H/A, Snellen visual acuity, or number of recurrences
- treatment for patients failing medical management (weight loss, Diamox...):
 - optic nerve sheath fenestration (ONSF) is best for visual loss without H/A
 - CSF shunt may be better than ONSF for H/A associated with visual loss

49

Pseudotumor cerebri (PTC) AKA idiopathic intracranial hypertension (IIH), AKA benign intracranial hypertension, (plus numerous other obsolete terms¹) is a condition (or perhaps a heterogeneous group of conditions) characterized by increased intracranial pressure with no evidence of intracranial mass, hydrocephalus, infection (e.g. meningitis, especially chronic ones such as fungal meningitis), or hypertensive encephalopathy. Some, but not all, authors exclude patients with intracranial hypertension in the presence of dural sinus thrombosis. PTC is thus a diagnosis of exclusion. There is a juvenile and an adult form.

In general, the term pseudotumor cerebri is preferred (what was old is new!) because it encompasses cases where there is a known etiology as well as idiopathic cases (the concept of “secondary idiopathic intracranial hypertension” is an oxymoron).

49.1.2 Epidemiology

1. female:male ratio reported ranges from 2:1 to 8:1 (no gender difference in juvenile form)
2. obesity is reported in 11–90% of cases, and is not as prevalent in men²
3. incidence among obese women of childbearing years^{3,4}: 19–21/100,000, (whereas incidence in general population¹: 1–2/100,000)
4. peak incidence in 3rd decade (range: 1–55 years). 37% of cases are in children, 90% of these are age 5–15 years. Very rare in infancy
5. frequently self limited (recurrence rate: 9–43%)
6. severe visual deficits develop in 4–12% unrelated to duration of symptoms, degree of papilledema, headache, visual obscuration, and number of recurrences.⁵ Perimetry is the best means to detect and follow visual loss

49.1.3 Pathogenesis

Pathogenesis is not fully understood. Increased cerebral edema & brain water content, increased venous pressure & cerebral blood volume, and reduced CSF absorption have all been demonstrated. Theories that also explain the high prevalence in obese females:

1. mechanical theory: obesity $\rightarrow$ $\uparrow$ intraabdominal pressure $\rightarrow$ $\uparrow$ central venous pressure $\rightarrow$ $\uparrow$ CSF resorption $\rightarrow$ $\uparrow$ ICP (however, other studies have indicated that elevated venous pressure may actually be an epiphenomenon to a primary increase in ICP⁶)
2. hormonal theory: adipocytes convert androstenedione $\rightarrow$ estrone $\rightarrow$ $\uparrow$ CSF production

Table 49.1 Modified Dandy’s criteria for PT
• signs & symptoms of increased ICP • no localizing signs other than Cr. N VI palsy^a in an otherwise awake and alert patient • increased CSF pressure without chemical or cytological abnormalities • normal to small ventricles and no intracranial mass
^a may result from ↑ ICP (p.567)

49.1.4 Diagnostic criteria

Modified Dandy’s criteria are shown in □ Table 49.1.
More specifically, four diagnostic criteria⁷:

1. CSF pressure: >20 cm H₂O (pressures >40 are not uncommon). Some recommend that the pressure should be >25 to exclude normals.⁸ Note: Diurnal variations in CSF pressure may occasionally cause a falsely low (i.e. normal) reading. □ If clinical suspicion is high, an LP at a different time of day or continuous ICP monitoring may be required.
2. CSF composition: normal glucose and cell count. Protein is normal, or in ≈ two thirds of cases it is low (<20 mg%)
3. symptoms & signs are those of elevated ICP alone (i.e. papilledema & H/A) with no focal findings. Allowed exception: abducens nerve palsy which may be due to increased ICP (p.567)
4. normal radiologic studies of the brain (CT or MRI) with the allowed exceptions of:
 - a) the occasionally seen slit ventricles (the incidence may be no higher in PTC than in age-matched controls⁹) or empty sella
 - b) infantile form may have generous ventricles and large fluid spaces over brain
 - c) intra-orbital abnormalities may be seen: see below

49.1.5 Clinical

Symptoms

See references.^{7,10}

1. classic (major) symptoms
 - a) H/A (the most common symptom): 94–99% Typically retro-ocular and pulsatile. May ↑ with eye movement. Severity does not correlate with degree of CSF pressure elevation. Occasionally worse in A.M.
 - b) nausea: 32%(actual vomiting is less common)
 - c) visual loss (PTC below):
 - transient visual obscuration (TVO)
 - permanent afferent visual pathway injury
 - d) diplopia (more common in adult, usually due to VI nerve palsy): 30%
2. minor symptoms¹¹
 - a) neck stiffness: 30–50%
 - b) tinnitus (the causal relationship with IIH has been demonstrated by resolution of these symptoms with reduction of CSF pressure): up to 60% Usually pulse synchronous. Described as rushing noise. May be unilateral (in these, may be reduced by ipsilateral jugular vein compression + ipsilateral head rotation)
 - c) ataxia: 4–11%
 - d) acral paresthesias: 25%
 - e) retrobulbar eye pain on eye movements
 - f) arthralgia: 11–18%
 - g) dizziness: 32%
 - h) fatigue
 - i) reduced olfactory acuity

Signs

Signs are generally restricted to visual system.
Conspicuously absent: altered level of consciousness in spite of high ICP

1. eye findings – also PTC below
 - a) papilledema:
 - present in almost ≈ 100%

- idiopathic intracranial hypertension without papilledema (IIHWOP)¹²: a variant of IIH. Visual loss tends not to occur
 - usually bilateral, occasionally unilateral¹³
 - may be mild (subtle nerve fiber elevation)
- b) abducens nerve (Cr. N. VI) palsy: 20% a false localizing sign (p.567). The esotropia ranges from <5 prism diopters dysconjugate angle in primary gaze to >50¹⁴
- c) visual acuity: relatively insensitive assessment of visual function
- d) visual field defect: 9%
- early changes: peripheral fields & nasal quadrant defect
 - enlarged blind spot (66%) and concentric constriction of peripheral fields (blindness is very rare at presentation)
2. infantile form may have only enlarging OFC, frequently self limited, usually requires only follow-up without specific treatment

□ Note. Worsening of any of the above symptoms with postural changes that increase ICP (bending over, Valsalva maneuver...) is characteristic in idiopathic intracranial hypertension.

Visual loss in PTC

General information

Quoted range of occurrence in PTC: 48–68%(lower numbers generally come from population based samples). A prospective study found changes by Goldman perimetry in 96%of 50 patients.¹⁵ The only parameter associated with worsening vision is recent weight gain.

Pathomechanics

Increased ICP is transmitted along optic nerve sheath → circumferential compression of the retinal ganglion cell axons at the level of the lamina cribrosa.¹⁴

Manifestations

1. transient visual obscurations (TVO): graying or blacking out of vision. Lasts ≈ 1 second. Uni- or bi-lateral. Typically occur with eye movement, bending over or valsalva maneuver. Directly proportional to severity of papilledema. Frequency of TVOs parallels ICP elevation, but doesn’t correlate with permanent visual loss
2. visual loss in PTC may occur early or late, may be sudden or gradually progressive, and is not reliably correlated to duration of symptoms, papilledema, H/A, Snellen visual acuity, or number of recurrences. It may escape detection until profound.
- a) early: usually constriction of fields and loss of color (□perimetry is the best test for following vision in PTC)
- b) late: central vision is affected. Findings include: concentric constrictions, enlargement of the blind spot, inferior nasal defects, arcuate defects, cecocentral scotomas...

49.1.6 Associated conditions

General information

Some cases of PTC are idiopathic (IIH). However, often what is considered “IIH” may actually be secondary to some other condition (e.g. transverse sinus thrombosis, see below). Many conditions cited as being associated with PTC may be coincidental. Four criteria suggested to establish a cause-effect relationship are shown in □ Table 49.2.¹⁰

□ Table 49.3 shows a scale¹⁶to rank the likelihood of association between various conditions and IIH based on the number of the criteria met in □ Table 49.2.

Other conditions not included in this list that meet minimal criteria but are unconfirmed in case-control studies¹ include:

1. other drugs: isotretinoin (Accutane®), trimethoprim-sulfamethoxazole, cimetidine, tamoxifen
2. systemic lupus erythematosus (SLE)

Table 49.2 Criteria for causality of PTC by another condition ¹⁰
• meets Dandy’s criteria (□ Table 49.1) • the condition should be proven to increase ICP • treatment of the condition should improve the IIH • properly controlled studies should show an association between the condition and IIH

Table 49.3 Conditions that may be associated with PTC ¹⁶
Proven association
Meets 4 criteria from □ Table 49.2
• obesity
Likely association
Meets 3 criteria from □ Table 49.2
• drugs: keprone, lindane • hypervitaminosis A
Probable association
Meets 2 criteria from □ Table 49.2
• steroid withdrawal^a • thyroid replacement in children • ketoprofen & indomethacin in Bartter syndrome • hypoparathyroidism • Addison’s disease^a • uremia • iron deficiency anemia • drugs: tetracycline, nalidixic acid, Danazol, lithium, amiodarone, phenytoin, nitrofurantoin, ciprofloxacin, nitroglycerin
Possible association
Meets 1 criterion from □ Table 49.2
• menstrual irregularity • oral contraceptive use^b • Cushing’s syndrome • Vitamin A deficiency • minor head trauma • Behçet syndrome
Unlikely association
Meets none of the criteria in □ Table 49.2
• hyperthyroidism • steroid use • immunization
Unsupported association
• pregnancy • menarche
^a may respond to steroids
^b may be associated with dural sinus thrombosis, see text

Conditions that may be related by virtue of increased pressure in the dural sinuses (see below):

1. otitis media with petrosal extension (so-called otitic hydrocephalus)
2. radical neck surgery with resection of the jugular vein
3. hypercoagulable states

Venous hypertension and sinovenous abnormalities

Venous hypertension has often been proposed as a unifying underlying cause of PTC. Abnormalities of the dural sinuses, including thrombosis, stenosis,¹⁷ obstruction, or elevated pressure (reaching levels as high as 40 mm Hg) have been demonstrated in a number of studies. While these findings may underlie a significant number of cases, they may in actuality be epiphenomena (e.g. venous hypertension may be due to compression of the transverse sinuses by elevated intracranial pressure⁶), and it is unlikely that such abnormalities will explain all cases.

Bilateral sinovenous stenosis was seen (using a sophisticated but sensitive test of elliptic-centric-ordered 3-D gadolinium-enhanced MRI venography) in 27 of 29 patients with PTC and in only 4 of 59 controls.¹⁷

49.1.7 Differential diagnosis

1. true mass lesions: tumor, cerebral abscess, subdural hematomas, rarely gliomatosis cerebri may be undetectable on CT and will be misdiagnosed as PTC
2. cranial venous outflow impairment (some authors consider these as IIH)¹⁸
 - a) dural sinus thrombosis (p.1308)
 - b) congestive heart failure
 - c) superior vena cava syndrome
 - d) unilateral or bilateral jugular vein or sigmoid sinus¹⁹ obstruction
 - e) hyperviscosity syndromes
 - f) Masson's vegetant intravascular hemangioendothelioma²⁰: an uncommon, usually benign lesion that may rarely involve the neuraxis (including intracranial occurrence). Not definitely neoplastic. Organizing thrombi develop endothelialized projections into the vessel lumen. Must be distinguished from other conditions such as angiosarcoma
3. Chiari I malformation (CIM): may produce findings similar to PTC. 6% of PTC patients have significant tonsillar ectopia, and $\approx 5\%$ of patient with CIM have papilledema¹⁴
4. infection (CSF will be abnormal in most of these): encephalitis, arachnoiditis, meningitis (especially basal meningitis or granulomatous infections, e.g. syphilitic meningitis, chronic cryptococcal meningitis), chronic brucellosis
5. inflammatory conditions: e.g. neurosarcoidosis (p.189), SLE
6. vasculitis: e.g. Behçet's syndrome
7. metabolic conditions: e.g. lead poisoning
8. pseudopapilledema (anomalous elevation of the optic nerve head) associated with hyperopia and drusen. Retinal venous pulsations are usually present. Especially deceptive when a patient with migraines has pseudopapilledema: treat the H/A
9. malignant hypertension: may produce H/A & bilateral optic disc edema which can be indistinguishable from papilledema. May also produce hypertensive encephalopathy (p.194). Check BP in all PTC suspects
10. meningeal carcinomatosis
11. Guillain-Barré syndrome (p.184): CSF protein is usually elevated
12. following head trauma

49.1.8 Evaluation recommendations

Overview

Most tests are intended to rule out conditions that may mimic (or produce) PTC:

1. cerebral imaging: cerebral CT or MRI (see below) scan without and with contrast
2. LP:
 - a) measure opening pressure (OP) with patient in lateral decubitus position
 - b) CSF analysis to rule-out infection (e.g. fungus, TB or Lyme disease), inflammation (e.g. sarcoidosis, SLE) or neoplasm (e.g. carcinomatous meningitis)
 - protein/glucose
 - cell count
 - routine & fungal cultures
 - cytology if suspicion of carcinomatous meningitis
3. routine labs: CBC, electrolytes, PT/PTT
4. W/U for sarcoidosis or SLE if other findings suggestive (e.g. cutaneous nodules, hypercoagulable state...)
5. neuro-ophthalmologic evaluation is recommended. Includes: visual field testing using quantitative perimetry, with evaluation of size of blind spot, slit-lamp examination $\pm$ fundus photographs
6. check BP to R/O malignant HTN $\rightarrow$ hypertensive encephalopathy (p.770)

CT

CT without and with IV contrast is usually adequate to R/O intracranial mass as a possible cause of intracranial hypertension, but may miss cases of dural sinus thrombosis. MRI & MRV are preferred.

MRI

Intracranial abnormalities are usually absent or minimal (slit ventricles, empty sella in 30–70%). However, intraorbital findings may be more substantial and include¹⁴:

1. flattening of the posterior sclera: occurs in 80%
2. enhancement of the prelaminar optic nerve: in 50%
3. distention of the perioptic subarachnoid space: in 45%
4. vertical tortuosity of the orbital optic nerve: in 40%
5. intraocular protrusion of the prelaminar optic nerve: in 30%

Venography

MR venography (MRV) has largely replaced conventional venography to rule out dural sinus or venous thrombosis.

49.1.9 Treatment and management

Natural history

Spontaneous resolution is common, sometimes within months, but usually after ≈ 1 year. Papilledema persists in $\approx 15\%$. Permanent visual loss occurs in 2–24% (depending on criteria used and degree to which it is sought). Persistent H/A may occur in some. Recurs in $\approx 10\%$ after initial resolution.¹⁴

Interventions

Overview

Studies are often difficult to interpret especially since spontaneous remission is common.

1. all patients must have repeated thorough ophthalmologic exams (see above)
2. stop possible offending drugs
3. weight loss: a weight loss of 6% usually results in complete resolution of papilledema.²¹ However, resolution may be too slow for acutely threatened vision. Weight loss is also associated with reduction of other health risks of obesity. Symptoms recur if the weight is regained
 - a) dieting: uncontrolled studies²² suggest that this is effective, but is it rarely accomplished or sustained
 - b) bariatric surgery: gastric bypass, laparoscopic banding...
4. treatment of asymptomatic PTC patients is controversial as there is no reliable predictor for visual loss. Close follow up with serial formal visual field evaluation is necessary. Intervention is recommended in unreliable patients, or whenever visual fields deteriorate. It is possible to lose vision without H/A or papilledema
5. most cases remit by 6–15 weeks, however relapse is common
6. medical treatment
 - a) fluid and salt restriction
 - b) diuretics (slows CSF production) – see below
 - c) if ineffective, add steroids (options: dexamethasone (Decadron®) 12 mg/day, prednisone 40–60 mg/day, or methylprednisolone 250 mg IV q 6 hrs). May $\uparrow$ CSF resorption in cases of inflammation or venous thrombosis. Can be used as temporizing agents for patients awaiting surgery. A reduction in symptoms should occur by 2 weeks, after which time the steroid should be tapered over 2 weeks. Long-term use is not recommended due to, among other things, associated weight gain
7. surgical therapy²³ (p 250–3) only for cases refractory to above, or where visual loss is progressive or is severe initially or unreliable patient:
 - a) serial LPs until remission (25% remit after 1st LP²⁴): remove up to 30 ml to halve OP, perform qod until OP < 20 cm H₂O, then decrease to q wk (no patient who had remission by 2nd LP had OP > 350 on 1st LP). Use a large gauge needle (e.g. 18 Ga) which may help promote a post-LP CSF leak into subcutaneous tissues. LPs may be difficult in obese patients. Revisions may be required in up to 50%. Side effects: include sciatica from nerve root irritation, acquired cerebellar tonsillar herniation (p.418), spinal H/A (from intracranial hypotension)
 - b) Shunts: see below
 - c) optic nerve sheath fenestration: see below
 - d) older treatment less commonly practiced today: subtemporal decompression (advocated by Dandy) or suboccipital decompression. Usually bilateral silver-dollar size craniectomies under

temporalis muscle to floor of middle fossa, open dura, cover brain with absorbable sponge, close fascia and muscle watertight, anticonvulsants were started due to risk of post-op seizures

8. interventional procedures: venous sinus stenting may be considered for refractory cases.²⁵
9. patients should be followed at least two years (with repeat imaging, e.g. MRI) to R/O occult tumor

Diuretics

1. carbonic anhydrase (CA) inhibitors:
 - a) acetazolamide (Diamox®): □ start at 125–250 mg PO q 8–12 hrs, or long acting Diamox Sequels® 500 mg PO BID. Increase by 250 mg/day until symptoms improve, side effects occur, or 2 gm/day reached. Side effects: (in high doses): acral paresthesias, nausea, metabolic acidosis, altered taste, renal calculi, drowsiness. Rare: Stevens-Johnson syndrome, toxic epidermal necrolysis, agranulocytosis. □ Contraindicated with allergy to sulfa or a history of renal calculi
 - b) methazolamide (Neptazane®): better tolerated but less effective. □ 50–100 mg PO BID-TID. This branded product is no longer on the market. Side effects: similar to acetazolamide
 - c) topiramate (Topamax®): anticonvulsant with secondary inhibition of CA. □ 200 mg PO BID. Side effects: Similar to acetazolamide, but can be used in sulfa allergic patients
2. furosemide (Lasix®)
 - a) start: 160 mg per day in adults, adjust per symptoms and eye exam (not to CSF pressure)
 - b) if ineffective, double (320 mg/day)
 - c) monitor K⁺ levels and supplement as needed

49

Shunts

1. lumbar shunt: usually lumboperitoneal; see insertion technique (p.1517). May be difficult in obese patient. May need a horizontal-vertical valve (p.418) to prevent H/A from intracranial hypotension. Alternative: lumbopleural shunt
2. other shunts may be used, especially when arachnoiditis precludes use of lumbar subarachnoid space, e.g.:
 - a) VP shunt: often difficult since the ventricles are frequently small or slit-like.²⁶ Stereotactic techniques may make this more technically feasible
 - b) cisterna magna shunt: may shunt to vascular system

Optic nerve sheath fenestration (ONSF)

See references.^{27,28,29}

Generally better for protection of vision and reversal of papilledema than for other symptoms (e.g. H/A). Performed via medial or less commonly a lateral orbitotomy or transconjunctival medial approach. May reverse or stabilize visual deterioration³⁰ and sometimes (but not always) lowers ICP (by continued CSF filtration) and may protect the contralateral eye (if not, contralateral ONSF must be performed). Has succeeded in cases where visual loss progressed after LP shunting,³¹ possibly due to poor communication between orbital and intracranial subarachnoid spaces. Side effects: potential adverse include: pupillary dysfunction, peripapillary hemorrhage, chemosis, chorioretinal scarring,³² diplopia (usually self-limited) from medial rectus disruption. Repeat fenestration is needed in 0–6%¹⁴

Management recommendations for specific situations

Weight loss should be attempted in all.

1. PTC patients with H/A and no visual loss: medical therapy to control ↑ ICP and H/A. ONSF not recommended. Shunting is an option if medical management fails
2. PTC with visual loss without H/A:
 - a) mild visual loss: acetazolamide 500–1500 mg/d, follow-up q 2 weeks
 - b) moderate visual loss: acetazolamide 2000–3000 mg/d, follow-up q week
 - c) severe visual loss, moderate visual loss that doesn't respond to acetazolamide, or optic disc at risk:
 - methylprednisolone 250 mg IV q 6 hrs + acetazolamide 1000 mg PO BID
 - if no improvement: ONSF. Consider shunt if ICP > 300 mm H₂O
 - d) PTC with visual loss AND H/A: for patients with surgical indications, either surgical procedure is appropriate. Shunting may relieve both problems simultaneously. ONSF may be more reliable to relieve the visual problems (the failure rate may be lower than the shunt malfunction rate) but is not as good for the H/A

3. IIHWOP: symptomatic treatment for H/A, diuretics
4. PTC in children and adolescents:
 - a) may be seen with withdrawal of steroids used for asthma
 - b) search for and correction of underlying etiology (offending drugs listed above, hypercalcemia, cancer...)
 - c) acetazolamide has been used with success
5. PTC in pregnancy:
 - a) women who first present with PTC during pregnancy: resolution of PTC following delivery is common
 - b) women who become pregnant during therapy:
 - 1st trimester: observation, limitation of weight gain, serial LPs. □ Acetazolamide should be avoided because of teratogenicity
 - 2nd & 3rd trimester: acetazolamide has been used safely, but involvement of high-risk obstetrician specialist is advised
6. pseudopapilledema (associated with drusen, etc., in the absence of intracranial hypertension): no interventions.¹⁴ Reassurance and H/A management are employed

49.2 Empty sella syndrome

49.2.1 General information

Empty sella syndrome (ESS) can be “primary” or “secondary.”

49.2.2 Primary empty sella syndrome

General information

Occurs in the absence of prior treatment of a pituitary tumor (medical, surgical or XRT). Herniation of the arachnoid membrane into the sella turcica³³ which can act as a mass, probably as a result of repeated CSF pulsation. The sella can become enlarged (see Sella turcica (p.216), for normal dimensions) and the pituitary gland may become compressed against the floor.

Frequent association: female sex (female:male ratio=5:1), obesity and HTN. The frequency of intrasellar arachnoid herniation is higher in patients with pituitary tumors and in those with increased intracranial pressure for any reason – including idiopathic intracranial hypertension (p.766) – than in the general population.

These patients usually present with symptoms that do not suggest an intrasellar abnormality including: headache (the most common symptom), dizziness, seizures... Occasionally patients may develop CSF rhinorrhea,³⁴ deterioration of vision (acuity or field deficit resulting from kinking of optic chiasm due to herniation into the sella), or amenorrhea-galactorrhea syndrome.

Clinically evident endocrine disturbances are rare with primary ESS, however up to 30% have abnormal pituitary function tests, most commonly reduced growth hormone secretion following stimulation. Mild elevation of prolactin (PRL) and reduction of ADH may occur, probably from compression of the stalk. These patients show a normal PRL rise with TRH stimulation (whereas patients with prolactinomas do not).

Treatment

Surgical treatment is usually not indicated, except in the case of CSF rhinorrhea. In this setting, it is necessary to determine if there is increased ICP, and if so, if there is an identifiable cause. Simple shunting for hydrocephalus runs the risk of producing tension pneumocephalus from air drawn in through the former leak site. This may necessitate transsphenoidal repair with simultaneous external lumbar drainage, to be converted to a permanent shunt shortly thereafter. Hyperprolactinemia may be treated e.g. with bromocriptine (p.740) if it interferes with gonadal function.

49.2.3 Secondary empty sella syndrome

Entities associated with secondary empty sella syndrome:

1. following trauma³⁵
2. after successful transsphenoidal removal or XRT for a pituitary tumor³⁵
3. any cause of increased intracranial pressure, including: idiopathic intracranial hypertension (pseudotumor cerebri), Chiari malformation

Often presents with visual deterioration due to herniation of the optic chiasm into the empty sella. There may be hypopituitarism from the underlying cause.

Visual deterioration may be treated with chiasm apexy (propping up the chiasm) usually by transphenoidal approach and packing the sella with fat, muscle or cartilage. May be done endoscopically.³⁶ Appears to be better for improving visual field deficits than loss of visual acuity.³⁷

References

[1] Radhakrishnan K, Ahlskog JE, Garrity JA, Kurland LT. Idiopathic Intracranial Hypertension. *Mayo Clin Proc.* 1994; 69:169–180

[2] Digre KB, Corbett JJ. Pseudotumor Cerebri in Men. *Arch Neurol.* 1988; 45:866–872

[3] Durcan FJ, Corbett JJ, Wall M. The Incidence of Pseudotumor Cerebri: Population Studies in Iowa and Louisiana. *Arch Neurol.* 1988; 45:875–877

[4] Radhakrishnan K, Ahlskog JE, Cross SA, Kurland LT, et al. Idiopathic Intracranial Hypertension (Pseudotumor Cerebri): Descriptive Epidemiology in Rochester, Minn, 1976 to 1990. *Arch Neurol.* 1993; 50:78–80

[5] Rush JA. Pseudotumor Cerebri: Clinical Profile and Visual Outcome in 63 Patients. *Mayo Clin Proc.* 1980; 55:541–546

[6] King JO, Mitchell PJ, Thomson KR, Tress BM. Manometry combined with cervical puncture in idiopathic intracranial hypertension. *Neurology.* 2002; 58:26–30

[7] Ahlskog JE, O'Neill BP. Pseudotumor Cerebri. *Ann Int Med.* 1982; 97:249–256

[8] Corbett JJ, Mehta MP. Cerebrospinal fluid pressure in normal obese subjects and patients with pseudotumor cerebri. *Neurology.* 1983; 33:1386–1388

[9] Jacobson DM, Karanjia PN, Olson KA, Warner JJ. Computed Tomography Ventricular Size has no Predictive Value in Diagnosing Pseudotumor Cerebri. *Neurology.* 1990; 40:1454–1455

[10] Giuseffi V, Wall M, Siegel PZ, Rojas PB. Symptoms and disease associations in idiopathic intracranial hypertension (pseudotumor cerebri): a case-control study. *Neurology.* 1991; 41:239–244

[11] Round R, Keane JR. The minor symptoms of increased intracranial hypertension: 101 patients with benign intracranial hypertension. *Neurology.* 1988; 38:1461–1464

[12] Wang SJ, Silberstein SD, Patterson S, Young WB. Idiopathic intracranial hypertension without papilledema: a case control study in a headache center. *Neurology.* 1998; 51:245–249

[13] Sher NA, Wirtschafter J, Shapiro SK, et al. Unilateral Papilledema in 'Benign' Intracranial Hypertension (Pseudotumor Cerebri). *JAMA.* 1983; 250:2346–2347

[14] Bejjani GK, Cockerham KP, Pless M, Rothfus WE. Idiopathic intracranial hypertension. *Contemp Neurosurg.* 2002; 24:1–8

[15] Wall M, George D. Idiopathic Intracranial Hypertension: A Prospective Study of 50 Patients. *Brain.* 1991; 114:155–180

[16] Digre KB. Epidemiology of idiopathic intracranial hypertension. 1992

[17] Farb RI, Vanek I, Scott JN, Mikulis DJ, Willinsky RA, Tomlinson G, TerBrugge KG. Idiopathic intracranial hypertension: The prevalence and morphology of sinovenous stenosis. *Neurology.* 2003; 60:1418–1424

[18] Johnston I, Hawke S, Halmagyi M, Teo C. The Pseudotumor Syndrome: Disorders of Cerebrospinal Fluid Circulation Causing Intracranial Hypertension Without Ventriculomegaly. *Arch Neurol.* 1991; 48:740–747

[19] Powers JM, Schnur JA, Baldree ME. Pseudotumor Cerebri due to Partial Obstruction of the Sigmoid Sinus by a Cholesteatoma. *Arch Neurol.* 1986; 43:519–521

[20] Wen DY, Hardten DR, Wirtschafter JD, et al. Elevated Intracranial Pressure from Cerebral Venous Obstruction by Masson's Vegetant Intravascular Hemangioendothelioma. *J Neurosurg.* 1991; 75:787–790

[21] Johnson LN, Krohel GB, Madsen RW, March GA, Jr. The role of weight loss and acetazolamide in the treatment of idiopathic intracranial hypertension (pseudotumor cerebri). *Ophthalmology.* 1998; 105:2313–2317

[22] Newberg B. Pseudotumor Cerebri Treated by Rice/Reduction Diet. *Arch Intern Med.* 1974; 133:802–807

[23] Wilkins RH, Rengachary SS. *Neurosurgery.* New York 1985

[24] Weisberg LA. *Benign Intracranial Hypertension.* Medicine (Baltimore). 1975; 54:197–207

[25] Higgins JN,owler BK, Cousins C, Pickard JD. Venous sinus stenting for refractory benign intracranial hypertension. *Lancet.* 2002; 359:228–230

[26] Hahn FJ, McWilliams FE. The Small Ventricle in Pseudotumor Cerebri: Demonstration of the Small Ventricle in Benign Intracranial Hypertension. *CT.* 1978; 2:249–253

[27] Brouman ND, Spoor TC, Ramocki JM. Optic Nerve Sheath Decompression for Pseudotumor Cerebri. *Arch Ophthalmol.* 1988; 106:1384–1390

[28] Sergott RC, Savino PJ, Bosley TM. Modified Optic Nerve Sheath Decompression Provides Long-Term Visual Improvement for Pseudotumor Cerebri. *Arch Ophthalmol.* 1988; 106:1384–1390

[29] Corbett JJ, Nerad JA, Tse D, et al. Optic Nerve Sheath Fenestration for Pseudotumor Cerebri: The Lateral Orbitotomy Approach. *Arch Ophthalmol.* 1988; 106:1391–1397

[30] Kelman SE, Heaps R, Wolf A, Elman MJ. Optic Nerve Decompression Surgery Improves Visual Function in Patients with Pseudotumor Cerebri. *Neurosurgery.* 1992; 30:391–395

[31] Kelman SE, Sergott RC, Cioffi GA, et al. Modified Optic Nerve Decompression in Patients with Functioning Lumboperitoneal Shunts and Progressive Visual Loss. *Ophthalmology.* 1991; 98:1449–1453

[32] Spoor TC, Ramocki JM, Madion MP, et al. Treatment of Pseudotumor Cerebri by Primary and Secondary Optic Nerve Sheath Decompression. *Am J Ophthalmol.* 1991; 112:177–185

[33] Kaufman B. The "empty" sella turcica - A manifestation of the intrasellar subarachnoid space. *Radiology.* 1968; 90:931–941

[34] Perani D, Scotti G, Colombo N, Sterzi R, Castelli A. Spontaneous CSF rhinorrhea through the lamina cribrosa associated with primary empty sella. *Ital J Neurol Sci.* 1984; 5:167–172

[35] Lee WM, Adams JE. The Empty Sella Syndrome. *J Neurosurg.* 1968; 28:351–356

[36] Alvarez Berastegui GR, Raza SM, Anand VK, Schwartz TH. Endonasal endoscopic transsphenoidal chiasm apexy using a clival cranial base cranioplasty for visual loss from massive empty sella following macroprolactinoma treatment with bromocriptine: case report. *J Neurosurg.* 2015:1–7

[37] Fouad W. Review of empty sella syndrome and its surgical management. *Alexandria Journal of Medicine.* 2011; 47:139–147

50 Tumors and Tumor-Like Lesions of the Skull

50.1 Skull tumors

50.1.1 General information

See Skull lesions (p.1376) for differential diagnosis and evaluation (including non-neoplastic lesions). Considering only tumors, the differential diagnosis includes:

1. benign tumors
 - a) osteoma: see below
 - b) hemangioma: see below
 - c) dermoid and epidermoid tumors: see below
 - d) chondroma: occur mainly in conjunction with the basal synchondroses
 - e) meningioma
 - f) aneurysmal bone cyst
2. malignant tumors: malignancy is suggested by a single large or multiple (>6) small osteolytic lesions with margins that are ragged, undermined and lacking sclerosis¹
 - a) bone metastases to the skull. Common ones include:
 - prostate
 - breast
 - lung
 - kidney
 - thyroid
 - lymphoma
 - multiple myeloma/plasmacytoma (p.714)
 - b) chondrosarcoma
 - c) osteogenic sarcoma
 - d) fibrosarcoma

50.1.2 Osteoma

General information

Osteomas are the most common primary bone tumor of the calvaria. They are benign, slow-growing lesions, that occur commonly in the cranial vault, mastoid and paranasal air sinuses, and the mandible. Lesions within air sinuses may present as recurrent sinusitis. More common in females, highest incidence is in 6th decade. Triad of Gardner's syndrome: multiple cranial osteomas (of calvaria, sinuses, and mandible), colonic polyposis, and soft-tissue tumors.

See Localized increased density or hyperostosis of the calvaria (p.1379) for differential diagnosis.

Pathology

Consists of osteoid tissue within osteoblastic tissue, surrounded by reactive bone. Difficult to distinguish from fibrous dysplasia.

Radiographic evaluation

Skull x-ray: round, sclerotic, well demarcated, homogeneous dense projection. Usually arise from outer table of skull (inner table less common). May be compact or spongy (spongy osteoma may be radiolucent). Unlike meningiomas, diploë are preserved and vascular channels are not increased.

Osteomas are "hot" on nuclear bone scan.

Treatment

Asymptomatic lesions may simply be followed. Surgery may be considered for cosmetic reasons, or if pressure on adjacent tissues produces discomfort. Lesions involving only the outer table may be removed leaving the inner table intact.

50.1.3 Hemangioma

General information

Comprise $\approx 7\%$ of skull tumors.¹ These benign tumors commonly occur in the skull (discussed here) and spine (p.794). Two types: cavernous (most common) and capillary (rare).

Radiographic evaluation

Skull x-ray: characteristically shows a circular lucency with honeycomb or trabecular pattern (seen in $\approx 50\%$ of cases) or radial trabeculations producing a sunburst pattern (seen in $\approx 11\%$ of cases).¹ Sclerotic margins are evident in only $\approx 33\%$

CT: hypodense lesion with sclerotic spaced trabeculations. Nonenhancing.

Bone scan: typically hot.

Treatment

Accessible lesions may be cured by en bloc excision or curettage. The gross appearance is of a hard, blue-domed mass beneath the pericranium. Radiation may be considered for inaccessible tumors.

50.1.4 Epidermoid and dermoid tumors of the skull

General information

See also epidermoids and dermoids in general (p.776).

Dermoids and epidermoids are benign inclusion cysts of ectoderm that may involve skull and underlying dural venous structures or brain. They may become infected. Primary skull involvement is rare and occurs when ectodermal rests are entrapped in the developing skull which causes these tumors to arise within the diploë and expand both inner and outer tables. Because they are not neoplastic, they grow at a linear rate (instead of exponential). Usually midline.

Epidermoid tumors contain only the outer layer of skin, and are therefore lined with stratified squamous epithelium and the resultant byproduct, keratin.

Dermoid tumors contain all elements of skin including hair follicles (which may produce hair in the tumor) sweat glands (sebaceous glands (apocrine) and sweat glands (eccrine)).²

Teratomas are true neoplasms and may also contain bone, cartilage, teeth and nails.

Presentation

These lesions may present as a result of mass effect from continued growth.

They may rupture (more common with dermoids than epidermoids), and can cause chemical meningitis (from the irritating properties of fat and or keratin), or, if infected, bacterial meningitis.

Radiographic evaluation

1. skull x-ray: these osteolytic lesions have well-defined, sclerotic margins
2. some imaging is required to evaluate possible intracranial involvement
 - a) CT: the lesions are hypodense (keratin contains fats), and non-enhancing
 - b) MRI: like CSF they are low intensity on T1WI and high signal on T2WI, but unlike CSF they are high signal on DWI > MRI (p.762)

Treatment

Treatment is surgical. Radiation and chemotherapy are not indicated.

When possible, the goal is to avoid rupture during removal in order to avoid chemical and/or bacterial meningitis.

Bone margins are curetted. Search must be made for a tract leading to the intracranial cavity which must be followed if found. Preparation for dural sinus repair must be made for lesions overlying the sagittal sinus (including torcular Herophili).

Endoscopic surgery may be an option for some skull base lesions.

50.1.5 Langerhans cell histiocytosis

General information

Histiocytic disorders may be classified as follows:

1. malignant (true histiocytic lymphoma)
2. reactive (benign histiocytosis)
3. Langerhans cell histiocytosis (LCH)
 - a) unifocal: nee eosinophilic granuloma: rare (1/1200 new cases/yr in US). More common in children. Slowly progressing disease. May occur in bone, skin, lungs or stomach.
 - b) multifocal unisystem: seen mostly in children. Fever, bone & skin lesions
 - c) multifocal multisystem: nee Letterer-Siwe disease (a fulminant, malignant lymphoma of infancy).³ Hand-Schüller-Christian triad: DI (from invasion of pituitary stalk), exophthalmos (from intraorbital tumor) and lytic bone lesions (particularly of cranium).

This section deals with unifocal Langerhans cell histiocytosis, formerly AKA eosinophilic granuloma.

Clinical

Generally a condition of youth, 70% of patients are <20 yrs age. In a series of 26 patients,³ age range was 18 mos-49 yrs (mean: 16 yrs).

Most common presenting symptom: tender, enlarging skull mass (>90%). May be asymptomatic and incidentally discovered on skull x-ray obtained for other reasons. Blood tests were normal in all except 1 who had eosinophilia of 23%

Parietal bone was the most common site (42%), frontal bone next (31%)³ (some series show frontal bone was the most common).

Evaluation

Skull x-rays

Classic radiographic finding: round or oval non-sclerotic punched out skull lesion with sharply defined margins, involving both inner and outer tables (the disease begins in diploic space), often with beveled edges. A central bone density is occasionally noted (rare, but diagnostic). No abnormal vascularity of adjacent bone. No periosteal reaction. Differentiate from hemangioma by absence of sunburst appearance.

CTscan

Characteristic appearance of a soft tissue mass within area of bony destruction having a central density.⁴ Differentiate from epidermoid which has dense surrounding sclerosis.

Pathology

Gross: pinkish gray to purple lesion extending out of bone and involving pericranium. Dural involvement occurs in only 1 of 26 patients, but with no dural penetration.

Microscopic: numerous histiocytes, eosinophils, and multinucleated cells in a reticulin fiber network. No evidence that this is a result of an infection.

Treatment

Tendency toward spontaneous regression, however, most single lesions are treated by curettage. Multiple lesions are usually associated with extracalvarial bony involvement and are often treated with chemotherapy and/or low dose radiation therapy. Very radiosensitive.

Outcome

After a mean 8 years follow-up, 8 patients (31%) developed additional lesions, 5 of these were ≤ 3 yrs age (all of 5 patients <3 yrs age)³ (may suggest a form of multifocal LCH, thus young patients should be followed closely). Recurrences were local in one case, and in others involved other bones (including the skull, femur, lumbar spine) or brain (including the hypothalamus, presenting with diabetes insipidus and growth delay).

50.1.6 Chordoma

General information

Key concepts

- primary malignant tumor, usually of clivus or sacrum, with high recurrence rate
- histology: characteristic physaliphorous cells (containing intracellular mucin)
- generally slow-growing and radioresistant
- treatment of choice: wide en bloc resection when possible (piecemeal removal carries risk of inducing metastases), proton-beam radiation may help

Rare tumors (incidence of ≈ 0.51 cases/million) of the remnant of the primitive notochord (which normally differentiates into the nucleus pulposus of the intervertebral disks). Can arise anywhere along the neuraxis where there is remnant of notochord, however, cases tend to cluster at the two ends of the primitive notochord: 35%cranially⁵ in the spheno-occipital region (clivus), and 53%⁵ in the spine at the sacrococcygeal region.⁶ Less commonly, they may occur in the spine above the sacrum.⁷ They represent less than 1%of intracranial tumors and 3%of primary spine tumors.⁸ The meta-static rate is low (5–20%),⁹ but there is a high recurrence rate of 85%following surgery, and therefore aggressive RTX is usually employed post-op.

Pathology

Histologically, these tumors are considered low-grade malignancies. However, their behavior is more malignant because of the difficulty of total removal, a high recurrence rate, and the fact that they can metastasize (usually late). They are slow growing, locally aggressive and osseodestructive. Metasta-ses occur in about 10%of sacral tumors, usually late and after multiple resections, and most often to lung, liver and bone. Malignant transformation into fibrosarcoma or malignant fibrous histiocytoma is rare. Physaliphorous cells are distinctive, vacuolated cells on histology that probably represent cytoplasmic mucus vacuoles seen ultrastructurally.

Radiographic appearance

Usually lytic with frequent calcifications.¹⁰ Enhances on CT with contrast.¹⁰ Rarely, may appear as a sclerotic vertebra¹¹ (“ivory vertebra”).

Cranial chordomas

Peak incidence of cranial chordomas is 50–60 years of age. These tumors are rare in patients <30 years of age.¹² Male:female distribution is $\approx$ equal.

Differential diagnosis: Primarily between other cartilaginous tumors of the skull base; see differ-ential diagnosis of other foramen magnum region tumors (p.1367):

1. chondrosarcomas
2. chondromas

Presentation: Usually produces cranial nerve palsies (usually oculomotor or abducens nerve).

Spinal chordomas

General information

Occur primarily in the sacrococcygeal region. Unlike cranial chordomas, sacrococcygeal chordomas show a male predominance,⁵ and these patients tend to be older. May also arise in C2. Chordomas constitute over 50%of primary bone tumors of the sacrum. May produce pain, sphincter disturbance or nerve root symptoms from local nerve root compression. It may occasionally extend cephalad into the lumbar spinal canal. It is usually confined anteriorly by the presacral fascia, and only rarely invades the wall of the rec-tum.¹³ A firm fixed mass may be palpable between the rectum and the sacrum on rectal exam.

Evaluation

Characteristic radiographic findings: centrally located destruction of several sacral segments, with an anterior soft-tissue mass that occasionally has small calcifications. CT and MRI show the bony destruction. This is usually difficult to see on plain x-rays. MRI also shows the soft-tissue mass.

Open or CT guided percutaneous posterior biopsy can confirm the diagnosis. Transrectal biopsy should be avoided because of the potential of rectal spread of tumor.¹⁴

Chest CT and bone scan: to R/O mets for staging purposes.

Treatment

Surgery

Wide en-bloc excision with postoperative radiation is usually the best option, although this may also be only temporarily effective. Decompression is best avoided since entering the mass serves to spread tumor (surgically induced metastases) which will then regrow. Chordomas located in C2 are usually not amenable to en bloc resection.¹⁵

Sacral chordomas: The particulars of the surgical procedure are highly dependent on the extent of the lesion. These tumors may spread through the gluteal musculature, and if significant muscular excision is required, then a pedicle based rectus abdominis flap may be employed. A diverting colostomy may be required if it is necessary to resect the rectum or if a cephalic sacral resection is anticipated.¹⁶

For chordomas caudal to the third sacral segment, most agree that a posterior approach is satisfactory. For more rostral lesions, some advocate a combined anterior-posterior approach. However, a posterior approach has been also been used for these.¹⁶

Adverse effects of sacrectomy: if S2 nerve roots are the most caudal nerve roots spared, there is $\approx 50\%$ chance of normal bladder and bowel control.¹⁶ If S1 or more cephalic roots are the most caudal nerve roots spared, most will have impaired bladder control and bowel problems.¹⁶

Radiation therapy (XRT)

Best results were obtained with en bloc excision (even if marginal), sometimes combined with high-dose XRT^{7,17} (conventional XRT did not prevent recurrence when incorporated with palliative or debulking surgery⁷, but it did lengthen the interval to recurrence¹⁷). Early radiation was associated with longer survival.¹⁸ Higher XRT doses can be used in the sacrococcygeal region (4500–8000 rads) than in the cervical spine (4500–5500 rads) because of concerns of radiation injury to the spinal cord. IMRT and stereotactic radiosurgery have also been used.¹⁵

Proton beam therapy, alone⁹ or combined with high-energy x-ray (photon) therapy^{19,20} may be more effective than conventional XRT alone. However, proton beam therapy requires travel to one of a very limited number of facilities with a cyclotron (in the U.S.: Boston, or Loma Linda, California) which may be difficult to arrange for what is typically ≈ 7 weeks of fractionated treatments.

Chemotherapy

Imatinib (Gleevec®) (a tyrosine kinase inhibitor) has some antitumor effect in chordoma.²¹

Outcome

Median survival is 6.3 years.¹⁵

50.2 Non-neoplastic skull lesions

50.2.1 General information

Includes:

1. osteopetrosis (p.1401)
2. Paget's disease of the skull
3. hyperostosis frontalis interna (see below)
4. fibrous dysplasia (p.780)

50.2.2 Hyperostosis frontalis interna

General information

See differential diagnosis (p.1379). Hyperostosis frontalis interna (HFI) is a benign irregular nodular thickening of the inner table of the frontal bone that is almost always bilateral. The midline is spared at the insertion of the falx. Unilateral cases have been reported,²² and in these cases one must R/O other etiologies such as meningioma, calcified epidural hematoma, osteoma, fibrous dysplasia, an epidural fibrous tumor,²³ or Paget's disease.

Epidemiology

The incidence of HFI in the general population is $\approx 1.4\text{--}5\%$ ²² HFI is more common in women (female: male ratio may be as high as 9:1) with an incidence of 15–72% in elderly women. A number of possible associated conditions have been described (most are unproven), the majority of which are metabolic, earning it the alias of metabolic craniopathy. Associated conditions include:

1. Morgagni's syndrome (AKA Morgagni-Stewart-Morel syndrome): headache, obesity, virilism and neuropsychiatric disorders (including mental retardation)
2. endocrinologic abnormalities
 - a) acromegaly (p. 725)²⁴ (elevated growth hormone levels)
 - b) hyperprolactinemia²⁴
3. metabolic abnormalities
 - a) hyperphosphatemia
 - b) obesity
4. diffuse idiopathic skeletal hyperostosis (DISH) (p. 1129)

Clinical

HFI may present without symptoms as an incidental finding on radiographic evaluation for other reasons. Many signs and symptoms have been attributed to HFI including: hypertension, seizures, headache, cranial nerve deficits, dementia, irritability, depression, hysteria, fatigability and mental dullness. The incidence of headache may be statistically higher in patients with HFI than in the general population.²⁵

Evaluation

Blood tests to R/O some of the above noted conditions may be indicated in appropriate cases: check growth hormone, prolactin, phosphate, alkaline phosphatase (to R/O Paget's disease).

Plain skull x-ray shows thickening of the frontal bone with characteristic sparing of the midline. Spread to parietal and occipital bone occasionally occurs.

CT demonstrates the lesion which usually causes 5–10 mm of bone thickening, but as much as 4 cm has been reported.

Bone scan: usually shows moderate uptake in HFI (generally not as intense as with bone mets). Also, indium-111 leukocyte scan (commonly used to detect occult infection) will show accumulation in HFI (a false positive).^{26,27}

Treatment

In spite of a large number of published descriptive works in the medical literature primarily in the early and mid-20th century, little has been written about treatment of cases where symptoms are suspected to be due to HFI. In one report, removal of the thickened bone was accomplished without evidence of dural adhesions, and with improvement in the presenting hysteria.²²

Surgical technique

One technique described consists of using the craniotome to excise the thickened portion of the bone (a plain skull x-ray may be used to make a template), and then the thickened bone is thinned down with a high-speed drill, and the bone flap is then replaced. Alternatively, a cranioplasty with methylmethacrylate or custom-made implant fabricated using CT data may be performed.

50.2.3 Fibrous dysplasia

General information

Usually a benign condition in which normal bone is replaced by fibrous connective tissue (malignant transformation occurs in $<1\%$). Does not appear to be heritable. Most lesions occur in the ribs or craniofacial bones, especially the maxilla.

Patterns of involvement

1. monostotic: most common
2. polyostotic: 25% with this form have $>50\%$ of the skeleton involved with associated fractures and skeletal deformities

3. as part of McCune-Albright syndrome (endocrine dysfunction, café au lait spots which tend to occur on one side of the midline and tend to be more jagged than those seen in neurofibromatosis (p.604), fibrous dysplasia, and precocious puberty primarily in females) and its variants

Clinical

Clinical manifestations of the fibrous dysplasia (FD) lesions include:

1. incidental finding (i.e. asymptomatic)
2. local pain
3. local swelling (rarely marked distortion resembling aneurysmal bone cyst may occur) or deformity
4. may predispose to pathologic fractures when they occur in long bones
5. cranial nerve involvement: including loss of hearing when the temporal bone is involved as a result of obliteration of the external auditory canal
6. seizures
7. serum alkaline phosphatase is elevated in about 33% calcium levels are normal
8. darkened hair pigmentation overlying skull lesions
9. spontaneous scalp hemorrhages
10. rarely associated with Cushing's syndrome, acromegaly

3 forms of the FD lesions:

1. cystic (the lesions are not actually cysts in the strict sense): widening of the diploë usually with thinning of the outer table and little involvement of the inner table. Typically occurs high in calvaria
2. sclerotic: usually involves skull base (especially sphenoid bone) and facial bones
3. mixed: appearance is similar to cystic type with patches of increased density within the lucent lesions

Ground glass appearance on x-rays is due to the thin spicules of woven bone.

Treatment

There is no cure for FD. Local procedures (mostly orthopedic) are used for deformities or bone pain that is refractory to other treatment. Neurosurgical involvement may be required for skull lesions producing refractory pain or neurologic symptoms. Calvarial lesions may be treated with curettage and cranioplasty. Calcitonin may be used for widespread lesions with bone pain and/or high serum alkaline phosphatase levels.

References

- [1] Thomas JE, Baker HL. Assessment of Roentgenographic Lucencies of the Skull: A Systematic Approach. *Neurology*. 1975; 25:99–106
- [2] Smirniotopoulos JG, Chiechi MV. Teratomas, dermoids, and epidermoids of the head and neck. *Radiographics*. 1995; 15:1437–1455
- [3] Rawlings CE, Wilkins RH. Solitary Eosinophilic Granuloma of the Skull. *Neurosurgery*. 1984; 15:155–161
- [4] Mitnick JS, Pinto RS. CT in the Diagnosis of Eosinophilic Granuloma. *J Comput Assist Tomogr*. 1980; 4:791–793
- [5] O'Neill P, Bell BA, Miller JD, Jacobson I, Guthrie W. Fifty Years of Experience with Chordomas in Southeast Scotland. *Neurosurgery*. 1985; 16:166–170
- [6] Heffelfinger MJ, Dahlin DC, MacCarty CS, et al. Chordomas and Cartilaginous Tumors at the Skull Base. *Cancer*. 1973; 32:410–420
- [7] Boriani S, Chevalley F, Weinstein JN, et al. Chordoma of the Spine Above the Sacrum. Treatment and Outcome in 21 Cases. *Spine*. 1996; 21:1569–1577
- [8] Wright D. Nasopharyngeal and Cervical Chordoma – Some Aspects of the Development and Treatment. *J Laryngol Otol*. 1967; 81:1335–1337
- [9] Hug EB, Loredó LN, Slater JD, et al. Proton Radiation Therapy for Chordomas and Chondrosarcomas of the Skull Base. *J Neurosurg*. 1999; 91:432–439
- [10] Meyer JE, Lepke RA, Lindfors KK, et al. Chordomas: Their CT Appearance in the Cervical, Thoracic and Lumbar Spine. *Radiology*. 1984; 153:693–696
- [11] Schwarz SS, Fisher WS, Pulliam MW, Weinstein ZR. Thoracic Chordoma in a Patient with Paraparesis and Ivory Vertebral Body. *Neurosurgery*. 1985; 16:100–102
- [12] Wold LE, Laws ER. Cranial Chordomas in Children and Young Adults. *J Neurosurg*. 1983; 59:1043–1047
- [13] Azzarelli A, Quagliuolo V, Cerasoli S, et al. Chordoma: Natural History and Treatment Results in 33 Cases. *J Surg Oncol*. 1988; 37:185–191
- [14] Mindell ER. Current Concepts Review. Chordoma. *J Bone Joint Surg*. 1981; 63A:501–505
- [15] Jiang L, Liu ZJ, Liu XG, Ma QJ, Wei F, Lv Y, Dang GT. Upper cervical spine chordoma of C2-C3. *Eur Spine J*. 2009; 18:293–298; discussion 298–300
- [16] Samson IR, Springfield DS, Suit HD, Mankin HJ. Operative Treatment of Sacrococcygeal Chordoma. A Review of Twenty-One Cases. *J Bone Joint Surg*. 1993; 75:1476–1484
- [17] Klekamp J, Samii M. Spinal Chordomas - Results of Treatment Over a 17-Year Period. *Acta Neurochir (Wien)*. 1996; 138:514–519
- [18] Cheng EY, Özerdemoglu RA, Transfeldt EE, Thompson RC. Lumbosacral Chordoma. Prognostic Factors and Treatment. *Spine*. 1999; 24:1639–1645

- [19] Suit HD, Goitein M, Munzenrider J, et al. Definitive Radiation Therapy for Chordoma and Chondrosarcoma of Base of Skull and Cervical Spine. *J Neurosurg*. 1982; 56:377–385
- [20] Rich TA, Schiller A, Mankin HJ. Clinical and Pathologic Review of 48 Cases of Chordoma. *Cancer*. 1985; 56:182–187
- [21] Magenau JM, Schuetze SM. New targets for therapy of sarcoma. *Curr Opin Oncol*. 2008; 20:400–406
- [22] Hasegawa T, Ito H, Yamamoto S, et al. Unilateral Hyperostosis Frontalis Interna: Case Report. *J Neurosurg*. 1983; 59:710–713
- [23] Willison CD, Schochet SS, Voelker JL. Cranial Epidural Fibrous Tumor Associated with Hyperostosis: A Case Report. *Surg Neurol*. 1993; 40:508–511
- [24] Fulton JD, Shand J, Ritchie D, McGhee J. Hyperostosis frontalis interna, acromegaly and hyperprolactinemia. *Postgrad Med J*. 1990; 66:16–19
- [25] Bavazzano A, Del Bianco PL, Del Bene E, Leoni V. A statistical evaluation of the relationships between headache and internal frontal hyperostosis. *Res Clin Stud Headache*. 1970; 3:191–197
- [26] Floyd JL, Jackson DE, Carretta R. Appearance of Hyperostosis Frontalis Interna on Indium-111 Leukocyte Scans: Potential Diagnostic Pitfall. *J Nucl Med*. 1986; 27:495–497
- [27] Oates E. Spectrum of Appearance of Hyperostosis Frontalis Interna on In-111 Leukocyte Scans. *Clin Nucl Med*. 1988; 13:922–923

51 Tumors of the Spine and Spinal Cord

51.1 General information

15% of primary CNS tumors are intraspinal (the intracranial:spinal ratio for astrocytomas is 10:1; for ependymomas it's 3–20:1).¹ There is disagreement over the prevalence, prognosis, and optimal treatment. Most primary CNS spinal tumors are benign (unlike the case with intracranial tumors). Most present by compression rather than invasion.²

51.2 Compartmental locations of spinal tumors

May be classified into 3 groups based on the compartment involved. Although metastases may be found in each area, they are most commonly extradural. Frequencies quoted below are from a general hospital, extradural lesions are less common in neurosurgical clinics because many of these tumors are managed by oncologists without requiring neurosurgical involvement.

1. extradural (ED) (55%): arise outside cord in vertebral bodies or epidural tissues
2. intradural extramedullary (ID-EM) (40%): arise in leptomeninges or roots. Primarily meningiomas and neurofibromas (together = 55% of ID-EM tumors)
3. intramedullary spinal cord tumors (IMSCT) (p.787), 5% arise in SC substance. Invade and destroy tracts and grey matter

51.2.1 Spinal lymphoma

Lymphoma may occur in any of or all 3 compartments.

1. epidural
 - a) metastatic or secondary lymphoma: the most common form of spinal lymphoma. Spinal involvement occurs in 0.1–10% of patients with non-Hodgkin's lymphoma
 - b) primary spinal epidural non-Hodgkin's lymphoma: rare. Completely epidural with no bony involvement. The existence of this entity is controversial, and some investigators feel that it represents extension of undetected retroperitoneal or vertebral body lymphoma. May have a better prognosis than secondary lymphoma³
2. intramedullary
 - a) secondary (p.789)
 - b) primary: very rare (see below)

51.3 Differential diagnosis: spine and spinal cord tumors

51.3.1 General information

See also Myelopathy (p.1407) for a list including not only tumors but also nonneoplastic causes of spinal cord dysfunction (e.g. spinal meningeal cyst, epidural hematoma, transverse myelitis...).

51.3.2 Extradural spinal cord tumors (55%)

Arise in vertebral bodies or epidural tissues

1. metastatic: comprise the majority of ED tumors
 - a) most are osteolytic (cause bony destruction): see Spinal epidural metastases (p.814). Common ones include:
 - lymphoma: most cases represent spread of systemic disease (secondary lymphoma); some cases may be primary (see below)
 - lung
 - breast
 - prostate
 - b) metastases that may be osteoblastic:
 - in men: prostate Ca is the most common
 - in women: breast Ca is the most common
2. primary spinal tumors (very rare)
 - a) chordomas (p.778)
 - b) osteoid osteoma (p.792)
 - c) osteblastoma (p.792)

- d) aneurysmal bone cyst (ABC): an expansile tumor-like osteolytic lesion consisting of a highly vascular honeycomb of blood-filled cavities separated by connective tissue septa, surrounded by a thin cortical bone shell which may expand. Comprise 15% of spine tumors.⁴ Etiology is controversial. May arise from preexisting tumor (including: osteoblastoma, giant cell tumor, fibrous dysplasia, chondrosarcoma) or following acute fracture. In spine, there is a tendency to involve primarily the posterior elements. Peak incidence is in 2nd decade of life. Treatment usually consists of intralesional curettage. High recurrence rate (25–50%) if not completely excised
- e) chondrosarcoma: a malignant tumor of cartilage. Lobulated tumors with calcified areas
- f) osteochondroma (chondroma): benign tumors of bone that arise from mature hyaline cartilage. Most common during adolescence. An enchondroma is a similar tumor arising within the medullary cavity
- g) vertebral hemangioma (p. 794)
- h) giant cell tumors (GCT) of bone: AKA osteoclastoma (p. 797)
- i) giant cell (reparative) granuloma: AKA solid variant of ABC.⁵ Related to GCT. Occurs primarily in mandible, maxilla, hands and feet, but there are case reports of spine involvement.^{5,6} Not a true neoplasm – more of a reactive process. Treatment: curettage. Recurrence rate: 22–50% treated with re-excision
- j) brown tumor of hyperparathyroidism
- k) osteogenic sarcoma: rare in spine
- 3. miscellaneous
 - a) plasmacytoma (p. 716)
 - b) multiple myeloma (p. 714)
 - c) unifocal Langerhans cell histiocytosis (LHC), nee eosinophilic granuloma: osteolytic defect with progressive vertebral collapse; LHC is one cause of vertebra plana (p. 1392). C-spine is the most commonly affected region. Individual LHCs associated with systemic conditions (Letterer-Siwe or Hand-Schüller-Christian disease) are treated with biopsy and immobilization. Collapse or neurologic deficit from compression may require decompression and/or fusion. Low-dose RTX may also be effective^{7,8}
 - d) Ewing's sarcoma: aggressive malignant tumor with a peak incidence during 2nd decade of life. Spine mets are more common than primary spine lesions. Treatment is mostly palliative: radical excision followed by RTX (very radiosensitive) and chemotherapy⁹
 - e) chloroma: focal infiltration of leukemic cells
 - f) angiolipoma: ≈ 60 cases reported in literature
 - g) neurofibromas (p. 786): most are intradural, but some are extradural, usually dilate neural foramen (dumbbell tumors)
 - h) Masson's vegetant intravascular hemangioendothelioma (p. 770)¹⁰

51.3.3 Intradural extramedullary spinal cord tumors (40%)

1. meningiomas: usually intradural, but may be partly or, in 15% wholly extradural see below
2. neurofibromas: usually intradural, but may be partly or wholly extradural
3. many lipomas are extramedullary with intramedullary extension
4. miscellaneous: only ≈ 4% of spinal metastases involve this compartment

51.3.4 Intramedullary spinal cord tumors (5%)

1. astrocytoma (p. 789): 30%
2. ependymoma (p. 788): 30% including myxopapillary ependymoma (p. 789)
3. miscellaneous: 30% includes:
 - a) malignant glioblastoma
 - b) dermoid. In addition to the general population, dermoids present in a delayed fashion following ≈ 16% of myelomeningocele (MM) closures.¹¹ An iatrogenic etiology has been debated,¹² however a case of a congenital dermoid in a newborn with MM¹³ indicates that the origin is not always from incompletely excised dermal elements at the time of MM closure
 - c) epidermoid
 - d) teratoma
 - e) lipoma
 - f) hemangioblastoma (p. 789)
 - g) neuroma (very rare intramedullary)
 - h) syringomyelia (not neoplastic)
4. extremely rare tumors
 - a) lymphoma

- b) oligodendroglioma
- c) cholesteatoma
- d) intramedullary metastases: comprises only $\approx 2\%$ of spinal mets
- e) solitary fibrous tumors of the spinal cord: recognized in 1996. Probable mesenchymal origin. May also occur extramedullary (less common). Treatment is complete surgical excision. Prognosis is unclear.¹⁴

51.4 Intradural extramedullary spinal cord tumors

51.4.1 Spinal meningiomas

See reference.¹⁵

Epidemiology

Peak age: 40–70 years. Female:male ratio = 4:1 overall, but the ratio is 1:1 in the lumbar region. 82% thoracic, 15% cervical, 2% lumbar. 90% are completely intradural, 5% are extradural, and 5% both intra- and extra-dural. 68% are lateral to the spinal cord, 18% posterior, 15% anterior. Multiple spinal meningiomas occur rarely.

Clinical

Symptoms

Signs prior to surgery (only 1 of 174 patients was intact)¹⁵:

- 1. motor
 - a) pyramidal signs only: 26%
 - b) walks with aid: 41%
 - c) antigravity strength: 17%
 - d) flexion-extension with gravity removed: 6%
 - e) paralysis: 9%
- 2. sensory
 - a) radicular: 7%
 - b) long tract: 90%
- 3. sphincter deficit: 51%

Outcome

Recurrence rate with complete excision is 7% with a minimum of 6 years follow-up (relapses occurred from 4 to 17 years post-op).¹⁵

51.4.2 Spinal schwannomas

General information

Key concepts

- slow growing benign tumors
- most (75%) arise from the dorsal (sensory) rootlets
- early symptoms are often radicular
- recurrence is rare after total excision (except in neurofibromatosis)

Table 51.1 Symptoms of spinal meningiomas		
	At onset	At time of first surgery
local or radicular pain	42%	53%
motor deficits	33%	92%
sensory symptoms	25%	61%
sphincter disturbance		50%

Incidence: 0.3–0.4/100,000/yr. Most occur sporadically and are solitary, but they may also be associated with neurofibromatosis (p.603) primarily type 2 (NF2), but can occur with type 1.

Configurations

Most are entirely intradural, but 8–32% may be completely extradural,^{16,17} 1–19% are a combination, 6–23% are dumbbell, and 1% are intramedullary.

Dumbbell tumors. Definition: tumors that develop an “hourglass” shape as a result of an anatomic barrier encountered during growth. Not all dumbbell tumors are Schwannomas, e.g. neuroblastoma (p.603). Most have a contiguous intraspinal, foraminal (usually narrower) and extraforaminal components (widening of the neural foramen is a characteristic finding, can be recognized even on plain films, and speaks to the longstanding benign nature of the lesion). The waist may also be due to a dural constriction.

Asazuma et al.¹⁸ classification system for dumbbell spinal Schwannomas is shown in □ Fig. 51.1.

Type I tumors are intradural and extradural and are restricted to the spinal canal. The constriction occurs at the dura.

Type II are all extradural, and are subclassified as: IIa do not extend beyond the neural foramen, IIb = inside spinal canal + paravertebral, IIc = foraminal + paravertebral.

Type IIIa are intradural and extradural foraminal, IIIb are intradural and extradural paravertebral.

Type IV are extradural and intravertebral. Type V are extradural and extralaminar with laminar invasion. Type VI show multidirectional bone erosion.

Craniocaudal spread: IF & TF designate the number of intervertebral foramina and transverse foramina involved, respectively (e.g. IF stage 2 = 2 foramina).

Schwannomas involving C1 & C2: May involve vertebral arteries and require additional caution.

Clinical

Patients typically present with local pain.

Neurologic deficits develop late. Tumors may cause radiculopathy (from nerve root compression), myelopathy (from spinal cord compression), radiculomyelopathy (from compression of both), or cauda equina syndrome (for tumors below conus medullaris).

Pathology

Composed of Antoni A (compact, interwoven bundles of long, spindly Schwann cells) and Antoni B tissue (sparse areas of Schwann cells in a loose eosinophilic matrix).

Surgical approaches

See reference.¹⁹

Posterior approaches: Types I, IIa IIIa, some upper cervical IIIb and some VI are generally amenable to a posterior approach. IIa & IIIa usually require total facetectomy for complete removal.¹⁸ Reconstruction with instrumentation may be needed if substantial posterior disruption occurs.

Anterior and combined anterior/posterior approaches: Asazuma et al.¹⁸ recommend a combined approach for Type IIb, IIc and IIIb lesions where the extraforaminal extension is large (viz. beyond the vertebral arteries). Reconstruction with instrumentation was required for some tumors (≈ 10% of all patients treated) which were type IV (2 patients), IIIb (1 pt) and VI (1 pt).

Nerve sacrifice

It is usually possible to preserve some fascicles of the nerve root, although sometimes section of the entire nerve root is required. New deficits may not occur since involved fascicles are often nonfunctional, and adjacent roots may compensate. The risk for motor deficit is higher for schwannomas than for neurofibromas, for cervical vs. lumbar tumors, and for cervical tumors with extradural extension.

Outcome

Recurrence is rare following gross total excision, except in the setting of NF2.

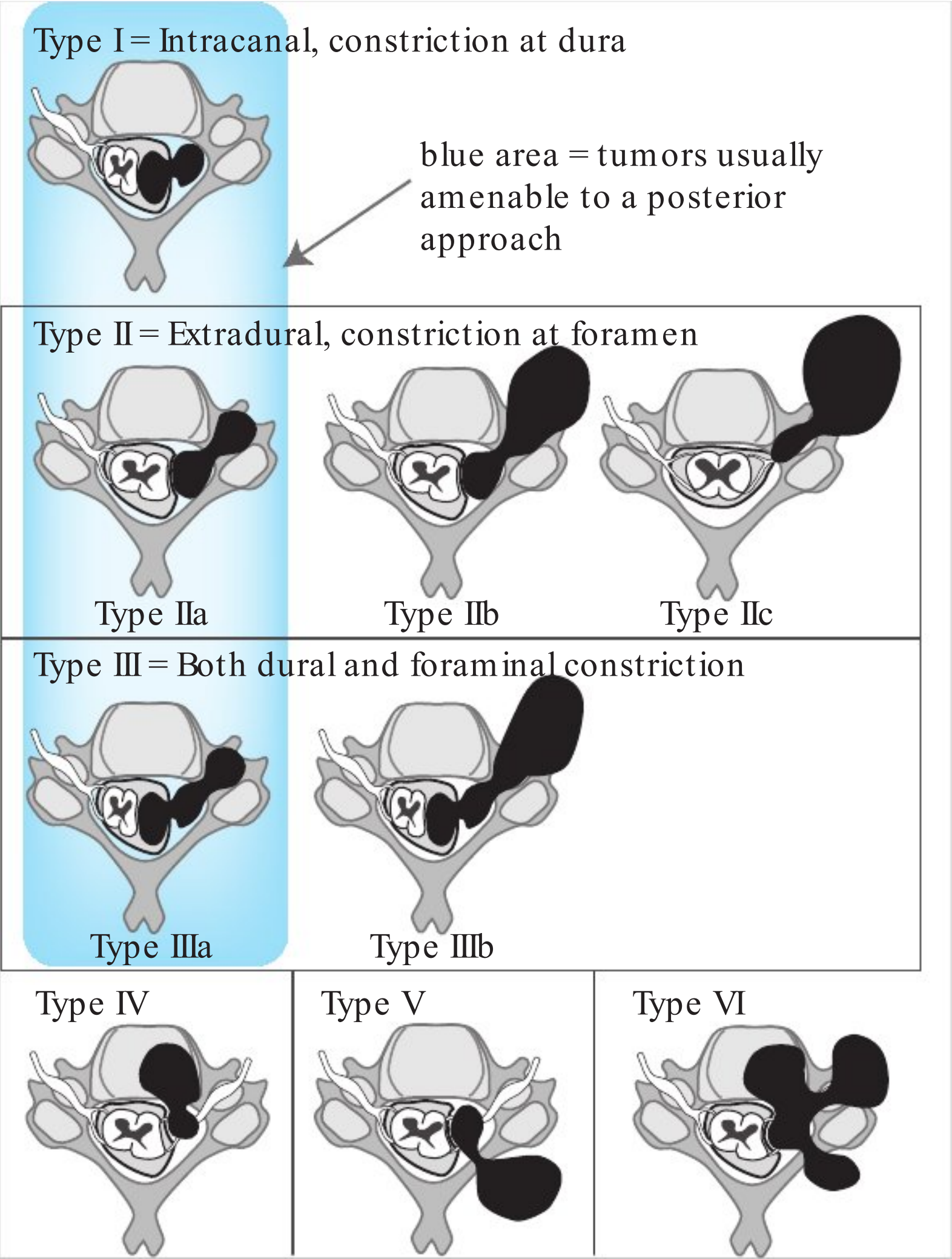


Fig. 51.1 Classification of dumbbell spinal tumors (Modified with permission from Asazuma T, Yoshiaki T, Hirofumi M, et al.: Surgical strategy for cervical dumbbell tumors based on a three-dimensional classification. Spine 29 (1): E10–4, 2003)

51.5 Intramedullary spinal cord tumors

51.5.1 Types of intramedullary spinal cord tumors

The following list excludes metastases (see below) and lipomas (of questionable neoplastic origin,²⁰ and most are actually extramedullary intradural, see below). Note: in pediatrics, astrocytoma and ependymoma constitute 90% of intramedullary spinal cord tumors (IMSCT).

- 1. astrocytoma (nonmalignant): 30%(the most common IMSC Toutside the filum terminale²) tend to be eccentric
- 2. ependymoma: 30% tend to be more central, more uniform dense enhancement
- 3. miscellaneous: 30% including:
 - a) malignant glioblastoma
 - b) dermoid
 - c) epidermoid (including iatrogenic from LP without stylet)^{21,22}
 - d) teratoma
 - e) hemangioblastoma (see below)
 - f) hemangioma
 - g) neuroma (very rarely intramedullary)
 - h) extremely rare tumors
 - primary lymphoma (only 6 case reports, all non-Hodgkin type²³)
 - oligodendroglioma, only 38 cases in world literature²⁴
 - cholesteatoma
 - paraganglioma
 - primary spinal embryonal tumor (“spinal PNET”) (p.663) ²⁵
 - pilomyxoid astrocytoma (p.632)
 - metastasis

51.5.2 Differential diagnosis

Also see DDX for Myelopathy (p. 1407).

- 1. neoplasm (tumor): (see above for list). Enhancement: 91%enhance²⁶; of the 9%that do not, most were astrocytomas, 1 was a subependymoma; enhancement did not correlate with grade
- 2. nonneoplastic lesions
 - a) vascular lesions (e.g. AVM): serpiginous linear flow-void. Spinal angiography may be useful²
 - b) demyelinating disease (e.g. multiple sclerosis):
 - usually does not extend >2 vertebral levels
 - cord lesions in MS are most common in the cervical region
 - c) inflammatory myelitis
 - d) paraneoplastic myelopathy
 - e) diseases causing pain over certain body segments (e.g. cholecystitis, pyelonephritis, intestinal pathology). To differentiate from these, look for dermatomal distribution, increase with Valsalva maneuver, and accompanying sensory and/or motor changes in LEs which suggest cord/radicular lesion. Radiographic studies are frequently required to differentiate
 - f) diseases of vertebral structures, e.g. Paget’s disease, giant cell tumors of bone (p.797), etc.

51.5.3 Specific types of intramedullary spinal cord tumors

Ependymoma

General information

Key concepts

- the most common glioma of lower cord, conus and filum (most ependymomas in conus and filum are myxopapillary ependymomas). More common in adults
- evaluation: includes imaging the entire neuraxis (usually with enhanced MRI: cervical, thoracic, lumbar & brain) because of potential for seeding through CSF
- associated cysts are common
- treatment: surgical excision (most are encapsulated)

The most common glioma of the lower spinal cord, conus and filum (below). Slow-growing. Benign. Slight male predominance; slight peak in 3rd to 6th decade. Over 50%in filum, next most common location is cervical. Histologically: papillary, cellular, epithelial, or mixed (in filum, myxopapillary ependymoma is most common, see below). Cystic degeneration in 46% May expand spinal canal in filum.²⁷ Usually encapsulated and minimally vascular (papillary: may be highly vascular; may cause SAH). Symptoms present > 1 yr prior to diagnosis in 82%of cases.²⁸

Myxopapillary ependymoma

Ependymomas of the conus medullaris and the filum terminale are usually of the myxopapillary subtype. WHO grade I. Usually solitary. Histology: papillary, with microcystic vacuoles, mucosubstance; connective tissue. No anaplasia, but CSF dissemination occurs rarely (can seed intracranially following removal of spinal tumor²⁹). Denovo intracranial lesions also occur rarely. Rare reports of systemic mets.¹ Outside the CNS, may occur in sacrococcygeal subcutaneous tissues from heterotopic rests of ependymal cells.³⁰

Surgical removal of filum tumors consists of coagulating and dividing the filum terminale just above and below the lesion – see Distinguishing features of the filum terminale intraoperatively (p.273) - and excising it in total. The filum is first cut above the lesion to prevent retraction upwards.

Astrocytoma

Uncommon in first year. Peak: 3rd – 5th decade. Male:female=1.5:1. The ratio of low-grade:high-grade=3:1 in all ages.²⁷ Occurs at all levels, thoracic most common, then cervical. 38% are cystic; cyst fluid usually has high protein.

Dermoid and epidermoid

Epidermoids are rare before late childhood. Slight female predominance. Cervical and upper thoracic rare; conus common. Usually ID-EM, but conus/cauda equina may have IM component (completely IM lesions rare).

Lipoma

May occur in conjunction with spinal dysraphism, see Lipomyeloschisis (p.269). The following considers lipomas that occur in the absence of spinal dysraphism.

Peak occurrence: 2nd, 3rd and 5th decade. Technically hamartomas. No sex predominance. Usually ID-EM (a sub-type is truly IM and essentially replaces the cord³¹), cervicothoracic region is the most common location. NB: unlike other IMSCT's, most common symptom is ascending mono- or para-paresis (c.f. pain). Sphincter disturbance is common with low lesions. Local subcutaneous masses or dimples are frequent. Malis recommends early subtotal removal at about 1 year age in asymptomatic patient.³¹ Superficial extrasacral removal is inadequate, as patients then develop dense scarring intraspinally leading to fairly rapid severe neurological damage with poor salvageability even after the definitive procedure.

Hemangioblastoma

Usually non-infiltrating, well demarcated, may have cystic caps. 33% of patients with spinal hemangioblastoma will have von Hippel-Lindau disease (p.703). Cannot incise nor core because of vascularity. Requires microsurgical approach similar to AVM, possibly with intraoperative hypotension.

Metastases

Most spinal mets are extradural, Intramedullary metastases are rare,³² accounting for 3.4% of symptomatic metastatic spinal cord lesions.³³ Primaries include: small-cell lung Ca,³⁴ breast Ca, malignant melanoma, lymphoma and colon Ca.^{33,35} Ca rarely presents first as an intramedullary spinal met.

51.5.4 Presentation

1. pain: the most common complaint. Almost always present in filum tumors (exception: lipomas).²¹ Possible pain patterns:
 - a) radicular: increases with Valsalva maneuver and spine movement. Suspect SCT if dermatome is unusual for disk herniation
 - b) local: stiff neck or back, Valsalva maneuver increases pain.
 - Pain during recumbency (“nocturnal pain”) is classic for SCT
 - c) medullary (as in syrinx): oppressive, burning, dysesthetic, non-radicular, often bilateral, unaffected by Valsalva maneuver
2. motor disturbances
 - a) weakness is 2nd or 3rd most common complaint. Usually follows sensory symptoms temporally
 - b) children present most frequently with gait disturbances

- c) syringomyelic syndrome: suggests IMSCT. Findings: UE segmental weakness, decreased DTR, dissociative anesthesia (see below)
- d) long-tract involvement → clumsiness and ataxia (distinct from weakness)
- e) atrophy, muscle twitches, fasciculations
- 3. non-painful sensory disturbances
 - a) dissociated sensory loss: decreased pain and temperature, preserved light touch, as in Brown-Séquard syndrome (p.947). There is disagreement whether this is common² or uncommon³⁶ in IMSCT. ± non-radicular dysesthesias (early), with upward extension³⁷
 - b) paresthesias: either radicular or “medullary” distribution
- 4. sphincter disturbances
 - a) usually urogenital (anal less common) → difficulty evacuating, retention, incontinence, and impotence. Early in conus/cauda equina lesions, especially lipomas (pain not prominent)
 - b) sphincter dysfunction common in age < 1 yr due to frequency of lumbosacral lesions (dermoids, epidermoids, etc.)
- 5. miscellaneous symptoms:
 - a) scoliosis or torticollis
 - b) SAH
 - c) visible mass over spine

Time course of symptoms

Onset usually insidious, but abruptness occurs (benign lesions in children occasionally progress in hours). The onset is often erroneously attributed to coincidental injury. Temporal progression has been divided into 4 stages³⁸:

1. pain only (neuralgic)
2. Brown-Séquard syndrome
3. incomplete transectional dysfunction
4. complete transectional dysfunction

51

Note: 78%(of 23) ependymomas, 74%(of 42) gliomas, all 7 dermoids, and 50%(of 8) lipomas reached the latter 2 stages before diagnosis (not affected by location in cross-sectional nor longitudinal dimension of SC (excludes conus lesions – more frequently diagnosed in 1st stage) (a pre-CT study).

51.5.5 Diagnosis

It is usually difficult to distinguish IMSCT, ID-EM and ED on clinical grounds.² Schwannomas often start with radicular symptoms that later progress to cord involvement. Most IMSCTs are located posteriorly in cord which may cause sensory findings to predominate early.²⁰

Diagnostic studies

MRI: mainstay of diagnosis. Ependymomas enhance intensely and are often associated with hemorrhage and cysts. Cord edema may mimic a cyst.

Plain radiographs: vertebral body destruction, enlarged intervertebral foramina, or increases in interpedicular distances suggests ED SCT.

Lumbar puncture: Elevated protein is the most common abnormality¹ seen in ≈ 95% The reported range with primary IMSCT's is 50–2,240 mg% Glucose is normal except with meningeal tumor. SCT can cause complete block, indicated by:

- Froin's syndrome: clotting (due to fibrinogen) and xanthochromia of CSF
- Queckenstedt's test (failure of jugular vein compression to increase CSF pressure, which it normally does in the absence of block)
- barrier to flow of myelographic contrast media

Myelography (p.818): classically shows fusiform cord widening (may be normal early). Distinct from ED tumors which produce hourglass deformity (with incomplete block) or paintbrush effect (with complete block), or ID-EM tumors which produce a capping effect with a sharp cutoff (meniscus sign).

CT: some IMSCTs enhance with IV contrast. Myelo-CT distinguishes IMSCT from ID-EM (poor in differentiating IMSCT subtypes).

Spinal angiography: rarely indicated, except in hemangioblastoma (may be suspected on myelography or MRI by linear serpiginous structures). MRI often obviates this test.

51.5.6 Management

General information

Asymptomatic lesions may be followed since there is significant risk of neurologic deficit with surgery. For symptomatic lesions, surgery should be performed as soon as possible (generally not as an emergency) after diagnosis since surgical results correlate with the preoperative neurologic condition, and it makes no sense to follow the patient as they develop progressive neurologic deficit³⁹ (some of which may be irreversible).

Astrocytomas: For low grade lesions, if a plane can be developed between the tumor and spinal cord (when it can, it usually consists of a thin gliotic layer traversed by small blood vessels and adhesions²⁰), an attempt at total excision is an option.⁴⁰ For high grade astrocytomas or for low-grade astrocytomas without a plane of separation, biopsy alone or biopsy plus limited excision is recommended.⁴⁰

For high-grade lesions, post-op RTX (± chemotherapy) is recommended.⁴⁰ RTX is not supported following radical resection of low grade gliomas.⁴⁰

Ependymomas: An attempt at gross total removal should be attempted. XRT is not recommended following gross total removal.⁴⁰

51.5.7 Technical surgical considerations

1. position: usually prone, well padded and also securely taped to avoid undesirable movement if MEP monitoring is to be used. Other options include: lateral oblique, sitting
2. if a cystic component is suspected, partial aspiration with a 25 Ga needle once the spinal cord is exposed will decrease the pressure (avoid total aspiration which makes it more difficult to locate the tumor).⁴¹ If the cyst forms a “cap” at either end of the tumor, the dura does not need to be opened over the cyst as drainage can be accomplished with removal of tumor
3. adjunctive options include:
 - a) intraoperative spinal cord monitoring (SSEP, and motor evoked potentials (MEPs)⁴²): SEPs almost always degrade with the initial myelotomy and do not correlate well with motor outcome⁴³ (which is critical)^{44,45} (e.g. it is not unusual for SEPs to be lost during the initial myelotomy without correlation with outcome) and postoperative motor deficit may occur in spite of unaltered intraoperative SEPs^{42,43} and conversely SEPs may be lost without motor deficit. However, proof of improved outcomes with MEP monitoring is also lacking⁴⁴
 - b) intraoperative ultrasound: also controversial,⁴⁵ favored by some experts. Astrocytomas are usually iso-echoic with spinal cord, whereas ependymomas are usually hyperechoic
4. a myelotomy is performed either in the midline or just to one side of the dorsal midline to avoid the posteromedian vein. Alternatively, if the tumor is known to be very superficial off the midline (which may be confirmed by ultrasound), entry may be made there. Tumors may cause distortion and displacement of the midline – look for dorsal root entry zones on both sides to identify the midline as the midpoint between root entry zones
5. 6–0 silk sutures are placed through the pial edge to gently retract the spinal cord open. Standard sized (i.e. non-micro) bayonet forceps can be used to gently spread tissues
6. copious irrigation is used whenever bipolar cautery is employed on the tumor/spinal cord, to minimize transference of heat to the spinal cord. Monopolar cautery should not be used⁴¹
7. either laser or ultrasonic aspiration (USA) are used to debulk tumor from within until the glial-tumor interface is reached. Charring from laser may make it more difficult to recognize the glial/tumor interface than USA, and the laser tends to be slower when debulking larger tumors
8. watertight dural closure is critical

51.5.8 Prognosis

No well designed studies give long term functional results with microsurgery, laser and radiotherapy. Better results occur with lesser initial deficits.²⁰ Recurrence depends on totality of removal, and on growth pattern of the specific tumor.

Table 51.2 Key concepts in surgical removal of IMSCT
• in almost all cases, IMSCTs should be debulked from within using ultrasonic aspirator or laser (to avoid manipulation of neural tissue), and no attempt should be made initially to develop a plane between tumor and spinal cord (even for ependymomas, which of the 3 most common IMSCTs is the only one that actually has such a plane) • if MEPs are monitored: it is suggested that tumor removal should be discontinued if the amplitudes drop to ≤50% of baseline

Ependymoma: total extirpation improves functional outcome, and myxopapillary ependymomas fare better than the “classic” type.²⁸ Best functional outcome occurs with modest initial deficits, symptoms <2 years duration,⁴⁶ and total removal. Survival is independent of extent of excision.

Astrocytomas: radical removal rarely possible (cleavage plane unusual even with microscope). Long term functional results poorer than ependymomas. There is 50% recurrence rate in 4–5 yrs.

51.6 Primary bone tumors of the spine

51.6.1 General information

Types of tumors.

1. metastatic: the most common malignancy of spine
 - a) common osteolytic metastatic tumors (p.814) include:
 - lung
 - breast
 - prostate
 - lymphoma: most cases represent spread of systemic disease (secondary lymphoma), however some may be primary (p.783)
 - plasmacytoma (p.714)
 - multiple myeloma (p.714)
 - Langerhans cell histiocytosis: see differentiating features (p.784)
 - b) metastases that may be osteoblastic:
 - in men: prostate Ca is the most common
 - in women: breast Ca is the most common
 - c) Ewing’s sarcoma (p.784)
 - d) chloroma: focal infiltration of leukemic cells
2. primary spinal tumors (very rare)
 - a) benign
 - vertebral hemangioma (p.784)
 - osteoid osteoma (p.792)
 - osteblastoma (p.792)
 - aneurysmal bone cyst (p.784): cavity of highly vascular honeycomb surrounded by a thin cortical shell which may expand
 - osteochondroma (chondroma) (p.784)
 - giant cell tumors of bone (p.797): AKA osteoclastoma. Almost always benign with pseudo-malignant behavior
 - b) malignant
 - chondrosarcoma (p.784)
 - chordomas (p.778)
 - osteogenic sarcoma: rare in spine

51.6.2 Osteoid osteoma and osteblastoma

General information

Key concepts

- both are benign bone tumors
- histologically identical, differentiation depends on size (≤ 1 cm = osteoid osteoma, > 1 cm = osteblastoma)
- can occur in the spine and may cause neurologic symptoms (esp. osteblastoma)
- high cure rate with complete excision

Two types of benign osteoblastic lesions of bone: osteoid osteoma (OO) and benign osteblastoma (BOB), see □ Table 51.3. They are indistinguishable histologically, and must be differentiated based on size and behavior.

Characteristically cause night pain and pain relieved by aspirin (see Clinical below).

Osteblastoma is a rare, benign, locally recurrent tumor with a predilection for spine, that may rarely undergo sarcomatous change (to osteosarcoma,⁴⁸ only a handful of known cases of this). More vascular than OO.⁴⁹

Table 51.3 Comparison of osteoid osteoma and benign osteoblastoma ⁴⁷		
	Osteoid osteoma	Benign osteoblastoma
percent of primary bone tumors	3.2%	
percent of primary vertebral tumors	1.4%	
percent that occur in spine	10%	35%
size limitations	≤ 1 cm	> 1 cm
growth pattern	confined, self limiting	more extensive, may extend into spinal canal
potential for malignant change?	no	rare
location within spine (83 patients)		
• %in cervical spine	27%	25%
• %in thoracic spine		35%
• %in lumbar region	59%	35%
location within vertebra (81 patients)		
• lamina only	33%	16%
• pedicle only	15%	32%
• articular facet only	19%	0
• vertebral body (VB) only	7%	5%
• transverse process only	6%	8%
• spinous process	5%	5%
• > 1 element of neural arch	6%	19%
• combined posterior elements & VB	0	11%

Differential diagnosis

Lesions with similar symptoms and increased uptake on radionuclide bone scan:

1. benign osteoblastoma
2. osteoid osteoma: more pronounced sclerosis of adjacent bone than BOB
3. osteogenic sarcoma: rare in spine
4. aneurysmal bone cyst (p.784): typically trabeculae in central, lucent region
5. unilateral pedicle/laminar necrosis

Clinical

See □ Table 51.4 for signs and symptoms. Tenderness confined to vicinity of the lesion occurs in ≈ 60% 28%of patients with BOB presented with myelopathy. OO presented with neurologic deficit in only 22%

Evaluation

Bone scans are a very sensitive means for detecting these lesions. Once localized, CT or MRI may better define the lesion in that region.

Caution regarding needle biopsy: if the lesion turns out to be osteosarcoma, the contaminated needle tract can result in worse prognosis.

Osteoid osteoma

Radiolucent area with or without surrounding density, often isolated to pedicle or facet. May not show up on tomograms.

Table 51.4 Signs and symptoms in 82 patients ⁴⁷		
Finding	Osteoid osteoma	Benign osteoblastoma
pain on presentation	100%	100%
pain increased by motion	49%	74%
pain increased by Valsalva	17%	36%
nocturnal pain	46%	36%
pain relieved by aspirin	40%	25%
radicular pain	50%	44%
scoliosis	66%	36%
neurologic abnormalities	22%	54%
myelopathy	0	28%
weakness	12%	51%
atrophy	9%	15%

Osteoblastoma
Most are expansile, destructive lesions, with 17% having moderate sclerosis. 31% have areas of ↑ density, 20% surrounded by calcified shell. Often a contralateral spondylolysis.⁴⁸

Treatment

In order to obtain a cure, these lesions must be completely excised. The role of radiation therapy is poorly defined in these lesions, but is probably ineffective.⁴⁸

Osteoid osteoma
Cortical bone may be hardened and thickened, with granulomatous mass in underlying cavity.

Osteoblastoma
Hemorrhagic, friable, red to purple mass well circumscribed from adjacent bone. Complete excision → complete pain relief in 93% Curettage only → pain relief, with more likely recurrence. Recurrence rate with total excision is ≈ 10%

51.6.3 Osteosarcoma

The most common primary bone cancer. More common in children, usually occurring near the ends of long bones, but also in the mandible, pelvis, and rarely in the spine.⁵⁰ Spinal osteosarcoma usually occurs in the lumbosacral region in males in their 40 s, sometimes arising from areas of osteoblastoma or Paget’s disease. If a percutaneous biopsy reveals osteosarcoma, the contaminated needle tract can increase the difficulty of subsequent surgery. Poor prognosis, median survival=10 months.⁵⁰

51.6.4 Vertebral hemangioma

General information

Key concepts

- the most common primary spine tumor. Benign
- rarely symptomatic (<1.2%), symptoms more commonly from compression fracture, disc herniation, and rarely neural compression from bone expansion

- MRI: small lesions are hyperintense on T1 WI and T2 WI. Larger ones may be hypointense. CT or X-ray: striations (corduroy pattern) or “honeycomb” appearance. Bone scan: usually do not have increased uptake
- treatment: incidental lesions require no routine follow-up. Biopsy when mets are a strong consideration. Treatment options (when indicated): XRT, embolization, vertebroplasty (better than kyphoplasty), surgery

Vertebral hemangiomas (VH), AKA spinal hemangioma, cavernous hemangioma, or hemangiomatous angioma. Benign lesions of the spine. The most common primary tumor of the spine (10–12% of primary spinal bone tumors). Estimated incidence: 9–12%^{51,52} 70% are solitary, 30% are multiple (up to 5 levels may be involved, often noncontiguous). Lumbar and lower thoracic spine are the most common locations, cervical and sacral lesions are rare. Lesions involve only the vertebral bodies in $\approx$ 25%, posterior spinal arch in $\approx$ 25% and both areas in $\approx$ 50%. Occasional cases of purely extradural lesions have been described.⁵³ Intramedullary lesions are even less common.⁵⁴ Typically found in post-pubertal females.

Malignant degeneration does not occur. Mature thin-walled blood vessels of varying sizes replace normal marrow, producing hypertrophic sclerotic bony trabeculations oriented in a rostral-caudal direction in one of two forms: cavernous (venous) or capillary (difference in subtype carries no prognostic significance).

Presentation

1. incidental: most VH are asymptomatic, these require no follow-up (see below)
2. symptomatic: only 0.9–1.2% are symptomatic. There may be a hormonal influence (unproven) that may cause symptoms to increase with pregnancy (could also be due to increased blood volume and/or venous pressure)⁵⁵ or to vary with the menstrual cycle and may explain why symptoms rarely occur before puberty
 - a) pain: occasionally VH may present with pain localized to the level of involvement with no radiculopathy. However, pain is more often due to other pathology (compression fracture, herniated disc, spinal stenosis...) rather than the VH itself
 - b) progressive neurologic deficit: this occurs rarely, and usually takes the form of thoracic myelopathy. Deficit may be caused by the following mechanisms
 - subperiosteal (epidural) growth of tumor into the spinal canal
 - expansion of the bone (cortical “blistering”) with widening of the pedicles and lamina producing a “bony” spinal stenosis
 - compression by vessels feeding or draining the lesion
 - compression fracture of the involved vertebra (very rare)⁵⁶
 - spontaneous hemorrhage producing spinal epidural hematoma⁵⁷ (also very rare)
 - spinal cord ischemia due to “steal”

Evaluation

Plain x-rays: classically show coarse vertically oriented striations (corduroy pattern) or a “honeycomb” appearance. At least $\approx$ one-third of the VB must be involved to produce these findings on plain x-ray (see □ Fig. 51.2 for demonstration of this finding on sagittal CT).

Bone scan: VH are usually not hot (unless a compression fracture has occurred), which may help distinguish VH from metastatic disease (which usually light up).

CT: diagnostic procedure of choice. “Polka-dot sign”⁵⁸: multiple high density dots within the bone represents cross-sections through thickened trabeculae (□ Fig. 51.3).

MRI: small hemangiomas are focal, round, and hyperintense on T1 WI and T2 WI. More extensive lesions can be hypointense. MRI may help distinguish lesions that tend not to evolve (mottled increased signal on T1 WI and T2 WI, possibly due to adipose tissue) from those that tend to be symptomatic (isointense on T1 WI, hyperintense on T2 WI).

Spinal angiography: also may help distinguish nonevolutive (normal or slight increased vascularity compared to adjacent bone) from symptomatic (moderate to marked hypervascularity) lesions. Therapeutic: if the feeding artery does not also supply the anterior spinal artery, it may be embolized preoperatively or sacrificed at surgery.

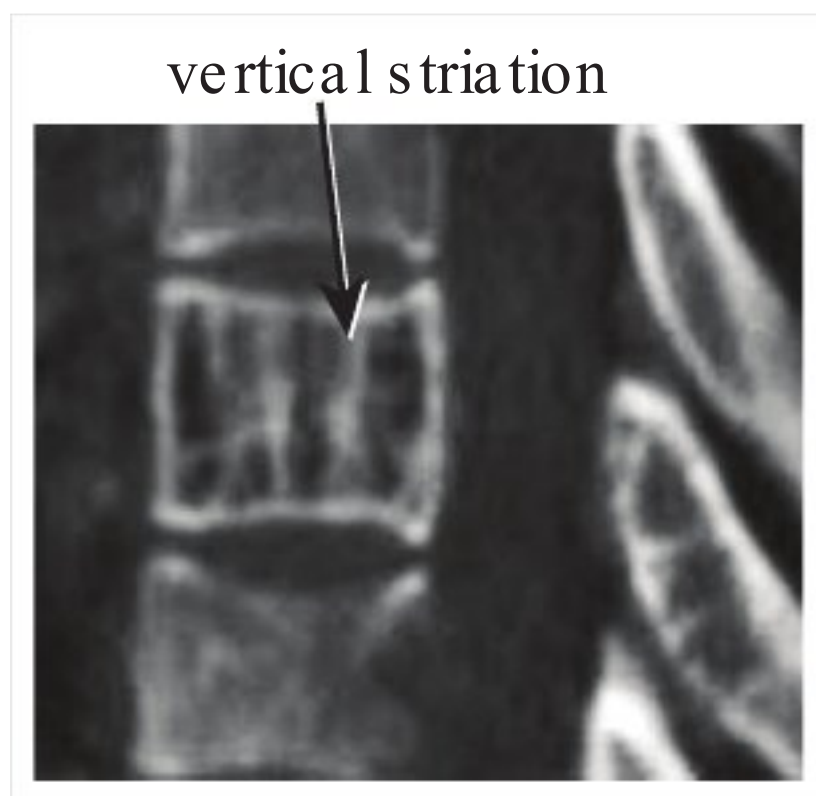


Fig. 51.2 Vertebral hemangioma. Vertical striations seen on sagittal CT reconstruction bone windows.

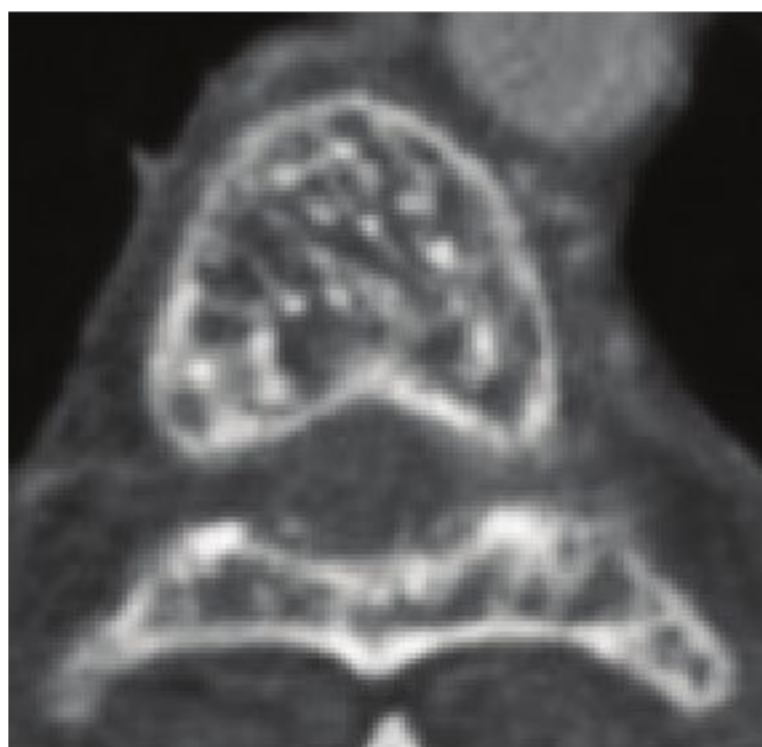


Fig. 51.3 Vertebral hemangioma. Axial CT bone windows demonstrating "polka-dot sign".

Treatment

See reference.⁵¹

Management guidelines:

1. asymptomatic VH require no routine follow-up or evaluation unless pain or neurologic deficit develop, which are rare occurrences in incidentally discovered VH
2. biopsy: may be indicated in cases where diagnosis is uncertain (e.g. when metastases are a strong consideration). In spite of highly vascular nature, there have been no reported bleeding complications with CT guided biopsy
3. those presenting with pain or neurologic deficit
 - a) radiation therapy: may be used alone for painful lesions, preoperatively as a surgical adjunct, or post-op following incomplete removal. VH are radiosensitive and undergo sclerotic obliteration. Total dosage should be ≤ 40 Gy to reduce risk of radiation myelopathy. Improvement in pain may take months to years, and no radiographic evidence of response may occur
 - b) embolization: provides more rapid relief of pain than RTX, can also be used pre-op as surgical adjunct. Risks spinal cord infarction if major radicular artery – e.g. artery of Adamkiewicz (p.87) – is embolized
 - c) vertebroplasty (p.1011): may be better than kyphoplasty for VH because kyphoplasty destroys the trabecular bone
 - d) surgery: for painful lesions that fail to respond to above measures, or for lesions with progressive neurologic deficit (see below)

Table 51.5 Recommendations for surgical management of VH ^a 51		
VH involvement	Approach	Post-op RTX?
posterior elements only	radical excision via posterior approach	not for total excision
VB involvement with anterior canal compression (with or without ST in canal)	anterior corpectomy with strut graft	
VB involved but no expansion, ST in lateral canal	laminectomy with removal of soft-tissue	follow serial CT, give RTX if VB expansion or ST expansion
extensive involvement of anterior and posterior vertebral elements with circumferential bone expansion, no ST compression	laminectomy	either RTX, or close follow-up with CT and RTX for ST recurrence or progressive VB expansion
extensive anterior and posterior involvement with ST in anterior canal	anterior corpectomy with strut graft	
^a abbreviations: VB=vertebral body, ST=soft-tissue component of VH, RTX=radiation treatment		

Surgical treatment

For surgical indications, see above. Recommendations for surgical management are shown in □ Table 51.5.

Major risks of surgery: blood loss, destabilization of the spine, neurologic deficit (during surgery, or postoperatively usually from epidural hematoma). Recurrence rate is 20–30%after subtotal resection, usually within 2 yrs. Patients with subtotal resection should have RTX which lowers recurrence rate to ≈ 7%

51.6.5 Giant cell tumors of bone

AKA osteoclastoma (cells arise from osteoclasts). In the same general category as aneurysmal bone cysts. Typically arise in adolescence. Most common in knees and wrists. Those that come to the attention of the neurosurgeon generally arise in the skull (especially the skull base, and in particular the sphenoid bone), or in the vertebral column (≈ 4%occur in sacrum).

Pathology

Lytic with bony collapse. Almost always benign with pseudomalignant behavior (recurrence is common, and pulmonary mets can occur).

Evaluation

Soft tissues are best evaluated with MRI. Spine CT is critical to assess degree of bony destruction and for surgical planning purposes.

Work-up includes chest CT because of possibility of pulmonary mets.

Treatment

Intratumoral curettage, possibly aided by pre-op embolization. Recurrence rate with this treatment (even if resection is subtotal) is only ≈ 20% Role of RTX is controversial⁷ because of the possibility of malignant degeneration (therefore use RTX only for non-resectable recurrence). Use of osteoclast inhibiting drugs – bisphosphonates, e.g. pamidronate (p.1122) -, has met with some success following subtotal resection.

For gross residual disease after resection, re-resection is a consideration.

Cryosurgery with liquid nitrogen has been employed in long bones. Its use is limited in neurosurgical cases because of risk of injury to adjacent neural structures (brain, spinal cord) and cryotherapy induced fractures, although it has been described for use in the sacrum.⁵⁹

Close follow-up is required due to propensity for recurrence. MRI or CT initially q 3 months is suggested.

References

- [1] Kopelson G, Linggood RM, Kleinman GM, et al. Management of Intramedullary Spinal Cord Tumors. *Radiology*. 1980; 135:473–479
- [2] Adams RD, Victor M. In: *Intraspinal Tumors. Principles of Neurology*. 2nd ed. New York: McGraw-Hill; 1981:638–641
- [3] Lyons MK, O'Neill BP, Kurtin PJ, Marsh WR. Diagnosis and Management of Primary Spinal Epidural Non-Hodgkin's Lymphoma. *Mayo Clin Proc*. 1996; 71:453–457
- [4] Liu JK, Brockmeyer DL, Dailey AT, Schmidt MH. Surgical management of aneurysmal bone cysts of the spine. *Neurosurg Focus*. 2003; 15
- [5] Suzuki M, Satoh T, Nishida J, Kato S, Toba T, Honda T, Masuda T. Solid variant of aneurysmal bone cyst of the cervical spine. *Spine*. 2004; 29:E376–E381
- [6] Neviasser JS, Eisenberg SH. Giant cell reparative granuloma of the cervical spine; case report. *Bull Hosp Joint Dis*. 1954; 15:73–78
- [7] Dunn EJ, Davidson RI, Desai S, The Cervical Spine Research Society Editorial Committee. In: *Diagnosis and Management of Tumors of the Cervical Spine. The Cervical Spine*. 2nd ed. Philadelphia: JB Lippincott; 1989:693–722
- [8] Menezes AH, Sato Y. Primary Tumors of the Spine in Children - Natural History and Management. *Concepts Pediatr Neurosurg*. 1990; 10:30–53
- [9] Grubb MR, Currier BL, Pritchard DJ, et al. Primary Ewing's Sarcoma of the Spine. *Spine*. 1994; 19:309–313
- [10] Porter DG, Martin AJ, Mallucci CL, et al. Spinal Cord Compression Due To Masson's Vegetant Intravascular Hemangioendothelioma: Case Report. *J Neurosurg*. 1995; 82:125–127
- [11] Scott RM, Wolpert SM, Bartoshesky LE, Zimble S, Klauber GT. Dermoid tumors occurring at the site of previous myelomeningocele repair. *J Neurosurg*. 1986; 65:779–783
- [12] Storrs BB. Are dermoid and epidermoid tumors preventable complications of myelomeningocele repair? *Pediatr Neurosurg*. 1994; 20:160–162
- [13] Ramos E, Marlin AE, Gaskill SJ. Congenital dermoid tumor in a child at initial myelomeningocele closure: an etiological discussion. *J Neurosurg Pediatrics*. 2008; 2:414–415
- [14] Metellus P, Bouvier C, Guyotat J, Fuentes S, Jouvet A, Vasiljevic A, Giorgi R, Dufour H, Grisoli F, Figarella-Branger D. Solitary fibrous tumors of the central nervous system: clinicopathological and therapeutic considerations of 18 cases. *Neurosurgery*. 2007; 60:715–22; discussion 722
- [15] Solero CL, Fornari M, Giombini S, Lasio G, Oliveri G, Cimino C, Pluchino F. Spinal meningiomas: review of 174 operated cases. *Neurosurgery*. 1989; 25:153–160
- [16] Seppala MT, Haltia MJ, Sankila RJ, Jaaskelainen JE, Heiskanen O. Long-term outcome after removal of spinal schwannoma: a clinicopathological study of 187 cases. *J Neurosurg*. 1995; 83:621–626
- [17] Conti P, Pansini G, Mouchaty H, Capuano C, Conti R. Spinal neurinomas: retrospective analysis and long-term outcome of 179 consecutively operated cases and review of the literature. *Surg Neurol*. 2004; 61:34–43; discussion 44
- [18] Asazuma T, Toyama Y, Maruiwa H, Fujimura Y, Hirabayashi K. Surgical strategy for cervical dumbbell tumors based on a three-dimensional classification. *Spine*. 2004; 29:E10–E14
- [19] Gottfried ON, Binning MJ, Schmidt MH. Surgical Approaches to Spinal Schwannomas. *Contemp Neurosurg*. 2005; 27:1–8
- [20] Stein B. Intramedullary Spinal Cord Tumors. *Clin Neurosurg*. 1983; 30:717–741
- [21] Stern WE. Localization and Diagnosis of Spinal Cord Tumors. *Clin Neurosurg*. 1977; 25:480–494
- [22] DeSousa AL, Kalsbeck JE, Mealey J, et al. Intraspinal Tumors in Children. A Review of 81 Cases. *J Neurosurg*. 1979; 51:437–445
- [23] Hautzer NW, Aiyesimoju A, Robitaille Y. Primary Spinal Intramedullary Lymphomas: A Review. *Ann Neurol*. 1983; 14:62–66
- [24] Alvisi C, Cerisoli M, Giuloni M. Intramedullary Spinal Gliomas: Long Term Results of Surgical Treatment. *Acta Neurochir*. 1984; 70:169–179
- [25] Kumar R, Reddy SJ, Wani AA, Pal L. Primary spinal primitive neuroectodermal tumor: case series and review of the literature. *Pediatr Neurosurg*. 2007; 43:1–6
- [26] White JB, Miller GM, Layton KF, Krauss WE. Nonenhancing tumors of the spinal cord. *J Neurosurg Spine*. 2007; 7:403–407
- [27] Dorwart RH, LaMasters DL, Watanabe TJ, Newton TH, Potts DG. In: *Tumors. Computed Tomography of the Spine and Spinal Cord*. San Anselmo: Clavadal Press; 1983:115–131
- [28] Mork SJ, Loken AC. Ependymoma: A Follow-Up Study of 101 Cases. *Cancer*. 1977; 40:907–915
- [29] Tzerakis N, Georgakoulis N, Kontogeorgos G, Mitsos A, Jenkins A, Orphanidis G. Intraparenchymal myxopapillary ependymoma: case report. *Neurosurgery*. 2004; 55
- [30] Helwig EB, Stern JB. Subcutaneous sacrococcygeal myxopapillary ependymoma. A clinicopathologic study of 32 cases. *Am J Clin Pathol*. 1984; 81:156–161
- [31] Malis LI. Intramedullary Spinal Cord Tumors. *Clin Neurosurg*. 1978; 25:512–539
- [32] Smaltino F, Bernini FP, Santoro S. Computerized Tomography in the Diagnosis of Intramedullary Metastases. *Acta Neurochir*. 1980; 52:299–303
- [33] Edelson RN, Deck MDF, Posner JB. Intramedullary Spinal Cord Metastases. *Neurology*. 1972; 22:1222–1231
- [34] Murphy KC, Feld R, Evans WK, et al. Intramedullary Spinal Cord Metastases from Small Cell Carcinoma of the Lung. *J Clin Onc*. 1983; 1:99–106
- [35] Jellinger K, Kothbauer P, Sunder-Plassmann, et al. Intramedullary Spinal Cord Metastases. *J Neurol*. 1979; 220:31–41
- [36] Stein B. Surgery of Intramedullary Spinal Cord Tumors. *Clin Neurosurg*. 1979; 26:473–479
- [37] Sebastian PR, Fisher M, Smith TW, et al. Intramedullary Spinal Cord Metastasis. *Surg Neurol*. 1981; 16:336–339
- [38] Nittner K, Olivecrona H, Tonniss W. *Handbuch der Neurochirurgie*. New York: Springer-Verlag; 1972:1–606
- [39] Post KD, Stein BM, Schmidek HH, Sweet WH. In: *Surgical Management of Spinal Cord Tumors and Arteriovenous Malformations. Operative Neurosurgical Techniques*. 3rd ed. Philadelphia: W.B. Saunders; 1995:2027–2048
- [40] Nadkarni TD, Rekate HL. Pediatric Intramedullary Spinal Cord Tumors: Critical Review of the Literature. *Childs Nerv Syst*. 1999; 15:17–28
- [41] Greenwood J. Surgical Removal of Intramedullary Tumors. *J Neurosurg*. 1967; 26:276–282
- [42] Morota N, Deletis V, Constantini S, et al. The Role of Motor Evoked Potentials During Surgery for Intramedullary Spinal Cord Tumors. *Neurosurgery*. 1997; 41:1327–1336
- [43] Kothbauer P, Deletis V, Epstein FJ. Intraoperative Spinal Cord Monitoring for Intramedullary Surgery: An Essential Adjunct. *Pediatric Neurosurgery*. 1997; 26:247–254
- [44] Albright AL. Intraoperative Spinal Cord Monitoring for Intramedullary Surgery: An Essential Adjunct? *Pediatric Neurosurgery*. 1998; 29
- [45] Albright AL. Pediatric Intramedullary Spinal Cord Tumors. *Childs Nerv Syst*. 1999; 15:436–437
- [46] Guidetti B, Mercuri S, Vagnozzi R. Long-Term Results of the Surgical Treatment of 129 Intramedullary Spinal Gliomas. *J Neurosurg*. 1981; 54:323–330
- [47] Janin Y, Epstein JA, Carras R, et al. Osteoid Osteomas and Osteoblastomas of the Spine. *Neurosurgery*. 1981; 8:31–38
- [48] Amacher AL, Eltomey A. Spinal Osteoblastoma in Children and Adolescents. *Childs Nerv Syst*. 1985; 1:29–32
- [49] Lichtenstein L, Sawyer WR. Benign Osteoblastoma. *J Bone Joint Surg*. 1964; 46A:755–765

- [50] Shives TC, Dahlin DC, Sim FH, Pritchard DJ, Earle JD. Osteosarcoma of the spine. *J Bone Joint Surg Am*. 1986; 68:660–668
- [51] Fox MW, Onofrio BM. The Natural History and Management of Symptomatic and Asymptomatic Vertebral Hemangiomas. *J Neurosurg*. 1993; 78:36–45
- [52] Healy M, Herz DA, Pearl L. Spinal Hemangiomas. *Neurosurgery*. 1983; 13:689–691
- [53] Richardson RR, Cerullo LJ. Spinal Epidural Cavernous Hemangioma. *Surg Neurol*. 1979; 12:266–268
- [54] Cosgrove GR, Bertrand G, Fontaine S, et al. Cavernous Angiomas of the Spinal Cord. *J Neurosurg*. 1988; 68:31–36
- [55] Tekkök IH, Açıkgöz B, Sağlam A, Önel B. Vertebral Hemangioma Symptomatic During Pregnancy - Report of a Case and Review of the Literature. *Neurosurgery*. 1993; 32:302–306
- [56] Graham JJ, Yang WC. Vertebral Hemangioma with Compression Fracture and Paraparesis Treated with Preoperative Embolization and Vertebral Resection. *Spine*. 1984; 9:97–101
- [57] Kosary IA, Braham J, Shacked I, Shacked R. Spinal Epidural Hematoma due to Hemangioma of Vertebra. *Surg Neurol*. 1977; 7:61–62
- [58] Persaud T. The polka-dot sign. *Radiology*. 2008; 246:980–981
- [59] Marcove RC, Sheth DS, Brien EW, Huvos AG, Healey JH. Conservative surgery for giant cell tumors of the sacrum. The role of cryosurgery as a supplement to curettage and partial excision. *Cancer*. 1994; 74:1253–1260

52 Cerebral Metastases

52.1 General information

Key concepts

- brain metastases are the most common brain tumor seen clinically
- at the time of onset of neurologic symptoms, 70% will be multiple on MRI
- with solitary brain lesions in a patient with a history of cancer, biopsy should almost always be done since 11% of these lesions will not be mets
- although median survival with maximal treatment is only 8 months (similar to GBM), long-term survivors do occur

52.2 Metastases to the brain

Cerebral metastases are the most common brain tumor seen clinically, comprising slightly more than half of brain tumors (if one considers only imaging studies, they comprise $\approx 30\%$). In the U.S., the annual incidence of new cases of metastases is up to 170,000,¹ compared to 17,000 for primary brain tumors. 15–30% of patients with cancer (Ca) develop cerebral mets.² In patients with no Ca history, a cerebral met was the presenting symptom in 15% of these, 43–60% will have an abnormal chest x-ray (CXR)^{3,4} (showing either a bronchogenic primary or other mets to lung).

In 9% of cases, a cerebral met is the only detectable site of spread. Cerebral mets occur in only 6% of pediatric cancers.

The route of metastatic spread to the brain is usually hematogenous, although local extension can also occur.

Solitary mets

- CT (p.806): at the time of neurologic diagnosis, 50% are solitary on CT^{5,6}
- MRI: if the same patients have an MRI, <30% will be solitary⁷
- on autopsy: mets are solitary in one-third of patients with brain mets, and 1–3% of solitary mets occur in the brain stem⁸

Increasing incidence of cerebral mets: May be due to a number of factors:

1. increasing length of survival of cancer patients⁹ as a result of improvements in treatment of systemic cancer
2. enhanced ability to diagnose CNS tumors due to availability of CT and/or MRI
3. many chemotherapeutic agents used systemically do not cross the blood-brain barrier (BBB) well, providing a “haven” for tumor growth there
4. some chemotherapeutic agents may transiently weaken the BBB and allow CNS seeding with tumor

52.3 Metastases of primary CNS tumors

52.3.1 Spread via CSF pathways

CNS tumors that more commonly spread via CSF pathways include the following (when these tumors spread to the spinal cord, they are often called “drop mets”):

1. high grade gliomas (p.619) (10–25%)
2. primitive neuroectodermal tumors (PNET), especially medulloblastoma (p.664)
3. ependymoma (p.642) (11%)
4. choroid plexus tumors (p.648)
5. pineal region tumors
 - a) germ cell tumors (p.659)
 - b) pineocytoma and pineoblastoma (p.659)
6. rarely:
 - a) oligodendrogliomas (p.638) ($\approx 1\%$)
 - b) hemangioblastomas (p.701)
 - c) primary CNS melanoma (p.701)

52.3.2 Extraneural spread

Although most CNS tumors do not spread systemically, there is some potential for extraneural spread with the following tumors:

- 1. medulloblastoma (cerebellar-PNET): the most common primary responsible for extraneural spread. May spread to lung, bone marrow, lymph nodes, abdomen
- 2. meningioma: rarely goes to heart or lungs
- 3. malignant astrocytomas rarely metastasize systemically
- 4. ependymomas
- 5. pineoblastomas
- 6. meningeal sarcomas
- 7. choroid plexus tumors
- 8. tumors that spread through CSF pathways (see above) may spread via a CSF shunt (e.g. to peritoneum with VP shunt or hematogenously with a VA shunt), however, this risk is probably quite small¹⁰

52.4 Location of cerebral mets

Intracranial metastases may be either parenchymal (≈ 75%) or may involve the leptomeninges in a carcinomatous meningitis (p.811). 80% of solitary metastases are located in the cerebral hemispheres.

The highest incidence of parenchymal mets is posterior to the Sylvian fissure near the junction of temporal, parietal, and occipital lobes (presumably due to embolic spread to terminal MCA branches).¹¹ Many tend to arise at the gray/white-matter interface.

The cerebellum is a common site of intracranial mets, and is the location in 16%of cases of solitary brain mets. It is the most common p-fossa tumor in adults, thus “a solitary lesion in the posterior fossa of an adult is considered a metastasis until proven otherwise.” Spread to the posterior fossa may be via the spinal epidural venous plexus (Batson’s plexus) and the vertebral veins.

52.5 Primary cancers in patients with cerebral metastases

52.5.1 General information

Accurately ascertaining the source of cerebral metastases in the U.S. is difficult because of lack of detailed coding.¹² In over 2,700 adults with a primary cancer undergoing autopsy at Sloan-Kettering, the sources of cerebral metastases are shown in □ Table 52.1. Sources of brain metastases in pediatrics is shown in □ Table 52.2.

In adults, lung and breast Ca together account for >50%of cerebral mets. In patients with a metastatic brain tumor as the initial presentation (i.e. undiagnosed primary) compared to patients with a known primary, there is about the same number of brain lesions, but there was an increased frequency of extracranial mets.¹⁴ In up to 26%of cases, the primary tumor was never identified.¹⁴

The autopsy incidence of cerebral mets for various types of primary cancers at Sloan-Kettering Cancer Center is shown in □ Table 52.3.

Table 52.1 Sources of cerebral mets in adults (autopsy data)	
Primary	%
lung Ca	44%
breast	10%
kidney (renal cell) ^a	7%
GI	6%
melanoma ^b	3%
undetermined	10%
^a a rare tumor that metastasizes frequently to brain (in 20–25%of cases)	
^b 16% in older series ¹³	

Table 52.2 Sources of cerebral mets in peds
neuroblastoma
rhabdomyosarcoma
Wilm’s tumor

Table 52.3 Autopsy incidence of cerebral mets for given primary cancers	
Primary	% with cerebral mets
lung	21%
breast	9%
melanoma	40%
lymphoma	1%
• Hodgkin’s	0
• non-Hodgkin’s	2%
GI	3%
• colon	5%
• gastric	0
• pancreatic	2%
GU	11%
• kidney (renal)	21%
• prostate ^a	0
• testes	46%
• cervix	5%
• ovary	5%
osteosarcoma	10%
neuroblastoma	5%
head and neck	6%
^a uncommon, but does occur	

52.5.2 Lung cancer

The lungs are the most common source of cerebral mets, and these are usually multiple. The lung primary may be so small as to render it occult.

Necropsy demonstrates cerebral mets in up to 50%of patients with small-cell lung Ca (SCLC) and non-squamous, non-small-cell lung Ca.¹⁵

Small-cell lung cancer (SCLC)

AKA “oat cell” Ca. A neuroendocrine tumor. 95%arise in proximal airways, usually in mainstem or lobar bronchi. Typically younger (27–66 years) than other lung Ca. Strongly associated with cigarette smoking. Median survival: 6–10 months. Considered a systemic disease. Staged in 1 of 2 categories:

1. limited: confined to an area of the chest that can be encompassed by a single radiation port
2. extensive: metastasis outside the thorax or intrathoracic disease that cannot be contained in a single radiation port

Although SCLC comprises only $\approx 20\%$ of primary lung cancers, it is more likely to produce cerebral mets than other bronchogenic cell types (brain mets are found in 80% of patients who survive 2 yrs after diagnosis of SCLC).⁹

Treatment

Very radiosensitive.

No identified brain mets: prophylactic cranial irradiation (PCI) with WBXRT reduces the incidence of symptomatic brain mets and increases survival (disease-free & overall).^{16,17} Typically 25 Gy in 10 fractions.

Brain mets: surgical resection considered for immediately life-threatening large lesions, XRT is used otherwise. Multiple SCLC brain lesions: XRT (initial treatment 30 Gy in 10 fractions) + chemotherapy.

Treatment of primary: usually not resected. Treated with chemotherapy $\pm$ XRT.

Recurrent brain mets after failure of initial treatment: 20 Gy in 10 fractions.

Non-small-cell lung cancer (NSCLC)

Includes: adenocarcinoma (the most common NSCLC), large cell, squamous cell, bronchoalveolar. Retrospective analysis of patients with NSCLC completely resected from lung found a 6.8% first recurrence rate in the brain.¹⁵ Staged with typical TNM system. Prognosis better than SCLC.

Treatment of lung primary:

1. grades I, II, IIIA: resection
2. higher grades (e.g. distal mets, excluding single brain met): XRT + chemotherapy

Staging studies for known lung primary

1. PET scan: can detect small malignancies. Useful in NSCLC to determine eligibility of resection of primary. Not useful in initial evaluation of SCLC
2. chest CT: usually includes adrenals and liver (thus abdomen and pelvis CT not necessary)
3. bone scan
4. brain: CT or MRI

When metastatic lung cancer is the suspected source of a newly diagnosed brain lesion, the lung lesion should be biopsied (if technically feasible) to rule out SCLC before obtaining tissue from the cerebral mass.

52

52.5.3 Melanoma

General information

Melanoma: the 5th most common cancer in men, 7th in women. Incidence is increasing. Most common sites of origin of melanoma metastases: skin, retina, brain – primary CNS melanoma (p.701) -, nail bed. The primary site cannot be identified in up to $\approx 14\%$ of cases.¹⁸ Extremely difficult to locate primary sites: intraocular, GI mucosa.

Brain mets are found in 10–70% of patients with metastatic melanoma in clinical studies, and in 70–90% on autopsy of patients who died from melanoma. Patients with melanoma who have neurosurgical lesions typically presented 14 months after primary lesion was identified. Once cerebral mets of melanoma are detected, median survival is ≤ 6 months^{19,20,21} and the mets contributed to the death in 94% of cases.²² A small group with survival > 3 yrs had a single surgically treated met in the absence of other visceral lesions.

Evaluation

Metastatic melanoma to the brain classically causes pia/arachnoid involvement on imaging. Hemorrhagic involvement is common.

CT: lesions may be slightly hyperdense to brain on unenhanced CT due to melanin. Enhancement is less constant than for other mets (e.g. bronchogenic Ca).

MRI: decreased signal on T2WI surrounded by intense halo of edema. Enhancing T1WI lesions in a patient with melanoma is highly suggestive of melanoma metastases.

Systemic work-up: systemic disease determines ultimate survival after treatment of melanoma mets to the brain in 70% of patients. $\square$ search for systemic mets should be done, including: CT of chest/abdomen/pelvis & bone scan. PET scan may be more sensitive for detecting metastatic spread

than CT when there are clinical signs that the tumor has spread²³; except for the brain, where brain MRI is more sensitive than CT or PET.

Treatment

□ Surgical indications

1. patients with 1-4 CNS metastases that can be completely resected when systemic disease is absent or slowly progressive: long-term survival is possible
2. patients with intracranial mets that cannot be completely removed or with uncontrolled systemic disease may be surgical candidates for the following:
 - a) for symptomatic relief: e.g. lesion causing painful pressure
 - b) life-threatening lesion: e.g. large p-fossa lesion with 4th ventricle compression
 - c) for hemorrhagic lesion causing symptoms by mass effect from the clot

□ Whole-brain radiation therapy (WBXRT). Melanoma is typically radioresistant. WBXRT provides 2–3 month survival benefit and may be considered for palliation in patients with multiple mets that preclude complete excision or SRS.

□ Stereotactic radiosurgery (SRS). Considered for ≤ 4 lesions all ≤ 3 cm diameter that are surgically inaccessible, with limited or quiescent systemic involvement. Relative contraindications: hemorrhagic lesions, lesions with significant mass effect surrounding edema.

□ Chemotherapy

1. alkylating agents:
 - a) dacarbazine formerly the gold-standard treatment for melanoma. About equally as effective as its newer orally administered analog temozolomide (Temodar®). Response rate: 10–20%
 - b) Fotemustine appeared promising in phase II trials but only 6% responded in phase III (vs. 0% for dacarbazine)²⁴
2. immunotherapy:
 - a) ipilimumab: monoclonal antibody against cytotoxic T lymphocyte antigen-4 (CTLA-4) antigen. More effective in patients who do not require corticosteroids
 - b) interleukin-2 (IL-2): has shown minimal activity in brain mets, and trials have usually excluded patients with untreated or uncontrolled brain mets due to risk of cerebral edema and hemorrhage from capillary leak^{25,26,27}
3. BRAF inhibitors (BRAFi): inhibits BRAF kinase (a protein that participates in regulation of cell division & differentiation) – useful in tumors with BRAF oncogene mutation (as opposed to BRAF wildtype) which is common in melanoma
 - a) dabrafenib: phase II trial (NCT01266967)²⁸
 - b) vemurafenib: promising results in heavily treated patients. Phase II trial (NCT01378975)²⁹
4. anti-PD-1 drug (monoclonal antibody to PD-1 programmed cell death receptor): pembrolizumab (Keytruda) approved for advanced or unresectable melanoma not responding to other drugs³⁰

Suggested algorithm for patients with metastatic melanoma to the brain (adapted,³¹ see □ Fig. 52.1). Patients with Karnofsky performance scale (KPS) score (p.1358) < 70 are likely to be poor surgical candidates.

Some key points:

1. patients with rapidly progressive systemic disease: treat the systemic disease first, before the brain mets
2. patients without systemic disease and 1-4 mets are candidates for surgery (based on the article by Bindal et al.³²) if they are all accessible and can all be removed. SRS is an alternative

Outcome

1. In a patient with a single brain met (any type) and good Karnofsky performance score (> 70) and no evidence of extracranial disease, surgery + XRT had a median survival of 40 weeks vs. 15 weeks for XRT alone.^{33,34}
2. For melanoma, retrospective studies have shown a benefit of treatment with either surgery or SRS only when all brain lesions are completely treated (selection bias possible in these studies)^{34,35,36,37}
3. Predictors of poor outcome in melanoma:
 - a) > 3 brain mets²⁰
 - b) development of brain mets after the diagnosis of extracranial disease²⁰
 - c) elevated lactate dehydrogenase $> 2 \times$ normal²¹

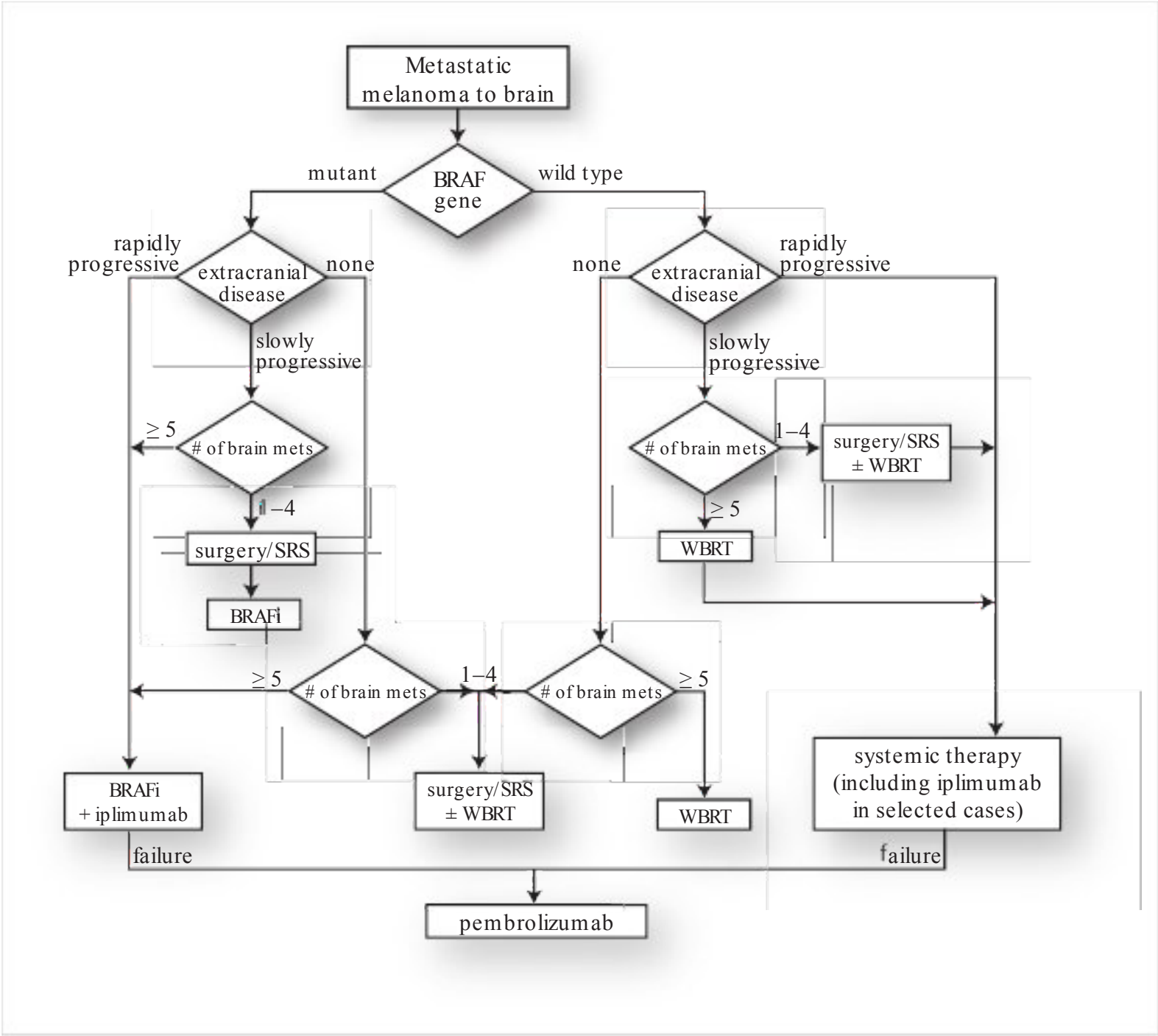


Fig. 52.1 Suggested algorithm for patients with metastatic melanoma to the brain (adapted³¹)

- d) presence of bone metastases²¹
- e) multiple brain mets and extensive visceral disease³⁸

52.5.4 Renal-cell carcinoma

AKA hypernephroma. Usually associated with spread to lungs, lymph nodes, liver, bone (high affinity for bone), adrenals, and contralateral kidney before invading the CNS (thus, this tumor rarely presents as isolated cerebral metastases). Look for hematuria, abdominal pain, and/or abdominal mass on palpation or CT. Response to XRT is only ≈ 10%

52.5.5 Esophageal cancer

Median survival is 4.2 months based on a review of 26 cases.³⁹ Solitary brain met with good Karnofsky score and surgical treatment may indicate a better prognosis.

52.6 Clinical presentation

As with most brain tumors, signs and symptoms are usually slowly progressive compared to those from vascular events (ischemic or hemorrhagic infarcts) which tend to be sudden in onset and slowly resolve, or electrical events (seizures) which tend to be sudden in onset and rapidly resolve. There are no findings that would allow differentiation of a metastatic tumor from a primary neoplasm on clinical grounds.

Signs and symptoms include:

1. those due to increased ICP from mass effect and/or blockage of CSF drainage (hydrocephalus):
 - a) headache (H/A): the most common presenting symptom, occurs in ≈ 50%

- b) nausea/vomiting
- 2. focal deficits:
 - a) due to compression of brain parenchyma by mass and/or peritumoral edema (e.g. monoparesis without sensory disturbance)
 - b) due to compression of cranial nerve
- 3. seizures: occur only in $\approx 15\%$ of cases
- 4. mental status changes: depression, lethargy, apathy, confusion
- 5. symptoms suggestive of a TIA (dubbed “tumor TIA”) or stroke, may be due to:
 - a) occlusion of a vessel by tumor cells
 - b) hemorrhage into the tumor, especially common with metastatic melanoma, choriocarcinoma, and renal-cell carcinoma⁴⁰; see Hemorrhagic brain tumors (p.1335). May also occur due to decreased platelet count

52.7 Evaluation

52.7.1 Imaging studies (CT or MRI)

Metastases usually appear as “non-complicated” masses (i.e. round, well circumscribed), often arising at the gray/white junction. Characteristically, profound white matter edema (“fingers of edema”) reach deep into brain from the tumor, usually more pronounced than that seen with primary (infiltrating) brain tumors. When multiple lesions are present (on CT or MRI of brains with multiple mets) Chamber’s rule applies: “Whoever counts the most mets is right.” Mets usually enhance, and must be considered in the differential diagnosis of a ring-enhancing lesion.

MRI is more sensitive than CT, especially in the posterior-fossa (including brain stem). Detects multiple mets in up to $\approx 20\%$ of patients who appear to have cases with single mets on CT.²

52.7.2 Lumbar puncture

Relatively contraindicated when there is a cerebral mass (may be indicated once mass lesion has been ruled out). May be most useful in diagnosing carcinomatous meningitis – see Carcinomatous meningitis (p.811) – and may be helpful in diagnosing lymphoma.

52.7.3 Metastatic work-up for patients with suspected brain metastases

When metastatic disease is suspected based on imaging or on surgical tissue, a search for a primary site and assessment for other lesions may be considered since it may provide alternative sites for tissue for histologic diagnosis, and it may guide treatment (e.g. widely disseminated metastases may preclude aggressive therapy). Metastatic work-up should include:

1. CT of the chest (more sensitive than CXR), abdomen and pelvis. Evaluates for primary and additional mets (to lung, adrenals, liver... CT has superseded CXR)
2. radionuclide bone scan: for patients with bone pain or bone lesions or for tumors that tend to produce osseous metastases (especially: prostate, breast, kidney, thyroid & lung)
3. mammogram in women
4. prostate specific antigen (PSA) in men
5. PET scan: can detect small malignancies

Cancer of unknown primary site (CUP) : If the metastatic work-up (see above) is negative, the pathology of a metastatic brain lesion as determined by biopsy may implicate specific primary sites.

Small-cell carcinoma metastatic to the brain is most likely from the lung. These tumors stain positive for neuroendocrine stains (p.600).

Adenocarcinoma: lung is the most common primary. Other sources: GI (mostly colon), breast. The primary site may remain occult even after extensive evaluation in up to 88%⁴¹ Immunostaining has been tried to identify the primary site but has not been found to be widely useful.

52.8 Management

52.8.1 General information

With optimal treatment, median survival of patients with cerebral mets is still only ≈ 26 –32 weeks, therefore management is mostly palliative. Also see Outcome (p.810) for comparison of various treatments.

52.8.2 Confirming the diagnosis

NB: 11% of patients with abnormalities on brain CT or MRI with a history of cancer (within past 5 yrs) do not have cerebral metastases.³³ Differential diagnoses include: primary brain tumor (glioblastoma, low grade astrocytoma), abscess, and nonspecific inflammatory reaction. If non-surgical treatment (e.g. chemotherapy or RTX) is being contemplated, the diagnosis should be confirmed by biopsy in almost all cases.

52.8.3 Management decisions

Prognostication

This is critical since many treatment decisions depend on overall prognosis.

RTOG RPA: Radiation Therapy Oncology Group recursive partitioning analysis classification⁴² (see □ Table 52.4; from 1200 patients with one or more brain metastases undergoing XRT). Conclusion: the specific tumor type, length of time since diagnosis, etc. are not as important prognostically as the Karnofsky Performance Scale (KPS) score (p.1358).

The applicability of RPA to melanoma mets to the brain is controversial (it has been both validated⁴³ and disputed³⁷).

RPA Class 3 patients have been shown to be unlikely to benefit from any of numerous treatment modalities studied. Class 1 are more likely to benefit. Most patients are Class 2, and benefit is unclear.

Management algorithm

□ Table 52.5 shows a summary of management suggestions (details appear in following sections).

Also, surgical excision may be considered for patients with completely resectable brain mets who are candidates for chemotherapy with interleukin-2 (IL-2) for systemic disease (e.g. for renal-cell Ca or melanoma) since this drug in some case reports produces significant cerebral edema if there are cerebral mets as well.

52.8.4 Medical management

Initial treatment

1. anticonvulsants: e.g. Keppra® (levetiracetam): start with 500 mg PO or IV q 12 hours. Generally not needed for posterior fossa lesions
2. corticosteroids: many symptoms are due to peritumoral edema (which is primarily vasogenic), and respond to steroids within 24–48 hrs. This improvement is not permanent, and prolonged steroid administration may produce side effects; see Possible deleterious side effects of steroids (p.594).
□ typical dose for a patient with significant symptoms who is not already on steroids: dexamethasone (Decadron®) 10–20 mg IV, followed by 6 mg IV q 6 hrs for 2–3 days, after which it is converted to ≈ 4 mg PO QID. Once symptoms are controlled, this is tapered to ≈ 2–4 mg PO TID as long as symptoms do not worsen
3. H2 antagonists (e.g. ranitidine 150 mg PO q 12 hrs) or proton pump inhibitor (omeprazole)

Table 52.4 RPA Classification for patients with brain mets		
RPA class	Description	Median survival (mos) ^a
1	• KPS^b ≥ 70 and • age <65 years and • controlled^c or absent primary tumor with the brain the only site of metastasis	7
2	• all others^d	4
3	• KPS <70	2
^a for patients undergoing XRT		
^b KPS=Karnofsky Performance Scale score (p.1358)		
^c controlled=stable disease over 3 months observation		
^d i.e. not RPA class 1 or 3		

Table 52.5 Management suggestions for cerebral metastases ^a		
Clinical situation		Management
unknown primary or unconfirmed diagnosis		stereotactic biopsy for $\approx$ all patients if surgical excision is not a consideration
uncontrolled widespread systemic cancer & obviously short life expectancy and/or poor performance status, Karnofsky (p.1358) ≤ 70		(biopsy as indicated above)+ WBXRT or no treatment
Stable systemic disease & KPS >70		
solitary met	symptomatic, large, or accessible lesion	surgical excision + WBXRT
	asymptomatic, small, or inaccessible lesion	WBXRT $\pm$ SRS boost
multiple mets	single large lesion that is life threatening or producing mass effect	surgery for the large lesion + WBXRT for the rest
	≤ 3 lesions: symptomatic & can all be removed	surgery+ WBXRT or SRS+ WBXRT
	≤ 3 lesions: cannot all be removed	WBXRT or SRS+ WBXRT
	>3 lesions: with no mass effect requiring surgery	WBXRT ⁴⁴
^a adapted. ⁴⁵ Abbreviations: WBXRT=whole brain radiation therapy, SRS=stereotactic radiosurgery		

Chemotherapy

See discussion of Limitations of chemotherapy in the brain (p.595). If multiple lesions of known small-cell Ca are detected on cerebral imaging, treatment of choice is radiation plus chemotherapy.

52.8.5 Radiation therapy

General information

NB: not all brain lesions in cancer patients are mets (see above).
In patients not considered for surgery, steroids and radiation may be palliative. H/A are usually reduced, and in $\approx 50\%$ of cases symptoms improve or completely resolve.⁴⁶ This does not result in local control for the majority of these patients and they frequently succumb from progressive brain disease.
“Radiosensitivity” of various metastatic tumors to whole brain radiation therapy (WBXRT) are shown in □ Table 52.6.
The usual dose is 30 Gy in 10 fractions given over 2 weeks. With this dose, 11% of 1-yr survivors and 50% of 2-yr survivors develop severe dementia.

Prophylactic cranial irradiation

Prophylactic cranial irradiation after resection of small-cell lung carcinoma (SCLC) reduces relapses in brain, but does not affect survival.⁴⁷

Post-op radiation therapy

WBXRT is usually recommended following craniotomy for metastatic disease,⁴⁸ especially with SCLC where “micro-metastases” are presumed to be present throughout brain. (Note: some centers do not routinely administer post-op WBXRT (except for very radiosensitive tumors such as SCLC) but instead follow patients with serial imaging studies and administer XRT only when metastases are documented.)
Optimal dose is controversial. Early reports recommended 30–39 Gy over 2–2.5 weeks (3 Gy fractions) with or without surgery.⁴⁹ This is acceptable in patients not expected to live long enough to get long-term radiation effects. Recent recommendations are for smaller daily fractions of 1.8–2.0 Gy to reduce neurotoxicity.⁵⁰ These low doses are also associated with a higher rate of recurrent brain metastases.⁵¹ Since 50 Gy are needed to achieve $>90\%$ control of micrometastases, some use 45–50 Gy WBXRT, plus a boost to the tumor bed to bring the total treatment up to 55 Gy, all with low fractions of 1.80–2.0 Gy.⁵²

Table 52.6 “Radiosensitivity” of brain metastases to WBXRT	
Radiosensitivity	Tumor
Radiosensitive ³³	• small-cell lung Ca • germ-cell tumors • lymphoma • leukemia • multiple myeloma
Moderately sensitive	• breast
Moderately resistant	• colon • non small-cell lung cancer
Highly resistant ^a	• thyroid • renal cell (10% respond) • malignant melanoma • sarcoma • adenocarcinoma
^a SRS may be better than WBXRT for these	

Stereotactic radiosurgery

Inconsistent in its ability to reduce tumor size. Some retrospective studies show results comparable to surgery.⁵³ Others do not.⁵⁴ Does not obtain tissue for histological analysis, and generally cannot be used for lesions > 3 cm. Also, see Stereotactic radiosurgery (p. 811).

52.8.6 Surgical management

Solitary lesions

Indications favoring surgical excision of a solitary lesion:

1. primary disease quiescent
2. lesion accessible
3. lesion is symptomatic or life-threatening
4. primary tumor known to be relatively radioresistant (excision is rarely indicated for untreated brain metastases from SCLC because of its radiosensitivity)
5. for recurrent SCLC following XRT
6. diagnosis unknown: alternatively consider biopsy, e.g. stereotactic biopsy

Surgical resection in patients with progressive systemic disease and/or significant neurologic deficit is probably unjustified.⁵⁵ Also, in newly diagnosed cancer patients, craniotomy may delay systemic treatment for weeks and the ramifications of this need to be considered.

Multiple lesions

Patients with multiple metastases generally have much worse survival than those with solitary lesions.⁵⁰ Multiple metastases are usually treated with XRT without surgery. However, if total excision of all mets is feasible, then even multiple mets may be removed with survival similar to those having a single met removed³² (also see □ Table 52.5 for summary). If only incomplete excision is possible (i.e. cannot remove all mets, or portions of 1 or more must be left behind) then there is no improvement in survival with surgery, and XRT alone is recommended. The mortality of removing > 1 met at a single sitting is not statistically significantly higher than removing a single met.

Situations where surgery may be indicated for multiple mets⁵⁶:

1. one particular and accessible lesion is clearly symptomatic and/or life threatening (life-threatening lesions include p-fossa and large temporal lobe lesions). This is palliative treatment to reduce the symptom/threat from that particular lesion
2. multiple lesions that can all be completely removed (see above)
3. no diagnosis (e.g. no identifiable primary): consider stereotactic biopsy

Stereotactic biopsy

- Considered for:
1. lesions not appropriate for surgery. Includes cases with no definite diagnosis and:
 - a) deep lesions
 - b) multiple small lesions
 2. patients not candidates for surgical resection
 - a) poor medical condition
 - b) poor neurologic condition
 - c) active or widespread systemic disease
 3. to ascertain a diagnosis
 - a) when another diagnosis is possible: e.g. no other sites of metastases, long interval between primary cancer and detection of brain mets...
 - b) especially if nonsurgical treatment modalities are planned (see above)

Intra-operative considerations for surgical removal

Most lesions present themselves on the surface of the brain or through the dura. For lesions not visible on the surface nor palpable immediately beneath the surface, intra-operative ultrasound or stereotactic techniques may be used to localize the lesion.

Metastases usually have a well defined border, thus a plane of separation from normal brain may be exploited, often allowing gross total removal.

52.9 Outcome

52.9.1 General information

□ Table 52.7 lists factors associated with better survival regardless of treatment. Also, the prognosis gets worse as the number of mets increases.⁴⁵ Median survival even with best treatment in some studies is only ≈ 6 months. To put this into perspective, this is worse than with glioblastoma.

52.9.2 Natural history

By the time that neurologic findings develop, median survival among untreated patients is ≈ 1 month.⁵⁷

52.9.3 Steroids

Using steroids alone (to control edema) doubles survival⁵⁸ to 2 mos (Note: this is based largely on pre-CT era data, and the tumors were therefore probably larger than in current studies⁵⁹).

52.9.4 Whole brain radiation therapy (WBXRT)

WBXRT+steroids increases survival to 3–6 mos.³² 50%of deaths are due to progression of intracranial disease.

52.9.5 Surgery ± WBXRT

Recurrence of tumor was significantly less frequent and more delayed with the use of post-op WBXRT.⁴⁸ Length of survival was unchanged with supplemental use of WBXRT. There is also an

Table 52.7 Factors associated with better prognosis for brain mets (with any treatment)
• Karnofsky score^a (KPS)>70 • age<60 yrs • metastases to brain only (no systemic mets) • absent or controlled primary disease • >1 yr since diagnosis of primary • the fewer the number of brain mets • female gender
^a the KPS (p. 1358) is probably the most important predictor; those with a score of 100 had median survival>150 weeks

additional loss of cognitive function in many cases, and patients are rarely independent after WBXRT.

In 33 patients treated with surgical resection of single mets and post-op WBXRT⁶⁰: median survival was 8 months; with 44% 1-yr survival. If no evidence of systemic Ca, 1-yr survival is 81%. If systemic Ca is present (active or inactive), 1-yr survival is 20%. Patients with solitary mets and no evidence of active systemic tumor have the best prognosis.^{46,55} With total removal, no recurrence nor new parenchymal mets occurred within 6 months, and the major cause of death was progression of Ca outside the CNS. A randomized trial verified the improved longevity and quality of survival of patients with solitary mets undergoing surgical excision plus WBXRT vs. WBXRT alone (40 weeks vs. 15 weeks median survival).³³ The surgical mortality was 4% ($\approx$ same as 30-day mortality in the RTX-only group). More patients treated with WBXRT alone die of their brain mets than those who underwent surgery. Following total removal and post-op WBXRT, 22% of patients will have recurrent brain tumor at 1 year.⁵⁰ This is better than surgery without XRT (with reported failure rates of 46%⁵⁰ and 85%⁶¹).

52.9.6 Stereotactic radiosurgery (SRS)

There has not been a randomized study to compare surgery to SRS. Retrospective studies suggest that SRS may be comparable to surgery.^{53,61} However, a prospective (non-randomized, retrospectively matched) study⁵⁴ found a median survival of 7.5 mos with SRS vs. 16.4 mos with surgery, and a higher mortality from cerebral disease in the SRS group (with the mortality due to the SRS treated lesions and not new lesions). A local control rate of $\approx$ 88% has been reported, with one study also recommending WBXRT following the SRS for better regional control.⁶²

Actuarial control rates at 1 year following SRS+WBXRT were 75–80% and appear to be similar to surgery+WBXRT.⁴⁵ However, SRS was unreliable in reducing tumor size.

52.9.7 Multiple mets

Patients with multiple mets that were totally removed have a survival that is similar to those having single mets surgically removed³² (see above).

52.10 Carcinomatous meningitis

52.10.1 General information

Carcinomatous meningitis (CM) AKA (lepto)meningeal carcinomatosis (LMC). Found in up to 8% of patients autopsied with systemic cancer. CM may be the presenting finding in up to 48% of patients with cancer (before the diagnosis cancer is known). Most common primaries: breast, lung, then melanoma.⁶³ (p. 610–2) Always include lymphomatous meningitis in the differential diagnosis; see CNS lymphoma (p. 710).

52.10.2 Clinical

Simultaneous onset of findings in multiple levels of neuraxis. Multiple cranial nerve findings are frequent (in up to 94% most common: VII, III, V & VI), usually progressive. Most frequent symptoms: H/A, mental status changes, lethargy, seizure, ataxia. Non-obstructive hydrocephalus is also common. Painful radiculopathies can occur with “drop mets.”

52.10.3 Diagnosis

Lumbar puncture

Perform only after mass lesion has been ruled out with cranial CT or MRI. Although the initial LP may be normal, CSF is eventually abnormal in >95%

CSF should be sent for:

1. cytology to look for malignant cells (requires $\approx$ 10 ml for adequate evaluation for CM). Repeat if negative (45% positive on first study, 81% eventually positive after up to 6 LPs). May need to pass CSF through a millipore filter
2. bacterial and fungal cultures (including unusual organisms, e.g. cryptococcus)
3. tumor markers: carcinoembryonic antigen, alpha-fetoprotein

4. protein/glucose: elevated protein is the most common abnormality. Glucose may be as low as $\approx$ 40 mg%in about a third of patients

MRI

Contrast enhanced MRI is more sensitive in showing meningeal enhancement.⁶⁴

CT

May show (mild) ventricular dilatation, enhancement of basal cisterns. Sulcal enhancement may also occur with involvement of the convexities.

Myelography

Spinal seeding (“drop mets”) will produce filling defects on myelography.

52.10.4 Survival

Untreated: <2 months. With radiation therapy + chemotherapy: median survival is 5.8 mos (range 1–29). Chemotherapy may be given intrathecally. About half of patients die of CNS involvement, and half die of systemic disease.

References

[1] Johnson JD, Young B. Demographics of brain metastasis. *Neurosurg Clin N Am.* 1996; 7:337–344

[2] Mintz AP, Cairncross JG. Treatment of a Single Brain Metastasis. The Role of Radiation Following Surgical Excision. *JAMA.* 1998; 280:1527–1529

[3] Voorhies RM, Sundaresan N, Thaler HT. The Single Supratentorial Lesion: An Evaluation of Preoperative Diagnosis. *JNeurosurg.* 1980; 53:364–368

[4] Patchell RA, Posner JB. Neurologic Complications of Systemic Cancer. *Neurol Clin.* 1985; 3:729–750

[5] Zimm S, Galen L, Wampler GL, et al. Intracerebral Metastases in Solid-Tumor Patients: Natural History and Results of Treatment. *Cancer.* 1981; 48:384–394

[6] DeAngelis LM. Management of Brain Metastases. *Cancer Invest.* 1994; 12:156–165

[7] Davis PC, Hudgins PA, Peterman SB, Hoffman JC. Diagnosis of Cerebral Metastases: Double-Dose Delayed CT versus Contrast-Enhanced MR Imaging. *AJNR.* 1991; 12:293–300

[8] Weiss HD, Richardson EP. Solitary Brainstem Metastasis. *Neurology.* 1978; 28:562–566

[9] Nugent JL, Bunn PA, Matthews MJ, et al. CNS Metastases in Small-Cell Bronchogenic Carcinoma: Increasing Frequency and Changing Pattern with Lengthening Survival. *Cancer.* 1979; 44:1885–1893

[10] Berger MS, Baumeister B, Geyer JR, Milstein J, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. *J Neurosurg.* 1991; 74:872–877

[11] Kindt GW. The Pattern of Location of Cerebral Metastatic Tumors. *JNeurosurg.* 1964; 21:54–57

[12] Gavrilovic IT, Posner JB. Brain metastases: epidemiology and pathophysiology. *J Neurooncol.* 2005; 75:5–14

[13] Vieth RG, Odom GL. Intracranial Metastases and their Neurosurgical Treatment. *J Neurosurg.* 1965; 23:375–383

[14] Agazzi S, Pampallona S, Pica A, Vernet O, Regli L, Porchet F, Villemure JG, Leyvraz S. The origin of brain metastases in patients with an undiagnosed primary tumour. *Acta Neurochir (Wien).* 2004; 146:153–157

[15] Figlin RA, Piantadosi S, Feld R, et al. Intracranial Recurrence of Carcinoma After Complete Resection of Stage I, II, and III Non-Small-Cell Lung Cancer. *N Engl J Med.* 1988; 318:1300–1305

[16] Auperin A, Arriagada R, Pignon JP, Le Pechoux C, Gregor A, Stephens RJ, Kristjansen PE, Johnson BE, Ueoka H, Wagner H, Aisner J. Prophylactic cranial irradiation for patients with small-cell lung cancer in complete remission. Prophylactic Cranial Irradiation Overview Collaborative Group. *N Engl J Med.* 1999; 341:476–484

[17] Slotman B, Faivre-Finn C, Kramer G, Rankin E, Snee M, Hatton M, Postmus P, Collette L, Musat E, Senan S. Prophylactic cranial irradiation in extensive small-cell lung cancer. *N Engl J Med.* 2007; 357:664–672

[18] Solis OJ, Davis KR, Adair LB, et al. Intracerebral Metastatic Melanoma: CT Evaluation. *Comput Tomogr.* 1977; 1:135–143

[19] Zakrzewski J, Geraghty LN, Rose AE, Christos PJ, Mazumdar M, Polsky D, Shapiro R, Berman R, Darvishian F, Hernando E, Pavlick A, Osman I. Clinical variables and primary tumor characteristics predictive of the development of melanoma brain metastases and post-brain metastases survival. *Cancer.* 2011; 117:1711–1720

[20] Davies MA, Liu P, McIntyre S, Kim KB, Papadopoulos N, Hwu WJ, Hwu P, Bedikian A. Prognostic factors for survival in melanoma patients with brain metastases. *Cancer.* 2011; 117:1687–1696

[21] Staudt M, Lasithiotakis K, Leiter U, Meier F, Eigentler T, Bamberg M, Tatagiba M, Brossart P, Garbe C. Determinants of survival in patients with brain metastases from cutaneous melanoma. *Br J Cancer.* 2010; 102:1213–1218

[22] Sampson JH, Carter JH, Friedman AH, Seigler HF. Demographics, Prognosis, and Therapy in 702 Patients with Brain Metastases from Malignant Melanoma. *JNeurosurg.* 1998; 88:11–20

[23] Swetter SM, Carroll LA, Johnson DL, Segall GM. Positron emission tomography is superior to computed tomography for metastatic detection in melanoma patients. *Ann Surg Oncol.* 2002; 9:646–653

[24] Avril MF, Aamdal S, Grob JJ, Hauschild A, Mohr P, Bonerandi JJ, Weichenthal M, Neuber K, Bieber T, Gilde K, Guillem Porta V, Fra J, Bonnetterre J, Saiag P, Kamanabrou D, Pehamberger H, Suflarsky J, Gonzalez Larriba JL, Scherrer A, Menu Y. Fotemustine compared with dacarbazine in patients with disseminated malignant melanoma: a phase III study. *J Clin Oncol.* 2004; 22:1118–1125

[25] Guirguis LM, Yang JC, White DE, Steinberg SM, Liewehr DJ, Rosenberg SA, Schwartzentruber DJ. Safety and efficacy of high-dose interleukin-2 therapy in patients with brain metastases. *J Immunother.* 2002; 25:82–87

[26] Lohead R, McKhann G, Hankinson T, et al. High dose systemic interleukin-2 for metastatic melanoma in patients with treated brain metastases. *J Immunother.* 2004; 27

- [27] Majer M, Jensen RL, Shrieve DC, Watson GA, Wang M, Leachman SA, Boucher KM, Samlowski WE. Biochemotherapy of metastatic melanoma in patients with or without recently diagnosed brain metastases. *Cancer*. 2007; 110:1329–1337
- [28] ClinicalTrials.gov identifier: NCT 01266967. A Study of GSK 2118436 in BRAF Mutant Metastatic Melanoma to the Brain (Break MB). 2014
- [29] ClinicalTrials.gov identifier: NCT 01378975. A Study of Vemurafenib in Metastatic Melanoma Patients With Brain Metastases. 2015
- [30] U.S. Food and Drug Administration (FDA), . FDA approves Keytruda for advanced melanoma. 2014
- [31] Carlino Matteo S, Fogarty Gerald B, Long Georgina V. Treatment of Melanoma Brain Metastases: A New Paradigm. *The Cancer Journal*. 2012; 18:208–212
- [32] Bindal RK, Sawaya R, Leavens ME, Lee JJ. Surgical Treatment of Multiple Brain Metastases. *J Neurosurg*. 1993; 79:210–216
- [33] Patchell RA, Tibbs PA, Walsh JW, Young B, et al. A Randomized Trial of Surgery in the Treatment of Single Brain Metastases to the Brain. *N Engl J Med*. 1990; 322:494–500
- [34] Vecht CJ, Haaxma-Reiche H, Noordijk EM, Padberg GW, Voormolen JH, Hoekstra FH, Tans JT, Lambooi N, Metsaars JA, Wattendorff AR, et al. Treatment of single brain metastasis: radiotherapy alone or combined with neurosurgery? *Ann Neurol*. 1993; 33:583–590
- [35] Sampson JH, Carter JH, Jr, Friedman AH, Seigler HF. Demographics, prognosis, and therapy in 702 patients with brain metastases from malignant melanoma. *J Neurosurg*. 1998; 88:11–20
- [36] Fife KM, Colman MH, Stevens GN, Firth IC, Moon D, Shannon KF, Harman R, Petersen-Schaefer K, Zacest AC, Besser M, Milton GW, McCarthy WH, Thompson JF. Determinants of outcome in melanoma patients with cerebral metastases. *J Clin Oncol*. 2004; 22:1293–1300
- [37] Eigentler TK, Figl A, Krex D, Mohr P, Mauch C, Rass K, Bostroem A, Heese O, Koelbl O, Garbe C, Schandendorf D. Number of metastases, serum lactate dehydrogenase level, and type of treatment are prognostic factors in patients with brain metastases of malignant melanoma. *Cancer*. 2011; 117:1697–1703
- [38] Gupta G, Robertson AG, MacKie RM. Cerebral metastases of cutaneous melanoma. *Br J Cancer*. 1997; 76:256–259
- [39] Song Z, Lin B, Shao L, Zhang Y. Brain metastases from esophageal cancer: clinical review of 26 cases. *World Neurosurg*. 2014; 81:131–135
- [40] Kondziolka D, Bernstein M, Resch L, et al. Significance of Hemorrhage into Brain Tumors: Clinicopathological Study. *J Neurosurg*. 1987; 67:852–857
- [41] Shildt RA, Kennedy PS, Chen TT, Athens JW, O'Bryan RM, Balcerzak SP. Management of patients with metastatic adenocarcinoma of unknown origin: a Southwest Oncology Group study. *Cancer Treat Rep*. 1983; 67:77–79
- [42] Gaspar L, Scott C, Rotman M, Asbell S, Phillips T, Wasserman T, McKenna WG, Byhardt R. Recursive partitioning analysis (RPA) of prognostic factors in three Radiation Therapy Oncology Group (RTOG) brain metastases trials. *Int J Radiat Oncol Biol Phys*. 1997; 37:745–751
- [43] Morris SL, Low SH, A'Hern RP, Eisen TG, Gore ME, Nutting CM, Harrington KJ. A prognostic index that predicts outcome following palliative whole brain radiotherapy for patients with metastatic malignant melanoma. *Br J Cancer*. 2004; 91:829–833
- [44] Nieder C, Andratschke N, Grosu AL, Molls M. Recursive partitioning analysis (RPA) class does not predict survival in patients with four or more brain metastases. *Strahlenther Onkol*. 2003; 179:16–20
- [45] Pollock BE. Management of Patients with Multiple Brain Metastases. *Contemp Neurosurg*. 1999; 21:1–6
- [46] Horton J. Treatment of Metastases to the Brain. 1984
- [47] Jackson DV, Richards F, Cooper MR, et al. Prophylactic Cranial Irradiation in Small Cell Carcinoma of the Lung: A Randomized Study. *JAMA*. 1977; 237:2730–2733
- [48] Patchell RA, Tibbs PA, Regine WF, Dempsey RJ, Mohiuddin M, Kryscio RJ, Markesbery WR, Foon KA, Young B. Postoperative radiotherapy in the treatment of single metastases to the brain: a randomized trial. *JAMA*. 1998; 280:1485–1489
- [49] Kramer S, Hendrickson F, Zelen M, et al. Therapeutic Trials in the Management of Metastatic Brain Tumors by Different Time/Dose Fraction Schemes. *Natl Cancer Inst Monogr*. 1977; 46:213–221
- [50] DeAngelis LM, Mandell LR, Thaler HT, et al. The Role of Postoperative Radiotherapy After Resection of Single Brain Metastases. *Neurosurgery*. 1989; 24:798–804
- [51] Smalley SR, Schray MF, Laws ER, O'Fallon JR. Adjuvant Radiation Therapy After Surgical Resection of Solitary Brain Metastasis: Association with Pattern of Failure and Survival. *Int J Radiation Oncology Biol Phys*. 1987; 13:1611–1616
- [52] Shaw E. Comment on DeAngelis LM, et al.: The Role of Postoperative Radiotherapy After Resection of Single Brain Metastases. *Neurosurgery*. 1989; 24:804–805
- [53] Sills AK. Current treatment approaches to surgery for brain metastases. *Neurosurgery*. 2005; 57:S24–32; discussion S1–4
- [54] Bindal AK, Bindal RK, Hess KR, Shiu A, Hassenbusch SJ, Shi WM, Sawaya R. Surgery versus Radiosurgery in the Treatment of Brain Metastasis. *J Neurosurg*. 1996; 84:748–754
- [55] Smalley SR, Laws ER, O'Fallon JR, Shaw EG, Schray MF. Resection for Solitary Brain Metastasis: Role of Adjuvant Radiation and Prognostic Variables in 229 Patients. *J Neurosurg*. 1992; 77:531–540
- [56] Tobler WD, Sawaya R, Tew JM. Successful Laser-assisted Excision of a Metastatic Midbrain Tumor. *Neurosurgery*. 1986; 18:795–797
- [57] Markesbery WR, Brooks WH, Gupta GD, et al. Treatment for Patients with Cerebral Metastases. *Arch Neurol*. 1978; 35:754–756
- [58] Ruderman NB, Hall TC. Use of Glucocorticoids in the Palliative Treatment of Metastatic Brain Tumors. *Cancer*. 1965; 18:298–306
- [59] Posner JB. Surgery for Metastases to the Brain. *N Engl J Med*. 1990; 322:544–545
- [60] Galicich JH, Sundaresan N, Thaler HT. Surgical Treatment of Single Brain Metastasis: Evaluation of Results by CT Scanning. *J Neurosurg*. 1980; 53:63–67
- [61] Alexander E, Moriarty TM, Davis RB, et al. Stereotactic Radiosurgery for the Definitive Noninvasive Treatment of Brain Metastases. *J Natl Cancer Inst*. 1995; 87:34–40
- [62] Fuller BG, Kaplan ID, Adler J, Cox RS, Bagshaw MA. Stereotactic Radiosurgery for Brain Metastases: The Importance of Adjuvant Whole Brain Irradiation. *Int J Radiation Oncology Biol Phys*. 1992; 23:413–418
- [63] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
- [64] Sze G, Soletsky S, Bronen R, Krol G. MR Imaging of the Cranial Meninges with Emphasis on Contrast Enhancement and Meningeal Carcinomatosis. *AJNR*. 1989; 10:965–975

53 Spinal Epidural Metastases

53.1 General information

Key concepts

- suspected in a cancer patient with back pain that persists in recumbency
- occurs in $\approx$ 10% of all cancer patients
- 80% of primary sites: lung, breast, GI, prostate, melanoma and lymphoma
- many treatments reduce pain. Surgery + XRT in selected cases increases chances of preserving ambulation & produces a modest improvement in survival
- if no neurologic compromise or bony instability, usual treatment: biopsy (CT- or fluoro-guided) followed by XRT (surgical indications $\square$ Table 53.4)
- surgery not helpful for: total paralysis > 8 hrs, loss of ambulation > 24 hrs, and not recommended for prognosis < 3–4 months survival, poor medical condition (poor PFTs...), or radiosensitive tumor

Spinal epidural metastases (SEM) occur in up to 10% of cancer patients at some time,¹ and are the most common spinal tumor. 5–10% of malignancies present initially with cord compression.² For other etiologies of spinal cord compression, see items marked with a dagger (†) under Myelopathy (p. 1407).

- Routes of metastasis to spine:
1. arterial
 2. venous: via spinal epidural veins (Batson’s plexus³)
 3. perinervous (direct spread)

The usual route of spread is hematogenous dissemination to the vertebral body with erosion back through pedicles and subsequent extension into the epidural space (i.e. anterior epicenter). Less commonly may initially metastasize to lateral or posterior aspect of canal. Most metastases (mets) are epidural, only 2–4% are intradural, and only 1–2% are intramedullary. Distribution between cervical, thoracic and lumbar spine is proportional to the length of the segment, thus the thoracic spine is the most common site (50–60%).

53

53.2 Primary tumors that metastasize to the spine

$\square$ Table 53.1 shows primary tumor types that give rise to SEM. The majority are common primaries that tend to metastasize to bone (lung, breast, prostate, renal-cell and thyroid). Rare tumors that may go to bone include the myxoid subtype of liposarcoma⁴ (17% of these patients develop bone mets, 5-year median survival is 16%).

53.3 Presentation

- Pain: the most common initial symptom. Occurs in up to 95% of patients with SEM.^{6,7} Types of pain:
1. local pain: typically aching, experienced at the level of involvement. Increased pain with recumbency (especially at night) is characteristic
 2. radicular: tends to be sharp or shooting, referred into dermatome of the involved nerve root. Commonly bilateral in thoracic region
 3. mechanical: usually exacerbated by movement

Neck-flexion, straight-leg-raising, coughing, sneezing, or straining may also aggravate the pain.

Motor or autonomic dysfunction: the second most common presentation. Up to 85% of patients have weakness at the time of diagnosis. Leg stiffness may be an early symptom. Bladder dysfunction (urinary urgency, hesitancy or retention) is the most common autonomic manifestation; others include constipation or impotence.

Sensory dysfunction: anesthesia, hypesthesia, or paresthesias usually occur with motor dysfunction. Cervical or thoracic cord involvement may produce a sensory level.

Table 53.1 Sources of spinal epidural metastases causing cord compression			
Site of primary	Series A	Series B ^a	Series C ^b
lung	17%	14%	31%
breast	16%	21%	24%
prostate	11%	19%	8%
kidney (renal-cell)	9%		1%
unknown site	9%	5%	2%
sarcoma	8%		2%
lymphoma	6%	12%	6%
GI tract	6%		9%
thyroid	6%		
melanoma	2%		4%
others (including multiple myeloma)	13%	29% ^c	13%
^a series B: retrospective study of 58 patients undergoing MRI evaluation for SEM ¹ ^b series C: 75 patients with SEM out of 140 patients evaluated prospectively for back pain ⁵ ^c in series B, “other” includes GI, GU, skin, ENT, CNS			

Other presentations: pathologic fracture. Bone metastases can sometimes produce hypercalcemia (a medical emergency).

The greater the neurologic deficit when treatment is initiated, the worse the chances for recovery of lost function. 76%of patients have weakness by the time of diagnosis.¹ 15%are paraplegic on initial presentation, and <5%of these can ambulate after treatment. Median time from onset of symptoms to diagnosis is 2 months.⁸

53.3.1 Metastases to the upper cervical spine

53

For differential diagnosis, see Foramen magnum lesions (p.1367), and Axis (C2) vertebra lesions (p.1391).

Metastases to the C1–2 region comprise only ≈ 0.5%of spinal mets.⁹ They typically present initially with suboccipital and posterior cervical pain, and as the lesion progresses patients develop a characteristic pain that makes it difficult to sit up (some will hold their heads in their hands to stabilize it). Possibly as a result of the capacious spinal canal at this level, only ≈11–15% of patients present with neurologic symptoms. 15%develop spinal cord compression,¹⁰ and quadriplegia from atlantoaxial subluxation occurred in ≈ 6%¹⁰

Anterior approaches for stabilization at this location are difficult. Pathologic fractures due to osteoblastic types of tumors (e.g. prostate, some breast) may heal with radiation treatment and immobilization. For others, good pain relief and stabilization may be achieved with radiation followed by posterior fusion.¹⁰

53.4 Evaluation and management of epidural spinal metastases

53.4.1 General information

There is no difference in outcome between lesions above or below the conus; thus spinal cord, conus medullaris, or cauda equina mets are considered together here as epidural spinal cord compression (ESCC). Features that help distinguish conus lesions from cauda equina are shown in □ Table 53.2

53.4.2 Grading function

There is prognostic significance in the presenting neurologic condition. Grading scales such as that of Brice and McKissock (□ Table 53.3) have been proposed, but are not widely used. The ASIA grading scale is more commonly applied.

Table 53.2 Features distinguishing conus lesions from cauda equina lesions with metastases ¹¹		
	Conus medullaris lesions	Cauda equina lesions
spontaneous pain	rare; when present, is usually bilateral & symmetric in perineum or thighs	may be most prominent symptom; severe; radicular type; in perineum, thighs & legs, back or bladder
sensory deficit	saddle; bilateral; usually symmetric; sensory dissociation	saddle; no sensory dissociation; may be unilateral & asymmetric
motor loss	symmetric; not marked; fasciculations may be present	asymmetric; more marked; atrophy may occur; fasciculations rare
autonomic symptoms (including bladder dysfunction, impotency...)	prominent early	late
reflexes	only ankle jerk absent (preserved knee jerk)	ankle jerk & knee jerk may be absent
onset	sudden and bilateral	gradual and unilateral

Table 53.3 Grading spinal cord function with spinal metastases (Brice & McKissock) ¹²		
Group	Grade	Description
1	mild	patient able to walk
2	moderate	able to move legs, but not antigravity
3	severe	slight residual motor and sensory function
4	complete	no motor, sensory, or sphincter function below level of lesion

53.4.3 Diagnostic tests

MRI in evaluating SEM

MRI without and with contrast is the diagnostic test of choice in most situations.

MRI findings in spinal epidural metastases:

1. vertebral mets are slightly hypointense compared to normal bone marrow on T1WI, and are slightly hyperintense on T2WI
2. axial cuts typically show lesion involving the posterior vertebral body with invasion into one or both pedicles
3. when myelopathy or radiculopathy are present, there is usually tumor extension into the spinal canal (may not occur in lesions presenting only with local pain)
4. DWI images may help differentiate osteoporotic compression fracture from pathologic fracture¹³

Plain x-rays

Most spinal mets are osteolytic, but at least 50% of the bone must be eroded before plain x-rays will be abnormal.¹⁴ Not very specific. Possible findings: pedicle erosion (defect in “owl’s eyes” AKA “winking owl sign” on LS or thoracic spine AP view) or widening, pathological compression fracture, vertebral body (VB) scalloping, VB sclerosis, osteoblastic changes (may occur with prostate Ca, Hodgkin’s disease, occasionally with breast Ca, and rarely with multiple myeloma)

Plain CT in evaluating SEM

Very good for bone detail. Often helpful for surgical planning. By itself, has low sensitivity for spinal cord compression by tumor. Sensitivity is increased with intrathecal contrast (CT-myelogram).

CT-myelogram (CT-myelo)

Indicated when MRI cannot be done (contraindications, unavailability...).

Advantages over MRI:

- Can obtain CSF (when performing LP to inject contrast) for cytological study
- Excellent bony detail
- Can be performed in patients with pacemaker/AICD, claustrophobia...

Disadvantages of myelography over MRI¹:

- Invasive
- May require second procedure (C1–2 puncture) if there is a complete block (providers proficient in this technique are becoming fewer)
- Risk of neurologic deterioration from LP in patient with complete block
- Cannot detect lesions that do not cause bony destruction or distortion of the spinal subarachnoid space
- Up to 20% of patients with SEM have at least two sites of cord compression, MRI can evaluate region between two complete blocks, myelography cannot
- Cannot demonstrate paraspinal lesions
- Does not image spinal cord parenchyma

Positron emission tomography (PET) scan

PET scan using [18F]-fluorodeoxyglucose may be used for whole-body work-up for bone mets in patients with known cancer.¹⁵ Sensitivity is high, but spatial resolution and specificity are low, so often must be used with CT and/or MRI.

Metastatic work-up for patients with suspected spine metastases

- CT of chest, abdomen and pelvis: assess tumor burden, staging, prognostication (which factors into decisions regarding surgery). Has superseded CXR to rule out lung lesion (primary or other mets)
- bone scan: looks for other sites of skeletal involvement
- serum prostate specific antigen (PSA) in males
- mammogram in females
- for multiple myeloma (p.715)
- careful physical exam of lymph nodes

53.4.4 Management algorithm

General information

Management is dependent on the degree and rapidity of neurologic involvement.¹¹ Patients may be categorized into one of the three groups that follows which outlines the subsequent steps. In a patient with suspected spine mets, the goals of management are:

- Assessment of neurologic involvement and timeline of neurologic changes.
- Delineate the degree of spinal involvement
- Determine a histologic diagnosis: this affects management
- Preserve or restore neurologic function
- Preserve or restore spinal stability
- Control pain

The tools that are employed in the assessment and stabilization phase are listed under diagnostic tests above. The section that follows discusses the rapidity with which they are implemented.

A metastatic work-up (p.817) is undertaken as time permits (a preliminary work-up, e.g. CXR and physical exam, may be all that can be initially obtained for patients in Group I, whereas more complete work-up can be done in others).

Group I – rapid progression or severe deficit

Signs / symptoms

Signs/symptoms of new or progressive (hours to days) cord compression (e.g. urinary urgency, ascending numbness). These patients have a high risk of rapid deterioration and require immediate evaluation.

Management

- 53
- dexamethasone (DMZ) (Decadron®): reduces pain in 85% may produce transient neurologic improvement. Optimal dose is not known. No difference was found comparing 100 mg IV bolus to 10 mg.¹⁶ Suggestion: 10 mg IV or PO q 6 hrs × 72 hrs, followed by lower dose of 4–6 mg q 6 hrs. Steroids may temporarily mask lymphoma (on imaging and at surgery), however, in this group the benefit of giving steroids usually outweighs this pitfall
 - radiographic evaluation
 - STAT MRI (above)
 - plain x-rays of entire spine: 67–85% will be abnormal (see above)
 - if time permits, plain CT scan through involved levels and at least 2 levels above and below to evaluate bone for surgical planning
 - emergency myelogram: indicated if MRI cannot be done (include possible C1–2 puncture on the consent). Start with a so-called “blockogram” to R/O complete block: instill small volume of contrast, e.g. iohexol (Omnipaque™) (p.219) via LP and run the dye all the way up the spinal column; CSF is usually xanthochromic with complete block, see Froin’s syndrome (p.790)
 - if there is not a complete block: withdraw 10 cc of CSF and send for cytology, protein & glucose. One may then inject more contrast to complete the study
 - if complete block: do not remove CSF (pressure shifts via LP caused neurologic deterioration in ≈ 14% of patients with complete block,¹⁷ whereas there was no deterioration after C1–2 puncture). In some cases, contrast can be “squeezed” past a “complete” block by injecting 5–10 ml of room air through a millipore filter,¹⁸ alternatively, perform a lateral C1–2 puncture (p.1511) and instill water soluble contrast to delineate the superior extent of the lesion
 - with myelography, epidural lesions classically produce hourglass deformity with smooth edges if block is incomplete, or paintbrush effect (feathered edges) if block is complete, unlike the sharp margins (capping or meniscus sign) of intradural extramedullary lesion, or fusiform cord widening of intramedullary tumors
 - bone scan if time permits. Abnormal in ≈ 66% of patients with spine mets
 - treatment based on results of radiographic evaluation
 - if no epidural mass: treat primary tumor (e.g. systemic chemotherapy). Local radiation therapy (XRT) to bony lesion if present. Analgesics for pain
 - if epidural lesion, either surgery or start XRT (usually 30–40 Gy in 10 treatments over 7–10 d with ports extending 2 levels above and below lesion). XRT is usually as effective as laminectomy with fewer complications; for further discussion see Treatment for SEM (p.819). Thus, surgery instead of XRT is considered only for the indications shown in □ Table 53.4
 - urgency of treatment (surgery or XRT) is based on degree of block and rapidity of deterioration:
 - if >80% block or rapid progression of deficit: emergency treatment ASAP (if treating with XRT instead of surgery, continue DMZ next day at 24 mg IV q 6 hrs x 2 days, then taper during XRT over 2 wks)
 - if <80% block: treatment on “routine” basis (for XRT, continue DMZ 4 mg IV q 6 hrs, taper during treatment as tolerated)

Table 53.4 Indications for surgery for spinal metastases
Indications
1. unknown primary and no tissue diagnosis (CT guided needle biopsy is an option for accessible lesions). NB: lesions such as spinal epidural abscess can be mistaken for metastases²¹ 2. spinal instability 3. deficit due to spinal deformity or compression by bone rather than by tumor (e.g. due to compression fracture with collapse and retropulsed bone) 4. radio-resistant tumors (e.g. renal-cell carcinoma, melanoma...) or progression during XRT (usual trial: at least 48 hrs, unless significant or rapid deterioration) 5. recurrence after maximal XRT 6. rapid neurologic deterioration
Relative contraindications
1. very radiosensitive tumors (multiple myeloma, lymphoma...) not previously radiated 2. total paralysis (Brice and McKissock group 4) >8 hours duration, or inability to walk (B&M group >1) for >24 hrs duration (after this, there is essentially no chance of recovery and surgery is not indicated) 3. expected survival: ≤3–4 months 4. multiple lesions at multiple levels 5. patient unable to tolerate surgery: for patients with lung lesions, check PFTs

Group II – mild and stable signs and symptoms

Signs / symptoms
Mild and stable signs/symptoms of cord compression (e.g. isolated Babinski), or either plexopathy or radiculopathy without evidence of cord compression. Admit and evaluate within 24 hrs.

Management

1. for suspected ESCC, manage as in Group I except on less emergent basis. Use low dose dexamethasone (DMZ) unless radiographic evaluation shows >80%block or if suspicion of lymphoma is high and tissue will be obtained relatively soon
2. for radiculopathy alone (radicular pain, weakness or reflex changes in one myotome or sensory changes in one dermatome): if plain x-rays show bony lesion then 70–88%will have ESCC on myelography. If the plain film is normal, only 9–25%will have ESCC. Obtain MRI or myelogram and manage as for suspected ESCC
3. for plexopathy (brachial or lumbosacral): pain is the most common early symptom, distribution not limited to single dermatome, commonly referred to elbow or ankle. May mask coexistent radiculopathy, distinguish by EMG (denervation of paraspinal muscles occurs in radiculopathy) or presence of proximal signs and symptoms (Horner’s syndrome in cervical region, ureteral obstruction in lumbar region). Management:
 - a) MRI is initial diagnostic procedure (CT if MRI unavailable): C4 through T4 for brachial plexopathy, L1 through pelvis for lumbosacral plexopathy
 - b) if CT shows bony lesion or paraspinal mass (with negative CT, plain films and bone scan are rarely helpful; however, if done, and plain x-ray shows malignant appearing bony lesion, or if bone scan shows vertebral abnormality, perform MRI or myelogram within 24 hrs) (give dexamethasone if ESCC suspected or MRI/myelogram delayed). Management as in Group I based on degree of block, XRT ports extended laterally to include any mass shown on CT
 - c) if no bony nor paraspinal lesion on MRI/CT, primary treatment of plexus tumor; analgesics for pain

Group III – pain without neurologic involvement

Signs / symptoms
Back pain without neuro signs/symptoms. Can be evaluated as outpatient over several days (modify based on ability of patient to travel, reliability, etc.).

53

53.4.5 Treatment for SEM

Treatment goals and outcome

No treatment for SEM significantly prolongs life. Treatment goals are palliative: pain control, preservation of spinal stability, and maintenance of sphincter control and ability to ambulate.

The most important factor affecting prognosis, regardless of treatment modality, is ability to walk at the time of initiation of therapy. Loss of sphincter control is a poor prognosticator and is usually irreversible.

The main decision is between surgery + post-op XRT, or XRT alone. As yet, no chemotherapy has been found to be useful for SEM (may help with primary). Surgery alone appears least effective for pain control (36% compared to 67% for surgery + XRT, and 76% for XRT alone).¹⁹ Surgery has the attendant complications of anesthetic risk, post-op pain, wound problems in 11%(further complicated by radiation),¹⁹ and mortality in 5–6%after laminectomy and 10%after anterior approach with stabilization.²⁰ Therefore, surgery appears best reserved for situations described in □ Table 53.4 .

Medical therapy

Chemotherapy is ineffective for SEM.

Bisphosphonates reduce the risk of vertebral compression fractures (VCF) by $\approx 50\%$ but the effect seems to abate after $\approx 2\text{--}3$ years.

Promising agents undergoing trials include: denosumab, a RANK ligand (RANKL) inhibitor (p.1011) that may counteract RANKL which is overexpressed in response to lytic bony metastases.²² The efficacy seems better than the bisphosphonates.

Vertebroplasty/kyphoplasty

Vertebroplasty/kyphoplasty (p.1011) reduces pain associated with pathologic fractures in up to 84%²³ with an associated increase in functional outcome.²⁴ Kyphoplasty appears to offer comparable pain relief to vertebroplasty with lower rates of cement leakage.²⁴

Relative contraindication: spinal cord compression. Unless the diagnosis has already been verified, a biopsy should be taken through one of the pedicles prior to injecting PMMA.

Radiation therapy

Radiosensitive tumors: □ Table 52.6 lists radiosensitivity of metastatic tumors (to brain or spine). Other radiosensitive tumors that metastasize to the spine include: myxoid liposarcoma.²⁵

Treatment²⁶: Dose: range =25–40 Gy. Typical plan: 30 Gy delivered in 3 Gy fractions over 10 days (2 working weeks) to ports extending at least 1 vertebral level above and below the extent of the lesion. Timing: for initial treatment, try to start XRT within 24 hours of diagnosis; for post-op XRT, within about 14 days following surgery.

There is a theoretical risk of radiation induced edema causing or accelerating neurologic deterioration. This has not been borne out by experimental studies with the usual small daily fractions utilized. Deterioration is more likely to be due to tumor progression.²⁷ The spinal cord is usually the dose limiting structure in treating SEM.

Increased doses are being made possible with the application of the added precision of stereotactic radiosurgery techniques to spinal metastases.²⁸

Surgical treatment

See □ Table 53.4 for indications for surgery.

Pre-op embolization by interventional radiologist may facilitate resection with less blood loss for highly vascular tumors such as: renal-cell, thyroid, and hepatocellular. Blood supply is through the intercostal arteries, and care must be taken to avoid embolizing vessels providing significant blood supply to the spinal cord, especially the artery of Adamkiewicz (p.87).

Approaches

Laminectomy alone is poor for spinal metastases when the pathology is anterior to the cord because of poor access to the tumor and the destabilizing effect of laminectomy when metastatic involvement of the vertebral body is significant.^{29,30}

Deterioration in one of the 3 major criteria (pain, continence, ambulation) occurred in 26% of patients treated with laminectomy alone, 20% of laminectomy + XRT, and 17% of XRT alone (roughly comparable). There is a 9% incidence of spinal instability¹⁹ following laminectomy without stabilization.

In a randomized controlled trial by Patchell et al.,³¹ approaches directed at the location of the tumor (e.g. costotransversectomy, transthoracic approach...) with stabilization where necessary, produced better results than simple laminectomy, and surgery + XRT was superior to XRT alone (see □ Table 53.5). This study found a modest increase in survival, but more significant maintenance or regaining of lost ambulation. However, operative mortality with anterior decompression and stabilization was $\approx$ double (10%) that of laminectomy with (5%) or without (6%) stabilization in a literature review.²⁰

Table 53.5 Comparing surgery + XRT to XRT alone ³¹		
Result	XRT	Surgery + XRT
Ambulatory after treatment	57%	84%
Days ambulatory after treatment	13	122
Ambulatory after treatment when nonambulatory before treatment	19%	62%
Mean survival (days)	100	126

Solitary spinal metastases with indolent tumors (e.g. renal cell Ca) may be candidates for attempted cure with en bloc resection (total spondylectomy).^{32,33}

Laminectomy is still appropriate with isolated involvement of the posterior elements. For anterior pathology, if the posterior elements are intact, a transthoracic approach with corpectomy and stabilization (e.g. with methylmethacrylate and Steinmann pins,³⁴ or with cage graft and lateral plate) followed by XRT improves neurologic function in $\approx 75\%$ and pain in $\approx 85\%$. A posterolateral approach (e.g. costotransversectomy) may be used for anterolateral tumor.³⁵ Combining a corpectomy and removal of the pedicle and posterior elements destabilizes the spine, therefore posterior instrumentation prior to performing the corpectomy is required, followed by cage graft.^{36,37,38,39,40,41,42} To access a VB via a costotransversectomy, the rib of the like numbered VB and the one below need to be removed.

References

- [1] Godersky JC, Smoker WRK, Knutzon R. Use of MRI in the Evaluation of Metastatic Spinal Disease. *Neurosurgery*. 1987; 21:676–680
- [2] Livingston KE, Perrin RG. The neurosurgical management of spinal metastases causing cord and cauda equina compression. *J Neurosurg*. 1978; 49:839–843
- [3] Batson OV. The Function of the Vertebral Veins and Their Role in the Spread of Metastases. *Ann Surg*. 1940; 112
- [4] Schwab JH, Boland P, Guo T, Brennan MF, Singer S, Healey JH, Antonescu CR. Skeletal metastases in myxoid liposarcoma: an unusual pattern of distant spread. *Ann Surg Oncol*. 2007; 14:1507–1514
- [5] Rodichok LD, Ruckdeschel JC, Harper GR, et al. Early Detection and Treatment of Spinal Epidural Metastases: The Role of Myelography. *Ann Neurol*. 1986; 20:696–702
- [6] Bach F, Larsen BH, Rhode K, et al. Metastatic spinal cord compression. Occurrence, symptoms, clinical presentations and prognosis in 398 patients with spinal cord compression. *Acta Neurochir (Wien)*. 1990; 107:37–43
- [7] Helwig-Larsen S, Sorensen PS. Symptoms and signs in metastatic spinal cord compression: a study from first symptom until diagnosis in 153 patients. *Eur J Cancer*. 1994; 30A:396–398
- [8] Levack P, Graham J, Collie D, et al. Don't wait for a sensory level: listen to the symptoms: a prospective audit of the delays in diagnosis of malignant cord compression. *Clin Oncol (R Coll Radiol)*. 2002; 14:472–480
- [9] Sherk HH. Lesions of the Atlas and Axis. *Clin Orthop*. 1975; 109:33–41
- [10] Nakamura M, Toyama Y, Suzuki N, Fujimura Y. Metastases to the upper cervical spine. *J Spinal Disord*. 1996; 9:195–201
- [11] Portenoy RK, Lipton RB, Foley KM. Back Pain in the Cancer Patient: An Algorithm for Evaluation and Management. *Neurology*. 1987; 37:134–138
- [12] Brice J, McKissock W. Surgical Treatment of Malignant Extradural Spinal Tumors. *Br Med J*. 1965; 1:1341–1344
- [13] Li KC, Poon PY. Sensitivity and specificity of MRI in detecting spinal cord compression and in distinguishing malignant from benign compression fractures of vertebrae. *Magn Reson Imaging*. 1988; 6:547–556
- [14] Gabriel K, Schi D. Metastatic spinal cord compression by solid tumors. *Semin Neurol*. 2004; 24:375–383
- [15] Francken AB, Hong AM, Fulham MJ, et al. Detection of unsuspected spinal cord compression in melanoma patients by 18F-fluorodeoxyglucose-positron emission tomography. *Eur J Surg Oncol*. 2005; 31:197–204
- [16] Vecht CJ, Haaxma-Reiche H, van Putten WL, et al. Initial bolus of conventional versus high-dose dexamethasone in metastatic spinal cord compression. *Neurology*. 1989; 39:1255–1257
- [17] Hollis PH, Malis LI, Zappulla RA. Neurological Deterioration After Lumbar Puncture Below Complete Spinal Subarachnoid Block. *J Neurosurg*. 1986; 64:253–256
- [18] Lee Y-Y, Glass JP, Wallace S. Myelography in Cancer Patients: Modified Technique. *AJR*. 1985; 145:791–795
- [19] Findlay GFG. Adverse Effects of the Management of Malignant Spinal Cord Compression. *J Neurol Neurosurg Psychiatry*. 1984; 47:761–768
- [20] Witham TF, Khavkin YA, Gallia GL, et al. Surgery insight: current management of epidural spinal cord compression from metastatic spine disease. *Nat Clin Pract Neurol*. 2006; 2:87–94
- [21] Danner RL, Hartman BJ. Update of Spinal Epidural Abscess: 35 Cases and Review of the Literature. *Rev Infect Dis*. 1987; 9:265–274
- [22] Mundy GR. Metastasis to bone: causes, consequences and therapeutic opportunities. *Nat Rev Cancer*. 2002; 2:584–593
- [23] Fourney DR, Schomer DF, Nader R, Chlan-Fourney J, Suki D, Ahrar K, Rhines LD, Gokaslan ZL. Percutaneous vertebroplasty and kyphoplasty for painful vertebral body fractures in cancer patients. *J Neurosurg*. 2003; 98:21–30
- [24] Bouza C, Lopez-Cuadrado T, Cediell P, Saz-Parkinson Z, Amate JM. Balloon kyphoplasty in malignant spinal fractures: a systematic review and meta-analysis. *BMC Palliat Care*. 2009; 8. DOI: 10.1186/1472-684X-8-12
- [25] Reitan JB, Kaalhus O. Radiotherapy of liposarcomas. *Br J Radiol*. 1980; 53:969–975
- [26] Faul CM, Flickinger JC. The use of radiation in the management of spinal metastases. *J Neurooncol*. 1995; 23:149–161
- [27] Rubin P. Extradural Spinal Cord Compression by Tumor: Part I. Experimental Production and Treatment Trials. *Radiology*. 1969; 93:1243–1248
- [28] Rock JP, Ryu S, Yin FF, Schreiber F, Abdulhak M. The evolving role of stereotactic radiosurgery and stereotactic radiation therapy for patients with spine tumors. *J Neurooncol*. 2004; 69:319–334
- [29] Onimus M, Schraub S, Bertin D, et al. Surgical Treatment of Vertebral Metastasis. *Spine*. 1986; 11:883–891
- [30] Cooper PR, Errico TJ, Martin R, Crawford B, DiBartolo T. A Systematic Approach to Spinal Reconstruction After Anterior Decompression for Neoplastic Disease of the Thoracic and Lumbar Spine. *Neurosurgery*. 1993; 32:1–8
- [31] Patchell RA, Tibbs PA, Regine WF, Payne R, Saris S, Kryscio RJ, Mohiuddin M, Young B. Direct decompressive surgical resection in the treatment of spinal cord compression caused by metastatic cancer: a randomized trial. *Lancet*. 2005; 366:643–648
- [32] Fourney DR, Abi-Said D, Rhines LD, et al. Simultaneous anterior-posterior approach to the thoracic and lumbar spine for the radical resection of tumors followed by reconstruction and stabilization. *J Neurosurg*. 2001; 94:232–244
- [33] Sakaura H, Hosono N, Mukai Y, et al. Outcome of total en bloc spondylectomy for solitary metastasis of the thoracolumbar spine. *J Spinal Disord*. 2004; 17:297–300

- [34] Sundaresan N, Galicich JH, Lane JM, et al. Treatment of Neoplastic Epidural Cord Compression by Vertebral Body Resection and Stabilization. *J Neurosurg.* 1985; 63:676–684
- [35] Overby MC, Rothman AS. Anterolateral Decompression for Metastatic Epidural Spinal Cord Tumors: Results of a Modified Costrotransversectomy Approach. *J Neurosurg.* 1985; 62:344–348
- [36] Shaw B, Mansfield FL, Borges L. One-Stage Posterolateral Decompression and Stabilization for Primary and Metastatic Vertebral Tumors in the Thoracic and Lumbar Spine. *J Neurosurg.* 1989; 70:405–410
- [37] Akeyson EW, McCutcheon IE. Single-stage posterior vertebrectomy and replacement combined with posterior instrumentation for spinal metastasis. *J Neurosurg.* 1996; 85:211–220
- [38] Fourney DR, Abi-Said D, Lang FF, et al. Use of pedicle screw fixation in the management of malignant spinal disease: experience in 100 consecutive cases. *J Neurosurg.* 2001; 94:25–37
- [39] Wang JC, Boland P, Mitra N, et al. Single-stage posterolateral transpedicular approach for resection of epidural metastatic spine tumors involving the vertebral body with circumferential reconstruction: results in 140 patients. *J Neurosurg Spine.* 2004; 1:287–298
- [40] Hunt T, Shen FH, Arlet V. Expandable cage placement via a posterolateral approach in lumbar spine reconstructions: technical note. *J Neurosurg Spine.* 2006; 5:271–274
- [41] Snell BE, Nasr FF, Wolfla CE. Single-stage thoracolumbar vertebrectomy with circumferential reconstruction and arthrodesis: surgical technique and results in 15 patients. *Neurosurgery (Operative Neurosurgery).* 2006; 58:263–269
- [42] Sciubba DM, Gallia GL, McGirt MJ, et al. Thoracic kyphotic deformity reduction with a distractible titanium cage via an entirely posterior approach. *Neurosurgery.* 2007; 60:223–231

XIV

Part XIV Head Trauma

54	General Information, Grading, Initial Management	824
55	Concussion, High-Altitude Cerebral Edema, Cerebrovascular Injuries	841
56	Neuromonitoring	856
57	Skull Fractures	882
58	Traumatic Hemorrhagic Conditions	891
59	Gunshot Wounds and Non-Missile Penetrating Brain Injuries	908
60	Pediatric Head Injury	913
61	Head Injury: Long-Term Management, Complications, Outcome	918

54 General Information, Grading, Initial Management

54.1 General information

54.1.1 Introduction

56–60% of patients with GCS score ≤ 8 have 1 or more other organ system injured.¹ 25% have “surgical” lesions. There is a 4–5% incidence of associated spine fractures with significant head injury (mostly C1 to C3).

When a detailed history is unavailable, remember: the loss of consciousness may have preceded (and possibly have caused) the trauma. Therefore, maintain an index of suspicion for e.g. aneurysmal SAH, hypoglycemia, etc. in the differential diagnosis of the causes of trauma and associated coma.

Brain injury from trauma results from two distinct processes:

1. primary brain injury: occurs at time of trauma (cortical contusions, lacerations, bone fragmentation, diffuse axonal injury, and brainstem contusion)
2. secondary injury: develops subsequent to the initial injury. Includes injuries from intracranial hematomas, edema, hypoxemia, ischemia (primarily due to elevated intracranial pressure (ICP) and/or shock), vasospasm

Since impact damage cannot be influenced by the treating neurosurgeon, intense interest has focused on reducing secondary injuries, which requires good general medical care and an understanding of intracranial pressure (p.856).

54.1.2 Delayed deterioration

≈ 15% of patients who do not initially exhibit signs of significant brain injury may deteriorate in a delayed fashion, sometimes referred to as patients who “talk and deteriorate” or when more lethal, patient who “talk and die”.² Etiologies:

1. ≈ 75% will exhibit an intracranial hematoma
 - a) may be present on initial evaluation and can then worsen
 - b) may develop in a delayed fashion
 - delayed epidural hematoma (EDH) (p.894)
 - delayed subdural (SDH) (p.898)
 - delayed traumatic contusions (p.892)
2. posttraumatic diffuse cerebral edema (p.848)
3. hydrocephalus
4. tension pneumocephalus
5. seizures
6. metabolic abnormalities, includes:
 - a) hyponatremia
 - b) hypoxia: etiologies include pneumothorax, MI, CHF...
 - c) hepatic encephalopathy
 - d) hypoglycemia: including insulin reaction
 - e) adrenal insufficiency
 - f) drug or alcohol withdrawal
7. vascular events
 - a) dural sinus thrombosis (p.1308)
 - b) carotid (or rarely, vertebral) artery dissection (p.1324)
 - c) SAH: due to rupture of aneurysm (spontaneous or posttraumatic) or carotid-cavernous fistula (CCF) (p.1256)
 - d) cerebral embolism: including fat embolism syndrome (p.835)
8. meningitis
9. hypotension (shock)

54.2 Grading

Despite many (valid) criticisms, the initial post-resuscitation Glasgow Coma Scale (GCS) score (□ Table 18.1) remains the most widely used and perhaps best replicated scale employed in for the assessment of head trauma. Problems with this type of scale is that it is an ordinal scale that is non parametric (i.e. does not represent precise measurements of discrete quantities), it is non-linear, and

it is not an interval scale, so that for example, a decrease of 2 points in one parameter is not necessarily equal to a decrease in 2 points of another.³ Thus, performing mathematical manipulations (e.g. adding components, or calculating mean values), while often done, is not statistically sound.⁴

There are a number schemes to stratify the severity of head injury. Any such categorization is arbitrary and will be imperfect. A simple system based only on GCS score is as follows:

- GCS 14–15 =mild
- GCS 9–13 =moderate
- GCS ≤8 =severe

An example of a more involved system³ incorporates other factors in addition to the GCS score as shown in □ Fig. 54.1.

54.3 Transfer of trauma patients

It is sometimes necessary for a neurosurgeon to accept a trauma patient in transfer from another institution that is not equipped to handle major neurologic injuries, or to transfer patients to other facilities for a variety of reasons. □ Table 54.1 lists factors that should be assessed and stabilized (if possible) prior to transfer. These items should also be evaluated in trauma patients on whom a neurosurgeon is consulted in his or her own E/R as well as in patients with other CNS abnormalities besides trauma (e.g. SAH).

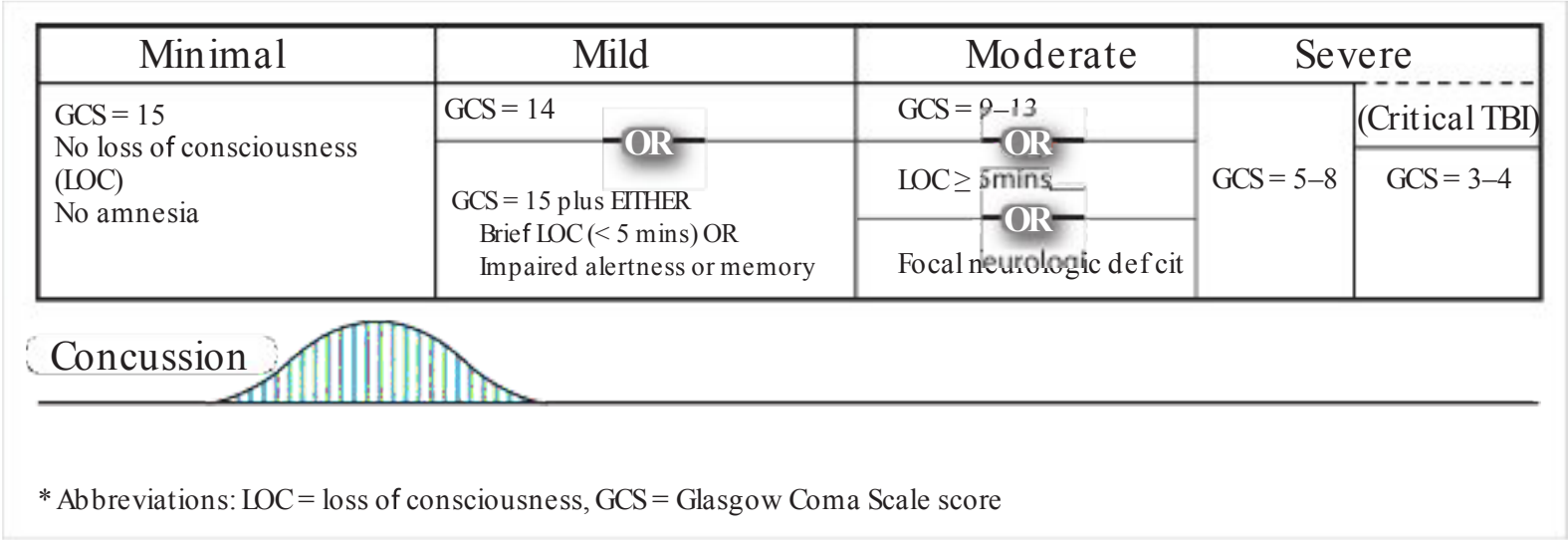


Fig. 54.1 Categorization of head injury severity

Table 54.1 Factors to assess in head injured patients		
Clinical concern	Items to check	Steps to remedy
hypoxia or hypoventilation	ABG, respiratory rate	intubate any patient who has hypercarbia, hypoxemia, or is not localizing
hypotension or hypertension	BP, Hgb/Hct	transfuse patients with significant loss of blood volume
anemia	Hgb/Hct	transfuse patients with significant anemia
seizures	electrolytes, AED levels	correct hyponatremia or hypoglycemia; administer AEDs when appropriate ^a
infection or hyperthermia	WBC, temperature	IP if meningitis is possible and no contraindications (p. 1504)
spinal stability	spine x-rays	spine immobilization (spine board, cervical collar & sandbags...); patients with locked facets should be reduced if possible before transfer
^a see Seizures (p.440), as well as Posttraumatic seizures (p. 462)		

54.4 Management in E/R

54.4.1 General measures

Blood pressure and oxygenation

Practice guideline: IBP and oxygenation

Level II⁵: monitor BP and avoid hypotension (SBP <90 mm Hg)
Level III⁵: monitor oxygenation and avoid hypoxia (PaO₂ <60 mm Hg or O₂ saturation <90%)

Hypotension

Hypotension (shock) is rarely attributable to head injury except:

- in terminal stages (i.e. with dysfunction of medulla and cardiovascular collapse)
- in infancy, where enough blood can be lost intracranially or into the subgaleal space to cause shock
- where enough blood has been lost from scalp wounds to cause hypovolemia (exsanguination)

Hypotension (defined as a single SBP <90 mm Hg) doubles mortality, hypoxia (apnea or cyanosis in the field, or PaO₂ <60 mm Hg on ABG) also increases mortality,⁶ and the combination of both triples mortality and increases the risk bad outcome. SBP <90 mm Hg may impair CBF and exacerbate brain injury and should be avoided (p.870).

Early use of paralytics and sedation (prior to ICP monitoring)

Practice guideline: Early sedation and paralysis

Level III⁷: sedation and neuromuscular blockade (NMB) can be helpful for transporting the head-injured patient, but they interfere with the neuro exam
Level III⁷: NMB should be used when sedation alone is inadequate

54

The routine use of sedatives and paralytics in neurotrauma patients may lead to a higher incidence of pneumonia, longer ICU stays, and possibly sepsis.⁸ These agents also impair neurologic assessment.^{7,9} Use should therefore be reserved for cases with clinical evidence of intracranial hypertension (see □ Table 54.2), for intubation, or where use is necessary for transport or to permit evaluation of the patient (e.g. to get a combative patient to hold still for a CTscan).¹⁰

Intubation and hyperventilation

Indications for intubation in trauma; also see Practice guideline: Intubation – indications (p.827):

1. depressed level of consciousness (patient cannot protect airway): usually GCS ≤ 7
2. need for hyperventilation (HPV): see below
3. severe maxillofacial trauma: patency of airway tenuous or concern for inability to maintain patency with further tissue swelling and/or bleeding
4. need for pharmacologic paralysis for evaluation or management

Table 54.2 Clinical signs of IC-HTN ^a
1. pupillary dilatation (unilateral or bilateral) 2. asymmetric pupillary reaction to light 3. decerebrate or decorticate posturing (usually contralateral to blown^b pupil) 4. progressive deterioration of the neurologic exam not attributable to extracranial factors
^aItems 1-3 represent clinical signs of herniation. The most convincing clinical evidence of IC-HTN is the witnessed evolution of 1 or more of these signs. IC-HTN may produce a bulging fontanelle in an infant. ^b “blown pupil”: fixed & dilated pupil

Practice guideline: Intubation – indications

Level III¹: secure the airway (usually by endotracheal intubation) in patients with $GCS \leq 8$ who are unable to maintain their airway or who remain hypoxic despite supplemental O_2

Cautions regarding intubation:

1. if basal skull fracture through cribriform plate is possible, avoid nasotracheal intubation (to avoid intracranial entry of tube). Use orotracheal intubation
2. prevents assessment of patient’s ability to verbalize⁹ e.g. for determining Glasgow Coma Scale score. This ability should be noted (none, unintelligible, inappropriate, confused, or oriented) prior to intubation
3. risk of pneumonia: see Practice guideline: Antibiotics for intubation (p.827) regarding antibiotics

Practice guideline: Antibiotics for intubation

Level II²: periprocedural antibiotics for endotracheal intubation reduce the risk of pneumonia, but do not alter length of stay or mortality

Hyperventilation (HPV)

Practice guideline: Early/prophylactic hyperventilation

Level II¹³: prophylactic hyperventilation ($PaCO_2 \leq 25$ mm Hg) is not recommended
Level III

- hyperventilation (HPV) before ICP monitoring is established should be reserved as a temporizing measure¹³ for patients with signs of transtentorial herniation (see □ Table 54.2) or progressive neurologic deterioration not attributable to extracranial causes⁷
- HPV should be avoided during the first 24 hrs after TBI (when CBF is often dangerously decreased)¹³

1. since HPV may exacerbate cerebral ischemia, HPV should not be used prophylactically (p.872)
2. prior to ICP monitoring, HPV should only be used briefly when CT or clinical signs of IC-HTN are present¹⁰ (see □ Table 54.2 for clinical signs)
 - a) when appropriate indications are met: HPV to $PaCO_2 = 30-35$ mm Hg
 - b) HPV should not be used to the point that $PaCO_2 < 30$ mm Hg (this further reduces CBF but does not necessarily reduce ICP)
3. acute alkalosis increases protein binding of calcium (decreases ionized Ca^{++}). Patients being hyperventilated may develop ionized hypocalcemia with tetany (despite normal total [Ca])

Mannitol in E/R

Practice guideline: Early use of mannitol

Level III^{7,14}: the use of mannitol before ICP monitoring is established should be reserved for patients who are adequately volume-resuscitated with signs of transtentorial herniation (see □ Table 54.2) or progressive neurologic deterioration not attributable to extracranial causes

Indications in E/R, see also more details (p.873):

1. evidence of intracranial hypertension (see □ Table 54.2)
2. evidence of mass effect (focal deficit, e.g. hemiparesis)

- 3. sudden deterioration prior to CT (including pupillary dilatation)
- 4. after CT, if a lesion that is associated with increased ICP is identified
- 5. after CT, if going to O.R.
- 6. to assess “salvageability”: in patient with no evidence of brainstem function, look for return of brainstem reflexes

Contraindications:

- 1. prophylactic administration is not recommended due to its volume-depleting effect. Use only for appropriate indications (see above)
- 2. hypotension or hypovolemia: hypotension can negatively influence outcome.¹⁰ Therefore, when intracranial hypertension (IC-HTN) is present, first utilize sedation and/or paralysis, and CSF drainage. If further measures are needed, fluid resuscitate the patient before administering mannitol. Use hyperventilation in hypovolemic patients until mannitol can be given
- 3. relative contraindication: mannitol may slightly impede normal coagulation
- 4. CHF: before causing diuresis, mannitol transiently increases intravascular volume. Use with caution in CHF, may need to pre-treat with furosemide (Lasix®)

□ bolus with 0.25–1 gm/kg over <20 min (for average adult: ≈ 350 ml of 20% solution). Peak effect occurs in ≈ 20 minutes (p.873) (for follow-up dosing).

Prophylactic antiepileptic drugs (AEDs)

Practice guideline: Prophylactic anticonvulsants after TBI

Level II^{15,16,17}: prophylactic phenytoin, carbamazepine, phenobarbital or valproate¹⁸ do not prevent late PTS

Level II: AEDs¹⁷ (e.g. phenytoin, valproate, or carbamazepine^{15,16,18}) may be used to decrease the incidence of early PTS (within 7 days of TBI) in patients at high risk of seizures after TBI (see □ Table 54.3), however, this does not improve outcome

Routine use of prophylactic antiepileptic drugs (AEDs) in traumatic brain injury (TBI) is ineffective in preventing the late development of posttraumatic seizures (PTS) i.e. epilepsy, and has been shown to not be useful except in certain circumstances.^{15,16}

See details on using (p.463) and discontinuing (p.463) prophylactic AEDs following TBI. □ Table 54.3 reiterates the markers for patients at increased risk of early PTSs.

54

54.4.2 Neurosurgical exam in trauma

General information

It is not possible to outline a physical exam that is universally applicable. Major trauma must be assessed rapidly, often under chaotic circumstances, and must be individualized based on patient’s medical stability, type of injury, degree of combativeness, use of pharmacologic paralytics (p.826), the needs of other caregivers attending to other organ injuries, the need to triage in the event of multiple patients requiring simultaneous attention...

The following describes some features that should be assessed under certain circumstances with the understanding that this must be individualized. This addresses only craniospinal injuries, and assumes that general systemic injuries (internal bleeding, myocardial and/or pulmonary

Table 54.3 Conditions with increased risk of posttraumatic seizures
1. acute subdural, epidural, or intracerebral hematoma SDH, EDH or ICH
2. open-depressed skull fracture with parenchymal injury
3. seizure within the first 24 hrs after injury
4. Glasgow Coma Scale score <10
5. penetrating brain injury
6. history of significant alcohol abuse
7. ± cortical (hemorrhagic) contusion on CT

contusion...) as well as orthopedic injuries (long bone and pelvic fractures...) will be treated by other members of a “trauma team.” Although organized here in outline form, the most efficient order of examination is usually dictated by circumstances unique to each situation.

General physical condition (oriented towards neuro assessment)

1. visual inspection of cranium:
 - a) evidence of basal skull fracture (p.884):
 - raccoon’s eyes: periorbital ecchymoses
 - Battle’s sign: postauricular ecchymoses (around mastoid air sinuses)
 - CSF rhinorrhea/otorrhea (p.387)
 - hemotympanum or laceration of external auditory canal
 - b) check for facial fractures
 - LeFort fractures (p.887): palpate for instability of facial bones, including zygomatic arch
 - orbital rim fracture: palpable step-off
 - c) periorbital edema, proptosis
2. cranio-cervical auscultation
 - a) auscultate over carotid arteries: bruit may be associated with indicate carotid dissection
 - b) auscultate over globe of eye: bruit may indicate traumatic carotid-cavernous fistula CCF; see Carotid-cavernous fistula (p.1256)
3. physical signs of trauma to spine: bruising, deformity
4. evidence of seizure: single, multiple, or continuing (status epilepticus)

Neurologic exam

1. cranial nerve exam
 - a) optic nerve function (p.836)
 - if conscious: serial quantitation of vision in each eye is important.¹⁹ A Rosenbaum near vision card is ideal (see inside back cover), otherwise use any printed material. If patient cannot see this, check if they can count fingers. Failing this, check for hand motion vision and lastly light perception. Children may develop transient cortical blindness lasting 1–2 days, usually after a blow to the back of the head
 - if unconscious: check for afferent pupillary defect (p.562), best demonstrated with swinging flashlight test (p.561). Indicates possible optic nerve injury
 - funduscopic exam: check for papilledema, pre-retinal hemorrhages, retinal detachment, or retinal abnormalities suggestive of anterior optic nerve injury. If a detailed exam is required, pharmacologic dilatation with mydriatics (p.563) may be employed, however, this precludes pupillary exam for a variable period of time, and should be undertaken advisedly
 - b) pupil: size in ambient light; reaction to light (direct & consensual)
 - c) VII: check for peripheral VII palsy (p.884) (facial asymmetry of unilateral upper and lower facial muscles)
 - d) VI: abducens palsy (p.567) following trauma may occur as a result of ↑ ICP or with clival fractures (p.884)
2. level of consciousness/mental status
 - a) Glasgow coma scale for quantitating level of consciousness in poorly responsive patient (see □ Table 18.1)
 - b) check orientation in patient able to communicate
3. motor exam (assesses motor tracts from motor cortex through spinal cord)
 - a) if patient is cooperative: check motor strength in all 4 extremities
 - b) if uncooperative: check for appropriate movement of all 4 extremities to noxious stimulus (differentiate voluntary movement from posturing or stereotypical spinal cord reflex). This also assesses sensation in an unresponsive patient
 - c) if any doubt about integrity of spinal cord: also check “resting” tone of anal sphincter on rectal exam, evaluate voluntary sphincter contraction if patient can cooperate, check anal wink with pinprick, and assess bulbocavernosus reflex (p.943) (see Neurological assessment, for details)
4. sensory exam
 - a) cooperative patient:
 - check pinprick on trunk and in all 4 extremities, touch on major dermatomes (C4, C6, C7, C8, T4, T6, T10, L2, L4, L5, S1, sacrococcygeal)
 - check posterior column function: joint position sense of LEs

- b) uncooperative patient: check for central response to noxious stimulus (e.g. grimace, vocalization..., as opposed to flexion-withdrawal which could be a spinal cord mediated reflex)
- 5. reflexes
 - a) muscle stretch (“deep tendon”) reflexes if patient is not thrashing: e.g. preserved reflex indicates that a flaccid limb is due to CNS injury and not nerve root injury (and vice versa)
 - b) check plantar reflex for upgoing toes (Babinski sign)
 - c) in suspected spinal cord injury: the anal wink and bulbocavernosus reflex are checked on the rectal exam (see above)

54.4.3 Indications for CT and admission criteria for TBI

General information

Numerous stratagems have been devised to determine what studies should be ordered for which patients. Patients with trivial injuries rarely need a CT scan, and those with severe head injuries obviously need one. Most of the effort centers around identifying the patient who appears to have a minor head injury but may be harboring or apt to develop a significant intracranial injury. An optimal protocol has not been developed, and a rigorous prospective application and further refinement of published systems is sadly lacking. In view of this state of affairs, the following is presented as a guideline.

Patients may be stratified into one of three groups based on the likelihood of intracranial injury as outlined in the following sections.^{20,21}

Category 1. Low risk for intracranial injury

Criteria

Possible findings are shown in □ Table 54.4.

In this group, there is an extremely low likelihood of intracranial injury (ICI) (incidence of ICI: ≤ 8.5 in 10,000 cases with 95% confidence level²⁰).

Management recommendations

CT scan is not usually indicated. Plain SXR's are not recommended: 99.6% of SXR's in this group are normal. Linear non-displaced skull fractures in this group require no treatment, although in-hospital observation (at least overnight) may be considered.

Patients in this group who meet Criteria for observation at home shown in □ Table 54.5 may be managed with observation at home with written head-injury discharge instructions, e.g. as illustrated □ Table 54.6.

Category 2. Moderate risk for intracranial injury

Criteria

Possible findings are shown in □ Table 54.7

Management recommendations

1. head CT scan (unenhanced): clinical grounds alone may miss important lesions in this group.²² 8–46% of patients with minor head injury (MHI) have an intracranial lesion (the most frequent finding was hemorrhagic contusion)²³
2. SXR (p.833): not recommended unless CT scan not available. Useless if normal. A SXR is helpful only if positive (a clinically unsuspected depressed skull fracture might be important)
3. observation
 - a) at home, if the patient meets the criteria outlined in □ Table 54.5. Provide caregiver with written head-injury discharge instructions (sometimes called “subdural precautions”), as shown in □ Table 54.6

Table 54.4 Findings with low risk of ICI
• asymptomatic • H/A • dizziness • scalp hematoma, laceration, contusion, or abrasion • no moderate nor high risk criteria (see □ Table 54.7 and □ Table 54.8, no loss of consciousness, etc.) • no history of loss of consciousness